GOVERNMENT OF MADHYA PRADESH

PUBLIC WORKS DEPARTMENT







SCHEDULE OF RATES (SOR)
For BUILDING WORK
(VOLUME-I)

In Force From January, 1st 2024

Issued by :-

Engineer-in-Chief (Building)

MP Public Works Department

BHOPAL



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FOREWORD

Schedule of Rates for Building is issued by the Engineer-in-Chief, (Building) Public Works Department, Building from time to time. Since its inception the Public Work Department has issued Schedule of Rates for Building works at several occasion. Last such exercise was done by the Engineer-in-Chief, (Building) PWD, on 01-01-2024. Since this date, various amendments were required to be issued over the time. A part from this introduction of new item using new technologies and newer materials have necessitated the issuance of this new revised SOR. Due to increased size SOR is being released in two volumes **(Volume I & II).** Volume -I containing chapters 1 to 13 and Volume-II containing chapters 14 to 31.

Due care has been taken to frame this Schedule of Rates as correctly as possible. It is however, possible that some errors might have crept in. In case any error or omission is noticed, the same may be brought to the noticed of this office.

In order to simplify understanding of SOR, item name to each Item has been assigned. It is advised that cost abstract of the estimate may include only item number, item name, unit, rate, quantity and amount. This will enable easy at a glance reading of the estimate. However it always required to include full description of items in detailed estimate & BOQ.

Schedule of Testing Charges for Building Material (Civil) has also been added, which includes rates of testing of material from Departmental/Government Laboratory/National Accreditation Board for Testing and Calibration Laboratories (NAB).

The effort of **Mr. S. L. Suryavanshi, Chief Engineer (Building), Bhopal, Mr. Anand Prakash Rane, Chief Engineer Planning, Mr. Praveen Nigam Executive Engineer (Design), Mr. Ajay Bain Executive Engineer, Mrs. Shireen Khan Executive Engineer, Ms. Shivani Budhauliya (A.E.), Mr. Rajeev Shrivastav, Sub Engineer,** is highly appreciated who have taken this challenge of bringing out this Schedule of Rates (SOR) into effect in view of modern technologies and invention in the fields of Building & related fields as per latest National Building Code & other Standards.

I also extend my gratitude and thanks to **Mr. V. K. Aarakh Chief Engineer (Building) Gwaliar, Mr. R. K. Ahirwar Chief Engineer (Building) Jabalpur, Mr. J.S. Baghel Chief Engineer (Building) Rewa, Mr. G.P. Verma Chief Engineer Indore, Mr. Surendra Rao Gourkhede, Chief Engineer (Building), Mr. Mukesh Manhar Chief Architect, Mr. R. D. Choudhary, Superintendent Engineer (Building).**

Suggestion, if any, towards further improvement to this schedule of Rates for Building works shall be highly appreciated.

This Schedule of Rates (SOR) is available on the departmental website www.mppwd.gov.in and shall be enforced **w.e.f. 01-01-2024**

Place: Bhopal
Date: **01-01-2024**

1/1/24

(Er. G.P. Mehra)
Engineer-in-Chief, (Building)
MP Public Works Department Building
Bhopal

GENERAL NOTES

1. Reference mentioned herein shall be applicable to all sections to the extent the context permits and are intended to supplement the provisions in the particular section. In case of any discrepancy/ deviation, the provisions in the particular section shall take precedence.
2. In absence of any stipulation to the contrary, unit rates for various items of works are for completing works to the requirements of the specifications including full compensation for all operations and inclusive of all labour, materials, royalties, lease rent, wastages, temporary work, plant, equipment over head charges and profit, unless specified otherwise.

3. Interpretations

3.1 Engineer-In-Chief (Building) Public Works Department MP PWD Government of M.P. shall be the sole deciding authority as to the meaning, interpretation and implications for various provisions in this publication. His decision shall be final and binding.

3.2 Wherever any reference is made to any Indian Standard, it shall be taken as reference to the latest edition with all amendments issued thereto.

3.3 In the event of any variation between the CPWD specifications (adopted) and theIndian Standard, the former shall take precedence over the later.

3.4 Precedence of specification in SOR: - In case of any contradiction in the provisions of the CPWD specifications (adopted) and this document the provisions of this document would take precedence.

3.5 Precedence of Rate in SOR: - The rates as given in this schedule for all items are final, binding and conclusive. In case of doubt and printing mistakes if any, the decision of the Engineer-In-Chief (Building) Public Works Department MP PWD, Government of M.P. shall be final and binding.

4. Definitions

The following terms and expressions in the SOR shall have the meaning or implicationhereby assigned to them unless otherwise specified elsewhere.

4.1 Specifications would refer to the Central Public Works Department (CPWD) specification as applicable as on date along with up to date amendments, addendums and errata.

4.2 Engineer-in-Charge: The "Engineer-in-Charge" means the engineer officer of Executive Engineer rank in Building and Road (B&R) Division, who shall supervise and In-charge of the works and who shall sign the contract on behalf of the Governor of M.P.

4.3 IS: The standards, specification and code of practices issued by the Bureau of Indian Standards along with up to date amendments.

4.4 Superintendent Engineer means officer in-charge of the circle.

4.5 Chief Engineer means officer in-charge of Zone.

4.6 Engineer-in-chief (Building) means head of the department, who heads Building wing of MPPWD.

4.7 Engineer-in-Chief means head of department who heads Building and Road wing of M.P. PWD.

4.8 Competent Authority i.e as per PWD Manual/ NIT / Agreement/ time to time instruction issued by government or higher authority.

4.9 Site: The 'site' shall mean the land/or other places on, in, into or through which the work is to be executed under the contract or any adjacent land, path or street through which the work is to be executed under the contract, or any adjacent land, path or street which may be allotted or used for the purpose of carrying out the contract.

4.10 Best: The word 'best' when used shall mean that in the opinion of the Engineer- in-Charge, there is no superior material/ article and workmanship obtainable in the market and trade respectively. As far as possible the standard required shall be specified in preference to the word 'best'.

4.11 Department: "Department" mean Public Works Department of Govt. of M.P.

5. Floor and Levels of Building

5.1 **Floor 1:-** It is the lowest floor above the ground level in the building unless otherwise specified in a particular case. The floors above floor 1 shall be numbered in sequence as floor 2, floor 3 and so on the number shall increase upwards.

5.2 **Floor Level :** - For floor 1, top level of finished floor shall be the floor level and for all other floors above floor 1, top level of the structural slabs shall be the floor level.

5.3 **Plinth level: -**Floor 1 level or 1.2 m above the ground level whichever is lower shall be the plinth level.

6. Floor Levels of Special Structures

For structures like retaining walls, wing walls, chimneys, reservoirs/ tanksand other elevated structures, where elevations/ heights above a defined datum level have not been specified and identification of floors cannot be done as in case of building. Level, at 1.2 m above the ground level shall bethe floor 1 level as well as plinth level.Level at a height of 3.6 m above floor1 level will be reckoned as floor 2 level and level at a height of 3.6 m above the floor 2 level will be floor 3 level and so on.Where the total height above floor 1 level is not a whole number multiple of 3.6 metre, top most floor level shall be the next in sequence to the floor level below even if the difference in height between the two upper most floor levels is less than 3.6 metres.

7. Foundation and Plinth

The work in foundation and plinth shall include:

7.1 For buildings: All works up to 1.2 metre above ground level or up to floor 1(one) level whichever is lower.

7.2 For abutments, piers and well staining: All works upto 1.2 metre above the bed level.

7.3 For retaining wall, wing walls, compound walls, chimneys, over head reservoirs/ tanks and other elevated structures: All works upto 1.2 metre above the ground level:

7.4 For reservoirs/tanks (other than overhead reservoirs /tanks): All works upto 1.2 metre above the ground level.

7.5 For basements: All works upto 1.2 metre above ground level or upto floor 1 (one) level whichever is lower.

7.6 For boundary walls: All works upto plinth beam or upto 1.2 metre above ground level whichever is lower.

Note: Specific provision shall be made in the estimate for such situations where the foundation level is more than 3 (three) metre depth from the plinth for all types of structures mentioned above.

8. Measurements

Mode of measurement shall be as per the provisions of the contract and as specified in B.O.Q.

8.1 The order of dimensions shall be consistent and in the sequence of length,width and height or depth or thickness (LBD).

8.2 Rounding off: Rounding off where required shall be done in accordance with IS:2-1960.

9. **Rates: -**The rates include elements of hire and running charges of all types of plants, machinery and equipment required to complete the work, unless specified otherwise. Due diligence for safety and health of workers shall be observed by thecontractor by observing all necessary protocol as per law; cost of which shall be deemed to have been included in rates of SOR. The rates provided in thisschedule of rates include cost of all materials lead, lift, water and electricity charges performance of all workmanship; providing erecting and removing centering, forms, scaffolding, benching ladders, templates, safety equipment tools and plants and all other applications etc, required for the proper execution of workunless otherwise specified inclusive charges of duty, royalty and insurance etc.as may be applicable but excluding GST, since all taxes such as VAT, entry taxes etc. have been subsumed now in GST, rate shall not include GST which shall be payable to the contractor at prevailing rate on gross amount of contractor's bill. Provisions for covering etc. necessary to protect the work/ structure from inclement weather etc. and damage arising from falling materials or other cause such as rain, fire etc. shall be the contractor's responsibility and cost of all such arrangements shall be deemed included in SOR rates. SOR rates are also inclusiveof curing wherever required including arrangement of water and its lead or lift whatsoever. If there is any source of water like tube well, hand pump, well etc. under Govt. custody and if water from this source is provided free of cost for construction activity by the contractor then water charge will be deducted at the rate of 1% of the amount paid to the contractor from the items involving use of water. SOR rates are also inclusive of testing charges and provision for laboratory at site as well SOR rate are also inclusive of any certification to be obtained from statutory authority before putting the work machineries equipment fitment to use.

10. **Materials**

10.1 **Only Approved Material: -** The contract shall get approval from Engineer-in-Charge for Samples of all materials to be used on the work. The approved samples duly authenticated and sealed shall be kept in the custody of the Engineer-in-Charge till the completion of the work. All materials to be provided by the contractor shall be brand new andas per the samples approved by the Engineer-in-Charge. Contractor shall bring to site for use in construction only from list of approved make issued by the department. Engineer-in-Charge shall be final authority to make final selection from amongst branded makes of item. Items for which list of approved are not declared brand make are not by the department that shall be selected by the CE/Engineer- in-Charge.

10.2 **Temporary Accommodation at site: -**For work, if the amount of contract exceeds Rs. 100.00 lacs, the contractor shall provide a temporary accommodation of at least 20 sqm within boundaries of site where all the approved material shall be kept for display under lock and key of Engineer-in-Charge the after completion of work this accommodation could be further used as maintenance office. If the contract amount is more than Rs.500.00 lacs additional temporary site accommodation of 20 sqm shall also be provided by the contractor for SQC (Services for Quality Control) office use related to work.

10.3 **Vigilance by CCTV : -** For work, if the amount of contract exceeds Rs. 1000.00 lacs contractor shall install a CCTV setup (Adequate number to cover whole site) before start of work, the access to the vision should be given to Engineer-in-Charge along with its supervision team. In case of non compliance, penalty at the rate of 1000 rupees per days will be imposed to contractor.

10.4 **Materials from approved source only:** -Materials obtained by the contractor from the sources approved by the department shall be subjected to the mandatory tests. Where such

materials do not conform to the relevantspecifications, the material shall be rejected not withstanding the fact that the material / make is the approved make by the department.

10.5 **Testing:-** The contractor shall at his own expense, provide a material for tests (routine field test of materials/ cube test /core cutting /slab deflection/ nondestructive test/ etc. for any component of building), at his cost, make all arrangements and shall provide all facilities as the Engineer-in-Charge may require for collecting and preparing the required number of samples for such tests at such time and to such place or places as may be directed by the Engineer in-Charge and bear all charges including testing charges. The Contractor shall not be eligible for any claim or compensation either arising out of any delay in the work or due to any corrective measures required to be taken on account of and as a result of testing of materials The Engineer -in- Charge or his authorized representative shall always have access to the works and to all workshops and places where work component is being prepared or from where materials, manufactured articles or machinery are being obtained for the works and the contractor shall afford every facility and every assistance in obtaining the right to such access.

10.6 **Protection of Material at site: -** Materials stored at site, depending upon the individual characteristics, shall be protected from atmospheric effects due to rain, sun, wind and moisture to avoid deterioration.

10.7 **Assistance for lease/permits: -** It will not be obligatory on the part of Engineer-In-Charge to provide any assistance in obtaining lease/permits forexcavation of minor minerals. The contractor shall not be entitled to any excuse, whatsoever on account of any delay in obtaining permit etc. for minor minerals to be used on the work.

10.8 **Warranty:-** Warranty card shall be given of all the items which covered under warranty to Engineer-in-Charge at the time of execution, payment shall not be made for such items if warranty card/certificate is not given. The warranty card/certificate shall be given to the concern department at the time of handover of the building.

1. Testing

1.1 **SOR Rates include testing charges:-** The rates also include the element oftesting of samples of various materials brought by the contractor for use on the work, as well as other necessary tests for items of work as stipulated in the specifications.

1.2 **Testing Frequency: -**Frequency of such tests to be carried out must not be less than the prescribed frequencies.

1.3 **Field Laboratory: -**The contractor shall have to establish a field laboratoryat the site of work, if the amount of contract exceeds Rs. 100.00 lacs.

1.4 **Testing in Laboratory other than field laboratory: -**In other cases, testing of construction materials should be got done from any of the testing laboratories of the government works departments, and technical institutes, engineering college, etc. In case the contract provides for testing of material through third party like NABL accredited lab or any other lab directed by the Engineer-In-Charge, the cost of testing shall be borne by thecontractor.

1.5 **Payment only after testing: -**The work shall not be accepted in any case, if the contractor fails to observe the instructions of the department, regarding testing of materials. Before making any payment, it will be the responsibility of the officers making payment to ensure that all testing asper prescribed frequencies, have been carried out, and shall certify that the test results are in conformity with the requirement. If tests are not conducted to the prescribed frequency, the Engineer-In-Charge should reject that part of the work. Duly certified (by the representative of Engineer-in-Charge) copies of registers, containing records of tests shall have to be presented along with running account bills. Register (in original) shall have to be

submitted along with the final bill. Tests shall have to be conducted by the contractor's engineer under the supervision of the Engineer-In-Charge or his authorized representatives.

2. **Acceptance at Reduced Rate: -**If any item of work is found not upto the prescribed standard but the Engineer-In-Charge is of the opinion that the same is structurally adequate and can be accepted at a reduced rate, then in such cases, the Engineer-In-Charge shall have to submit proposals for appropriate reduction of rates supported by rate analysis, in justification thereof, through a D.O. letter to the territorial Superintending Engineer or Chief Engineer to obtain his approval expeditiously (ordinarily within 15 days).

3. **The Site Order Book: -** The "Site Order Book" shall be in custody of departmental supervisory staff. In case of a work for which a separate agency for supervision of work has been engaged by the department, the site order book shallbe in possession of the field staff of SQC agency. The Engineer-In-Charge or his authorized representative may record his instructions in this book, which shall be noted by the contractor or his authorized representative for compliance.

4. **Cement at Site: -** A duplicate cement register as per format here under shall be maintained at site of work. Extract certified copies of the entries for each month/each running bill shall be submitted to the Engineer-in-Charge by the Contractor. The original register shall also be submitted to the Engineer-in- Charge on completion of the work by the Contractor.

S. No.	Date of receipt of cement	No. of bags	Name and address of firm, from whom purchased	Signature of contractor or his authorised representative	Signature of authorised representative of Engineer-In- Charge	Result of test for initial and final setting time	Result of tests for compressive strength of cement	Remarks

15. **Only Tested Cement: -**Where contract provides for cement to be arranged by the contractor himself, only I.S. marked cement of relevant I.S. specifications as per IS: 4031 PART- 5, IS 269-2013, IS 1489, IS 8112 . IS 12269 etc. shall be allowed to be used in the work.

NOTE : The arrangement for necessary equipment and testing shall have to be made by the contractor himself at site, to be decided by the Engineer-in-Charge. All testing expenses shall be borne by the contractor. Any lot of cement brought to site by the contractor, would be permitted to be used in the work only after the satisfactory results of the tests, under the supervision of the Engineer-in-Charge or his authorized representative. The record of the tests result shall be maintained in the register referred in subsequent para.

16. **Volumetric use of cement:-**When cement is used for mortar/volumetric mix etc. batch of one bag mix cement or its multiples shall be mixed and the measurement of box for metal, sand to be of 35 liters or multiples of it (convenient box size 40x35x25cm.). For design mix concrete batching shall be done by weight.

17. **Aggregates: -**The aggregates to be used for all cement concrete items shall be crushed in the mechanical crushers. All concrete shall have to be mixed in power driven mixers having hoppers and capacity of mixing concrete mix of at least one bag of cement.

18. **Sand: -**Nothing extra shall be payable on account of use of Narmada sand or sand brought from any other quarry/river. Sand should conform to the requirements of IS: 1542 and IS: 2166 for using it in construction work.

19. **Rubbles from excavation: -**Rubble available from excavation of hard rock shall be the property of the contractor subject to recovery of Rs. 150/- per cum. of the quantity of the rock excavated. This excavated material should be removed from site by contractor at its own cost.

If after the instruction issued by engineer in charge for shifting of material from site contractor has bind to shift this excavated material from site within 15 days otherwise after this penalty will imposed on contractor at the rate of Rs 50 per cum per day.

20. **Ban on Teak wood:** -Teak wood shall not be used in any building work unless specific sanction of the State Government is obtained. Provision of teak wood shall not be made **in** any of the estimate without prior sanction of the state Government. Item such as teak wood beading, veneering, style and rail of flush door shall not be banned.

21. **Approval of Non SOR item** - If any Non S.O.R. Item is to be executed in any proposed project or in ongoing project, the rate analysis with full justification thereof must be approved by the Superintending Engineer of respective circle (B & R) or Chief Engineer, (Building) .

22. **In case** of any difference in item description and rate between BOQ and SOR, the rate and description of BOQ shall be prevail, on the other hand if there is difference in between rates only while description being the same in BOQ and SOR, the rate of SOR with up to date amendment shall be prevail.

23. **Safety In Construction** - The contractor shall employ only such methods of construction, tools and plant as are appropriate for the type of work or as approved by Engineer-in-Charge in writing. The contractor shall take all precautions and measures to ensure safety of works and workman and shall be fully responsible for the same. Safety pertaining to construction works such as excavation, centering and shuttering, trenching, blasting, demolition, electric connections, scaffolds, ladders, working platforms, gangway, mixing of bituminous materials, electric and gas welding, use of hoisting and construction machinery shall be governed by CPWD safety code, relevant safety codes and the direction of Engineer-in-Charge.

The Contractor must ensure safety of workmen as well as safety for the general public during construction in and around work-site. He must follow the laws, codes and standards laid down in this regard. The work-men must be trained and provided protective gear, life-saving equipment like safety shoes , safety net ,safety jacket , safety helmet and appropriate tools for their jobs and also make ensure that all safety equipment are being used properly without any excuse. Special precautions must be used if hazardous chemicals are used or stored at workplace (lead, silica, asbestos and wood/stone that will be cut and generate dust, construction materials containing zinc, cadmium, beryllium and mercury).Besides protection from noise and environmental pollution, public must also be safeguarded from falling through dug-up area, electrocution, flooding, falling objects, bridge-span dropping/ failures, crane falling/ overturning and damage to building from vibrations/ cave-ins from construction activities. Engineer-in- charge must ensure that contractor does not adopt any short-cut in this regard. Contractor is only responsible to provide sufficient required barricading material and properly barricade the project site to avoid any kind of external hindrance at project site. Type of barricading will choose by project type and location by Engineer-in-charge and contractor has to provide same up to the satisfaction level of engineer-in-charge.

24. **Black cotton (BC) soil**- If BC soil having swelling index more than 50 % , that should not be used for refilling under plinth, and under floor filling same shall be replaced by good earth.

25. **Mix design approval:** Proposal of mix design has to be submitted by contractor at its own cost before execution well with in time to start of work and the same shall be approved by competent authority.

26. **Samples approval-** Sample of all materials to be used on the work shall be got approved by the contractor from the Competent Authority well in time by producing at least four sample. The approved samples duly authenticated and sealed shall be kept in the custody of the Engineer-in-Charge till the completion of the work. All materials to be provided by the contractor shall be brand new and as per the samples approved by the Competent Authority.

27. The Competent Authority / Engineer-in-Charge shall have full powers to require the removal from the premises of all materials which in his opinion are not in accordance with the specifications and in case of default, the Engineer-in-Charge shall be at liberty to employ at the expense of the contractor, other persons to remove the same without being answerable or accountable for any loss or damage that may happen or arise to such materials.

11/1/24

(Er. G.P. Mehra)
Engineer-in-Chief (Building)
MP Public Works Department (Building)
Bhopal

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CHAPTER – 1.0

CARRIAGE OF MATERIALS

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 4082	Stacking and storage of construction materials and components at site – Recommendations
2	IS 1141	Seasoning of Timber – Code of Practice

S.No.	Sub Item No.	Item Name	Materials	Unit	Rate (Rs.)								Remarks
					1 km	2 km	3 km	4 km	5 km	Beyond 5 km up to 10 km per km	Beyond 10 km up to 20 km per km	Beyond 20km per addl. Km.	
1			2	3	4	5	6	7	8	9	10	11	12
1.1		Transportation by mechanical means	Transportation By Mechanical Transport including loading, unloading and stacking.										
	1.1.1	Moorum,	Lime, moorum, building rubbish	cum	111.50	125.60	139.46	152.75	165.63	11.38	10.30	7.27	The rates will be applicable to net quantities after deduction of prescribed percentage for voids mentioned in the specification under subhead "Carriage of materials".
	1.1.2	Earth	Earth	cum	139.47	157.00	174.33	190.94	207.04	14.23	12.88	9.08	
	1.1.3	Manure	Manure of sludge	cum	121.20	136.52	151.59	166.03	180.03	12.38	11.20	7.90	
	1.1.4	Rock	Excavated rock	cum	223.16	251.21	270.93	305.50	331.27	22.77	20.61	14.54	
	1.1.5	Aggregate <40 mm	Sand stone aggregate below 40 mm nominal size	cum	111.50	125.60	139.46	152.75	165.63	11.38	10.30	7.27	
	1.1.6	Aggregate >40 mm	Stone aggregate 40 mm nominal size and above	cum	121.20	136.52	151.59	166.03	180.03	12.38	11.20	7.90	
	1.1.7	Stone	Stone soling	cum	101.43	114.10	126.79	138.86	150.57	10.35	9.37	6.60	
	1.1.8	Bricks	Bricks	1000 Nos.	297.55	334.95	371.92	407.34	441.69	30.36	27.48	19.38	
	1.1.9	Pipes	Cement, stone blocks, G.I...C.I., A.C., & C.C. pipes below 100 mm dia and other heavy materials	tonne	99.18	111.65	123.97	135.78	147.23	10.12	9.16	6.46	
	1.1.10	Steel	Steel	tonne	99.18	111.65	123.97	135.78	147.23	10.12	9.16	6.46	
	1.1.11	Timber	timber	cum	127.52	143.55	159.39	174.57	189.30	13.01	11.78	8.31	

Item No.	Sub Item No.	Item Name	Descriptions	Unit	Rate (In Rs.)
1	2	3	4	5	6
1.2		Transportation by manual means	Transportation by manual labour including loading, unloading and stacking for lead up to **50 metre.**		
	1.2.1	Moorum	Lime, moorum, building rubbish	cum	53.43
	1.2.2	Earth	Earth	cum	66.78
	1.2.3	Manure	Manure or sludge	cum	58.07
	1.2.4	Rock	Excavated rock	cum	106.85
	1.2.5	Aggregate <40 mm	Sand, stone aggregate below 40mm nominal size	cum	66.78
	1.2.6	Aggregate >40 mm	Stone aggregate 40mm nominal size and above	cum	72.20
	1.2.7	Stone	Stone soling	cum	78.57
	1.2.8	Bricks	Bricks	1000 Nos.	124.66
	1.2.9	S.W. Pipes	S.W. Pipes		
	1.2.9.1	100 mm dia	100 mm dia	100 m	96.94
	1.2.9.2	150 mm dia	150 mm dia	100 m	164.95
	1.2.9.3	200 mm dia	200 mm dia	100 m	230.57
	1.2.9.4	250 mm dia	250 mm dia	100 m	384.35
	1.2.9.5	300 mm dia	300 mm dia	100 m	549.07
	1.2.9.6	350 mm dia	350 mm dia	100 m	768.70
	1.2.9.7	400 mm dia	400 mm dia	100 m	960.87
	1.2.9.8	450 mm dia	450 mm dia	100 m	1164.70
	1.2.9.9	500 mm dia	500 mm dia	100 m	1423.52
	1.2.9.10	600 mm dia	600 mm dia	100 m	1747.05
	1.2.10	Other Pipes	R.C.C. pipes, Steel cylinder, R.C. pipes, C.I. pipes, and unreinforced cement pipes		

Item No.	Sub Item No.	Item Name	Descriptions	Unit	Rate (In Rs.)
1	2	3	4	5	6
	1.2.10.1	100 mm dia	100 mm dia	100 m	135.49
	1.2.10.2	125 mm dia	125 mm dia	100m	165.79
	1.2.10.3	150 mm dia	150 mm dia	100 m	190.91
	1.2.10.4	200 mm dia	200 mm dia	100 m	286.47
	1.2.10.5	250 mm dia	250 mm dia	100 m	503.52
	1.2.10.6	300 mm dia	300 mm dia	100 m	630.09
	1.2.10.7	350 mm dia	350 mm dia	100 m	900.82
	1.2.10.8	400 mm dia	400 mm dia	100 m	1048.23
	1.2.10.9	450 and 500 mm dia	450 and 500 mm dia	100 m	1397.64
	1.2.10.10	600, 700, 750 and 800 mm dia	600, 700, 750 and 800 mm dia	100 m	1537.41
1.3		Add extra for additional lead by manual means	Add extra for every additional lead of 50 metre or part thereof.		
	1.3.1	Moorum	Lime, moorum, building rubbish	cum	11.63
	1.3.2	Earth	Earth	cum	14.54
	1.3.3	Manure	Manure or sludge	cum	12.64
	1.3.4	Rock	Excavated rock	cum	23.27
	1.3.5	Aggregate <40 mm	Sand, stone aggregate below 40mm nominal size	cum	14.54
	1.3.6	Aggregate >40 mm	Stone aggregate 40mm nominal size and above	cum	15.72
	1.3.7	Stone	Stone soling	cum	17.11
	1.3.8	Bricks	Bricks	1000 Nos.	27.14
	1.3.9	S.W. Pipes	S.W. Pipes		
	1.3.9.1	100 mm dia	100 mm dia	100 m	14.21
	1.3.9.2	150 mm dia	150 mm dia	100 m	24.19
	1.3.9.3	200 mm dia	200 mm dia	100 m	33.81

Item No.	Sub Item No.	Item Name	Descriptions	Unit	Rate (In Rs.)
1	2	3	4	5	6
	1.3.9.4	250 mm dia	250 mm dia	100 m	56.36
	1.3.9.5	300 mm dia	300 mm dia	100 m	80.51
	1.3.9.6	350 mm dia	350 mm dia	100 m	112.71
	1.3.9.7	400 mm dia	400 mm dia	100 m	140.89
	1.3.9.8	450 mm dia	450 mm dia	100 m	170.78
	1.3.9.9	500 mm dia	500 mm dia	100 m	208.73
	1.3.9.10	600 mm dia	600 mm dia	100 m	256.17
	1.3.10	Other Pipes	R.C.C. pipes, Steel cylinder, R.C. pipes, C.I. pipes, and unreinforced cement pipes		
	1.3.10.1	100 mm dia	100 mm dia	100 m	19.87
	1.3.10.2	125 mm dia	125 mm dia	100m	24.31
	1.3.10.3	150 mm dia	150 mm dia	100 m	27.99
	1.3.10.4	200 mm dia	200 mm dia	100 m	42.00
	1.3.10.5	250 mm dia	250 mm dia	100 m	73.83
	1.3.10.6	300 mm dia	300 mm dia	100 m	92.39
	1.3.10.7	350 mm dia	350 mm dia	100 m	132.09
	1.3.10.8	400 mm dia	400 mm dia	100 m	153.70
	1.3.10.9	450 and 500 mm dia	450 and 500 mm dia	100 m	204.93
	1.3.10.10	600, 700, 750 and 800 mm dia	600, 700, 750 and 800 mm dia	100 m	225.43

CHAPTER – 2.0

EARTHWORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 632	Gamma – BHC (Lindane) emulsifiable concentrates
2	IS 1200 (Part-1)	Method of measurement of earth work
3	IS 1200 (Part-27)	Method of measurement of earth work (by Mechanical Appliances)
4	IS 4081	Safety code for Blasting and related drilling operation
5	IS 4988 (Part-IV)	Excavators
6	IS 6313 (Part-II)	Anti-Termite measures in buildings (pre -constructional)
7	IS 6313(Part-III)	Anti-Termite Measures in Buildings for existing buildings
8	IS 6940	Methods of test for pesticides and their formulations
9	IS 8944	Chlorpyrifos emulsifiable concentrates
10	IS 8963	Chlorpyrifos – Technical specifications
11	IS 12138	Earth moving Equipment

PREAMBLE
EARTHWORK
(CHAPTER: 2.0)

2.1 CLASSIFICATION OF EARTH WORK:

The earthwork shall be classified under the following categories and measured separately for each category:

(A) **All kind of soils:** Generally any strata, such as sand, gravel, loam, clay, mud, black cotton moorum, shingle, river or nallah bed boulders, siding of roads, paths etc. and hard core, macadam surface of any description (water bound, grouted tarmac etc.), lime concrete mud concrete and their mixtures which for excavation yields to application of picks, showels, jumper, sacrifiers, ripper and other manual digging implements.

(B) **Ordinary rock:** Generally any rock which can be excavated by splitting with crow bars or picks and does not require blasting, wedging or similar means for excavation such as lime stone, sand stone, hard laterite, hard conglomerate and un-reinforced cement concrete below ground level. If required light blasting may be resorted to for loosening the materials but this will not in any way entitle the material to be classified as 'Hard rock'.

(C) **Hard rock:** Generally any rock or boulder for the excavation of which blasting is required such as quartzite, granite, basalt, reinforced cement concrete (reinforcement to be cut through but not separated from concrete) below ground level and the like.

(D) **Hard rock (blasting prohibited):** Hard rock requiring blasting as described under (c) but where the blasting is prohibited for any reason and excavation has to be carried out by chiseling, wedging, use of rock hammers and cutters or any other agreed method.

2.2 FOUL CONDITION:

Filthy and unhygienic conditions where physical movements are hampered such as soil mixed with sewage or night soil.

2.3 LEAD:

All distances shall be measured over the shortest practical route and not necessarily the route actually taken. Route other than shortest practical route may be considered in cases of unavoidable circumstances and approved by Engineer-in-charge along with reasons in writing.

2.4 LIFT:

The vertical distance for removal with reference to the ground level. The excavation up to 1.5 metres depth below the ground level and depositing the excavated materials up to 1.5 metres above the ground level are included in the rate of earth work. Lifts inherent in the lead due to ground slope shall not be paid for.

2.5 PROTECTIONS:

Excavation where directed by the Engineer-in-Charge shall be securely barricaded and provided with proper caution signs, conspicuously displayed during the day and properly illuminated with red lights and/or written using fluorescent reflective paint as directed by engineer in charge during the night to avoid accident.

The Contractor shall take adequate protective measures to see that the excavation operations do not damage the adjoining structures or dislocate the services. Water supply pipes, sluice valve chambers, sewerage pipes, manholes, drainage pipes and chambers, communication cables, power supply cables etc. met within the course of excavation shall be properly

supported and adequately protected, so that these services remain functional. However, if any service is damaged during excavation shall be restored in reasonable time, for which no extra payment shall be admissible.

Excavation shall not be carried out below the foundation level of the adjacent buildings until underpinning; shoring etc. is done as per the directions of the Engineer-in-Charge for which payment shall be made separately.

Any damages done by the contractor to any existing work shall be made good by him at his own cost. Existing drains pipes, culverts, overhead wires, water supply lines and similar services encountered during the course of execution shall be protected against damage by the contractor. The contractor shall not store material or otherwise occupy any part of the site in manner likely to hinder the operations of such services.

2.6 EXCAVATION IN ALL KINDS OF SOIL:

All excavation operations manually or by mechanical means shall include excavation and 'getting out' the excavated materials. In case of excavation for trenches, basements, water tanks etc. 'getting out' shall include throwing the excavated materials at a distance of at least one metre or half the depth of excavation, whichever is more, clear off the edge of excavation. In all other cases 'getting out' shall include depositing the excavated materials as specified. The subsequent disposal of the excavated material shall be either stated as aseparate item or included with the items of excavation stating lead.

During the excavation the natural drainage of the area shall be maintained. Excavation shallbe done from top to bottom. Undermining or undercutting shall not be done.

In firm soils, the sides of the trenches shall be kept vertical up to a depth of 2 metres from the bottom. For greater depths, the excavation profiles shall be widened by allowing steps of 50 cm on either side after every 2 metres from the bottom. Alternatively, the excavation can be done so as to give slope of 1:4 (1 horizontal: 4 vertical). Where the soil is soft, loose or slushy, the width of steps shall be suitably increased or sides sloped or the soil shored up as directed by the Engineer-in- Charge. It shall be the responsibility of the contractor to take complete instructions in writing from the Engineer-in-Charge regarding the stepping, sloping or shoring to be done for excavation deeper than 2 metres.

The excavation shall be done true to levels, slope, shape and pattern indicated by the Engineer-in- Charge. Only the excavation shown on the drawings with additional allowances for centering and shuttering or as required by the Engineer-in-Charge shall be measured and recorded for payment.

In case of excavation for foundation in trenches or over areas, the bed of excavation shall be to the correct level or slope and consolidated by watering and ramming. If the excavation for foundation is done to a depth greater than that shown in the drawings or as required by the Engineer-in-Charge, the excess depth shall be made good by the contractor at his own cost with the concrete of the mix used for levelling/ bed concrete for foundations. Soft/defective spots at the bed of the foundations shall be dug out and filled with concrete (to be paid separately) as directed by the Engineer-in-Charge.

While carrying out the excavation for drain work care shall be taken to cut the side and bottom to the required shape, slope and gradient. The surface shall then be properly dressed. If the excavation is done to a depth greater than that shown on the drawing or as required by the Engineer-in-Charge, the excess depth shall be made good by the contractor at his own cost with stiff clay puddle at places where the drains are required to be pitched and with ordinary earth, properly watered and rammed, where the drains are not required to be pitched. In case

the drain is required is to be pitched, the back filling with clay puddle, if required, shall be done simultaneously as the pitching work proceeds. The brick pitched storm water drains should be avoided as far as possible in filled-up areas and loose soils.

In all other cases where the excavation is taken deeper by the contractor, it shall be brought to the required level by the contractor at his own cost by filling in with earth duly watered, consolidated and rammed.

In case the excavation is done wider than that shown on the drawings or as required by the Engineer-in-Charge, additional filling wherever required on the account shall be done by the contractor at his own cost.

The excavation shall be done manually or by mechanical means as directed by Engineer-in-charge considering feasibility, urgency of work, availability of labour /mechanical equipments and other factors involved. Contractor shall ensure every safety measures for the workers. Neither any deduction will be made nor will any extra payment be made on this account.

2.7 EXCAVATION IN ORDINARY/HARD ROCK

All excavation operations shall include excavation and 'getting out' the excavated matter. In case of excavation for trenches, basements, water tanks etc. 'getting out' shall include throwing the excavated materials at a distance of at least one metre or half the depth of excavation, whichever is more, clear off the edge or excavation. In all other cases 'getting out' shall include depositing the excavated materials as specified. The subsequent disposal of the excavated material shall be either stated as a separate item or included with the item of excavation stating lead.

During the excavation, the natural drainage of the area shall be maintained. Excavation shall be done from top to bottom. Undermining or under cutting shall not be done.

Where hard rock is met with and blasting operations are considered necessary, the contractor shall obtain the approval of the Engineer-in-Charge in writing for resorting to the blasting operations. Blasting operations shall be done as specified in subsequent para and chiseling shall be done to obtain correct levels, slopes, shape and pattern of excavation as per the drawings or as required by the Engineer-in-Charge and nothing extra shall be payable for chiseling.

Where blasting operations are prohibited or are not practicable, excavation in hard rock shall be done by chiseling.

In ordinary rock excavation shall be carried out by crowbars, pick axes or pneumatic drills and blasting operation shall not be generally adopted. Where blasting operations are not prohibited and it is practicable to resort to blasting for excavation in ordinary rock, contractor may do so with the permission of the Engineer-in-Charge in writing but nothing extra shall bepaid for this blasting. Blasting shall be done as specified.

If the excavation for foundations or drains is done to a depth greater than that shown in the Drawings or as required by the Engineer-in-Charge. The excess depth shall be made good by the contractor at his own cost with the concrete of the mix used for levelling/ bed concrete for foundations. Soft/ defective spots at the bed of foundations shall be dug out and filled with concrete (to be paid separately) as directed by the Engineer-in-Charge.

2.8 BLASTING

Where hard rock is met with and blasting operations are considered necessary, the contractor shall obtain the approval of the Engineer-in-Charge in writing for resorting to blasting operation.

Note: In ordinary rock blasting operations shall not be generally adopted. However, the contractor may resort to blasting with the permission of the Engineer-in-charge, but nothing extra shall be paid for such blasting operations. The contractor shall obtain license from the competent authority for undertaking blasting work as well as for obtaining and storing the explosive as per the Explosive Act, 1884 as amended up to date and the Explosive Rules, 1983. The contractor shall purchase the explosives fuses, detonators, etc. only from a licensed dealer. The contractor shall be responsible for the safe transportation, storage and custody as per explosive rules and proper accounting of the explosive materials. Fuses and detonators shallbe stored separately and away from the explosives. The Engineer-in-Charge or his authorized representative shall have the right to check the contractor's store and account of explosives. The contractor shall provide necessary facilities for this. The contractor shall be responsible for any damage arising out of accident to workmen, public or property due to storage, transportation and use of explosive during blasting operation.

Blasting operations shall be carried out under the supervision of a responsible authorized agent of the contractor (referred subsequently as agent only), during specified hours asapproved in writing by the Engineer-in-Charge. The agent shall be conversant with the rulesof blasting. In case of blasting with dynamite or any other high explosive, the position of all the bore holes to be drilled shall be marked in circles with white paint. These shall be inspected by the contractor's agent. Bore holes shall be of a size that the cartridge can easily pass down. After the drilling operation, the agent shall inspect the holes to ensure that drilling has been done only at the marked locations and no extra hole has been drilled. The agent shallthen prepare the necessary charge separately for each bore hole. The bore holes shall be thoroughly cleaned before a cartridge is inserted. Only cylindrical wooden tamping rods shall be used for tamping. Metal rods or rods having pointed ends shall never be used for tamping. One cartridge shall be placed in the bore hole and gently pressed but not rammed down. Othercartridges shall then be added as may be required to make up the necessary charge for thebore hole. The top most cartridges shall be connected to the detonator which shall in turn be connected to the safety fuses of required length. All fuses shall be cut to the length required before being inserted into the holes. Joints in fuses shall be avoided. Where joints areunavoidable a semi-circular nitch shall be cut in one piece of fuse about 2 cm deep from the end and the end of other piece inserted into the nitch. The two pieces shall then be wrapped together with string. All joints exposed to dampness shall be wrapped with rubber tape. The maximum of eight bore holes shall be loaded and fired at one occasion. The charges shall be fired successively and not simultaneously. Immediately before firing, warning shall be given and the agent shall see that all persons have retired to a place of safety. The safety fuses of thecharged holes shall be ignited in the presence of the agent, who shall see that all the fuses are properly ignited.

Careful count shall be kept by the agent and others of each blast as it explodes. In case all the charged bore holes have exploded, the agent shall inspect the site soon after the blast but in case of misfire the agent shall inspect the site after half an hour and mark red crosses (X) over the holes which have not exploded. During this interval of half an hour, nobody shall approach the misfired holes. No driller shall work near such bore until either of the following operations have been done by the agent for the misfired boreholes.

(a) The contractor's agent shall very carefully (when the tamping is of damp clay) extract the tamping with a wooden scraper and withdraw the fuse, primer and detonator. After this a fresh detonator, primer and fuse shall be placed in the misfired holes and fired, or

(b) The holes shall be cleaned for 30 cm of tamping and its direction ascertained byplacing a stick in the hole. Another hole shall then be drilled 15 cm away and parallelto it. This hole shall be charged and fired. The misfired holes shall also explode along with the new one. Before leaving the site of work, the agent of one shift shall inform the agent relieving him for the next shift, of any case of misfire and each such location shall be jointly inspected and the action to be taken in the matter shall be explained to the relieving agent. The Engineer-in-Charge shall also be informed by the agent of all cases of misfires, their causes and steps taken in that connection.

2.9 RATES

Rates for Earthwork shall include the following:

(a) Excavation and depositing excavated material as specified.
(b) Handing of antiquities and useful material as specified.
(c) Protection as specified.
(d) Site clearance as specified.
(e) Setting out and making profiles as specified.
(f) Bailing out or pumping of rain water from excavations.
(g) Initial lead of 50 m and lift of 1.5 m.
(h) Blasting operations for hard rock as specified.
(I) Lead beyond 50 metre shall be payable in accordance with chapter -1- Transportation.

Anti-termite: - Contractor shall procure the chemical of required concentration in sealed original containers directly from the reputed and authorized dealers; chemical shall be kept in the custody of the Engineer-in Charge or his authorized representatives and issued for use to meet the day's requirements. Empty containers after washing and concentrated chemical left unused at the end of the day's work shall be returned to the Engineer-in-Charge or his authorized representative.

2.10 ANTIQUITIES AND USEFUL MATERIALS

- Any finds of archaeological interest such as relics of antiquity, coins, fossils or other articles of value shall be delivered to the Engineer-in-Charge and shall be the property ofthe Government**.**

- Any material obtained from the excavation which in the opinion of the Engineer-in-Charge is useful shall be stacked separately in regular stacks as directed by the Engineer-in-Charge and shall be the property of the Government.

- Shoring in wells, foundations and trenches will not be payable without obtaining prior approval of concern Superintending Engineer.

- Black cotton soil shall not be permitted for back filling and or under floors.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.1		Surface dressing	Surface dressing of the ground including removing vegetation and in-equalities not exceeding 15 cm deep and disposal of rubbish, lead up to 50 m and lift up to 1.5 m. in all kinds of soil.	100 sqm	803.00
2.2		Jungle clearing	Clearing jungle including uprooting of rank vegetation, grass, brush wood,trees and saplings of girth up to 30 cm measured at a height of 1 m above ground level and removal of rubbish up to a distance of 50 m outside the periphery of the area cleared.	100 sqm	414.00
2.3		Grass clearing	Clearing grass and removal of the rubbish up to a distance of 50 m outside the periphery of the area cleared.	100 sqm	213.00
2.4		Trees felling	Felling trees of the girth (measured at a height of 1 m above ground level) including cutting of trunks and branches removing the roots and stacking of serviceable material and disposal of unserviceable material.		
	2.4.1	>30 cm < 60 cm dia	beyond 30 cm girth up to and including 60 cm girth	each	350.00
	2.4.2	>60 cm< 120 cm dia	beyond 60 cm girth up to and including 120 cm girth	each	1150.00
	2.4.3	>120 cm<240 cmdia	beyond 120 cm girth up to and including 240 cm girth	each	3500.00
	2.4.4	>240 cm dia	above 240 cm girth	each	6300.00
2.5		Excavation surface	Earth work in surface excavation not exceeding 30 cm in depth but exceeding 1.5 m in width as well as 10 sqm on plan including disposal of excavated earth up to 50 m lead and lift up to 1.5 m, disposed soil to be levelled and neatly dressed, All kinds of soil	sqm	43.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.6		Excavation soil	Earth work in excavation by mechanical means (Hydraulic excavator) / manual means over areas (exceeding 30cm in depth. 1.5m in width as well as 10 sqm on plan) including dressing of sides and ramming of bottom disposal of excavated earth, lead up to 50m and lift up to 1.5m, disposed earth to be levelled and neatly dressed. (No extra lift is payable if work is done by mechanical means) All kinds of soil	cum	147.00
2.7		Excavation rock	Earth work in excavation/ by mechanical means (Hydraulic Excavator)/ manual means over areas (exceeding 30 cm in depth, 1.5m in width as well as 10 sqm on plan) including dressing of sides and ramming of bottom disposal of excavated earth, Note:-Excavation of hard rock shall be the property of the contractor subject to recovery. excavated material has to shift from site within 15 days after instruction issued by Engineer-In-Charge otherwise after this penalty will imposed on contractor at the rate of Rs 50 per cum per day.		
	2.7.1	OR	Ordinary rock	cum	287.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	2.7.2	HR (blasting)	Hard rock (requiring blasting)	cum	519.00
	2.7.3	HR (blasting prohibited)	Hard rock (blasting prohibited)	cum	828.00
2.8		Excavation rock in narrow areas	Earth work in excavation by mechanical means (Hydraulic Excavator)/ manual means in foundation trenches or drains (not exceeding 1.5 m in width or 10 sqm on plan) including dressing of sides and ramming of bottoms, lift up to 1.5 m, including getting out the excavated soil and disposal of surplus excavated soil as directed, (No extra lift is payable if work is done by mechanical means) Note:-Excavation of hard rock shall be the property of the contractor subject to recovery. excavated material has to shift from site within 15 days after instruction issued by Engineer-In-Charge otherwise after this penalty will imposed on contractor at the rate of Rs 50 per cum per day.		
	2.8.1	OR	Ordinary rock	cum	216.00
	2.8.2	HR	Hard rock (requiring blasting)	cum	452.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	2.8.3	HR (blasting prohibited)	Hard rock (blasting prohibited)	cum	593.00
2.9		Excavation trenches soil up to 1.5 depth	Excavating trenches of required width for pipes, cables, etc. including excavation for sockets, and dressing of sides, ramming of bottoms, depth up to 1.5 m including getting out theexcavated soil, and then returning the soil as required, in layers not exceeding 20 cm in depth including consolidating each deposited layer by ramming, watering, etc. and disposing of surplus excavated soil as directed, within a lead of 50 m : (No extra lift is payable if work is done by mechanical means) in all kinds of soil		
	2.9.1	<80mm dia	Pipes, cables etc., not exceeding 80 mm dia.	metre	96.00
	2.9.2	>80mm dia/ <300 mmdia	Pipes, cables etc. exceeding 80 mm dia. but not exceeding 300 mm dia.	metre	156.00
	2.9.3	>300mm dia	Pipes, cables etc. exceeding 300 mm dia but not exceeding 600 mm.	metre	244.00
2.10		Extra for excavation trenches soil 1.5 to 3.0 m depth	Extra for excavating trenches for pipes, cables etc. in all kinds of soil for depth exceeding 1.5 m, but not exceeding 3 m. (Rate is over corresponding basic item for depth up to 1.5 metre).	metre	134%
2.11		Extra for excavation trenches 3mto 4.5 m depth	Extra for excavating trenches for pipes, cables, etc., in all kinds of soil for depth exceeding 3 m in depth, but not exceeding 4.5 m. (Rate is over corresponding basic item for depth up to 1.5 metre.)	metre	345%
2.12		Excavation trenches rock up to 1.5 m depth	Excavating trenches of required width for pipes, cables, etc., including excavation for sockets, depth up to 1.5 m including getting out the excavated materials, refiling the soil as required in layers not exceeding 20 cm in depth including consolidating each deposited layers by ramming, watering etc. stacking serviceable material for		

Item No.	SubItem No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			measurements and disposal of unserviceable material as directed, within a lead of 50m :		
	2.12.1	OR	Ordinary rock :		
	2.12.1.1	up to 80mm dia	Pipes, cables etc. not exceeding 80 mm dia.	metre	139.00
	2.12.1.2	>80mm to 300 mm dia	Pipes, cables etc. exceeding 80 mm dia but not exceeding 300 mm dia.	metre	344.00
	2.12.1.3	>300mm dia	Pipes, cables exceeding 300 mm dia but not exceeding 600mm dia	metre	395.00
	2.12.2	HR (blasting)	Hard rock (requiring blasting)		
	2.12.2.1	up to 80mm dia	Pipes, cables etc. not exceeding 80 mm dia.	metre	251.00
	2.12.2.2	>80 mm to 300mm dia	Pipes, cables etc. exceeding 80 mm dia. but not exceeding 300 mm dia.	metre	621.00
	2.12.2.3	>300mm to 600 mm dia	Pipes, cables etc. exceeding 300 mm dia but not exceeding 600mm dia	metre	715.00
	2.12.3	HR (blasting prohibited)	Hard rock (blasting prohibited)		
	2.12.3.1	up to 80mm dia	Pipes, cables etc. not exceeding 80 mm dia.	metre	270.00
	2.12.3.2	>80 to 300mm dia	Pipes, cables etc. exceeding 80 mm dia. but not exceeding 300 mm dia.	metre	670.00
	2.12.3.3	>300mm to 600 mm dia	Pipes, cables etc. exceeding 300 mm dia but not exceeding 600mm dia.	metre	770.00
2.13		Extra for excavation trenches in rock from 1.5 m to 3.0m	Extra for excavating trenches for pipes, cables, etc. in ordinary/hard rock exceeding 1.5 m in depth but not exceeding 3 m. (Rate is over corresponding basic item for depth up to 1.5 metre.)	metre	102%
2.14		Extra for excavation trenches in rock from 3.0 m to 4.5m	Extra for excavating trenches for pipes, cables, etc. in ordinary/hard rock exceeding 3m in depth but not exceeding 4.5 m. (Rate is over corresponding basic item for depth up to 1.5 metre.)	metre	252%

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.15		Close timbering trenches	Close timbering in trenches including strutting, shoring and packing cavities (wherever required) complete. (Measurements to be taken of the face area timbered).		
	2.15.1	Upto 1.5 m depth	Depth not exceeding 1.5 m.	sqm	118.00
	2.15.2	>1.5m to 3.0m depth	Depth exceeding 1.5 m but not exceeding 3 m.	sqm	122.00
	2.15.3	>3.0m to 4.5 m depth	Depth exceeding 3 m but not exceeding 4.5 m.	sqm	130.00
2.16		Close timbering shafts	Close timbering in case of shafts, wells, cesspits, manholes and the like including strutting, shoring and packing cavities (wherever required) etc. complete. (Measurements to be taken of the face area timbered).		
	2.16.1	Upto 1.5 m depth	Depth not exceeding 1.5 m.	sqm	120.00
	2.16.2	>1.5m to 3.0m depth	Depth exceeding 1.5 m but not exceeding 3 m.	sqm	128.00
	2.16.3	>3.0m to 4.5 m depth	Depth exceeding 3 m but not exceeding 4.5 m.	sqm	136.00
2.17		Close timbering over areas	Close timbering over areas including strutting, shoring and packing, cavities (wherever required) etc. complete. (Measurements to be taken of the face area timbered) :		
	2.17.1	Up to 1.5 m depth	Depth not exceeding 1.5 m.	sqm	105.00
	2.17.2	>1.5m to 3.0m depth	Depth exceeding 1.5 m but not exceeding 3 m.	sqm	109.00
	2.17.3	>3.0 m to 4.5 m depth	Depth exceeding 3 m but not exceeding 4.5 m.	sqm	114.00
2.18		Extra for leaving permanently in position in trenches	Extra for planking, strutting and packing materials for cavities (in close timbering) if required to be left permanently in position. (Face area of timber permanently left to be measured	sqm	1558.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.19		Open timbering trenches	Open timbering in trenches including strutting and shoring complete (measurements to be taken of the face area timbered):		
	2.19.1	Up to 1.5 m depth	Depth not exceeding 1.5 m.	sqm	59.00
	2.19.2	>1.5m to 3.0m depth	Depth exceeding 1.5 m but not exceeding 3 m.	sqm	62.00.
	2.19.3	>3.0m to 4.5 m depth	Depth exceeding 3 m but not exceeding 4.5 m.	sqm	66.00
2.20		Open timbering shafts	Open timbering in case of shafts, wells, cesspits, manholes and the like including strutting and shoring complete (Measurements to be taken of the face area timbered):		
	2.20.1	Upto 1.5 m depth	Depth not exceeding 1.5 m.	sqm	55.00
	2.20.2	>1.5m to 3.0m depth	Depth exceeding 1.5 m but not exceeding 3 m.	sqm	59.00
	2.20.3	>3.0m to 4.5 m depth	Depth exceeding 3 m but not exceeding 4.5 m.	sqm	53.00
2.21		Open timbering over areas	Open timbering over areas including strutting, shoring etc. complete. (Measurements to be taken of the face area timbered):		
	2.21.1	Upto 1.5 m depth	Depth not exceeding 1.5 m.	sqm	32.00
	2.21.2	>1.5m to 3.0m depth	Depth exceeding 1.5 m but not exceeding 3 m.	sqm	35.00
	2.21.3	>3.0m to 4.5 m depth	Depth exceeding 3 m but not exceeding 4.5 m.	sqm	39.00
2.22		Extra for leaving planking	Extra for planking and strutting in open timbering if required to be left permanently in position. (Face area of	sqm	784.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		permanently in position	the timber permanently left to be measured)		
2.23		Extra for under water and foul position	Extra rates for quantities of works, executed:		
	2.23.1	Under water	In or under water and/or liquid mud, including pumping out water as required.	metre depth	20%
	2.23.2	In foul condition	In or under foul position, including pumping out water as required.	metre depth	25%
			Note: - The extra percentage rate is applicable in respect of each item but limited to quantities of work executed in these difficult conditions. The unit, namely, metre depth, to be considered for payment, shall be the depthmeasured from the sub soil water level up to the Centre of gravity of the quantity executed in difficult conditions. The depth shall be reckoned correct to 0.1 metre. 0.05 metre or more shall be taken as 0.1 metre and less than 0.05 metre ignored.		
2.24		Filling available excavated earth	Filling available excavated earth (excluding hard rock/Ordinary rock) in trenches, plinth, sides of foundations etc. in layers not exceeding 20cm in depth, consolidating each deposited layer by ramming and watering, lead up to 50 m and lift upto 1.5 m.	cum	95.00
2.25		Extra for additional lift	Extra for every additional lift of 1.5 m or part thereof in excavation. (No extra lift is payable if work is done by mechanical means)		
	2.25.1	Soil	All kinds of soil.	cum	30.00
	2.25.2	Rock	Ordinary or hard rock.	cum	53.00
2.26		Under floor filling stone dust	Supplying and filling in plinth with crusher stone dust / coarse sand under floors including, watering, ramming and compacting in layers not exceeding	cum	672.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			20cm in depth and dressing complete **Note: -** Maximum thickness of this layer shall be 20 cm.		
2.27		Under floor filling	Supplying and filling in plinth under floors including watering ramming consolidating in layers not exceeding 20cm in depth and dressing complete.		
	2.27.1	Hard muram hard copra	With hard muram having CBR >12 % at under floors including watering, ramming and compacting (minimum compaction 95% of MDD) in layers not exceeding 20cm in thickness and dressing complete. (**Note:-** maximum thickness of this layer to be provided shall be 30cm)	cum	393.00
	2.27.2	Selected soil	Supplying and filing in plinth with soil (other then BC soil) having MDD not less than 1.85 t/m3 under floors including watering, ramming and compacting (minimum compaction 95% of MDD) in layers not exceeding 15cm depth and dressing complete.	cum	331.00
2.28		Excavating holes	Excavating holes more than 0.10 cum & upto 0.5 cum including getting out the excavated soil, then refiling the soil as required in layers not exceeding 20 cm in depth, including consolidating each deposited layer by ramming, watering etc., disposing of surplus excavated soil; as directed within a lead of 50 m and lift upto 1.5m.		
	2.28.1	Soil	All kinds of soil.	each	42.00
	2.28.2	OR	Ordinary rock	each	68.00
	2.28.3	HR (blasting)	Hard rock (requiring blasting)	each	138.00
	2.28.4	HR (blasting prohibited)	Hard rock (blasting prohibited)	each	151.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.29		Supplying chemical emulsion	Supplying chemical emulsion in sealed containers including delivery as specified. (Chlorpyriphos/ Lindane emulsifiable concentrate of 20%)	liter	174.00
2.30		Post-constructional anti-termite treatment	Diluting and injecting chemical emulsion for POST-CONSTRUCTIONAL anti-termite treatment (excluding the cost of chemical emulsion) :		
	2.30.1	Trenching along existing external wall	Along external wall where the apron is not provided using chemical emulsion@ 7.5 liters / sqm of the vertical surfaceof the substructure to a depth of 300mm including excavation channel along the wall and rodding etc. complete, with Chlorpyriphos/ Lindane E.C. 20% with 1% concentration.	metre	36.00
	2.30.2	Drilling plugging holes along existing external wall	Along the external wall below concrete or masonry apron using chemical emulsion @ 2.25 litres per linear metre including drilling and plugging holes etc. With Chlorpyriphos/ Lindane E.C. 20% with 1% concentration.	metre	42.00
	2.30.3	Existing floor	Treatment of soil under existing floors using chemical emulsion @ one litre per hole, 300 mm apart including drilling 12 mm diametre holes and plugging with cement mortar 1 :2 (1 cement : 2 sand) to match the existing floor, With Chlorpyriphos/ Lindane E.C. 20% with 1% concentration.	sqm	199.00
	2.30.4	Existing masonry	Treatment of existing masonry using chemical emulsion @ one litre per hole at 300 mm interval including drilling 10mm dia and 200mm deep holes at 45 degree and plugging them with cement mortar 1:2 (1 cement : 2 sand) to the full depth of the hole, with Chlorpyriphos/ Lindane E.C. 20% with 1% concentration.	metre	50.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	2.30.5	Existing wood work	Treatment at points of contact of wood work by chemical emulsion Chlorpyriphos/ Lindane (in oil or kerosene based solution) @ 0.5 litresper hole by drilling 6 mm dia holes at downward angle of 45 degree at 150mm centre to centre and sealing the same.	metre	201.00
2.31		Deduct for not levelled soil	Deduct for disposed soil not levelled and neatly dressed.	cum	22.00
2.32		Preconstruction antitermite treatment	Preconstruction antitermite treatment to the building under construction by providing.	sqm	136.00
			Stage 1: Treating the bottom and the sides (upto 30 cm height of the excavated trench @ 5 liters per sqm of the surface area.		
			Stage 2: After masonry/RCC work, the backfill in the immediate contact with the foundation structure treatment @ 7.5 liters per sqm. Of the vertical surface of the substructure for each side.		
			Stage 3: surface treatment by spreading emulsion over the plinth area before laying the base concrete under floors @ 5.0 litres/Sqm.		
			Stage 4: Pumping the emulsion in plinth masonry on filling side at floor junction @ 7.5 litres/Sqm.		
			Stage 5: Pumping the emulsion from outer side of the plinth below ground around the masonry @ 5.0 litres/Sqm as per I.S. 8944 Emulsion. (1Chlorpyrifos: 19 water) with five years service guarantee (Measurements to be taken for plinth area)		
			Note – Treatment should start when foundation trenches and pits are ready to take bed concrete or leveling course in foundations. Lying of bed concrete should start when the chemical emulsion has been absorbed by the soil and the surface is reasonably dry. Treatment should not be carried out when it is		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			raining or when soil is wet with rain or subsoil water. This also applies to fill up soil within the plinth area before laying the subgrade for flooring.		
2.33		Bore holes for dowel bars	Making bore holes for providing dowel bars at designed depth in rock foundation by drilling 40 mm dia bore hole in rock including necessary bending hooking, in position and grouting with cement slurry etc. complete as per drawing and specification. (Steel to be paid separately).	metre	205.00
2.34		Excavation & loading by hydraulic excavator	Excavation in soil using hydraulic excavator and including cutting and loading in tippers, trimming bottom and side slopes in accordance with requirements of lines, grades and cross-section, and transporting to the appropriate ground contour, location and levelling and dressing earth neatly withall lifts and lead upto 1000 metre.	cum	55.00
2.35		Rough excavation banking/filing excavated earth	Earth work in rough excavation by any means and banking\filling excavated earth in layers not exceeding 20cm in depth, breaking clods, watering rolling each layer with 1/2 tonne roller or wooden or steel rammers etc. and rolling every 3rd and top-most layer with power roller of minimum 8 tonnes and dressing up, in embankments for roads, flood banks, marginal bank and guide bank of filling up ground depression within campus, lead upto 50m and all lift. All kinds of soil.	cum	130.00
2.36		Banking/filling excavated earth in layers	Banking\filling excavated earth in layers not exceeding 20cm in depth, breaking clods, watering rolling each layer with 1/2 tonne roller or wooden or steel rammers etc. and rolling every 3rd and top-must layer with power roller of minimum 8 tonnes and dressing up, in embankments for roads, flood banks, marginal bank and guide bank etc. lead upto 50m and liftupto 1.5m. All kinds of soil.	cum	80.00
2.37		Deduct for not rolling	Deduct for not rolling with power roller of minimum 8 tonnes for banking\filling excavated earth in layers not exceeding 20cm in depth.	cum	19.00
2.38		Deduct for not watering	Deduct for not watering with the excavated earth for banking\filling	cum	15.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.39		Anti-termite pipe system	Providing and laying of permanent piping technology anti-termite treatment before flooring work by installing LLDP (Low linear density polyethylene) tube of 8 mm O.D. & 6.4 mm I.D.with inbuilt pressurecompensation chip at every 30 cminterval in the tube, having working pressure of 2 kg/cm2 and release rate of 1.9 ltr/hour fixed by P-clips and nails. The LLDP pipe shall be installed at the entire periphery of the building and at internal network of building at a depth of 20 to 200 mm under floor at every 2 to 3 mtr. c/c distance (adjusted as per building layout) & Ends of loop pass through a PVC elbow of minimum 32mm ID at junction box of wall and floor level, entering into a steel reinforced grooved flexible pipe of minimum 22 mm ID leading into Junction box and the loops shall terminate in junction boxes & test every junction during injecting chemicals for termite control treatment. The anti-termite chemical Imidacloprid 30.5% SC mix as per IS-6313, (part III) shall be injected by the pressure pump diluted with water @ 10.5ml/5 ltr of water at the rate of 2 kg/sq.cm. @ 5Ltr/SMT.The contractor shall submit approved line plan for piping system and junction boxes dully approved by Engineer-in-charge with bond of 5 year warranty.		
	2.39.1	Pipe system	Permanent piping system with Junction boxes	sqm	113.00
	2.39.2	Chemical injection	Anti-termite chemical injected area	sqm	53.00
2.40		Ploughing of existing ground	Ploughing the existing ground to a depth of 15 cm to 25 cm and watering the same. (All kinds of soil)	sqm	13.00
2.41		Brick bats 250 mm thick,	Providing , laying, spreading, and compacting, brick bats in uniform thickness 250 mm with hand packing, rolling, with vibratery roller 8-10 tonnes in stages to proper grade, watering and compacting to required density.	sqm	897.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
2.42		Ground and embankment filling	Supplying and filling in Ground and embankment including watering ramming consolidating in layers not exceeding 20cm in depth and dressing complete.		
	2.42.1	Selected soil	Supplying and filing in plinth with soil (other then BC soil) having MDD not less than 1.85 t/m3 under floors including watering, ramming and compacting (minimum compaction 95% of MDD) in layers not exceeding 15cm depth and dressing complete.	cum	331.00

CHAPTER – 3.0

MORTARS

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 269	Specification for 33 grade ordinary Portland cement
2	IS 383	Specification for coarse and fine aggregate from natural Source for concrete.
3	IS 455	Specification for Portland slag cement.
4	IS 460 (Part-I)	Specification for test sieves: wire cloth test sieves.
5	IS 650	Specification for standard sand for testing of cement
6	IS 1269	Specification for 53 grade ordinary Portland cement
7	IS 1344	Specification for calcined clay Pozzolana.
8	IS 1489	Specification for Portland pozzolana cement
9	IS 1542	Specification for sand for plaster
10	IS 1727	Methods of Test for Pozzolanic materials
11	IS 2116	Specification for sand for masonry mortar.
12	IS 2250	Code of practice for preparation and use of masonry Mortar.
13	IS 2386 (Part-I)	Method of test for aggregate for concrete (Particle size and shape)
14	IS 2386 (Part -II)	-Do- Estimation of deleterious materials and organic impurities.
15	IS 2386 (Part -III)	-Do- Specific gravity, density, voids, absorption and bulking.
16	IS 3025	Method of sampling and test for water
17	IS 3406	Specification for masonry cement.
18	IS 3812 (Part-I)	Specification for fly ash for use as pozzolana in cement mortar and concrete
19	IS 3812 (Part-II)	Specification for fly ash for use as admixture in cement mortar and concrete
20	IS 4031 (Part-I) to (Part-XIII)	Method of Physical test for hydraulic cement
21	IS 4032	Method of chemical analysis of Hydraulic cement.
22	IS 8041	Rapid hardening Portland cement.
23	IS 8042	Specification for white cement
24	IS 8043	Hydrophobic Portland cement
25	IS 8112	Specification for 43 grade ordinary Portland cement
26	IS 11652	Woven HDPE sacks for packing cement
27	IS 11653	Woven polypropylene sacks for packing cement
28	IS 12174	Jute synthetic union bags for packing cement

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
1	2	3	4	5	6	7
Water	3.1.1	(i) pH Value (ii) Limits of Acidity (iii) Limits of Alkality (iv) Percentage of solids (a) Chlorides (b) Suspended matter (c) Sulphates (d) Inorganic solids (e) Organic solids	Lab Lab Lab Lab Lab Lab Lab Lab Lab	IS 3025	-	Water from each source shall be got tested before the commencement of work and thereafter once in every three months till the completion of the work. Water from municipal source need be tested only once in six months. Number of Tests for each source shall be 3
Cement	3.1.2	(a) Physical requirement (i) Fineness (ii) Soundness (iii) Setting time (Initial and Final) (iv) Compressive strength (v) Consistency of standard cement paste	Lab Lab Lab Lab Lab	IS 4031 (Part II) IS 4031 (Part III) IS 4031 (Part V) IS 4031 (Part VI) IS 4031 (Part VI)	Each lot	Every 50 tonnes or part Thereof. Each brand of cement brought to site shall be tested as per This frequency.
Sand	3.1.3.1	Organic Impurities	Field	Appendix 'A'	20 cum	Every 20 cum or part thereof or more frequently as decided By Engineer-in-Charge.
	3.1.3.2	Silt Content	Field	Appendix 'A'	20 cum	-do-
	3.1.3.4	Particle size distribution a,b,c,d and e	Field or Laboratory as decided by the Engineer-Incharge	Appendix B	40 cum	40 cum or part thereof
Sand	3.1.3.5	Bulking of Sand	Field	Appendix D	20 cum	Every 20 cum or part thereof or more frequently as decided By Engineer-in-Charge.

PREAMBLE
MORTAR

(CHAPTER: 3.0)

3.1 Water

Water used for mixing and curing shall be clean and free from injurious quantities of alkalies, acids, oils, salts, sugar, organic materials, vegetable growth or other substance that may be deleterious to bricks, stone, concrete or steel. Potable water is generally considered satisfactory for mixing. The Ph value of water shall be not less than 6.

Water found satisfactory for mixing is also suitable for curing. However, water used for curing shall not produce any objectionable stain or unsightly deposit on the surface.Sea water shall not be used for mixing or curing.

Water from each source shall be tested before the commencement of the work and thereafter once in every three months till the completion of the work. In case of ground water, testing shall also be done for different points of drawdown. Water from each source shall be got tested during the dry season before monsoon and again after monsoon.

3.2 Cement

The cement used shall be any of the following grade and the type selected should be appropriate for the intended use.

(a) 33 grade ordinary Portland cement conforming to IS 269.
(b) 43 grade ordinary Portland cement conforming to IS 8112.
(c) 53 grade ordinary Portland cement conforming to IS 12269.
(d) Rapid hardening Portland cement conforming to IS 8041.
(e) Portland slag cement conforming to IS 455.
(f) Portland Pozzolana cement (fly ash based) conforming to IS 1489 (Part 1).
(g) Portland Pozzolona cement (calcined clay based) conforming to IS 1489 (part 2).
(h) Hydrophobic cement conforming to IS 8043
(i) Low heat Portland cement conforming to IS 12600.
(j) Sulphate resisting Portland cement conforming to IS 12330
(k) White cement conforming to IS 8042

Different types of cement shall not be mixed together. In case more than one type of cement is used in any work, a record shall be kept showing the location and the types of cement used.
Caution in Use of Cement Grade 53 in Construction: Because of the faster hydration process, the concrete releases heat of hydration at a much faster rate initially and release of heat is the higher in case of Grade. 53. The heat of hydration being higher, the chances of micro-cracking of concrete is much greater. Thus, during initial setting period of concrete, the higher heat of hydration can lead to damaging micro-cracking within the concrete which may not be visible at surface. This cracking is different from shrinkage cracks which occurs due to faster drying of concrete in windy conditions. The situation can be worse when we tend to increase the quantity of the cement in the concrete with a belief that such increases are better for both strength and durability of concrete. Thus, it is very essential to be forewarned that higher grade cement specially grade 53 should be used only where such use is warranted for making higher strength concrete and also where good Quality Assurance measures are in place, by which proper precaution are taken to relieve the higher heat of hydration through chilling of aggregates or by proper curing of concrete. There are

instances where higher grade cement is being used even for low strength concrete, as, mortar or even for plastering. This can lead to unnecessary cracking of concrete/ surfaces. Another issue to be cautioned against is the tendency of the manufacturers to project Grade 53 cement as stronger cement, whereas Grade 33 or 43 are enough to produce the concrete of desired characteristic strength. The scenario of method of production of cement by various manufacturers should also be kept in mind while ordering various grades of cement.The ability to produce cements of particular fineness get fixed by the machinery installed bythe manufacturers, and thus the ability to produce other various grades of cement by a particular manufacturer also gets limited. Whereas tendency today is to supply the consumer what he orders for by the manufacturers by simply stamping such grades on the bags. Thus, it is often observed that cement bags marked as grade 33 or 43 may really be containing cements of much higher grade.

Compressive Strength: Compressive strength requirement of each type of cement for various grades when tested in accordance with IS 4031 (part 6) shall be as under:

Sample	**Strength in N/mm2 not less than for**		
Age at testing	Gr. 33	Gr. 43	Gr. 53
72 + 1 hr	16	23	27
168 + 2 hrs	22	33	37
672 + 4 hrs	33	43	53

Setting Time: Setting time of cement of any type of any grade when tested by Vicat apparatus method described in IS 4031 shall conform to the following requirement:

(a) Initial setting time: Not less than 30 minutes.

(b) Final setting time: Not more than 600 minutes.

3.3 Fine Aggregate

Aggregate most of which passes through 4.75 mm IS sieve is known as fine aggregate. Fine aggregate shall consist of natural sand, crushed stone sand, crushed gravel sand stone dust or marble dust, fly ash. It shall be hard, durable, chemically inert, clean and free from adherent coatings, organic matter etc. and shall not contain any appreciable amount of clay balls or pellets and harmful impurities e.g. iron pyrites, alkalies, salts, coal, mica, shale or similar laminated materials in such form or in such quantities as to cause corrosion of metal or affect adversely the hardening, the strength, the durability or the appearance of mortar, plaster or concrete. The sum of the percentages of all deleterious material shall not exceed 5%. Fine aggregate must be checked for organic impurities such as decayed vegetation humps, coal dust etc. in accordance with the procedure prescribed in Appendix ‘A’

Silt Content: The maximum quantity of silt in sand as determined by the method prescribed in Appendix ‘C’ shall not exceed 8%. Fine aggregate containing more than allowable percentage of silt shall be washed as many times as directed by Engineer-in- charge so as to bring the silt content within allowable limits for which nothing extra shall bepaid.

Grading: On the basis of particle size, fine aggregate is graded in to four zones. The grading when determined in accordance with the procedure prescribed in Appendix ‘B’shall be within the limits given in Table 3.1 below. Where the grading falls outside thelimits of any particular grading zone of sieves, other than 600 micron IS sieve, by a total amount not exceeding 5 per cent, it shall be regarded as falling within that grading zone.

TABLE 3.1
Fine Aggregates

IS Sieve	*Percentage passing for*			
	Grading Zone I	*Grading Zone II*	*Grading Zone III*	*Grading Zone IV*
10 mm	100	100	100	100
4.75 mm	90-100	90-100	90-100	95-100
2.36 mm	60-95	75-100	85-100	95-100
1.18 mm	30-70	55-90	75-100	90-100
600 microns	15-34	35-59	60-70	80-100
300 microns	5-20	8-30	12-40	15-50
150 microns	0-10	0-10	0-10	0-15

Marble dust: Marble dust shall be obtained by crushing marble. Its grading shall fall within the limits of Grading Zone IV of Table 3.1. Grading of Marble dust for use in Mortar shall be as per following table.

Table 3.2
Grading of Marble Dust

IS Sieve	*Percentage Passing*
10 mm	100
4.75 mm	95-100
2.36 mm	95-100
1.18 mm	90-100
600 micron	80-100
300 micron	15-50
150 micron	0-15

Bulking: Fine aggregate, when dry or saturated, has almost the same Volume but dampness causes increase in volume. In case fine aggregate is damp at the time of proportioning the ingredients for mortar or concrete, its quantity shall be increased suitably to allow for bulkage, which shall be determined by the method prescribed in Appendix 'D' Table 3.3 gives the relation between moisture content and percentage of bulking for guidance only.

Table 3.3
Variation of Bulking with Moisture Content

Moisture content % age	*Bulking % age (by volume)*
2	15
3	20
4	25
5	30

3.4 Mixing

The mixing of mortar shall be done in mechanical mixers operated manually or by power as decided by Engineer-in-Charge. The Engineer-in-Charge may, however, permit hand mixing at his discretion taking into account the nature, magnitude and location of the work and practicability of the use of mechanical mixers or where item involving small quantities are to be done or if in his opinion the use of mechanical mixer is not feasible. In cases,

where mechanical mixers are not to be used, the contractor shall take permission of the Engineer-in-Charge in writing before the commencement of the work.

(a) Mechanical Mixing: Cement and sand in the specified proportions shall be mixed dry thoroughly in a mixer. Water shall then be added gradually and wet mixing continued for at least three minutes. Only the required quantity of water shall be added which will produce mortar of workable consistency but not stiff paste. Only the quantity of mortar, which can be used within 30 minutes of its mixing, shall be prepared at a time. Mixer shall be cleaned with water each time before suspending the work.

(b) Hand Mixing: The measured quantity of sand shall be leveled on a clean masonry platform and cement bags emptied on top. The cement and sand shall be thoroughly mixed dry by being turned over and over, backwards and forwards, several times till the mixture is of a uniform colour. The quantity of dry mix which can be used within 30 minutes shall then be mixed in a masonry trough with just sufficient quantity of water tobring the mortar to a stiff paste of necessary working consistency.

Precautions: Mortar shall be used as soon as possible after mixing and before it begins to set, and in any case within half hour, after the water is added to the dry mixture.

Item No.	Sub Item No	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
3.1		Cement mortar 1:1	Cement mortar 1:1 (1 cement: 1 sand)	cum	7539.00
3.2		Cement mortar 1:2	Cement mortar 1:2 (1 cement: 2 sand).	cum	5929.00
3.3		Cement mortar 1:3	Cement mortar 1:3 (1 cement: 3 sand).	cum	5127.00
3.4		Cement mortar 1:4	Cement mortar 1:4 (1 cement: 4 sand).	cum	4361.00
3.5		Cement mortar 1:5	Cement mortar 1:5 (1 cement: 5 sand).	cum	3948.00
3.6		Cement mortar 1:6	Cement mortar 1:6 (1 cement: 6 sand).	cum	3595.00
3.7		Cement mortar 1:2	Cement mortar 1:2 (1 cement: 2 stone dust).	cum	5497.00
3.8		Cement mortar 1:2	Cement mortar 1:2 (1 cement: 2 marble dust).	cum	5516.00
3.9		White cement mortar 1:2	White cement mortar 1:2 (1 white cement: 2 marble dust).	cum	9189.00
3.10		Cement mortar 1:5	Cement mortar 1:5 (1 cement: 5 marble dust).	cum	3637.00
3.11		White cement mortar 1:3	White cement mortar 1:3 (1 white cement: 3 marble dust).	cum	7416.00
3.12		White cement mortar 1:5	White cement mortar 1:5 (1 white cement: 5 marble dust).	cum	5158.00
3.13		Mortar in lime, surkhi, marble	Mortar in lime , surkhi (50% red and 50% light yellow) and marble dust 1:1.5:0.5	cum	4024.00

CHAPTER – 4.0

PLAIN CEMENT CONCRETE

LIST OF BUREAU OF INDIAN STANDARD CODES

S. No.	I.S. No.	Subject
1	IS 383	Specification for coarse and fine aggregate from natural Source for concrete.
2	IS 456	Plain and reinforced concrete - Code of practice
3	IS 516	Method of test for strength of concrete
4	IS 1199	Method of sampling and analysis of concrete
5	IS 1200 (Part-II)	Method of measurement of building and civil engineering work (concrete work)
6	IS 1322	Specification for bitumen felt for water proofing and damp proofing.
7	IS 1791	General requirements for batch type concrete mixers
8	IS 2386	Method of test for aggregates for concrete
		(a) Part I - Particle size and shape
		(b) Part II - Estimation of deleterious materials and organic impurities
		(c) Part III - Specific gravity, density, voids absorption and bulking.
		(d) Part IV - Mechanical properties.
		(e) Part V – Soundness
9	IS 2505	General requirements for concrete vibrators - immersion type
10	IS 2506	General requirements for concrete vibrators - screed board concrete vibrators
11	IS 2645	Specification for integral water proofing compounds for cement mortar and concrete
12	IS 3068	Specification for broken brick (burnt clay) coarse aggregate for use in concrete.
13	IS 3812	Specification for fly ash for use as pozzolana and admixture in cement mortar and concrete.
14	IS 4656	Specification for form vibrators for concrete.
15	IS 7861 (Part-I)	Code of practice for extreme weather concreting (Part-I) recommended practice for hot weather concreting.
16	IS 7861 (Part–II)	Code of practice for extreme weather concreting (Part-II) recommended
17	IS 9103	Specification for concrete admixtures

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
1	2	3	4	5	6	7
Stone Aggregate	4.1.2.2	(a) Percentage of soft or deleterious material	Field or Laboratory-Test as Required	IS 2386-Part II	As required by Engineer in-charge	For all quantities
	4.1.2.3	Particle size	Field/ Lab	Appendix 'E'	45 cum	For every 45 cum or part thereof for RCC work only. For rest of items as decided by Engineer-in charge
	4.1.2.5	(a) Estimation of organic impurities	Field/ Lab	IS 2386-Part II	10 cum	For every 40 cum or part thereof
		(b) Surface moisture	Field/ Lab	IS 2386	10 cum	For every 40 cum or part thereof
		(c) Determination of 10% fine value	Field/ Lab	IS 2386	10 cum	For every 40 cum or part thereof
		(d) Specific gravity	Field/ Lab	IS 2386	10 cum	For every 40 cum or part thereof
		(e) Bulk density	Field/ Lab	IS 2386	10 cum	For every 40 cum or part thereof
		(f) Aggregate crushing strength	Field/ Lab	IS 2386	10 cum	For every 40 cum or part thereof
		(g) Aggregate impact value	Field/ Lab	IS 2386	10 cum	For every 40 cum or part thereof
Concrete	4.2.2	Slump test	Field	Appendix 'F'	10 cum	15 cum or part thereof

PREAMBLE
PLAIN CEMENT CONCRETE
(CHAPTER: 4.0)

4.1 Material

Water, cement, fine aggregate or sand shall be as specified in Chapter 3.0 – Mortar.

4.2 Coarse Aggregate

General: Aggregate most of which is retained on 4.75 mm IS Sieve and contains only as much fine material as is permitted in IS 383 for various sizes and grading is known as coarse aggregate. Coarse aggregate shall be specified as stone aggregate and it shall be obtained from approved/ authorized sources.

(a) Stone Aggregate: It shall consist of naturally occurring (uncrushed, crushed or broken) stones. It shall be hard, strong, dense, durable and clean. It shall be free from veins, adherent coating, and injurious amounts of disintegrated pieces, alkali, vegetable matter and other deleterious substances. It shall be roughly cubical in shape.Flaky and elongated pieces shall be avoided. It shall conform to IS 383 unless otherwise specified.

Deleterious Material: Course aggregate shall not contain any deleterious material, such as pyrites, coal, lignite, mica, shale or similar laminated material, clay, alkali, soft fragments, sea shells and organic impurities in such quantity as to affect the strength or durability ofthe concrete. Coarse aggregate to be used for reinforced cement concrete shall not contain any material liable to attack the steel reinforcement. Aggregates which are chemically reactive with alkalies of cement shall not be used. The maximum quantity of deleterious material shall not be more than 5% of the weight of coarse aggregate when determined in accordance with IS 2386.

Size and Grading

Stone aggregate: It shall be either graded or single sized as specified. Nominal size and grading shall be as under:-

(a) Nominal sizes of graded stone aggregate or gravel shall be 40, 20, 16, or 12.5 mm as specified. For any one of the nominal sizes, the proportion of other sizes shall be in accordance with Table 4.1.

TABLE 4.1
Graded Stone Aggregate or Gravel

IS Sieve Designation	***Percentage passing (by weight) for nominal size of***			
	40 mm	20mm	16mm	12.5mm
80mm	100	-	-	-
63mm	-	-	-	-
40 mm	95 to 100	100	-	-
20 mm	30 to 70	95 to 100	100	100
16 mm	-	-	90 to 100	-
12.5 mm	-	-	-	90 to 100
10 mm	10 to 35	25 to 55	30 to 70	40 to 85
4.75 mm	0 to 5	0 to 10	0 to 10	0 to 10

(b) Nominal sizes of single sized stone aggregate or gravel shall be 63, 40, 20, 16, 12.5 or 10 mm as specified. For any one of the nominal size, the proportion of other sizes as determined by the method prescribed.

TABLE 4.2
Single Sized (Ungraded) Stone Aggregate or Gravel

IS Sieve Designation	***Percentage passing (by weight) for nominal size of***					
	63 mm	*40 mm*	20mm	16mm	12.5mm	10mm
80mm	100	-	-	-	-	-
63mm	85-100	100	-	-	-	-
40 mm	0-30	85-100	100	-	-	-
20 mm	0-5	0-20	85-100	100	-	-
16 mm	-	-	-	85-100	100	-
12.5 mm	-	-	-	-	85-100	100
10 mm	0-5	0-5	0-20	0-30	0-45	85-100
4.75 mm	-	-	0-5	0-5	0-10	0-20
2.36 mm	-	-	-	-	-	0-5

4.3 Fly Ash

Fly ash is finely divided residue resulting from the combustion of pulverized coal in boilers. Fly ash is the pulverized fuel ash extracted from the flue gases by any suitable process such as cyclone separation or electrostatic precipitation. The ash collected from the bottom of boilers is termed as bottom ash. Fly ash is finer than bottom ash. Siliceous fly ash (ASTM Class F) containing calcium oxide less than 10% by mass is normally produced from burning anthracite or bituminous coal and possesses pozzolanic properties. Calcareous fly ash (ASTM Class C) is poruduced by burning lignite or sub-bituminous coal and contains calcium oxide more than 10% by mass; the content could be as high as 25%. This fly ashhas both hydraulic and pozzolanic properties. It shall be clean and free from any contamination of bottom ash, grit or small pieces of pebbles. It is obligatory on the part of supplier/ manufacture that the fly ash conforms to the requirements if mutually agreed upon and shall furnish a certificate to this effect to the purchaser or his representative.

Characteristics: The physical requirements of fly-ash shall be as specified in Appendix 'G'. The chemical properties of fly ash shall be as per IS 3812 (part 1 and 2) depending on the usage.

4.4 Cement Concrete

4.5 Grades of Cement Concrete

The concrete shall be in grade designated as under:

TABLE 4.3
(As per BIS 456-2000)

Grades of Concrete

Group	Grade Designation	Specified characteristic compressive strength of 150 mm Cube at 28 Days In N/mm2
1	**2**	**3**
Ordinary Concrete	M-10	10
	M-15	15
	M-20	20
Standard Concrete	M-25	25
	M-30	30
	M-35	35
	M-40	40
	M-45	45
	M-50	50
	M-55	55
High Strength Concrete	M-60	60
	M-65	65
	M-70	70
	M-75	75
	M-80	80

4.6 Mixing

Concrete shall be mixed in mechanical batch type concrete mixers conforming to IS 1791 having two blades and fitted with power loader (lifting hopper type). Half bag mixers and mixers without lifting hoppers shall not be used for mixing concrete. In exceptional circumstances, such as mechanical breakdown of mixer, work in remote areas or power breakdown and when the quantity of concrete work is very small, hand mixing may be done with the specific prior permission of the Engineer-in-Charge in writing subject to adding 10% extra cement. When hand mixing is permitted, it shall be carried out on a water tight platform and care shall be taken to ensure that mixing is continued until the concrete is uniform in colour and consistency. Stone aggregate shall be washed with water to remove, dirt, dust and other foreign materials. For guidance, the mixing time may be $1^1/2$ to 2 minutes, for hydrophobic cement it may be taken as $2^1/2$ to 3 minutes.

Machine Mixing: The mixer drum shall be flushed clean with water. Measured quantity of coarse aggregate shall be placed first in the hopper. This shall be followed with measured quantity of fine aggregate and then cement. In case fine aggregate is damp, half the required quantity of coarse aggregate shall be placed in the hopper, followed by fine aggregate and cement. Finally the balance quantity of coarse aggregate shall be fed in the hopper, and then the dry materials are slipped into the drum by raising the hopper. The dry material shall be mixed for at least four turns of the drum. While the drum is rotating, water shall be added gradually to achieve the water cement ratio as specified or as required by the Engineer-in-

Charge. After adding water, the mixing shall be continued until concrete of uniform colour, uniformly distributed material and consistency is obtained. Mixing shall be done for at least two minutes after adding water. If there is segregation after unloading from the mixer, the concrete should be remixed.

The drum shall be emptied before recharging. When the mixer is closed down for the day or at any time exceeding 20 minutes, the drum shall be flushed cleaned with water.

Transportation and Handling: Concrete shall be transported from the mixer to the placeof lying as rapidly as possible by methods which will prevent the segregation or loss of any of the ingredients and maintaining the required workability. During hot or cold weather, concrete shall be transported in deep containers, other suitable methods to reduce the loss ofwater by evaporation in hot weather and heat loss in cold weather may also be adopted.

4.7 Placing

The concrete shall be deposited as nearly as practicable in its final position to avoid rehandling. It shall be laid gently (not thrown) and shall be thoroughly vibrated and compacted before setting commences and should not be subsequently disturbed. Method of placing shall be such as to preclude segregation. Care shall be taken to avoid displacement of reinforcement or movement of form work and damage due to rains. As a generalguidance, the maximum free fall of concrete may be taken as 1.5 metre.

4.8 Compaction

Concrete shall be thoroughly compacted and fully worked around embedded fixtures and into corners of the form work. Compaction shall be done by mechanical vibrator of appropriate type till a dense concrete is obtained. The mechanical vibrators shall conform toIS 2505, IS 2506, IS 2514 and IS 4656. To prevent segregation, over vibration shall be avoided.

Compaction shall be completed before the initial setting starts. For the items where mechanical vibrators are not to be used, the contractor shall take permission of the Engineer-in-Charge in writing before the start of the work. After compaction the top surfaceshall be finished even and smooth with wooden trowel before the concrete begins to set.

4.9 Construction Joints

Concreting shall be carried out continuously upto construction joints. The position and arrangement of construction joints shall be as shown in the structural drawings or asdirected by the Engineer-in- Charge. Number of such joints shall be kept minimum. Joints shall be kept as straight as possible. Construction joints should comply with IS 11817.

When the work has to be resumed on a surface which has hardened, such surface shall be roughened. It shall then be swept clean and thoroughly wetted. For vertical joints, neat cement slurry, of workable consistency by using 2 kg of cement per sqm shall be applied onthe surface before it is dry. For horizontal joints, the surface shall be covered with a layer of cement mortar 1:1 about 10-15 mm thick composed of cement and sand in the same ratio as the cement and sand in concrete mix. This layer of cement slurry of mortar shall be freshly mixed and applied immediately before placing of the concrete.

Where the concrete has not fully hardened, all laitance shall be removed by scrubbing the wet surface with wire or bristle brushes, care being taken to avoid dislodgement of particles of coarse aggregate. The surface shall be thoroughly wetted and all free water removed. The surface shall then be coated with neat cement slurry @ 2 kgs of cement per sqm. On this surface, a layer of concrete not exceeding 150 mm in thickness shall first be placed and shall be well rammed against old work particular attention being paid to corners and close spots; work, thereafter, shall proceed in the normal way.

4.10 Concreting under Special Conditions

Work in Extreme Weather Conditions: During hot and cold weather, the concreting shall be done as per the procedure set out in IS 7861 (Part–I)-1975 and IS 7861 (Part II)-1981 respectively. Concreting shall not be done when the temperature falls below 4.5ºC. In cold weather, the concrete placed shall be protected against frost. During hot weather, it shall be ensured that the temperature of wet concrete does not exceed 38ºC.

Under Water Concreting: Concrete shall not be deposited under water if it is practicable to dewater the area and place concrete in the regular manner. When it is necessary to deposit concrete under water, the methods, equipment, materials and proportions of the mix to be used shall be submitted to and approved by the Engineer-in-Charge before the work is started. Under-water concrete should have a slump recommended. The water-cement ratio shall not exceed 0.6 and may need to be smaller, depending on the grade of concrete or the type of chemical attack. For aggregates of 40 mm maximum particle size, the cement content shall be at least 350 kg/m3 of concrete.

4.11 Curing

Curing is the process of preventing loss of moisture from the concrete. The following methods shall be employed for effecting curing.

Moist Curing: Exposed surfaces of concrete shall be kept continuously in a damp or wet condition by ponding or by covering with a layer of sacking, canvas, Hessian or similar materials and kept constantly wet for at least 7 days from the date of placing concrete in case of ordinary Portland cement and at least 10 days where mineral admixtures or blended cements are used. The period of curing shall not be less than 10 days for concrete exposed to dry and hot weather conditions. In the case of concrete where mineral admixtures or blended cements are used, it is recommended that above minimum periods may be extended to 14 days.

Membrane Curing: Approved curing compounds may be used in lieu of moist curing with the permission of the Engineer-in-Charge. Such compound shall be applied to all exposed surfaces of the concrete as soon as possible after the concrete has set. Impermeable membrane such as polythene sheet covering the concrete surface may also be used to provide effective barrier against the evaporation.

Freshly laid concrete shall be protected from rain by suitable covering.

Over the foundation concrete, the masonry work may be started after 48 hours of its compaction but the curing of exposed surfaces of cement concrete shall be continued along with the masonry work for at least 7 days. And where cement concrete is used as base concrete for flooring, the flooring may be commenced before the curing period of base concrete is over but the curing of base concrete shall be continued along with top layer of flooring for a minimum period of 7 days.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
4.1		C.C. upto plinth level	Providing and laying in position cement concrete of specified grade excluding the cost of centering and shuttering - All work upto plinth level :		
	4.1.1	With 20mm metal	With 20mm nominal size graded stone aggregate.		
	4.1.1.1	M-25	M-25 – Grade concrete	cum	6338.00
	4.1.1.2	M-20	M-20 – Grade concrete	cum	4933.00
	4.1.1.3	M-15	M-15 – Grade concrete	cum	4471.00
	4.1.1.4	M-10	M-10Nominal (Mix-1 Cement : 3 sand : 6 graded stone aggregate)	cum	3890.00
	4.1.2	With 40 mm metal	With 40mm nominal size graded stone aggregate.		
	4.1.2.1	M-15	M-15 - Grade concrete	cum	4403.00
	4.1.2.2	M-10	M-10 Nominal (Mix -1 Cement : 3 sand : 6 graded stone aggregate)	cum	3808.00
	4.1.2.3	M-7.5	M-7.5 Nominal Mix -1 Cement : 4 sand : 8 graded stone aggregate	cum	3474.00
	4.1.3	With fly ash	With fly ash.		
	4.1.3.1	1:2:3½:9	1:2:3½:9 (1 ordinary Portland cement : 2 Fly ash : 3½ sand: 9 graded stone aggregate 40 mm nominal size)	Cum	3459.00
	4.1.3.2	1:2½:4:11	1:2½:4:11 (1 ordinary Portland cement : 2½ fly ash : 4 sand : 11 graded stone aggregate 40 mm nominal size)	Cum	3150.00
			Note: No fly ash is to be added to Portland Pozzolona cement in any case which itself contains fly ash.		
4.2		C.C. upto floor two level	Providing and laying cement concrete in retaining walls, return walls, walls (any thickness) including attached pilasters, columns, pillars, posts, struts, buttresses, string or lacing courses, parapets, coping, bed blocks, anchor blocks, plain window sills, fillets etc. upto floor two level, excluding the cost of centering, shuttering and finishing :		
	4.2.1	With 20mm metal	With 20mm nominal size graded stone aggregate		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	4.2.1.1	M-25	M-25 – Grade concrete	Cum	6582.00
	4.2.1.2	M-20	M 20 – Grade concrete	Cum	5177.00
	4.2.1.3	M-15	M 15 – Grade concrete	Cum	4714.00
	4.2.1.4	M-10	M-10 Nominal Mix -1 Cement : 3 sand : 6 graded stone aggregate	Cum	4134.00
	4.2.2	With 40mm metal	With 40mm nominal size graded stone aggregate		
	4.2.2.1	M-15	M-15 - Grade concrete	cum	4647.00
	4.2.2.2	M-10	M-10 Nominal Mix -1 Cement : 3 sand : 6 graded stone aggregate	cum	4052.00
4.3		Centering and shuttering	Centering and shuttering including strutting, propping etc. and removal of form work for :		
	4.3.1	Upto plinth	Upto plinth: Foundations, footings, bases for columns, plinth beams, curtain walls, columns below plinth.	Sqm	169.00
	4.3.2	Above plinth	above plinth: Retaining walls, return walls, walls (any thickness) including attached pilasters, buttresses, and string courses fillets etc.	sqm	267.00
	4.3.3	Above plinth in columns	Above plinth: Columns, pillars, posts and struts.	Sqm	356.00
4.4		C.C. kerbs, steps	Providing and laying cement concrete in kerbs, steps and the like at or near ground level excluding the cost of centering, shuttering and finishing.		
	4.4.1	M-15	M-15 (With 20mm nominal size graded stone aggregate)	cum	4471.00
	4.4.2	M-10	M-10 (With 20mm nominal size graded stone aggregate) Nominal Mix -1 Cement : 3 sand : 6 graded stone aggregate	cum	3890.00
4.5		Precast C.C. in kerbs & edgings M-15	Providing and fixing at or near ground level precast cement concrete in kerbs, edgings etc. as per approved pattern and setting in position with cement mortar 1:3 (1 Cement : 3 sand) including the cost of required centering, shuttering and finishing smooth with 6mm thick cement plaster 1:3 (1 cement : 3 sand) on exposed surfaces complete. (With 20mm nominal size graded stone aggregate)	cum	4702.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
4.6		Precast C.C. upto floor two level i/c shuttering	Providing and fixing upto floor two level precast cement concrete string or lacing courses, copings, bed plates, anchor blocks, plain window sills, shelves, louvers, steps, stair cases, etc. Including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 sand), including cost of required centering, shuttering and finishing smooth with 6mm thick cement plaster 1:3 (1Cement : 3 sand) on exposed surfaces complete.		
	4.6.1	M-25	M-25 (With 20mm nominal size graded stone aggregate)	cum	5095.00
	4.6.2	M-20	M-20 (With 20mm nominal size graded stone aggregate)	cum	5558.00
	4.6.3	M-15	M-15 (With 20mm nominal size graded stone aggregate)	cum	5095.00
	4.6.4	M-10	M-10 (With 20mm nominal size graded stone aggregate) Nominal Mix -1 Cement: 3 sand: 6 graded stone aggregate.	cum	4515.00
4.7		Precast C.C. solid block	Providing and fixing upto floor two level precast cement concrete solid block including hoistingand setting in position with cement mortar 1:3 (1 cement : 3 sand), cost of required centering, shuttering and finishing smooth with 6mm thick cement plaster 1:3 (1 cement : 3 sand) on exposed surfaces complete :		
	4.7.1	M-15	M-15 (With 20mm nominal size graded stone aggregate)	cum	7057. 00
	4.7.2	M-10	M-10 (M 10) (With 20mm nominal size graded stone aggregate) Nominal Mix -1 Cement : 3 sand : 6 graded stone aggregate	cum	6477.00
4.8		Precast C.C. hollow block	Providing and Fixing upto floor two level precast concrete hollow block including hoisting and setting in position with cement mortar 1:3 (1 cement : 3 sand), cost of required centering, shuttering and finishing smooth with 6mm thick cement plaster 1:3 (1 cement : 3 sand) on exposed surfaces complete.		
	4.8.1	M-15	M-15 (With 20mm nominal size graded stone aggregate)	cum	5339.00
	4.8.2	M-10	M-10 (With 20mm nominal size graded stone aggregate) Nominal Mix -1 Cement: 3 sand: 6 graded stone aggregate.	cum	5072.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
4.9		Extra for C.C. above floor II	Add extra for item no. 4.6, 4.7 and 4.8 above floor two level for every additional floor or part there of	cum	1%
4.10		DPC 40mm thick	Providing and laying damp-proof course 40mm thick with cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 12.5mm nominal size).	Sqm	200.00
4.11		DPC 50mm thick	Providing and laying damp-proof course 50mm thick with cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 20mm nominal size).	sqm	247.00
4.12		Extra for water proofing in c.c	Extra for providing and mixing water proofing material as per IS standard in cement concrete work quantity as prescribed by the manufacturer.	Per bag of 50 kg of cement	44.00
4.13		Bitumen coat for DPC	Applying a coat of residual petroleum bitumen of penetration 80/100 of approved quality using 1.7kg per square metre on damp proof courseafter cleaning the surface with brushes and finallywith a piece of cloth lightly soaked in kerosene oil.	Sqm	124.00
4.14		Extra for C.C. work in superstructure	Extra for concrete work in superstructure above floor two level for each floors or part thereof	cum	1%
4.15		Extra for C.C. in or under water	Extra for laying concrete in or under water and/or liquid mud including cost of pumping or bailing out water and removing slush etc. complete. **Note for item No. 4.15:-** The quantity will be calculated by multiplying the depth measured from the sub-soil water level upto centre of gravity of concrete under sub-soil water level with quantity of concrete in cum executed under sub-soil water. The depth of centre of gravityshall be reckoned correct to 0.1m. 0.05m or more shall be taken as 0.1m and less than 0.05m ignored.	cum per meter depth	297.00
4.16		Extra for c.c. under foul positions	Extra for laying concrete in or under foul positions.	cum	109.00
4.17		Plinth protection by C.C.	Making plinth protection 50mm thick of cement concrete grade M10 over 75mm bed of dry metal ballast 40mm nominal size well rammed and consolidated and filled with sand including finishing the top smooth (with 20mm nominalsize graded stone aggregate.)	sqm	351.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
4.18		Extra for fibre in C.C.	Extra for addition of synthetic polyestertriangular fiber of length 12 mm, effectivediameter 10-40 microns and specific gravity of 1.34 to 1.40 in cement concrete/RCC/Flooring/ Water retaining structures by using 125 gms of synthetic polyester triangular fibre for 50 kgs cement used as per directions of Engineer-in-Charge.	per bag of 50 kg of cement	45.00
4.19		Precast bollards	Precasting and placing in position 125 mm dia Bollards 600 mm high of required shape including providing M.S. Pipe Sleeve 50 mm dia 300 mm long in the Bollard and M.S. Pipes 40 mm dia and 450 mm long with 150x150x6mm M.S. plate welded at bottom and embedded 150mm in cement concrete 1:3:6 (1 Cement : 3 sand : 6 graded stone aggregate 20 mm nominal size) including necessary excavation of size 250x250x450mm deep for the same in bitumen/concrete pavement at specified spacing.	each	485.00
4.20		Precast string/lacing courses, copings etc M-20	Providing and fixing up to floor two level precast cement concrete string or lacing courses, copings, bed plates, anchor blocks, plain window sills, shelves, louvers, steps, stair cases, etc., including hoisting and setting in position with cement mortar 1:3 (1 Cement : 3 coarse sand), cost of required Centering complete. 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) : 3 graded stone aggregate 20mm nominal size)	cum	6583.00
4.21		Precast solid block M-20	Providing and fixing up to floor two level precast cement concrete solid block, including hoisting and setting in position with cement mortar 1:3 (1 cement : 3 coarse sand), cost of required centering, shuttering complete . 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) : 3 graded stone aggregate 20mm nominal size)	cum	9843.00
4.22		Precast hollow block M-20	Providing and fixing up to floor two level precast cement concrete hollow block, including hoisting and setting in position with cement mortar 1:3 (1 cement : 3 coarse sand), cost of required centering, shuttering complete. 1:1.5:3 (1 cement : 1.5 coarse sand(zone-III) : 3 graded stone aggregate 20mm nominal size)	cum	7469.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
4.23		PCC with RMC (OPC with flyash)	Providing and laying in position ready mixed plain cement concrete, using fly ash and cement content as per approved design mix and manufactured in fully automatic batching plant and transported to site of work in transit mixerfor all leads, having continuous agitated mixer, manufactured as per mix design of specified grade for plain cement concrete work, including pumping of R.M.C. from transit mixer to site of laying and curing, excluding the cost of centering, shuttering and finishing, including costof curing, admixtures in recommended proportions as per IS : 9103 to accelerate/ retard setting of concrete, improve workability without impairing strength and durability as per direction of the Engineer-in-charge. Note : (1) Fly ash conforming to grade I of IS 3812 (Part-1) only be used as part replacement of OPC as per IS : 456. Uniform blending with cement is to be ensured in accordance with clauses 5.2 and 5.2.1 of IS: 456 -2000 in theitems of BMC and RMC.		
	4.23.1	Upto plinth	All works up to plinth level:		
	4.23.1.1	M-20	M-20 grade plain cement concrete (cement content considered @ 280kg/cum)	cum	5810.00
	4.23.1.2	M-15	M-15 grade plain cement concrete (cement content considered @ 240 kg/cum)	cum	5720.00
	4.23.1.3	M-10	M-10 grade plain cement concrete (cement content considered @ 220 kg/cum)	cum	5592.00
	4.23.2	Above plinth	All works above plinth and upto floor two level		
	4.23.2.1	M-20	M-20 grade plain cement concrete (cement content considered @ 280kg/cum)	cum	5930.00
	4.23.2.2	M-15	M-15 grade plain cement concrete (cement content considered @ 240 kg/cum)	cum	5886.00
	4.23.2.3	M-10	M-10 grade plain cement concrete (cement content considered @ 220 kg/cum)	cum	5758.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
4.24		PCC with RMC (using PPC Cement)	Providing and laying in position ready mixed plain cement concrete, with cement content as per approved design mix and manufactured in fully automatic batching plant and transported to site of work in transit mixer for all leads, having continuous agitated mixer, manufactured as per mix design of specified grade for plain cement concrete work, including pumping of R.M.C. from transit mixer to site of laying and curing, excluding the cost of centering, shuttering and finishing, including cost of curing, admixtures in recommended proportions as per IS : 9103 to accelerate/ retard setting of concrete, improve workability without impairing strength and durability as per direction of the Engineer-in-charge.		
	4.24.1	Upto plinth	All works up to plinth level:		
	4.24.1.1	M-20	M-20 grade plain cement concrete (cement content considered @ 280kg/cum)	cum	5910.00
	4.24.1.2	M-15	M-15 grade plain cement concrete (cement content considered @ 240 kg/cum)	cum	5869.00
	4.24.1.3	M-10	M-10 grade plain cement concrete (cement content considered @ 220 kg/cum)	cum	5740.00
	4.24.2	Above plinth	All works above plinth and upto floor two level		
	4.24.2.1	M-20	M-20 grade plain cement concrete (cement content considered @ 280kg/cum)	cum	6315.00
	4.24.2.2	M-15	M-15 grade plain cement concrete (cement content considered @ 240 kg/cum)	cum	6235.00
	4.24.2.3	M-10	M-10 grade plain cement concrete (cement content considered @ 220 kg/cum)	cum	6106.00
4.25		Plum Cement concrete	Plum Cement concrete 1:2:4 with 75% graded metal of maximum size 40 mm and 25% plum of 150mm size.	cum	3509.00
4.26		PCC for sunken floor	Providing and laying cement concrete in sunken floors of WC, Toilets, Wash Areas etc with 1:5:10 (1 cement:5 coarse sand (zone-III) :10 graded stone aggregate 40 mm nominal size) excluding the cost of centering and shuttering.	cum	3140.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	**2**	**3**	**4**	**5**	**6**
4.27		Dewatering by Trimix process	Labour charge for Trimix vacuum dewatering process for concrete and cost of fixing, removing of "C" Channel formwork including using vacuum dewatering machine machine finished as directed including light brooming and curing including making construction joint by cutting of joint of by using of concrete cutter machine.		
	4.27.1	100 mm	For 100 mm thickness.	sqm	115.00
	4.27.2	150 mm	For 150 mm thickness.	sqm	121.00

CHAPTER – 5.0

REINFORCED CEMENT CONCRETE

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 226	Structural Steel
2	IS 432 (Part I)	Specification for mild steel and medium tensile steel bars and hard drawn steel wire for concrete reinforcement part-I mild steel and medium tensile steel bars.
3	IS 432 (Part II)	Specification for mild steel and medium tensile steel bars and hard drawn steel wire for concrete reinforcement – Part-II hard drawn steel wire.
4	IS 456	Code of Practices for plain and Reinforced concrete.
5	IS 516	Method of test for strength of concrete.
6	IS 716	Specification for pentachlorophenol
7	IS 1199	Method of sampling and analysis of concrete.
8	IS 1200 (Part II)	Method of measurement of building and civil engineering work – concrete work
9	IS 1200 (Part V)	Method of measurement of building and civil engineering work – concrete work (Part 5- Form work)
10	IS 1566	Specification for hard drawn steel wire fabric for concrete requirement
11	IS 1599	Method for bend test
12	IS 1343	Code of Practice for Prestressed Concrete
13	IS 1608	Method for tensile testing of steel products
14	IS 1786	Specification for high strength deformed steel and wires for concrete reinforcement.
15	IS 1791	Specification for batch type concrete mixes
16	IS 2502	Code of practice for bending and fixing of bars for concrete Reinforcement.
17	IS 2751	Recommended practice for welding of mild steel plain and deformed bars for reinforced construction
18	IS 4925	Batch plants specification for concrete batching and mixing plant
19	IS 4926	Ready – Mixed Concrete
20	IS 6523	Specification for precast reinforced concrete door, window frames
21	IS 10262	Recommended guidelines for concrete mix design
22	IS 13311 (Part I)	Indian standard for non-destructive testing of concrete. Method of test for ultrasonic pulse velocity
23	IS 13311 (Part II)	Indian standard for non-destructive testing of concrete. Method of testing by rebound hammer.

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
1	2	3	4	5	6	7
Reinforced cement concrete (Nominal Mix)	5.4.1	(a) Slump Test	Field/ Lab	Appendix 'F'	(i) 5 cum in case of column (ii) 20 cum for slabs, beams and connected columns (iii) 20 cum for other R.C.C. work for all other small items and where R.C.C. done in a day is less than 5 cum test may be carried out as required by Engineer in-Charge	(i) Every 5 cum ofpart thereof (ii) Every 20 cumor part thereof (iii) -Do-
	5.4.9.1	(b) Cube Test	Lab	Appendix 'H'	(i) 5 cum in case of column (ii) 20 cum for slabs, beams and Connected columns (iii) 20 cum for other R.C.C. work for all other small items and where R.C.C. done in a day is less than 5 cum test may be carried out As required by Engineer in-Charge.	(i) Every 5 cum orpart thereof (ii) Every 20cum or part thereof (iii) -Do

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
1	2	3	4	5	6	7
Reinforced Cement Concrete (Design Mix)	Coarse Aggregates				50 cum or part thereof and also on each change of source	
	Fine Aggregates				50 cum or part thereof and also on each change of source	
	Cement				50 MT or on each change of source	
	Fresh Concrete	(a) Slump Test	Field	Appendix 'F'	10 cum	50 cum for R.C.C. work including in all other small location. R.C.C. done in a day isless than 50 cum test may be carried out as required byEngineer-in-Charge
	Fresh Concrete	(b) Cube Test	Lab	Appendix 'H'	10 cum or part Thereof	50 cum or 10Batches of 5-7 cum each for R.C.C. work in all location takentogether.R.C.C. done in a day is less than 50 cum test may be carried out asrequired by Engineer-in-Charge
Reinforced Cement Concrete (Ready Mix)	Coarse Aggregates				50 cum or part thereof and also on each change of source	
	Fine Aggregates				50 cum or part thereof and also on each change of source	
	Cement				50 MT or on each change of source	

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing	
1	2	3	4	5	6	7	
	Fresh Concrete	(a) Slump Test	Field/ Lab	Appendix 'F'	10 cum	50 cum for R.C.C. work including in all other small location. R.C.C. done in a day isless than 50 cum test may becarried out as requiredbyEngineer-in-Charge	
	Fresh Concrete	(b) Cube Test	Lab	Appendix 'H'	10 cum or partThereof	50 cum or 10 batches of5-7 cum each forR.C.C. work in all location – taken together. R.C.C. donein a day is less than50 cum test may be carried out asrequiredby Engineer-in-Charge	
Steel for Reinforced cement concrete	5.1.3	(A) Physical Test and chemical tests				(a) For consignment below 100 tonnes (i) under 10 mm dia, one sample for each 25 tonnes or part thereof (ii) 10 mm to 16 mm dia one sample for each 35 tonnes or part thereof (iii) over 16 mm dia one sample for each 45 tonnes or part thereof	(b) For consignment over 100tonnes (i) Under10 mm dia, one sample for each 40tonnesor part thereof (ii)10mmto 16mm, onesample for each 45tonnesor part thereof (iii) over16 mm dia, one samplefor each 50tonnes or part thereof

PREAMBLE
REINFORCED CEMENT CONCRETE
(CHAPTER: 5.0)

5.1 General

IS 456- 2000 Code of Practice for Plain and Reinforced Concrete (as amended up to date) shall be followed in regard to Concrete Mix Proportion and its production as under:

(a) The concrete mix design shall be done as "Design Mix Concrete" as prescribed in clause-9 of IS 456 mentioned above.

(b) Concrete shall be manufactured in accordance with clause 10 of above mentioned IS 456 covering quality assurance measures both technical and organizational, which shall also necessarily require a qualified Concrete Technologist to be available during manufacture of concrete for certification of quality of concrete.

Minimum M -20 grade of concrete shall be used in all structural elements made with RCC both in load bearing and framed structure.

5.2 Materials

Water, cement, fine and coarse aggregate shall be as specified under respective clauses of chapter 03 mortars and chapter 04 concrete work as applicable.

5.3 Steel for Reinforcement

The steel used for reinforcement shall be any of the following types:

(a) Mild steel and medium tensile bars conforming to IS 432 (Part I)

(b) High strength deformed steel bars conforming to IS 1786

(c) Hard drawn steel wire fabric conforming to IS 1566

(d) Structural steel conforming to Grade aof IS 2062

(e) Thermo-mechanically treated (TMT) Bars/Thermax poweredTMX Bars.

Elongation percent on gauge length is 5.65 $\sqrt{A}$ where A is the cross sectional areas of the test piece.

Mild steel is not recommended for the use in structures located in earthquake zone subjected to severe damage and for structures subjected to dynamic loading (other than wind loading) such as railway and highway bridges.

Welding of reinforcement bars covered in this SOR shall be done in accordance with the requirements of IS 2751.

Nominal mass/weight : The tolerance on mass/ weight for round and square bars shall be the percentage given in Table 5.1 of the mass/ weight calculated on the basis that the massesof the bar/ wire of nominal diameter and of density 7.85 kg/ cm^3 or 0.00785 kg/mm^3. Or 7850 kg per m^3

TABLE 5.1

Tolerance on Nominal Mass/ Weight

Nominal size in mm	Tolerance on the Nominal Mass per cent		
	Batch	Individual sample +	Individual sample for coil (x)
(a) Upto and including 10	±7	-8	±8
(b) Over 10, upto and including 16	±5	-6	±6
(c) Over 16	±3	-4	±4

A. High strength deformed bars and wires shall conform to IS 1786. The physical properties for all sizes of steel bars are mentioned below.

TABLE 5.2
Physical Properties of Bars and Wires as per IS 1786

Sl. No	Property	Fe 415	Fe 415 D	Fe 500 D	Fe 550 D
1	0.2 Per cent Proof stress/ yield stress, Min, N/mm2	415.0	415.0	500.0	550.0
2	Elongation, per cent, Min. on gauge length 5.65 √ A, where A is the cross-sectional area of the test piece.	14.5	18.0	16.0	14.5
3	Tensile strength, Min	10 Per cent more than the actual 0.2 per cent proof stress/ yield stress but not less than 485.0 N/mm2	12 Per cent more than the actual 0.2 per cent proof stress/yield stress but not less than 500.0 N/mm2	10 Per cent more than the actual 0.2 per cent proof stress/ yield stress but not less than 565.0 N/mm2	8 Per cent more than the actual 0.2 per cent proof stress/yield stress but not less than 600.0 N/mm2
4	Total elongation at maximum force, percent, Min on gauge length 5.65 √A , where A is the cross-sectional Area of the test piece.	-	5	5	5

B. ThermoMechanically Treated Reinforcement Bars:

(a) There is no BIS code for TMT bars. The available code BIS 1786 pertains to HSD Bars. Therefore there should be no stipulation that TMT bars should conform to relevant BIS code.

(b) The TMT/TMX bars are being produced under valid license from either of the firms namely Temporal, ThermexEvcon Turbo and Turbo Quench. These firms have acquired patents and are giving licenses to various producers to produce TMT Bars.

(c) The TMT bars shall conform to IS 1786 pertaining to Fe 415 D or Fe 500 D or Fe 550 grade of steel as specified.

(d) In design and construction of reinforced concrete building in seismic zone III and above, steel reinforcement of Grade Fe 415 D shall be used. However, high strength deformed steel bars, produced by thermomechanical treatment process of grade Fe 415, Fe 500 and Fe 550 having elongation more than 14.5. % and conform to other requirements of Fe 415 D, Fe 500 D and Fe 550 D respectively of IS 1786 may alsobe used for reinforcement

All reinforcement shall be free from loose mill scales, loose rust and coats of paints, oil, mud or other coatings which may destroy or reduce bond.

The Contractor shall have to produce Test Certificate in the proforma prescribed approved by B.I.S. from the manufacturer for every batch of steel brought to site of work.

Table 5.3

Mechanical Properties of High Yield Strength Deformed (HYSD) Bars and Wires and compression with recommended value in relevant IS Code

SI No.	Property	Fe 415	Fe 415S	Fe 500	Fe 500D	Fe 500S	Fe 550	Fe 550 D	Fe 600
(1)	(2)	(3)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
i)	0.2 percent proof stress/yield stress, Min, N/mm^2								
ii)	0.2 percent proof stress/yield stress,Max, N/mm^2								
iii)	TS/YS ratio[1)], N/mm^2								
iv)	Elongation, percent, min. on gauge length $5.65\sqrt{A}$, where A is the cross-sectional area of the test piece (see **3.9**)[2)]								

Table 5.4
Chemical Composition of High Yield Strength Deformed (HYSD) Bars and Wires

Constituent	Percent, Maximum						
	Fe 415	Fe 415D	Fe 500	Fe 500D	Fe 550	Fe 550D	Fe 600
Carbon							
Sulphur							
Phosphorus							
Sulphur and phosphorus							

Table 5.5
Chemical Composition of High Yield Strength Deformed (HYSD) Bars and Wires

Particulars Chemical Requirements	Test Results	Requirements as per relevant IS Code for cement
1. $\frac{Cao - 0.7SO_3}{2.8\ SIO_2 + 1.2Al_2O_3 + 0.65\ Fe_2O_3}$		
2. Al_2O_3 / Fe_2O_3		
3. Insoluble Residue (% by mass)		
4. Magnesia (% by mass)		
5. Sulphuric Anhydride (% by mass)		
6. Total Loss on lgnition (% by mass)		
7. Total Chlorides (% by mass)		
PHYSICAL REQUIREMENTS 1. Fineness (m^2 /kg) 2. Standard Consistency (%) 3. Setting Time (minutes) a. Initial b. Final 4. Soundness a. Le-Chat Expansion (mm) b. Autoclave Expansion (%) 5. Compressive Strength (MPa) a. 72+/- 1hr. (3 days) b. 168+/- 2hr. (7 days) c. 672+/- 4hr. (28 days) 6. Performance Improver (%) a. Limestone b. Fly Ash c. Granulated Slag The above cement complies with the requirements of IS**: 12269-1987** for Grade Ordinary Portland cement.		

Before commencement of use of steel, from any batch brought to site of work by the contractor, the Engineer-in Charge shall arrange to get samples tested for nominal mass, tensile strength, bend test and rebend test from any Laboratory of his choice at the cost of Contractor. The selection of test specimens and frequency shall be as per relevant I.S. specification and of steel used.

5.4 Form Work

Form work shall include all temporary or permanent forms or moulds required for forming the concrete which is cast-in-situ, together with all temporary construction required for their support.

5.5 Material for Form Work

Propping and Centering: All propping and centering should be either of steel tubes with extension pieces or built up sections of rolled steel.

Centering/Staging : Staging should be as designed with required extension pieces as approved by Engineer-in-Charge to ensure proper slopes, as per design for slabs/ beams etc. and as per levels as shown in drawing. All the staging to be either of Tubular steel structure with adequate bracings as approved or made of built up structural sections made form rolled structural steel sections.

Shuttering: Shuttering used shall be of sufficient stiffness to avoid excessive deflection and Joints shall be tightly butted to avoid leakage of slurry. If required, rubberized lining of material as approved by the Engineer-in-Charge shall be provided in the joints. Steel shuttering used or concreting should be sufficiently stiffened. The steel shuttering should also be properly repaired before use and properly cleaned to avoid stains, honey combing, seepage of slurry through joints etc.

(a) Runner Joists: RSJ, MS Channel or any other suitable section of the required size shall be used as runners.
(b) Assembly of beam head over props. Beam head is an adopter that fits snugly on the head plates of props to provide wider support under beam bottoms.
(c) Only steel shuttering shall be used, except for unavoidable portions and very small works for which 12 mm thick water proofing ply of approved quality may be used.

Form work shall be properly designed for self-weight, weight of reinforcement, weight of fresh concrete, and in addition, the various live loads likely to be imposed during the construction process (such as workmen, materials and equipment). In case the height of centering exceeds 3.50 metres, the prop may be provided in multi-stages. Form work shall be checked by an officer of rank not below Assistant Engineer before permitting lying of concrete.

5.6 Design and Tolerance in Constructionfor Form work

Form work shall be designed and constructed to the shapes, lines and dimensions shown on the drawings with the tolerance given below.

Table 5.6
Tolerance in Construction of Form Work

(a)	Deviation from specified dimension of cross section of columns and beams	+12 mm - 6mm
(b)	Deviation from dimensions of footings	
	(i) Dimension in Plan	+ 50 mm -12 mm
	(ii) Eccentricity in plan	0.02 times the width of the footing in the direction of deviation but not more than 50 mm.
	(iii) Thickness	+ 0.05 times the specified thickness.

Note- These tolerances apply to concrete dimensions only, and not to positioning of vertical steel or dowels.

5.7 GeneralRequirementfor Form work

It shall be strong enough to withstand the dead and live for form workloads and forces caused by ramming and vibrations of concrete and other incidental loads, imposed upon it during and after casting of concrete. It shall be made sufficiently rigid by using adequate number of ties and braces, screw jacks or hard board wedges where required shall be provided to make up any settlement in the form work either before or during the placing of concrete.

Form shall be so constructed as to be removable in sections in the desired sequence, without damaging the surface of concrete or disturbing other sections, care shall be taken to see that no piece is keyed into the concrete.

(a) In case of structures with two or more floors, the weight of concrete, centering and shuttering of any upper floor being cast shall be suitably supported on one floor below the top most floor already cast.

(b) Form work and concreting of upper floor shall not be done until concrete of lower floor has set at least for 14 days.

Camber: Suitable camber shall be provided in horizontal members of structure, especially in cantilever spans to counteract the effect of deflection. The form work shall be so assembled as to provide for camber. The camber for beams and slabs shall be 4 mm per metre (1 to 250) or as directed by the Engineer-in-Charge, so as to offset the subsequent deflection, for cantilevers the camber at free end shall be 1/50th of the projected length or asdirected by the Engineer-in-Charge.

Removal of Form work (Stripping Time): In normal circumstance and where various types of cements are used, forms, may generally be removed after the expiry of the following periods:

Table 5.7
Removal of Form Work

Type of Form work	Minimum period Before Striking Form work for OPC 33 grade	Minimum period Before Striking Form work for OPC 43 grade	Minimum period Before Striking Form work for PPC
(a) Vertical form work to columns, walls, beams	16-24 h	16-24 h	24-36 h
(b) Soffit form work to slabs(Props to be refixed immediately after removal of formwork)	3 days	3 days	4 days
(c) Soffit form work to beams (Props to be refixed immediately after removal of formwork	7 days	7 days	10 days
(d) Props to slabs: (1) Spanning upto 4.5m (2) Spanning over 4.5m	 7 days 14 days	 7 days 14 days	 10 days 20 days
(e) Props to beams and arches: (1) Spanning upto 6m (2) Spanning over 6m	 14 days 21 days	 14 days 21 days	 20 days 30 days

5.8 Placing of Concrete

Concreting shall be commenced only after Engineer-in-Charge has inspected the centering, shuttering and reinforcement as placed and passed the same. Shuttering shall be clean and free from all shavings, saw dust, pieces of wood, or other foreign material and surfaces shall be treated with oil or lubricant as prescribed.

In case of concreting of slab and beams, wooden plank or cat walks of chequerred MS plated or bamboo chalies or any other suitable material supported directly on the centering by means of wooden blocks or lugs shall be provided to convey the concrete to the place of deposition without disturbing the reinforcement in any way. Labour shall not be allowed to walk over the reinforcement.

In case of columns and wall, it is desirable to place concrete without construction joints. The progress of concreting in the vertical direction shall be restricted to one metre per hour.

The concrete shall be deposited in its final position in a manner to preclude segregation of ingredients. In deep trenches and footings concrete shall be placed through chutes or as directed by the Engineer-in-Charge. In case of columns and walls, the shuttering shall be so adjusted that the vertical drop of concrete is not more than 1.5 metres at a time.

During cold weather, concreting shall not be done when the temperature falls below 4.5^0C. The concrete placed shall be protected against frost by suitable covering. Concrete damaged by frost shall be removed and work redone.

During hot weather precaution shall be taken to see that the temperature of wet concrete does not exceed 38ºC. No concrete shall be laid within half an hour of the closing time of the day, unless permitted by the Engineer-in-Charge.

It is necessary that the time between mixing and placing of concrete shall not exceed 30 minutes so that the initial setting process is not interfered with.

5.9 Compaction

It shall be as specified in chapter IV sub-head of Concrete Work of this S.O.R.

5.10 Finishing

In case of roof slabs the top surface shall be finished even and smooth with wooden trowel, before the concrete begins to set. Sprinkling of dry cement while finishing shall not be resorted to.

Immediately on removal of forms, the R.C.C. work shall be examined by the Engineer-in-Charge, before any defects are made good.

(a) The work that has sagged or contains honey combing to an extent detrimental to structural safety or architectural concept shall be rejected.

(b) Surface defects of minor nature may be accepted. On acceptance of such a work by the Engineer-in-Charge, the same shall be rectified as follows:
Surface defects which require repair when forms are removed, usually consist of bulged due to movement of forms, ridges at form joints, honey-combed areas, damage resulting from the stripping of forms and bolt holes, bulges and ridges are removed by careful chipping or tooling and the surface is then rubbed with a grinding stone. Honey-combed and other defective areas must be chipped out, the edges being cut as straightas possible and perpendicularly to the surface, or preferably slightly under cut toprovide a key at the edge of the patch.

(c) The surface which is to receive plaster or where it is to be joined with brick masonry wall, shall be properly roughened immediately after the shuttering is removed, taking care to remove the laitance completely without disturbing the concrete. The roughening shall be done by hacking. Before the surface is plastered, it shall be cleaned and wetted so as to give bond between concrete and plaster. RCC work shall be done carefully so that the thickness of plaster required for finishing the surface is not more than 6 mm.

(d) The surface of RCC slab on which the cement concrete or mosaic floor is to be laid shall be roughened with brushes while the concrete is green. This shall be done without disturbing the concrete.

5.11 Strength of Concrete

The compressive strength on the work tests for different mixed shall be as given in Table 5.6

5.12 Ready Mixed Concrete (as per IS 4926) RMC

Materials

- **Selection and Approval of Materials:**Materials used should satisfy the requirements for the safety, structural performance durability and appearance of the finished structure, takingfull account of the environment to which it will be subjected. The selection and use of materials shall be in accordance with IS 456. Materials used shall conform to the relevant Indian Standards applicable. Where materials are used which are not covered by the provisions of the relevant Indian Standard, there should be satisfactory data on their

suitability and assurance of quality control. Records and details of performance of such materials should be maintained. Account should be taken of possible interactions and compatibility between IS 4926 and materials used. Also, prior permission of the purchaser shall be obtained before use of such materials.

Cement: Cement used for concrete shall be in accordance with the requirements of IS 456.
Mineral Admixtures: Use of mineral admixtures shall be permitted in accordance with the provisions of IS 456.
Aggregates: Aggregates used for concrete shall be in accordance with the requirement of IS 456. Unless otherwise agreed testing frequencies for aggregates in plant shall be as given IS 4926.

Chemical Admixtures

(i) Use of chemical admixtures shall be permitted in accordance, with the provisions of IS 456 and IS 9103.
(ii) It shall be the responsibility of the producer to establish compatibility and suitability of any admixture with the other ingredients of the mix and todetermine the dosage required to give the desired effect.
(iii) Admixtures should be stored in a manner that prevents degradation of the product and consumed within the time period indicated by the admixture supplier. Anyvessel containing an admixture in the plant or taken to site by the producer shall be clearly marked as to its content.
(iv) When offering or delivering a mix to a purchaser it should be indicated if such a mix contains an admixture or combination of admixtures or not. The admixtures may be identified generically and should be declared on the delivery ticket.
(v) The amount of admixture added to mix shall be recorded in the production record. In special circumstances, if necessary, additional dose of admixture may be added at project site to regain the workability of concrete with the mutual agreement between the producer and the purchaser.

Water: Water used shall be in accordance with the requirement of IS 456. The use of re-cycled water is encouraged as long as concrete of satisfactory performance can be produced and steps are taken to monitor the build up of chlorides in any recirculated water and that any subsequent adjustments to the mix design are made to ensure that any overall limit on chloride contents is satisfied. The addition of any recycled water shall be monitored and controlled to meet these requirements. The total amount of water added to the mix shall be recorded in the production record. The water content of concrete shall be regulated by controlling its workability or by measuring and adjusting the moisture contents of itsconstituent materials. The producer's production staff and truck-mixer, drivers shall be made aware of the appropriate responses to variations in concrete consistency of a particularmix caused by normal variations in aggregate moisture content or grading.

5.13 General Requirements for RMC

Basis of Supply: Ready-mixed concrete shall be supplied having the quality and thequantity in accordance with the requirement agreed with the purchaser or his agent. Notwithstanding this, the concrete supplied shall generally comply with requirements of IS
456. All concrete will be supplied and invoiced in terms of cubic metres (full or part) of compacted fresh concrete. All proportioning is to be carried out by mass except water and admixture, which may be measured by volume.

Transport of Concrete: Ready-mixed concrete shall be transported from the mixer to the point of placing as rapidly as practicable by methods that will maintain the required workability and will prevent segregation, loss of any constituents or ingress of foreign matter or water. The concrete shall be placed as soon as possible after delivery, as close asis practicable to its final position to avoid re-handling or moving the concrete horizontallyby vibration. If required by the purchaser the producer can utilize. Admixtures to slowdown the rate of workability loss; however this does not remove the need for the purchaser to place the concrete as rapidly as possible. The purchaser should plan his arrangements so as to enable a full load of concrete to be discharged within 30 minutes of arrival on site. Concrete shall be transported in a truck-mixer unless the purchaser agrees to the use of non-agitating vehicles. When non-agitating vehicles are used, the mixed concrete shall be protected from gain or loss of water.

Time in Transport: The general requirement is that concrete shall be discharged from the truck-mixer within 2 h of the time of loading. However, a longer period may be permitted if retarding admixtures are used or in cool humid weather or when chilled concrete is produced. The time of loading shall start from adding the mixing water to the dry mix of cement and aggregate or of adding the cement to the wet aggregate whichever is applicable. Ready-mixed concrete plant shall have test facilities at its premises to carry out routine testsas per the requirement of the standard. Information to be made available by the purchaser tothe RMC supplier: following information shall be necessarily made available by the purchaser to RMC supplier:

TABLE 5.8
CONCRETE MIX INFORMATION TO BE SUPPLIED BY THE PURCHASER

RMC: ……..
Contractor: …….
Site: …….

MIX CODE						
Grade (N/mm2) (Characteristic strength)						
Minimum Cement Content (kg./m3)						
Mineral Additives (Pulverized fuel ash/Slag/Others) (kg/m3)						
Maximum Free water Cement Ratio						
Nominal Maximum Aggregate size						
Cement Type and Grade (if preferred)						
Target workability (slump) (mm)						
Target workability at site						
Maximum Temperature of Concrete at the time of placing						
Class of sulphate Resistance (if applicable)						
Exposure condition (if applicable)						

Class of finish (if applicable)						
Mix Application						
Method of Placing						
Any other requirements (early strength workability retention, permeability testing, chloride content restriction, maximum cement content, etc.)						
Concrete Testing (Frequency)						
Material's Testing (any non-routine requirements)						
Alternatives to be offered: Yes/No						
Method of Curing to be used by contractor						
Quantity (m^3)						

- **Note:** Additional proforma for further information may be used, such as for specific test rates to be achieved for concrete or raw materials, exact method statements of the contractors proposed site practice.

DELIVERY TICKET INFORMATION for RMC

1) The following information shall be included in the delivery ticket to accompany the load to the purchaser:

(a) Name or number of the ready-mixed concrete depot

(b) Serial number of the ticket

(c) Date

(d) Truck number

(e) Name of the Purchaser

(f) Name and location of site

(g) Grade or mix description of the concrete

(h) Specified target workability

(i) Minimum cement content (if specified)

(j) Type of cement and grade (if specified)

(k) Maximum free water-cement ratio (if specified)

(l) Nominal maximum size of aggregate

(m) Generic Type or name of any chemical and mineral admixtures included.

(n) Quantity of concrete in m3

(o) Time of loading

(p) Signature of the plant operator

(q) A statement warning the purchaser of the precautions needed to be taken when working with cement and wet concrete.

2) On site the following information will be added. :

(a) Time of arrival on site.
(b) Time when discharge was completed.
(c) Any water/admixture added by the supplier to meet the specified workability.
(d) Any extra water /admixture added at the request of the purchaser of the concrete, or his Representative, and his signature.
(e) Pouring location.
(f) Signature of the purchaser or his representative conforming discharge of the load.

5.14 Sampling and Testing of Ready-Mixed Concrete

Point and Time of Sampling : For the assessment of compliance of ready-mixed concrete, the point and time of sampling shall be at discharge from the producer's delivery vehicle or from the mixer to the site or when delivered into the purchaser's vehicle. It is critical that the sampling procedure and equipment used enables as representative a sample as possibleto be taken of the quantity of concrete delivered. The sampling may be carried out jointly bythe purchaser and the supplier with its frequency mutually agreed upon. However, it willnot absolve the supplier of his responsibility from supplying in concrete as per the requirement given in this standard or otherwise agreed to where so permitted in the standard.

5.15 Workability

The workability shall be within the following limits on the specified value as appropriate:
Slump ± 25 mm or 1/3 of the specified value, whichever is less.
Compacting factor: ± 0.03, where the specified value is 0.90 or greater,
± 0.04, where the specified value is less than 0.90 but more than 0.80,
± 0.05, where the specified value is 0.80 or less.
Flow table test may be specified for concrete, for very high workability (see IS 9103) Acceptance criteria for spread (flow) are to be established between the supplier and the purchaser.

5.16 Specified Strength

Compliance shall be assessed against the requirements of IS 456 or other agreed Indian Standard. The purchaser may perform his sampling and testing or may enter into an arrangement with the producer to provide his testing requirements.

Unless otherwise agreed between the parties involved, the minimum testing frequency to be applied by the producer in the absence of a recognized ready- mixed concrete industry method of production control should be one sample for every 50 m^3 of production or every 50 batches, whichever is the greater frequency. Three test specimens shall be made up for each sample for testing at 28 days (see also IS 456). In order to get a relatively quicker idea of the quality of concrete, optional test on beams for modulus of rupture at 72 ± 2 h or at 7 days or compressive strength test at 7 days may be carried out in addition to 28 days compressive strength test. For this purpose the value should be arrived at based on actual testing. In all cases 28 days compressive strength shall alone be the criteria for acceptanceor rejection of the concrete.

The purchaser shall inform the producer if his requirements for sampling and testing are higher than one sample every 50 m^3 or 50 batches, whichever is the greater frequency.

5.17 Sampling and Testing for Quality Control of Hardened Concrete

Test on cube crushing strength of concrete in accordance and compliance with IS 456 and IS 516 shall do as under:-

Sample of fresh concrete shall be taken from concrete at central batch plant mixer while loading delivery trucks or other transport and also from concrete transported to placement location.

Test on specimens made form samples collected at placement location shall be considered as field test specimens and results therefrom shall be the criterion of concrete strength. Test in specimens made from samples at the batch plant shall only be taken as guidelines test. Only in the case of doubtful result, the Engineer-in-Charge may refer to such guideline results for deciding on the quality of concrete.

For truck mix concrete and shrink mix concrete guideline test specimens shall be made from samples collected at discharge location from mixing trucks. For this purpose first and last 15% of the load shall be omitted while collecting samples.

SAMPLING OF CONCRETE (RMC)

After the truck mixer has re-mixed its delivery on site, allow at least the first one-third of a m3 of concrete to be discharged prior to taking any samples. Take at least 4 incremental samples from the remainder of the load avoiding sampling the last cubic metre of concrete. Thoroughly re-mix this composite sample either on a mixing tray or in the sampling bucket and proceed with the required testing.

Frequency of sampling shall be as given below in Table 5.5 for each grade of concrete of different workability's and for each type of specimens (field test specimens and guideline test specimens) for conducting 28 days crushing strength tests.

TABLE 5.9
Sampling of Concrete

Quantity of concrete Delivered (cum)	Number of samples
Less than 5	1
6 to 15	2
31 to 50	3
51 and above sample for each	4 plus one additional 50 cum or part thereof

Each sample shall be of adequate quantity so that a minimum of 3 specimen cubes can be made test of the sample in accordance with IS 516.
All test specimens shall be made compacted cured and tested in compliance with IS 516 and test result interpreted in accordance with IS 456 for acceptance of concrete strength, field specimens test results shall not be less than values as prescribed.

In addition to 28 day crushing strength test on specimens made at frequencies specified above, early strength tests at 7 days shall also be conducted on field specimens as well as guideline test specimens. Frequency of sampling for this set of test shall also be same as those specified above. 7 day strength shall conform to values given in Table as under. But these test results even if conforming to specified values shall only be taken a guideline

values for projecting concrete strength and shall not be construed as conforming to specifications.

TABLE 5.10
Compressive Strength of Concrete

Concrete Mix (Nominal Mix on Volume basis)	Compressive Strength in (Kg/ sq cm)	
	7 days'	28 days'
1:1:2	210	315
1:1.5:3	175	265
1:2:4	140	210

For each grade of concrete and for all workability conditions with different water – cement ratios and compositions of admixtures, preliminary test shall be conducted for crushing strength on finalization to design mix for each type of concrete. Such test shall be conducted both at 7 days and 28 days under laboratory conditions. Six test specimens shall be made for 7 days test and six test specimens shall be made for 28 days test. Average ofthe six test results of different periods shall not be less than those specified.

Crushing strengths on cubes shall also be conducted during the process of finalization of concrete design mix. Frequency and number of such tests shall be as per Mix of requirements of concrete manufacturer.

All test specimens for conducting crushing strength shall be properly labeled for identification indicating:-

(i) Date of making specimen
(ii) Grade of concrete
(iii) Placement location exact
(iv) Purchasers order number

In addition to crushing strength test on concrete, the Engineer-in-Charge may call for other tests on hardend concrete. The placement contractor and the manufacturer of concrete shall comply with all such instructions.

5.18 Non-destructive Tests

When the 28 days crushing values on field specimens and/ or specimens and/or specimens made for guideline test fall short of specified values, or in case of doubtful placement of concrete, the Engineer-in-Charge shall call for non-destruction tests on the structure. Such tests may be any one or a combination of the following:-

(i) Rebound hammer test
(ii) Windsor Penetration Probe test
(iii) Pulse velocity (sonic or Ultrasonic) test
(iv) Core test
(v) Load test

Interpretation of rebound hammer, Windsor Probe and Pulse velocity test results shall rest with the Engineer-in-Charge.

Core test, if ordered by the Engineer-in-Charge, shall be done in accordance with IS: 516. Samples for such test shall be taken from locations to be identified by the Engineer-in- Charge and such samples shall be collected in compliance with IS: 1199.

If felt necessary, the Engineer-in-Charge may instruct load testing for any part of the structure based on doubtful concrete strengths. Such test shall be carried out as per details tobe provided by the Engineer-in-Charge in consultation with the structural consultants.

The concrete manufacturer/ concrete placement contractor shall arrange for all test to be conducted in accordance with these specifications, including all necessary tools, plants, equipment and material, and shall be responsible for conducting all test at his cost.

All test conducted at the field laboratory shall be carried out by qualified technicians employed by the concrete manufacturer/ concrete placement contractor, in presence of authorized representative of the Engineer-in-Charge. All test reports and observation reports shall be jointly signed by the Engineer-in-Charge authorized representative and the technician conducting such test.

Engineer-in-Charge shall alone decide where such tests are to be conducted. He mayinstruct tests to be conducted at laboratories other than the field laboratory and suchinstructions shall be followed without claiming extra charges on this account.

The Concrete Manufacturer/ Placement contractor shall set up a laboratory at this own expense which shall have facilities, for conducting all necessary field test on materials and field and laboratory test on concrete.

TEST FOR ORGANIC IMPURITIES

- The aggregate must also be checked for organic impurities such as decayed vegetation humus, coal dust etc. What is called the colour test is reliable indicator of the presence of harmful organic matter in aggregate, except in the area where there are deposits of lignite.

Test Procedure:

- Fill a 350 ml clear glass medicine bottle upto 70 ml mark with a 3% solution of caustic soda or sodium hydroxide.
- The sand is next added gradually until the volume measured by the sandy layer is 125 ml. The volume is then made upto 200 ml by addition of more of solution.
- The bottle is then stoppered and shaken vigorously and allowed to stand for 24 hours.
- At the end of this period, the colour of the liquid will indicate whether the sand contains a dangerous amount of matter. A colourless liquid indicates a clean sand free from organic matter. A straw coloured solution indicates some organic matter but not enough to beseriously objectionable. Darker colour means that the sand contains injurious amounts and should not be used unless it is washed, and a retest shows that it is satisfactory.
- Add 2.5 ml of two per cent solution of tannic acid in 10 per cent alcohol, to 97.5 ml of three per cent sodium hydroxide solution. Place in a 350 ml bottle, fix the stopper, shake vigorously and allow to stand for 24 hours before comparison with the solution above the sand.

    - ✓ **Note:** A three per cent solution of caustic soda is made by dissolving 3 g of sodium hydroxide in 100 ml of water, preferably distilled. The solution should be kept in a glass of bottle tightly closed with a rubber stopper. Handling sodium hydroxide with moist hands may result in serious burns. Care should be taken not to spill the solution for it is highly injurious to clothing, leather, and other materials.

TEST FOR PARTICLE SIZE (SIEVE ANALYSIS)

- **Apparatus:** Perforated plate sieves of designation 10 mm, 4.75 mm and fine mesh sieve of designation 2.36 mm, 1.18 mm, 600 micron, 300 micron and 150 micron should be used. The balance or scale shall be such that it is readable and accurate to 0.1 per cent of the weight of the test sample.

- **Sample:** The weight of sample available shall not be less than the weight given in the table below. The sample of sieving shall be prepared from the larger sample either by quartering or by means of a sample divider.

TABLE 5.11
Showing Minimum Weights for Sampling

Maximum size present in substantial proportions (mm)	Minimum weight of sample for sieving (Kg)
10	**0.5**
4.75	**0.2**
2.36	**0.1**

Test Procedure:

- The sample shall be brought to an air-dry condition before weighing and sieving. This may be achieved either by drying at room temperature or by heating at a temperature of 100 degree to 110 degree centigrade.
- The air dry sample shall be weighed and sieved successively on the appropriate sieves starting with the largest. Care shall be taken to ensure that the sieves are clean before use. Each sieve shall be shaken separately over a clean tray until not more than a trace passes,but in any case for a period of not less than two minutes. The shaking shall be done with a varied motion, backwards and forwards, left to right, circular clockwise and anti-clockwise, and with frequent jarring, so that the material is kept moving over the sieve surface in frequently changing directions.
- Materials shall not be forced through the sieve by hand pressure, but on sieves coarser than 20 mm, placing of particles is permitted, Lumps of fine material, if present may be broken by gentle pressure with fingers against the side of the sieve. Light brushing of under side of the sieve with a soft brush may be used to clear the sieve openings. Light brushing with a fine camel hair brush may be used on the 150 micron IS sieve to prevent segregation of powder and blinding of apertures. Stiff or worn out brushes shall not be used for thispurpose and pressure shall not be applied to the surface of the sieve to force particles through the mesh.
- On completion of sieving the material retained on each sieve, together with any material cleaned from the mesh, shall be weighed.

- ✓ **Reporting of Results:** The results shall be calculated and reported as:
  (i) The cumulative percentage by weight of the total sample passing each of the sieves, to the nearest whole number:
  Or

  (ii) The percentage by weight of the total sample passing one sieve and retained on the next smaller sieve, to the nearest 0.1 percent.

TEST FOR SILT CONTENT

- A sample of sand to be tested shall be placed without drying in a 200 ml measuring cylinder. The volume of the sample shall be such that it fills the cylinder upto 100 ml mark

- Clean water shall be added upto 150 ml mark. Dissolve a little salt in the water in the proportion one tea spoon to half a litre. The mixture shall be shaken vigorously, the last few shakes being sidewise direction to level off the sand and the contents allowed to settle for three hours.

- The height of the silt visible as settled layer above the sand shall be expressed as apercentage of the height of sand below. The sand containing more than the above allowable percentage of silt, shall be washed so as to bring the silt contents within allowable limits.

✓The sand shall not contain more than 8% of silt.

BULKING OF FINE AGGREGATES/SAND (FIELD METHODS)

- Two methods are suggested for determining the bulking of sand/fine aggregate. The procedure may be suitably varied, if necessary. Both depend on the fact that the volume of inundated sand/fine aggregate is the same if the sand/fine aggregate were dry.

Method -1:

- Put sufficient quantity of sand loosely into a container until it is about two-third full. Level off the top of the sand and push a steel rule vertically down through the sand at the middle to bottom, measure the height. Suppose this is 'X' cm.

- Empty the sand out of the container into another container where none of it is lost. Half fill the first container with water. Put back about half the sand and rod it with a steel rod, about 6 mm in diameter, so that its volume is reduced to a minimum. Then add the remainder and level the top surface of the inundated sand. Measure its depth at the middle with the steel rule. Suppose this is 'Y' cm. The percentage of bulking of the sand due to moisture shall be calculated from the formula:

- Percentage bulking = (X/Y -1) X 100

Method-2:

- In a 250 ml measuring cylinder, pour the damp sand, consolidate it by staking until it reached the 200 ml mark. Then fill the cylinder with the water and stir the sand well (the water shall be sufficient to submerge the sand completely). It will be seen that the sand surface is now below its original level. Suppose the surface is at the mark of Yml, the percentage of bulking of sand due to moisture shall be calculated from the formula.

- Percentage bulking= (200/Y – 1) x 100

DETERMINATION OF PARTICLE SIZE

The apparatus, sample size and test procedure shall be same as specified in sub-head 'MORTARS'. In order that the sieves shall not be overloaded, care must be taken to ensure that the maximum sieve loads shown in Table A-4.1 (below) are not exceeded at the completion of sieving.

TABLE 5.12
Particle Size Specification

I.S. Sieve Designation	Maximum weight for	
	45 cm dia sieve **Kg**	**30 cm dia sieve** **kg**
45 mm	**10**	4.5
40 mm	8	3.5
31.5 mm or 22.1 mm	6	2.5
20 mm	4	2.0
16 mm or 12.5 mm	3	1.5
10 mm	2	1.0
5.6 mm	1.5	0.75
4.75 mm	1.0	0.50
3.35 mm	-	0.30

Test Procedure:

- The sample weight taken will thus normally require several operations on each sieve. Each sieve should be taken separately over a clean tray or receiver until no more than a trace passes, but in any case for not less than two minutes.
- Materials should not be forced through the apertures but hand placing is permitted. A light brush should be used with fine sieves.
- The cumulative weight passing each sieve should be calculated as percentage of the total sample weight to the nearest whole number.

SLUMP TEST
Apparatus:

Mould shall consist of a metal frustum of cone having the following internal dimensions:

Bottom diameter ..... 20 cm
Top diameter..... 10 cm
Height… ..... 30 cm

The mould shall be of a metal other than brass and aluminium of at least 1.6 mm (or 16 BG) thickness. The top and bottom shall be open and at right angles to the axis of the cone. The mould shall have a smooth internal surface. It shall be provided with suitable foot pieces and handles to facilitate lifting it from the moulded concrete test specimen in a vertical direction as required by the test. A mould provided with a suitable guide attachment may be

used. Tamping rod shall be of steel or other suitable material 16 mm in diameter 60 mm long and rounded at one end.

Procedure:

- The internal surface of the mould shall be thoroughly cleaned and free form superfluous moisture and any set concrete before commencing the test.
- The mould shall be placed on a smooth horizontal, rigid and non-absorbent surface viz. levelled metal plate. The operator shall hold the mould firmly in place while it is being filled with test specimen of concrete.
- The mould shall be filled in four layers, each approximately one quarter of height of mould. Each layer shall be tamped with twenty five strikes of the rounded end of the tamping rod. The strokes shall be distributed in a uniform manner over the cross section of the mould and for the second and subsequent layers shall penetrate into the underlying layer. The bottom layer shall be tamped through out its depth.
- After the top layer has been rodded, the concrete shall be struck off level with trowel or the tamping rod, so that the mould is exactly filled. Any mortar which shall leak out between the mould and the base plate shall be cleaned away.
- The mould shall be removed from the concrete immediately after filling by raising it slowly and carefully in a vertical direction.
- The moulded concrete shall then be allowed to subside and the slump shall be measured immediately by determining the difference between the height of the mould and that of the highest point of specimen.
- The above operations shall be carried out at a place free from vibration or shock, and within a period of two minutes after sampling.

Result:

The slump shall be recorded in terms of millimeters of subsidence of the specimen during the test. Any slump specimen which collapses or shears off laterally give incorrect result. If this occurs, the test shall be repeated with another sample.

✓The slump test shall not be used for very dry mixes as the results obtained are not accurate.

TABLE 5.13

Physical Requirements of Fly Ash

Sl. No	*Characteristics*	*Requirement of Fly Ash*	
		For use as Pozzolana	*For use as Pozzolana*
I	Fineness- Specific surface in m2/kg by Blaine's permeability method, min	320	200
II	Lime reactivity – average compressive strength in N/mm2 Min	4.5	-
III	Compressive strength at 28 days in N/ mm2	Not less than 80 per cent of the strength of corresponding Mortar cubes.	-

IV	Soundness of autoclave test expansion of specimens, per cent, max	0.8	0.8
V	Particles retained on 45 micron IS sieve (wet sieving) in percent maximum	34	50

CUBE TEST FOR COMPRESSIVE STRENGTH OF CONCRETE – MANDATORY LAB TEST

One sample (consisting of six cubes 15x15x15 cm shall be taken for every 20 cum or part thereof concrete work ignoring any part less than 5cum or as often as considered necessary by the Engineerin- Charge. The test of concrete cubes shall be carried out in accordance with the procedure as described below. A register of cubes shall be maintained at the site of work in Appendix C. The casting of cubes, concrete used for cubes and all other incidental charge, such are curing, carriage to the testing laboratory shall be borne by the contractors. The testing fee for the cubes, if any, shall be borne by the department.

Test Procedure:

Mould

- The mould shall be of size 15cm x15cmx15cm for the maximum nominal size of aggregate not exceeding 40 mm. For concrete with aggregate size more than 40 mm size of mould shall be specified by the Engineer-in-charge, keeping in view the fact that the length of size of mould should be about four times the size of aggregate. The moulds for test specimens shall be made of non-absorbent material and shall be substantially strong enough to hold their form during the moulding of test specimens. They shall not vary from the standard dimensions by more than one percent. The moulds shall be so constructed that there is no leakage of water from the test specimen during moulding. All the cube moulds for particular site should, prior to use, be checked for accuracy in dimensions and geometric form and such test should at least be made once a year. Each mould shall be provided with a base plate having a plane surface and made ofnon-absorbent material. This plate shall be large enough in diameter to support the moulds properly without leakage. Glass plates not less than 6.5mm thick or plain metal not less than 12mm thick shall be used for this purpose. A similar plate shall be provided for covering thetop surface of the test specimen when moulded.

- **Note:** Satisfactory moulds can be made from machine or steel castings, rolled metal plates or galvanized.

Sample of Concrete

- Sample of concrete for test specimen shall be taken at the mixer or in the case of ready mixed concrete from the transportation vehicle discharge or as directed by Engineer-in- Charge. Such samples shall be obtained by repeatedly passing a scoop or pail through the discharge stream of concrete.
- The sampling operation should be spread over evenly to the entire discharging operation. The samples thus obtained shall be transported to the place of moulding of the specimen to counteract segregation. The concrete shall be mixed with a shovel until it is uniform in appearance. The location in the work of the batch of concrete this sampled shall be noted for further reference. In case of paving concrete, samples shall be taken from the batch immediately after deposition of the sub grade. At least five samples shall be taken from

different portion of the pile and these samples shall be thoroughly mixed before being used to form the test specimen. The sampling shall be spread as evenly as possible throughout the day. When wide changes occur during concreting, additional sample shall be taken if so desired by the Engineer-in-Charge.

Preparation of Test Specimens

- The interior surfaces of the mould and base plate shall be lightly oiled before the concrete is placed in the mould. The samples of concrete obtained as described under the test specimen shall be immediately moulded by one of the following methods as indicated below:-

- When the job concrete is compacted by manual methods, the test specimen shall be moulded by placing the fresh concrete in the mould in three layers, each approximately one third of the volume of the mould. In placing each scoopful of concrete the scoop shall be moved around the top edge do the mould as the concrete there sided from it, in order to ensure a uniform distribution of concrete within the mould. Each layer shall be rodded 35 times with 16 mm rod, 60 cm in length, bullet pointed at the lower end. The strokes shall be distributed in uniform manner over the cross section of the mould and shall penetrate into underlying layer. The bottom layer shall be rodded through its depth. After the top layer hasbeen rodded, the surface of the concrete shall be struck off with a trowel and covered with a glass plate at least 6.5 mm thick or a machined plate. The whole process of moulding shall be carried out in such a manner as to preclude the change of the water cement ratio of the concrete, by loss of water either by leakage from the bottom or over flow from the top ofthe mould. When the job concrete is placed by vibration and the consistency of the concrete is such that the test specimens cannot be properly moulded by hand rolling as described above, the specimens shall be vibrated to give a compaction corresponding to that of the jobconcrete. The fresh concrete shall be placed in mould in two layers, each approximately halfthe volume of the mould. In placing each scoopful of concrete the scoop shall be moved around the top edge of the mould as the concrete there slides from it, in order to ensure a symmetrical distribution of concrete within the mould. Either internal or external vibrators may be used. The vibration of each layer shall not be continued longer than is necessary to secure the required density. Internal vibrators shall only be used when the concrete is required to be compacted in layers. In compacting the first layer, the vibrators shall not be allowed to rest on the bottom of the mould. In placing the concrete for top extent that there will be no mortar loss during vibrations. After vibrating the second layer enough concrete shall be added to bring level above the top of the mould. The surface of the concrete shall then the struck off with a trowel and covered with a glass or steel plate as specified above. The whole process of moulding shall be carried out in such a manner as to preclude the alteration of water-cement ratio of the concrete by loss of water, either by leakage for the bottom or over flow from the top of the mould.

Curing and Storage of Test Specimen

- In order to ensure reasonably uniform temperature and moisture conditions during the first 24 hours for curing the specimen and to protect them from damage, moulds shall be covered with wet straw or gunny sacking and placed a storage box so constructed and kept on the work site that its air temperature when containing concrete specimens shall remain 22ºC to 33ºC. Other suitable means which provide such a temperature and moisture conditions may be used.

- ✓ **Note: -**It is suggested that the storage box be made of 25 mm dressed tongued and grooved

timber, well braced with battens to avoid warping. The box should be well painted inside and outside and should be provided with a hinged cover and padlock. The test specimen shall be removed from the moulds at the end of 24 hours and stored in a moist condition at a temperature within 24ºC to 30ºC until the time of test. If storage in water is desired, asaturated lime solution shall be used.

Testing Procedure:

- The specimens shall be tested in accordance with procedure as described below:

- The tests shall be made at an age of concrete corresponding to that for which the strengths are specified.

- Compression tests shall be made immediately upon removal of the concrete test specimen from the curing room i.e. the test specimen shall be loaded in damp condition. The dimensions of the test specimens shall be measured in mm accurate to 0.5 mm.
- The metal bearing plates of the testing machine shall be placed in contact with the ends of the test specimens. Cushioning materials shall not be used. In the case of cubes, the test specimen shall be placed in the machine in such a manner that the load is applied to sides ofthe specimens as cast. An adjustable bearing block shall be used to transmit the load to the test specimen. The size of the bearing block shall be the same or slightly larger than that of test specimen. The upper or lower section of the bearing block shall be kept in motion as thehead of the testing machine is brought to a bearing on the test specimen.

- The load shall be applied axially without shock at the rate of approximately 140 kg. per sq.cm. per minute. The total load indicated by the testing machine at failure of test specimen shall be recorded and the unit compressive strength is calculated in kg per sq. cm. using the area computed from the measured dimension of the test specimen. The type of failure and Appearance of the concrete shall be noted.

ADDITIONAL TESTS FOR CONCRETE

- In case the concrete fails when tested as per the method prescribed in Appendix A, one or more of the following check tests may be carried out at the discretion of Engineer-in-Charge to satisfy the strength of the concrete laid. All testing expenditure shall be borne by the contractor, the number of additional tests to be carried out shall be determined by the Engineer-in-Charge. He shall be the final authority for interpreting the results of additional test and shall decide upon the acceptance or otherwise. His decision in this regard shall be final and binding. For the purpose of payment, the Hammering test results only shall be the criteria. Some of the tests are outlined below:-

REBOUND HAMMER TEST

- If a rebound hammer is regularly used by trained personnel in accordance with procedure described in IS 13311 (part II) and a continuously maintained individual charts are kept showing a large number of reading and the relation between the reading and strength of concrete cubes made from the same batch of concrete, such charts may be used in conjunction with hammer readings to obtain an approximate indication of the strength of concrete in a structure for element. If calibration charts are available from manufactures, it can be used. When making rebound hammer test each result should be the average of at least 12 readings. Reading should not be taken within 20mm of the edge of concrete members and it may be necessary to distinguish between readings taken on a troweled face and those on a moulded face. When making the tests on a precast unit, special care shouldbe taken to bed them firmly against the impact of the hammer.

CUTTING CORES

- This method involves drilling and testing cores from the concrete for determination of compressive strength. In suitable circumstances, the compressive strength of the concrete in the structure may be assessed by drilling cores from the concrete and testing. The procedure used shall comply with the requirements of IS 1199 and IS 516. The points from which cores shall be taken shall be representative of the whole concrete and at least three cores shall be obtained and tested. If the average of the strength of all cores cut from the structure is less than the specified strength, the concrete represented by the cores shall be liable to rejection and shall be rejected if a static load test (B-5) either cannot be carried out or is not permitted by the Engineer-in-Charge.

ULTRASONIC TEST

- If an ultrasonic apparatus is regularly used by trained personnel in accordance with IS 13311 (part I) and continuously maintained individual charts are kept showing a large number of readings & the relation between the reading and strength of cubes made from the same batch of concrete, such charts may be used to obtain approximate indications of the strength of concrete in the structures. In cases of suspected lack of compaction or low cube strength the results obtained from the ultrasonic test results on adjacent acceptable section of the structures may be used for the purpose of assessing the strength of concrete in the suspected portion.

LOAD TESTS ON INDIVIDUAL PRECAST UNITS

- The load tests described in this clause are intended as check on the quality of the units and should not be used as substitute for normal design procedure. Where members require special testing. Such special testing procedures shall be in an accordance with the specification. Test loads shall be applied and removed incrementally.

Non Destructive Tests

- The unit shall be supported at its designed point of support and loaded for five minutes with a load equal to the sum of the characteristic dead load plus one and a quarter time the characteristic imposed load. The deflection is then recorded. The maximum deflection after application of the load shall be in accordance with the requirements defined by the Engineer-in-Charge. The recovery is measured five minutes after the removal of the load and the load then reimposed. The percentage recovery after the second loading shall be not less than that after the first loading nor less than 90% of the deflection recorded during the second loading. At no time during the tests, shall the unit show any sign of weakness or faulty construction as defined by the Engineer-in-Charge in the light of reasonable interpretation of relevant data.

CALIBRATION AND WEIGHING EQUIPMENT ACCURACY

- **The following limits shall apply to all ready-mixed concrete plants :**

(a) The accuracy, sensitivity and arrangement of the weighing devices shall be such as to enable the materials to be batched within the following tolerances:

(1) Cement, mineral Admixtures	Within + 2 percent of the quantity of the constituent being measured
(2) Aggregate, chemical admixtures and water	Within + 3 percent of the quantity of the constituent being measured.

(b) Analogue scales shall have scale increments not exceeding 5 kg. for cement and mineral admixtures, 25 kg. for aggregate and 2 kg. for water.

(c) Preset controls shall be calibrated in increments not exceeding 5 kg. for cement and mineral admixtures, 10 kg. for aggregate and 2 kg. for water.

(d) For continuous mixer plants calibration shall be in increments not exceeding 10 kg. /m3 for cement and mineral admixtures, 25 kg. /m3 for aggregates and 10 l/m3 for water.

(e) Digital readous shall have scale increments not exceeding 2 kg. for cement and mineral admixtures, 10 kg. for aggregate and 10 for water.

(f) At the time of installation, or reconditional the accuracy of the indicated mass at any point on the scale shall be within 0.25 percent of the full scale reading.

(g) Any other time during the masonry operation the accuracy shall be within 0.50 percent of the full scale reading.

(h) Chemical Admixture dispensers shall have scale increment for exceeding.

TABLE 5.14
Admixtures Quantity Specifications

Ranger of scale in kg/l	Scale increment in Kg/l
0.1 – 0.5	0.01
0.5 - 1.0	0.02
1.0 – 10.0	0.2
more than 10.0	0.4

(i) All weighing and measuring equipment shall be tested and calibrated over its full working Range at the following intervals:

(1) Mechanical /knife edge systems	At least once every two month
(2) Electrical /load cell systems	At least once every three months

Adequate and identified facilities shall be provided for the application of the test loads.
(j) In the case of batch weighing systems testing and calibration shall be based on the application test loads to the weigh hoppers.

(k) Checks on continuous weigh systems shall be based on comparison of preset quantities with those actually produced.

(l) To achieve the required accuracy of calibration, a minimum of 500 kg. Of stamped weights are required, except that for low capacity scale an acceptable limit on the total mass of calibration weight would be 20 percent of the scale capacity.

(m) When calibration of weighing equipment is carried out all personnel involved should be competent and fully trained, the procedures should be fully documented, and special attention should be paid to the health and safety aspects of the procedure.

TABLE 5.15
Register of Work Test of Concrete (by volume)

(a) Name of work	Concrete mix, (by volume)	Compressive strength in kg/cm2 on 7 days
(b) Name of Contractor	1:1:2	210
(c) Agreement No.	1:1.5:3	175
(d) Sample No.	1:2:4	140
(e) Identification mark		
(f) Portion of work any quantity represented by sample		
(g) Date and time of casting cube		
(h) Proportion of mix/ grade of concrete		

6 Days' Test

(1) Due date of test	Cube No.
(2) Actual date of test	
(3) (a) Minimum strength	
(b) Maximum strength	
(c) Average strength of three cubes	
(d) Difference between 3a and 3b	
(e) Difference in % age in terms of Average strength i.e.$\frac{3b - 3a \times 100}{3c}$	
(4) Specified compressive strength of concrete mix used	
(5) (a) If 3(e) is more than 30%	Sample is not acceptable, then 28 days Strength test shall be carried out.
(b) If 3(e) is equal to or less than 30% then proceed as below:-	
(i) Difference between column 4 specified compressive strength and column 3 (c) i.e., actual average is higher, it will be denoted (+) and (-) if it is less	
(ii) Difference in column 5(b) (i) terms of % age of specified strength	$\frac{3(c) - (4) \times 100\%}{4}$
(iii) If the difference in column 5(b) (i) Is +ve and the same in terms of % age of specified strength (4) i.e., value of col. 5(b) (ii) is within (+15% range)	Acceptable & strength is considered in order

TABLE 5.15
Register of Work Test for Concrete

(a) Name of work	Concrete mix, (by volume)	Compressive strength in kg/cm2 on 7 days
(b) Name of Contractor	1:1:2	315
(c) Agreement No.	1:1.5:3	265
(d) Sample No.	1:2:4	210
(e) Identification mark		
(f) Portion of work any quantity represented by sample		
(g) Date and time of casting cube		
(h) Proportion of mix/ grade of concrete		

7 Days' Test

(1) Due date of test	Cube No.
(2) Actual date of test	
(3) Actual compressive strength of cubes (min. no. of cubes to be tested –three) (a) Minimum strength	
(b) Maximum strength	
(c) Average strength of three cubes	
(d) Specified compressive strength of concrete mix used	
(e) 70% specified strength	i.e. 70% of 3(d)
(f) 130% of specified strength	i.e. 130% of 3(d)
(4) If 3(b) = 3(f) and 3(a) > 3(e)	Value of 3(c) shall be compressive strength of sample
(5) If 3 (c) is more than 3(f)	EE may order further investigation
(6) If any test value exceeds 3(f)	It should be restricted to 3(f) for computation of strength
(7) If 3 (c) > 3(d) but < 3(f)	Strength is in order and concrete accepted at full rates.
(8) If 3 (c) < 3(d) and > 3(e)	Concrete may be accepted at reduced rates in accordance with para 5.4.13.2
(9) If 3(c) < 3(e)	Work represented by this sample shall be rejected and action taken as prescribed in clause 5.4.10.4

Item No.	SubItem No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.1		RCC upto plinth	Providing and laying in position specified grade of reinforced cement concrete (nominal / design mix with 20mm nominal size graded stone aggregate) excluding the cost of centering,shuttering, finishing and reinforcement -All work up to plinth level :		
	5.1.1	M-20	M-20 Grade Concrete	cum	6153.00
	5.1.2	M-15	M-15 Grade Concrete	cum	5397.00
5.2		RCC in walls, columns, pillars etc	Reinforced cement concrete work (with 20 mm nominal size graded stone aggregate) in walls (any thickness), including attached pilasters, buttresses, plinth and string courses, fillets, columns, pillars, posts and struts etc. above plinth level and upto floor two level excluding cost of centering, shuttering, finishing and reinforcement:		
	5.2.1	M-20	M-20 Grade Concrete	cum	6231.00
5.3		RCC in beams, floors, staircases etc	Reinforced cement concrete work (with 20 mm nominal size graded stone aggregate)in beams, suspended floors, roofs havingslope of any degree landings, balconies, shelves, chajjas, lintels, bands, plain window sills, staircases and spiralstaircases above plinth level and upto floor two level excluding the cost of centering, shuttering, finishing and reinforcement inconcrete grade.		
	5.3.1	M-20	M-20 Grade Concrete	cum	6541.00
5.4		RCC in kerbs, steps	Providing and laying upto floor two level reinforced cement concrete (with20 mm nominal size graded stone aggregate) in kerbs, steps and the like excluding the cost of centering, shuttering, finishing and reinforcement with Concrete grade.		
	5.4.1	M-20	M-20 Grade Concrete	cum	5542.00
5.5		RCC in arches, domes, shells etc.	Reinforced cement concrete work (with 20 mm nominal size graded stone aggregate) in arches, archribs, domes, vaults, shells, folded plate above plinth level and upto floor two level excluding the cost of centering, shuttering, finishing and reinforcement in Concrete grade.		
	5.5.1	M-20	M-20 Grade Concrete	cum	6906.00

Item No.	SubItem No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.6		RCC in chimneys and shafts etc.	Reinforced cement concrete work (with 20 mm nominal size graded stone aggregate) in chimneys, shafts, upto floor two level excluding the cost of centring, shuttering, finishing and reinforcement with		
	5.6.1	M-20	M-20 Grade Concrete	cum	6337.00
5.7		RCC in well steining	Reinforced cement concrete work (with 20 mm nominal size graded stone aggregate) in well-steining excluding the cost of centering, shuttering, finishing and reinforcement with		
	5.7.1	M-20	M-20 Grade Concrete	cum	5448.00
5.8		RCC in fins	Reinforced cement concrete work (with 20 mm nominal size graded stone aggregate) in vertical and horizontal fins individually or forming box louvers, facias and eaves boards up to floor two level excluding the cost of centering, shuttering, finishing and reinforcement in M 20 –Grade Concrete	cum	6518.00
5.9		Centering and Shuttering	Centering and shuttering in cluding strutting, propping etc. and removal of form for:		
	5.9.1	Upto plinth	Foundations, footings, bases of columns, etc. for concrete upto plinth level	sqm	190.60
	5.9.2	Walls	Walls (any thickness) in cluding attached pilasters butter esses, plinth beams and string courses etc.		
	5.9.2.1	Upto plinth level	Uptoplinthlevel.	sqm	219.00
	5.9.2.2	Above plinth	Above plinth upto 3.6 m height from plinth level	sqm	330.00
	5.9.2.3	Extra for additional height in centering, shuttering	Extra for additional height in centering, shuttering where ever required with adequate bracing, propping etc. including cost of de-shuttering and decentering at all levels, over a height of 3.6 m, for every additional height of 1 metre or part thereof.	sqm	40.00
	5.9.3	Suspended floors, roofs etc	Suspended floors, roofs, landings, balconies and access platform.	sqm	439.60

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	5.9.4	Shelves	Shelves(Cast in situ)	sqm	290.00
	5.9.5	Lintels, Beams and Centilevers	Lintels, beams, girders, bressumers and cantilevers	sqm	356.00
	5.9.6	Columns	Columns, Pillars, Piers, Abutments, Posts and Struts.	sqm	471.20
	5.9.7	Staircase	Stairs, (excluding landings) except spiral-Staircases.	sqm	435.70
	5.9.8	Spiral Staircase	Spiral staircases (including landing).	sqm	450.00
	5.9.9	Arches	Arches, domes,vaults upto 6m span	sqm	1141.60
	5.9.10	Extra for arches >6m	Extra for arches, domes, vaults exceeding 6 m span	sqm	451.30
	5.9.11	Chimneys and shafts	Chimneys and shafts	sqm	294.00
	5.9.12	Coffer slab	Coffer/waffle slab of any size or shape as shown in the drawing.	sqm	1481.00
	5.9.13	Well steining	Well steining	sqm	264.00
	5.9.14	Fins	Vertical and horizontal fins individually or forming Box louvers band, facias and eaves boards.	sqm	643.40
	5.9.15	Extra for circular or other geometrical shape	Extra for shuttering in circular work or any other geometrical shape (20% ofrespective centering and shuttering items).	sqm	20%
	5.9.16	Lintels	Small lintels not exceeding 1.5m clear span, mouldingas incornices, window, sills, courses, bands, copings, bed plates, anchor blocks and the like.	sqm	191.00
	5.9.17	Cornices and mouldings	Cornices and mouldings	sqm	565.00
	5.9.18	Small surfaces	Small surfaces such as cantilever ends, brackets and ends of steps, caps and bases to pilasters and columns and the like.	sqm	529.60
	5.9.19	Weather shade and chajjas	Weather shade, Chajjas, corbels etc., including Edges.	sqm	534.20

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	5.9.20	Suspended floors with ply	Suspended floors, roofs, landings, balconies and Access platform. With water proof ply 12 mm thick. (item to be executed with prior permission of C.E./S.E. in case of B & R works .	sqm	575.35
	5.9.21	Lintels, beams, columns etc with ply	Lintels, beams, columns, girders, bressumers and cantilevers with waterproof ply 12 mm thick (item to be executed with prior permission of C.E./S.E. in case of B & R works .	sqm	521.20
5.10		Extra for additional height in centering, shuttering	Extra for additional height in centering, shuttering where ever required with adequate bracing, propping etc. including cost of de-shuttering and decentering at all levels, over a height of 3.6 m, for every additional height of 1 metre or part thereof (Plan area to be measured)		
	5.10.1	Suspended floors	Suspended floors, roofs, landing, beams and balconies (Plan area to be measured).	sqm	212.00
5.11		Precast RCC M-15	Providing, hoisting and fixing up to floor two level precast reinforced cement concrete in lintels, manhole cover, shelves and like including setting in cement mortar 1:3 (1 cement : 3 sand), cost of required centering, shuttering and finishing with neat cement punning on exposed surfaces but excluding the cost of reinforcement with M 15 –Grade Concrete.	cum	7830.00
5.12		Precast RCC Jali 1:2:4	Providing precast cement concrete Jali 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 6mm nominal size) reinforced with 1.6 mm dia mild steel wire including centering and shuttering, roughening cleaning, fixing and finishing in cement mortar 1:3 (1 cement: 3 sand) etc. complete excluding plastering ofthe jambs, sills and soffits.		
	5.12.1	50mm thick	50 mm thick	sqm	790.00
	5.12.2	40mm thick	40 mm thick	sqm	689.00
	5.12.3	25mm thick	25 mm thick	sqm	598.00
5.13		Encasing RCC in rolled steel sections	Encasing rolled steel sections, in beams and columns, with cement concrete M 15 (1 cement: 2 sand: 4 graded stone aggregate 20mm nominal size) including centering and shuttering complete but excluding cost of reinforcement.	cum	6934.00

Item No.	SubItem No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.14		Encasing RCC in grillages	Encasing rolled steel section in grillages with cement concrete M 15 (1 cement: 2 sand: 4 graded stone aggregate 20 mm nominal size) including centering and shuttering but excluding cost of expanded metal and hangers.	cum	4835.00
5.15		Expanded metal for encasing rolled steel sections	Extra for providing and fixing expanded metal mesh of size 20x60mm and strands 3.25mm wide 1.6mm thick weighing 3.64 kg per sqm for encasing of rolled steel sections in beams, columns and grillages excluding cost of hangers.	sqm	408.00
5.16		Reinforcement	Reinforcement for R.C.C. work including straightening, cutting, bending, placing in position and binding including cost of binding wire all wastages and overlaps, couplers (overlaps shall be provided as per requirement of IS 13920 for ductile detalingIS 456 RCC Design & SP 34 Reinforcement detailing) etc. complete . (**Note:-** 1. Spacer bars (chairs) shall be paid separately as shown in the drawing and as per direction of engineer in charge. 2. No extra payment shall be done for overlaps and/or couplers. 3. Couplers shall be conforming to IS code on "Reinforcement Couplers for Mechanical Splices of Bars for Concrete Reinforcement - Specification")		
	5.16.1	Mild and medium	Mild steel and Medium Tensile steel bars.	kg	78 .00
	5.16.2	Hard	Hard drawn steel wire	kg	75.00
	5.16.3	Cold twisted	Cold twisted bars (CTD)	kg	81.00
	5.16.4	Hot rolled	Hot rolled deformed bars	kg	81.00
	5.16.5	Hard drawn	Hard drawn steel wire fabric	kg	76.00
	5.16.6	TMT (FE 500D or more)	Thermo-Mechanically Treated bars.FE 500 D or more conforming to IS 1786	kg	80.00
	5.16.7	TMT (FE 500 or more)	Thermo-Mechanically Treated bars. FE 500 conforming to IS 1786.	kg	77.00
	5.16.8	CRS bars	Corrosion resistance Steel bar Grade Fe-500 D & above , use with prior permission of C.E.	Kg	98.00
5.17		Reinforcement above floor two level	Add extra for providing reinforcement above Floor two level for every additional floor or part thereof.	kg	0.20

Item No.	SubItem No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.18		Copper plate for expansion joints	Providing and fixing in position copper plate as per design for expansion joints.	Kg	472.00
5.19		Blown bitumen	Providing and filling in position, blown bitumen in expansion joints.	per cum	45353.00
5.20		Bitumen mix filler	Providing and filling in position bitumen mix filler of Proportion 80 kg. Of hot bitumen, 1 kg. Of cement and 0.25 cubic metre of sand for expansion joints.	per cum	13863.00
5.21		Silicon sealant	Providing and filling gap in expansion joint between RCC/ Stone/Ceramic tiles/Granite/ Marble by providing silicon sealant over backer rod of approved quality weather sealant as per drawing and direction of engineer in charge up to 25mm width & 12 mm depth.	Per cm depth per cm width one m long.	104.54
5.22		Bitumen impregnated fibre board	Providing and fixing in position bitumen impregnated fibre board conforming to IS: 1838 including cost of primer, sealing compound in expansion joints.		
	5. 22.1	12 mm thick	12 mm thick	sqm	402.00
	5. 22.2	25 mm thick	25 mm thick	sqm	574.00
	5. 22.3	50 mm thick	50 mm thick	sqm	674.00
5.23		Sheet cover over expansion joints	Providing and fixing sheet covering over expansion joints with iron screws as per design.		
	5. 23.1	Fibre cement	Non-asbestos fibre cement board 6 mm thick as per IS: 14862.		
	5. 23.1.1	150mm	150mm wide.	RM	102.00
	5. 23.1.2	200mm	200mm wide.	RM	137.00
	5. 23.1.3	250mm	250mm wide.	RM	153.00
	5. 23.2	Aluminium	Aluminium fluted strips 3.15mm thick.		

Item No.	SubItem No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	5. 23.2.1	150 mm	150 mm wide.	RM	483.00
	5. 23.2.2	200 mm	200 mm wide.	RM	646.00
	5. 23.2.3	250 mm	250 mm wide.	RM	725.00
	5. 23.3	SS -304	Stainless steel Grade 304 plate-1.0 mm thick		
	5. 23.3.1	150 mm	150 mm wide.	RM	367.00
	5. 23.3.2	200 mm	200 mm wide.	RM	443.00
	5. 23.3.3	250 mm	250 mm wide.	RM	555.00
5.24		Plaster drip course / groove	Providing for plaster drip course/ groove in plastered surface or moulding to R.C.C. projections.		
	5.24.1		Preparation of grooves	RM	18.50
	5.24.2		Preparation of Drip course	RM	30.00
5.25		Extra RCC for in or under water	Extra for laying reinforced cement concrete in or under water and/ or liquid mud including cost of pumping or bailing out water and removing slush etc., complete. **Note-** For Item No. 5.25:-The quantity will be calculated by multiplying the depth measured from the subsoil water level up to the centre of gravity of the R.C.C. under ubsoil water with the quantity of R.C.C. in cubic metre executed under subsoil water. The depth of centre of gravity shall be reckoned correct to 0.1 m. 0.05 m or more shall be taken as 0.1 m and less than 0.05 m ignored. No extra payment shall be made for placing reinforcement or centering & shuttering under sub – soil water condition	cum	297.00
5.26		Extra for RCC in or under foul position	Extra for laying reinforced cement concrete in or under foul positions.	cum	109.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.27		BMC	Providing and laying in position machine batched, machine mixed cement concrete (BMC) of M-20 grade for reinforced cement concrete work using cement content as per approved mix design including use of admixtures in recommended proportions as per IS 9103 to accelerate/retard setting, improve workability of concrete without impairing strength and durability as per direction of Engineer-in charge and pumping of concrete to site of laying and finishing but excluding the cost of centering, shuttering, and reinforcement. (Note: - Minimum cement content is 320 kg/cum) and no extra payment for extra cement used as per mix design.		
	5.27.1	Up to plinth	All works up to plinth level.	cum	6100.00
	5.27.2	Above plinth up to floor 5	All works above plinth level and up to floorfive level.	cum	6581.00
	5.27.3	Above floor 5	Add extra for each floor for all works above floor five level.	cum	72.00
5.28		Extra for richer BMC mix/ RMC	Extra for providing richer mixes of BMC/RMC design mix as per item no 5.27&5.30 up to plinth level and at all floor levels. (No extra payment for using extra cement as per requirement of mix design.)		
	5. 28.1	M-25	Providing M-25 grade concrete instead of M-20 grade. (Note : minimum cement content is @ 330 kg/cum)	cum	110.00
	5. 28.2	M-30	Providing M-30 grade concrete instead of M-20 grade. (Note : minimum cement content is @ 340 kg/cum)	cum	140.00
	5. 28.3	M-35	Providing M-35 grade concrete instead of M-20 grade.(Note: : minimum cement content is @ 350kg/cum)	cum	280.00
	5. 28.4	M-40	Providing M-40 grade concrete instead of M-20 grade.(Note: minimum cement content is @ 360kg/cum)	cum	421.00
5.29		Precast RCC waffle unit	Providing and placing in position precast reinforced cement concrete waffle units square or rectangular as per design and shape for floors and roofs in M-20 –Grade Concrete (with graded stone aggregate 10mm nominal size) including flush or deep ruled pointing at	cum	13479.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			joints in Cement mortar 1:2 (1 Cement : 2 sand), making necessary holes of required sizes for carrying through service lines etc. including provision of steel hooks for lifting and form work, handling, hoisting, centering and erection complete for all floor levels but excluding the cost of reinforcement.		
5.30		RMC M-20	Providing and laying in position ready mixed concrete (RMC) of M-20 grade for reinforced cement concrete work, using cement content as per approved design mix and manufacturedin fully automatic batching plant andtransported to site of work in transit mixer for all leads, having continuous agitated mixer, manufactured as per mix design of specified grade for reinforced cement concrete work, including pumping of R.M.C. from transit mixer to site of laying, excluding the cost of centering, shuttering, finishing and reinforcement, including cost of admixtures in recommended proportions as per IS : 9103 to accelerate / retard setting of concrete, improve workability without impairing strength and durability as per direction of the Engineer - in - charge. **Note:** Minimum cement content is @ 320 kg/cum. (No extra payment for using extra cement as per requirement of mix design.)		
	5.30.1	Upto plinth	All works upto plinth level.	cum	6332.00
	5.30.2	Upto floor V level	All works above plinth level and upto floor V level.	cum	6814.00
	5.30.3	Extra for Additional floor	Extra for each additional floor for all works above floor five level	cum	72.00
5.31		Extra for R.C.C. above floor two level	Extra for R.C.C. work above floor two level for each floors or part thereof. (except items 5.27 and 5.30)	cum	1% of respective item
5.32		Water based concrete curing compound	Supplying and applying pre tested and approved water based concrete curing compound to concrete/ masonry surface, allas per manufacturer's specification and direction of Engineer-in-charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
	5.32.1	Non pigmented	Non pigmented wet curing compound	sqm	93.00
	5.32.2	Pigmented	Pigmented wet curing compound.	sqm	100.00
5.33		Doors windows RCC frames	Providing and fixing in position factory made precast RCC M-40 doors, windows and ventilator frames having excellent smooth finish, as per IS: 6523 with 6mm dia main reinforcement (3 nos for single rebate and 4 nos for double rebate) tied with 3mm mild steel stirrups @ 200mm c/c, with high destinyvirgin polymer blocks of size 25X25X125 mm, to receive hinges (3 to 4 for door and window and at least 2 for ventilator), each polymer block having at least four holes for passing one of the main reinforcement bar, providing specially designed M.S. galvanised sleeves for accommodating 6mm dia. fullythreaded hold fast bolts, and providing hole cap for receiving 16mm dia sliding door bolts (L-drop), providing virgin high density polymer blocks for fixing socket of tower boltwindow frame provided with holes between or adjacent rebates to hold fast grill or steel bar, ventilator frame provided with grooves for louvers at an angel of 45 degree at spacingto block direct vision etc. complete, as per the direction of Engineer-in-charge. The cost of hold fast and cement concrete block of 1:3: 6 (1 Cement: 3 sand: 6 Graded stone aggregate 20mm nominal size) mix is also included in the item.		
	5. 33.1.	Double rebate	Frame 125 mmx 60 mm (double rebate)	RM	366.00
	5. 33.2.	Single rebate	Frame 100mmx 60 mm (Single rebate)	RM	339.00
	5. 33.3	Single rebate	Frame 85mmx 60 mm (Single rebate)	RM	325.00
5.34		6 mm plaster (cm1.3) over RCC	Smooth finishing of the exposed surface of RCC work with 6mm thick cement mortar 1:3 (1 cement: 3 sand).	sqm	144.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.35		Compressible filler in expansion joints.	Providing and fixing in position pre moulded compressible filler board made from polymer base in black color approved make conforming to the specifications having minimum density of 95 kg / cum, non- staining, with less than 1 percent waterabsorption and compression recovery of 93 percent minimum including cost of double sided adhesive tape, sealant etc. in expansion joints.		
	5. 35.1	12 mm thick	12 mm thick	sqm	441.00
	5. 35.2	25 mm thick	25 mm thick.	sqm	894.00
	5. 35.3	50 mm thick	50 mm thick.	sqm	1194.00
5.36		Expansion joints with aluminum section	Providing and sealing the retaining wall expansion Joints with anodized aluminum extruded section of 135 mm × 3.1 mm confirm to alloy of 6063 T5/T6including fixing of an imported EPDM moisture Barrierconform to (American Society for Testing and Materials) 2240 & ASTM - G53 & 84 fixing with Epoxy adhesive confirm to ASTMC-881 complete in all respect as per manufacturer Drawing and specifications. With 5 year guarantee.	metre	3234.00
5.37		Expansion joints cover	Providing and fixing sheet covering over expansion joints with iron screws as per design. Non- asbestos cement bonded particle board 6 mm thick as IS: 14276		
	5.37.1	150mm	150mm wide	meter	96.00
	5.37.2	200mm	200mm wide	meter	130.00
	5.37.3	250mm	250mm wide	meter	160.00
5.38		High tensile steel wires	Providing & laying in position Prestressing steel strands (low relaxation) on hollow core bed by using mechanical pulling arrangement like Rabbit/ Bed master including all accessories for Stressing &destressing operations as per approved make conforming to IS1343 & grade FY-1860 etc, complete as per drawings and direction of Engineer -in-charge.	Kg	168.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.39		Expansion Joint for Floor	Providing and fixing of expansion joint system related with floor location as per drawings and direction of Engineer-In- Charge. The joints system will be of extruded aluminum base members, self aligning / self centering arrangement and support plates etc. as per ASTM B221-02. The system shall be such that it provides floor to floor /floor to wall expansion control system for various vertical localtion in load application areas that accommodates multi directional seismic movement without stress to it's components. System shall consist of metal profiles with a universal aluminum base member designed to accommodate various project conditions and finish floor treatments. The cover plate shall be designed of width and thickness requiredto satisfy projects movement and loading requirements and secured to base members by utilizing manufacturer's pre-engineered self-centering arrangement that freely rotates / moves in all directions. The Self - centering arrangement shall exhibit circular sphere ends that lock and slide inside the corresponding aluminum extrusion cavity to allow freedom of movement and flexure in all directions including vertical displacement. Provision of Moisture Barrier Membrane in the Joint System to have watertight joint is mandatory requirement all as per the manufactures design and as approved by Engineer - inCharge. (Material shall confirm to ASTM 6063).		
	5. 39.1	100 mm	Floor Joint of 100 mm gap	metre	4857.00
	5. 39.2	150 mm	Floor Joint of 150 mm gap	metre	5946.00
	5. 39.3	200 mm	Floor Joint of 200 mm gap	metre	7616.00
	5. 39.4	250 mm	Floor Joint of 250 mm gap	metre	9450.00
	5. 39.5	300 mm	Floor Joint of 300 mm gap	metre	11090.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
5.40		Expansion Joint for walls	Providing and fixing of expansion joint system related with wall joint (internal/ external) location as per drawings and direction of Engineer-In- Charge. The joints shall be of extruded aluminum base members, self aligning / centering arrangement and support plates as per ASTM B221- 02. The material shall be such that it provides an Expansion Joints System suitable for vertical wall to wall/ wall to corner application, both new and existing construction in office Buildings & complexes with no slipping down tendency amongst the components ofthe Joint System. The Joint System shall utilize light weight aluminum profiles exhibiting minimal exposed aluminumsurfaces mechanically snap locking the multicellular to facilitate movement.(Material shall confirm to ASTM 6063).		
	5. 40.1	100 mm	Wall Joint of 100 mm gap	metre	4044.00
	5. 40.2	150 mm	Wall Joint of 150 mm gap	metre	4552.00
	5. 40.3	200 mm	Wall Joint of 200 mm gap	metre	5293.00
	5. 40.4	250 mm	Floor Joint of 250 mm gap	metre	6034.00
	5. 40.5	300 mm	Floor Joint of 300 mm gap	metre	6775.00
5.41		Expansion joint for roofs	Providing and fixing of expansion joint system of approved make and manufactures for various roof locations as per approved drawings and direction of Engineer-In-Charge. The joints shall be of extruded aluminum base members with, self aligning and self centeringarragement support plates asper ASTM B221-02. The system shall be such that it provides watertight roof to roof/roof to corner joint cover expansion control system that is capable of accommodating multidirectional seismic movement without stress to its components. System shall consist of metal profile that incorporates a universal aluminum base member designed to accommodate various project conditions and roof treatments. The cover plate shall be designed of width and thickness required to satisfy movement and loading requirements and secured to base members by utilizing manufacturer's pre-engineered self-centering arrangement that freely rotates/moves in all directions. The Self		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			centering arrangement shall exhibit circular sphere ends that lock and slide inside the corresponding aluminum extrusion cavity to allow freedom of movement and flexure in all directions including vertical displacement.The Joint System shall resists damage or deterioration from the impact of falling ice, exposure to UV, airborne contaminants and occasional foot traffic from maintenance personnel. Provision of Moisture Barrier Membrane in the Joint System to have water tight joint is mandatory requirement. (Material shall confirm to ASTM 6063.)		
	5.41.1	100 mm	Roof Joint for 100 mm gap	metre	4509.00
	5.41.2	150 mm	Roof Joint for 150 mm gap	metre	5017.00
	5.41.3	200 mm	Roof Joint for 200 mm gap	metre	5950.00
5.42		Scaffolding net	Providing and fixing Scaffolding net of required width made of high density Polyethylene UV stabilized knitted on warp knitting machines having density 100gm/sqm and shading coefficient minimum 75% around the construction site/ for vertical extension as per requirement including fastening/tying with building/scaffoldingpipes or with any other fixtures etc. complete as per direction of Engineer-in-Charge. (One time payment shall be made for providing Scaffolding net from start of work till completion of work including shifting if any. The Scaffolding net shall be the property of the contractor on completion of the work)	sqm	20.00
5.43		Glass Reinforced Concrete (G.R.C) Screens/ Jali	Providing and fixing Glass Reinforced Concrete (G.R.C) Screens/ Jali customized as per design, dimensions as per drawings in approved shade in overall thickness of 40mm, or inner design membrane between 30-35mm and borders 50mm, casted with layering technique Power Spray methodology have weight approximately between 3.75 -4.5 Kg per Sq. Ft. The screens casting shall take place with layering methodology using- Direct Power Spray machines. The screens shall be casted from 53 grade White Portland Cement manufactured by Fine graded, sieved Quartz & Silica Sand, reinforced with Alkali Resistant Glass Fiber manufactured between 4-5% by having Zirconia content at least 16%, Super	sqm	4721.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Plasticizers and UV resistant Synthetic inorganic pigments should be used for pigmentation		
5.44		Cement concrete Block jali	Providing and laying Cement concrete Block jali made out by white cement , (Size 200X200X75mm thick), approximate weight of each block 2.5 kg and blocks jointed with approved block laying polymer modified adhesive mortar applied in uniform thickness of 2 to 3 mm in horizontal and verticals joints . Block should be gently pressed down using a rubber mallet to expel out entrapped air and the masonry shall be cured two days. 1st layer of CC block shall be placed on perfectly leveled surface, Rates include all materials including steel bars, wastages, scaffolding and all labour etc. complete. as per architectural drawing. The rates are inclusive of all lead & lifts and All necessaries to complete the item in good condition, nothing shall be paid extra separately unless specified.	sqm	3660.00
5.45		Teracotta jali	Providing and fixing Eco Friendly Teracotta jali size length 210 x width 210x thick 70 mm including m.s. frame , all joints, matching pigment etc. complete with compressive strength 12-15 mpa. Density of material 1200-1300 kg per m^3 and water absorption range 12-15 %. As per drawing and as directed by Engineer in charge.	sqm	1961.00
5.46		Steel fibers in cement concrete for flooring/pa cement/CC roads (All ground supported slab)/undergrou nd structure/precast items (Like Drains, Manhole covers, Walls, Paver blocks, pipes, Tunnel Segments) etc.	Steel Fiber Reinforcement for RCC for in Flooring pavement, parking area and for heavy duty roads, Supplying & providing and continuously deformed undulating steel fibers with aspect ratio between 50 to 80 (Diameter between 0.45 to 01mm and length between 12.5 to 60mm) manufactured from cold drawn wire and having a minimum tensile strength of 1100N/Sq. mm, as per CE Certification label mark, non-glued/glued, non-balling, conforming to ASTM A 820/A 820 M-06, compatible with all construction chemicals used for the batching, laying and curing of concrete. The fibers shall be added to the concrete at specified dosage and shall be mixed with concrete for sufficient time to ensure uniform distribution and pumping concrete. Addition of steel fibers in cement.	Kg	98.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
			concrete supported slab)/underground structure/precast items (like drains, manhole covers, walls, paver blocks, pipes, tunnel segments) etc. doses as per approved designs and directions of Engineer-in-charge		
5.47		Glass Fiber for RCC work	Glass Fiber Reinforcement of size 12mm for Reinforced cement concrete for pavement and Flooring work; Supplying and providing Glass fibers chopped strands of 12mm length dosage at the rate of 900 grams per/m^3 for mixing in RCC work to gain benefit of anti-cracking. Glass Fibers conform to properties as per ASTM c1666@-M-08 Density:2.68+-0.3g/m^3 Tensile strength:1-1.7 MPA Filament Diameter:31x10^{-5} -118x10^{-5} inch AR Value: As per EN 14649 Dose of Fiber shall be as per IRC SP: 46-2013 update Amendment.	Kg	250.00
5.48		Extra rate for internal plastering work in upper floors (Above Ground Floor).	Extra rate in internal plastering work for every additional floors above ground floor (due to lifting of material, ,scaffolding and labour etc. complete) Note:- Applicable to item No. 13.1 to item No. 13.10	sqm	5%

CHAPTER – 6.0

MASONRY WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 712	Specification for building limes.
2	IS 1077	Common burnt clay building bricks.
3	IS 1200 (Part 3)	Method of measurements of brick works.
4	IS 2212	Code of practice for brick work. (1st Revision)
5	IS 2222	Specification for burnt clay perforated building bricks.
6	IS 2849	Specification for non-load bearing gypsum partition blocks, (Solid and hollow types)
7	IS 3495	Method of test for burnt clay building bricks.
8	IS 3812	Specification for fly ash for use as pozzolana and admixture.
9	IS 4139	Specification of calcium silicate bricks.
10	IS4885	Specification for sewer brick.
11	IS 5454	Methods of sampling of clay building bricks.
12	IS 12894	Pulverized fuel ash lime bricks specification.
13	IS 13757	Specification of burnt clay fly ash bricks.
14	IS 2185 (part-3)	Autoclaved cellular aerated concrete
15	IS 6073	Autoclaved Reinforced cellular concrete floor and roof slabs saticifications
16	IS 6072	Autoclaved Reinforced cellular concrete wall slabs
17	IS 1893	Criteria for earth quake

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Frequency of testing
1	**2**	**3**	**4**	**5**	**6**
1	Bricks/ brick/ Tiles	6.1.3, 6.1.4, 6.1.5,	Testing of Bricks/Brick Tiles for dimensions, Compressive strength, Water absorption and efflorescence	Laboratory	As per Table 6.3 and 6.4
2	Sewer Bricks	6.1.4	Dimensions, Compressive strength, Water absorption and Efflorescence	Laboratory	
3	Burnt clay Perforated building bricks	6.1.5	Dimensions, Compressive strength, Water absorption and Efflorescence	Laboratory	As per Table 6.3 and 6.4
4	AAC blocks		Dimensions, Laboratory, Compressive, strength, Block density Thermal conductivity Drying shrinkage		

PREAMBLE

BRICK WORK
(CHAPTER: 6.0)

6.1 Bricks used in the masonry may be of the following type.

The Common Burnt Clay Bricks shall conform to IS: 1077 and shall be hand moulded or machine moulded. They shall be free from nodules of free lime, visible cracks, flaws warpage and organic matter, have a frog 100 mm in length 40 mm in width and 10 mm to 20 mm deep on one of its flat sides. Each brick shall be marked (in the frog where provided) with the manufacturer's identification mark or initials. The common burnt bricks classified as under:

(a) Class 40 TM chimney brick/grog or ghol brick: - For this item either selected chimney burnt bricks or ghol bricks are used and superior workmanship than the following varieties is required. The crushing strength when thoroughly soaked in water shall not be less than 40 kg/sq.cm.
(b) Class 25 TM chimney brick masonry: - The crushing strength when thoroughly soaked in water shall not be less than 25 kg/sq.cm.
(c) Class 25 TM open bhatta or pajawa burnt brick: - As is clear the only difference between (b) and (c) varies in the method of burning bricks. The crushing strength when thoroughly soaked in water shall not be less than 25 kg/sq.cm.

Fly Ash Lime Bricks (FALG Bricks): The Fly Ash Lime Bricks (FALG Bricks) shall conform to IS 12894. Visually the bricks shall be sound, compact and uniform in shape free from visible cracks, warpage, flaws and organic matter. The bricks shall be solid and with or without frog on one of its flat side.

Fly Ash: Fly ash shall conform to IS 3812.

Note: FLAG Bricks will be used only for load bearing structure upto 2 storeys and for non- load bearing walls 20 cms thick for multi-storeyed buildings. Bottom ash used as replacement of sand shall not have more than 12% loss on ignition when tested.

Sand: Deleterious materials, such as clay and silt in the sand shall preferably be less than 5%.
Lime: Lime shall conform to class 'C' hydrated lime of IS 712.
Additives: Any suitable additive considered not detrimental to the durability of bricks may be used.

Clay Fly Ash Bricks: The clay fly ash bricks shall conform to IS 13757. The bricks shall be sound, compact and uniform in shape and colour. Bricks shall have smooth rectangular faces with sharp and square corners. The bricks shall be free from visible cracks, flaws, warpage, nodules of free lime and organic matter, the bricks shall be hand or machine moulded. The bricks shall have frog of 100 mm in length 40 mm width and 10 to 20 mm deep on one of its flat sides. If made by extrusion process may not be provided with frogs. Fly Ash shall conformto grade I or grade II of IS 3812.

Calcium Silicate Bricks: The bricks shall conform to IS 4139. The Calcium silicate bricks shall be sound, compact and uniform in shape. Bricks shall be free from visible cracks, warpage, organic matter, large pebbles and nodules of free lime. Bricks shall be solid and with or without frog. The bricks shall be made of finely grounded sand siliceous rock and lime. In addition limited quantity of fly ash conforming to IS 3812 may be used in the mix. These bricks are also known as Fly Ash Sand Lime bricks in the construction industry.

Mechanized Autoclave Fly Ash Lime Brick: These bricks shall be machine moulded and prepared in plant by appropriate proportion of fly ash and lime. The autoclave fly ash bricks shall conform to IS 12894. Visually, the bricks shall be sound, compact and uniform shape, free from visible cracks, warpage and organic matters. The brick shall be solid with or without frog, and of 100/80 mm in length, 40 mm width and 10 to 20 mm deep one of its flat side asper IS 12894. The brick shall have smooth rectangular faces with sharp corners and shall be uniform in shape and colour. Fly ash shall conform to IS 3812 and lime shall conform to class 'C' hydrated lime of IS 712.

6.2 Dimensions

The brick may be modular bricks/tiles shall be as per Table 6.1. While use of modularbricks/tiles is recommended, sizes shall be as mentioned in Table 6.1 specified.

Table 6.1

Dimensions of Bricks

Type of Bricks/ Tiles	Nominal Size mm	Actual Size mm
Modular Bricks	200 × 100 × 100 mm	190 × 90 × 90 mm
Modular tile bricks	200 × 100 × 40 mm	190 × 90 × 40 mm

6.3 Classification

Bricks/Brick tiles shall be classified on the basis of their minimum compressive strength as given below:

Table 6.2

Classification of Bricks

Class Designation	Average compressive strength			
	N/mm2	(kgf/cm2)	N/mm2	(kgf/cm2)
Designation 40	4.0	40	5.0	50
Designation 25	2.5	25	3.0	30

The bricks shall have smooth rectangular faces with sharp corner and shall be uniform in colour and emit clear ringing sound when struck.

Note: Upper limits specified in Table 6.2 are for calculating the average compressive strength.

6.4 Sampling and Tests

Samples of bricks shall be subjected to the following tests:

(a) Dimensional tolerance.
(b) Water absorption.
(c) Efflorescence.
(d) Compressive strength.

Sampling: For carrying out compressive strength, water absorption, efflorescence and dimensional tests, the samples of bricks shall be taken at random from the lot given in Table 6.3. The sample thus taken shall be stored in a dry place until tests are made. For the purpose of sampling, the following definition shall apply.

Lot:A collection of bricks of same class and size, manufactured under relatively similar conditions of production. For the purpose of sampling a lot shall contain a maximum, of 50,000 bricks. In case a consignment has bricks more than 50,000 of the same classification and size and manufactured under relatively similar conditions of production, it shall be dividedinto lots of 50,000 bricks or part thereof.

Sample: A collection of bricks selected for inspection and/or testing from a lot to reach the Decision regarding the acceptance or rejection of the lot.

Defective:A brick failing to meet one or more of the specified requirements.

The samples shall be taken as below:
Sampling from a Stack: When it is necessary to take a sample from a stack, the stack shall be divided into a number of real or imaginary sections and the required number of bricks drawn from each section. For this purpose bricks in the upper layers of the stack shall be removed to enable units to be sampled from places within the stack.

Scale of sampling and criteria for conformity for visual and dimensional characteristics:

Visual characteristics: The bricks shall be selected and inspected for ascertaining their conformity to the requirements of the relevant specification.

The number of bricks to be selected from a lot shall depend on the size of lot and shall be in accordance of Col. 1 and 2 of Table 6.3 for visual characteristics in all cases and dimensional characteristics if specified for individual bricks.

Visual Characteristics: All the bricks selected above in accordance with Col. 1 and 2 of Table 6.3 shall be examined for visual characteristics. If the number of defective bricks found in the sample is less than or equal to the corresponding number as specified in Col. 3 of Table 6.3 the lot shall be considered as satisfying the requirements of visual characteristics, otherwise the lot shall be deemed as not having met the visual requirements.
Dimensional Characteristics: The number of bricks to be selected for inspecting the dimensions and tolerance shall be in accordance with Col. 1 and 4 of Table 6.3. These bricks will be divided into groups of 20 bricks at random and each of the group of 20 bricks thus formed will be tested for all the dimensions and tolerances. A lot shall be considered having found meeting the requirements of dimensions and tolerance if none of the groups of bricks inspected fails to meet the specified requirements.

TABLE 6.3

Scale of Sampling and Permissible Number of Defectives for Visual and Dimensional Characteristics.

No. of bricks in the lot	For characteristics specified for individual bricks		For dimensional characteristics for group of 20 bricks
	No. of bricks to be selected	Permissible no. of defective in the sample	No. of bricks to be selected
1	2	3	4
2001—10000	20	1	40
10001—35000	32	2	60
35001—50000	50	3	80

Note: In case the lot contains 2000 or less bricks the sampling shall be as per decision of the Engineer-in-Charge.

Scale of Sampling and Criteria for Physical Characteristics: The lot which has been found satisfactory in respect of visual and dimensional requirements shall be next tested for physical characteristics like compressive strength, water absorption, efflorescence as specified in relevant material specification. The bricks for this purpose shall be taken at random from thosealready selected above. The number of bricks to be selected for each of these characteristics shall be in accordance with relevant columns of Table 6.4.

Table 6.4

Scale of Sampling for Physical Characteristics

Lot size	Sample size for compressive strength, water absorption and efflorescence	Permissible No. of defectives for efflorescence	Warpage	
			Sample size	Permissible No of defects
1	2	3	4	5
2001—10000	5	0	10	0
10001—35000	10	0	20	1
35001—50000	15	1	30	2

Note: In case the lot contains 2000 or less bricks, the sampling shall be as per decision of Engineer-in-Charge.

A lot shall be considered having satisfied the requirements of physical characteristics if the condition stipulated here in are all satisfied.

From the test results for compressive strength, the average shall be calculated and shall satisfy the requirements specified in relevant material specification.

Note: In case any of the test results for compressive strength exceeds the upper limit for the class of bricks, the same shall be limited to the upper limit of the class for the purpose of ave

Wherever specified in the material specification, the compressive strength of any individual bricks tested in the sample shall not fall below the minimum average compressive strength specified for the corresponding class of brick by more than 20 per cent.

From the test results for water absorption, the average for the bricks in the sample shall be calculated and shall satisfy the relevant requirements specification in material specification.

The number of bricks failing to satisfy the requirements of the efflorescence specified in the relevant specification should not be more than the permissible no. of defectives given in Col. of Table 6.4.

Dimensional Tolerances: The dimensions of, specifications modular bricks when tested as described above as per procedure described in Appendix A of Chapter 6 shall be within the following limits per 20 bricks or locally available size as approved by Engineer-in-charge.

For modular size

Length 7320 to 3880 mm (3800 ± 80 mm)

Width 1760 to 1840 mm (1800 ± 40 mm)

Height 1760 to 1840 mm (1800 ± 40 mm) for 90 mm high bricks

60 to 840 mm (800 ± 40 mm) for 40 mm high bricks

6.5 BRICK WORK

Classification

The brick work shall be classified according to the class designation of bricks used.

6.6 Mortar

The mortar for the brick work shall be as specified, and conform to accepted standards. Lime shall not be used where reinforcement is provided in brick work.

6.7 Soaking of Bricks

Bricks shall be soaked in water before use for a period for the water to just penetrate the whole depth of the bricks. Alternatively bricks may be adequately soaked in stacks by profusely spraying with clean water at regular intervals for a period not less than six hours. When the bricks are soaked they shall be removed from the tank sufficiently early so that at the time of lying they are skin-dry. Such soaked bricks shall be stacked on a clean place where they are not again spoiled by dirt earth etc.

Note I: The period of soaking may be easily found at site by a field test in which the bricks are soaked in water for different periods and then broken to find the extent of water penetration. The least period that corresponds to complete soaking will be the one to be allowed for in construction work.

Note II: If the bricks are soaked for the required time in water that is frequently changed the soluble salt in the bricks will be leached out, and subsequently efflorescence will be reduced.

6.8 Laying

Bricks shall be laid in English Bond, unless otherwise specified. For brick work in half brick wall, bricks shall be laid in stretcher bond. Half or cut bricks shall not be used except as closer where necessary to complete the bond. Closers in such cases, shall be cut to the required size and used near the ends of the wall. Header bond shall be used preferably in all courses in curved plan for ensuring better alignment.

Note: Header bond shall also be used in foundation footings unless thickness of walls (width of footing) makes the use of headers impracticable. Where thickness of footing is uniform for a

number of courses, the top course of footing shall be headers.

All loose materials, dirt and set lumps of mortar which may be lying over the surface on which brick work is to be freshly started, shall be removed with a wire brush and surface wetted. Bricks shall be laid on a full bed of mortar, when lying, each brick shall, be properly bedded and set in position by gently pressing with the handle of a trowel. It's inside face shall be buttered with mortar before the next brick is laid and pressed against it. Joints shall be fully filled and packed with mortar such that no hollow space is left inside the joints.

The walls shall be taken up truly in plumb or true to the required batter where specified. All courses shall be laid truly horizontal and all vertical joints shall be truly vertical. Vertical joints in the alternate course shall come directly one over the other. Quoin, Jambs and other angles shall be properly plumbed as the work proceeds. Care shall be taken to keep the perpends properly aligned within following maximum permissible tolerances:

(i) Deviation from vertical within a storey shall not exceed 6 mm per 3 m height.
(ii) Deviation in verticality in total height of any wall of building more than one storey in height shall not exceed 12.5 mm.
(iii) Deviation from position shown on plan of any brick work shall not exceed 12.5 mm.
(iv) Relative displacement between load bearing wall in adjacent storeys intended to be vertical alignments shall not exceed 6 mm.
(v) A set of tools comprising of wooden straight edge, masonic spirit levels, square, 1 metre rule line and plumb shall be kept on the site of work for every 3 masons for proper check during the progress of work.

All quoins shall be accurately constructed and the height of brick courses shall be keptuniform. This will be checked using graduated wooden straight edge or storey rod indicating height of each course including thickness of joints. The position of damp proof course,window sills, bottom of lintels, top of the wall etc. along the height of the wall shall be markedon the graduated straight edge or storey rod. Acute and obtuse quoins shall be bonded, where practicable in the same way as square quoins. Obtuse quoins shall be formed with squint showing three quarters brick on one face and quarter brick on the other.

The brick work shall be built in uniform layers. No part of the wall during its construction shall rise more than one metre above the general construction level. Parts of wall left at different levels shall be raked back at an angle of 45 degrees or less with the horizontal. Toothing shall not be permitted as an alternative to raking back. For half brick partition to be keyed into main walls, indents shall be left in the main walls.

All pipe fittings and specials, spouts, hold fasts and other fixtures which are required to be built into the walls shall be embedded, as specified, in their correct position as the work proceeds unless otherwise directed by the Engineer-in-Charge.

Top courses of all plinths, parapets, steps and top of walls below floor and roof slabs shall be laid with brick on edge, unless specified otherwise. Brick on edge laid in the top courses at corner of walls shall be properly radiated and keyed into position to form cut corners. Where bricks cannot be cut to the required shape to form cut corners, cement concrete 1:2:4
(1 cement:2 sand: 4 graded stone aggregate 20 mm nominal size) equal to thickness of course shall be provided in lieu of cut bricks.

Bricks shall be laid with frog (where provided) up. However, when top course is exposed, bricks shall be laid with frog down. For the bricks to be laid with frog down, the frog shall be filled with mortar before placing the brick in position.

In case of walls one brick thick and under, one face shall be kept even and in proper plane, while the other face may be slightly rough. In case of walls more than one brick thick, both thefaces shall be kept even and in proper plane.

To facilitate taking service lines later without excessive cutting of completed work, sleeves (to be paid separately) shall be provided, where specified, while raising the brick work. Such sleeves in external walls shall be sloped down outward so as to avoid passage of water inside.

Top of the brickwork in coping and sills in external walls shall be slightly tilted. Where brick coping and sills are projecting beyond the face of the wall, drip course/throating (to be paid separately) shall be provided where indicated.

Care shall be taken during construction that edges of jambs, sills and projections are not damaged in case of rain. New built work shall be covered with gunny bags or tarpaulin so as toprevent the mortar from being washed away. Damage, if any, shall be made good to the satisfaction of the Engineer-in-Charge.

Vertical reinforcement in the form of bars (MS or high strength deformed bars or thermo mechanically treated bars as per direction of Engineer-in-Charge)), considered necessary at the corners and junction of walls and jamb opening doors, windows etc. shall be encased with cement mortar not leaner than 1:4 (1 cement : 4 sand), or cement concrete mix as specified. The reinforcement shall be suitably tied, properly embedded in the foundation and at roof level. The dia. of bars shall not be less than 8 mm and concrete grade shall be minimum 1:3:6 (1 cement : 3 sand : 6 graded stone aggregate 20 mm nominal size).

In retaining walls and the like, where water is likely to accumulate, weep holes, 50 to 75 mm square shall be provided at 2 m vertically and horizontally unless otherwise specified. The lowest weep hole shall be at about 30 cm above the ground level. All weep holes shall be surrounded by loose stones and shall have sufficient fall to drain out the water quickly.

Note: Work of providing loose stone will be payable extra.

Work of cutting chases, wherever required to be made in the walls for housing G.I. pipe, CI pipe or any other fixtures shall be carried out in various locations as per guidelines given below:

Cutting of chases in one brick thick and above load bearing walls.

(i) As far as possible services should be planned with the help of vertical chases. Horizontal chases should be avoided.

(ii) The depths of vertical chases and horizontal chases shall not exceed one-third and one-sixth of the thickness of the masonry respectively.

(iii) When narrow stretches of masonry (or short length of walls) such as between doors and windows, cannot be avoided they should not be pierced with openings for soil pipes or waste pipes or timber joints, etc. Where there is a possibility of load concentration such narrow lengths of walls shall be checked for stresses and high strength bricks in mortar or concrete walls provided, if required.

(iv) Horizontalchases when unavoidable should be located in the upper or lower one-third of height of storey and not more than three chases should be permitted in any stretch of a wall. No continuous horizontal chase shall exceed one metre in length. Where unavoidable, stresses in the affected area should be checked and kept within the permissible limits.

(v) Vertical chases should not be closer than 2 m in any stretch of a wall. These shall be kept away from bearings of beams and lintels. If unavoidable, stresses in the affected area should be checked and kept within permissible limits.

(vi) Masonry directly above a recess, if wider than 30 cm horizontal dimension) should be supported on lintel. Holes in masonry may be provided upto 30 cm width and 30 cm height without any lintel. In the case of circular holes in the masonry, no lintel need be provided for holes upto 40 cm in diameter.

Cutting of chases in half brick load bearing walls. No chase shall be permitted in half brick load bearing walls and as such no recessed conduits and concealed pipes shall be provided with half brick thick load bearing walls.

Cutting of chases in half brick non-load bearing wall: Services should be planned with the help of vertical chases. Horizontal chase should be provided only when unavoidable.

6.9 Joints

The thickness of all types of joints including brick wall joints and cross joints shall be such that four course and three joints taken consecutively shall measure as follows:

(i) In case of modular bricks conforming to IS 1077 specification for common burnt clay buildings bricks, equal to 39 cm.

Note: Specified thickness of joints shall be of 1 cm. Deviation from the specified thickness of all joints shall not exceed one-fifth of specified thickness.

Finishing of Joints: The face of brick work may be finished flush or by pointing. In flush finishing either the face joints of the mortar shall be worked out while still green to give a finished surface flush with the face of the brick work or the joints shall be squarely raked out to a depth of 1 cm while the mortar is still green for subsequently plastering. The faces of brick work shall be cleaned with wire brush so as to remove any splashes of mortar during the course of raising the brick work. In pointing, the joints shall be squarely raked out to a depth of 1.5 cm while the mortar is still green and raked joints shall be brushed to remove dust and loose particles and well wetted, and shall be later refilled with mortar to give ruled finish. Some such finishes are 'flush', 'weathered', ruled, etc.

6.10 Curing

The brick work shall be constantly kept moist on all faces for a minimum period of seven days. Brick work done during the day shall be suitably marked indicating the date on which the work is done so as to keep a watch on the curing period.

6.11 Scaffolding

Scaffolding shall be strong to withstand all dead, live and impact loads which are likely to come on them. Scaffolding shall be provided to allow easy approach to every part of the work.

Single Scaffolding: Where plastering, pointing or any other finishing has been indicated for brick work, single scaffolding may be provided, unless otherwise specified. In single scaffolding, one end of the put-logs/pole shall rest in the hole provided in the header course of brick masonry. Not more than one header for each put-log/pole shall be left out. Such holes shall not be allowed in the case of pillars, brick work less than one metre in length between the openings or near the skew backs of arches or immediately under or near the structural member supported by the walls. The holes for putlogs/poles shall be made good with brick work and wall finishing as specified.

Double Scaffolding: Where the brick work or tile work is to be exposed and not to be finished with plastering etc. double scaffolding having two independent supports, clear of the work, shall be provided.

6.12 HALF BRICK WORK

Brick work in half brick walls shall be done in the same manner as described earlier except that the bricks shall be laid in stretcher bond. When the half brick work is to be reinforced, 2 Nos. M.S. bars of 8 mm dia., shall be embedded in every third course as given in the item. These shall be securely anchored at their end where the partitions end. The free ends of the reinforcement shall be keyed into the mortar of the main brick work to which the half brick work is joined. The mortar used for reinforced brick work shall be rich dense cement mortar ofmix 1:4 (1 cement: 4 sand). Lime mortar shall not be used. Over laps in reinforcement, if any shall not be less than 30 cm. The mortar interposed between the reinforcement bars and the brick shall not be less than 5 mm. The mortar covering in the direction of joints shall not beless than 15 mm.

TEST FOR COMPRESSIVE STRENGTH OF BRICKS

Specimen

- Five whole bricks shall be taken from the samples as specimens for this test. Length and width of each specimen shall be measured correct to 1 mm.

Apparatus

- The apparatus consists of compression testing machine, the compression plate of which shall have a ball seating in the form of portion of a sphere the centre of which shall coincidewith the centre of the plate.

Procedure:

- Pre-conditioning: The specimen shall be immersed in the water for 24 hours at 25º to 29ºC. Any surplus moisture shall be allowed to drain at room temperature. The frog of the bricks should be filled flush with mortar 1:3 (1 cement: 3 sand of grade 3 mm and down) and shall be kept under damp jute bags for 24 hours, after that these shall be immersed in clean water for three days. After removal from water, the bricks shall be wiped out of any traces of moisture.
- Actual Testing: Specimen shall be placed with flat faces horizontal and mortar filled face upward between three 3 ply plywood sheets each of thickness 3 mm and carefully centred between plates of the testing machine. Plaster of Paris can also be used in place of plywood sheets to ensure a uniform surface. Load shall be applied carefully axially at uniform rate of 14 N/mm2 per minute till the failure of the specimen occurs.

Reporting the Test Results

- The compressive strength of each specimen shall be calculated in N/mm^2 as under :

  $$\text{Compressive Strength} = \frac{\text{Maximum load at failure (in N)}}{\text{Area of Specimen (in sq mm)}}$$

  In case the compressive strength of any individual brick tested exceeds the upper limit of the average compressive strength specified for the corresponding class of brick, the same shall be limited to the upper limit of the class for the purpose of calculating the average compressive strength. Compressive strength of all the individual bricks comprising the sample shall be averaged and reported.

Criteria for Conformity

- A lot shall be considered having satisfied the requirements of average compressive strength if the average compressive strength specified for the corresponding class of brick tested is not below the minimum average compressive strength specified for the corresponding class of bricks by more than 20 per cent.

TEST FOR WATER ABSORPTION (BRICKS)

No. of Specimen

- Five whole bricks shall be taken from samples as specimen for this test.

Apparatus

- A balance required for this test shall be sensitive to weigh 0.1 percent of the weight of the specimen.

Procedure

- Pre-conditioning: The specimen shall be allowed to dry in a ventilated oven at a 110°C to 115°C till it attains a substantially constant weight. If the specimen is known to be relatively dry, this would be accomplished in 48 hours, if the specimen is wet, several additional hours may be required to attain a constant weight. It shall be allowed to cool at room temperature. In a ventilated room, properly separated bricks will require four hours for cooling, unless electric fan passes air over them continuously in which case two hours may suffice. The cooled specimen shall be weigh (W1) a warm specimen shall not be used for this purpose.

- Actual Testing: Specimen shall be completely dried before immersion in the water. It shall be kept in clean water at a temperature of 27°C ± 2°C for 24 hours. Specimen shall be wiped out of the traces of water with a damp cloth after removing from the water and then shall be weighed within three minutes after removing from water (W2).

Reporting the Test Results

- The water absorption of each specimen shall be calculated as follows and the average of five tests shall be reported.

  Water Absorption $= \frac{W_2 - W_1}{W_1} \times 100$

Criteria for Conformity

- A lot shall be considered having satisfied the requirements of water absorption if the average water absorption is not more than 20% by weight.

TEST FOR EFFLORESCENCE (BRICKS)

No. of Specimen

- Five whole bricks shall be taken as specimen for this test.

Apparatus

- Apparatus required for this test shall be a shallow flat bottom dish containing distilled water.

Procedure (actual testing)

- The brick shall be placed vertically in the dish with 2.5 cm immersed in the water. The room shall be warm (18ºC to 30ºC) and well ventilated. The bricks should not be removed until it absorbs whole water. When the whole water is absorbed and the brick appears to be dry, place a similar quantity of water in that dish and allow it to evaporate as before. The brick shall be examined after the second evaporation.

Reporting the Test Results

- The rating to efflorescence in ascending order shall be reported as 'NIL', 'SLIGHT', 'MODERATE', 'HEAVY' or 'SERIOUS' in accordance with the following :
  (i) NIL: When there is no perceptible deposit of efflorescence.
  (ii) SLIGHT: When not more than 10 per cent of the area of the brick is covered with a thin deposit of salts.
  (iii) MODERATE: When there is heavier deposit and covering upto 50% of the area of the brick surface but unaccompanied by powdering or flaking of the surface.
  (iv) HEAVY: When there is a heavy deposit of salts covering 50% or more of the brick surface but unaccompanied by powdering or flaking of the surface.
  (v) SERIOUS: When there is heavy deposit of salts, accompanied powdering and/or flaking of the surface and tending to increase in the repeated wetting of the specimen.

Criteria for Conformity

- A lot be considered having satisfied the requirements of efflorescence if for 4 out of the specimen of 5 bricks, the rating of efflorescence is not beyond "Moderate"

6.14 AUTOCLAVED AERATED CONCRETE BLOCK MASONRY WORK

6.14.1 Terminology

For the purpose of, Autoclave Aerated Concrete Block masonry work, the following definitions shall apply

1. Autoclaved -Steam curing of concrete Products, sand lime bricks, asbestos cement products, hydrous calcium silicate insulation Products, or cement in an autoclave at maximum ambient temperatures generally between 1700C to 2150C.

2. Block - A concrete masonry unit, any one of the external dimensions of which is greater than the corresponding dimension of a brick as specified in IS : 3952-1978 and of such size and mass as to permit it to be handled by one man. Furthermore, to avoid confusion with slabs and Panels, the height of the block shall not exceed either its length or six times its width.

3. Block Density - The density calculated by dividing the mass of a block by the overall volume,including holes or cavities and end recesses..

4. Drying Shrinkage - The difference between the length of specimen which has been immersed in water and then subsequently dried to constant length, all under specified conditions; expressed as a percentage of the dry length of the specimen.

5. Gross Area-The total area occupied by a block on its loading face, including areas of the cavities and end recesses.

6. Height -The vertical dimension of the exposed face of a block, excluding any tongue or other device designed to provide mechanical keying

7. Length - The horizontal dimension of the exposed face of a block excluding any tongue or other device designed to provide mechanical keying.

8. Width -The external dimension of a block at the bedding plane, measured at right angles to the length and height of the block.

6.14.2 Dimensions & Tolerances:

Autoclave Aerated Concrete Block shall be made in sizes and shapes to fit different concrete needs. They include stretcher, corner, double corner or pier, jamb, header, bull nose, partition block and concrete floor units.

Autoclave Aerated Concrete Block shall be referred to by its normal dimension the term'normal' means that the dimension includes the thickness of the mortar joints. The actual dimension shall be 10mm short of the normal dimension (or 6mm short in special areas finer joints as specified).

6.14.2.1 The normal dimension of the concrete block shall be as follows:-

Length : 400, 500 or 600 mm
Height : 200, 250 or 300 mm
Width : 100, 150, 200 or 250 mm

In addition, Autoclave Aerated Concrete Block shall be manufactured in half length of 200, 250 or 300 mm
correspond to the full lengths.

6.14.2.2 The nominal dimensions of the units are so designed that taking account of the thickness of mortar joints, they will produce wall length sand heights which will conform to the principles of modular co-ordination.

6.14.2.3 Block of sizes other than those specified above, may also be used if so specified in the case of special Autoclave Aerated Concrete Block such as jallie or screen wall and ornamental block ,the specified size may not necessarily apply.

6.14.2.4 The maximum variation in the length of the Autoclave Aerated Concrete Block shall not be more than plus/minus 5mm and maximum variation in the height and width of Autoclave Aerated Concrete Block, not more than plus/minus 3mm.

6.14.2.5 The faces of Autoclave Aerated Concrete Block shall be flat & Rectangular, opposite faces shall be parallel and all arises shall be square. The bedding surfaces shall be at right angle to the face of the Blocks.

6.14.2.6 The Autoclave Aerated Concrete Block with special faces shall be manufactured and supplied if so specified.

6.14.3 Materials

6.14.3.1 Cement complying with any of the Indian Standard may be used as per the direction of the manufacturer.

6.14.3.2 Use of Fly ash conforming to IS 3812-1981 may be permitted to a limit of 20% in cement conforming to IS 269-1976.

6.14.3.3 The lime shall satisfy the requirement for class C lime specified as IS 712-1973.

6.14.3.4 The aggregate used for the manufacture of Autoclave Aerated Concrete Block shall conform to the following requirements

(a) Sand-Conforming to IS 383-1970 except for the grading which may be made to suit the product and silica content shall not be less than 80%.

(b) Fly ash – Conforming to IS 3812-1981 with loss on ignition not more than 6%.

6.14.3.5 The water used in the manufacture of Autoclave Aerated Concrete Block shall be free from matter harmful to concrete or reinforcement or matter likely to cause efflorescence in theblock and shall meet the requirements of IS 456-2000.

6.14.3.6 Additives and Admixtures may be added either as additives to the cement during manufacturing or as additive or admixtures to the concrete mix. Additive or admixtures used in the manufacture of concrete block may be

(a) Accelerating , water reducing and air –entraining admixtures conforming to IS 9103-1979

(b) Water proofing agent conforming to IS 2645-1975

(c) Colouring pigments

6.14.4 Physical requirements

6.14.4.1 All Autoclave Aerated Concrete Block shall be sound, free of cracks or other defects which interfere with the proper placing of block units, impair the strength or performance of the construction.

6.14.4.2 Where block units are to be used in exposed wall construction, the face or faces that are to be exposed shall be free of chips, cracks or other imperfections except that if not more than 5% of a consignment contains slight cracks or small chippings not larger than 25mm, this shall not be deemed grounds for rejection.

6.14.4.3 Dimensions- The overall dimension of the block units when measured shall be in accordance with para 6.14.2.1 subjected to the tolerances mentioned in para 6.14.2.4

6.14.4.4 Block Density - The Block density shall conform to the requirements specified in table of para 6.14.3, when tested accordance with para 6.14.6 (1)

6.14.4.5 Compressive Strength - The min. compressive strength being the average of twelve block units shall be as prescribed in table of para 6.14.3, when tested accordance with para 6.14.6(2)

6.14.4.6 Thermal Conductivity - The thermal conductivity shall be not exceed the values specified in table of para 6.14.3 when tested accordance with para 6.14.6(3)

6.14.4.7 Drying Shrinkage – the drying shrinkage shall be not more than 0 .05% for grade –1 block and 0.10% for grade-2 block when tested accordance with para 6.14.6(4)

6.14.5 Tests

1. Block Density- The block density shall be determined in the manner described in IS 6441 (part-1) -1972

2. Compressive Strength- The compressive strength of block shall be determined in accordance with IS 6441 (part-5) -1972

3. Thermal Conductivity- The thermal conductivity of block shall be determined in accordance with IS 3346 -1980

4. Drying Shrinkages-The drying shrinkage of block shall be determined in the manner described in IS 6441 (part-2) -1972

6.14.6 Sampling

6.14.6.1 **Lot** - In any consignment, all the blocks of the same size and from the same batch of manufacture shall be grouped together into a minimum number of groups of 10000 blocks or less. Each such group shall constitute a lot.

6.14.6.2 From each lot, a sample of 24 blocks shall be selected at random. The required numbers of Blocks shall be taken at regular intervals during the loading of the vehicle or unloading the vehicles depending on whether sample is taken before delivery or after delivery. When this is not practicable, sample shall be taken from the stack in which case the required number of blocks shall be taken at random from across the top of the stacks, the sides accessible and from the interior of the stacks by opening trenches from the top.

6.14.6.3 The sample of blocks shall be marked for future identification of the consignment it represents. The blocks shall be kept under cover and protected from extreme conditions of temperature, relative humidity and wind until they are required for test. The tests shall be undertaken as soon as practicable after the sample has been taken.

6.14.7 Number of tests

6.14.7.1 All the 24 Blocks shall be checked for dimensions and inspected for visual defects.

6.14.7.2 Out of the 24 blocks, 12 blocks shall be subjected to the test for compressive strength, 3 blocks to the test for density, 3 blocks to the test for thermal conductivity and 3 blocks to the test for drying shrinkage. The remaining 3 blocks shall be reserved for re-test for drying shrinkage if a need arises.

6.14.7.3 The samples of AAC blocks (each sample consisting of 6 specimen) shall be chosen randomly from the lot procured and tested for various parameters specified in para 6 above. One samples shall be tested for every **100 cum** or part thereof. However, minimum one sample shall be tested from each lot received at site if the quantity procured in the lot is less than 100 cum. If required, Engineer-in-Charge or his authorized representative shall inspect the factory during production of the material for this work and also collect samples (of materials used for making AAC blocks and precast AAC blocks) from the factory itself. The contractor shall consider this contingency also while placing the order with one of the approved firms. Nothing extra shall be payable on this account.

6.14.8 – Criteria for conformity

6.14.8.1 The number of blocks with dimensions outside the tolerance limit and or with visual defects, among those inspected, shall not be more than two.

6.14.8.2 For density, the mean value shall be within the range specified in Table of para3

6.14.8.3 For compressive strength, the mean value, say X shall be determined. The test results shall be grouped into groups of 4, individual values of ranges shall be determined, the average range a calculated from these values and shall satisfy the following condition: X- 0.6 R> minimum value specified in Table of para3.

6.14.8.4 For thermal conductivity, the mean value shall be equal to or less than the value specified in Table of para3.

6.14.8.5 For drying shrinkage, all the test specimens shall satisfy the requirements of the test. If one or more specimens fail to satisfy the requirements, the remaining 3 blocks shall be subjected to these tests. All these blocks shall satisfy the requirements.

6.14.9 Manufacturer's Certificate

6.14.9.1 The manufacturer shall satisfy himself that the masonry units conform to the requirements of this specification and, if requested, shall supply a certificate to this effect to the purchaser or his representative.

6.14.10 Independent Tests

6.14.10.1 If the purchaser or his representative requires independent tests, the samples shall be taken before or immediately after delivery, at the option of the purchaser or his representative and the tests shall be carried out in accordance with this specification.

6.14.10.2 The manufacturer shall supply free of charge the units required for testing.

6.14.11 Storage

6.14.11.1 General requirements of storage of autoclaved cellular (aerated) concrete blocks shall be as described in IS : 4082-1977*.

6.14.12 Marking

6.14.12.1 Each lot of concrete masonry units manufactured in accordance with this specification shall be suitably marked with information-

(i) The identification of the manufacture

(ii) The grade and block density of the unit

(iii) The month and year of manufacturing

Each block may also be marked with the ISI Certification mark.

6.14.13 The R.C C bend shall be provided on **150mm /230mm/300mm** thick masonry to increase the strength and compatibility . The RCC bend shall be provided at sill level and at lintel level over throughout the wall. This thickness of the bend shall be approved by the Engineer in charge or as specified in drawing. The payment of RCC bend and reinforcement shall be paid separately.

Autoclave Aerated Concrete Block masonry shall be provided with polymer modified adhesive mortar.

The polymer modified adhesive mortar shall be provided @ 30 kg per cum or with cement mortar 1:4 (1 cement : 4 coarse sand).

6.14.14. Autoclave Aerated Concrete Block with **100 mm thick** masonry shall be provided with two number 6mm dia reinforcement steel bar at every third course. The payment of reinforcement shall be paid separately.

6.14.15. Autoclaved Aerated Concrete Block confirming the IS Code – 2185 (Part-3) 1984 (Reaffirmed 2005)

6.14.16 Measurements

6.14.16.1 Autoclave Aerated Concrete Block Masonry shall be measured in cubic metres unless otherwise specified. Any extra work over the specified dimensions shall be ignored. Dimensions shall be measured correct to the nearest 0.01 metre. ie. 1 cm. Areas shall be calculated to the nearest 0.01 sqm and the cubic contents shall be worked out to the nearest 0.01 cubic metres.

Note : (i) Autoclave Aerated Concrete Block work in parapet walls, mumty, lift machine room and water tanks constructed on the roof upto 1.2 m height above roof shall be measured together with the corresponding work of the floor next below.

6.14.16.2 No deductions or additions shall be done and no extra payment made for the following:

Note: Where minimum area is defined for deduction of an opening, void or both, such areas shall refer only to

opening or void within the space measured.

(a) Ends of dissimilar materials (that is, joists, beams, lintels, posts, girders, rafters, purlins, trusses, corbels, steps etc.); up to 0.1 m2 in section;

(b) Opening up to 0.1 m2 in area (see Note);

(c) Wall plates, bed plates, and bearing of slabs, chajjas and the like, where thickness does not exceed 10 cm and bearing does not extend over the full thickness of wall;

(d) Cement concrete blocks as for hold fasts and holding down bolts;

(e) Iron fixtures, such as wall ties, pipes upto 300 mm diameter and hold fasts for doors and windows;

(f) Chases of section not exceeding 50 cm in girth; and

(g) Bearing portion of drip course, bearing of moulding and cornice.

Note: In calculating area of an opening, any separate lintel or sills shall be included with the size of the opening but end portions of lintel shall be excluded. Extra width of rebated reveals, if any, shall also be excluded.

6.14.16.3 String courses, projecting pilasters, aprons, sills and other projections shall be fully described and measured separately in running metres stating dimensions of each projection.

6.14.16.4 Square or rectangular pillars shall be measured separately in cubic metres

6.14.16.5 Circular pillars shall be measured separately in cubic metres as per actual dimensions.

6.14.16.6 Autoclave Aerated Concrete Block work curved on plan shall be measured like the block work in straight walls and shall include all cutting and wastage of blocks, tapered vertical joints and use of extra mortar, if any. Block work curved on plan to a mean radius not exceeding six metres shall be measured separately and extra shall be payable over the rates for block work in straight walls. Nothing extra shall be payable if the mean radius of the block work curved in plan exceeds six metres.

6.14.16.7 Tapered walls shall be measured net as walls and extra payment shall be allowed for making tapered surface for block work in walls.

6.14.17 Rate

The rate shall include the cost of materials and labour required for all the operations described. The rate shall also include the following:

(a) Raking out joints or finishing joints flush as the work proceeds;

(b) Preparing tops of existing walls and the like for raising further new block work.

(c) Rough cutting and waste for forming gables, splays at eaves and the like.

(d) Leaving holes for pipes upto 150 mm dia. and encasing hold fasts etc.

(e) Rough cutting and waste for block work curved in plan and for backing to stone or other types off acing.

(f) Embedding in ends of beams, joists, slabs, lintels, sills, trusses etc.

(g) Bedding wall plates, lintels, sills, roof tiles, corrugated sheets, etc. in or on walls if not covered in respective items

(h) Leaving chases of section not exceeding 50 cm in girth or 350 sq cm in cross-section; and

(i) Block on edge courses, cut brick corners, splays reveals, cavity walls, brick works curved on plan to a mean radius exceeding six metres.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
6.1		Brick work with 40kg/cm^{2}chimney bricks up to plinth	Brick work with well burnt chimney bricks in bulls pattern trench kiln manufactured by ghol process, crushing strength not less than 40kg /cm^2 and water absorption not more than 15% in foundation and plinth i/c curing etc. complete.		
	6.1.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5818.00
	6.1.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	5626.00
6.2		Brick work with 40kg /cm^{2}chimney bricks in superstructure	Brick work with well burnt chimney bricks in bulls patent trench kiln manufactured by ghol process ,crushing strength not less than 40kg /cm^2 and water absorption not more than 15% in superstructure above plinth level and up to floor two level i/c curing etc. complete.		
	6.2.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	6012.00
	6.2.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	5821.00
6.3		Brick work with 25kg /cm^2 chimney bricks in foundation and plinth	Brick work with well burnt chimney bricks in bulls pattern trench kiln, crushing strength not less than 25kg /cm^2 and water absorption not more than 20% in foundation and plinth i/c curing etc complete.		
	6.3.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5252.00
	6.3.2	CM 1:6	Cement Mortar 1:6 (1 cement:6 sand).	cum	5061.00
6.4		Brick work 25kg /cm^2 chimney bricks in superstructure	Brick work with well burnt chimney bricks in bulls pattern trench kiln, crushing strength not less than 25kg /cm^2 and water absorption not more than 20% in superstructure above plinth level and up to floor two level i/c curing etc complete.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	6.4.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5446.00
	6.4.2	CM 1:6	Cement Mortar 1:6 (1 cement:6 sand).	cum	5255.00
6.5		Brick work with bhatta, bricks in foundation	Brick work with well brunt open bhatta, bricks, crushing strength not less than 25kg /cm^2 and water absorption not more than 20% in foundation and plinth i/c curing etc complete.		
	6.5.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	4347.00
	6.5.2	CM 1:6	Cement Mortar 1:6 (1 cement: 6 sand).	cum	4155.00
6.6		Brick work with bhatta, bricks in superstructure	Brick work with well brunt open bhatta, bricks, crushing strength not less than 25kg / cm^2 and water absorption not more than 20% in superstructure above plinth level and up to floor two level i/c curing etc complete.		
	6.6.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	4541.00
	6.6.2	CM 1:6	Cement Mortar 1:6 (1 cement : 6 sand)	cum	4350.00
6.7		Brick work fly ash bricks in superstructure	Brick work with fly ash lime bricks (FALG Bricks) conforming to IS:12894-2002, in super structure above plinth level up to floor two level in :		
	6.7.1	100 kg / cm^2	Having 100 kg / cm^2 average compressive strength.		
	6.7.1.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5384.00
	6.7.1.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	5162.00
	6.7.2	75 kg / cm^2	Having 75 kg / cm^2 average compressive strength.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	6.7.2.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5271.00
	6.7.2.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	5049.00
	6.7.3	50 kg / cm^2	Having 50 kg / cm^2 average compressive strength.		
	6.7.3.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5158.00
	6.7.3.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	4936.00
6.8		Brick work with calcium silicate bricks	Brick work with calcium silicate bricks machine moulded conforming to IS:4139, having 100 kg/cm^2 average compressive strength in super structure above plinth level upto floor II level in :		
	6.8.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5287.00
	6.8.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	5095.00
6.9		Brick work sewer bricks in foundation	Brick work with extruded burnt fly ash clay sewer bricks (Conforming to IS: 4885) in foundation and plinth :	cum	5082.00
			Cement Mortar 1:4 (1 cement : 4 sand)		
6.10		Brick work sewer bricks in arches	Brick work with extruded burnt fly ash clay sewer bricks (conforming to IS: 4885) in archesin foundation and plinth including centering and shuttering complete in cement mortar 1:3 (1 cement: 3 sand).	cum	7655.00
6.11		Extra for brick work above floor two level	Extra for brick work in superstructure above floor two levels for each additional floor or part thereof respective item.	cum	2%

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
6.12		Half brick masonry ghol brick 40 kg /cm^2	Providing half brick masonry with well burnt chimney bricks in bull patent trench kiln manufactured by ghol process crushing strength not less than 40 kg /cm^2 and water absorption not more than 15% in superstructure above plinth level and up to floor two level.		
	6.12.1	CM 1:3	Cement mortar 1:3 (1 cement : 3 sand)	sqm	697.00
	6.12.2	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	sqm	673.00
6.13		Half brick masonry chimney 25 kg /cm^2	Providing half brick masonry with well burnt chimney bricks in bull pattern trench kiln crushing strength not less than 25 kg /cm^2 and water absorption not more than 20% in superstructure above plinth level and up to floor two level.		
	6.13.1	CM 1:3	Cement mortar 1:3 (1 cement : 3 sand)	sqm	639.00
	6.13.2	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	sqm	615.00
6.14		Half brick masonry open bhatta bricks	Providing half brick masonry with well burnt open bhatta brickscrushing strength not less than 25 kg /cm^2 and water absorption not more than 20% in superstructure above plinth level and up to floor two level		
	6.14.1	CM 1:3	Cement mortar 1:3 (1 cement : 3 sand)	sqm	546.00
	6.14.2	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	sqm	522.00
6.15		Half brick masonry fly ash lime bricks	Half brick masonry with fly ash lime Gypsum bricks (FALG bricks) conforming to IS: 12894, in super structure above plinth and upto floor two level.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	6.15.1	100 kg/cm^2	Having 100 kg/cm^2 average compressive strength.		
	6.15.1.1	CM 1:3	Cement mortar 1 : 3 (1 cement : 3 sand)	sqm	570.00
	6.15.1.2	CM 1:4	Cement mortar 1 : 4 (1 cement : 4 sand)	sqm	551.00
	6.15.2	75 kg/cm^2	Having 75 kg/cm^2 average compressive strength.		
	6.15.2.1	CM 1:3	Cement mortar 1 : 3 (1 cement : 3 sand)	sqm	524.00
	6.15.2.2	CM 1:4	Cement mortar 1 : 4 (1 cement : 4 sand)	sqm	505.00
	6.15.3	50 kg/cm^2	Having 50 kg/cm^2 average compressive strength.		
	6.15.3.1	CM 1:3	Cement mortar 1 : 3 (1 cement : 3 sand)	sqm	479.00
	6.15.3.2	CM 1:4	Cement mortar 1 : 4 (1 cement : 4 sand)	sqm	459.00
6.16		Extra for half brick masonry above floor two level	Extra for half brick masonry in superstructure, above floor two level for every floor or part thereof respective item.	sqm	11.00
6.17		Extra for M.S. bars in half brick masonry	Extra for providing and placing in position 2 Nos. 8mm dia. M.S. bars at every third course of in half brick masonry.	sqm	127.00
6.18		Honey-comb brick work	Honey-comb brick work 10 cm thick with well burnt chimney bricks manufactured by ghol process crushing strength not less than 40kg /sqcm and water absorption not more than 15% in cement mortar 1:4 (1 cement : 4 sand).	sqm	456.00
6.19		Extra for forming cavity in wall	Extra for forming cavity 5cm to 7.5cm wide in cavity walls with necessary weep and vent holes including use of cores and cost of providing and fixing bitumastic coated M .S. Ties 300mm long of	sqm	75.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			25x3mm section at not less than 3 ties per sqm as per approved design.		
6.20		Brick work in plain arches	Brick work in plain arches in superstructure including centering and shuttering complete with chimney bricks having crushing strength not less than 40kg /cm^2 and water absorption not more than 15% in cement mortar 1:3 (1 cement : 3 sand).	cum	8507.00
6.21		Brick work in gauged arches	Brick work in gauged arches in superstructure in cement mortar 1:3 (1 cement : 3 sand) including centering and shuttering complete, with chimney bricks having crushing strength not less than 40kg /cm^2 and water absorption not more than 15%	cum	9691.00
6.22		Half brick masonry for closing cavity	Providing half brick masonry with kiln burnt clay bricks of 40kg /cm^2 and water absorption not more than 15% in cement mortar 1:3 (1Cement : 3 sand) in superstructure for closing cavity 5 to 7.5 cm widein cavity wall complete with 10 cm wide bitumen felt type 3 grade 1.	metre	210.00
6.23		Extra for brick work under water	Extra for laying brick work in or under water and/or liquid mud including cost of pumping or bailing out water and removing slush etc. complete. **NOTE: -** The quantity will be calculated by multiplying the depth measured from sub - soil water level up to the center of gravity of brick work under sub - water with the quantity of brick work in cum executed under the sub - soil water. The depth of center of gravity shall be reckoned correct to 0.1 m. 0.05 mor more shall be taken as 0.1 m and less than 0.05 m ignored.	Cum per metre depth	297.00
6.24		Extra for brick work in foul position	Extra for laying brick work in or under foul position.	cum	109.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
6.25		Cellular concrete block masonry	Providing and laying cellular concrete blocks masonry having density ranging from 800kg/m^3 in super structure above plinth level up to floor two level in cement mortar 1:4 (1 cement: 4 sand) The nominal dimensions of the concrete blocks shall be as follows in additional block shall be of half lengths of 200, 250 or 300 mm to correspond to the full lengths. The size of the blocks specified other than those specified may also be used.	cum	5052.00
6.26		Autoclaved Aerated Concrete (AAC) blocks masonry	Providing and laying Autoclaved Aerated Concrete (AAC) blocks masonry, using AAC blocks (Size length 300, 400, 500, 600 mm Height 200, 250, 300 mm Width 100, 150, 200, 250 mm) conforming to IS 2185 PART 3, having compressive strength not less than 4 MPa (Grade II) in super structure, blocks jointed with approved block laying polymer modified adhesive mortar as per ASTM 1660-09,applied in uniform thickness of 2 to 3 mm in horizontal and verticals joints using special trowel, polymer modified mortar having minimum compressive strength and minimum splitting tensile strength of not less than 5MPa and 0.34 MPa respectively as per ASTM international standard including raking out joints. The surface of masonry units (AAC-Block) should be slightly moist prior to application of jointing mortar. Block should be gently pressed down using a rubber mallet to expel out entrapped air and the masonry shall be cured two days. 1st layer of masonry block shall be placed on perfectly leveled surface, which if required, shall be leveled with M-15 cement concrete at plinth		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			or floor level. Masonry shall have 75mm thick RCC band with M-15 cement concrete having 2 bars horizontal of 8 mm dia with 8mm dia U- shaped single stirrup @ 300mm C/C. Such horizontal bands shall be placed at sill level and lintel level where there is a provision of windows in walls otherwise placedat vertical interval of 1.8 meter in vertical direction shall be spaced at 4.5 meter in horizontal direction. Mullion,Vertical bands, shall be placed along jams (vertical member of chowkhats) and extend upto lintel levels to arrest cracking at corner openings. RCC bands shall be measured with AAC block masonry. Rates include all materials including steel bars, wastages, RCC, Centering scaffolding and all labour etc. complete.		
	6.26.1	Polymer modified mortar	AAC Masonry in polymer modified mortar	cum	5500.00
6.27		Extra for AAC work above floor two level	Extra for AAC masonry work in superstructure above floor two level for each additional floor or part thereof	cum	1%
6.28		Precast Reinforced AAC roof slabs, lintels and chhajjas	Providing hoisting & fixing precast Autoclaved Aerated Concrete (AAC) reinforced roof slabs, lintels and chhajjas as per IS: 6073 -2006 grade D specifications (dry density, without steel, 550 kg/m3 to 650 kg/m3 and minimum compressive strength 3.5 N/mm2) Reinforcement shall be designed & spot welded as per ACI 523.4R-09 and shall haveanti corrosive protective coating as per IS: 6441 (Part -4). The roof slabs shall have tongue and groove arrangement, jointed with polymer modified mortar, as per ASTMC 1660-09, groove provided with one bar of 10mm dia and filled with M20grade cement concrete. Both ends of bar in groove shall be tied with bond beam of minimum 80 mm	cum	5700.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			width and depth equal to slab/panel thickness running at center of load bearing wall/ beam/ supports and on other support as well to achieve diaphragm action as per IS 1893:2000. The Peripheral bond beam shall have minimum 2 bars of 12 mm dia as longitudinal barswhich shall be held down by hook made in 12 mm dia vertical bars coming out from structural beam. However Concrete in groove and bond beam shall not be paid separately but shall be measured with AAC reinforced slab. Precast orlintel and chhajja shall be jointed with cm 1:6 or 2 to 3 mm thick polymer modified cement mortar as per ASTMC 1660-09 to AAC Block masonry. Structural steel if used for providing support to the lintels (suchas cleat of angle iron, C-section anchored/hold fasted to column) at column end towards (where door / window opening is adjacent to the column) shall not be paid separately. Rate includes all handling, transportation, hire charges of machineries scaffolding and anti-corrosive protective coating on reinforcement inside AAC slab / lintel/chhajja etc. complete up to floor two levels. All reinforcementof peripheral bond beam, grooves and inside AAC slab, lintel, and chhajja shall be paid @ SOR rate(including tender percentage) separately. Quantity of reinforcement steel provided inside AAC slab, lintel and chhajja shall be ascertained on the basis of manufacturer's certificate and also by random destructive testing done by engineer-in-charge or his representative for which cost of samples and testing charges shall be borne by the contractor. Deflection & ultimate load test for AAC reinforced roof slab and lintel shall		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			be as per annex. B of IS: 6073 -2006 and criteria for conformity and sampling shall be as per clause-13 of IS: 6073 -2006.		
6.29		Precast Reinforced AAC wallpanels	Precast autoclaved aerated concrete (AAC) reinforced wall panels:- Providing hoisting & fixing precast Autoclaved Aerated Concrete (AAC) reinforced wall panels as per IS:6072 -1972 grade D (dry density, without steel, 550 kg/m3 to 650 kg/m3 and minimum compressive strength (3.5 N/mm2). Wall panels shall have tongue and groove arrangement, jointed with polymer modified mortar as per ASTMC 1660-09. Galvanized iron connector/ cleats, nails & screw and all accessories as per manufactures designed shall not be paid separately. All reinforcement inside panels, as per manufacturer's design & spot welded as per ACI 523.4R-09, shall have anti corrosive protective coating as per IS: 6441 (Part -4), shall be paid separately. Rate includes all handling, transportation, hire charges of machineries scaffolding and anti-corrosive protective coating on reinforcement inside AAC wall panels etc. complete up to floor two level but does not include cost of reinforcement inside AAC panels. Quantity of reinforcement steel in AAC panels shall be ascertained on the basis of certificate of the manufacturer and also by random destructive testing by engineer-in-charge or his representative for which cost of samples and testing charges shall be borne by the contractor. Cost of reinforcement inside AAC panels shall be paid separately.	cum	5800.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
6.30		Extra for reinforced AAC work above floor two level	Add extra for AAC wall panels, roof slabs, lintels and chhajjas abovefloor two level for each additionalfloor as per item no 6.29 and 6.30.	cum	1%
6.31		AAC blocks masonry Jali (honey comb)	Providing and laying Autoclaved Aerated Concrete (AAC) blocks masonry Jali work, using AAC blocks (Size length 300, 400, 500, 600 mm Height 200, 250, 300 mm Width 100, 150, 200, 250 mm) conforming to IS 2185 PART 3, having compressive strength not less than 4 MPa (Grade II) in super structure, blocks jointed with approved block laying polymer modified adhesive mortar as per ASTM 1660-09,applied in uniform thickness of 2 to 3 mm in horizontal and verticals joints using special trowel, polymer modified mortar having minimum compressive strength and minimum splitting tensile strength of not less than 5MPa and 0.34 MPa respectively as per ASTM international standard including raking out joints. The surface of masonry units (AAC-Block) should be slightly moist prior to application of jointing mortar. Block should be gently pressed down using a rubber mallet to expel out entrapped air and the masonry shall be cured two days. 1st layer of masonry block shall be placed on perfectly leveled surface, which if required, shall be leveled with M-15 cement concrete at plinth or floor level. . Rates include all materials including steel bars, wastages, scaffolding and all labour etc. complete.as per architectural drawing. The rates are inclusive of all lead & lifts and All necessaries to complete the item in good condition, nothing shall be paid extra separately unless specified.	cum	4237.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
6.32		Exposed non modular brick work of class designation 7.5	Brick work with common burnt clay selected F.P.S. (non modular) bricks of class designation 7.5 in exposed brick work including making horizontal and vertical grooves 10 mm wide 12 mm deep complete in cement mortar 1:6 (1 cement : 6 coarse sand)		
	6.32.1	Upto plinth	From ground level upto plinth level	cum	4689.00
	6.32.2	Above plinth upto floor two level	Above plinth level upto floor II level	cum	5236.00
6.33		Exposed modular brick work of class designation 7.5	Brick work with modular bricks of class designation 7.5 in exposedbrick work including making horizontal and vertical grooves 10mm wide 12mm deep complete from ground level up to plinth level in cement mortar 1:6 (1 cement : 6 coarse sand)		
	6.33.1	Upto plinth	From ground level upto plinth level	cum	4038.00
	6.33.2	Above plinth upto floor two level	Above plinth level upto floor II level	cum	4563.00
6.34		Gypsum panel partitions	Providing and laying Gypsum panel partitions 100 mm thick with water proof Gypsum panels of size 666x500x100 mm, made of calcite phosphor Gypsum fixed with tongue and groove, jointed with bonding plaster as per manufacturer's specifications in superstructure above plinth level up to floor two level. Gypsum blocks will have a minimum compressive strength of 9.3 kg/cm2	sqm	653.00
6.35		Extra for above floor two level	Extra for Gypsum panel Partitions in superstructure above floor two level for every four floors or part thereof.	sqm	70.00
6.36		Brick work on exposed wall	Providing and laying Red Brick work of size 200X100x50 mm on exposed wall in cement morter 1:3 including racking and jointing rubbing and filler polishing 2 coat as per direction by Engineer in Charge.	sqm	1838.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	**2**	**3**	**4**	**5**	**6**
6.37		Brick work with fly ash bricks up to plinth	Brick work with fly ash lime bricks (FALG Bricks) conforming to IS:12894-2002,		
	6.37.1	100 kg / cm2	Having100 kg / cm2 average compressive strength.		
	6.37.1.1	CM 1:4	Cement mortar 1:4 (1 cement : 4 sand)	cum	5115.00
	6.37.1.2	CM 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	4904.00
	6.37.2	75 kg / cm2	Having75 kg / cm2 average compressive strength.		
	6.37.2.1	CM 1:4	Cement mortar 1:4 (1 cement : 4sand)	cum	5007.00
	6.37.2.2	CM 1:6	Cement mortar 1:6 (1 cement : 6sand)	cum	4797.00
	6.37.3	50 kg / cm^2	Having50 kg / cm^2 averagecompressive strength.		
	6.37.3.1	CM 1:4	Cement mortar 1:4 (1 cement : 4sand)	cum	4900.00
	6.37.3.2	CM 1:6	Cement mortar 1:6 (1 cement : 6sand)	cum	4689.00

CHAPTER – 7.0

STONE WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 737	Specifications for wrought aluminium and aluminium alloy, steel and strip for general engineering purpose.
2	IS 1121 - (Part-I)	Methods of determination of properties and strengths of natural building stones (Part-I compressive strength).
3	IS 1122	Methods for determination of specific gravity of natural building stone
4	IS 1123	Methods of identification of natural building stones.
5	IS 1124	Methods of test of determination of water absorption, apparent, specific gravity and porosity of natural building stones.
6	IS 1125	Methods of test of determination of weathering of natural building stone
7	IS 1126	Methods of test for determination of durability of natural building stone
8	IS 1128	Specification for Lime stone (Slab and Tiles).
9	IS 1129	Recommendations for dressing of natural building stones.
10	IS 1200 (Part-IV)	Methods of measurements of building and Civil engineering works stone Masonry.
11	IS 1197 (Part-I)	Code of practice for construction of rubble stone masonry
12	IS 1597 (Part-II)	Code of practice for construction of ashlar stone masonry
13	IS 1805	Glossary of terms relating to stones, quarrying and dressing
14	IS 3620	Specification for laterite stone block for masonry
15	IS 3622	Sand stone (Slab and Tiles)
16	IS 4104 (Part-I)	Code of practice for external facings and veneers (Part I-Stone facing)
17	IS 4101 (Part-II)	Code of practice for external facing and veneers: (Part II-Cement Concrete facing).

LIST OF MANDATORY TESTS

Material	Clause	Test		Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
Stone	7.1.1 7.4 7.8 7.9	(i) Water Absorption	Not more than 2.5% by mass for sand stone and as specified in IS 1123 for Other stones.	Laboratory	IS 1124	50 sqm. for slabs and 10 cum in stone masonry	100 sqm/20 cum or part thereof or change of source as per direction of Engineer in-Charge
		(ii) Transverse Strength	Not less than 7 N/mm2 (70 Kgf/cm2) for sand stone and as specified in IS 1123 for Other stones.	Laboratory	IS 1121 Part II	-do-	-do-
		(iii) Resistance to wear	Not greater than 2 mm on the average and 2.5 mm for any individual specimen for sand stone and as specified in IS 1123 for other stones.	Laboratory	IS 1706	-do-	-do-
		(iv) Durability	Shall not develop signs of spalling, disintegration or cracks for sand stone and as specified in IS 1123 for Other stones.	Laboratory	IS 1126	-do-	-do-

PREAMBLE
STONE WORK
(CHAPTER: 7.0)

7.1 RANDOM RUBBLE STONE MASONRY

- **Material**
- **Stone**

The stone shall be of the type specified such as granite, trap, limestone, sand stone, quartzite, etc. and shall be obtained from the quarries, approved by the Engineer-in-Charge. Stone shall be hard, sound, durable and free from weathering decay and defects like cavities, cracks, flaws, sand holes, injurious veins, patches of loose or soft materials and other similar defects that may adversely affect its strength and appearance. As far as possible stones shall be of uniform colour, quality or texture. Generally stone shall not contain crypts crystalline silica, mica and other deleterious materials like iron-oxide organic impurities etc. Stones withround surface shall not be used. The compressive strength of common types of stones shallbe as per Table 7.1 and the percentage of water absorption shall generally not exceed 5% for stones other than specified in Table 7.1. For laterite this percentage is 12%.

Table 7.1
Types of Stones

Type of stone	Maximum Water Absorption Percentage by weight	Minimum Compressive Strength kg./sq.cm.
Granite	0.5	1000
Basalt	0.5	400
Lime stone (Slab and Tiles)	0.15	200
Sand stone (Slab and Tiles)	2.5	300
Marble	0.40	500
Quartzite	0.40	800
Late rite (Block)	12	35

Note 1: Test for compressive strength shall be carried out as laid down in IS 1121 (Part I).
Note 2: Test for water absorption shall be carried out as laid down in IS 1124.

7.2 Size of Stones

Normally stones used should be small enough to be lifted and placed by hand. Unless otherwise indicated, the length of stones for stone masonry shall not exceed three times the height and the breadth on base shall not be greater than three-fourth of the thickness of wall, or not less than 150 mm. The height of stone for rubble masonry may be upto 300 mm. The selection and grading of stones for rubble masonry is largely done at site and the smaller stones are used in the hearting of wall.

Random Rubble Masonry shall be uncoursed or brought to courses as specified. Uncoursed random rubble masonry shall be constructed with stones of sizes as specified and shapes picked up random from the stones brought from the approved quarry. Stones having sharp corners or round surfaces shall, however, not be used.

Random rubble masonry brought to the course is similar to uncoursed random rubble masonry except that the courses are roughly levelled at intervals varying from 300 mm to 900 mm in height according to the size of stones used.

7.3 Dressing

Hammer Dressed Surface

A hammer dressed stone shall have no sharp and irregular corners and shall have a comparatively even surface so as to fit well in masonry. Hammer dressed stone is also known as hammer faced, quarry faced and rustic faced. The bushing from the general wall face shall not be more than 40 mm on exposed face and 10 mm on faces to be plastered.

Each stone shall be hammer dressed on the face, the sides and the beds. Hammer dressing shall enable the stones to be laid close to neighboring stones such that the bushing in the faceshall not project more than 40 mm on the exposed face.

(i) **Face stone:** At least 25% stones shall be headers tailing into the work at least 2/3rd the thickness of wall in super structure masonry. Such stones shall not be less than 200 sq. cm in cross sections.

(ii) **Hearting Stones:** The hearting or interior filling of a wall face shall consist of rubble stones not less than 150 mm in any direction, carefully laid, hammered down with a wooden mallet into position and solidly bedded in mortar. The hearting should be laid nearly level with facing and backing.

(iii) **Quoin Stone:** Quoin stone shall be less than 0.03 cum in volume.

(iv) **Jamb stones:** The jambs shall not be made with stones specified for quoins except that the stones which were required to be provided at 1 metre centre to centre on both the exposed faces shall here be provided only on the jamb and the length shall be equal to the thickness of the wall for wall upto 60 cm and a line of headers shall be provided for walls thicker than 60 cm as specified for bond.

7.4 Courses

The masonry shall be carried out in regular courses of height not exceeding 50 cm and masonry on any day will not be raised more than 60 cm in height when using mortars having compressive strength less than 20 kg. /sq. cm at 28 days and 100 cm when using mortars exceeding this strength.

7.5 Thickness of Joints

The joint thickness shall not exceed 30 mm at any point on the face. Chips of the stone and spalls shall be wedged into seating bed of face stones to avoid excessive bed thickness. No pinning shall be allowed to avoid excessive joint thickness.

7.6 Mortar

The mortar used for joining shall be as specified.

7.7 Laying

Stone shall be laid on their natural bed and shall be solidly bedded full in mortar with close joints, chips of stone spalls be wedged into the work wherever necessary. No dry work or hollow spaces shall be allowed and every stone whether large or small shall be carefully

selected to fit snugly the interstices between the large stones. Masonry shall be built breaking joints in all the three directions. Bond stone and headers shall be properly laid into the work and shall be marked by the contractor with white lead paint. The bond stones shall be provided as specified. The masonry work in wall shall be carried up true to plumb or to specified batter. Random rubble masonry shall be brought to the level courses at plinth, window sills, and lintel and roof levels. Leveling shall be done with concrete comprising of one part of the mortar as used for masonry and two parts of graded stone aggregate of 20 mmnominal size. The masonry in structure shall be carried uniformly. Where the masonry of onepart is to be delayed, the work shall be raked back at an angle not steeper than 45°.

7.8 Raking out joints

All the joints on the faces to be pointed or plastered shall be racked out with racking tool to a depth of 20mm while the mortar is still green.

7.9 Bond Stones

Though bond stones shall be provided in walls upto 600 mm thickness, a set of two or more bond stones overlapping each other by at least 150 mm shall be provided in a line from face to back. In case of highly absorbent types of stones (porous lime stone and sand stone etc.) the bond stone shall extend about two-third into the wall, as through stones in such walls a set of two or more bond stones overlapping each other by at least 150 mm shall be provided. Each bond stone or a set of bond stones shall be provided for every 0.5 m^2 of the wallsurface and shall be provided at 1.5 m to 1.8 m apart clear in every course. In case of highly absorbent types of stones (porous lime stone and sand stone etc.) single piece bond stones may give rise to dampness. For all thicknesses of such walls a set of two or more bond stonesoverlapping each other by at least 15 cm shall be provided. Length of each such bond stone shall not be less than two-third of the thickness of the wall. Where bond stones of suitable lengths are not available pre-cast cement concrete block of 1:3:6 mix (1 cement : 3 sand: 6 graded stone aggregate 20 mm nominal size) of cross section not less than 225 square centimeters and length equal to the thickness. At least one bond stone or a set of bond stones shall be provided at 1.5 m to 1.8 m apart clear in every course. (Bond stones shall be marked suitably with paint as directed by the Engineer-in-Charge).

7.10 Quoin and Jamb Stones

The quoin and jamb stones shall be of selected stones neatly dressed with hammer or chisel to form the required angle. Quoin stones shall not be less than 0.01 cum in volume. Height of quoins and jamb stones shall not be less than 15 cm. Quoins shall be laid header and stretcher alternatively.

7.11 Joints

Stones shall be so laid that all joints are fully packed with mortar and chips. Face joints shall not be more than 20 mm thick. The joints shall be struck flush and finished at the time of laying when plastering or pointing is not to be done. For the surfaces to be plastered or pointed, the joints shall be raked to a minimum depth of 20 mm when the mortar is still green.

7.12 Scaffolding

Single scaffolding having one set of vertical support shall be allowed. The supports shall be sound and strong, tied together by horizontal pieces, over which the scaffolding planks shall be fixed. The inner end of the horizontal scaffolding member may rest in a hole provided in the masonry. Such holes, however, shall not be allowed in pillars under one metre in width or near the skew back of arches. The holes left in masonry work for supporting scaffolding shall be filled and made good with cement concrete 1 : 3 : 6 (1 cement : 3 sand : 6 stone aggregate 20 mm nominal size).

COURSED RUBBLE MASONRY - FIRST SORT

7.13 Dressing

Face stones shall be hammer dressed on all beds, and joints so as to give them approximately rectangular block shape. These shall be squared on all joints and beds. The bed joint shall be rough chisel dressed for at least 80 mm back from the face, and side joints for at least 40 mm such that no portion of the dressed surface is more than 6 mm from a straight edge placed on it The remaining unexposed portion of the stone shall not project beyond the surface of bed and side joint. The bushing on the face shall not project more than 40 mm as an exposed face and 10 mm on a face to be plastered. The hammer dressed stone shall also have a rough tooling for minimum width of 25 mm along the four edges of the face of the stone, when stone work is exposed.

7.14 Mortar

The mortar for jointing shall be as specified.

7.15 Laying

All stones shall be wetted before use. The walls shall be carried up truly plumb or to specified batter. All courses shall be laid truly horizontal and all vertical joints shall be truly vertical. The height of each course shall not be less than 15 cm nor shall more than 30 cm. Face stones be laid alternate headers and stretchers. No pinning shall be allowed on the face. No face stone shall be less in breadth than its height and at least one third of the stones shall tail into the work for length not less than twice their height. The hearting or the interior filling of the wall shall consist of stones carefully laid on their proper beds in mortar; chips and spalls of stone being used where necessary to avoid thick beds of joints of mortar and at the same time ensuring that no hollow spaces are left anywhere in the masonry. The chips shall not be used below the hearting stone to bring these upto the level of face stones. Theuse of chips shall be restricted to the filling of interstices between the adjacent stones in hearting and these shall not exceed 10% of the quantity of stone masonry. The masonry in a structure shall be carried up uniformly but where breaks are unavoidable, the joints shall be raked back at angle not steeper than 45°. Toothing shall not be allowed.

7.16 Bond Stones

Shall be as specified except that a bond stone or a set of bond stones shall be inserted 1.5 to 1.8 metres apart, in every course.

7.17 Quoins

The quoins shall be of the same height as the course in which these occur. These shall be at least 450 mm long and shall be laid stretchers and headers alternatively. These shall be laid square on the beds, which shall be rough-chisel dressed to a depth of at least 100 mm. In case of exposed work, these stones shall have a minimum of 25 mm wide chisel drafts at four edges, all the edges being in the same plane.

7.18 Joints

All bed joints shall be horizontal and all side joints vertical. All joints shall be fully packed with mortar, face joints shall not be more than one cm thick. When plastering or pointing is not required to be done, the joints shall be struck flush and finished at the time of laying. Otherwise, joints shall be raked to a minimum depth of 20 mm by raking tool during the progress of work, when the mortar is still green.

COURSED RUBBLE MASONRY - SECOND SORT

7.19 Dressing:

Shall be as specified except that no portion of dressed surface of joints shall show a depth of gap more than 10 mm from a straight edge placed on it and use of chips shall not exceed 15 percent of the quantity of stone masonry.

7.20 Mortar

The mortar for jointing shall be as specified.

7.21 Laying

Shall be as specified as above except that the use of chips shall not exceed 15% of the quantity of stone masonry and stone, in each course need not be of the same height but not more than two stones shall be used in the height of a course.

7.22 Joints

All bed joints shall be horizontal and all side vertical. All joints shall be fully packed with mortar, face joints shall not be more than 20 mm thick. When plastering or pointing is not required to be done, the joints shall be struck flush and finished at the time of laying. Otherwise, the joints shall be raked to a minimum depth of 20 mm by raking tool during progress of work, where the mortar is still green.

7.23 Plain Ashlar Masonry

Stone shall be of the type specified. It shall be hard, sound, durable and tough, free from cracks, decay and weathering and defects like cavities, cracks, flaws, sand holes, veins,patches of soft or loose materials etc. before starting the work, the contractor shall get the stones approved by Engineer-in-Charge.

7.24 Kota Stone for Veneering

Kota stone shall be of selected quality, hard, sound, dense and homogeneous in texture free from cracks, decay, weathering and flaws. They shall be machine cut to requisite size and thickness. They shall be of colour indicated in the drawings or as instructed by the Engineer-in-Charge. The stone shall have the top (exposed) face polished before being brought to site unless otherwise specified. Before starting the work, the contractor shall get the samples of Kota stone approved from the Engineer-in- Charge.

7.25 Dressing

Every stone shall be cut to the required size and shape and fine machine dressed to the full depth so that a straight edge laid along the side of stone shall be in full contact with it. The thickness of the slab after it is dressed shall be 20, 25, 30 or 40 mm as specified in the item. Tolerance of ± 2 mm shall be allowed for the thickness.

7.26 Red Sand Stone and White Sand Stone Ashlar Masonry

The stone shall be red or white as specified in the description of item. The stone shall be hard, sound, tough, and free from cracks, decay and weathering. In case of red sand stone, white patches or streaks shall not be allowed. However scattered spots upto 10 mm diameter will be permitted. Before starting the work the contractor shall get samples of stone approved by the Engineer-in-Charge.

7.27 Size of Stone

Normally stones used should be small enough to be lifted and placed by hand. The length of the stone shall not exceed three times the height and the breadth on base shall not be greater than three fourth of the thickness of wall nor less than 15 cm. The height of stone may be upto 30 cm.

7.28 Dressing

Every stone shall be cut to the required size and shape chisel dressed on all beds and joints so as to be free from waviness and to give truly vertical and horizontal joints. In exposed masonry, the faces that are to remain exposed in the final position and the adjoining faces to a depth of 6 mm shall be the fine chisel dressed so that when checked with 60 cm straight edge, no point varies from it by more than 1 mm. The top and bottom faces that are to form the bed joints shall be chisel dressed so that variation from 60 cm straight edge at no point exceeds 3 mm. Faces which are to form the vertical joints should be chisel dressed so that variation at any point with 60 cm straight edge does not exceed 6 mm. Any vertical face that is to come against backing of masonry shall be dressed such that variation from straight edge does not exceed 10 mm. All angles and edges that are to remain exposed in the final position shall be true, square and free from chippings. A sample of dressed stone shall be prepared for approval of Engineer-in-Charge. It shall be kept at the worksite as a sample after being approved.

7.29 Mortar

The mortar for jointing shall be as specified.

7.30 Laying

All stones shall be wetted before placing in position. These shall be floated on mortar and bedded properly in position with wooden mallets without the use of chips or under pinning of any sort. The walls and pillars shall be carried up truly plumb or battered as shown in drawings. All courses shall be laid truly horizontal and all vertical joints shall be truly vertical. In case of ashlar work without backing of brick work or coursed rubble masonry, face stone shall be laid headers and stretchers alternately unless otherwise directed. The headers shall be arranged to come as nearly as possible in the middle of stretchers aboveand below. Stone shall be laid in regular courses of not less than 30 cm in height and all the courses shall be of same height, unless otherwise specified. For ashlar facing with backingof brick work or coursed rubble masonry face stone shall be laid in alternate courses of headers and stretchers unless otherwise directed. Face stone and bond stone course shall be maintained throughout. All connected masonry in a structure shall be carried up nearly at one uniform level throughout, but where breaks are avoidable, the joint shall be made in good long steps so as to prevent cracks developing between new and old work. Bond stone provided in the masonry shall be payable in the item of Ashlar masonry. Neither any deduction will be made from the brick masonry for embedding the bond stone in the backing nor any extra payment shall be made for any extra labour involved in making holes in brick masonry backing. When necessary, jib crane or other mechanical appliances shall be used to hoist the heavy pieces of stones and place these into correct positions, care being taken that the corners of the stone are not damaged. Stone shall be covered with gunnybags, before tying chain or rope is passed over it, and it shall be handled carefully. No piece which has been damaged shall be used in work.

7.31 Joints

All joints shall be full of mortar. These shall be not more than 6 mm thick. Face joints shall be uniform throughout and a uniform recess of 20 mm depth from face shall be left with the help of the steel plate during the progress of work.

7.32 Pointing

All exposed joints shall be pointed with mortar as specified. The pointing when finished shall be sunk from stone face by 5 mm or as specified. The depth of mortar in pointing workshall not be less than 15 mm.

7.33 Curing

Masonry work in cement or composite mortar shall be kept constantly moist on all faces for a minimum period of seven days. In case of masonry with fat lime mortar, curing shall commence two days after laying of masonry and shall continue for at least seven days thereafter.

7.34 Protections

Green work shall be protected from rain by suitable covering. The work shall also be suitably protected from damage, mortar dropping and rain during construction.

7.35 Punched Ashlar (Ordinary) Masonry

Stone: Shall be as specified above In case of red or white sand stone, stone shall be red or white as specified in the item. In red sand stone, white patches or streaks shall not be allowed. However, scattered spots upto 10 mm diameter will be permitted.

Dressing: Shall be as previously described for ashlar masonry except that the faces exposed in view shall have a fine dressed chisel draft 2.5 cm wide all-round the edges and shall be rough tooled between the drafts, such that the dressed surface shall not be more than 3 mm from a straight edge placed over it.

7.36 Other Details

The specifications for mortars, laying and fixing, bond stone, joints, pointing, curing, protections, scaffolding, measurements and rates shall be same as specified.

7.37 Stone Veneering Work

Stone lining upto 8 cm shall be treated as veneering work and lining of greater thickness as plain Ashlar Masonry.

The stone shall be gang saw cut into slabs of required thickness along the planes parallel to the natural bed of stone.

7.38 Dressing

Shall be as specified in earlier paras except that dressing at the back shall not be done, so as to ensure better grip with the hearting or backing. The dressed slabs shall be of the thickness as specified, with permissible tolerance of ± 2 mm.

7.39 Mortar

Mortar for fixing shall be as specified.

7.40 Laying

- The stone shall be wetted before laying. They shall then be fixed with mortar in position without the use of chips or underpinning of any sort.

- Where so desired, the adjoining stones shall be secured to each other by means of copper pins 75 mm long and 6 mm diameter or as specified.

- Further the stones shall be secured to the backing by means of cramps. The material for cramps shall have high resistance to corrosion under conditions of dampness and against the chemical action of mortar or concrete in which cramps are usually embedded. Cramps shall be of 25 mm x 6 mm and 30 cm long in case of backing of stone masonry walls and brick masonry walls thicker than 200 mm. In case of backing with brick masonry walls 200 mm or less thick or RCC members, cramps shall be of 25 x 6 mm and length as per requirement made out of stainless steel or any other metal specified. Generally the outer length of cramp

in half brick work backing shall be 115 mm and in one brick work backing it shall be 150 mm. Cramps shall be spaced not more than 60 cm apart horizontally. Alternatively the stone may be secured to the backing by means of stone dowels 10 x5 x2.5 cm and the adjoining stone secured to each other by means of stainless steel cramps or copper pins of thespecified size. Minimum one cramp/stone dowel shall be used to secure one slab to the backing.

- Cramps may be attached to its sides or top and bottom or sides, top and bottom. The actual number of cramps and their sections, however, shall be as per requirements of design to carry the loads.

- Where cramps are used to hold the unit in position only, the facings shall be provided with a continuous support on which the stones rest at the ground level and other storey levels, the support being in the form of projection from or recess into the concrete floor slab, or a beam between the columns or a metal angle attached to the floor slab or beams. These supports shall preferably be at vertical intervals not more than 3.5 m apart and also over the heads of all openings. Such supports shall also be provided where there is transition from thin facings below to thick facings above.

- Alternatively cramps may be used to hold the units in position and in addition to support the units thus transferring the weight of the units to the backing. Such cramp should be properly designed as per IS 4101 (Part 1).

- The cramps shall be of copper alloyed with zinc or nickel or of stainless steel of grade 304.

- The pins, cramps and dowels shall be laid in cement mortar 1:2 (1 cement: 2 sand) and their samples got approved by the Engineer-in-Charge and kept at site.

- The walls shall be carried up truly plumb. All courses shall be laid truly horizontal and all vertical joints truly vertical. The stone shall break joints on the face for at least half the height of the course, unless otherwise shown in the drawings. The stone shall be laid in regular courses not less than 20 cm height and all the stones shall be of the same height unless otherwise specified. No stone shall be less in length than one and a half times its height unless otherwise specified.

- As far as possible the backing shall be carried up simultaneously with the face work. In case of reinforced cement concrete backing, the lining shall be secured to the backing after it has set and got cured. The cramps shall be fixed in concrete at the required positions, while laying.

7.41 Joints

The joints shall be done with cement mortar 1: 3 (1 cement: 3 sand). All joints shall be full of mortar. Special care shall be taken to see that the groundings for veneer work are full of mortar. If any hollow grounding is detected by taping the face stones, these shall be taken out and relaid. The thickness of joints shall be as small as possible, not exceeding 5 mm.For a close butt jointed facing the thickness shall not exceed 1.5 mm. The face joints shall be uniform throughout. Where joint filler or compound is to be used, the joints shall beraked out to a depth of at least 25 mm after the mortar in the joints has set sufficiently and the filler or compound applied. The joints may be subsequently finished with a mortar suited to the appearance of the work. It is preferable to use joint sealing compounds where the facings are exposed to heavy rainfall and winds and their selections would depend upon local experience and availability of joint sealing compounds. In their absence only masonry mortars 1:3 (1 cement: 3 sand) which are proved to be successful from local exposure conditions shall be used.

7.42 Curing

Masonry work in cement or composite mortar shall be kept constantly moist on all faces for a minimum period of seven days. In case of masonry with fat lime mortar curing shall commence two days after laying of masonry and shall continue for at least seven days thereafter.

7.43 Protection

Green work shall be protected from rain by suitable covering. The work shall also be suitably protected from damage, mortar dropping and rain during construction.

7.44 Rate

The rate shall include the cost of materials and labour required for all the operations described above and shall include the following:

(i) Raking out joints for plastering or pointing done as a separate item, or finishing flush as the work proceeds.

(ii) Preparing tops and sides of existing walls for raising and extending.

(iii) Rough cutting and waste for forming gables cores, skew backs or spandrels of arches, splays at eaves and all rough cutting in the body of walling unless otherwise specified.

(iv) Bond stones or cement concrete bond blocks.

(v) Leading and making holes for pipes etc.

(vi) Bedding and pointing wall plates, lintels, sills etc. in or on walls, bedding roof tiles and corrugated sheets in or on walls.

(vii) Building in ends of joists, beams, lintels etc.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
7.1		RR masonry in foundation and plinth	Random rubble masonry with hard stone in foundation and plinth including leveling up with cement concrete 1:6:12 (1 cement :6 sand : 12 graded stoneaggregate 20mm nominal size) at plinth level with :		
	7.1.1	Cement mortar 1:6	Cement mortar 1:6 (1 cement : 6 sand)	cum	4147.00
7.2		RR masonry in superstructure upto floor two	Random rubble masonry with hard stone in superstructure aboveplinth level and upto floor two level, including leveling up with cement concrete 1:6:12 (1 cement : 6 sand : 12 graded stone aggregate 20mm nominal size) at window sills, ceiling level and the like.		
	7.2.1	Cement mortar 1:6	Cement mortar 1:6 (1cement: 6 sand).	cum	4351.00
7.3		Extra for RR masonry above floor two	Extra for random rubble masonry with hard stone in superstructure above floor two level for every floors or part thereof	cum	2.5%
7.4		Extra for RR masonry pillar	Extra for random rubble masonry with hard stone in :		
	7.4.1	Square	Square or rectangular pillars	cum	213.00
	7.4.2	Circular	Circular pillars.	cum	894.00
7.5		Extra for RR masonry curve	Extra for random rubble masonry with hard stone curved on plan for a mean radius not exceeding 6 m.	cum	354.00
7.6		CR masonry upto plinth	Coursed rubble masonry (1st or 2nd sort) with hard stone in foundation and plinth.		
	7.6.1	1st sort	Cement mortar 1:6 (1 cement : 6 sand)	cum	4603.00
	7.6.2	2nd sort	Cement mortar 1:6 (1 cement : 6 sand)	cum	4342.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
7.7		CR masonry above plinth & upto floor two	Coursed rubble masonry with hard stone (first or second sort) in superstructure above plinth level and upto floor two level.		
	7.7.1	1st sort	Masonry work (first sort) in cement mortar 1:6 (1 cement : 6 sand)	cum	4803.00
	7.7.2	2nd sort	Masonry work (second sort) , in cement mortar 1:6 (1 cement : 6 sand)	cum	4541.00
7.8		Extra for CR masonry above floor two level	Extra for coursed rubble masonry with hard stone (first or second sort) in superstructure above floor two level for every floors or part thereof	cum	2.5%
7.9		Extra for CR masonry	Extra for coursed rubble masonry with hard stone (first or second sort) in :		
	7.9.1	Square or Rectangular	Square or rectangular pillars	cum	235.00
	7.9.2	Circular	Circular pillars.	cum	1003.00
	7.9.3	Curved in plan	Extra for coursed rubble masonry with hard stone (first or second sort) curved on plan for a mean radius not exceeding 6 m.	cum	384.00
7.10		Ashlar masonry	Stone work in plain ashlar masonry in super structure upto floor two level in cement mortar 1:6 (1 cement : 6 sand) including pointing with cement mortar 1:2(1 white cement : 2 marble dust) with an admixture of pigment matching the sand stone shade :		
	7.10.1	One face dressed	One face dressed.		
	7.10.1.1	Red sand stone	Red sand stone	cum	12905.00
	7.10.1.2	White sand stone	White sand stone	cum	12998.00
	7.10.2	Both face dressed	Both face dressed.		
	7.10.2.1	Red sand stone.	Red sand stone.	cum	18003.00
	7.10.2.2	White sand stone	White sand stone	cum	18080.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
7.11		Ashlar masonry in arches	Stone work plain ashlar masonry in arches in super structure upto floor two level in cement mortar 1:3 (1 cement: 3 sand) including centering, shuttering and pointing with white cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment matching the sand stone shade.		
	7.11.1	One face dressed	One face dressed.		
	7.11.1.1	Red sand stone	Red sand stone	cum	17182.00
	7.11.1.2	White sand stone	White sand stone	cum	17207.00
	7.11.2	Both face dressed	Both face dressed.		
	7.11.2.1	Red sand stone.	Red sand stone.	cum	22264.00
	7.11.2.2	White sand stone	White sand stone	cum	22341.00
7.12		Ashlar masonry in domes	Stone work plain ashlar masonry in domes, in super structure upto floor two level in cement mortar 1:3 (1 cement: 3 sand) including centering, shuttering and pointing with white cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment matching the sand stone shade.		
	7.12.1	One face dressed	One face dressed		
	7.12.1.1	Red sand stone	Red sand stone	cum	27834.00
	7.12.1.2	White sand stone	White sand stone	cum	27912.00
	7.12.2	Both face dressed.	Both face dressed.		
	7.12.2.1	Red sand stone	Red sand stone.	cum	39693.00
	7.12.2.2	White sand stone.	White sand stone.	cum	39770.00
7.13		Ashlar masonry punched	Stone work ashlar masonry punched (ordinary) in superstructure above plinth level and upto floor two level in cement mortar 1:6 (1 white cement: 6 sand) including pointing with		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			cement mortar 1:2 (1 whitecement: 2 marble dust) with an admixture of pigment matching the sand stone shade.		
	7.13.1	Red sand stone one face	Red sand stone one face punched.	cum	12190.00
	7.13.2	Red stone double face	Red sand stone both face punched	cum	12267.00
	7.13.3	White sand stone single face	White sand stone one face punched.	cum	1654100
	7.13.4	White stone double face	White sand stone double face punched	cum	16618.00
7.14		Extra for Ashlar or ashlar punched above floor-II	Extra for stone work, plain ashlar or ashlar punched above floor two level for every floor or part thereof.	cum	123.00
7.15		Extra for plain ashlar in pillar	Extra for plain ashlar or ashlar punched in :		
	7.15.1	Square or rectangular pillars	Square or rectangular pillars	cum	1771.00
7.16		Extra for ashlar or ashlar punched in curved	Extra for stone work; plain ashlar or ashlar punched in curved on plan with a mean radius not exceeding 6 m.	cum	1232.00
7.17		Extra for centering for arches	Extra for additional cost of centering for arches exceeding 6m span including all strutting,bolting, wedging etc. And removal (area of soffit to be measured).	sqm	503.00
7.18		Sunk or mouldedor sunk and moulded	Stone work sunk or moulded or sunk and moulded upto floor Two level in cement mortar 1:6 (1cement : 6 sand) including pointing with white cement mortar 1:2 (1 white : 2 marble dust) with an admixture of pigment matching the stone shade :		
	7.18.1	Red stone	Red sand stone	cum	20629.00
	7.18.2	White stone	White sand stone	cum	19706.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
7.19		Extra for sunk or moulded or sunk and moulded	Extra for stone work sunk or moulded or sunk and moulded or carved in :		
	7.19.1	Triangular /square rectangular	Triangular or Square or rectangular pillars	cum	2310.00
	7.19.2	Circular /polygon	Circular or polygonal pillars	cum	6545.00
7.20		Extra for stone work sunk or moulded in cornices	Extra for stone work sunk or moulded in cornices.	per metre per cm girth	14.00
7.21		Stone work for wall lining	Stone work (machine cut edges)for wall lining etc. (veneer work) up to 10 meter height backing filled with a grout of 12mm thick cement mortar 1:3 (1 cement : 3 sand) including pointing in white cement mortar 1:2 (1 white cement : 2 marble dust) with an admixture of pigment matching the stone shade : (To be secured tothe backing by means of clamps which shall be paid for separately):		
	7.21.1	Red stone face dressed face	Red sand stone - exposed face fine dressed with rough backing.		
	7.21.1.1	70 mm	70 mm thick.	sqm	1321.00
	7.21.1.2	60 mm	60 mm thick.	sqm	1299.00
	7.21.1.3	50 mm	50 mm thick	sqm	1276.00
	7.21.1.4	40 mm	40 mm thick.	sqm	1257.00
	7.21.1.5	30 mm	30 mm thick.	sqm	1230.00
	7.21.2	Red stone machine cut face	Red sand stone - Exposed face machine cut and table rubbed with rough backing.		
	7.21.2.1	70 mm	70 mm thick.	sqm	1808.00
	7.21.2.2	60 mm	60 mm thick.	sqm	1785.00
	7.21.2.3	50 mm	50 mm thick	sqm	1762.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	7.21.2.4	40 mm	40 mm thick.	sqm	1739.00
	7.21.2.5	30 mm	30 mm thick.	sqm	1717.00
	7.21.3	White stone fine dress face	White sand stone - exposed face fine dressed with rough backing.		
	7.21.3.1	70 mm	70 mm thick.	sqm	1327.00
	7.21.3.2	60 mm	60 mm thick.	sqm	1303.00
	7.21.3.3	50 mm	50 mm thick	sqm	1280.00
	7.21.3.4	40 mm	40 mm thick.	sqm	1256.00
	7.21.3.5	30 mm	30 mm thick.	sqm	1233.00
	7.21.4	White stone machine cut stone	White sand stone - Exposed face machine cut and table rubbed with rough backing.		
	7.21.4.1	70 mm	70 mm thick.	sqm	1813.00
	7.21.4.2	60 mm	60 mm thick.	sqm	1790.00
	7.21.4.3	50 mm	50 mm thick	sqm	1766.00
	7.21.4.4	40 mm	40 mm thick.	sqm	1743.00
	7.21.4.5	30 mm	30 mm thick.	sqm	1719.00
7.22		Extra for stone work curved on plan	Extra for stone work (veneer work) curved on plan with a mean radius not exceeding 6 m.	cum	1771.00
7.23		Cramps for wall lining stone	Providing and fixing stainless steel cramps of required size and shape for anchoring stone wall lining to the backing or securing adjacent stones in stone wall lining in cement mortar 1:2 (1 cement :2 sand) including making the necessary chases in stone and holes in walls wherever required.	kg	631.00
7.24		Stone dowels	Providing and fixing stone dowels of size 10x5x2.5cm cut to double wedge shape as per design in cement mortar 1:2 (1 cement : 2 sand) including making the necessary chases	each	35.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
7.25		Copper pins for securing adjacent stones	Providing and fixing copper pins 7.5 cm long 6 mm diameter for securing adjacent stones in stone wall lining in cement mortar 1:2 (1 cement : 2 sand) including making the necessary chases.	each	35.00
7.26		Sloping chajja of stone	Providing and fixing sloping chajja of stone 40 mm thick and upto 80 cm wide beyond the wall as measured along the slope in cement mortar 1:4 (1 cement : 4 sand) with 12mm diameter anchoring steel bar 45 cm long fixed in each stone and supported on and including bricks cove in cement mortar 1:4 (1 cement :4 sand) including pointing in cement mortar 1:2 (1 whitecement: 2 marble dust) with an admixture of pigment matching the stone shade:		
	7.26.1	Red stone	Red sand stone:		
			With common burnt clay bricks of 25 kg/cm^2 compressive strength.	sqm	1196.00
	7.26.2	White stone	White sand stone		
			With common burnt clay bricks of 25 kg/cm^2 compressive strength.	sqm	1196.00
7.27		Horizontal chajja of stone	Providing and fixing horizontal chajja of stone 40 mm thick and upto 80 cm projection in cement mortar 1:4 (1 cement : 4 sand) including pointing in whitecement mortar 1:2 (1 white cement : 2 marble dust) with an admixture of pigment matching the stone shade:		
	7.27.1	Red stone	Red sand stone	sqm	744.00
	7.27.2	White stone	White sand stone	sqm	744.00
7.28		Sand stone sun-shade	30mm red sand stone sun-shade (chisel-dressed) supported on red sand stone brackets, fixed in walls with cement mortar 1:4 (1 cement: 4 sand) including finishing complete.	sqm	891.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
7.29		Red sand stone brackets	Providing and fixing red sandstone brackets 55x22.5x45cm sunk and moulded including providing and fixing with 4 Nos. gun metal cramp 25x6mm 30 cm long and dowel bars 7.5 cm long 6mm dia as per design.	each	1672.00
7.30		Stone work in copings, cornices, string courses and plinth courses	Stone work, plain in copings, cornices, string courses and plinth courses, upto 75 mm thick in Cement mortar 1:6 (1 cement : 6 sand) including pointing with white cement mortar 1:2 (1 white cement : 2 stone dust) with an admixture of pigment matching the stone shade.		
	7.30.1	Red stone	Red sand stone	cum	13187.00
	7.30.2	White stone	White sand stone	cum	13249.00
7.31		Stone jali	Providing and fixing stone jali 40mm thick throughout in cement mortar 1:3 (1cement :3 sand) including pointing in whitecement mortar 1:2 (1white cement: 2stone dust) with an admixture of pigment , matching the stone shade, jali slab without any chamfers etc.		
	7.31.1	Red stone	Red sand stone	sqm	5797.00
	7.31.2	White stone	White sand stone	sqm	5797.00
7.32		Extra for stone work under water	Extra for laying stone work in or under water and/or liquid mud including cost of pumping or bailing out water and removing slush etc. complete. **NOTE: -** The quantity will be calculated by multiplying the depth measured from sub - soil water level up to the centre of gravity of stone work under sub - water with the quantity of stone work in cum executed under the sub - soil water. The depth of centre of gravity shall be reckoned	cum/mtr depth	296.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			correct to 0.1 m. 0.05 m or more shall be taken as 0.1 m and less than 0.05 m ignored.		
7.33		Extra for stone work in foul position	Extra for laying stone work in or under foul position.	cum	109.00
7.34		Stone wall lining butch work	Wall lining butch work upto 10m height with red/ white sand stone 40 mm thick rough facing on the exposed surface with stone strips of minimum length 300 mm and required width including embedding every tenth layer and bottom most layer in masonry or concrete after making necessary chases of size 75x75mm and by providing layer of 75mm thick strips i/c 12mm thick bed of cement mortar 1:3 (1 Cement : 3 sand) i/c ruled pointing in cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment to match the shade of stone complete as per direction of Engineer-in-charge.	sqm	1543.00
7.35		Stone wall lining	Stone work (machine cut edges) for wall lining upto 10 m height etc. (Veneer work) backing filled with a grout of 12mm thick cement mortar 1:3 (1 Cement : 3 sand) and jointed with Cement mortar 1:2 (1 cement : 2 marble dust) including rubbing and polishing complete. (To besecured to the backing by means of cramps which shall be paid for separately)		
	7.35.1	Kota stone dressed rubbed	25mm thick Kota stone slabs exposed face dressed and rubbed.	sqm	1222.00
	7.35.2	Dholpur shivpuri dressed and rubbed stone	25mm thick Dholpur stone slabs exposed face dressed and rubbed.	sqm	1469.00
	7.35.3	Kota stone not dressed	20 to 25mm thick kota stone slabsexposed slab not dressed and rubbed.	sqm	900.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	7.35.4	Dholpur shivpuri stone note dressed	20 to 25mm thick Dholpur stone/shivpuri stone not dressed and rubbed.	sqm	1122.00
7.36		Stone wall lining tile machine cut & polished upto 10 m	Stone tile work with 8mm thick mirror polished and machine cut edge for wall lining upto 10 m height with special adhesive over 12 mm thick bed of cement mortar 1:3 (1 cement : 3 sand), including pointing in white cement with an admixture of pigment to match the stone shade.		
	7.36.1	Granite	Granite stone of any colour and shade	sqm	1524.00
	7.36.2	Marble	Raj Nagar plain white marble/ Udaipur green marble/ Zebra black marble	sqm	1119.00
7.37		Extra for extra height for stone wall lining	Extra for stone work for wall lining on exterior walls of height more than 10 m from ground level for every additional height of 3 m or part thereof.	sqm	67.00
7.38		Dry stone cladding upto 10m height	Providing and fixing dry stone cladding upto 10 metre height with 30mm thick gang saw cut stone with (machine cut edges) of uniform colour and size upto 1mx1m, fixed to structural steel frame work and/ or with the help of cramps, pins etc. And sealing the joints with approved weather sealant as per Architectural drawing and direction of Engineer-in-charge. (The steel frame work, stainless steel cramps and pins. shall be paid for separately.)		
	7.38.1	Red sand stone	Red sand stone.	sqm	1639.00
	7.38.2	White sand stone	White sand stone	sqm	1706.00
7.39		Structural steel for dry stone cladding	Providing and fixing structural steel frame (for dry stone claddingwith 30 mm thick gang saw cut with machine cut edges sand stone) on walls at all heights using M.S. square/rectangular tube in the required pattern as per	kg	138.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			architectural drawing including cost of cutting, bending, welding etc. The frame work shall besupported in wall with the help of MS brackets/lugs of angle iron/ flats etc. which shall be welded to the frame and embedded in brick wall with cement concrete block 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 20mm nominal size) of size 300x200x300mm including cost of necessary centering and shuttering and with approved expansion hold fasteners on CC/RCC surface includingdrilling necessary holes. Approved cramps/ pins etc. shall be welded to the frame work to support stone cladding the steel work will be given a priming coat of Zinc primer as approved by Engineer-in-charge and painted with two or more coats of epoxy paint (Shop drawings shall besubmitted by the contractor to the Engineer-in-charge for approval before execution).The frame work shall be fixed in true horizontaland vertical lines/ planes. (Onlystructural steel frame work shallbe measured for the purpose of payment, stainless steel clamps shall be paid for separately and nothing extra shall be paid.)		
7.40		S.S. clamps for dry stone cladding	Providing and fixing adjustable stainless steel clamps of approved quality and of required shape and size adjustable with stainless steel nuts bolts and washer (total weight not less than 260 gms) for dry stone cladding fixed on frame work at suitable location including making necessary recesses in stone slab, drilling required holes etc. complete as per direction of the Engineer-in-Charge.	each	208.00

CHAPTER – 8.0

ALUMINIUM STRUCTURAL GLAZING COMPOSITE PANEL AND SPIDER GLAZING

PREAMBLE
STRUCTURAL GLAZING AND ALUMINUM COMPOSITE PANEL
(CHAPTER 8.0)

8.1 Materials (General)

Materials and components used shall be of the first / superior quality and suitable for the purpose. All materials shall be free from any defects that may impair the strength, functioning/ performance or appearance of the curtain wall or adjacent construction.

Fasteners The type, size, alloy, quantity and spacing of all anchor fasteners and/or anchorage devices shall be as required for the specified performance standards.

Bolts anchors and other fastening devices like screws, nuts, washers etc. shall be of approved types as required for the strength of the connections, shall be self-locking, unless otherwise specified. These shall be of austenitic stainless steel of specified grade and shall be torque tightened, wherever required, to achieve the maximum torque tension relationship in the fasteners. Washers, nuts and all accessory items shall be of the same material as fasteners. The rivets/ nuts, bolts and washers for fixing insulation layer to the shadow box or with fire-stops (barriers)cum-smoke seal shall be stainless steel of approved grade. The anchor fasteners shall not be provided using PVC sleeves. Only expandable type self locking fasteners shall be provided.

Aluminum extrusions

In general aluminum alloy for extrusions shall be 6063 T5 or T6 grade as per B.S.1474. However, the grade and tempering specifications shall be as recommended by the supplier for each application.

All extruded aluminum sections shall be anodized in approved colour to a minimum thickness of 20 microns or shall be PVDF coil / spray coated in approved colour and shade with metallic coloursto a minimum thickness of 35 microns. The colour and the finish shall be uniform and free of streaks. The aluminum sections, before coating, shall be suitably cleaned, rinsed, buffed properly and sealed and protected after anodizing / PVDF coating, till the completion of the work.

All surfaces of the aluminum sections designed to receive the sealants shall be finished properlyto match the finish of the parent section as used for initial testing of sealant and aluminum surface adhesion. Further, it shall be ensured that the entire aluminum surface has adequate sealant contactand adhesion.

Sill sheets, plates and extrusions shall be visually flat under all lighting conditions.

The members of aluminum extruded sections for mullions, transoms, members of sub frames & sash frames shall be in single piece and not be splice jointed in the panel length and height.

Aluminum flashing

All flashings shall be made from 1.0 mm thick solid aluminum sheet transparent anodized to a minimum thickness of 10 microns. It shall be fixed using stainless steel screws dipped in weather silicone sealant.

Aluminum composite panels

The soffits of the canopies / walkways / entrance canopies etc., required as per the architectural drawings, shall be covered with aluminum composite panel material. The top of the canopies / walkways / entrance canopies shall be covered with zincalume sheets. The aluminum composite panel and zincalume sheets shall be bent to the required profile and fixed as per the approved shop drawings prepared on the basis of architectural drawings.

Brackets

The brackets shall be fixed with high degree of accuracy to achieve the elevation as per the architectural drawing. The brackets shall have suitable lengths and sections to align curtain glazing in one face, as required as per the architectural drawings. Nothing extra shall be payable on this account. The brackets shall be fabricated from M.S rolled sections / plates to have the design strength. The quality of the weld shall also be ensured as per the standards. These shall be provided corrosion protection treatment by Hot Dipped Galvanizing. The mass of the zinc coating to be not less than 610 gm. per sqm of steel area to be galvanized. Slots of elliptical or circular shape in the brackets shall be pre-drilled / machine punched and not flame cut and it shall be done before galvanizing. The surface of the brackets shall be serrated for additional grip before galvanizing. Washers made of serrated plates of the corresponding material shall also be provided for additional grip. The directions of the serration and the slot shall be such that they allow movements as per the design requirement and at the same time prevent any movement in the other direction. Each bracketshall be fixed to the R.C.C using anchor fasteners of suitable capacities and in numbers as required as per the design requirements. The brackets shall be fixed to the structural steel members of the building using stainless steel bolts & nuts / fasteners of required capacity and in numbers as per the design requirement. The holes of the required sizes shall be pre- drilled in RCC/ structural steel for fixing anchor fasteners/ bolts etc. Nothing extra shall be payable on this account.

Fittings

All hardware and fittings such as patch fittings, handles, locks, stay-arms, floor springs, friction stays etc. for doors , windows and open able panels shall be heavy duty and of approved make as specified. Hinges for open able panels shall be heavy duty top hung stainless steel friction hinges selected for specified wind load and dead loads. All fittings and locks shall be as specified. Each open able panel of the Curtain glazing shall be provided with the fittings as specified in item nomenclature.

8.2 Sealants & Gaskets

Selection of sealants

- The compatibility and sequence of installation for all sealants must be carefully considered in all proposals in order to ensure the required curing and performance.

- Sealants must not degrade and / or fail under any or all design conditions including wind, thermal and seismic movements, exposure to water and humidity, ultraviolet exposure and / or other adverse environmental conditions.

- The designations of sealant types specified herein are intended for general design guidance only.

- Final selection for the sealant types shall be based on their conformity with the Performance Requirements specified herein and as per the recommendations of the sealant manufacturer. It may use sealant of equivalent grade and characteristics, manufactured by the manufacturer other than those specified herein, based on recommendations of those sealant manufacturers for specified use but with the prior approval of the Engineer-in-Charge. The contractor shall submit the documentary evidence in this regard.

- All precautions shall be taken during design of structural silicone bite and also during fabrication of the curtain glazing system to prevent failure of sealant during the guarantee period of 10 years after the date of completion of work and even beyond, upto the expected service life of the curtain wall.

- Sealants and gaskets shall not leach, discolour, stain or dry.

Structural silicone sealant

The sealant manufacturer shall design the silicone bite for the design loads as specified and likely to come during the life of the curtain glazing system for arriving at bite size of the structural silicone sealant.

The structural silicone sealant bite as designed by the sealant manufacturer and as per the approved shop drawings shall be provided.

The Structural sealant shall be two part pump-filled Silicone sealant DC 983 of Dow Corning or equivalent recommended by manufacturer. The weather silicone sealant shall be one-part Silicone sealant DC 795 of Dow Corning or equivalent of other approved brand as per the list of approved materials.

The structural sealant to be used as specified for all exposed and concealed metal to metal(including tight or butt type metal to metal assembly prior to assembly) or glass to glass shall be 2- part silicone sealant, conforming to the manufacturer's recommendations for the specific uses and performance criteria. The sealant shall be applied using two-part pump for the same. All the sealingshall be done in a clean and controlled environment as specified by the silicone sealant manufacturer.

Weather silicone sealant

The grade of weather silicone sealants wherever required like for concealed metal to metal, metal to glass and metal to concrete/ masonry such as embedment and lapping of flashings etc. where elements are to be installed or embedded, the weather sealant shall be of grade 795 of Dow Corningor equivalent for the other approved brand, as per the recommendations of the sealant manufacturers. Also, the gap between the aluminum sections and the glass, if so required, shall be filled with weather sealant as specified above including providing and fixing backer rod wherever required as per the approved shop drawings. The weather silicone sealant shall be of approved colour and shade. The weather silicone sealant for fixing the butt jointed glass for the fixed partitions shall be transparent in colour DC 791 of Dow Corning or equivalent of other approved brands.

Compatibility

All sealants must be non-staining and compatible with adjoining sealants, backup materials, substrate materials and their respective finishes and/or applied colour coatings. Care shall be taken to ensure that two different types of sealant should not come in contact with each other unless compatibility is satisfied as per manufacturer's specifications.

Caulking compound

Dow Corning weather silicone sealant – 795 or equivalent as approved by the Engineer-in-Charge, (of approved colour and shade to match adjacent material wherever exposed and visible) for use around frame/ flashings or between frame/flashing and RCC/ masonry surface.

Gaskets

Gaskets and seals shall be of approved quality compatible with substrates, finishes and other components they are in contact with. All gaskets exposed directly on the exterior face shall be silicon gaskets, which are UV resistant. They shall not degenerate, discolour or leach on exposureto solar radiations/ rains/ pollutants etc.

Manufacturers' test Certificate shall be submitted as specified.

8.3 Glass

General

All glass and glazing materials shall be as specified. Vision and spandrel glass shall have characteristics as specified.

The performance characteristics of glass panels, have to be ensured within the constraints of aesthetic requirements like colour, shade, reflectivity etc. And performance requirements like light transmission, U value, shading coefficient, relative heat gain etc. as specified. Minor variations in the characteristics of glass on superior side may be allowed, but without any extra cost to the Department on this account.

Installation

Install glass panels and carry out glazing work as indicated on the drawings and as specified herein. All glass panels shall be of accurate sizes as required.

All glass panels shall have clear undamaged edges and surfaces, which are not disfigured.

Any glass panel that does not fit in the curtain glazing system shall be rejected. Therefore, all care and precautions shall be taken while procuring the glass panels from the manufacturer / processors of the glass. No claims of any kind or any hindrance shall be entertained from the contractor on this account. Glass panels shall not be in direct contact with the aluminum framework.

Identification

Permanent identification marking on glass shall be accomplished by a technique selected by the manufacturer. The location of the marking shall be proposed by the Manufacturer and approved by the Engineer – in - Charge. All glass shall be delivered to site with the manufacturer's / processor's label of identification attached.

Selection of Glass

Each type of glass shall be obtained from only one manufacturer and preferably in one lot.

Insulated Glazed Units (IGUs) in the vision panels

Insulated glazed units shall be obtained only from approved manufacturers/ processors as per the approved list.

Insulated units shall be factory assembled, with multiple panes, hermetically sealed, separated by and sealed to spacer tubes perforated on inner exposed face forming airtight dehydrated airspace inside the insulated units. The IGUs shall be assembled (prepared) by the manufacturer/ processor of the glass as per the approved list, in their dedicated workshops/ factory.

Laminating units

The glasses shall be laminated with interlayer of Polyvinyl butyral (PVB) sheet of specified thickness

The interlayer material (PVB) shall be clear or as specified with no tendency to bubble, discolour or lose physical and mechanical properties after laminating glasses.

The laminated panels shall be free of foreign substances, air or glass pockets and shall not delaminate at edges.

Precaution in storing and handling glasses

The glass manufacturer/ processor shall take necessary precautions as stated below besides any other precautions not specifically mentioned herein:

The reflective/ low E coating on the glass shall be protected against scratches, surface corrosion, staining and/ or any other abrasion. The glasses shall not be stored without a clean inter-leaving material. Also they should not slide against each other. The glass shall be protected from weld or grinding splatter. The reflective/ low E coating shall be protected against contact with acids or strong alkalies. The cleaners to be used for cleaning the surface shall be as per the manufacturer's recommendations. The glass shall be protected against moisture from humidity, which can stain glass as well as coating. Reflective/ low E coating shall also be protected against splashes from paints etc.

8.4 Metal Coatings

Anodizing / PVDF coating

Aluminum extruded sections shall be satin finish colour anodized to minimum 20 microns thickness, as per the approved colour and shade or PVDF coil / spray coated to approved metallic colour and shade to minimum 35 microns thickness.

Galvanizing

The brackets for the curtain glazing system shall be hot dipped galvanized. The mass of the zinc coating to be not less than 610 gm. per sqm of steel area to be galvanized.

Samples

Three samples shall be prepared, which shall define the colour and gloss of anodizing and submit them for approval.

Matching of finish

Wherever the same colour finish is specified for extruded aluminum sections and composite aluminum sheets. It shall be ensured that the colour of both is matched as closely as possible.

Standards

In general, it shall be followed either of the latest Indian/ International Standards as applicable for this sub head.

Design

- Architectural drawings and specifications only indicate the required basic dimensions, and performance criteria. It shall be ensured that proper structural analysis and design forvarious load cases and their combination. This shall include designing and proper sizing of all sections meeting structural and architectural requirements. The anchor assemblies shall meet the performance and design requirements including installation of all inserts, fasteners, clips, bracing and framework as required for the proper anchorage to the structure, unless otherwise specified.
- Design of the curtain glazing system shall comply with all Government codes and regulations. The Contractor shall design the entire curtain glazing system for dead loads, wind loads, seismic loads, storm, air pollution, thermal stresses, building movements and consequent deflections without compromising the performance characteristic. Further, the individual members of the structural framing shall not deflect beyond permissible limits as specified. The design shall comply with the requirements of the relevant National Building Code and Indian Standard Code/ International Standards, unless specified otherwise.
- The curtain glazing system and its elements shall not sustain permanent deformation or failure under loading equivalent to 1.5 times the design wind pressure.
- The specified deflections must be reduced if they are in any way detrimental to curtain glazing system and building.

- It shall be ensured that the elevations are strictly as per the architectural drawings and that the intent of the architectural design is retained. Visual appearance shall be a key consideration for acceptance of work.

8.5 Shop Drawings

Submittals

The contractor shall prepare shop drawings based on approved design and submit the same to the Engineer –in-charge for approval. The review of the shop drawings shall be limited to their conformity to the architectural and structural design concept & specifications. No fabrication shall be taken up until the shop drawings and all other related submittals, documentation, certification, samples and the mock-up for that work have been reviewed and approved by the Engineer-in- charge.

Scope of shop drawings

Shop drawings shall incorporate scaled and dimensioned plans, elevations, sections and complete size details for all the works. The shop drawings shall indicate the required dimensional profiles and modules, function, design and performance standards and in general cover all dimensions and details required to fabricate and install the curtain glazing system. The contractor shall verify and co-ordinate the shop drawings with all applicable and inter-related trades, drawings and specifications. All dimensions / modules, etc., shall be field checked and the drawings shall be modified, if required, based on actual measurements at site. Details shall show and specify all metal sections, types of finishes, areas to be sealed and sealant materials, gaskets, applicable construction materialsincluding fasteners and welds, all anchorage assemblies and components, fabrication and erection tolerances for the work. All details shall be subject to the approval of the Engineer-in-Charge, after incorporating all the modifications suggested by the Engineer-in-Charge.

Section profiles

Profile adjustments, if required as per the site conditions may be allowed by the Engineer-in- Charge subject to meeting the architectural / performance requirements. However, this shall be carried out only with the written approval of the Engineer-in-charge provided that the general design and intent of the drawings and specifications are also maintained. Also, if any new / non- standard aluminum extruded section is required as per the site requirement and / or the architectural drawings for functional and / or aesthetic reasons, the contractor shall procure the same from the approved manufacturers for the aluminum sections, even if it entails preparing new die, etc. Nothing extra shall be payable on this account.

8.6 Documentation and Certification

The manufacturer's certificate for compliance of the various components/ materials for the works (under this sub head) as per the manufacturer's specifications for the various characteristics. A copy of the manufacturer's test report for each lot of material procured and supplied for the work shall also be obtained from the respective manufacturers and submitted to the Engineer-in-Charge for the record.

Glass and glazing documentation

Before taking up the work, the glass manufacturer / processor shall submit written certification for the review of the Engineer-in-Charge and record, stating that all glass (properties as specified such as U value, shading coefficient, light transmission, solar factor, relative heat gain etc.) and glazing requirements (including heat strengthening/ toughening, reflective soft coating, low E coating, lamination, fabrication of IGUs including sealants) as per the shop drawings are recommended by them for use related to their specific applications and design parameters and that they are in conformity with the specifications. Tests shall be carried out for glass, including properties after processing, for each lot supplied, by the glass manufacturer / processor in his factory /laboratory or any other accredited laboratory and the copies of the test results shall be obtained by the contractor and submitted to the Engineer-in-Charge for the record.

Sealant Documentation : All sealant applications must be clearly designated on shop drawings.

Quality control documentation : The methodology and quality assurance statement shall be submitted for quality control procedures for the review and approval of the Engineer-in-Charge before taking up the work to ensure the design integrity and performance of the curtain glazing, aluminum composite panel cladding and aluminum work (PVDF coated). The Engineer-in-Charge or his authorized representatives may visit the plant / workshop / factory to inspect material, fabrication and quality assurance procedures.

8.7 Samples and Mock-Up at Site

Submittals The samples of the following materials together with detailed technical data / catalogues shall be submitted for review and approval of the Engineer-in-Charge along with the shop drawings..

Aluminum Composite Panel : Each type and thickness 600mmx600mm

Aluminum extrusions : Each section: 500mm long.

Glass : Each type 600 mm x 600 mm.

Gaskets, separators, glass setting blocks / spacer tape, etc : Each section or unit, backer rods, 300mm long or unit Bracket, fasteners and Connecting devices : Each type and size.
Finish samples : After approval of the final finish coating, the Engineer-inCharge shall be provided with three approved samples. Ironmongery and accessories, as applicable.
Finished flashing samples : Finished samples of shadow boxes, fire stop (barrier)-cum smoke seals Structural and weather silicone sealant

Mock-up at site Before the fabrication and site installation is taken up and after the approval of shop drawings by the Engineer-in-Charge, a mock-up shall be prepared of his proposed curtain glazing system for a size of panel not exceeding 6 sqm. The mock-up shall be essentially put up at site for final approval of all materials and installation details by the Engineer-in-Charge. The mock up shall not form part of the work and shall not be paid for. It shall be dismantled and taken away by the contractor at his own cost, with the prior permission of the Engineer-in-Charge. Nothing extra shall be payable on this account.

8.8 Storage, Protection and Programme

A schedule of procedure shall be submitted for inspection during installation so as to control and assure quality on the job site.

A detailed method statement shall be submitted for the protection of the surface of the curtain glazing, aluminum composite panel cladding and aluminum work (PVDF coated), etc during delivery and erection, with description as to when the protection can be removed. The protection paper shall be kept for a period as recommended by Engineer-in-Charge and shall be replaced with the fresh protection paper, if so required. Further, it shall not have acid content, which in any manner may affect the substrate.

Delivery and Storage of Materials:

All materials delivered to site shall be stored in allocated spaces where the stored materials shall not get exposed to rainwater, moisture or damage, and shall permit easy access to and handling of the materials. Materials shall be stored neatly and properly stacked.

Factory made glazing units and/or their components shall be transported, handled and stored in a manner to preclude damage of any nature. Necessary materials, required for erection at the site shall be delivered in labeled containers by the manufacturer / supplier.

All units or components, which are cracked, bent, chipped, scratched or otherwise defective and unsuitable for installation shall be removed and replaced by the contractor. Nothing extra shall be payable on this account.

8.9 Performance Requirements

All components, assemblies and completed work shall conform to the various performance standards as applicable in respect of thermal movement of the curtain glazing, allowance for vertical and horizontal expansion and building movement and related building tolerance etc. The design and installation of the curtain glazing system shall accommodate all inherent building movements and deflections and the fabrication and installation tolerances of all related work not involved in this section without the loss of, or any detrimental effect to, the performance requirements herein specified. The contractor shall verify and coordinate all such movements and tolerances with the Engineer-in-Charge before designing the components of the curtain glazing system so that movements and deflections in the structure do not at any time affect the integrity andsafety of curtain glazing system and vice versa.

Thermal property

All insulation materials, fire-stops (barriers)-cum-smoke seal shall comply with the current requirements of the Fire Officer, MIDC and other authorities.

Structural properties

The curtain glazing system shall be anchored to the R.C.C floor through serrated Hot Dipped Galvanized M.S brackets. As far as possible, the contractor shall take all precautions to avoid cutting through any reinforcement bars while fixing the brackets. The cost includes provision of sleeves/ leave slots at appropriate locations during casting of the concrete itself for making provision for fixing brackets for the curtain glazing system and to avoid chipping/ dismantling of concrete. The slot shall be filled up with concrete of the same grade in a workman like manner,after fixing the brackets. Any defect in alignment/ plumb in the building face shall be rectified by chipping/ dismantling of the concrete/ masonry and repairing the same as specified to achieve the required alignment of the curtain wall as specified. Any change in lengths of bracket/s required on this account and the consequent requirement of their sections and sizes shall be carried out. Nothing extra shall be payable on this account.

No holes shall be burned, filed or drilled in any structural steel/ RCC members unless expressly approved by the Engineer – in – Charge.

Member shapes and/or profiles if schematically shown on the Architectural drawings are not necessarily the exact shapes required or best suited for the particular conditions. Final shapes and locations shall be as designed by the contractor and are subject to the review and approval of the Engineer – in – Charge.

All framing members shall be shop fabricated and finished as specified.

Concrete tolerances

While fixing the brackets for curtain glazing system, the contractor shall take into account the variation in the concrete and the masonry faces to which the structural framework of the curtain glazing system is to be fixed and such variations shall be adjusted in the lengths of brackets to align them in perfect plumb. The bracket shall be designed accordingly. Nothing extra shall be payable on this account.

Fire stops (barriers)-cum-smoke seal and interface with building

Gaps between the building face and the curtain glazing system at soffit level between the successive floors shall be closed as specified with fire-stops (barriers)-cum-smoke seal. It shallhave the required fire resistance to be approved by Fire Officer. Suitable gap for accommodating deflections of the aluminum framing of curtain glazing system as per the approved shop drawings shall be maintained between the fire-stops (barriers)-cum-smoke seal and the curtain glazing system. This smoke seal shall however be provided using backer rod and weather silicone sealant as specified and as approved by the Engineer-in-Charge.

The fire-stops (barriers)-cum-smoke seal shall consist of 1 mm thick plain G.I. sheet tray with 100 mm thick layer of non inflammable heat insulating material, rock wool, having density of minimum 64 Kg. per Cum. of the make as approved by the Engineer-in-Charge. The rock wool layer shall be attached to G.I sheet using stainless steel rivets/ nuts, bolts and washers. The tray shall be fixed to the RCC / Masonry surface by using stainless steel screws dipped in weather silicone sealant as per the approved shop drawings. Screws with plastic sleeves shall not be allowed to be used for the above fixing.

An aluminum flashing of 1.0 mm thick shall be permitted transparent anodized (10 micronthickness) solid aluminum sheet of the approved design and profile at the window sill level and also fill the gap between the aluminum flashing and the curtain glazing using weather silicone sealant as specified and as approved by the Engineer-in-Charge. Also, the fasteners/ screws to be used for fixing flashing shall be dipped in weather silicone sealant before using.

Acoustics

Gaps between the mullions and the partitions of the cabins shall be suitably closed by double skin partition as directed by the Engineer-in-Charge including allowing for permissible deflections of mullions as per design requirements but without affecting the partitions and the curtain glazing system. The payment for this partition work shall be made under relevant item. Provisions shall also be made to prevent metal to metal rubbing, any rattling, noise due to thermal changes and wind pressure by using Teflon separators and shims.

Visual appearance

It shall be ensured that the elevations are strictly as per the Architectural drawings and that the intent of the architectural design is retained. Visual appearance shall be a key consideration for acceptance of work.

8.10 Curtain Glazing and Aluminum Composite Panel Cladding Systems

General

Movement of building components to which the curtain glazing system is attached including long term and short term movements due to thermal effect, structural effect, wind pressure, seismic forces, erection or dead loads, creep, column shortening, deflection, torsion and vibrations etc shall be free and noiseless. This shall be achieved without any strain or stress being transferred to the glass, without buckling of any components, without excessive stress to any members or assemblies and without compromising on any of the performance requirement of the curtain wall.

Waterproofing

Following precautions shall be taken by the contractor to ensure that the curtain glazing system is completely water tight during its guarantee period as well as expected service life besides any other precautions not specifically mentioned herein:

A drainage system must be incorporated into the curtain glazing system. The curtain glazing system shall have provision for air pressure equalization (all the internal spaces shall be vented by acceptable means to ensure air pressure equalization) so that water leakage and condensation, ifany shall be drained or discharged to exterior face of the curtain glazing.

Care should be taken that the sections of the aluminum extrusions used for structural framing of curtain glazing provide for proper drainage of water that infilters into the system by gravity and for this the section should have proper slope and weep holes as required. These shall be clearly indicated on the shop drawings.

Movement of water on exposed faces must be controlled to ensure that water is not retained and that elements will not be damaged or corroded by water and to minimize the potential for algae and fungal growth as a result of standing or trapped water.

EPDM gaskets of the quality as specified and of required size and thickness shall be provided at all required locations to prevent ingress of water or moisture. The same shall be indicated on the shop drawings also.

EPDM gaskets of the quality as specified and of required size and thickness shall be provided at all

required locations to prevent ingress of water or moisture. The same shall be indicated on the shop drawings also.

Aluminum sheet flashing using 1.0mm thick transparent anodized (10 microns) aluminum sheet wherever required shall be provided including sealing the gap between the flashing and the other material like RCC, masonry, aluminum etc. by using weather silicone sealant as specified.

Mullions and transoms

The sections of mullions and transoms shall be designed to restrict deflection under dead loads, wind load, seismic loads etc. as specified and shall be rigid and stable enough to support and retain the in-fill panels in position under all conditions. The mullions and transoms shall also be designed for additional horizontal loads from the cleaning equipment and process besides horizontal live loads as specified.

Spandrel units

Spandrel shall be of glass having same colour matching with vision areas after using a shadow box as specified.

Structural spandrel wall, fins, slab or beam, aluminum frame work, anchor fasteners, brackets , shadow boxes, fire stop(barrier)-cum-smoke seals and other construction shall not be visible through the glass in the spandrel portion of the curtain glazing from the exterior and shall be fully concealed behind the shadow box.

A shadow box shall be provided at a distance of minimum 50 mm behind the spandrel glass panel to ensure that the insulation panel material does not come in contact with the soft coating of the spandrel glass to prevent any damage to the coating on account of any chemical reaction or otherwise. It shall consist of an approved black fibre glass non-woven tissue stuck on surface #1of 50 mm thick semi-rigid fibre glass wool insulation panel of minimum density of 48 kg per cum., and 1.5 mm thick transparent anodized (10 microns) solid aluminum sheet tray, on surface #2 by using suitable stainless steel rivets/ nuts, bolts and washers to hold the insulation panel in position. The periphery shall be properly sealed. Surface #1 shall be adequately protected against damage until spandrel glazing is done. Further, care shall be taken that the aluminum sheet backing of the shadow box does not heave or warp due to thermal stresses and/or its self-weight. Proper gaps atthe edges of the tray shall be provided to accommodate movements on account of thermal stresses besides making elliptical slots if required to facilitate movements. The shadow box shall be fixed tothe structural framing of the curtain glazing by using stainless steel screws. The fixing arrangement shall be as per the approved shop drawings.

Ventilators, open able windows and doors

Ventilators, openable windows and doors shall be provided at positions as shown on the architectural drawings. The openable panels when in closed position shall remain watertight under all weather conditions and pass the water tightness tests as specified. Besides, the openable panels shall appear similar to the fixed ones from outside.

All hardware and accessories shall be provided and fixed by the contractor and shall be asspecified.

Coping and soffit trimmer

All coping and soffit panels shall have aluminum structural frame fixed rigidly to the structure. Effective drainage system shall be provided to drain out the water that may penetrate through the joints, on to the exterior face of the curtain glazing. Coping and soffits shall be visibly flat in all lighting conditions.

8.11 Measurements

All the aluminum sections including snap beadings fixed in place shall be measured in running meter along the outer periphery of composite section correct to a millimeter. The weight calculated on the basis of actual average (average of five samples) weight of composite section in kilogram correct to the second place of decimal shall be taken for payment (weight shall be taken after anodizing). The weight of cleat shall be added for payment. Neither any deduction nor anything extra shall be paid for skew cuts

The height and width of double glazed/single glazed unit (the area of glass unit outside the snap beading shall only be measured) as fixed in place shall be measured correct to one centimeter and area calculated in sqm. correct to second place of decimal shall be taken for payment.

8.12 Rate

Rate shall includes cost of all inputs of labour, materials including wastages, T &P, equipments , other enabling temporary structures and services and all other incidental charges, if any, not specifically mentioned here, but as required for complete design, proof checking, engineering, fabrication, assembling, delivery, anchorage, installation, protection of curtain glazing, aluminum composite panel cladding and aluminum work (PVDF coated), etc. and making the system water tight (wherever specified), all complete, all in accordance with the true intent and meaning of the specifications and the drawings taken together, regardless of whether the same may or may not be

particularly shown on the drawings and / or described in the specifications , provided that the same can be reasonably inferred there from. The curtain glazing, aluminum composite panel cladding and aluminum work (PVDF coated) shall have framing which shall be structurally and mechanically designed to achieve the architectural elevations as well as performance parameters specified herein. Anchorage shall include all supporting brackets & anchor fasteners, as required to rigidly secure the structural framing to the RCC / Masonry / structural steel members of the building.

The curtain glazing, aluminum composite panel cladding and aluminum work (PVDF coated), etc. work shall include but will not necessarily be limited to the following:

- Frames, fixed glazed / vision panels, spandrels, hard wares, open able panels, as in the drawings inclusive of all accessories and fittings. Glass wool Insulation panel (shadow box), fire stop(barrier) - cum - smoke seals, splice plates, connectors, sleeves, anti-buckling clips etc.
- Anodized aluminum work for framing of curtain glazing as well as other aluminum work PVDF coated) for trellis, louvers, fins, box sections, capping, strip etc wherever indicated in the schedule of quantities and drawings. Glazes doors.
- Structural, weather and other silicone sealants within and all round the perimeter of all the work under this sub head for fabricating IGUs, holding the glass to the aluminum & glass to glass and to provide water tightness to the curtain glazing.
- EPDM / silicone gaskets, trims, shims, setting blocks, double sided spacer tape, spacer blocks, weathering strips etc. All sealing and flashings including sealing at junctions with the building members.
- All brackets, anchor fasteners, screws, inserts, nuts, bolts & washers, and attachments required for complete installation and fixing to the RCC, masonry and/or the structural steel members of the building.
- All accessories, fasteners, screws, nuts and bolts, toggles, rivets etc. and other items implied in the drawings and the specifications though are not specifically indicated. Isolation of all dissimilar metal surfaces as well as moving surfaces by use of TEFLON (PTFE) separators.
- Engineering proposals, design, drawings and Architectural data. Shop drawings, engineering data and structural calculations (analysis & design) of all systems including aluminum structural framing, fasteners, sealants etc. Scheduling and monitoring of the work.
- Cost of all samples of the individual components, mock-ups at site and field tests.

Coordination with work of other agencies. Protection during storage and construction until handing over the building for occupation etc.

- All final exterior and interior cleaning of the curtain glazing, aluminum composite panel cladding and aluminum work (PVDF coated) etc. Before handing over the building for occupation. Hoisting, staging, scaffolding and temporary enabling structural work/services, cranes and cradles etc. Specified tests, inclusive of necessary records, reports, logbook etc. Design and performance guarantee in the enclosed format.
- Construction monitoring for regular quality control and technical inspection to ensure the work conforms to the approved shop drawings and details (including any modifications made after field testing) and acceptable standards of quality including monitoring theprogress of the work.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
8.1		Aluminum sections frame work powder coated	Providing and supplying aluminum extruded tubular and other aluminum sections frame work as per the architectural drawings and approved shop drawings, the aluminum quality as per grade 6063 T5 or T6 as per BS 1474,including super durable powder coated of 60-80 microns conforming to AAMA 2604 of required colour and shade as approved by the Engineer-in-Charge. (The item includes cost of material such as cleats, sleeves, screws etc. necessary for fabrication of extruded aluminum frame work. Nothing extra shall be paid on this account).	kg	416.00
8.2		Aluminum Structural glazing	Designing, fabricating, testing, protection, installing and fixing in position semi (grid) unitized system of aluminum structural glazing (with open joints) for linear as well as curvilinear portions of the building for all heights and all levels. **(A)** Structural analysis and design and preparation of shop drawings for the specified design loads conforming to IS 875 part III (the system must passed the proof test at 1.5 times design wind pressure without any failure), including functional design of the aluminum sections for fixing glazing panels of various thicknesses, aluminum cleats, sleeves and splice plates etc. gaskets, screws, toggles, nuts, bolts, clamps etc., structural and weather silicone sealants, flashings, fire stop (barrier)-cum-smoke seals, microwave cured EPDM gaskets for water tightness, pressure equalisation and drainage and protection against fire hazard including:	sqm	2674.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			(B) Fabricating and supplying serrated M.S. hot dip galvanised / Aluminum alloy of 6005 T5 brackets of required sizes, sections and profiles etc. to accommodate 3 Dimentional movement for achieving perfect verticality and fixing structural glazing system rigidly to the RCC/ masonry/structural steel framework of building structure using stainless steel anchor fasteners/ bolts, nylon seperator to prevent bimetallic contacts with nuts and washers etc. of stainless steel grade 316, of the required capacity and in required numbers. **(C)** Providing and filling, two part pump filled, structural silicone sealant and one part weather silicone sealant compatible with the structural silicone sealant of required bite size in a clean and controlled factory / work shop environment, including double sided spacer tape, setting blocks and backer rod, all of approved grade, brand and manufacture, as per the approved sealant design, within and all around the perimeter for holding glass. **(D)** Providing and fixing in position flashings of solid aluminum sheet 1 mm thick and of sizes, shapes and profiles, as required as per the site conditions, to seal the gap between the building structure and all its interfaces with curtain glazing to make it watertight. **(E)** Making provision for drainage of moisture/ water that enters the curtain glazing system to make it watertight, by incorporating principles of pressure equalization, providing suitable gutter profiles at bottom (if required), making necessary		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			holes of required sizes and of required numbers etc. complete. This item includes cost of all inputs of designing, labour for fabricating and installation of aluminum grid, installation of glazed units, TandP, scaffolding and other incidental charges including wastages etc., enabling temporary structures and services, cranes or cradles etc. as described above and as specified. The item includes the cost of getting all the structural and functional design including shop drawings checked by a structural designer, dully approved by Engineer-in-charge. The item also includes the cost of all mock ups at site, cost of all samples of the individual components for testing in an approved laboratory, field testson the assembled working structural glazing as specified, cleaning and protection till thehanding over of the building for occupation. In the end, the Contractor shall provide a water tight structural glazing having all the performance characteristics etc. all complete as required, as per the Architectural drawings, as per item description, as specified, as per the approved shop drawings and as directed by the Engineerin- Charge. **Note: - 1.** The cost of providing extruded aluminum frames, shadow boxes, extruded aluminum section capping for fixing in the grooves of the curtain glazing and vermin proof stainless steel wire mesh shall be paid for separately under relevant items under this sub-head. However, for the purpose of payment, only the actual area of structural glazing (including width of grooves) on the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			external face shall be measured in sqm. Upto two decimalplaces. **Note: 2.** The following performance test are to be conducted on structural glazing system if area of structural glazing exceeds 2500 Sqm from the certified laboratories accreditated by NABL (National Accreditation Board for Testing and Calibration Laboratories), Department of Science and Technologies, India. Cost of testing is inclusive in the rates. Performance Testing of Structural glazing system Teststo be conducted in the NABL Certified laboratories (1) Performance Laboratory Test for Air Leakage Test (-50pa to – 300pa) and (+50pa to +300pa) as per ASTM E-283-04 testing method for a range of testing limit 1 to 200 mVhr" (2) Static Water Penetration Test. (50pa to 1500pa) as per ASTME- 331-09 testing method for a range upto 2000 ml." (3) Dynamic Water Penetration (50pa to 1500pa) as per AAMA 501.01-05 testing method for a range upto 2000 ml" (4) Structural Performance Deflection and deformation by static air pressure test (1.5 times design wind pressure without any failure) as per ASTME-330-10 testing method for arange upto 50 mm" (5) Seismic Movement Test (upto 30 mm) as per AAMA 501.4-09 testing method for Qualitative test" Tests to be conducted on site		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			(6) Onsite Test for WaterLeakage for a pressurerange 50 kpa to 240 kpa (35psi) upto 2000ml"		
8.3		Hermetically-sealed vision glass panels	Providing, assembling and supplying vision glass panels (IGUs) comprising of hermetically-sealed 6-12- 6 mm insulated glass (double glazed) vision panel units of size and shape as required and specified, comprising of an outer heat strengthened float glass 6mm thick, of approved colour and shade with reflective soft coating on surface # 2 of approvedcolour and shade, an inner Heat strengthned clear float glass 6mm thick, spacer tube 12mm wide, dessicants, including primary seal and secondary seal (structural silicone sealant) etc.all complete for the required performances, as per the Architectural drawings, as per the approved shop drawings, as specified and as directed by the Engineer-in-Charge. The IGUs shall be assembled in the factory/ workshop of the glassprocessor. (Payment for fixing of IGU Panels in the curtain glazing is included in cost of item No.8.2) For payment, only the actual area of glass on face # 1 of the glass panels (excluding the areas of the grooves and weather silicone sealant) provided and fixed in position, shall be measured in sqm. (i) Coloured tinted float glass 6mm thick substrate with reflective soft coating on face # 2, + 12mm Airgap + 6mm Heat Strengthened clear Glass of approved make having properties as visible Light transmittance (VLT) of 25 to 35	sqm	3200.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			%, Light reflection internal 10 to 15%, light reflection external 10 to 20 %, shading coefficient (0.25- 0.28) and U value of 3.0 to 3.3 W/m^2 degree K etc. The properties of performance glass shall be decided by technical sanctioning authority as per the site requirement.		
8.4		Extra for openable side / top hung vision glass panels	Extra for openable side / top hung vision glass panels (IGUs) including providing and supplying at site all accessories and hardwares for the openable panels as specified and of the approved make such as heavy duty stainless steel friction hinges, min 4 -point cremone locking sets with stainless steel plates, handles, buffers etc. including necessary stainless steel screws/ fasteners, nuts, bolts, washers etc. all complete as per the Architectural drawings, as per the approved shop drawings, as specified and as directed by the Engineerin-Charge.	sqm	3009.00
8.5		Shadow box for spandrel portion	Providing, fabricating and supplying shadow box of required size and shape, for fixing in the spandrel portion of the structural glazing, in linearas well as curvilinear portions of the building by providing semi - rigid, inorganic, non-combustible fibre glass wool insulation 50 mm thick having density 48 kg/m^3, conforming to IS: 8183 and BS: 3958 Part 5. The insulation layer shall have facing (factory bonded on surface # 1 of the fibre glass insulation layer), of black non-woven fibre glass tissue of nominal thickness 0.5 mm and nominal mass not less than 60 gm/sqm, made of randomly	sqm	1761.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			oriented glass fibres distributed in a binder by a wet-lay process including fixing 1.5 mm thick solid aluminum sheet backing using, 6 mm thick cement board including SS rivets, nuts, bolts, washers etc complete.		
8.6		Spandrel glass panels	Providing and supplying Spandrel Glass Panels comprising of 6 mm thick heat strengthened monolithic float glass of approved colour and shade with reflective soft coating on surface # 2 of approvedcolour and shade so as to match the colour and shade of the IGUsin the vision panels etc., all complete for the required performances as specified, as per the Architectural drawings, as per the approved shop drawings, as specified, and as directed by the Engineer- in- Charge. For payment, only the actual area of glass on face # 1 of the glass panels (but excluding the area of grooves and weather silicone sealant) provided and fixed in position, shall be measured in sqm.(Payment for fixing of Spandrel Glass Panels in the curtain glazing is included in cost of relevent Item). (i) Coloured tinted float glass 6mm thick substrate with reflective soft coating on face # 2, having properties as visible Light transmittance (VLT) of 25 to 35 %, Light reflection internal 10 to 15%, light reflection external 10 to 20 %, shading coefficient (0.25- 0.28) and U value of 3.0 to 3.3 W/m^2 degree K etc. The properties of performance glass shall be decided by technical sanctioning authority as per the site requirement.	sqm	1706.00
8.7		Aluminum composite panel	Designing, fabricating, testing, installing and fixing in position		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
		(ACP)	Curtain Wall with Aluminum Composite Panel (ACP) Cladding, with open grooves for linear as well as curvilinear portions of the building, for all heights and all levels etc. including: **a)** Structural analysis and design and preparation of shopdrawings for pressure equalisation or rain screen principle as required, proper drainage of water to make it watertight including checking of all the structural and functional design. **b)** Providing, fabricating and supplying and fixing panels of aluminum composite panel cladding in pan shape in metalic colour of approved shades.The aluminum composite panel cladding sheet shall be coil coated, with kynar 500 based PVDF / fluoropolymer resin coating of approved colour and shade on face # 1 and polymer (Service) coating on face # 2 as specified using stainless steel screws, nuts, bolts, washers, cleats, weather silicone sealant, backer rods etc. **c)** The fastening brackets of Aluminum alloy 6005 T5 / MS with Hot Dip Galvanised with serrations and serrated washersto arrest the wind load movement, fasteners, SS 316Pins and anchor bolts of approved make in SS 316, Nylonseparators to prevent bi-metallic contacts all complete required to perform as per specification and drawing The item includes cost of all material and labour component, the cost of all mock ups at site, cost of all samples of the individual components for testing in an approved laboratory, field tests on the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			assembled working curtain wall with aluminum composite panel cladding, cleaning and protection of the curtain wall with aluminum composite panel cladding till the handing over of the building for occupation. Base frame work for ACP cladding is payable under the relevant aluminum items. The Contractor shall provide curtain wall with aluminum composite panel cladding, having all the performance characteristics all complete, as per theArchitectural drawings, as per item description, as specified, as per the approved shop drawings and as directed by the Engineer- in-Charge. However, for the purpose of payment, only the actual area on the external faceof the curtain wall with Aluminum Composite Panel Cladding (including width of groove) shall be measured in sqm. upto two decimal places.		
	8.7.1	4 mm (0.5mm sheet)	4 mm thick aluminum composite pannel material consisting of 3 mm thick FR grade mineral core sandwiched between two aluminum sheets (each 0.5mm thick)	sqm	3800.00
	8.7.2	4 mm (0.25mm sheet)	4 mm thick aluminum composite pannel material consisting of 3.5 mm thick FR grade mineral core sandwiched between two aluminum sheets (each 0.25mm thick)	sqm	3556.00
	8.7.3	3 mm (0.25mm sheet)	3 mm thick aluminum composite pannel material consisting of 2.5 mm thick FR grade mineral core sandwiched between two aluminum sheets (each 0.25mm thick)	sqm	3312.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
8.8		Spider glazing system	"Design, supply and installation of suspended Spider Glazing system designed to withstand the wind pressure as pr IS 875 (Part-III). The Suspended System held with Spider Fittings of SS-316 Grade Steel of approved manufacturer with glass panel having 12 mm thick clear toughened glass held together with SS- 316 Grade Stainless steel Spider and bolt assembly with laminated glass fins 21 mm thick. The Glass fins and glass panel assembly shall beconnected to Slab/beams by means of SS-316 Grade stainless steel brackets and Anchor bolts and at the bottom using SS channel of 50x25x2mm using fastener and anchor bolts, non staining weather sealants of approved make, Teflon/ nylon bushes and separators to prevent bi-metallic contacts, all completeto perform as per specificationand approved drawings. The complete system to be designed to accommodate thermal expansion and seismic movements etc. The joints between glass panels (6 to 8 mm) and gaps at the perimeter and in U channel of the assembly to be filled with none staining weather sealant, so as to make the entire system fully air tight, water proof and dust proof. The rate shall include all design, Engineering and shop drawing including approval fromstructural designer, labour,TandP, scaffolding, other incidental charges including wastage, enabling temporary services all fitting fixers nutbolts, washer, Buffer plates, fastener, anchors, SS channel laminated glass etc. all complete. For the purpose of payment,	Sqm	5983.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			actual elevation area of Glazing including thickness of joints and the portion of Glass panel inside the SS channel shall bemeasured.		
8.9		Aluminum magnesium alloy wall cladding	Providing and fixing aluminum magnesium alloy (FACADE) wall cladding system consisting of panels 200mm wide x 17mm deep x 0.5mm thick of maximum length of 6.0 mtr. coil coated on a continuous paint line double baked and roll formed from enameled corrosionresistant GI pre-coated alloy for higher strength and roll forming characteristics, panel shall be clipped to baked enamel panel stringers of 34.5mm wide 48mm deep x 0.5mm thick GI alloy with cut-outs to hold the panel horizontally on the stringer in a module of 200mm centre tocentre at a distance of maximum 1.20 mtr. The stringer shall be fixed to a suitable rigid structure. All panels shall be achromatised for maximum bond between metal and paint, enameled twice under high temperature one side with full primer and the other side (inner side) with a primercoat on a continuous paint line, and fixed as per the direction of the Engineer-in-Charge.	sqm	4004.00
8.10		Alloy sun louvres	Providing and fixing Aluminum Magnesium alloy Plain PanelSun Louvres of approved colour consisting of panel 84 mm wide x 16mm deep x 0.6mm thick with round edges panel length upto 6 mtrs coil coated on a continuos paint line double	sqm	3883.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			baked and roll formed from enameled corrosion Resistance Aluminum Alloy AA3005 (Al.Mg.) for higher strength and good Roll forming characteristics. Panel shall be clipped to baked enameled Aluminum panel stringer of 33mm wide and 86mm deep x 0.95 mm thick in a standard length of 5mtrs made of double baked enameled AluminumAlloy AA5050 (Al.Mg.) black with cut-outs to hold the panel ina module of 86mm center to center at a distance of 1mtr. The carrier shall be fixed to a suitable structure by means of rigid fixing details. **Paint Finish:** Aluminum panels shall be chromatised for maximum bond between metal and paint, enameled twice under high temperature one side with full primer and finish, the other side (inner side) with a primer coat and Skin Coat on a Continues Paint Line. **Note: -** 1. The cost of providing extruded aluminum frames, shadow boxes, extruded aluminum section capping for fixing in the grooves of the curtain glazing shall be paid for separately under relevant items under this subhead. However, for the purpose of payment, only the actual area of structural glazing (including width of grooves) on the external face shall be measured in sqm. upto two decimal places.		

CHAPTER – 9.0

WOOD WORK, PVC WORK AND UPVC WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 204 (Part I)	Specification for tower bolts (ferrous bolt)
2	IS 204 (Part II)	Specification for tower bolts (non ferrous metals)
3	IS 205	Specification for non ferrous metal butt hinges
4	IS206	Specification for Tee and strap hinges
5	IS 207	Specification for Gate and shutter hook and eye
6	IS 208	Specification for door handles
7	IS 281	Specification for mild steel door bolts for use with pad locks
8	IS 287	Recommendations for maximum permissible moisture contents of timber used for different purpose
9	IS 303	Specification for plywood for general purposes
10	IS 362	Specification for parliament hinges
11	IS 363	Specification for hasps and stapple
12	IS 364	Specification for fan light catch
13	IS 401	Code of practice for preservation of timber
14	IS 419	Putty for use on window frames
15	IS 451	Technical supply condition for wood screws
16	IS 452	Specification for door spring rat tail type
17	IS 453	Specification for double acting spring hinge
18	IS 707	Glossary of terms applicable to timber technology and utilization
19	IS 710	Specifications for Marine Plywood
20	IS 723	Specification for steel counter sunk head wire nails.
21	IS 729	Specification for drawer lock, cup board lock and box locks
22	IS 848	Specification for synthetic resin adhesive for plywood (phenoic and amino plastic)
23	IS 851	Specification for synthetic resin adhesive for const. work (non structural in wood)
24	IS 852	Specification for animal glue for general wood work purpose
25	IS 1003 (Part I)	Specification for timber panelled and glazed shutter Part I (door shutters)
26	IS 1003 (Part II)	Specification for timber panelled and glazed shutter Part II (window and ventilator shutter)
27	IS 1141	Specification for code of practice for seasoning of timber
28	IS 1200 Part XIV	Method of measurement of building and civil engg work glazing.

29	IS 1200 Part XII	Wood work and joinery
30	IS 1328	Specification for veneered decorative plywood
31	IS 1341	Specification for steel butt hinges
32	IS 1378	Specification for oxidized copper finishes
33	IS 1566	Specification for hard drawn steel wire fabric
34	IS 1568	Specification for wire cloth for general purpose
35	IS 1658	Specification for hard drawn steel wire fabric
36	IS 1659	Specification for block boards
37	IS 1708 (Part 1 to 18)	Testing of small clear specimen for timber
38	IS 1734	Determination of density and moisture content.
39	IS 1823	Specification for floor door stopper
40	IS 1868	Specification for anodic coating on aluminium and its alloy -do-
41	IS 2046	Decorative thermosetting synthetic resin bonded laminated sheet
42	IS 2095	Specification for gypsum plaster board
43	IS 2096	Specification for asbestos cement flat sheet.
44	IS 2202 (Part I)	Specification for wooden flush door shutter, solid core type (plywood face panels)
45	IS 2202 (Part II)	-do- (Particle boards and hard board face panels)
46	IS 2209	Specification for mortice lock (Vertical Type)
47	IS 2380	Method of test for wood particle board and board for lignocelluloses material
48	IS 2547	Specification for gypsum plaster
49	IS 2753	Method for estimation of preservatives in treated timber and in treating solutions.
50	IS 2681	Specification for non-ferrous metal sliding door bolts use with pad locks
51	IS 3087	Specification for wood particle boards (Medium density) for general purpose.
52	IS 3097	Specification for veneered particle board
53	IS 3828	Specification for ventilator chain
54	IS 3400 (Part II)	Method of test for Vulcanized rubber (hardness)
55	IS 3400 (Part IV)	Method of test for Vulcanized rubber (Accelerated aging)
56	IS 3400 (Part IX)	Method of test for Vulcanized rubber (Density)
57	IS 3564	Specification for door closer (hydraulically regulated)

58	IS 3618	Phosphate treatment of iron and steel for protection against corrosion
59	IS 3813	“C” hooks for use with swivels
60	IS 3818	Specification for continuous (Piano) hinges
61	IS 3847	Specification for mortice night latch
62	IS 4835	Specification for polyvinyl acetate dispersion based adhesive for wood
63	IS 4948	Specification for welded steel wire fabric for general use
64	IS 4992	Specification for rebated mortice lock
65	IS 5187	Specification for flush bolts
66	IS 5509	Specification for Fire Retardant Plywood
67	IS 5930	Specification for mortice latch
68	IS 6318	Specification for plastic wire window fastners
69	IS 6607	Specification for rebated mortice lock (Vertical type)
70	IS 6760	Specification for sloted counter sunk head wood screws.
71	IS 7196	Specification for hold fast
72	IS 7534	Specification for sliding locking bolts for use with pad lock
73	IS 7638	Wood/lignocellulosic based panel products - Method for sampling
74	IS 8756	Specification for mortice ball catch for use in wooden almirah
75	IS 9308 (Part II)	Specification for mechanically extracted coir fibres. (Mattress coir fibres)
76	IS 9308 (Part III)	-do- Decorated coir fibre
77	IS 11215	Moisture content of timber and timber products method of determination
78	IS 12049	Dimensions and tolerance relating to wood based panel materials
79	IS 12406	Specification for medium density fibre board
80	IS 12817	Specification for stainless steel Butt Hinges
81	IS 12823	Specification for wood products -Prelaminated particle Boards
82	IS 14616	Specifications for laminated veneer lumber
83	IS 14842	Specification for coir veneer board for general purposes
84	IS 14856	Specification for glass fibre reinforced plastic (FRP) panel type door
85	IS 14900	Specifications for transparent float glass

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
1	**2**	**3**	**4**	**5**	**6**	**7**
Timber	9.1.6	Moisture content	Field (by moisture meter) laboratory test as required by Engineer-in-Charge	Appendix 'C'	1 cum	Every one cum or partthereof.
Flush door	9.7.10	End immersion Test knife test Adhesion Test	Laboratory	IS 2202 Appendix 'F'	26 shutters	As per sampling and testiNg Specified in lause 9.7.11
Mortice Locks	9.15.13	Testing of spring	Laboratory	IS 2209-Appendix 'G'	50 Nos	100 or part thereof.

PREAMBLE
WOODWORK AND P.V.C. WORK(CHAPTER: 9.0)

9.1 Door, Window and Ventilator Frames

Timber for door, window and ventilators frames shall be as specified. Timber shall be sawn in the direction of the grains. All members of a frame shall be of the same species of timber and shall be straight without any warp or bow. Frames shall have smooth, well-planed (wrought) surfaces except the surfaces touching the walls, lintels, sill etc., which may beleft clean sawn. Rebates, rounding or moulding shall be done before the members are jointed into frames. The depth of the rebate for housing the shutters shall be 15 mm, and the width of the rebates shall be equal to the thickness of the shutters. A tolerance of ± 2 mm shall be permitted in the specified finished dimensions of timber sections in frames.

9.2 Joints

The Jamb posts shall be through tenoned in to the mortise of the transoms to the full thickness of the transoms and the thickness of the tenon shall be not less than 2.5 cm. The tenons shall closely fit into the mortise without any wedging or filling. The contact surface of tenon and mortise before putting together shall be glued with polyvinyl acetate dispersion based adhesive conforming to IS 4835 or adhesive conforming IS 851 and pinned with 10 mm dia hard wood dowels, or bamboo pins or star shaped metal pins. The joints shall be at right angles when checked from the inside surfaces of the respective members. The joints shall be pressed in position. Each assembled door frame shall be fitted with a temporary stretcher and a temporary diagonal brace on the rebated faces.

9.3 Fixing of Frames

The frames shall be got approved by the Engineer-in-Charge before being painted, oiled or otherwise treated and before fixing in position. The surface of the frames abutting masonry or concrete and the portions of the frames embedded in floors shall be given a coating of coal tar. Frames shall be fixed to the abutting masonry or concrete with holdfasts or metallic fasteners as specified. After fixing, the jamb posts of the frames shall be plugged suitably and finished neat. Vertical members of the door frames shall be embedded in the floor for the full thickness of the floor finish and shall be suitably strutted and wedged in order to prevent warping during construction. A minimum of three hold fasts shall be fixed on each side of door and window frames one at centre point and other two at 30 cm from the top and bottom of the frames. In case of window and ventilator frames of less than 1 m in height two hold fasts shall be fixed on each side at quarter point of the frames. Hold fasts and metallic fasteners shall be measured and paid for separately.

9.4 Panelled Glazed or Panelled and Glazed Shutters

Panelled or glazed shutters for doors, windows, ventilators and cupboards shall be constructed in the form of timber frame work of stiles and rails with panel inserts of timber, plywood, block board, veneered particle board, fibre board wire gauze or float glass. The shutters may be single or multipanelled, as directed by the Engineer-in-Charge. Timber for frame work, material for panel inserts and thickness of shutters shall be as specified. All members of the shutters shall be straight without any warp or bow and shall have smooth well planed face at right angles to each other. Any warp or bow shall not exceed 1.5 mm for door shutter and 1 mm for window and ventilator shutters the right angle for the shutter shall be checked by measuring the diagonals and the difference between the two diagonals should not be more than 3 mm. Generally panelled glazed or panelled and glazed shutter shall conform to IS 1003 (Pt. 1 and 2).

9.5 Frame Work

Timber for stiles and rails shall be of the same species and shall be sawn in the directions of grains. Sawing shall be truly straight and square. The timber shall be planed smooth and accurate to the required dimensions. The stiles and rails shall be joined to each other by plain or haunched mortise and tenon joints and the rails shall be inserted 25 mm short of the width of the stiles. The bottom rails shall have double tenon joints and for other rails single tenon joints shall be provided. The lock rails of door shutter shall have its centre line at a height of 800 mm from the bottom of the shutters unless otherwise specified. The thickness of each tenon shall be approximately one-third the finished thickness of the members and the width of each tenon shall not exceed three times its thickness.

Gluing of Joints : The contact surfaces of tenon and mortise shall be treated, before putting together, with bulk type synthetic resin adhesive conforming to IS 851 suitable for construction in wood or synthetic resin adhesive (Phenolic and aminoplastic) conforming to IS 848 or polyvinyl acetate dispersion based adheshive conforming to IS 4835 and pinned with 10 mm dia hardwood dowels or bamboo pins or star shaped metal pins; after the frames are put together and pressed in position by means of press.

Stiles and bottom rail shall be made out of one piece of timber only. Intermediate rail exceeding 200 mm in width may be of one or more pieces of timber. The width of each piece shall be not less than 75 mm. Where more than one piece of timber is used for rails, they shall be joined with a continuous tongued and grooved joint glued together and reinforced with metal dowels at regular intervals not exceeding 200 mm.

9.6 Window and Ventilator

ShuttersRebating

The shutters shall be single-leaf or double leaved as directed by the Engineer-in-Charge. In case of double leaved shutters, the meeting of the stiles shall be rebated by one third the thickness of the shutter. The rebating shall be either splayed or square type.

Panelling

The panel inserts shall be either framed into the grooves or housed in the rebate of stiles and rails. Timber, plywood, and particle board panels shall be fixed only with grooves. The depth of the groove shall be 12 mm and its width shall accommodate the panel inserts such that the faces are closely fitted to the sides of the groove. Panel inserts shall be framed into the grooves of stiles and rails to the full depth of the groove leaving space of 1.5 mm. Width and depth of the rebate shall be equal to half the thickness of stiles and rails. Glass panels, asbestos panels wire gauze panels and panel inserts of cupboard shutters shall be housed in the rebates of stiles and rails.

Timber Panels: Timber panels shall be preferably made of timber of large width, the minimum width and thickness of the panel shall be 100 mm, and 15 mm respectively. When made from more than one piece, the pieces shall be jointed with a continuous tongued and grooved joint glued together and reinforced with headless nails at regular intervals not exceeding 100 mm. Depth and thickness of such joint shall be equal to one-third of thickness of panel. The panels shall be designed such that no single panel exceeds 0.5 square metre in area. The grains of timber panels shall run along the longer dimensions of the panels. All panels shall be of the same species of timber unless otherwise specified.

Plywood Panels: Plywood boards used for panelling of shutters shall be BWP type or grade as specified. Each panels shall be a single piece of thickness, 9 mm for two or more panel construction and 12 mm thickness for single panel construction unless otherwise specified.

Veneered Particle Board Panels: Veneered Particle board used for panelling of shutters shall be Exterior Grade bonded with BWP type synthetic resin adhesive as specified. Each panel shall be a single piece of thickness 12 mm unless otherwise specified.

Fibre Board Panels: Fibre board used for panelling of shutters shall be Exterior Grade bonded with BWP type synthetic resin adhesive each fibre board panel shall be a single piece unless otherwise specified.

Wire Gauze Panels: Wire Gauze used for panelling of shutters shall be woven with 0.63 mm dia galvanised mild steel wire to form average aperture size of 1.40 mm as specified. Wire gauze shall be securely housed into the rebates of stiles and rails by giving right angles bend turned back and fixed by means of suitable staples at intervals of 75 mm and over this wooden beading shall be fixed. The space between the rebate and the beading shall be fixed with putty to give a neat finish. Each wire gauze panel shall be a single piece, and the panels shall be so designed that no single panels exceeds 0.5 sqm in area. However, care shall be taken to prevent sagging of wire gauge, of panel by providing and fixing 20 x 20 mm square or equivalent beading to the external face to the required patterns as decided by the Engineer-in-Charge.

Glass Panels: Glass panelling (Glazing) shall be done as specified. Glazing in the shutters of doors, windows and ventilators of bath, WC and Lavatories shall be provided withfrosted glass the weight of which shall be not less than 10 kg/sqm. Frosted glass panes shall be fixed with frosted face on the inside. Glass panels shall be fixed by providing a thin layerof putty conforming to IS 419 applied between glass pane and all along the length of the rebate and also between glass panes and wooden beading.

Putty can be prepared by mixing one part of white lead with three parts of finely powdered chalk and then adding boiled linseed oil to the mixture to form a stiff paste and adding varnish to the paste at the rate of 1 litre of varnish to 18 kg of paste. Fixing of glass panes without beading shall not be permitted. Glazing shall be done after the shutters have been primed and prepared for painting, so that wood may not draw oil out of putty.

Finish: Panels of shutters shall be flat and well sanded to a smooth and level surface.

9.7 Beading

Beadings in panelled shutter shall be provided where specified in architectural drawings or directed by the Engineer-in-Charge. Each length of beading shall be single piece. Joints at the corners shall be mitred and exposed edges shall be rounded. Beading shall be fixed with headless nails at 75 mm intervals. For external shutters, the beading shall be fixed on the outside face.

9.8 Machine/Factory made Shutters

Machine made shutters, where specified, shall be procured from an approved factory. For machine made shutters, operations like sawing, planning, making tongue and tenons, cutting grooves, mortises and rebates, drilling holes and pressing of joints shall be done by suitable machines. Machines made shutters shall be brought to the site fully assembled but without any priming coat. Panel inserts of sheet glass and wire gauze may, however, be fixed at site.

9.9 Fixing of Shutters

For side hung shutters of height upto 1.2 m, each leaf shall be hung on two hinges at quarter

points and for shutter of height more than 1.2 m, each leaf shall be hung on three hinges one at the centre and the other two at 200 mm from the top and bottom of the shutters. Top hung and bottom hung shutters shall be hung on two hinges fixed at quarter points of top rail or bottom rail. Centre hung shutter shall be suspended on a suitable pivot in the centre of the frame. Size and type of hinges and pivots shall be as specified. Flap of hinges shall beneatly counter sunk into the recesses cut to the exact dimensions of flap. Screws for fixing the hinges shall be screwed in with screw driver and not hammered in. Unless otherwise specified, shutters of height more than 1.2 mm shall be hung on butt hinges of size 100 mm and for all other shutters of lesser height butt hinges of size 75 mm shall be used. For shutter of more than 40 mm thickness butt hinges of size 125 × 90 × 4 mm shall be used. Continuous (piano) hinges shall be used for fixing cup-board shutters where specified.

9.10 Flush Door Shutters

Flush door shutters shall have a solid core and may be of the decorative or non-decorative (Paintable type as per IS 2202 (Part I). Nominal thickness of shutters may be 25, 30 or 35 mm. Thickness and type of shutters shall be as specified.

Width and height of the shutters shall be as shown in the drawings or as indicated by the Engineer-in- Charge. All four edges of the shutters shall be square. The shutter shall be free from twist or warp in its plane. The moisture content in timbers used in the manufacture of flush door shutters shall be not more than 12 per cent when tested according to IS 1708.

9.11 Core

The core of the flush door shutters shall be a block board having wooden strips held in a frame constructed of stiles and rails. Each stile and rail shall be a single piece without any joint. The width of the stiles and rails including lipping, where provided shall not be less than 45 mm and not more than 75 mm. The width of each wooden strip shall not exceed 30 mm. Stiles, rails and wooden strips forming the core of a shutter shall be of equal and uniform thickness. Wooden strips shall be parallel to the stiles. End joints of the pieces of wooden strips of small lengths shall be staggered. In a shutter, stiles and rails shall be of one species of timber. Wooden strips shall also be of one species only but it may or may not be of the same species as that of the stiles and rails. Any species of timber may be used for core of flush door. However, any non-coniferous (Hard wood) timber shall be used for stiles, rails and lipping.

9.12 Face Panel

The face panel shall be formed by gluing, by the hot-press process on both faces of the core, either plywood or cross-bands and face veneers. The thickness of the cross bands as such or in the plywood shall be between 1.0 mm and 3.0 mm. The thickness of the face veneers as such or in the plywood shall be between 0.5 mm and 1.5 mm for commercial veneers and between 0.4 mm and 1.0 mm for decorative veneers, provided that the combined thickness of both is not less than 2.2 mm. The direction of the veneers adjacent to the core shall be at right angles to the direction of the wooden strips. Finished faces shall be sanded to smooth even texture. Commercial face veneers shall conform to marine grade plywood and decorative face veneers shall conform to type I decorative plywood in IS 1328.

9.13 Tests

Samples of flush door shutters shall be subjected to the following tests:

(i) End Immersion Test
(ii) Knife Test
(iii) Glue Adhesion Test

One end of each sample shutter shall be tested for End Immersion Test. Two specimens of 150 x150 mm size shall be cut from the two corners at the other end of each sample shutter

for carrying out Glue Adhesion Test. Knife Test shall be done on the remaining portion of each sample shutter. Test shall be done as laid down in Appendix P

9.14 Sample Size

Shutters of decorative and non-decorative type from each manufacturer, irrespective of their thickness, shall be grouped separately and each group shall constitute a lot. The number of shutters (sample size) to be selected at random from each lot for testing shall be as specified in Table 9.1 If The total number of shutters of each type in a work (and not the lot) is less than twenty five, testing may be done at the discretion of the Engineer-in-Charge and in such cases extra payment shall be made for the sample shutter provided the sample does not fail in any of the test. For knife test, glue adhesive test, slamming test, the end immersion test, the number of shutters shall be as per col. 4 of Table 9.1

TABLE 9.1

Sample Size and Criteria for Conformity

Lot Size	Sample Size	Permissible no. of Defective	Sub. Sample size
1	**2**	**3**	**4**
Upto 26 to 50	8	0	1
51 – 100	13	1	2
101 – 150	20	1	2
151 – 300	32	1	3
301 – 500	50	2	4
501 and above	80	2	5

9.15 Wire Gauze Fly Proof Shutters

Specified timber shall be used, and it shall be sawn in the direction of the grains. Sawing shall be truly straight and square. The timber shall be planed smooth and accurate to the full dimensions, rebates, roundings and mouldings as shown in the drawings made, before assembly. Patching or plugging of any kind shall not be permitted except as provided.

9.16 Stile and Rails

The stiles and rails shall be given a rebate to receive the wire gauze which shall form the panels.

9.17 Wire Gauze

The wire gauze shall be bent at right angles in the rebates of stiles and rails, turned back and fixed tight with blue tacks at about 75 mm centres, fixed alternately in the two faces of the rebates. Over this, wooden beading shall be fixed with brads or small screws at about 75mm centres. The space between the beading and rebates, where the wire gauze is bent, shall be neatly finished with putty, so that the end of the wire gauze may not be visible.

9.18 Moisture Content

The average moisture content of three test specimens, when determined in accordance with IS 1734 (Part 1) shall be between 5 to 15%.

Tests

- The tests as per Table-1 of IS 14616 shall be carried out by the manufacturer on the LVL(Laminated Veneer Lumber) sections on each batch.
- The manufacturer shall get the tests done on at least three samples of each batch by the standard method of test to ensure quality and performance of the material as per para of IS 14616.

- The manufacturer shall provide a certificate with the delivery challan indicating that the material conforms to IS 14616 along with the copy of the test report of the relevant batch.

9.19 Laminated Veneer Lumber (LVL) Door Shutters

This specification lays down requirements regarding types, sizes, material, construction, workmanship and finish, performance evaluation, sampling, measurements, rates and testing of Laminated Veneer Lumber (LVL) door shutter for use in domestic buildings, offices, schools, hospitals etc. This specification does not cover large size door shutters for industrial and special buildings such as workshops, garages, godowns etc.

The material of each lot shall be supported by a certificate to that effect: Each lot of LVL materials shall be accompanied by the test reports. Fabricator shall take up manufacturing of shutters only if provisions of IS 14616 are fulfilled; failing which, shutters so manufactured are liable for rejection.

9.20 Panelling Materials

Plain Particle Board: Plain particle boards used for panels shall be FPT-1 conforming to IS 3087 and shall have been bonded with BWP type of synthetic resin adhesive as per IS 848.

Pre-laminated Particle Board: Pre-laminated particle boards used for panels shall conform to IS 12823. The plain particle boards used in pre-laminated particle boards conform to specifications.

Medium Density Fibre Board: Medium density fibre board used for panels shall conform to exterior grade as per IS 12406 made from agro-forest products or agricultural wastes or natural fibers.

Pre-laminated Medium Density Fibre Board: Pre-lamination in pre-laminated medium density fiber board shall conform to the requirements such as Abrasion Resistance, Resistance to Steam, Crack Resistance, Resistance to Cigarette Burn and Resistance to Stain as specified in IS 12823.

Glass: Glass for glazing shall conform to IS 2835 or IS 2553. The use of other types of glass, such as frosted glass, wired glass and coloured glass may also be specified by the Engineer-in-Charge.

Wire Gauze: Wire gauze shall generally conform to IS 1568 and shall be regularly woven with equally spaced galvanized mild steel wires of 0.63 mm nominal diameter in both warp and weft directions to form aperture of average width 1.40 mm.

9.21 UPVC Door Frame

UPVC door frame shall be made of PVC material conforming to IS 10151. The door frame shall be made from extruded UPVC section having overall dimensions of 48 x 40 mm or 42 x 50 mm having wall thickness of 2.0 mm + 0.2 mm. Corners of the door frame to be jointed by M.S. galvanized brackets. Joints mitred and plastic welded. The hinge side vertical outer frames shall be reinforced by galvanized M.S. Tube of size 19 x 19 mm of wall thickness 1 mm + 0.1 mm and a tie rod shall be provided at the bottom of the frame. The frame shall be fabricated in factory as per nomenclature of the item and directions of Engineer-in-Charge.

9.22 Fixing of Frames

The frames are to be fixed in prepared openings in the walls. All civil work and tiling should be completed before the fixing of the frames. The frames are to be fixed directly on

the plastered wall. In case tiling is to be done in the place the frames are to be fitted, a 50 mm strip should be left untiled at the location where the frames are to be fitted. The frames are erected in the prepared opening such that the vertical members of the door frame are embedded 50 mm in the floor. The frame shall be fitted truly in plumb. A minimum of three anchor bolts or screws of size 65/100 shall be used to fix each vertical member. One bolt shall be fixed at 200 mm from the top member and one bolt shall be fixed at 200 mm from the floor. The third anchor bolt shall be fixed in the center. The top horizontal member shall be fixed using two 65/100 size anchor bolts or screws at a distance of 200 mm from both the corners.

9.23 PVC Door Shutters

The shutters shall be fabricated at factory as per nomenclature of the item and directions of Engineer-in-Charge. Shutter shall be made of PVC material conforming to IS 10151.

24 mm thick PVC Door Shutter
30 mm Thick PVC Door
Shutters
Sampling and Criteria for Conformity

9.24 General Precautions

The test specimens shall not have been exposed to a temperature below 40°C for 24 hours immediately preceding the test and shall be free from all visible moisture. The specimen shall be inspected and any specimen with visible flaws shall be discarded.

If any test specimen fails because of mechanical reason, such as failure of testing equipmentor improper specimen preparation, it shall be discarded and another specimen taken.

9.25 Test

The door shutters shall be subjected to the following tests in accordance with IS 4020 (Part 1 to 16).

(i) Dimension and Squareness Test: Door shutters when tested in accordance with IS 4020 (Part 2) the dimensions of nominal width and height will be within a limit of + 5 mm. The door shutter shall not deviate by more than 1 mm on a length of 500 mm. The thickness of the door shutter shall be uniform throughout with the permissible variation of not more than 0.8 mm between any two points. The nominal thickness of the shutter shall be within a limit of + 1.5 mm.

(ii) General Flatness Test: Door shutter, when tested in accordance with IS 4020 (Part 3) the twist, cupping and warping shall not exceed 6 mm.

(iii) Local Planeness Test: Door shutters, when tested in accordance with IS 4020 (Part 4), the depth of deviation measured at any point shall not be more than 0.5 mm.

(iv) Impact Indentation Test: Door shutters, when tested in accordance with IS 4020 (Part 5),shall have no defects such as cracking, tearing or delamination and the depth of indentation shall not be more than 0.2 mm.

(v) Edge Loading Test: Door shutters, when tested in accordance with IS 4020 (Part 7) the deflection of the edge at the maximum load shall not be more than 5 mm. On removal ofthe loads, the residual deflection shall not be more than 0.5 mm, failing which the test may be repeated on the other edge in the reverse direction. Also there shall be no lateral

buckling by more than 2 mm during loaded condition and no residual lateral buckling after removal of the load.

(vi) Shock Resistance Test: Door shutters, when tested in accordance with 2.1 of IS 4020 (Part 8), there shall be no visible damage in any part of the door after twenty five blows on each end.

(vii) Buckling Test: Door shutters, when tested in accordance with IS 4020 (Part 9), shall not show any deterioration and any residual deformation more than 5 mm after 15 min. of unloading and the initial deflection also shall not be more than 50 mm.

(viii) Slamming Test: Door shutters, when tested in accordance with 2.1 of IS 4020 (Part 10), shall not have any damage in any part of the door at the end of successive impacts. Door shutters, when tested in accordance with 3.1 of IS 4020 (Part 10), shall not have any visible damage in part of the door at the end of 100 successive impacts.

(ix) Misuse Test : Door shutters, when tested in accordance with IS 4020 (Part 11), there shall not be any permanent deformation of the fixing or any other part of the door set in hindering its normal working after the test.

(x) Screw Holding Test: Door shutters, when tested in accordance with IS 4020- Part 16, the load shall not be less than 1000 N.

(xi) End Immersion Test: Door shutters, when tested in accordance with IS 4020- Part 13, the shutter shall not show any delamination.

(xii) Knife Test: Door shutter, when tested in accordance with IS 4020 – Part 14, the grading shall be standard and excellent.

(xiii) Glue Adhesion Test: Door shutters shall be tested in accordance with IS 4020 – Part 15. There should be no delamination.

9.26 Gypsum Partition Panels

The material shall conform to IS: 2849

9.27 Dimensions

As per the item nomenclature.

9.28 Laying

(i) Panels are stored in a dry place and water should not come in contact with panels during or after construction. If the panels get wet, they should be dried before use.

(ii) The floor should be perfectly level before laying the first course. All panels must be properly aligned to the plumb. Successive layer of panels must be alternatively staggered so that vertical joints are not in the same line.

(iii) The recommended quantity of Gypsum Bonding Plaster must be used for joints and filling the grooves made for conduits, pipelines, etc. Excess Bonding Plaster must be scooped and removed, so that the joints and the places where the grooves are filled in are flush and even.

(iv) The walls should be dry and sanding done properly especially at joints before the primer is applied so that the surface is even and joints will not be visible after painting. Avoid

chasing with chisel and hammer. Use electrical saw or grooving tools for conduiting etc.

(v) The recommended span of walls is maximum 6 meters and maximum height is 4.5 meters.

(vi) Gypsum panel can easily be cut with coarse tooth hand saw, electric jigsaw, etc. The panels can be cut, sawn, drilled, milled or dowelled on the job. For concealed piping and conduit, the depth of groove should not exceed 50 mm. Hammer and chisel techniques to form chases must be avoided.

9.29 Sanding:

This application is to make the surface level without undulations. To make the gypsum wall surface level (in particular at joints, where there is excess bonding plaster), do sanding with sand paper at joints and other places, wherever you find uneven surface, otherwise joints will be visible after painting. It is important to sand all joints uniformly.

9.30 Primer Application:

The purpose of the primer is to give a better adhesion to the paint and also to reduce consumption of paint on the wall. Water thinable primers shall be used only.

TABLE 9.2

TABLE FOR PERMISSIBLE DEFECTS FOR VARIOUS GRADES OF TIMBERS

Sl. No.	Defects	First Grade	Second Grade
1	2	3	4
(i)	Cross-grain	Not steeper than 1 in 15	Not steeper than 1 in 10
(ii)	Sound knots and live knots	(i) Stiles and Rails (a) Short Exposed Face: Not more than 15 mm size and not more than 1 knot/metre (b) Long Exposed Face- Not more than 15 mm size and not more than 1 knot/m. No knot shall occur within 20 mm of the edges (ii) Panels – Not more than 20 mm size and not more than 2 knots/m^2. No knot shall occur on edge of any component of a panel.	(i) Stiles and Rails (a) Short exposed Face – Not more than 15 mm size and not more than 3 knots per stile and 1 knot per rail (b) Long Exposed Face- Not more than 20 mm size and not more than 3 knots per stile and 1 knot per rail (ii) Panels – Not more than 20 mm size and not more the 4 knots/m^2. No knots shall occur on edge of any component of a panel.

(iii)	Dead and loose knots (plugged)	(i) Stils and Rails – Not more than 10 mm size – centrally located and not more than 1 knot / m (ii) Panels – Not more than 15 mm size and not more than 2 knots/m^2. No knot shall occur on edges of any component of a panel.	(i) Stiles and Rails – Not more than 10 mm size, centrally located and not more than 3 knots per stile and 1 knot per rail. (ii) Panels – Not more than 15 mm size and not more than 4 knots/m^2. No knot shall occur on edge of any Component of a panel.
(iv)	Pitch pockets or Streaks	None	Permissible except on exposed edges provided that they are clean and filled up with suitable putty or filler when pitch pockets or streaks
			are located on the exposed edges of the core, they shall be cut out and filled with piece of wood of Similar species with grain running in the same direction. The piece shall be well glued.
(v)	Sapwood	Total not exceeding 5 mm wide and 150 mm long per metre. (This restriction applies only to super group species).	Total not exceeding 10 mm wide and 300 mm long per metre. (This restriction applies only to super group species).
(vi)	Pin holes	Permitted provided they are not in Cluster	Permitted.
(vii)	Worm holes	None	Permitted provided they are not more than 10 mm in diameter and not more thanone per metre and provided such worm holes are plugged with similar timber in such a manner that the pluging merges with the surrounding area both as to colour and grain.

Note :

(i) Dead and loose knots are permitted only if they are suitably plugged.
(ii) Knot shall not occur where hinges or locks are to be fixed.

TABLE 9.3
PHYSICAL AND MECHANICAL PROPERTIES OF PLYWOOD

Moisture content 5 to 15 per cent.

Shear Strength

	Dry State	**Resistance to Moisture**
Average	135	100
Individual	110	80

TABLE 9.4

PHYSICAL AND MECHANICAL PROPERTIES OF FPT OR GRADED PARTICLE BOARD

Sl. No.	**Properties**	**Flat pressed Three layer multi layer and graded**	
		Grade- I	**Grade- II**
(i)	Density variation, percent	± 10	± 10
(ii)	Water absorption, percent (a) 2 h soaking (b) 24 h soaking	 10 20	 40 80
(iii)	Linear expansion (swelling in water). 2 h soaking-percent (a) Length (b) Width	0.5 0.5	0.5 0.5
(iv)	Thickness, swelling, percent 2 h soaking	8	12
(v)	Swelling in thickness due to surface absorption, Percent	6	9
(vi)	Modulus of rupture, N/mm^2 (a) Average (b) Minimum	15 13	11 10
(vii)	Modulus of elasticity, N/mm^2 (a) Average (b) Minimum	2500 2250	2000 1800
(viii)	Tensile strength perpendicular to surface, N/mm^2 (a) Upto 20 mm thickness (b) Above 20 mm thickness	0.45 0.45	0.3 0.3
(ix)	Tensile strength perpendicular to surface, N/mm^2 (a) After cyclic test (b) Accelerated water resistance test 2)	0.2 0.15	- -

(x)	Screw withdrawal strength, N/mm^2 (a) Face (b) Edge (for thickness >12mm)	1250 850	1250 700

1) Cyclic Test— Specimen are immersed in water at 27 ± 2°C for a period of 72 h, followed by drying in air at 27 ± 2°C for 24 H and then heating in dry air at 70°C for 72 h. Three such cycles are to be followed and then the specimens are tested for tensile strength perpendicular to surface.

2) Accelerated Water Resistance Test— Specimen are immersed in water at 27 ± 2°C and water is brought to boiling and kept at boiling temperature for 2 h. Specimen are then cooled in water to 27
± 2°C and then tested for tensile strength perpendicular to surface.

TESTS FOR FLUSH DOOR SHUTTERS

END IMMERSION TEST

Door shutters shall be tested for resistance of their base to immersion in water as follows:

- The door shutter shall be immersed vertically to a height of 30 cm in water at roomtemperature for 24 hours
- Then allowed to dry for 24 hours at 27 ± 2°C and relative humidity of 65 ± 5 per cent.
- The cycle shall be repeated eight times.
- There shall be no delamination at the end of the test.

KNIFE TEST

Apparatus: The type of knife required to be used in the test is given in Fig. below. It may be made from a 250 x 25 mm file. The cutting edge should be kept chiselsharp. The test shall be carried out on a stout table to which a wooden batten is screwed against which the edge of test piece is placed.

KNIFE FOR TESTING PLYWOOD FOR ADHESION OF PLIES

Procedure:

- The knife is inserted with its cutting edge parallel to the grain of the outer veneer andworked into, or if possible along a glue line and the veneer is prised upwards.
- A hard and dense specie of plywood requires considerable force to effect entry and to priseand veneer.
- In a soft timber the knife tends to follow an easy course through the wood and in this case itis essential that the knife be firmly guided along the glue line.
- The bond should just pass the requirement, it is judged by the relative amount of wood fibreleft on the core veneer, and the area prised off.
- The grading is assessed chiefly on the appearance of the break.
- The force needed to effect separation is also an accompanying requirement. The bond is ‘excellent’, when it is difficult to find the glue line and impossible to keep the tool within it for more than 6 mm without cutting adjacent wood.
- On prising upwards, the veneer usually breaks off over a width slightly greater than that of the tool.
- The bond is ‘poor’ when knife meets little opposition in the glue line and the prise results inthe easy removal of almost all the veneers from one side of the test piece. The separated veneers are usually almost free from adherent fibre.
- Reporting of test results: The results shall be reported as ‘pass standard’ ‘excellent’ or ‘poor’.

GLUE ADHESION TEST

- Four square sections, 150 x 150 mm shall be cut from the corners of the door.
- These four cornersections as cut from the door shall be immersed in boiling water for 4hours, then dried at 27 ± 2°C and relative humidity of 65 ± 5 per cent for 24 hours.
- At the end of the drying period, the samples shall be examined for delamination.
- In the case of the glue lines in the plywood, all the four exposed edges of the plywood on both faces of a specimen shall be examined for delamination. A specimen shall be considered to have passed the test if no delamination has occured in the glue lines in the plywood and if no single delamination more than 50 mm in length and more than 3 mm in depth has occured in the assembly glue lines between the plywood faces and the stile and rail.
- Delamination at the corner shall be measured continuously around the corner. Delamination at a knot, knot hole, a pitch pocket and worm hole or other permissible wood defects shall not be considered in assessing the sample. A door shall be deemed to have passed this test ifthree of the four specimens tested pass the test.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.1		Wood work in frames of doors, windows	Providing wood work in frames of doors, windows, clerestory windows and other frames, wrought framed and fixed in position with hold fast lugs or with dash fasteners of required dia and length (hold fast lugs or dash fastener shall be paid for separately).		
	9.1.1	2nd class teak wood	Second class teak wood	cum	90618.00
	9.1.2	Sal wood	Sal wood	cum	71309.00
	9.1.3	Kiln seasoned hard wood finger jointed	Kiln Head wood seasoned and chemically treated hard wood (Haldu, Kail, Bija wood.)	cum	72744.00
	9.1.4	Kiln seasoned hard wood	Factory made Kiln seasoned, chemically treated, finger jointed at spacing not closer than 50 cm, 2nd class hard wood.		
	9.1.4.1	Frame 100mmX60 mm	Frame of size 100mmX60mm single or double rebate	meter	389.00
	9.1.4.2	Frame 120mmX60 mm	Frame of size 120mmX60mm double rebate.	meter	457.00
9.2		Laminated veneer lumber frames	Providing laminated veneer lumber conforming to IS:14616 and TAD -15: 2001(Part-B) in factory made frames of doors, windows, clerestory windows and other frames, wrought framed and fixing in position with hold fast lugs or with dash fasteners of required dia and length (hold fast lugs or dash fastener shall be paid for separately).	cum	92195.00
9.3		Wood work in frames of false ceiling	Providing wood work in frames of false ceiling, partitions etc. sawn and fixed in position (with necessary stainless steel screw)		
	9.3.1	Sal wood	Sal wood	cum	76229.00
	9.3.2	Hard wood	Kiln seasoned and chemically treated hardwood Haldu, Kail, Bija wood.	cum	77538.00
	9.3.3	2nd class, hard wood finger jointed	Factory made Kiln seasoned, chemically treated, finger jointed at spacing not closer than 50 cm, 2nd class hard wood.	cum	64833.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.4		Extra for circular shape	Extra for additional labour wood work in circular shape, such as in frames of fan light :		
	9.4.1	2nd class teak wood	Second class teak wood	cum	9062.00
	9.4.2	Sal wood	Sal wood	cum	7131.00
	9.4.3	Hard wood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	cum	7275.00
	9.4.4	2nd class hardwood finger jointed	Factory made Kiln seasoned, chemically treated, finger jointed at spacing not closer than 50 cm, 2nd class hard wood.	cum	6030.00
9.5		Paneled wooden shutters excluding, panelling/ glazing	Providing and fixing panelled or panelled and glazed shutters for doors, windows and clerestory windows including ISI marked M.S. pressed butt hinges bright finished of required size with necessary screws excluding, panelling/glazing which will be paid for separately, all complete as per direction of Engineer-in-charge.		
	9.5.1	2nd class teak wood	Second class teak wood		
	9.5.1.1	35 mm	35 mm thick shutters	sqm	2776.00
	9.5.1.2	30 mm	30 mm thick shutters	sqm	2448.00
	9.5.2	Hard wood	Kiln seasoned and chemically treated hardwood Haldu, Kail, Bija wood.		
	9.5.2.1	35 mm	35 mm thick shutters	sqm	2205.00
	9.5.2.2	30 mm	30 mm thick shutters	sqm	1956.00
	9.5.3	Sheesham wood	Kiln seasoned selected planks of sheesham wood.		
	9.5.3.1	35 mm	35 mm thick shutters	sqm	2060.00
	9.5.3.2	30 mm	30 mm thick shutters	sqm	1816.00
	9.5.4	2nd class hard wood finger jointed	Factory made Kiln seasoned chemically treated and finger jointed at spacing not closer than 50cm, 2nd class hard wood shutters		
	9.5.4.1	35 mm	35 mm thick	sqm	1089.00
	9.5.4.2	30 mm	30 mm thick	sqm	969.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.6		Panelling, glazing and wire mesh in shutters	Providing and fixing panelling or panelling and glazing in panelled or panelled and glazed shutters for doors, windows and clerestory windows (Area of opening for panel inserts excluding portion inside grooves or rebates to be measured). Panelling for panelled or panelled and glazed shutters 25 mm to 40 mm thick:		
	9.6.1	2nd class teak wood	Second class teak wood	sqm	2215.00
	9.6.2	Hard wood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood	sqm	1763.00
	9.6.3	2nd class hard wood finger jointed	15mm thick Factory made Kiln seasoned, chemically treated and finger jointed at spacing not closer than 50cm, 2nd class hard wood.	sqm	971.00
	9.6.4	Ply wood (5 ply)	Ply wood 5 ply, 9 mm thick :		
	9.6.4.1	Decorative both side	Decorative plywood both side decorative veneer (Type - I) conforming to IS 1328 BWR type.	sqm	1439.00
	9.6.4.2	Decorative one side	Decorative plywood one side decorative veneer and commercial veneer on other face (Type 1) conforming to IS 1328 BWR Type	sqm	1369.00
	9.6.5	7 ply	Ply wood 7 ply, 9 mm thick :		
	9.6.5.1	Decorative both side	Decorative plywood both side decorative veneer (Type- I) conforming to IS 1328 BWR type.	sqm	1861.00
	9.6.5.2	Decorative one side veneer on other face	Decorative plywood one side decorative veneer and commercial veneer on other face (Type-1) conforming to IS 1328 BWR Type	sqm	1650.00
	9.6.6	Particle board	Particle Board 12 mm thick		
	9.6.6.1	Plain	Plain particle board flat pressed, 3 layer or graded wood particle board medium density Grade I, IS: 3087 marked.	sqm	791.00
	9.6.6.2	Veneered flat	Veneered flat pressed three layer or graded wood particle board with commercial veneering on both sides conforming to IS: 3097, grade I.	sqm	1087.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.6.6.3	Pre-laminated one side	Pre-laminated particle board with decorative lamination on one side and balancing lamination on other side, Grade I, Type II IS: 12823 marked.	sqm	1333.00
	9.6.6.4	Pre-laminated both sides	Pre-laminated particle board with decorative lamination on both sides, Grade I, Type II, IS: 12823 marked.	sqm	1390.00
	9.6.6.5	Coir veneer board	Coir Veneer Board (conforming to IS 14842) 12 mm thick	sqm	1298.00
	9.6.7	Float glass	Float glass panes		
	9.6.7.1	4 mm	4 mm thick glass pane (weight not less than 10 kg/sqm)	sqm	973.00
	9.6.7.2	5 mm	5 mm thick glass pane (weight not less than 12.5 kg/sqm)	sqm	1046.00
	9.6.8	Wire gauge	Fly proof stainless steel grade 304 wire gauge with 0.5 mm dia. Wire and 1.4 mm wide aperture with matching wood beading	sqm	1532.00
	9.6.8.1	PVC sheet 4mm to 5mm thick	4mm to 5mm thick integral PVC form sheet with Celluka finish surface on both side with density not less than 600 kg/cum	sqm	580.00
	9.6.8.2	laminated PVC sheet 4mm to 5mm thick	4mm to 5mm thick laminated PVC form sheet with density not less than 600 kg/cum, laminated on both sides by 0.12 mm thick laminated sheet with density not less than 1400 kg/cum	sqm	754.00
9.7		Factory made laminated veneer lumber door shutter	Providing and fixing 35 mm thick factory made laminated veneer lumber door shutter conforming to IS : 14616 and TADS 15:2001 (Part B) including ISI marked M.S. pressed butt hinges bright finished of required size with necessary screws, all complete as per directions of Engineer-in-charge and panelling with panels of:		
	9.7.1	Plain particle board	12mm thick plain grade - 1, medium density flat pressed three layer particle board FPT - I or graded wood particle board FPT - I IS : 3087 marked bonded with BWP type synthetic resin adhesive as per IS : 848 :	sqm	2717.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.7.2	Particle board pre-lamination decorative both sides	12 mm thick pre-laminated particle board (decorative lamination on both sides) grade - 1, medium density flat pressed, three layer particle board FPT - I or graded wood particle board FPT - I, conforming toIS : 3087 bonded with BWP type synthetic resin adhesive as per IS : 848 and pre-lamination conforming to IS : 12823 Grade1, Type - II marked:	sqm	2868.00
	9.7.3	Particle board pre-lamination on one side	12 mm thick one side pre-laminated particle board (decorative lamination on one side and other sides balancing lamination) grade - 1 medium density flat pressed, three layer particle board FPT - I or graded wood particle board FPT-1 conforming to IS : 3087 bonded with BWP type synthetic resin adhesive as perIS : 848 and prelamination conforming to IS : 12823 Grade -1, Type II marked:	sqm	2775.00
9.8		Glazed shutters using 4 mm thick float glass	Providing and fixing glazed shutters for doors, windows and clerestory windows using 4 mm thick float glass panes including ISI marked M.S. pressed butt hinges bright finished of required size with necessary screws.		
	9.8.1	2nd class teak wood	Second class teak wood		
	9.8.1.1	35 mm	35 mm thick	sqm	3408.00
	9.8.1.2	30 mm	30 mm thick	sqm	3033.00
	9.8.2	hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood		
	9.8.2.1	35 mm	35 mm thick	sqm	2758.00
	9.8.2.2	30 mm	30 mm thick	sqm	2474.00
	9.8.3	sheesham wood plank	Kiln seasoned selected planks of sheesham wood.		
	9.8.3.1	35 mm	35 mm thick	sqm	2594.00
	9.8.3.2	30 mm	30 mm thick	sqm	2331.00
	9.8.4	2nd class hardwood finger jointed	Factory made Kiln seasoned chemically treated and finger jointed at spacing not closer than 50cm, 2nd class hard wood		
	9. 8.4.1	30 mm	30 mm thick shutters	sqm	1469.00
	9. 8.4.2	35 mm	35 mm thick shutters	sqm	1589.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.9		Factory made laminated veneer lumber glazed shutter 4mm thick float glass	Providing and fixing factory made laminated veneer lumber glazed shutter conforming to IS: 14616 and TADS 15:2001 (Part B) , using 4mm thick float glass panes for doors, windows and clerestory windows including ISI marked fixing with M.S. pressed butt hinges bright finished of required size with necessary screws as per directions of Engineer-in-charge. 30 mm thick shutters.	Sqm	2311.00
9.10		Extra for heavy glass panes	Extra for providing heavy sheet float glass panes instead of ordinary float glass in glazed doors, windows and clerestory window shutters. (Area of opening for glass panes excluding portion inside rebate shall be measured)		
	9.10.1	5 mm float glass	5 mm thick instead of 4 mm thick.	Sqm	73.00
	9.10.2	10 to 12 mm float glass	10 to 12 mm float glass	Sqm	132.00
	9.10.3	5.5 mm safety glass	5.5 mm safety glass	Sqm	400.00
	9.10.4	10-12mm safety glass	10-12mm safety glass	Sqm	960.00
9.11		Extra for frosted glass panes	Extra for providing frosted glass panes 4 mm thick instead of ordinary float glass panes 4 mm thick in doors, windows and clerestory window shutters. (Area of opening for glass panes excluding portion inside rebate shall be measured).	Sqm	236.00
9.12		Deduct for pin headed glass	Deduct for providing pin headed glass panes instead of ordinary float glass panes 4 mm thick in doors, windows and clerestory windows, shutters (Area of opening for glass panes excluding portion inside rebate shall be measured).	Sqm	15.00
9.13		Extra for S.S. butt hinges	Extra for S.S, hinges providing ISI marked Stainless Steel butt hinges instead of M.S. pressed butt hinges bright finished of required size with necessary screws. (Shutter area to be measured).	Sqm	117.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.14		Deduct for not providing hinges	Deduct for not providing hinges in doors, windows or clerestory window shutters with :		
	9.14.1	S.S. hinges	Stainless steel butt hinges with stainless steel screws for 2nd class teak wood and other class of wood shutters.	Sqm	170.00
	9.14.2	M.S. hinges	ISI marked M.S. pressed butt hinges bright finished of required size For 2nd class teak wood and other class of wood shutters.	Sqm	53.00
9.15		Shutters for cup board	Providing and fixing 25 mm thick Second class teak wood shutters using 4mm float glass for cup board etc. :		
	9.15.1	Panelled shutters	Panelled or panelled and glazed shutters :		
	9.15.1.1	Aluminium butt hinges	Using ISI marked anodised aluminium butt hinges with necessary screws.	Sqm	2736.00
	9.15.1.2	M.S. piano hinges	Using ISI marked nickel plated bright finished M.S. piano hinges with necessary screws.	Sqm	2753.00
	9.15.2	Glazed shutters	Glazed shutters :		
	9.15.2.1	2 nd class teak shutters butt hinges	Second class teak wood including ISI marked anodised aluminium butt hinges with necessary screws.	Sqm	2815.00
	9.15.2.2	2 nd class teak shutters piano hinges	2nd class teak wood including ISI marked nickel plated bright finished M.S. piano hinges with necessary screws.	Sqm	2759.00
9.16		Particle board to frames, backing or studding	Providing and fixing flat pressed 3 layer particle board medium density exterior grade (Grade I) or graded wood particle board IS : 3087 marked to frame, backing or studding with screws etc. complete (Frames, backing or studding to be paid separately) :		
	9.16.1	12 mm	12 mm thick	Sqm	555.00
	9.16.2	18 mm	18 mm thick	Sqm	654.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.17		Pre-laminated particle board in shelves	Providing and fixing Pre-laminated flat pressed 3 layer (medium density) particle board or graded wood particle board IS : 3087 marked with one side decorative and other side balancing lamination Grade I, Type II exterior grade IS : 12823 marked in shelves with screws and fittings wherever required, edges to be painted with polyurethane primer (fittings to be paid separately).		
	9.17.1	18 mm	18 mm thick	Sqm	1270.00
	9.17.2	25 mm	25 mm thick	Sqm	1383.00
9.18		Shutters decorative type 1st class hard wood and well matched teak 3 ply	Providing and fixing ISI marked flush door shutters conforming to IS: 2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutter.		
	9.18.1	35 mm butt hinges	35 mm thick including ISI marked Stainless Steel butt hinges with necessary screws.	Sqm	2823.00
	9.18.2	30 mm butt hinges	30 mm thick including ISI marked Stainless Steel butt hinges with necessary screws.	Sqm	2648.00
	9.18.3	25 mm Piano hinges	25 mm thick (for cupboard) including ISI marked nickel plated bright finished M.S. Piano hinges IS: 3818 marked with necessary screws.	Sqm	2465.00
9.19		Flush door shutters non-decorative type 1st class hard wood and well matched teak 3 ply	Providing and fixing ISI marked flush door shutters conforming to IS: 2202 (Part I)non-decorative type, core of block board construction with frame of 1st class hard wood and well matched commercial 3 ply veneering with vertical grains or cross bands and face veneers on both faces of shutters.		
	9.19.1	35 mm butt hinges	35 mm thick including ISI marked Stainless Steel butt hinges with necessary screws.	Sqm	1870.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.19.2	30 mm butt hinges	30 mm thick including ISI marked Stainless Steel butt hinges with necessary screws.	Sqm	1696.00
	9.19.3	25 mm Piano hinges	25mm thick (for cupboard) including ISI marked nickel plated bright finished M.S. piano hinges with necessary screws.	Sqm	1623.00
9.20		Extra for decorative veneering one side	Extra for Providing and fixing flush doors with decorative veneering instead of non decorative ISI marked flush door shutters conforming to IS: 2202 (Part I) On one side only.	Sqm	440.00
9.21		Extra for lipping	Extra for providing lipping with 2nd class teak wood battens 25 mm minimum depth on all edges of flush door shutters (over all area of door shutter to be measured).	Sqm	401.00
9.22		Extra for vision panel	Extra for providing vision panel not exceeding 0.1 sqm in all type of flush doors (cost of glass excluded) (overall area of door shutter to be measured) :		
	9.22.1	Rectangular or square.	Rectangular or square.	Sqm	157.00
	9.22.2	Circular.	Circular.	Sqm	244.00
9.23		Extra for louvers	Extra if louvers (not exceeding 0.2 sqm) are provided in flush door shutters (overall area of door shutters to be measured).	Sqm	354.00
9.24		Extra for rebate	Extra for cutting rebate in flush door shutters (Total area of the shutter to be measured).	Sqm	93.00
9.25		Wire gauze shutters	Providing and fixing wire gauze shutters using galvanised M.S. wire gauze of average width of aperture 1.4mm in both direction with wire of dia. 0.63 mm for doors, windows and clerestory windows with hinges and necessary screws:		
	9.25.1	35 mm thick shutter	35 mm thick shutters with ISI marked M.S. pressed butt hinges bright finished of required size.		
	9.25.1.1	2nd class teak wood	Second class teak wood.	Sqm	3174.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.25.1.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	Sqm	2599.00
	9.25.1.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	Sqm	2452.00
	9.25.2	35 mm ss hinges	35 mm thick shutters with ISI marked stainless steel butt hinges of required size.		
	9.25.2.1	2nd class teak wood	Second class teak wood.	Sqm	3291.00
	9.25.2.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	Sqm	2716.00
	9.25.2.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	Sqm	2569.00
	9.25.3	30 mm thick shutters	30 mm thick shutters with ISI marked M.S. pressed butt hinges bright finished of required size.		
	9.25.3.1	2nd class teak wood	Second class teak wood.	Sqm	2797.00
	9.25.3.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	Sqm	2306.00
	9.25.3.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	Sqm	2179.00
	9.25.4	30 mm ss butt hinges	30 mm thick shutters with ISI marked stainless steel butt hinges of required size.		
	9.25.4.1	2nd class teak wood	Second class teak wood.	Sqm	2914.00
	9.25.4.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	Sqm	2423.00
	9.25.4.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	Sqm	2296.00
9.26		Wire gauge laminated veneer lumber shutter	Providing and fixing wire gauge laminated veneer lumber shutters conforming to IS : 14616, and as per TADS 15 :2001 (Part B) using galvanised wire gauge with average width of aperture 1.4 mm in both directions with wire of dia 0.63 mm as per IS :1568 for doors, windows and clerestory windows including ISI marked M.S. pressed fixing with butt hinges bright		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			finished of required size with necessary screws as per directions of Engineer-in-charge:		
	9.26.1	35 mm	35 mm thick shutters	Sqm	2427.00
	9.26.2	30 mm	30 mm thick shutters	Sqm	2102.00
9.27		Teak wood plugs	Providing 50x50x50mm 2nd class teak wood plugs including cutting brick work and fixing in cement mortar 1:3 (1 cement : 3 sand) and making good the walls etc.	Each	18.00
9.28		Expandable fasteners	Providing and fixing expandable fasteners of specified size with necessary plastic sleeves and galvanised M.S. screws including drilling holes in masonry work /CC/ R.C.C. and making good etc. complete.		
	9.28.1	25 mm	25 mm long	Sqm	16.00
	9.28.2	32 mm	32 mm long	Sqm	18.00
	9.28.3	40 mm	40 mm long	Sqm	21.00
	9.28.4	50 mm	50 mm long	Sqm	25.00
	9.28.5	60 mm	60 mm long	Sqm	28.00
9.29		Wall lining teak wood tongued and grooved	Providing and fixing 2nd class teak wood plain lining tongued and grooved including wooden plugs complete with necessary screws and priming coat on unexposed surface.		
	9.29.1	40 mm	40 mm thick.	Sqm	5407.00
	9.29.2	25 mm	25 mm thick.	Sqm	3469.00
	9.29.3	20 mm	20 mm thick	Sqm	2802.00
	9.29.4	12 mm	12 mm thick	Sqm	1768.00
9.30		Wall lining particle board	Providing and fixing in wall lining flat pressed three layer (medium density) particle board or graded wood Pre-laminated one side decorative lamination on other side balancing lamination Grade I, Type II, IS : 12823 marked including priming coat on unexposed surface, with necessary fixing arrangement and screws etc. complete :		
	9.30.1	12 mm	12 mm thick	Sqm	1174.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.30.2	18 mm	18 mm thick	Sqm	1410.00
	9.30.3	25 mm	25 mm thick	Sqm	1525.00
9.31		Wood frame work battens 50x25mm	Providing and fixing specified wood frame work consisting of battens 50x25mm fixed with rawl plug and drilling necessary holes for rawl plug etc. including priming coat complete. Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	cum	96415.00
9.32		Plain lining / cladding with plywood 4mm thick cladding	Providing and fixing plywood 4 mm thick one side decorative veneer conforming to IS: 1328 (type-1) for plain lining / cladding with necessary screws, priming coat on unexposed surface with: Decorative veneer facings of approved manufacture.	Sqm	1068.00
9.33		Plain lining with coir veneer board	Providing and fixing 4mm thick coir veneer board, ISI marked IS: 14842, plain lining with necessary screws, priming coat on unexposed surface etc., complete.	Sqm	818.00
9.34		Skirting of pre-laminated particle board	Providing and fixing skirting of Pre-laminated with (one side decorative and other side balancing lamination) flat pressed, 3 layer or graded particle board (medium density) Grade I, Type II, IS :12823 marked, with necessary fixing arrangements and screws including drilling necessary holes for rawl plugs etc. and priming coat on unexposed surface complete :		
	9.34.1	18 mm	18 mm thick	Sqm	1626.00
	9.34.2	25 mm	25 mm thick	Sqm	1741.00
9.35		Wooden moulded beading to door and window frames	Providing and fixing wooden moulded beading to door and window frames with iron screws, plugs and priming coat on unexposed surface etc. complete :		
	9.35.1	2nd class teak wood	2nd class teak wood		
	9.35.1.1	50x12 mm	50x12 mm	metre	102.00
	9.35.1.2	50 x 20 mm	50 x 20 mm	metre	147.00
	9.35.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.35.2.1	50x12 mm	50x12 mm	metre	85.00
	9.35.2.2	50x20 mm	50x20 mm	metre	119.00
9.36		Plain jaffri wooden	Providing and fixing plain jaffri of 35x10 mm laths placed 35 mm apart (frames to be paid separately) including fixing 50x12 mm beading complete with : Second class teak wood.	Sqm	1773.00
9.37		Pelmet of wood particle board	Providing and fixing 18 mm thick, 150 mm wide pelmet of flat pressed 3 layer or graded wood particle board medium density grade I, IS : 3087 marked, including top cover of 6 mm commercial ply wood conforming to IS: 303 BWR grade, nickel plated M.S. pipe 20 mm dia (heavy type) curtain rod with nickel plated brackets, including fixing with 25x3 mm M.S. flat 10 cm long fixed to pelmet with Haldu, Kail, Bija wood cleats of size 100 mm x 40 mm x 40 mm on both inner side of pelmet and rawl plugs 75 mm long etc. all complete.	metre	333.00
9.38		Pelmet of coir veneer board	Providing and fixing 18 mm thick, 150 mm wide pelmet of coir veneer board ISI marked IS: 14842, including top cover of 6 mm coir veneer board, nickle plated M.S. Pipe 20 mm dia. (heavy type) curtain rod with nickel plated brackets, including fixing with 25x3 mm M.S. Flat 10 cm long fixed to pelmet with Haldu, Kail, Bija wood cleats of size 100 mm x 40 mm x 40 mm on both inner side of pelmet and rawl plugs 75 mm long etc. all complete.	metre	466.00
9.39		Extra for veneered pelmet	Extra for using veneered particle board conforming to IS 3097 Grade I, in item of pelmet 18mm thick 150mm wide.		
	9.39.1	Non Decorative	Non decorative veneer on both sides.	metre	34.00
	9.39.2	Decorative	Particle board with decorative veneering on both sides.	metre	102.00
9.40		Lipping in pelmet	Providing and fixing teak wood lipping of size 25x3 mm in pelmet.	metre	41.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.41		Curtain rod chromium plated brass plate	Providing and fixing of 1.25 mm thick chromium plated brass plate, with two chromium plated curtain rods brass brackets fixed with C.P. brass screws and wooden plugs, etc., wherever necessary complete:		
	9.41.1	12 mm	12 mm dia.	metre	289.00
	9.41.2	20 mm	20 mm dia.	metre	377.00
	9.41.3	25 mm	25 mm dia.	metre	382.00
9.42		Curtain rods nickel plated curtain rods M.S. pipe	Providing and fixing nickel plated curtain rods M.S. pipe with nickel plated brackets :		
	9.42.1	20 mm	20 mm dia (heavy type)	metre	127.00
	9.42.2	25 mm	25 mm dia (heavy type)	metre	141.00
9.43		M.S. grills and windows	Providing and fixing M.S. grills of required pattern in frames of windows etc. with M.S. flats, square or round bars etc. including priming coat with approved steel primer all complete.		
	9.43.1	Fixed to steel windows	Fixed to steel windows by welding.	Kg	119.00
	9.43.2	Fixed to openings /wooden frames	Fixed to openings /wooden frames with rawl plugs screws etc.	Kg	128.00
9.44		Expanded metal for windows	Providing and fixing expanded metal 20x60 mm strands 3.25 mm wide and 1.6 mm thick for windows etc. including 62 x19 mm beading of II nd class teak wood and priming coat with approved steel primer all complete.	Sqm	952.00
9.45		Steel wire fabric to window	Providing and fixing hard drawn steel wire fabric 75x25 mm mesh of weight not less than 7.75 Kg per sqm to window frames etc. including 62x19 mm beading of second class teak wood and priming coat with approved steel primer all complete.	Sqm	1224.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.46		M.S. wire mesh	Providing and fixing fly proof galvanized M.S. wire gauge to windows and clerestory windows using wire gauge with average width of aperture 1.4 mm in both directions with wire of dia. 0.63 mm all complete.		
	9.46.1	With 2nd class teak wood beading	With 2nd class teak wood beading 62x19 mm.	Sqm	1000.00
	9.46.2	With mild steel beading	With 12 mm mild steel U beading.	Sqm	681.00
9.47		Deduct for Steel wire instead of glass	Deduct for fixing 75x25 mm hard drawn steel wire fabric of weight not less than 7.75 Kg. per sqm in panelled and glazed door and window shutter instead of glass sheet 4 mm thick.	Sqm	166.00
9.48		Flat iron hold fast	Providing 40x5 mm flat (iron hold fast 40 cm long) (0.628 kg) including fixing to frame with 10 mm diameter bolts, nuts and wooden plugs and embeddings in cement concrete block 30x10x15cm 1:3:6 mix (1 cement : 3 sand : 6 graded stone aggregate 20mm nominal size)	Each	95.00
9.49		Wooden beams	Providing wooden beams including hoisting, fixing in position and applying wood preservative for the unexposed surfaces, etc. complete with :		
	9.49.1	Sal	Sal wood.	cum	72600.00
	9.49.2	Hardwood	Haldu, Kail, Bija wood.	cum	72550.00
9.50		M.S. Butt hinges	Providing and fixing ISI marked M.S. pressed butt hinges bright finished with necessary screws etc. complete :		
	9.50.1	125 mm	125x65x2.12 mm	Each	30.00
	9.50.2	100 mm	100x58x1.90 mm	Each	23.00
	9.50.3	75 mm	75x47x1.70 mm	Each	18.00
	9.50.4	50 mm	50x37x1.50 mm	Each	10.00
9.51		Heavy weight butt hinges	Providing and fixing IS : 1341 marked M.S. heavy weight butt hinges with necessary screws etc. complete :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.51.1	125 mm	125x90x4.00 mm	Each	53.00
	9.51.2	100 mm	100x75x3.50 mm	Each	35.00
	9.51.3	75 mm	75x60x3.10 mm	Each	21.00
	9.51.4	50 mm	50x40x2.50 mm	Each	13.00
9.52		Oxidised butt hinges	Providing and fixing ISI marked oxidised M.S. pressed butt hinges with necessary screws etc. complete.	Each	
	9.52.1	125 mm	125x65x2.12 mm	Each	32.00
	9.52.2	100 mm	100x58x1.90 mm	Each	23.00
	9.52.3	75 mm	75x47x1.70 mm	Each	17.00
	9.52.4	50 mm	50x37x1.50 mm	Each	10.00
9.53		Parliamentary hinges	Providing and fixing ISI marked oxidised M.S. pressed Parliamentary hinges with necessary screws etc. complete :	Each	
	9.53.1	150 mm	150x125x27x2.80 mm	Each	58.00
	9.53.2	125 mm	125x125x27x2.80 mm	Each	55.00
	9.53.3	100 mm	100x125x27x2.80 mm	Each	46.00
	9.53.4	75 mm	75x100x20x2.24 mm	Each	39.00
9.54		Single acting spring hinges	Providing and fixing ISI marked oxidised M.S. single acting spring hinges with necessary screws etc. complete :	Each	
	9.54.1	150 mm	150 mm	Each	169.00
	9.54.2	125 mm	125 mm	Each	163.00
	9.54.3	100 mm	100 mm	Each	131.00
9.55		Double acting spring hinges	Providing and fixing oxidised M.S. double acting spring hinges with necessary screws etc. complete.	Each	
	9.55.1	150 mm	150 mm	Each	174.00
	9.55.2	125 mm	125 mm	Each	163.00
	9.55.3	100 mm	100 mm	Each	137.00
9.56		Piano hinges	Providing M.S. Piano hinges ISI marked IS: 3818 finished with nickel plating and fixing with necessary screws etc., complete.		
	9.56.1	35 mm	Overall width 35 mm.	metre	133.00
	9.56.2	50 mm	Overall width 50 mm.	metre	138.00
	9.56.3	65 mm	Overall width 65 mm.	metre	141.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.57		Sliding door bolts	Providing and fixing ISI marked oxidised M.S. sliding door bolts with nuts and screws etc. complete:		
	9.57.1	300 mm	300x16 mm	Each	170.00
	9.57.2	250 mm	250x16 mm	Each	157.00
9.58		Tower bolt	Providing and fixing ISI marked oxidised M.S. tower bolt black finish, (Barrel type) with necessary screws etc. complete :	Each	
	9.58.1	250 mm	250x10 mm	Each	68.00
	9.58.2	200 mm	200x10 mm	Each	60.00
	9.58.3	150 mm	150x10 mm	Each	42.00
	9.58.4	100 mm	100x10 mm	Each	35.00
9.59		Pull bolt lock	Providing and fixing ISI marked 85x42mm oxidised M.S. pull bolt lock conforming to IS: 7534 with necessary screws bolts, nut and washers etc. complete	Each	71.00
9.60		Door latches	Providing and fixing ISI marked oxidised M.S. door latches conforming to IS:5930 with screws etc. complete :	Each	
	9.60.1	300 mm	300x20x6 mm	Each	72.00
	9.60.2	250mm	250x20x6 mm	Each	67.00
9.61		Oxidised M.S. handles	Providing and fixing ISI marked oxidised M.S. handles conforming to IS:4992 with necessary screws etc. complete :		
	9.61.1	125mm	125 mm	Each	27.00
	9.61.2	100mm	100 mm	Each	25.00
	9.61.3	75mm	75 mm	Each	24.00
9.62		Oxidised M.S. hasp and staple	Providing and fixing oxidised M.S. hasp and staple (safety type) conforming to IS : 363 with necessary screws etc. complete :	Each	
	9.62.1	150 mm	150 mm	Each	23.00
	9.62.2	115 mm	115 mm	Each	20.00
	9.62.3	90 mm	90 mm	Each	17.00
9.63		Oxidised M.S. casement stays	Providing and fixing oxidised M.S. casement stays (straight peg type) with necessary screws etc. complete.		
	9.63.1	300 mm	300 mm weighing not less than 200 gms.	Each	40.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.63.2	250 mm	250 mm weighing not less than 150 gms.	Each	34.00
	9.63.3	200 mm	200 mm weighing not less than 120 gms.	Each	28.00
9.64		Oxidised M.S. safety chain	Providing and fixing oxidised M.S. Safety chain with necessary fixtures for doors. (Weighing not less than 450 gms.)	Each	93.00
9.65		S.S. sliding door	Providing and fixing ISI marked stainless steel sliding door bolts with nuts and screws etc. complete:		
	9.65.1	300 mm	300x16 mm	Each	389.00
	9.65.2	250 mm	250x16 mm	Each	326.00
	9.65.3	400 mm	400x16mm	Each	601.00
9.66		S.S. tower bolt	Providing and fixing ISI marked stainless steel tower bolt black finish, (Barrel type) with necessary screws etc. complete :		
	9.66.1	250 mm	250x10 mm	Each	148.00
	9.66.2	200 mm	200x10 mm	Each	93.00
	9.66.3	150 mm	150x10 mm	Each	71.00
	9.66.4	100 mm	100x10 mm	Each	50.00
9.67		S.S. pull bolt lock	Providing and fixing ISI marked 85x42mm stainless steel pull bolt lock with necessary screws bolts, nut and washers etc. complete.	Each	86.00
9.68		S.S. door latches	Providing and fixing ISI marked stainless steel door latches with screws etc. complete :		
	9.68.1	300 mm	300x20x6 mm	Each	211.00
	9.68.2	250mm	250x20x6 mm	Each	173.00
9.69		S.S. handles	Providing and fixing ISI marked stainless steel handles with necessary screws etc. complete :	Each	
	9.69.1	125 mm	125 mm	Each	60.00
	9.69.2	100 mm	100 mm	Each	54.00
	9.69.3	75 mm	75 mm	Each	48.00
9.70		S.S. hasp and staple	Providing and fixing stainless steel hasp and staple (safety type) conforming to IS specification with necessary screws etc. complete :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.70.1	150 mm	150 mm	each	37.00
	9.70.2	115 mm	115 mm	each	33.00
	9.70.3	90 mm	90 mm	each	26.00
9.71		S.S. casement stays straight peg type	Providing and fixing stainless steel casement stays (straight peg type) conforming to IS specification with necessary screws etc. complete.		
	9.71.1	300 mm	300 mm weighing not less than 200 gms.	each	79.00
	9.71.2	250 mm	250 mm weighing not less than 150 gms.	each	61.00
	9.71.3	200 mm	200 mm weighing not less than 120 gms.	each	49.00
9.72		S.S. safety chain	Providing and fixing stainless steel safety chain with necessary fixtures for doors. Conforming to IS specification (Weighing not less than 450 gms.)	each	272.00
9.73		S.S. butt hinges	Providing and fixing IS : 12817 marked stainless steel butt hinges with stainless steel screws etc. complete :		
	9.73.1	125mm	125x64x1.90 mm	each	67.00
	9.73.2	100 mm	100X58X1.90 mm	each	59.00
	9.73.3	75 mm	75x47x1.80 mm	each	39.00
	9.73.4	50 mm	50x37x1.50 mm	each	25.00
9.74		S.S. butt hinges (heavy weight)	Providing and fixing IS : 12817 marked stainless steel butt hinges (heavy weight) with stainless steel screws etc. complete :		
	9.74.1	125 mm	125x64x2.50 mm	each	76.00
	9.74.2	100 mm	100x60x2.50 mm	each	60.00
	9.74.3	75 mm	75x50x2.50 mm	each	46.00
9.75		Knob	Providing and fixing 50 mm Stainless steel cup board or wardrobe knob with necessary screws (best make of approved quality)	each	62.00
9.76		Stainless steel hanging type floor	Providing and fixing Stainless steel hanging type floor door stopper with necessary screws, etc. complete.	each	103.00
9.77		Brass butt hinges	Providing and fixing bright finished brass butt hinges with necessary screws etc. complete :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.77.1	125x5.5 mm	125x85x5.5 mm (heavy type)	each	456.00
	9.77.2	125x4 mm	125x70x4 mm (ordinary type)	each	148.00
	9.77.3	100x5.5 mm	100x85x5.5 mm (heavy type)	each	387.00
	9.77.4	100x4 mm	100x70x4 mm (ordinary type)	each	117.00
	9.77.5	75x4 mm	75x65x4 mm (heavy type)	each	145.00
	9.77.6	75x2.5 mm	75x40x2.5 mm (ordinary type)	each	74.00
	9.77.7	50x2.5 mm	50x40x2.5 mm (ordinary type)	each	32.00
9.78		Brass parliamentary hinges	Providing and fixing bright finished brass parliamentary hinges with necessary screws etc. complete :		
	9.78.1	150mm	150x125x27x5 mm	each	385.00
	9.78.2	125 mm	125x125x27x5 mm	each	345.00
	9.78.3	100mm	100x125x27x5 mm	each	304.00
	9.78.4	75mm	75x100x20x3.2 mm	each	239.00
9.79		Brass tower bolts	Providing and fixing bright finished brass tower bolts (barrel type) with necessary screws etc. complete :		
	9.79.1	400mm	400x10 mm	each	400.00
	9.79.2	250mm	250x10 mm	each	329.00
	9.79.3	200mm	200x10 mm	each	261.00
	9.79.4	150mm	150x10 mm	each	221.00
	9.79.5	100mm	100x10 mm	each	142.00
9.80		Brass door latch	Providing and fixing bright finished brass door latch with necessary screws etc. complete.		
	9.80.1	300 mm	300x16x5 mm	each	237.00
	9.80.2	250mm	250x16x5 mm	each	225.00
9.81		Brass handles	Providing and fixing bright finished brass handles with necessary screws etc. complete.		
	9.81.1	125 mm	125 mm	each	205.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.81.2	100 mm	100 mm	each	181.00
	9.81.3	75 mm	75 mm	each	141.00
9.82		Brass sliding door bolts	Providing and fixing bright finished brass sliding door bolts with nuts and screws etc complete.		
	9.82.1	300mm	300x16mm	each	407.00
	9.82.2	250mm	250x16mm	each	374.00
9.83		Brass mortice latch and lock lever handles	Providing and fixing bright finished brass 100 mm mortice latch and lock with 6 levers and a pair of lever handles with necessary screws etc. complete (best make of approved quality).	each	539.00
9.84		Brass mortice latch and lock aluminium lever handles	Providing and fixing bright finished brass 100 mm mortice latch and lock ISI marked with six levers and a pair of anodised (anodic coating not less than grade AC 10 as per IS : 1868) aluminium lever handles with necessary screws etc. complete (Best make of approved quality).	each	820.00
9.85		Brass mortice latch with one dead bolt	Providing and fixing bright finished brass 100 mm mortice latch with one dead bolt and a pair of lever handles with necessary screws etc. complete (best make of approved quality).	each	475.00
9.86		Brass night latch	Providing and fixing bright finished brass night latch including necessary screws etc. complete (best make of approved quality).	each	678.00
9.87		Special quality brass cupboard or wardrobe locks	Providing and fixing special quality bright finished brass cupboard or wardrobe locks with four levers including necessary screws etc. complete (best make of approved quality) :		
	9.87.1	40 mm	40 mm	each	144.00
	9.87.2	50 mm	50 mm	each	150.00
	9.87.3	65 mm	65 mm	each	162.00
	9.87.4	75 mm	75 mm	each	185.00
9.88		Brass cup board or wardrobe knob	Providing and fixing 50 mm bright finished brass cup board or wardrobe knob with necessary screws (best make of approved quality)	each	51.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.89		Brass hanging type floor door stopper	Providing and fixing bright finished brass hanging type floor door stopper with necessary screws, etc. complete.	each	91.00
9.90		Brass casement window window fastener	Providing and fixing bright finished brass casement window fastener with necessary screws etc. complete.	each	72.00
9.91		Brass casement stays (straight peg type)	Providing and fixing bright finished brass casement stays (straight peg type) with necessary screws etc. complete :		
	9.91.1	300 mm	300 mm weighing not less than 330 gms	each	184.00
	9.91.2	250 mm	250 mm weighing not less than 280 gms	each	147.00
	9.91.3	200 mm	200 mm weighing not less than 240 gms	each	136.00
9.92		Brass hasp and staple (safety type)	Providing and fixing bright finished brass hasp and staple (safety type) with necessary screws etc. complete:		
	9.92.1	150 mm	150 mm	each	115.00
	9.92.2	115 mm	115 mm	each	94.00
	9.92.3	90 mm	90 mm	each	80.00
9.93		C.P. brass hanging type floor door stopper	Providing and fixing Cromium plated brass hanging type floor door stopper with necessary screws, etc. complete.	each	92.00
9.94		Aluminium hydraulic door closer door weight upto 35 kg	Providing and fixing aluminium die cast body tubular type universal hydraulic door closer (having brand logo with ISI, IS : 3564, embossed on the body, door weight upto 35 kg and door width upto 700 mm) with necessary accessories and screws etc. complete.	each	780.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.95		Aluminium hydraulic door closer door weight upto 36 kg to 80 kg	Providing and fixing aluminium extruded section body tubular type universal hydraulic door closer (having brand logo with ISI, IS : 3564, embossed on the body, door weight upto 36 kg to 80 kg and door width from 701 mm to 1000 mm) with double speed adjustment with necessary accessories and screws etc. complete.	each	1186.00
9.96		C.P. brass mortice latch	Providing and fixing chromium plated brass 100 mm mortice latch and lock with 6 levers and a pair of lever handles with necessary screws etc. complete (best make of approved quality).	each	777.00
9.97		C.P. brass mortice latch with one dead bolt	Providing and fixing chromium plated brass 100 mm mortice latch with one dead bolt and a pair of lever handles with necessary screws etc. complete (best make of approved quality).	each	835.00
9.98		C.P. brass night latch	Providing and fixing chromium plated brass night latch including necessary screws etc. complete (Best make of approved quality).	each	766.00
9.99		Special quality C.P cupboard brass locks	Providing and fixing special quality chromium plated brass cupboard locks with six levers including necessary screws etc. complete (Best make of approved quality) of :		
	9.99.1	40 mm	Size 40 mm	each	144.00
	9.99.2	50mm	Size 50 mm	each	156.00
	9.99.3	65mm	Size 65 mm	each	167.00
	9.99.4	75 mm	Size 75 mm	each	196.00
9.100		C.P. brass cupboard knobs	Providing and fixing chromium plated brass 50 mm cupboard or wardrobe knobs with nuts complete.	each	68.00
9.101		C.P.Brass sliding door bolts	Providing and fixing chromium plated brass sliding door bolts ISI marked with nuts and screws etc complete.		
	9.101.1	300 mm	300x16mm	each	268.00
	9.101.2	250mm	250x16mm	each	245.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.102		C.P. brass tower bolts	Providing and fixing chromium plated brass tower bolts ISI marked with nuts and screws etc complete.		
	9.102.1	400mm	400X10mm	each	260.00
	9.102.2	300 mm	300x10mm	each	198.00
	9.102.3	250mm	250x10mm	each	164.00
	9.102.4	200mm	200x10 mm	each	113.00
	9.102.5	150mm	150x10 mm	each	91.00
	9.102.6	100mm	100x10 mm	each	73.00
9.103		Chromium plated brass door latch	Providing and fixing chromium plated brass door latch ISI marked with necessary screws etc complete.		
	9.103.1	300 mm	300x16x5mm	each	255.00
	9.103.2	250 mm	250x16x5mm	each	243.00
9.104		Chromium plated brass night latch	Providing and fixing chromium plated brass night latch ISI marked with necessary screws etc complete.	each	249.00
9.105		Chromium plated brass handles	Providing and fixing chromium plated brass handles with necessary screws etc. complete:		
	9.105.1	125 mm	125 mm	each	241.00
	9.105.2	100 mm	100 mm	each	206.00
	9.105.3	75 mm	75 mm	each	182.00
9.106		Chromium plated brass casement window fastener	Providing and fixing chromium plated brass casement window fastener with necessary screws etc. complete.	each	137.00
9.107		Chromium plated brass casement stays (straight peg type)	Providing and fixing chromium plated brass casement stays (straight peg type) with necessary screws etc. complete :		
	9.107.1	300 mm	300 mm weighing not less than 330 gms	each	207.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.107.2	250mm	250 mm weighing not less than 280 gms	each	178.00
	9.107.3	200mm	200 mm weighing not less than 240 gms	each	149.00
9.108		Aluminium butt hinges	Providing and fixing ISI markedaluminium butt hinges anodised (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to required colour or shade with necessary screws etc. complete: Note:- Aluminum hinges shall not be used in wooden shutters.		
	9.108.1	125x75mm	125x75x4 mm	each	115.00
	9.108.2	125x63 mm	125x63x4 mm	each	108.00
	9.108.3	100x75 mm	100x75x4 mm	each	95.00
	9.108.4	100x63x4 mm	100x63x4 mm	each	93.00
	9.108.5	100x63x3.2 mm	100x63x3.2 mm	each	90.00
	9.108.6	75x63x4 mm	75x63x4 mm	each	77.00
	9.108.7	75x63x3.2 mm	75x63x3.2 mm	each	75.00
	9.108.8	75x45x3.2 mm	75x45x3.2 mm	each	67.00
9.109		Aluminium sliding door bolts	Providing and fixing aluminium sliding door bolts ISI marked anodised (anodic coating not less than grade AC 10 as per IS: 1868) transparent or dyed to required colour or shade with nuts and screws etc. complete:		
	9.109.1	300mm	300x16 mm	each	241.00
	9.109.2	250mm	250x16 mm	each	195.00
9.110		Aluminium tower bolts	Providing and fixing aluminium towerbolts ISI marked anodised (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to required colour or shade with necessary screws etc. complete:		
	9.110.1	400mm	400X10mm	each	123.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.110.2	300mm	300x10 mm	each	114.00
	9.110.3	250mm	250x10 mm	each	102.00
	9.110.4	200mm	200x10 mm	each	82.00
	9.110.5	150mm	150x10 mm	each	69.00
	9.110.6	100mm	100x10 mm	each	55.00
9.111		Aluminium pull bolt lock	Providing and fixing aluminium pull bolt lock anodised ISI marked (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to requiredcolour and shade with necessary screws bolts, nut and washers etc. complete.	each	74.00
9.112		Aluminium kicking plate	Providing and fixing 50cm long aluminium kicking plate 100x3.15 mm anodised (anodic coating not less than grade AC 10 as per IS :1868) transparent or dyed to required colour or shade with necessary screws etc. complete.	each	167.00
9.113		Aluminium handles	Providing and fixing aluminium handles ISI marked anodised (anodic coating not less than grade AC 10 as per IS : 1868) transparent or dyed to required colour or shade with necessary screws etc. complete:		
	9.113.1	125 mm	125 mm	each	75.00
	9.113.2	100 mm	100 mm	each	55.00
	9.113.3	75 mm	75 mm	each	46.00
9.114		Aluminium hanging floor door stopper	Providing and fixing aluminium hanging floor door stopper ISI marked anodised (anodic coating not less than grade AC 10 as per IS: 1868) transparent or dyed to required colour and shade with necessary screws etc. complete.		
	9.114.1	Single	Single rubber stopper	each	35.00
	9.114.2	Double	Twin rubber stopper	each	86.00
9.115		Aluminium casement stays	Providing and fixing aluminium casement window stays ISI marked anodised (anodic coating not less than grade AC 10 as per IS: 1868) transparent or dyed to required colour and shade with necessary screws etc. complete.	each	67.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.116		Aluminium tee channels	Providing and fixing aluminium tee channels (heavy duty) with rollers, stop end in pelmets as curtain rod.	metre	68.00
9.117		Aluminum U beading to flush door shutter	Providing and fixing aluminum U beading of required size to Pre-laminated /flush door shutter including fixing etc. complete as per direction of Engineer-in-charge.	kg	441.00
9.118		PTMT handles	Providing and fixing PTMT handles with necessary screws etc. complete.		
	9.118.1	125mm	125x34x24 mm weighing not less than 23 gms.	each	44.00
	9.118.2	150mm	150x34x24 mm weighing not less than 26 gms.	each	48.00
9.119		PTMT Butt hinges	Providing and fixing PTMT Butt hinges with necessary screws etc. complete.		
	9.119.1	75mm	75x60x10 mm fitted with 5.5 mm dia M.S. Bright Bar Rod weighing not less than 34 gms.	each	62.00
	9.119.2	100mm	100x75x10 mm fitted with 5.5 mm dia MS Bright Bar Rod weighing not less than 53 gms.	each	82.00
9.120		PTMT tower bolts	Providing and fixing PTMT Tower Bolts with 12 mm one piece rod inside and necessary screws etc., complete.		
	9.120.1	152mm	152x42x18 mm weighing not less than 60 gms.	each	87.00
	9.120.2	202mm	202x42x18 mm weighing not less than 78 gms.	each	106.00
9.121		PTMT door catcher	Providing and fixing PTMT door catcher of length 72 mm and dia. of 42 mm with suitable washers weighing not less than 33 gms.	each	37.00
9.122		PTMT hanging type floor door stoper	Providing and fixing PTMT hanging type floor door stoper with necessary screws etc complete.	each	35.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.123		Wire gauge shutters using s.s. 304 wire gauge for doors and windows	Providing and fixing wire gauge shutters using stainless steel grade 304 wire gauge with wire of dia 0.5 mm and average width of aperture 1.4 mm in both directions for doors, windows and clerestory windows with necessary screws:		
	9.123.1	35mm M.S. butt hinges	35 mm thick shutters with ISI marked M.S. pressed butt hinges bright finished of required size.		
	9.123.1.1	2nd class teak wood	Second class teak wood.	sqm	3583.00
	9.123.1.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	sqm	3008.00
	9.123.1.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	sqm	2862.00
	9.123.2	35 mm s.s. butt hinges	35 mm thick shutters with ISI marked stainless steel butt hinges of required size.		
	9.123.2.1	2nd class teak wood	Second class teak wood.	sqm	3700.00
	9.123.2.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	sqm	3125.00
	9.123.2.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	sqm	2979.00
	9.123.3	30 mm M.S. butt hinges	30 mm thick shutters with ISI marked M.S. pressed butt hinges bright finished of required size.		
	9.123.3.1	2nd class teak wood	Second class teak wood.	sqm	3206.00
	9.123.3.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	sqm	2715.00
	9.123.3.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	sqm	2589.00
	9.123.4	30 mm s.s. butt hinges	30 mm thick shutters with ISI marked stainless steel butt hinges of required size.		
	9.123.4.1	2nd class teak wood	Second class teak wood.	sqm	3323.00
	9.123.4.2	Hardwood	Kiln seasoned and chemically treated Haldu, Kail, Bija wood.	sqm	2832.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.123.4.3	Sheesham wood	Kiln seasoned selected class of sheesham wood.	sqm	2706.00
9.124		Fly proof stainless steel wire gauge, to windows	Providing and fixing fly proof stainless steel grade 304 wire gauge, to windows and clerestory windows using wire gauge with average width of aperture 1.4 mm in both directions with wire of dia. 0.50 mm all complete.		
	9.124.1	T.W. beading	With 2nd class teak wood beading 62x19 mm.	sqm	1689.00
	9.124.2	M.S. beading	With 12 mm mild steel U beading.	sqm	1370.00
9.125		Bamboo jaffery/ fencing	Providing and fixing Bamboo jaffery/ fencing consisting of superior quality 25mm dia (Average) half cut bamboo placed vertically and fixed together with three numbers horizontal running members of Haldu, Kail, Bija wood in scantling of section 50x25mm fixed with nails and G.I wire on existing support complete as per direction of Engineer-in-charge.	sqm	379.00
9.126		Corner beading of triangular shape	Providing and fixing wooden moulded corner beading of triangular shape to the junction of panelling etc. with iron screws, plugs and priming coat on unexposed surface etc. complete 2nd class teak wood. 50x50mm (base and height).	metre	182.00
9.127		Lipping/ moulded beading	Providing and fixing 2nd class teak wood lipping/ moulded beading or Taj beading of size 18x5mm fixed with wooden adhesive of approved quality and screws/ nails on the edges of the Pre-laminated particle board as per direction of Engineer-in-charge.	metre	47.00
9.128		Mortice lock for aluminum door	Providing and fixing bright finished 100mm mortice lock with 6 levers without pair of handles for aluminium door with necessary screws etc complete (Best make of approved quality) as per direction of Engineer-in-charge.	each	480.00
9.129		Magnetic catcher in cupboard /ward robe shutters	Providing and fixing magnetic catcher in cupboard / ward robe shutters including fixing with necessary screws etc. complete (Best make of approved quality).		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.129.1	Triple	Triple strip vertical type.	each	28.00
	9.129.2	Double	Double strip (horizontal type).	each	23.00
9.130		Magic eye	Providing and fixing magic eye in door shutters including fixing. Complete (Best make of approved quality).	each	329.00
9.131		Telescopic drawer channels	Providing and fixing powder coated telescopic drawer channels with necessary screws etc. complete as per directions of Engineer-in-charge.		
	9.131.1	300mm	300mm long	each set	209.00
	9.131.2	450mm	450mm long	each set	256.00
	9.131.3	500mm	500mm long	each set	302.00
9.132		Sliding rollers in racks/ cupboards/cabinets	Providing and fixing sliding arrangement in racks/ cupboards/cabinets shutter by P/F stainless steel rollers to run inside C or E aluminium channel section (The payment of C or E channel shall be made separately)	each	14.00
9.133		Ply wood Paneling for doors, windows	Providing and fixing 12mm thick panelling or panelling and glazing in panelled or panelled and glazed shutters for doors, windows and clerestory windows (area of opening for panel inserts excluding portion inside grooves or rebates to be measured). Panelling for panelled or panelled and glazed shutters 25mm to 40mm thick.		
	9.133.1	Marine ply	Marine plywood conforming to IS: 710	sqm	1591.00
	9.133.2	Fire retardant ply	Fire retardant plywood conforming to IS: 5509.	sqm	1660.00
9.134		Laminated sheet	Providing and Fixing decorative high pressure laminated sheet of plain / wood grain in gloss / matt / suede finish with high density protective surface layer and reverse side of adhesive bonding qualityconforming to IS : 2046 Type S including cost of adhesive of approved quality.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.134.1	1.5 mm	1.5 mm thick.	sqm	851.00
	9.134.2	1.0 mm	1.0 mm thick.	sqm	691.00
9.135		Cup board shutters 25mm thick with Pre-laminated particle board	Providing and fixing cup board shutters 25mm thick, with Pre-laminated flat pressed three layer particle board or graded wood particle board IS: 12823 marked exterior grade (Grade l Type ll) having one side decorative lamination and other side balancing lamination including IInd class teak wood lipping of 25mm wide x12 mm thick with necessary screws and bright finished stainless steel piano hinges complete as per direction of the Engineer-in-Charge.	sqm	1688.00
9.136		Cup board shutters with veneered particle board	Providing and fixing cup board shutters with 25mm thick veneered particle board IS : 3097 marked exterior grade (Grade I) of approved make including IInd class teak wood lipping of 25mm wide x 12 mm thick with necessary screws and bright finished stainless steel piano hingescomplete as per direction of Engineer-in- Charge.		
	9.136.1	One side	With decorative veneering on one side and commercial veneering on other side.	sqm	1433.00
	9.136.2	Both side	With non decorative veneering on both sides.	sqm	1322.00
9.137		Factory made flush door shutters of Pre-laminated particle board as per architectural drawing	Providing and fixing factory made shutters of 25 mm thick Pre-laminated particle board flat pressed three layer or graded wood particle board with one side decorative finish and other side balancing lamination conforming to IS: 12823 Gradel Type ll, of approved design, and edges sealed with water resistant paint and lipped with aluminium 'U' type edge beading all-round the shutter, including fixing with angle cleat, grip strip, cadmium plated steel screws including fixing of aluminium hinges 100x63x4 mm etc. complete as per architectural drawing and direction of Engineer-in-Charge (Cost of 'U' beading and hinges will be paid for separately).	sqm	1412.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.138		Wall lining with Pre-laminated fibre board	Providing and fixing in wall lining MDF board medium density fibre board IS: 14587:1998 marked, Pre-laminated one side decorative lamination and other side balancing lamination, with necessary fixing arrangement and screws etc. complete.		
	9.138.1	12 mm	12 mm thick.	sqm	919.00
	9.138.2	18 mm	18 mm thick.	sqm	1161.00
	9.138.3	25 mm	25 mm thick.	sqm	1230.00
9.139		Pre-laminated cup board shutters Fibre board piano hinges	Providing and fixing 25mm thick pre-laminated medium density fibre board exterior grade(Grade-I) IS:14587:1998 marked one side decorative and other side balancing lamination for cupboard shutters edges to be sealed with PVC edge bending tape 2.00 mm thick of approved brand including ISI marked nickeled plated bright finishing M.S. piano hinges IS:3818 marked with necessary screwed complete	sqm	1726.00
9.140		Skirting with pre-laminated fibre board	Providing and fixing skirting with Pre-laminated medium density fibre board exterior grade (Grade-I) IS: 14587:1998 marked, (one side decorative and other side balancing lamination) with necessary fixing arrangements and screws, including drilling necessary holes for rawl plugs etc. and edges to be sealed with PVC edge bending tape 2.00 mm thick of approved brand complete.		
	9.140.1	18 mm	18 mm thick	sqm	1493.00
	9.140.2	25 mm	25 mm thick	sqm	2017.00
9.141		Partition wall with G.I. frame and required board	Providing and fixing partition upto ceiling height consisting of G.I. frame and required board including providing and fixing of frame work made of special section power pressed/ roll form G.I. sheet with zinc coating of 120 gms/sqm(both side inclusive), consisting of floor and ceiling channel 50mm wide having equal flanges of 32 mm and 0.50 mm thick, fixedto the floor and ceiling at the spacing of 610 mm centre to centre with dash fastenerof 12.5 mm dia meter 50 mm length or		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			suitable anchor fastener or metal screws with nylon plugs and the studs 48 mm wide having one flange of 34 mm and other flange 36 mm and 0.50 mm thick fixed vertically within flanges of floor and ceiling channel and placed at a spacing of 610mm centre to centre by 6 mm dia bolts and nuts, including fixing of studs along both ends of partition fixed flush to wall with suitable anchor fastener or metal screws with nylon plugs at spacing of 450 mm centre to centre, and fixing of boardsto both side of frame work by 25 mm long dry wall screws on studs, floor and ceiling channels at the spacing of 300 mm centre to centre. The boards are to be fixed to the frame work with joints staggered to avoid through cracks, Galvanized M.S. fixing channel of 99 mm width (0.9 mm thick having two flanges of 9.5 mm each with zinc coating of 120 gm both side inside) to be provided at the horizontal joints of two boards, fixed to the studs using metal to metal flat head screws, including jointing and finishing to a flush finish with recommended jointing compound, jointing tape, angle beads at corner (25 mm x 25 mm x 0.5 mm), joint finisher and two coatsof primer suitable for board as per manufacture's specification and direction of engineer in charge all complete.		
	9.141.1	75 mm with GRG board	75mm overall thickness partition with 12.5 mm thick double skin fire rated glass reinforced gypsum (GRG) plaster board conforming to IS: 2095: part 3 1996 Board with BIS certification marks.	sqm	1212.00
	9.141.2	75 mm with gypsum board	75mm overall thickness partition with 12.5 mm thick double skin tapered edged plain Gypsum plaster board conforming to IS:2095: part I 1996 Board with BIS certification marks.	sqm	905.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.141.3	66 mm with calcium silicate board	66mm overall thickness Partition with 8mm thick double skin Calcium Silicate Board made with Calcareous and Siliceous materials reinforced with cellulose fiber manufactured through autoclaving process with Compressive Strength 225 kg/sq.cm, Bending Strength 100 kg./sq.cm.	sqm	1491.00
	9.141.4	66 mm with cement board	66mm overall thickness partition using 8mm thick double skin non- asbestos multipurpose cement board reinforced with cellulose fibre manufactured through autoclaving process (High pressure steam cured) as per IS: 14862 with suitable fibre cement screw.	sqm	1212.00
	9.141.5	68mm with heavy duty fiber cement board	68mm overall thickness partition using 9mm thick Heavy Duty Fibre Cement Board double skin non asbestos, reinforced with cellulose fibre,manufactured through autoclaving proces(High presssure steam cured)Type A, Category 4 conforming to IS 14862:2000. (for wet areas)	sqm	1450.00
	9.141.6	70 mm with cement bonded particle board	70 mm overall thickness Partition using 10mm thick double skin non-asbestost multipurpose termite, fire & moisure resistance (acoustic 30db) cement bonded particle board as per IS: 14276 withsuitable cement bonded board screw.	sqm	1359.00
9.142		Wall paneling using boards	Providing and fixing, in position concealed G.I. section for wall paneling using board of required thickness fixed on the 'W' profile (0.55mm thick) having a knurled web of 51.55mm and two flanges of 26mm each with lips of 10.55 mm, placed @ 610mmC/C in perimeter channel having one flange of 20mm and another flange of 30mm with thickness of 0.55mm and web of length 27mm. Perimeter channel is fixed on the floor and the ceiling with the nylon sleeves @ 610mm C/C with fully threaded self- tapping dry wall screws. Board is fixed tothe 'W' profile with 25 mm countersunk ribbed head screws @ 200mm C/C., all complete as per the drawing and directions of engineer-in-charge the joints of the boards are finished with specially formulated jointing compound and 48mm wide jointing tape to provide seamless finish.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.142.1	Calcium silicate board	10 mm thick tapered edge calcium silicate board made with calcareous and siliceous materials reinforced with cellulose fibre manufactured through autoclaving process to give stable crystalline structure, with compressive strength 225 kg/sq.cm, Bending strength 100 kg/sq.cm.	sqm	834.00
	9.142.2	Gypsum board	12 mm thick Gypsum board conforming to IS: 2095 Part-I.	sqm	550.00
	9.142.3	Cement bonded particle board	8 mm thick non-asbestost multipurpose termite, fire & moisure resistance cement bonded particle board as per IS: 14276 with suitable cement bonded board screw.	sqm	675.00
	9.142.4	Cement bonded prelamiante d particle board	8mm Both side Factory Prelamianted cement bonded particle board as per IS: 15786.	sqm	1072.00
	9.142.5	Heavy duty fiber cement board	9mm thick non asbestos,heavy duty fibre cement board reinforced with cellulose fibre manufactured through autoclaving process(High pressure steam cured) Type A,Category 4, conforming to IS 14862.	sqm	760.00
9.143		Frame work for partitions/ wall lining of hollow MS tubes	Providing and fixing frame work for partitions/ wall lining etc. made of 50x50x1.6mm hollow MS tube, placed along the walls, ceiling and floor in a grid pattern with spacing @ 60 cm centre to centre both ways (vertically and horizontally) or at required spacing near opening, with necessary welding at junctions and fixing the frame to wall/ ceiling/ floors with steel dash fasteners of 8 mm dia, 75 mm long bolt, includingmaking provision for opening for doors,windows, electrical conduits, switch boards etc., including providing with two coats of approved steel primer etc. complete, all as per direction of Engineer- in-charge.	kg	101.00
9.144		Panelling prelamiante d MDF board	Providing and fixing panelling or paneling and glazing in panelled or panelled and glazed shutters for doors, windows and clerestory windows (area of opening for panel inserts excluding portion inside grooves or rebated to be measured). Panelling for panelled and glazed shutters	sqm	1087.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			25mm to 40mm thick: Pre-laminated with decorative lamination on both side exterior Grade - I MDF Board 12 mm thick confirming to IS:14587		
9.145		Fibre board to frame	Providing and fixing Pre -laminated medium density fibre board exterior grade (Grade-I) IS: 14587:1998 marked, to frame, backing or studding with screws etc. complete (Frames, backing or studdingto be paid separately).		
	9.145.1	12 mm	Board 12 mm thick	sqm	811.00
	9.145.2	18 mm	Board 18 mm thick	sqm	943.00
9.146		Pre-laminated fibre board for shelves	Providing and fixing Pre-laminated medium density fibre board IS: 14587:1998 marked, with one side decorative lamination other side balancing lamination Grade-I(exterior grade) in shelves with screws and fittings wherever required, edges to be sealed with PVC edge bending tape 2.00 mm thick of approved brand (fittings to be paid separately).		
	9.146.1	18 mm	18 mm thick	sqm	1055.00
	9.146.2	25 mm	25 mm thick	sqm	1420.00
9.147		Acoustical wall panelling fabric finish	Providing and Installation of acoustical wall panelling fabric finish hardened square edge, FR grade fabric (colour as approved by Egnineer in charge) wrapped soft fibre core (glass fibre) panel having metal framing, of size 600x2100x25mm having density 100-120Kgs/ m3, weight 3.0kg/m2 NRC up to 0.9 installed by using impaling clip. Impaling clips are attached directly to the wall/frame work with the points extending outward. Then panels shall positioned and pressed into place.	sqm	7566.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.148		Acoustical wall panelling with perforated HDF board	Providing and installation of acoustical wall panelling wood finish Slats made out of HDF board , Melamine / veneer laminated finish, perforated wooden grooved slats (2mm grooves @ 8mm centers) / (2mm Slats @16mm pitch) / (2mm grooves @ 32mm centers) / (2mm grooves @ 64mm centers), backlined with black acoustical fleece, tongue-groove edge for a seamless look, FR grade, of lineal dimension size 128mm x 2440mm x 16mm thick having density 1000Kg /m^3, weight 12.8Kgs/m^2 installed by using GI strut system. The GI strut system includes GI Cross channel having thickness 0.45mm, length 3600mm, knurled web 40mm, depth 10mm and equal flanges 15mm is fastened vertically/ horizontally at every 600mm centers. Aluminium core cross channel having thickness 0.5mm, length 2400mm, web 15mm and 27mm, depth 18mm and flanges of 7mm with suitable edge and centre brackets is then fixed perpendicular to the cross channel with the help of fasteners at every 400mm centers. Contractor to Provide expansion joints of 3mm at every 5mts bothways.	sqm	5179.00
9.149		Acoustic wall panelling paintble consisting magnesite bonded wood fiber wt. 8.20 kg/m^2	Providing and installation of paintble acoustic wall panelling consisting of strand kerfed edge paint finished as approved by Engineer in charge. magnesite bondedwood fiber panels of size 600x2400x20mmhaving density 400kg/m^3, weight 8-20kg/m^2 which is fixed to GI strut system. The GI strut work includes Cross channel having thickness 0.45mm, length 3600mm,knurled web 40, depth 10mm and equal flanges 15mm is fastened to wall positioned horizontally in a regular mannerat 600mm centers. PVC Vinyl core UV treated H-Spline having thickness 2mm and length 2400mm to be fixed perpendicular to the Cross channel at 600mm centers. The Kerfed edge panels shall be then inserted into the H-spline against wall to perfect fit.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.149.1	Wt. 8.20 kg/m^2	Wt. 8.20 kg/m^2	sqm	1427.00
	9.149.2	Wt. 4.2 kg/m^2	Wt. 4.2 kg/m^2	sqm	1303.00
9.150		Plywood lining with ply on wooden /steel frame	Providing and fixing plain lining with necessary screws/nuts and bolts/ nails including a coat of approved primer on one face and fixed on wooden /steel frame work complete as per direction of Engineer-in-charge. (Frame work shall be paid for separately)		
	9.150.1		12 mm thick commercial ply conforming to IS: 1328 BWR type.	Sqm	1210.00
	9.150.2		18 mm thick commercial ply conforming to IS: 1328 BWR type.	Sqm	1320.00
9.151		Acoustic wall panelling soak coard of fabric wrapped wood fiber panels inserted into the H-spline	Providing and Installation of acoustic wall panelling consisting of Soak Coard of selected colours as approved by Engineer in charge. Kerf edge Fabric wrapped wood fiber panels of size 600x2400x20mm having density 400Kgs/m3, weight 4-20kg/m^2 installed by using GI strutsystem. The GI strut works includes Cross channel having thickness 0.45mm, length 3600mm, knurled web 50, depth 50mm and equal flanges of 13mm is fastened to Wall positioned horizontally in a regular manner at 600mm centers. PVC Vinyl core UV treated H-Spline having thickness 2mm and length 2400mm to be fixed perpendicular to the Cross channels at 600mm centers. Kerfed edge Fabricwrapped panels shall be then inserted into the H-spline against wall to perfect fit.	sqm	3716.00
9.152		Acoustic wall panelling soak coard of fabric wrapped wood fiber panels	Providing and Installation of acoustic wall panelling consisting of Soak Cord of selected colours as approved by Engineer in charge, square edge Fabric wrapped wood fiber panels of size 600x1200x10mm having density 400Kgs/m3, weight 4-20kg/m2 installed by using GI struts,Z bars. The GI strut systems includes Cross channel having thickness 0.45mm, length 3600mm, knurled web 40, depth 10mm and equal flanges 15mm is fastened to wallpositioned vertically in a regular manner at 600mm c/c.GI Z bars having 40mm height,	sqm	3358.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			thickness 1.2mm is first fixed behind the panels by identifying resin spots by using suitable fasteners.Clean exposed surfacesof wall and ensure walls are in plumb line before installing on wall (providenecessary packing shims if required). Soak cord square edge panels shall be slided intothe Z bars fixed on wall.		
9.153		Acoustic wall panelling fabric panels	Providing and installation of acoustic wall panelling consisting of Stretch NRC by using FR Grade NRC fabric of size 1700mm width, shall be stretched by using wooden base 10mm thick, rigid vinylStretch Tracks half wrap/ full wrap 25mm and Midseam 25mm , strand, synth PF with requisite accessories and tools.Woodenbase 10mm is first installed on the wall along the marking lines with metal fasteners at 300mm centers. Tracks to be installed on wooden base, by using Base tape (by removing the protective cover) fora true and continuous secure grip, and heavy-duty fasteners at 15mm centers on one/both sides of Stretch Tracks. Strand square edge magnesite bonded wood fiber panels of size 600x1200x15mm having density 400kg/m3, weight 4-8kg which is fixed to wall. Synth PF 10x10mm thickadhered on strand panel by using Stick 7 adhesive. Fabric of width 1700mm is then aligned to the Stretch tracks by following the fabric thread line or pattern. The fabric is stuck on the Align Tape (by removingthe protective cover) at the same time as being stretched and tucked into the Stretch tracks and secure into the locking jaws so that it will be smooth, free of wrinkles with the tucking tools. Note -minimum 50mm additional fabric is required for tucking hence maximum module width would be 1600mm.	sqm	3348.00
9.154		Wall panelling of magnesia core panels	Providing and Installation of wall panelling consisting of Square edge Magnesia core panels of size	sqm	771.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			1100x2400x8mmmm having density 1000kgs/m3, weight 8kg/m2 installed by using GI Strut system. The GI struct system includes Cross channel having thickness 0.45mm, length 3600mm,knurled web 40, depth 10mm and equalflanges of 15mm is fastened to wallpositioned horizontally in a regular mannerat 600mm c/c. The panels to be fixed onthe cross channel with suitable fasteners.		
9.155		Drywall partition magnesia block 66 mm thick	Supplying and installation of magnesia block 66 mm thick drywall partition including Framework includes Floor/ Head GI channels having thickness 0.5mm, length 3600mm, equal flanges of 32mmand web of 50mm fixed to floor and ceiling with suitable fasteners at 300mm centers staggered. Noise and fire rated silicon bead sealant to be caulked along theperimeter of the partition frame before fixing channels. Then Stud channel having thickness 0.45mm, length 3600mm, unequal flanges of 34/36mm and web of 48mm should be placed into the floor/head channel positioned vertically at 600mm centers. Extra reinforcement to be provided at openings (doors, windows, etc.,) The first layer of magenesia oxide block 8mm thick should be fixed on one side of the Stud framework in a staggered pattern by using suitable metal fasteners at 300mmcenters. Second layer of magenesia oxide block 8mm thick should be fixed other sideof the frame work similar fashion as the first layer. 3mm gap should be maintained between boards.3 mm gap shall be filled with jointing compound. Magnesia oxide block shall have following physical charecterstics. • Core - Magnesia• Fire Rating – upto 150mins • Acoustics – STC upto 44-50 • Climate (OC RH) – 50, 99 • Moisture Absorption - 6.8% after 2hrs		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			and 11.3% after 24hrs soaking • Wet Expansion - < 0.02% from ambient to saturation • Dry Contraction - ≤ 0.02% from evaporation • Moisture Movement - 0.02% • Light reflectance – 80 %• Green (RC %) – 30 • Strength - Antisag • Impact - 4kg hammer 175mm		
	9.155.1	66mm thick dry well	Using 8mm thick magnesia block	sqm	1578.00
	9.155.2	70mm thick dry wall	Using10mm thick magnesia block	sqm	1868.00
	9.155.3	74mm thick dry wall	Using12mm thick magnesia block	sqm	1984.00
	9.155.4	94 mm thick dry wall	Using12mm thick magnesia block	sqm	3320.00
9.156		Extra for thermal insulation to the dry wall	Add extra for providing and fixing Thermal insulation by providing 50mm thick synth PF having dencity 1000 GSM within the cavity held in position by using chicken wire mesh/ cross bracing/ pasting to the wall.	sqm	768.00
9.157		Gypsum panel partitions	Providing and laying Gypsum panel partitions 100mm thick with water proof Gypsum panels of size 666x500x100mm, made of calcite phosphor Gypsum fixed with tongue and groove, jointed with bonding plaster as per manufacturers specifications in superstructure aboveplinth level up to floor II level. Gypsum blocks will have a minimum compressive strength of 9.3 kg/cm2.	sqm	810.00
9.158		UPVC door frame wall thickness 2.0mm	Providing and fixing factory made UPVC door frame made of UPVC extruded sections having an overall dimension as below (tolerance ±1mm), with wall thickness 2.0 mm (± 0.2mm), corners ofthe door frame to be Jointed with galvanized brackets and stainless steel screws, joints mitred and Plastic welded. The hinge side vertical of the frame reinforced by galvanized M.S. tube of size 19 X 19mm and 1mm (± 0.1 mm) wallthickness and 3 nos. stainless steel hinges		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			fixed to the frame complete as per manufacturers specification and direction of Engineer-in-charge:		
	9.158.1	48x40 mm.	Extruded section Profile size 48x40 mm.	metre	203.00
	9.158.2	42x50 mm.	Extruded section Profile size 42x50 mm.	metre	208.00
9.159		Panelled UPVC door shutter 24 mm thick	Providing and fixing Panelled UPVC door shutter to existing door frames. 24 mm thick factory made UPVC door shutters made of styles and rails of a UPVC hollow section of size 59x24 mm and wall thickness 2 mm (± 0.2 mm) with inbuilt edging on both sides. The styles and rails mitred and joint at the corners by means of M.S. galvanised/plastic brackets of size 75x220 mm having wall thickness 1.0 mm and stainless steel screws. The styles of the shutter reinforced by inserting galvanised M.S. tube of size 20x20 mm and 1 mm (± 0.1 mm) wall thickness. The lock rail made up of a UPVC hollow 'H' section, section of size 100x24 mm and 2 mm (± 0.2 mm) wall thickness, fixed to the shutter styles by means of plastic/galvanised M.S. 'U' cleats. The shutter frame filled with a UPVC multi-chambered single panel of size not less than 620 mm, having over all thickness of 20 mm and 1 mm (± 0.1 mm) wall thickness. The panels filled vertically and tie bar at two places by inserting horizontally 6 mm galvanised M.S. rod and fastened with nuts and washers, complete as per manufacturer's specification and direction of Engineer-in-charge. (For W.C. and bathroom door shutter).	sqm	1750.00
9.160		Panelled door shutter UPVC 30 mm thick	Providing and fixing UPVC panelled door shutter UPVC 30 mm thick factory made PVC or UPVC door shutter made of styles and rails of a UPVC hollow section of size 60x30 mm and wall thickness 2 mm (± 0.2 mm), with inbuilt decorative moulding edging on one side. The styles and rails mitred and joint at the corners by means of M.S. galvanised/plastic brackets of size 75x220 mm having wall thickness 1.0 mm and stainless steel screws. The styles of the	sqm	2026.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			shutter reinforced by inserting galvanised M.S. tube of size 25x20 mm and 1 mm (± 0.1 mm) wall thickness. The lock rail made up of 'H' section, a UPVC hollow section of size 100x30 mm and 2 mm (± 0.2 mm) wall thickness fixed to the shutter styles by means of plastic/ galvanised M.S.'U' cleats. The shutter frame filled with a UPVC multi-chambered single panel of size not less than 620 mm, having over all thickness of 20 mm and 1 mm (± 0.1 mm) wall thickness. The panels filled vertically and tie bar at two places by inserting horizontally 6 mm galvanised M.S. rod andfastened with nuts and washers, complete as per manufacturer's specification and direction of Engineer-in-charge.		
9.161		Flush door shutter PVC, 25mm	Providing and fixing 25mm thick PVC flush door shutters made out of a one pieceMulti chamber extruded PVC section ofthe size of 762mmx25mm or less as per requirement with an average wall thicknessof 1mm (± 0.3mm). PVC foam end cap of size 23x10mm are provided on both vertical edges to ensure the overall thickness of 25mm. An M.S. tube having dimensions 19mm x 19mm and 1.0 mm (± 0.1mm) is inserted along the hinge side of the door. Core of the door shutter should be filled with High Density Polyurethane foam. The Top and Bottom edges of the shutter are covered with an end-cap of the size 25mmx11mm. Door shutter shall be reinforced with special polymeric reinforcements as per requiredspecification and direction of Engineer-in- charge to take up necessary hardware and fixtures. Stickers indicating the locations of hardware will be pasted at appropriate places.	sqm	1968.00
9.162		Panelled door shutter factory made PVC 37 mm	Providing and fixing 37 mm thick factory made PVC door shutter, styles and rails made of PVC hollow section of size 100mmx37mm with wall thickness 2mm (±0.2mm), with inbuilt bead on one side, styles and rails mitered cut and joint at the corners by means of 2 nos. of plastic brackets of	sqm	2142.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			size 75mm x 220 mm at each corner and stainless steel screws, reinforcing the hinge side of style by inserting PVC profile of size 28mm x 30mm,with wall thickness 2 mm (±0.2mm). Lockrail of size 100mmx37mm, wall thickness 2mm (+0.2mm) will be fixed to the vertical styles. Providing with PVC snapfit beads and panel of size 100mmx20 mm, and inserting 2 nos.tie bar of 6 mm dia and fastening with nuts and washers complete, all as per manufacturer's specification and direction of Engineer-in-charge.		
9.163		Door frame PVC size 75mmx53 mm	Providing and Fixing, factory made, PVC door frame made of PVC extruded sections of size 75mmx53 mm, having wall thickness 2.0mm (±0.2mm). Both verticals sides of the frame reinforced with PVC profile of cross section size 28mm x 30mmx2mm thickness (± 0.2 mm) and 75mmx200 mm long, including reinforcing both ends of the top frame with PVC profile. PVC Door Frame and PVC reinforcement profile to be mitred cut jointed and fusion welded together, including providing andfixing 3 nos. of 125mm long stainless steel hinges to frame, fixing the frame with jamb with required nos. and sizes of anchor dash fastener, all complete as per manufacturer's specification and direction of engineer-in-charge.	metre	342.00
9.164		PVC door shutter	Providing and fixing 37 mm thick factory made PVC Door shutter, styles and rails made of PVC hollow extruded printed and laminated section having overall dimension 115mm x 37mm with wall thickness 2 mm (±0.2mm) with inbuilt beading on one side, the styles and rails mitred cut and joint at corners by inserting 2 nos. PVC profile reinforcement of size 75 mm x 200 mm long with cross section size of 28mm x 30mm having wall thickness 2 mm (±0.2mm). Styles, rails and reinforcements to be fusion welded together. Only hinge side vertical style to be reinforced with PVC profile	sqm	2153.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			reinforcement in full length. Printed and laminated PVC lock rail of size 110mm x 37mm having wall thickness 2 mm (±0.2mm) to be welded horizontally with the vertical styles after inserting PVC profile reinforcement as in styles and rails, providing with PVC snap fit beading, panels of 100 x 20 mm printed andlaminated and inserting 2 nos. 6 mm dia. bright steel rod horizontally with both side threaded and tightened with check nuts and washers complete, all as per manufacturer's specification and direction of Engineer-in-Charge.		
9.165		P.V.C. door frame size 50x47mm wall thickness 5mm	Providing and fixing factory made P.V.C. door frame of size 50x47mm with a wall thickness of 5mm, made out of extruded 5mm rigid PVC foam sheet, mitred at corners and joined with 2 Nos. of 150mm long brackets of 15x15mm M.S. square tube, the vertical door frame profiles to be reinforced with 19x19mm M.S. square tube of 19 gauge, EPDM rubber gasket weather seal to be provided through out theframe. The door frame to be fixed to the wall using M.S. screws of 65/100mm size, complete as per manufacturer's specification and direction of Engineer-in- Charge.	metre	371.00
9.166		Panelled PVC door shutter	Providing and fixing factory made 30 mm thick PVC door shutter consisting of frame made out of M.S. tubes of 19 gauge thickness and size of 19mm x 19mm for styles and 15x15mm for top and bottom rails. M.S. frame shall have a coat of steel primers of approved make and manufacture. M.S. frame covered with 5mm thick heat moulded PVC 'C' channel of size 30mm thickness, 70mm width out of which 50mm shall be flat and 20mm shall be tapered in 45degree angle on both side forming styles; and 5mm thick, 95mm wide PVC sheet out of which 75mm shall be flat and 20mm shall be tapered in 45 degree on the inner side to form top and bottom rail and 115mm wide PVC sheet out of which 75mm shall be flat and 20mmshall be tapered on both sides to form lock rail. Top, bottom and lock rails shall be		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			provided both side of the panel. 10mm (5mmx2) thick, 20mm wide cross PVC sheet be provided as gap insert for top rail and bottom rail. paneling of 5mm thickboth side PVC sheet to be fitted in the M.S. frame welded/ sealed to the styles and rails with 7mm (5mm+2mm) thick x 15mm wide PVC sheet beading on inner side, and joined together with solvent cement adhesive. An additional 5mm thick PVC strip of 20mm width is to be stuck on the interior side of the 'C' Channel using PVC solvent adhesive etc. complete as per direction of Engineer-in-charge. Manufacturer's specification and drawing.		
	9.166.1	Plain PVC	Plain PVC door shutters.	sqm	2026.00
	9.166.2	Pre-laminated PVC	Pre laminated PVC door shutters.	sqm	2490.00
9.167		Door frame (single rebate) extruded solid PVC foam profile	Providing and fixing factory made door frame (single rebate) made out of single piece extruded solid PVC foam profile with homogenous fine cellular structure having smooth outer integral skin having62 mm width and 32 mm thickness, frame will be mitred and Jointed with self driven self tapping screws of size 38 mm x 4 mm and PVC solvent cement, including fixing the frame to wall with suitable dia and length anchor fastener as per manufacturer's specification and direction of Engineer-in-charge.	metre	313.00
9.168		Door shutter(panelled) 30mm thick solid PVC foam profile	Providing and fixing factory made 30 mm thick door shutter made of solid PVC foam profile. The styles and rails shall be of size 75 mmx30 mm having wall thickness 5mm. The styles, top and bottom rails shall have one side wall thickness of 15 mmintegrally extruded on the hinge side of theprofile for better screw holding power. Thestyles and rails shall be reinforced with M.S. tubes of size 33 mmx17 mmx 1 mm, painted with primer, all four corners of reinforcement to be welded or sealed. Solid PVC extruded bidding (push fit type) will be set inside the styles and the rails with a cavity, to receive single piece extruded 5		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			mm PVC sheet as panel. The styles andrails will be mitred cut and joint with the help of PVC solvent cement and self driven self tapping screws. Single piece extruded solid PVC lock rail of size 100 mm x 30 mm with wall thickness 5 mmand 15 mm integrally extruded in the middle of the lock rail and fixed with styles with the help of PVC solvent cement and self driven self tapping screws of size 100 mmx8 mm complete as per manufacturer's specifications and direction of Engineer-in-charge:		
	9.168.1	Non decorative	Non decorative finish.	sqm	2026.00
	9.168.2	Decorative	Decorative finish (both side wood grained finish).	sqm	2432.00
9.169		Factory made solid wood plastic composite (WPC) doorframes	Providing and fixing factory made single extruded WPC(Wood Polymer Composite) solid door/window/ceosetory windows and other frames/chowkhat comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (woodpowder/ rice .husk/wheat husk) and non toxic additives (maximum toxicity indexof 12 for 100 gms) fabricated with miter joints after applying PVC solvent cement and screwed with full body threaded star headed S.S. screws having minimum density of 750 kg/m^3 and screw withdrawal strength of 2200 N (Face) & 900 N (Edge), minimum compressive strength 58 N/mm^2, modulus of elasticity 900 N/ mm^2 and resistance to spread of flame of Class A category with property of being termite/borer proof, water/moisture proof and fire retardant and fixed in position with M.S. hold fast/lugs/S.S.dash fasteners of required dia and length complete as per direction of Engineer-in-Charge. (M.S. hold fast/lugs or S.S. dash fasteners shall be paid separately)		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.169.1	75 mm X 50 mm	Single rebate (75 mm X 50 mm)	RM	459.00
	9.169.2	100 mm X50 mm	Single rebate (100 mm X50 mm)	RM	720.00
	9.169.3	125 mm X 65 mm	Single or double rebate (125 mm X 65 mm)	RM	923.00
9.170		WPC flush door (plain)	Providing and fixing factory made single extruded WPC (Wood Polymer Composite) solid plain flush door shutter of required size comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (wood powder/ rice .husk/wheat husk) and non toxic additives (maximum toxicity index of 12 for 100 gms) having minimum density of 650 kg/cum and screw withdrawal strength of 1800 N (Face) & 900 N (Edge), minimum compressive strength 50 N/mm2, modulus of elasticity 850 N/mm2 and resistance to spread of flame of Class A category with property of being termite/borer proof, water/moisture proof and fire retardant and fixing with stainless steel butt hinges of required size with necessary full body threaded star headed counter sunk S.S screws, all as per direction of Engineer-InCharge. (Note: stainless steel butt hinges and necessary S.S screws shall be paid separately)		
	9.170.1	24 mm thick	24 mm thick WPC (Wood polymer composite) solid plain flush door shutter	sqm	2250.00
	9.170.2	28 mm thick	28 mm thick WPC (Wood polymer composite) solid plain flush door shutter	sqm	3250.00
	9.170.3	30 mm thick	30 mm thick WPC (Wood polymer composite) solid plain flush door shutter	sqm	3693.00
	9.170.4	35 mm thick	35 mm thick WPC (Wood polymer composite) solid plain flush door shutter	sqm	4135.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.171		WPC flush door (decorative type)	Providing and fixing factory made single extruded WPC (Wood Polymer Composite) solid decorative type flush door shutter of required size comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (wood powder/ rice .husk/wheat husk) and non toxic additives (maximum toxicity index of 12 for 100 gms) having minimum density of 650 kg/cum and screw withdrawal strength of 1800 N (Face) & 900 N (Edge), minimum compressive strength 50 N/mm2, modulus of elasticity 850 N/mm2 and resistance to spread of flame of Class A category with property of being termite/borer proof, water/moisture proof and fire retardant. WPC to be laminated with PVC foil of minimum 14 microns thick of approved design pasted with hot melt adhesive on both faces of shutter and fixing with stainless steel butt hinges of required size with necessary full body threaded star headed counter sunk S.S screws, all as per direction of Engineer-InCharge. (Note: stainless steel butt hinges and necessary S.S screws shall be paid separately)		
	9.171.1	30 mm thick	30 mm thick WPC (Wood Polymer Composite) solid decorative type flush door shutter	sqm	3858
	9.171.2	35 mm thick	35 mm thick WPC (Wood Polymer Composite) solid decorative type flush door shutter	sqm	4417

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.172		WPC solid board	Providing and fixing factory made single extruded WPC (Wood Polymer Composite) solid board one side white colour and other side of board laminated with PVC foil of minimum 14 micron thickness of approved design pasted with hot melt adhesive for cup boards, work stations and bathroom/ kitchen cabinet etc. of required sizes comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (wood powder/ rice husk/wheat husk) and non toxic additives (maximum toxicity index of 12 for 100 gms) having minimum density of 650 kg/cum and screw withdrawal strength of 1800 N (Face) & 900 N(Edge), minimum compressive strength 50 N/mm^2, modulus of elasticity 850 N/mm2 and resistance to spread of flame of Class A category with property of being termite/borer proof, water/moistureproof and fire retardant and fixing with stainless steel piano hinges/soft close clip on concealed hinges of required size with necessary full body threaded star headed counter sunk S.S screws, all as per direction of Engineer-In-Charge. (Note: stainless steel piano hinges/soft close clip on concealed hinges and necessary S.S screws shall be paid separately)		
	9.172.1		18 mm thick	sqm	2423.65
	9.172.2		25 mm thick	sqm	3083.95
9.173		WPC solid board for cupboard	Providing and fixing factory made single extruded WPC (Wood polymer composite solid plain white color board for backing of cupboards and bathrooms/ kitchen cabinets etc. of required size comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and naturalfibers (wood powder/rice husk/wheat husk) and non toxic additives (maximum toxicity index of 12 for 100 gms.), having minimum density of 650 kg/cum and screwwithdrawal strength of 1800 N (Face) & 900 N (Edge), minimum compressivestrength 50 N/mm2, modulus of elasticity 850 N/mm2 and resistance to spread of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			flame of Class A category with property of being termite/borer proof, water/moisture proof and fire retardant and fixing with stainless steel screws etc. all as per direction of Engineer-in-Charge. (Note:stainless steel screws shall be paid separately)		
	9.173.1	6 mm	6 mm thick	sqm	800.00
	9.173.2	12 mm	12 mm thick	sqm	1300.00
9.174		18 mm thick WPC Jali	Providing and fixing factory made 18 mm thick single extruded WPC (Wood Polymer Composite) solid plain white colour board Jali CNC (Computer numeric control) routed of approved design by Engineer-in -charge which are machine cut for duct/shaft covering, partitions and facades comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (wood powder/ rice husk/wheat husk) and non toxic additives(maximum toxicity index of 12 for 100 gms) having minimum density of 650 kg/cum and screw withdrawalstrength of 1800 N (Face) minimum compressive strength 50 N/mm2, modulus of elasticity 850 N/mm2 and resistance to spread of flame of Class A category with properties of being termite/borer proof, water/ moisture proofand fire i etardant and fixing on M.S (mild steel) frame made of 25 x 25 x 1.5 mm square hollow box section including applying a priming coat of approved steel primer, placed at grid made at 1.0 x 1.0 mor as per requirement at site with necessary stainless steel fasteners and SS screws etc., all complete as per direction of Engineer-In-Charge. (Note: M.S (mild steel) framework with priming coat and necessary SS fasteners and SS screws shall be paid separately.	sqm	2700.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.175		Door frame PVC single piece extruded PVC profile	Providing and fixing PVC Door Frame of size 50x47mm with a wall thickness of 5 mm (±0.2 mm) made out of single piece extruded PVC profile, with mitred cut joints and joint with 2nos of PVC bracket of size 190 mm x 100 mm long arms of cross section size 35x15mm and self driven self taping screws, the vertical door profiles to be reinforced with 40x20mm M.S. rectangular tube of 0.8 mm , including providing EPDM rubber gasket weather seal throughout the frame, including jointing 5mm PVC frame strip with PVC solvent cement on the back ofthe profile. The door frame to be fixed to the wall using 8x100mm long anchor fasteners complete, all as per manufacturer's specification and direction of Engineer -in- charge.	metre	517.00
9.176		Door shutter (panelled) 35mm thick made out of single piece extruded soild PVC	35 mm thick factory made Solid panel PVC Door shutter made out of single piece extruded soild PVC profiles, 5mm (±0.2mm) thick, having styles and rails (except lock rail) of size 95 mmx 35 mm x 5 mm, out of which 75 mm shall be flat and 20 mm shall be tapered (on both side), having one side thickness of 15 mm integrally extruded on the hinge side of the profile for better screw holding power, including reinforcing with MS tube of size 40 mm x 20 mm x 1 mm, joints of styles and rails to be mitered cut and joint with the help of PVC solvent cement, self driven self tapping screws and M.S. rectangular pipes bracket of size 190 mm x100 mm of cross section size 35 mm x 17 mm x 1 mm at each corner. Single piece extruded 5mm thick solid PVC Lock rail of size 115 mm x 35 mm x 35 mm, out of which 95 mm to be flat and 20 mm to be tapered at both ends, having 15 mm solid core in middle of rail section integrally extruded, fixing the styles and rails with the help of solvent and self driven self tapping screws of 125 mm x11 mm, including providing 5mm Single piece solid PVC extruded sheet inserted in the door as panel, all complete as per manufacturer's specification and direction of Engineer-in-charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.176.1	Non decorative	Non decorative finish (matt finish).	sqm	2037.00
	9.176.2	Decorative	Decorative finish (wood grained finish).	sqm	2142.00
9.177		PVC rigid foam sheet 1mm thick on existing door	Providing and fixing PVC rigid foam sheet 1mm thick on existing door shutters (bathroom and W.C. doors) using synthetic rubber based adhesive.	sqm	313.00
9.178		Door frames fibre glass reinforced plastic (FRP)	Providing and fixing Fibre Glass Reinforced plastic (FRP) Door Frames of cross-section 90mm x 45mm having single rebate of 32mmx15mm to receive shutter of 30mm thickness .The laminate shall be moulded with fire resistant grade unsaturated polyester resin and chopped mat .Doorframe laminate shall be 2mm thick and shall be filled with suitable wooden block in all the three legs. The frame shall be covered with fibreglass from all sides. MS stat shall be provided	metre	535.00
9.179		Door shutter glass fibre reinforced plastic (FRP) panelled 30mm	Providing and fixing to existing doorframes. 30 mm thick glass Fibre Reinforced Plastic (FRP) panelled door shutter of required colour and approved brand and manufacture, made with fire - retardant grade unsaturated polyester resin, moulded to 3 mm thick FRP laminate for forming hollow rails and styles, with wooden frame and suitable blocks of seasoned wood inside at required placesfor fixing of fittings, cast monolithically with 5mm thick FRP laminate for panels conforming to IS: 14856 including fixing to frames.	sqm	2574.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.180		Existing door frame door frame FRP 30mm	Providing and fixing door frame FRP30mm thick fiberglass reinforced Plastic (F.R.P.) flush door shutter in different plain and wood inish made with fire retardant grade unsaturated polyester resin,moulded to 3mm thick FRP laminate all around, with suitable wooden blocks inside at required places for fixing of fittings and polyurethane foam (PUF)/Polystyrenefoam to be used as filler material throughout the hollow panel, casted monolithically with testing parameters of F.R.P. laminate conforming to table - 3 of IS: 14856, complete as per direction of Engineer-in-charge.	sqm	2806.00
9.181		Fibreglass reinforced plastics chajja	Providing and fixing factory made Fibreglass Reinforced plastics (F.R.P.) chajja 4mm thick of required colour, size and design made by Resin Transfer Moulding (RTM) Machine Technology, resulting in void free compact laminate in single piece, having smooth gradual slope curvature for easy drainage of water and duly reinforced by 2nos. vertically and 1nos. horizontally 50x2mm thick M.S. flat with 12mm in built hole for grouting on the existing wall along with the 50mm flanges duly inserted and sealed in the wall complete in one single piece casted monolithically, including all necessary fittings . The FRP Chajja should be manufactured using unsaturated Polyester resin as per IS: 6746 duly reinforced with fibre glass chopped strand mat (CSM) as per IS: 11551 complete with protective Gelcoat U/V coating on Top for complete resistance from the extreme of temperature, weather and sunlight.	sqm	4625.00
9.182		Door frame fire resistant made out of 16 SWG G.I.sheet (1 hr fire rating)	Providing and fixing fire resistant door frame of section 143x57 mm having built in rebate made out of 16 SWG G.I.sheet (zinc coating not less than 120 gm/sqm) duly filled with vermuculite based concrete mix, suitable for mounting 60 minutes fire rated door shutters. The frame is fitted with intumuscent fire seal strip of size 10x4 mm (minimum) alround the frame and fixing with dash fastener of approved size and	metre	1191.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			make, including applying a coat of approved brand fire resistant primer etc. complete as per direction of Engineer-in-charge (Dash fastener to be paid for separately).		
9.183		Door shutters fire resistant made out of 16 SWG G.I.sheet (1 hr fire rating)	Providing and fixing 50 mm thick glazed fire resistant door shutters of 60 minutes fire rating conforming to IS:3614 (Part-II) tested and certified as per laboratory approved by Engineer-in-charge with suitable mounting on door frame,consisting of vertical styles, lock rail, top rail 100mm wide, bottom rail 200mm wide, made out of 16 SWG G.I.sheet (zinc coating not less than 120 gm/m2) duly filled FR insulation material and fixing with necessary stainless steel ball bearing hinges of approved make, includingapplying a coat of approved fire resistant primer etc. all complete as per direction of Engineer-in-charge (panneling to be paid for separately).	sqm	5850.00
9.184		Door shutters glazing in fire resistant (1 hr fire rating)	Providing and fixing glazing in fire resistant door shutters, fixed panels,ventilators and partitions etc., with G.I. beading of appropriate size made out of 20 SWG G.I.sheet (zinc coating not less than 120 gm/m2), fire resistant sealant, including applying a coat of approved fire resistant primer on G.I. beading etc. complete all as per direction of Engineer-in-charge. With clear fire resistant glass panes 6mm thick of approved brand, having minimum 60 minutes fire resistance.	sqm	27797.00
9.185		Panic bar / latch (double point)	Providing and fixing panic bar / latch (Double point) fitted with a single body, Trim Latch and Lock on back side of the Panic Latch of reputed brand and manufacture to be approved by the Engineer- in- charge, all complete.	Nos.	6091.00
9.186		Door frame UPVC size	Providing and Fixing factory made UPVC door frame made of UPVC exturded	metre	435.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		65 mmx55	sections, of size 65 mmx55 mm with wall thickness 2.0 mm (±0.2 mm), corners of the door frame to be mitred cut and jointed with plastic brackets and stainless steel screws, reinforcing hinge side vertical of the frames with PVC profile of Size 28mmx30 mm having wall thickness 2 mm (±0.2mm), including providing and fixing 3 nos. of 125 mm long stainless steel hinges to the frame, fixing the frame with jamb with required number and size of anchor dash fastners, all complete as per manufacturer's specification and direction of Engineer-in-charge.		
9.187		UPVC white opnable casement / sliding window and door	Providing and Fixing, UPVC factory made, white colour casement/sliding window and door made of extruded profiles. Profiles of frames and sash will be mitered cut and fusion welded at allcorners including drilling of holes for fixing hardware and drainage of water etc.,making arrangement for fixing ofhardware, EPDM gasket, 1.2 ± 0.2mm thick galvanised steel profile to be inserted in required profile, frame will be fixed to the wall with 8 mmx100 mm long fasteners, all complete as per direction of Engineer-in-charge. (Glazing, hardware hinges and fitting etc. to be paid separately.)		
	9.187.1	Casement window (outward openable)	Casement Window (Outward/Inward opening) with hinge System.		
	9.187.1.1	Frame	Frame (50 mm x 50mm)	metre	422.00
	9.187.1.2	Sash	Sash (Style and Rail) (62 mm x 34 mm)	metre	403.00
	9.187.1.3	Mullion	Mullion (intermediate Section) (66mm x 50 mm)	metre	434.00
	9.187.1.4	‘T' Profile	‘T' Profile (one vertical length in between two shutters) (24mm x 34.5mm)	metre	99.00
	9.187.1.5	Glazing bead	Glazing bead (12mm x 18mm)	metre	130.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.187.2	Casement window (outward openable)	Casement Window (With friction hinge and outward opening)		
	9.187.2.1	Casement Frame	Casement Frame (67mm x 62mm)	metre	495.00
	9.187.2.2	Sash/Mullion	Casement Window Sash/Mullion (67mm x75mm) (style, rail and intermediate section)	metre	519.00
	9.187.2.3	Glazing bead	Casement Glazing bead (35mm x 18 mm)	metre	178.00
	9.187.3	Sliding window 2 track	Sliding Window (Two Track, 2/4 Shutters)		
	9.187.3.1	Frame	Two Track Sliding Frame (67mm x 52mm)	metre	566.00
	9.187.3.2	Sash	Sliding window Sash (60mm x 44mm)	metre	529.00
	9.187.3.3	Interlock	Sliding Interlock for Window (one vertical length in each shutter)(45.5 mm x 28 mm)	metre	139.00
	9.187.3.4	Glazing bead	Sliding Glazing bead (35 mm x 18 mm)	metre	178.00
	9.187.4	Sliding door 2 track	Sliding Door (Two Track, 2/4 Shutters)		
	9.187.4.1	Frame	Two Track Sliding Frame (67mm x 52mm)	metre	539.00
	9.187.4.2	Sash	Sliding Door Sash (80mm x 44mm)	metre	653.00
	9.187.4.3	Interlock	Sliding Interlock for Door (one vertical length in each shutter) (45.5 mm x 28 mm)	metre	139.00
	9.187.4.4	Glazing bead	Sliding Glazing bead (35 mm x 18 mm)	metre	178.00
9.188		Friction hinges to the side/ top hung UPVC windows	Providing and fixing stainless steel (SS- 304 grade) friction hinges to the side/ top hung UPVC windows, of approved quality with necessary stainless steel screws etc. as per direction of Engineer-in-charge.		
	9.188.1	200 mm	200 x 19 x 1.9 mm	each	321.00
	9.188.2	250 mm	250 x 19 x 1.9 mm	each	369.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.188.3	300 mm	300 x 19 x 1.9 mm	each	462.00
	9.188.4	350 mm	350 x 19 x 1.9 mm	each	526.00
	9.188.5	400 mm	400 x 19 x 1.9 mm	each	636.00
9.189		Handle for UPVC casement windows	Providing and fixing casement handle made of zinc alloyed (white powder coated) for UPVC casement window with necessary screws etc. complete.	each	214.00
9.190		Lock for UPVC sliding window	Providing and fixing zinc alloyed (white powder coated) touch lock for UPVC sliding window with necessary screws etc. complete.	each	144.00
9.191		Steel roller for UPVC sliding window	Providing and fixing steel roller for UPVC sliding window with necessary screws etc. complete.	each	74.00
9.192		Steel roller for UPVC sliding door	Providing and fixing steel roller for UPVC sliding door with necessary screws etc. complete.	each	115.00
9.193		Steel crescent lock for UPVC sliding window/ door	Providing and fixing steel (white powder coated) crescent lock for UPVC sliding window/ door with necessary screws etc. complete.	each	127.00
9.194		UPVC white colour casement	Providing and fixing factory made UPVC white colour casement/casement cum fixed glazed windows comprising of UPVC multi-chambered frame, sash and mullion (where ever required) extruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made from roll forming process of required length (shape & size accordingto UPVC profile), UPVC extruded glazing beads of appropriate dimension EPDM gasket, stainless steel (SS 304 grade) friction hinges, zinc alloy (white powder coated) casement handles, G.I fasteners 100 x 8 mm size for fixing frame to finished wall, plastic packers,plastic caps and necessary stainless steel screws etc. Profile of frame & sash shall be mitred cut and fusion welded at allcorners, mullion (if required) shall be also fusion welded including drilling of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			holes for fixing hardware's and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealant over backer rod of required size and of approved quality, all complete as per approved drawing & direction of Engineer-in-Charge. (Single / double glass panes and silicon sealant shall be paid separately) Note: For UPVC frame, sash and mullion extruded profiles minus 5% tolerancein dimension i.e. in depth & width of profile shall be acceptable. Variation in profile dimension in higher side shall be accepted but no extra payment on thisaccount shall be made.		
	9.194.1	Single panel upto 0.75 sqm	Casement window single panel with S.S. friction hinges (300 x 19 x 1.9 mm), made of (small series) frame 47 x 50 mm & sash 47 x 68 mm both having wall thickness of 1.9 ± 0.2 mm and single glass pane glazing bead of appropriate dimension. (Area of window upto 0.75 sqm.)	sqm	6798.00
	9.194.2	Double panels 0.75 sqm upto 1.50 sqm	Casement window double panels with S.S. friction hinges (300 x 19 x 1.9 mm) made of (small series) frame 47 x 50 mm, sash 47 x 68 mm & mullion 47 x 68 mm all having wall thickness of 1.9 ± 0.2 mm and single glazing bead of appropriate dimension. (Area of window above 0.75 sqm upto 1.50 sqm).	sqm	6221.00
	9.194.3	Double panels top fixed upto 2.50 sqm	Casement window double panels with top fixed with S.S. friction hinges (350 x 19 x 1.9 mm) made of (small series) frame 47 x 50 mm, sash 47 x 68 mm & mullion 47 x 68 mm all having wall thickness of 1.9 ± 0.2 mm and single glazing bead of appropriate dimension. (Area of window upto 2.50 sqm).	sqm	4842.00
	9.194.4	Single panel big series	Casement window single panel with S.S. friction hinges (400 x 19 x 1.9 mm) made of (big series) frame 67 x 60 mm & sash67 x 80 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead /	sqm	6681.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			double glazing bead of appropriate dimension. (Area of window above 0.75 sqm.)		
	9.194.5	Duble panel big series	Casement window duble panel with S.S. friction hinges (400 x 19 x 1.9 mm) made of (big series) frame 67 x 60 mm & sash 67 x 80 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead / double glazing bead of appropriate dimension. (Area of window above 1.50 sqm.)	sqm	6942.00
	9.194.6	Both and single casement panel	Casement cum fixed panel window having both and single casement panel, middle fixed panels and at top completely fixed ventilator with S.S. friction hinges (350 x 19 x 1.9 mm) made of (big series) frame 67 x 60 mm & sash 67 x 80 mm & mullion67 x 80 all having wall thickness of 2.3 ± 0.2 mm and single glazing bead / double glazing bead of appropriate dimension. (Area of window above 3.00 sqm. Upto 5.00 sqm.)	sqm	5290.00
9.195		UPVC white colour fixed glazed windows/ventilators	Providing and fixing factory made UPVC white colour fixed glazed windows/ ventilators comprising of UPVC multi-chambered frame and mullion (where ever required) extruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made from roll forming process of required length (shape & size according to UPVC profile), UPVC extruded glazing beads of appropriate dimension, EPDM gasket, G.I fasteners 100 x 8 mm size for fixing frame to finished wall, plastic packers, plastic caps and necessary stainless steel screws etc. Profile of frame shall be mitred cut and fusion welded at all corners, mullion (ifrequired) shall be also fusion welded including drilling of holes for fixing hardware's and drainage of water etc. Afterfixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealant over backer rod of required size and of approved quality, all complete as per approved drawing & direction of Engineer-in-		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Charge. (Single / double glass panes and silicon sealant shall be paid separately). **Note: -** For UPVC frame, sash and mullion extruede profiles minus 5% tolerance in dimension i.e. in depth & width of profile shall be acceptable. Variation in profile dimension in higher side shall be accepted but no extra payment on this account shall be made.		
	9.195.1	Window / ventilator	Fixed window / ventilator made of (small series) frame 47 x 50 mm & mullion 47 x 68 mm both having wall thickness of 1.9 ± 0.2 mm and single glazing bead of appropriate dimension. (Area upto 0.75 sqm.)	sqm	4860.00
9.196		Casement/ casement cum fixed glazed door	Providing and fixing made UPVC white colour casement/ casement cum fixed glazed door comprising of UPVC multi-chambered frame, sash and mullion (where ever required) extruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made fromroll forming process of required length(shape & size according to UPVC profile),UPVC extruded glazing beads of appropriate dimension, EPDM gasket, zinc alloy (white powder coated) 3D hinges and one handle on each side of panels along with zinc plated mild steel multi point locking having transmission gear, cylinder with keeps and one side key, G.I fasteners 100 x 8 mm size for fixing frame to finished wall and necessary stainless steel screws, etc. Profile of frame & sash shall be mitred cut and fusion welded at all corners, mullion (if required) shall be also fusion welded including drilling of holes for fixing hardware's and drainage of wateretc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealent over backer rod of required size and of approved quality, all complete as per approved drawing & direction of Engineer-in-Charge. (Single / double glass panes andsilicon sealent shall be paid separately). Note: - For UPVC frame, sash and mullion extruede profiles minus 5% tolerance in		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			dimension i.e. in depth & width of profile shall be acceptable. Variation in profile dimension in higher side shall be accepted but no extra payment on this account shall be made.		
	9.196.1	Casement door with 3D hinges	Casement door with 3D hinges made of (big series) frame 67 x 64 mm & sash 67 x 110 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead / double glazing bead of appropriate dimension. (Area of door upto 2.00 sqm).	sqm	7085.00
	9.196.2	Casement door with top hung ventilator	Casement door with top hung ventilator with 3D and S.S. friction hinges (400 x 19 x 1.9 mm) made of (big series) frame 67 x 64 mm, sash 67 x 110 mm & mullion 67 x 80 mm all having wall thickness of 2.3 ±. 0.2 mm and single glazing bead /double glazing bead of appropriate dimension. (Area of door upto 2.50 sqm).	sqm	7063.00
9.197		UPVC white colour sliding glazed window	Providing and fixing factory made UPVC white colour sliding glazed window upto 1.50 m in height dimension comprising of UPVC multi-chambered frame with in-built roller track and sash extruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made from roll forming process of required length (shape & size according to UPVC profile), appropriate dimension of UPVC extruded glazing beads and UPVC extruded interlocks, EPDM gasket, wool pile, zinc alloy (white powder coated) touch locks with hook, zinc alloy body with single nylon rollers (weight bearing capacity to be 40 kg), G.I fasteners 100 x 8 mm size for fixing frame to finished wall and necessary stainless steel screws etc. Profileof frame & sash shall be mitred cut and fusion welded at all corners, including drilling of holes for fixing hardware's and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealent over backer rod of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			required size and of approved quality, all complete as per approved drawing & direction of Engineer-in-Charge. (Single / double glass panes, wire mesh and silicon sealent shall be paid separately) **Note: -** For UPVC frame, sash and mullion extruede profiles minus 5% tolerance in dimension i.e. in depth & width of profile shall be acceptable. Variation in profile dimension in higher side shall be accepted but no extra payment on this account shall be made.		
	9.197.1	Two track two panels sliding window	Two track two panels sliding window made of frame 52 x 44 mm &sash 32 x 60 mm both having wall thickness of 1.9 ± 0.2mm and single glazing bead of appropriate dimension. (Area of window upto 1.75 sqm)	sqm	5060.00
	9.197.2	Three track three panels sliding window with fly proof SS wire mesh	Three track three panels sliding window with fly proof SS wire mesh (Two nos. glazed & one no. wire mesh panels) made of frame 92 x 44 mm & sash 32 x 60 mm both having wall thickness of 1.9 ± 0.2 mm and single glazing bead of appropriate dimension (Area of window upto 1.75 sqm).	sqm	6880.00
	9.197.3	Two track two panels sliding window made of (big series)	Two track two panels sliding window made of frame 67 x 50 mm & sash 46 x 62 mm both having wall thickness of 2.3 ± 0.2mm and single glazing bead / double glazing bead of appropriate dimension (Area of window above 1.75 sqm upto 2.50 sqm).	sqm	5134.00
	9.197.4	Three track three panels sliding window with fly proof S.S wire mesh (big series)	Three track three panels sliding window with fly proof S.S wire mesh (Two nos. glazed & one no. wire mesh panels) made of frame 116 x 45 mm & sash 46 x 62 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead/double glazing bead of appropriate dimension. (Area ofwindow above 1.75 sqm).	sqm	6371.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.197.5	Three track three panels sliding window made of (big series)	Three track three panels sliding window made of frame 116 x 45 mm & sash 46 x 62 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead / double glazing bead of appropriate dimension. (Area of window above 1.75 sqm)	sqm	5913.00
9.198		UPVC white colour sliding glazed window	Providing and fixing factory made UPVC white colour sliding glazed window above 1.50 m in height dimension comprising of UPVC multi-chambered frame with in-built roller track and sash extruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made from roll forming process of required length (shape & size according to UPVC profile), appropriate dimension of UPVC extruded glazing beads, UPVC extruded interlocks and UPVC extruded Inline sash adaptor (if required), EPDM gasket, wool pile, zinc alloy (white powder coated) handle on one side of extreme panel along with zinc plated mild steel multi point locking having transmission gear with keeps, zinc alloy (white powder coated) touch lock with hook (if required for wire mesh panel), stainless steel (SS 304 grade) body with adjustable double nylon rollers (weight bearing capacity to be 120 kg), G.I fasteners 100 x 8 mm size for fixing frame to finished wall and necessary stainless steel screws etc. Profile of frame & sash shall be mitred cut and fusion welded at all corners. Including drilling of holes for fixing hardware's and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealent over backer rod of required size and of approved quality, all complete as per approved drawing & direction of Engineer-in-Charge. (Single / double glass panes, wire mesh and silicon sealent shall be paid separately). **Note: -** For UPVC frame, sash and mullion extruede profiles minus 5% tolerance in dimension i.e. in depth & width of profile shall be acceptable. Variation in profile dimension in higher side shall be accepted but no extra payment on this account shall be made.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.198.1	Two track two panels sliding window made of (big series)	Two track two panels sliding window made of frame 67 x 50 mm & sash 46 x 62 mm both having wall thickness of 2.3 ± 0.2mm and single glazing bead / double glazing bead of appropriate dimension. (Area of window above 2.50 sqm upto 4.00 sqm).	sqm	4332.00
	9.198.2	Two track four panels sliding window made of (big series)	Two track four panels sliding window made of frame 67 x 50 mm & sash 46 x 62 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead / double glazing bead of appropriate dimension. (Area of window above 4.00 sqm upto 8.00 sqm).	sqm	3532.00
9.199		UPVC white colour sliding glazed door	Providing and fixing factory made UPVC white colour sliding glazed door comprising of UPVC multi-chambered frame with in-built roller track and sashextruded profiles duly reinforced with 1.60 ± 0.2 mm thick galvanized mild steel section made from roll forming process of required length (shape & size according to UPVC profile), appropriate dimension UPVC extruded glazing beads, UPVC extruded interlock and UPVC extruded Inline sash adaptor (if required), EPDM gasket, wool pile, zinc alloy (white powder coated) handle with key on one side of extreme panels along with zinc plated mild steel multi point locking having transmission gear with keeps, zinc alloy (white powder coated) cresent lock (if required), stainless steel (SS 304 grade) body with adjustable double nylon rollers (weight bearing capacity to be 120 kg), G.I fasteners 100 x 8 mm size for fixing frame to finished wall and necessary stainless steel screws etc. Profile of frame & sash shall be mitred cut and fusion welded at allcorners, including drilling of holes for fixing hardware's and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealent over backer rod of required size and ofapproved quality, all complete as per approved drawing & direction of Engineer-in-Charge. (Single / double glass panes,		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			wire mesh and silicon sealent shall be paid separately). **Note: -** For UPVC frame and sash extruded profiles minus 5% tolerance in dimension i.e. in depth & width of profile shall be acceptable. Variation in profile dimension in higher side shall be accepted but no extra payment on this account shall be made.		
	9.199.1	Two track two panels sliding door made of (big series	Two track two panels sliding door made of frame 67 x 50 mm & sash 46 x 82 mm both having wall thickness of 2.3 ± 0.2 mmand single glazing bead / double glazing bead of appropriate dimension. (Area ofdoor above 2.00 sqm upto 10.00 sqm).	sqm	3676.00
	9.199.2	Two track four panels sliding door made of (big series)	Two track four panels sliding door made of frame 67 x 50 mm & sash 46 x 82 mm both having wall thickness of 2.3 ± 0.2 mmand single glazing bead / double glazing bead of appropriate dimension. (Area ofdoor above 8.00 sqm upto10.00 sqm).	sqm	3810.00
	9.199.3	Three track three panels sliding door made of (big series)	Three track three panels sliding door made of frame 116 x 45 mm & sash 46 x 82 mm both having wall thickness of 2.3 ± 0.2 mm and single glazing bead/ double glazing bead of appropriate dimension. (Area of door above 5.00 sqm)	sqm	4361.00
	9.199.4	Three track three panels sliding door with fly proof S.S wire mesh	Three track three panels sliding door with fly proof S.S wire mesh (Two nos. glazed & one no. wire mesh panels) made of frame 116 x 45 mm & sash 46 x 82 mm both having wall thickness of 2.3 ± 0.2 mmand single glazing bead / double glazing bead of appropriate dimension. (Area ofdoor above 2.00 sqm upto 5.00 sqm)	sqm	5617.00
9.200		Door shutter solid panel pre-laminated PVC	Providing and fixing 30mm thick factory made solid panel prelaminated PVC internal door shutter consisting of frame made out of M.S. tubes of 19 gauge thickness and size of 40mm x 20mm for	sqm	3908.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			styles top, lock and bottom rails. M.S. frame shall have a coat of steel primers of approved make and manufacture.		
			M.S. frame shall be covered with 5mm thick prelaminated heat moulded PVC 'C' channel of size 30mmx100mm forming stiles and 5mm thick prelaminated 125mm wide PVC sheets for top rail, lock rail and bottom rail on eigher side of panel and 15mm (5mmx3) thick, 20mm wide cross PVC sheet as gap insert for top and bottom rail. Panelling of 10mm thick prelaminated PVC sheet to be fitted in the M.S. Frame welded/sealed to the stiles and rails with 30mm wideX5mm thick prelaminated PVC sheet beading on eigher side and additional 30mm wide PVC beading as gap insert below top and bottom rail joined together using solvent cement adhesive. An additional 5mm thick Prelaminated PVC strip of 20mm width is to be stuck on the interior side of the 'C' channel using PVC solvent cement adhesive etc. complete as per direction of Engineer-in- charge. Manufacturer's specification and drawing. 30 mm thick pre laminated PVC door shutters.		
9.201		Factory made integral PVC flush door shutter	Providing and fixing factory made integral PVC flush door shutter made from rigid PVC foam sheet (Single Extruded) with density not less than 600 kg/cum, conforming to IS: 2380 part 14 for screw holding strength with Celluka finish surfaces on both sides at the manufacturing process itself, manufactured by an ISO : 9001 -2000 certified company.		
	9.201.1	24 to 25 mm	24 to 25 mm thick plain solid PVC flush door shutter	sqm	2250.00
	9.201.2	28 to 30 mm	28 to 30 mm thick plain solid PVC flush door shutter	sqm	3229.00
9.202		Laminated solid PVC flush door co-extruded	Providing and fixing factory made pre laminated solid PVC flush door shutters 28 to 30 mm thick made from co-extruded		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		three layer rigid PVC foam sheet	three layer rigid PVC foam sheet with density not less than 600 kg/cum, conforming to IS: 2380 part 14, for screw		
			holding strength and outer layers laminated on both sides by laminated sheet with density not less than 1400 kg/cum, lamination shall be at the manufacturing process itself and manufactured by an ISO: 9001-2000 certified company.		
	9.202.1	1mm thick lamination	1mm thick laminated sheet	sqm	3693.00
	9.202.2	0.12 mm thick lamination	0.12 mm thick laminated sheet	sqm	3461.00
9.203		Laminated flush door shutters core of block board construction frame of finger jointed 1st class hard wood	Providing and fixing ISI marked factory made flush door shutters conforming to IS: 2202(Part-I) laminated type, core of block board construction with kiln seasoned and chemically treated 2nd class hard wood battens (strips) and frame of kiln seasoned and chemically treated, finger jointed 1st class hard wood of width not less than 65mm for style, top rail, bottom rail and 125mm for lock rail. Block board shall be provided with well matched 3mm thick commercial 3ply veneering consisting of vertical grains, cross bands and commercial grade face veneers on both outer faces of shutters including kiln seasoned and chemically treated, finger jointed 2nd class teakwood internal lipping battens fixed in jointed 2nd class teakwoodinternal ipping (battens) fixed in factory by using peneumatic gun, of depth not lessthan 25 mm on all edges of shutters. Spacing of finger joints in wood as above shall not be less than 50cm. Both surfaces of shutters to be provided with 1.00 mm thick decorative high pressure laminated sheet on plain/ wood grain in gloss/matt/suede finsh with high density protective surface. Adhesive on revers sideof lamination sheet shall be as per IS: 2046type S of approved (opening) withing frame of door/cupboards and shall bestamped by the manufacturer to display name of manufacturer, batch number of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			product and conformity with IS: 2202. The rate shall be inclusive of all material and labour complete and as per specification and dirction of Engineer-in-Charge.		
	9.203.1	35mm 4 hinges	35mm thick including 4 nos. 100mm long, ISI marked heavy weight (2.5mm thick) Stainless steel butt hinges with necessary screws.	sqm	3339.00
	9.203.2	30mm 3 hinges	30mm thick including 3 nos. 100mm long, ISI marked heavy weight (2.5mm thick) Stainless steel butt hinges with necessary screws.	sqm	3171.00
	9.203.3	25mm 3 hinges (cupboard)	25mm thick (for cupboard) including 3 nos. 75mm long, ISI marked heavy weight (2.5mm thick) nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	3005.00
9.204		Flush door non decorative type, core of block board frame of finger jointed 1st class hard wood	Providing and fixing ISI marked factory made flush door shutters conforming to IS: 2202(Part-I) non decorative type, core of block board construction with kiln seasoned and chemically treated 2nd class hard wood battens (strips) and frame of kiln seasoned and chemically treated, finger jointed 1st class hard wood of width not less than 65mm for style, top rail, bottom rail and 125mm for lock rail. Block board shall be provided with well matched 3mm thick commercial 3ply veneering consisting of vertical grains, cross bands and commercial grade face veneers on both outer faces of shutters including kiln seasoned and chemically treated, finger jointed 2nd class teak wood external lipping (battens) fixed in factory by using peneumatic gun, of depth not less than 25 mm on all edges of shutters including allmaterial and labour complete as perspecification and direction of engineer-in- charge. Spacing of finger joints in wood as above shall not be less than 50cm. Each shutter shall be factory made to the exact opening size within frame of doors/cupboards and shall be stamped by the manufacturer to display name of manufacturer, batch number of product and confirmity with IS:2202.		
	9.204.1	35mm	35mm thick including 4 nos. 100mm long, ISI marked heavy weight (2.5mm thick) Stainless steel butt hinges with necessary.	sqm	1953.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.204.2	30mm	30mm thick including 3 nos. 100mm long, ISI marked heavy weight (2.5mm thick) Stainless steel butt hinges with necessary screws.	sqm	1772.00
	9.204.3	25mm	25mm thick including 3 nos. 100mm long, ISI marked heavy weight (2.5mm thick) Stainless steel butt hinges with necessary screws.	sqm	1532.00
9.205		Deduct for not provi - ding 2nd class teak wood atten lipping	Deduct for not providing 2nd class teak wood batten lipping in factory made shutters i.e. item no 9.203 and 9.204.	sqm	401.00
9.206		Acrylic solid surface	Providing & fixing acrylic solid surface in thickness 4/6/12 mm of approved colour made out of non-porous surfacing material homogeneously composed of ± 1/3 acrylic resin (also known as PolyMethyl Methacrylate -PMMA), and ± 2/3 Aluminium Trihydrate (ATH). The sheet size should be 2.44 mts X 0.76 mts. Thematerial should have been manufactured by using high-tech and computerized controlled processes divided into various units such as mixing of raw materials, casting process on 4th generation steel belt (double belt) production technology with online quality testing process. The material should conform to National Science Foundation (NSF), Society General Surveillance (SGS), Green Guard, Korea Testing & Research Institute (KTRI), Central Institute of plastics Engineering &v FCWS) norms. The installation should be done as per the Standard Operating Procedure (SOP) & finished installation should be supported with conformation card issued by the company Technology (CIPET) teng standard and Fabricatrion Components Warranty System (with installation photographs. Suggested base substrates are water proof plywood of minimum 10-12 mm thickness, finished kota stone, fully levelled plastered wall or any hard Substrate which prevents water & moisture seepage through it. Cost of substrate to be paid seperatly. The solid surface sheet should be applied on the substrate with natural silicon sealant, water clear polyurethane sealant, PVC fixit, or other material depending upon application / usage of solid surface sheet.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Final finish must be obtained as per FCWS process and norms. Thermoforming (banding) process must be as per standard FCWS guidelines and fabrication manual. The joint / groove between sheets should be filled with “Solid Surface Adhesive" with same shade of sheet, manufactured by the same manufacturer. NO extra shall be paid for joint adhesive, silicon or related accessories and for transportation. Item should cover 5 years, commercial warranty of product as per FCWS norms. The product should be UV protected for exterior applications & LGA hyginene tested anti bacterial for health care application areas. Area of application are Table Tops, Exterior facade cladding, health care areas like modular operation theater hospital scrubs, wash basins, partition & paneling, laboratory commercial show room, hotels, office & residential areas.		
	9.206.1	Snow white	Acrylic Solid Surface sheet – Glacier white designer white		
	9.206.1.1	6.0 mm	6 mm thick sheet	sqm	5920.00
	9.206.1.2	12.0 mm	12 mm thick sheet	sqm	7212.00
	9.206.2	Any Solid colours	Acrylic Solid Surface sheet - Any Solid colours/Hospital use Medi Blue		
	9.206.2.1	6.0 mm	6 mm thick sheet	sqm	6512.00
	9.206.2.2	12.0 mm	12 mm thick sheet	sqm	8826.00
	9.206.3	Any Solid colours	Acrylic Solid Surface sheet - Any other		
	9.206.3.1	6.0 mm	6 mm thick sheet	sqm	7550.00
	9.206.3.2	12.0 mm	12 mm thick sheet	sqm	9960.00
9.207		Wall cladding/ façade with 8 mm thick decorative non asbestos	Providing and fixing of 8 mm thick non asbestos multipurpose board reinforcedwith cellulose fibers for wall cladding will consist of following operations: 1. Fixing of a layer of 6mm Moisture/fire Resistance Heavy Duty fibre cement Boards (High Pressure Steam Cured) Type	sqm	1769.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		fiber cement board	A,Category 4 conforming to IS 14862:2000, fixed over steel framework of Aluminium/GI section/MS Section.The Aluminium section /GI section/MS Section of minimum SHS 50mm x 25mm x 2.6mm thick.the Main vertical members are placed @ 610mm c/c and secured to the existing wall, floor and soffit with the help of MS/Aluminium cleat fixed @ 610mm c/c vertically.The main member should be fixed horizontally @ 1220mm c/c. with the help of 25mm self drilling, self tapping Fibre Cement Screws. 2. Fixing of 8mm thick pre-painted, self-embossed, reinforced cement board Type B,Category 3 conforming to IS 14862:2000 , of size 3000 mm x 150mm in overlap (maximum 25 mm) or groove manner (the grooves are to be sealed with any exterior grade sealant) over above mentioned 6 mm heavy duty fibre cement board with the help of 25mm self drilling, self tapping Fibre Cement screws @ 300mm c/c. All external corners are to be protected with the use of suitable corner profile of GI or PVC. All above operation is to be done in order and as directed by Engineer-in-Charge. (frame work to be paid separately).		
9.208		HDF veneer decorative pine wood flush door:	Providing and fixing IS marked factory made veneer High Density Fibre (HDF) flush door shutters conformity to IS: 15380(2003), eco-friendly, termiteresistant, borer free, anti warping phenol formaldehyde (PF), decorative designe moulded or plain construction with kiln seasoned chemically treated 1st class pine wood frame 65 mm style rails, top rail, bottom rail and 125 mm for lock rail, particle board shall be used as a core material between the style and rail 3mm thick HDF ply fixed on face and at back side in facing using PF ply pasting glue, having density between 1000-1200 Kg/m^3. Each shutter shall be factory made to the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			exact opening size within frame of door/capboards and shall be stamped & hologramed Test report by the manufacturer displaying name and batch number of product and conformity with IS:15380(2003) shall be produced. The rate shall be inclusive of all material and labour complete as per specifications and direction of engineer-in-charge. Material supply company' gujcon, century, green shall be registered by Bureau of Indian Standards, Member of Indian GreenBuilding Council, ISO 9001, ISO 14001 (Environment management system), National small industries corporation(NSIC), SME Rating Agency of Indian Ltd.(SMERA), Dun and Bradstreet (D&B).		
	9.208.1	35 mm	35 mm thick including 4 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	3366.00
	9.208.2	30 mm	30 mm thick including 3 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	3268.00
	9.208.3	25 mm	25 mm thick ISI marked heavy weight nickel plated bright finished M.S. piano hinges with screws.	sqm	3056.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.209		HDF non decorative pine wood flush door with primer	Providing and fixing IS marked factory made veneer High Density Fibre (HDF) flush door shutters conformity to IS:15380(2003), eco-friendly, termite resistant borer free, anti warping phenol formaldehyde (PF), non decorative designe moulded or plain construction with kiln seasoned chemically treated 1st class pine wood frame 65 mm style rails, top rail, bottom rail and 125 mm for lock rail, particle board shall be used as a core material between the style and rail 3mm thick HDF ply fixed on face and at back side in facing using PF ply pasting glue, having density between 1000-1200 Kg/m^3. Each shutter shall be factory made to the exact opening size within frame of doorand shall be stamped & hologramed and Test report by the manufacturer displaying name and batch number of product and conformity with IS:15380(2003) shall beproduced. The rate shall be inclusive of all material and labor complete as per specifications and direction of engineer-in- charge. Material supply company shall be registered by Bureau of Indian Standards, Member of Indian Green Building Council,ISO 9001, ISO 14001 (Environment management system), National small industries corporation(NSIC), SME Rating Agency of Indian Ltd.(SMERA), Dun and Bradstreet (D&B).		
	9.209.1	35 mm	35 mm thick including 4 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	2175.00
	9.209.2	30 mm	30 mm thick including 3 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws	sqm	2075.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.209.3	25 mm	25 mm thick ISI marked heavy weight nickel plated bright finished M.S. piano hinges with screws.	sqm	1975.00
9.210		HDF non decorative pine wood panel door with glass/wiremesh and primer	Providing and fixing IS marked factory made High Density Fibre (HDF) flush door shutters conformity to IS:15380(2003), eco-friendly, termite resistant borer free, anti warping phenol formaldehyde (PF), 2 and 4 Panel designe moulded or plain construction with kiln seasoned chemically treated 1st class pine wood frame 100 mm style rails, top rail, bottom rail and 125 mm for lock rail, particle board shall be used as a core material between the style and rail 3mm thick HDF ply fixed on face and at back side in facing using PF ply pasting glue, having density between 1000-1200 Kg/m^3. SS grade 304 wire gouge dia 0.5mm and average width of aperture 1.4mm in both directions for door. Each shutter shall be factory made to the exact opening size within frame of door and shall be stamped & hologramed Test report by the manufacturer displaying name andbatch number of product and conformity with IS:15380(2003) shall be produced. The rate shall be inclusive of all material and labour complete as per specifications and direction of engineer-in-charge. Material supply company shall be registered by Bureau of Indian Standards, Member of Indian Green Building Council, ISO 9001, ISO 14001 (Environment management system), National small industries corporation(NSIC), SME Rating Agency of Indian Ltd.(SMERA), Dun and Bradstreet (D&B).		
	9.210.1	35 mm	35 mm thick including 4 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	3789.00
	9.210.2	30 mm	30 mm thick including 3 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	3643.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.211		HDF decorative pre-laminated pipe wood door glass / wiremash	HDF decorative prelaminated pine wood panel door with glass / wiremash:- Providing and fixing IS marked factory made veneer High Density Fibre (HDF) flush door shutters conformity to IS:15380(2003), eco-friendly, termite resistant borer free, anti warping phenol formaldehyde (PF), decorative designe moulded or plain construction with kiln seasoned chemically treated 1st class pine wood frame 100 mm style rails, top rail, bottom rail and 125 mm for lock rail, particle board shall be used as a core material between the style and rail 3mm thick HDF ply fixed on face and at back side in facing using PF ply pasting glue, having density between 1000-1200 Kg/m^3complete without any coating on finish product. Each shutter shall be factory made to the exact opening size within frame of door and shall be stamped & holograned and Test report by the manufacturer displaying name batch number of product and conformity with IS:15380(2003) shall be produced. The rate shall be inclusive of all material and labour complete as per specifications and direction of engineer-in-charge. Material supply company shall be registered by Bureau of Indian Standards, Member of Indian Green Building Council, ISO 9001, ISO 14001 (Environment management system), National small industries corporation(NSIC), SME Rating Agency of Indian Ltd.(SMERA), Dun and Bradstreet (D&B).		
	9.211.1	35 mm	35 mm thick including 4 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	4483.00
	9.211.2	30 mm	30 mm thick including 3 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	4269.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.212		HDF decorative pre-laminated pipe wood decorative flush door	Providing and fixing IS marked factory made veneer High Density Fibre (HDF) flush door shutters conformity to IS:15380(2003), eco-friendly, termite resistant borer free, anti warping phenol formaldehyde (PF), decorative designe moulded or plain construction with kiln seasoned chemically treated 1st class pine wood frame 65 mm style rails, top rail, bottom rail and 125 mm for lock rail, particle board shall be used as a core material between the style and rail 3mm thick HDF ply fixed on face and at back side in facing using PF ply pasting glue, having density between 1000-1200 Kg/m^3complete without any coating on finish product. Each shutter shall be factory made to the exact opening size within frame of door and shall be stamped & holograned Test report by the manufacturer displaying name and batch number of product and conformity with IS:15380(2003) shall be produced. The rate shall be inclusive of all material and labour complete as per specifications and direction of engineer-in-charge. Material supply company shall be registered by Bureau of Indian Standards, Member of Indian Green Building Council, ISO 9001, ISO 14001 (Environment management system), National small industries corporation(NSIC), SME Rating Agency of Indian Ltd.(SMERA), Dun and Bradstreet (D&B).		
	9.212.1	35 mm	35 mm thick including 4 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	4350.00
	9.212.2	30 mm	30 mm thick including 3 nos 100 mm long ISI marked heavy weight (2.5 mm thick) Stainless Steel butt hinges with necessary screws.	sqm	4150.00
	9.212.3	25 mm	25 mm thick ISI marked heavy weight nickel plated bright finished M.S. piano hinges with screws.	sqm	3950.00
9.213		Non decorative	Non Decorative Pine Wood Flush Door:- Providing and fixing ISI marked factory		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		pine wood flush door	made flush door shutters conforming to IS: 2202 (Part I) non decorative type, core of block board construction with kiln seasoned and chemically treated 1st class pine wood battens (strips) and frame ofkiln seasoned and chemically treated, finger jointed 1st class pine wood of width not less than 65mm for style, top rail, bottom rail and 125mm for lock rail. Blockboard shall be provided with well matched 3mm thick commercial 3 ply veneering consisting of vertical grains, cross bands and commercial grade face veneers on both outer faces of shutters including kiln seasoned and chemically treated, finger jointed all material and labour complete as per specification and direction of engineer-in-charge. Spacing of finger joints in wood as above shall not be less than 50 cm. Each shutter shall be factory made to the exact opening size within frame of doors/ cupboards and shall be stamped by the manufacturer to display name of manufacturer, batch number of product and conformity with IS:2202.		
	9.213.1	35 mm	35 mm thick including 4 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) Stainless Steel butt hinges with necessary screws.	sqm	2775.00
	9.213.2	30 mm	30 mm thick including 3 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) Stainless Steel butt hinges with necessary screws.	sqm	2675.00
	9.213.3	25mm	25mm thick (for cupboard) including 3 nos. 75 mm long, ISI marked heavy weight (2.5mm thick) nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	2575.00
9.214		Laminated pine wood flush door	Providing and fixing ISI marked factory made flush door shutters conforming to IS: 2202 (Part I) laminated type, core of block board construction with kiln seasoned and chemically treated 1st class pine wood battens (strips) and frame of kiln seasoned and chemically treated, finger jointed 1st class pine wood of width not less than 65mm for style, top rail, bottom rail and 125mm for lock rail. Block board shall be provided with well matched 3mm thick		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			commercial 3 ply veneering consisting of vertical grains, cross bands and commercial grade face veneers on both outer faces of shutters including kiln seasoned and chemically treated, finger jointed. Spacing of finger joints in wood as above shall not be less than 50 cm. Both surfaces of shutters to beprovided with 1.00mm thick decorative high pressure laminated sheet on plain/wood grain in gloss/matt/suede finish with high density protective surface layer of make thereby not requiring polish, putty or painting work. Adhesive on reverse side of lamination sheet shall be as per IS: 2046 type S of approved quality. Each shutter shall be factory made to the exact size (opening) within frame of doors/ cupboards and shall be stamped by the manufacturer to display name of manufacturer, batch number of product and conformity with IS:2202. The rate shall be inclusive of all material and labour complete and as per specification and direction of engineer-in-charge.		
	9.214.1	35 mm	35 mm thick including 4 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) Stainless Steel butt hinges with necessary screws.	sqm	2775.00
	9.214.2	30mm	30 mm thick including 3 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) Stainless Steel butt hinges with necessary screws.	sqm	2675.00
	9.214.3	25mm	25mm thick (for cupboard) including 3 nos. 75 mm long, ISI marked heavy weight (2.5mm thick) nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	2575.00
9.215		Laminated pine wood flush door	Providing and fixing ISI marked factory made flush door shutters conforming to IS: 2202 (Part I) laminated type, core of block board construction with kiln seasoned and chemically treated 1st class pine wood battens (strips) and frame of kiln seasoned and chemically treated, finger jointed 1st class pine wood of width not less than 65mm for style, top rail, bottom rail and 125mm for lock rail. Block board shall be provided with well matched 3mm thick commercial 3 ply veneering consisting of vertical grains, cross bands and commercial grade face veneers on both outer faces of shutters including kiln seasoned and chemically treated, finger jointed. Spacing of finger joints in wood as above shall not be less than 50 cm. Both surfaces of shutters to beprovided with 1.00mm thick decorative high pressure laminated sheet on plain/wood grain		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			in gloss/matt/suede finish with high density protective surface layer thereby not requiring polish, putty or painting work. Adhesive on reverse side of lamination sheet shall be as per IS: 2046 type S of approved quality. Each shutter shall be factory made to the exact size (opening) within frame of doors/ cupboards and shall be stamped by the manufacturer to display name of manufacturer, batch numberof product and conformity with IS:2202. The rate shall be inclusive of all material and labour complete and as per specification and direction of engineer-in-charge.		
	9.215.1	35 mm thick	35 mm thick including 4 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) tainless Steel butt hinges with necessary screws.	sqm	3650.00
	9.215.2	30 mm thick	30 mm thick including 3 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) Stainless Steel butt hinges with necessary screws.	sqm	3550.00
	9.215.3	25mm thick	25mm thick (for cupboard) including 3 nos. 75 mm long ISI marked heavy weight (2.5mm thick) nickel plated bright finished M.S. piano hinges with necessary screws.	sqm	3450.00
9.216		Veneer decorative pine wood flash door	Providing and fixing ISI marked factory made flush door shutters conforming to IS: 2202 (Part I) veneer decorative type, core of block board construction with kiln seasoned and chemically treated 1st class pine wood battens (strips) and frame of kiln seasoned and chemically treated, finger jointed 1st class pine wood of width not less than 65mm for style, top rail, bottom rail and 125mm for lock rail. Block board shall be provided with well matched 3mm thick commercial 3 plyveneering consisting of vertical grains, cross bands and commercial grade face veneers on both outer faces of shutters including kiln seasoned and chemically treated, finger jointed all material and labour complete as per specification and direction of engineer-in- charge. Spacing of finger joints in wood asabove shall not be less than 50 cm. Each shutter shall be factory made to the exact opening size within frame of doors/ cupboards and shall be stamped by the manufacturer to display name of manufacturer, batch numberof product and conformity with IS:2202.		
	9.216.1	35 mm thick	35 mm thick including 4 nos. 100 mm long, ISI marked heavy weight (2.5mm thick) Stainless Steel butt hinges with necessary screws.	sqm	3121.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	9.216.2	30 mm thick	30 mm thick including 3 nos. 100 mm long, ISI marked heavy weight (2.5mm thick)Stainless Steel butt hinges with necessaryscrews.	sqm	3021.00
	9.216.3	25mm thick	25mm thick (for cupboard) including 3 nos. 75mm long, ISI marked heavy weight (2.5mm thick) nickel plated bright finished M.S. pianohinges with necessary screws.	sqm	2921.00
9.217		UPVC kitchen cabinetframe	Providing and fixing factory made solid Foam uPVC profile for kitchen cabinet frame (45 x 20 mm) of approved shade, quality and make. The profile shall be laminated on both sides, made from rigid foam sheets (Single extruded) having density 600 Kg/cum and the exposed edges sealed with PVC edge beading of same shade and colour. The frame shall be fire retardent with necessary screw holding capacity. Frame shall be fixed to wall using Expendable Fastne with necesary stainless steel screws, all complete as per direction of Engineer-in-charge.	metre	238.00
9.218		UPVC kitchen cabinet shutter/ partition	Providing and fixing factory made Kitchen Cabinet Shutter/Partition 20 mm nominal thickness of approved shade, quality and make,made from rigid foam sheets (Single extruded) having density 600 Kg/cum and laminated onboth side by laminate Sheet/PVC foil lamination. The exposed edges shall be sealed with PVC edge beading of same shade and colour. The shutter shall be fire retardent having necessary screw holding capacity. Shutter shall be fixed to frame using approved hinges with necessary stainless steel screws, all complete as per direction of Engineer-in-charge.	sqm	2742.00
9.219		Fire resistant door frame (2 hr fire rating)	Providing and fixing fire resistant door frame of section 50 x 60 mm on horizontal side & 35 x 60 mm on vertical sides having built in rebate made out of 1.6 mm thick GI sheet (Zink coating not less than 120 gm/m2) suitable for mounting 120 min fire Rated Glazed Door Shutters. The frame shall be filled with Mineral wool insulation provision of GI. Anchor fasters 14 nos (5 each on vertical style & 4 on horizontal style of size M10 x 80) suitable for fixing in the opening along with factory made Template for ss Ball Bearing Hinges of size 100x89x3mm for fixing of fire rated glazer	Meter	1260.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			shutter. The frame shall be finished with a approved fire resistant primer or Powder coating of not less than 30 micron in desired shade as per the direction of Engineer-in-charge.(Coast of ss ball beating hinges is excluded).		
9.220		Fly proof wire gauze to UPVC/ Aluminum windows and doors.	Providing and fixing fly proof wire gauze to windows, clerestory windows and doors with u clip, EPDM rubber / neoprene gasket complete.		
	9.220.1	S.S.	Stainless steel (grade 304) wire gauze of 0.5 mm dia wire and 1.4 mm aperture on both sides.	sqm	1040.00
9.221		Fire resistant door frame (2 hr fire rating)	Providing and fixing non load bearing fixed frame for fire resistant glazed Partition for 120 minutes Fire Rating, made out to a profile of dimension 60mm x 70 mm of 1.6 mm thick galvanised steel sheet as per test evidence suitable for fixing fire rated glass for 120 minutes of both integrity & radiation control (EW120) & minimum 20 minutes of insulation (EI20).The profile has to be fixed to the supporting construction by means of anchor fasteners of size M10 x 80, every 150 mm from the edges and every 500 mm (approx) c/c. Linear measurement of frame shall be measured for payment. The frame shall be filled with mineral wool insulation of density min 96kg/ m^3. and finished with a approved fire resistant primer or Powder coating of not less than 30 micron in desired shade as per NBC 2016, IS 16231 (Part 3):2016 and directions of Engineer - in-charge.	meter	1260.00
9.222		Fire resistant door glazing (2 hr fire rating)	Providing and fixing glazing in fire resistant door shutters, fixed panels & partitions etc., with G.I. beading made out of 1.6 mm thick G.I. sheet (zinc coating not less than 120 gm/m^2) of size 20 x 33 mm screwed with M4 x 38 mm SS screws at distance 75 mm from the edges and 150 mm c/c , including applying a coat of approved fire resistant primer/powder coating of not less than 30 micron on G.I. beading, & special ceramic tape of 5 x 20 mm size etc complete in all respect as per NBC 2016, IS 16231 (Part 3):2016 and as per direction of Engineer-in-	sqm	28929.0

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			charge with glass of required thickness having 120 minutes of fire resistance both integrity & radiation control (EW120) and minimum 20 minutes of insulation (EI20). The manufacturer have to give test report/certification of fire glass and the glass should have the stamp showing the value of E, EW & EI. The glass shall be tested in approved NABL accredited lab or by any other accreditation body which operates in accordance with ISO/IEC 17011 and accredits labs as per ISO/IEC 17025 for testing and calibration scopes shall be eligible. The maximum glazing size shall not be more than 1100x2200 mm (w x h) or 2.42 sqm		
9.223		GI Metal Fire Door (Uninsulated)	Providing and fixing of ISI marked Un-insulated fire rated GI Sheet doors conforming to FD 120 classification of IS 3614 :2021 with fire rating of 120 minutes integrity. The factory manufactured door frame and shutter assembly shall be made out of minimum 1.20 mm thick GI Sheet with minimum 120 GSM coating as per IS: 277. Door and frame should be finished with powder coating in standard RAL shade finish having DFT of minimum 30 microns.The door should carry a minimum 500 Hrs of salt spray test. All accessories like hardware, vision panel, fire & smoke seal etc to be supplied by Fire door manufacturer only as per the guidelines laid down in IS 3614:2021. The Door frame shall be single rebate grooved profile of size minimum 125mm x 55mm made out of 1.50mm (18gauge) minimum thick galvanized steel sheet. Frames shall be butt / mitred joint and field assembled with bolting system. Frames to have inbuilt grooved to accommodate Intumescent fire seal strip of size 10x4mm. All provisions should be mortised & provided with necessary inbuilt reinforcement to receive appropriate hardware. Frames should be provided with back plate bracket for anchor fastener with 8-10 mm dia and minimum 80 mm long HDG fasteners, with Polyamide sleeves (Payable in extra item) for installation of frame in finished plastered masonry wall opening.	sqm	12719.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Door leaf should be minimum 60mm thick fully flush double skin door. Door leaf must be manufactured from 1.2mm (18gauge) thick galvanised steel sheet. The infill material shall be proprietary insulation material. All doors shall be factory prepared for receiving appropriate hardware and provided with necessary reinforcement for hinges, locks, and door closers. The edges should be interlocked with a bending radius of 1.4mm. For double leaf doors integrated astragals has to be provided on the meeting stile for both active and inactive leaf. The shutters shall be mounted to the frame with SS 304 grade ball bearing type hinges of size 100x89x3 mm along with all suitable hardware). Vision lite size 185 *285 or 300x400 mm (max area 0.12sqm) wherever applicable should be provided with single layered clear fire rated glass panes as per guidelines laid down under IS 3614 2021 with a clipon arrangement. The glass will be provided with fire rated ceramic tape of min. 2 mm thickness with min. 15mm width all-around on either face.		
9.224		Minimum 50mm thick door shutter with both side 4 mm thick burma teak veneer (Wooden Fire Door)	Providing and fixing of Fully Insulated Non Metallic Wooden doors as per IS 3614:2021, for 120 min integrity & 120 min insulation. The door is made out of thoroughly seasoned hard wood. The moisture content of any material used in the construction of wooden composite fire door sets comprises of both frame and shutter shall not be more than 15% approx. in accordance with IS 287. The fire door must have earlier test evidence as per IS/ISO 3008-1. It must have successfully tested from CBRI Roorkee/ NABL accrediated lab. Door and frame shall be finished with fire retardent primer FR880. All accressesories like hardware, fire seal etc to be supplied by Fire door manufacturer only as per the guidelines laid down in IS 3614:2021. Every door must carry a label confirming to type of door & its fire rating. Door frame shall be Single rebate profile of section 120 x 60 mm made out of Hard wood - Red meranti / Ivory Coast (2nd class) or equivalent having density of 550-600 kg/m3 approx. with factory made groove of	sqm	16886.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			suitable size so as to accomodate intumuscent Fire seal of size 20x4mm . An EPDM seal also to be provided in the frame to check spread of smoke. The frame must be primed with Fire retardant primer FR880 . The frame to be fixed to fire exit opening with minimum 10 nos. 10 mm dia and minimum 120 mm long HDG (hot dip galavanized) headed screws with polyamide sleeves. Wooden shutter must be of minimum 50 mm of thickness finished with commercial ply on both faces. If required, the Commercial ply may be replaced with teak ply or may be pasted with 1mm thick Laminate on both faces in desired regular SF grade . The composite thickness of the core insulation will be 37mm consists of minimum 10mm thick Calcium Silicate boards and treated wool Insulation material . The calcium silicate board must have Density not more than 90kg/m3 , having thermal conductivity 0.15W/mK . The shutter should be provided with heat activated intumescent fire seal strip of size 20mm x 4 mm mounted in the suitable grooves all three sides except bottom provided on Exposed OR concealed wooden lipping .The shutters shall be mounted to the frame with SS 304 grade ball bearing type hinges along with all suitable hardware. Vision lite size 200x300 mm or 150 x 400mm(max area 0.06sqm) wherever applicable should be provided with single layered clear fire rated glass panes as per guidelines laid down under IS 3614 : 2021. Every Glass pane must carry a permanent marking of its glass manufacturer.		
9.225		Cement bonded particle board in wall panel	Supply and installation of termite proof, moisture/fire resistant 8mm thick cement bonded particle board confirming to IS: 14276: 1995 in wall panel on structure to form wall/partition wall on external face of frame work made of special sections, power pressed from M.S. sheets and galvanized with zinc coating of 125 gms/sqm (both side inclusive) as per IS : 277 and consisting of angle cleats of size 25 mm wide x 1.6 mm thick with flanges of 27 mm and 37mm, at 1200 mm centre to	sqm	1429.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			centre, one flange fixed to the ceiling with dash fastener 12.5 mm dia x 50mm long with 6mm dia bolts,other flange of cleat fixed to the angle hangers of 25x10x0.50 mm of required length with nuts & bolts of required size and other end of angle hanger fixed with intermediate G.I. channels 45x15x0.9 mm running at the spacing of 1200 mm centre to centre, to which the ceiling section 0.5 mm thick bottom wedge of 80 mm with tapered flanges of 26 mm each having lips of 10.5 mm, at 450 mm centre to centre fixed the self-drilling/tapping screws/fasteners of approved make. as per the directions of Engineer-in-Charge.		
9.226		Acoustic Sound Proof Door	Providing and fixing 2 Hrs fire rated acoustic wooden Door of approved make, the door and whole system and criteria shall be comply with IS: 3614 – 2021, also the door shall be acoustic suitable to achieve minimum 35db sound reducing properties, the system shall comprising of door frame, shutter with hardware and other accessories & ancillary material as per approved manufacturer specification, as given in description/technical specification and as directed Engineer-In-Charge. Door Frame: The grooved Frames for Fire resistant door shall be Hardwood (Red merandi), densified and pressure treated with fire retardant chemicals in vacuum impregnation vessel as per IS: 401 and kiln seasoned to moisture level, the moisture content of any material used in the construction of wooden fire rated door, shall not be more than 15 percent, in accordance with IS 287. Wood dry density shall be above 450kgs/cum, The size of frame shall be min. 140 X 90 mm or as per approved manufacturer, with approved intumescent strip of size 20mmx4mm or as approved manufacturer to be concealed in the groove of the Frame for fire and smoke sealing, etc. Frame rebate shall be as per manufacturer standard, the frame to be fixed with anchor fastener / dash fasteners or as per approved manufacturer and as approved by EIC. Complete all as per direction of Engineer in Charge.Shutter: The thickness of shutter shall be 77-85mm (ASBESTOS	sqm	26456.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			FREE) with commercial ply on both faces. Acoustic Door Shutter shall be complying with fire performance - FD 120 as per IS: 3614-2021. The frame work to make shutter shall be hardwood made (Red merandi) , densified and pressure treated with fire retardant chemicals in vacuum impregnation vessel as per IS:401 and seasoned to moisture level, the moisture content of any material used in the construction of wooden fire rated door, shall not be more than 15 percent, in accordance with IS 287. Wood dry density shall be above 450kgs/cum, The shutter system shall be comprising of 2 x 12 mm thick noncombustible Calcium Silicate boards, approved in filled material with appropriate density as per IS 3614, other approved infill material as acoustic insulation filler, interlayer Rubber sheet, Fire retardant or approved plywood, internally lipped with hardwood beading and all around shutter etc. shall be as approved by EIC and as per approved manufacturer. The frame work, lipping shall be fire retardant primer/paint finish. The Intumescent seals shall 20 mm x 4 mm, the seal affixed on periphery for fire and smoke sealing in the slit on all edges of shutter except bottom. Rebate shall be as per manufacturer standard, The shall be finished with 1 mm thick approved laminate on both side, All complete as per approved manufacturer standard and as directed by EIC.		
9.227		Non-Woven Polyester Fibre Acoustic wall Panels	Supplying & Installation of Non-Woven 100% Polyester Fibre Acoustic wall Panels made from polyester short fibre. It is prepared by needle punching on multi folded polyester fabric sheets is lightweight and flexible along with excellent tension/compression stiffness. The panel can be easily cut into any desired shape pattern and V-Grove pattern (arranged by carding process). The final material consists of a three-dimensional network structure with breathable open cells. fixed with Rubber adhesive. Physical Properties 1. Eco-friendly 2. 100% recyclable	sqm	3364.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			3. Moisture proof,Humidity Resistance 4. Flammability tested as per IS 11871 A 5. Thickness 10 mm 6. NRC: High Performance 0.60 to 0.90 7. Density 180-200kg/m3		
9.228		PVC wall panelling system	Providing & Fixing PVC ceiling system profile panel with click system interlock panel size 4000mmx340mmx1mm resistance to weather perfect finish in wood grain & solid colour maintenance free Resistance to weather truly anti bacterial. with edge profile J-trim on periphery & center profile T-trim in panel joint Complete System to be fixed on G.I. framework of ceiling section at 450 mm distance with enter section suspended by	sqm	3178.00
			roof by G.I. angle hanger in proper line & level etc complete PRODUCT SPECIFICATIONS: Meets or exceeds the standard specification of (ASTM D-4477) and its referenced documents. Impact resistance: No deformation observed. (Tested at 50 LBS) Tensile strength, MPa: >37.0 Tensile modulus, MPa: >2000 Surface distortion: None at 120F Squareness: <1/8" of square Length: Within ¼" of specification Lock control: Complies Color Change: <2 grayscales (16 hrs under 1200W/m2 UV radiation at 50 degrees)		
9.229		Stretch Fabriq Acoustical Wall Paneling	Providing & Fixing Fabric Acoustic Wall Paneling System of size 1200mm.maximum in width & Length as required as per site Condition with 25mm thick stretch half / full wrap rigid vinyl tracks to be fix on existing surface with PVC insert and screws as required pattern in panel as Per required module. 25mm thick acoustic rockwool pad of 100kg density Fire Class A-1 as per EN 13501, NRC <0.70 to be fill in panel fixed by rubber adhesive. Approved fabric to be grip in stretch track by tucking tools to make panel in proper line and level with wrinkle free finish. The stretch track create V groove / Butt joint in each panel. Complete as per satisfaction by Engineer in charge.	Sqm	5505.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.230		Laminate Wooden Louvers Acoustical wall paneling system	Providing & Fixing Laminated wood finish 19 mm thick Acoustical Louvers wall panel system made from 17mm thick & 2440mm Length. MDF Perforated grooved Panel with 2-4mm thick grooved & 10-13mm thick Pitch. with 2mm thick backing of Polyesterfibre FR grade Acoustical Felt Panel to be fixed with Tongue and groove arrangement fixed on aluminum channel at 750mm distance perpendicular to panel with G.I. clips. Aluminum channel to be fix on G.I. Section of size 45mmx12mmx0.90 mm thick intermediate channel. 50 mm thick & 48 kg density Rockwool Fire class A-1 as per EN13501 NRC 0.88 to be fill in air gap the total acoustical paneling system finished in proper line & level. as per satisfaction by Engineer in charge. mm thick intermediate channel. 50 mm thick & 48 kg density Rockwool Fire class A-1 as per EN13501 NRC 0.88 to be fill in air gap the total acoustical paneling system finished in proper line & level. as per satisfaction by Engineer in charge.	Sqm	5773.00
9.231		ACOUSTICAL WOODWOOL WALL PANELLING SYSTEM	ACOUSTICAL WOODWOOL WALL PANELLING: Providing & fixing of 100% Natural Material- Magnesite (Hygroscopic & biologically sound binding agent) bonded woodwool Heradesign planks of 600mm X 1200mm X 25mm thickness made from wood wool fiber of width 1mm fiber thickness mixed with natural water & magnesite (No-Cement) with open & smooth surface texture, weight of panel to be 11.3 Kg/m2. A completely harmless environment& human friendly & also a disposable or recyclable Material & can last for more than 80 Years in building interiors. As it has long lifespan & suitable for Indoor conditions with relative air humidity of up to 90%. Magnesite protects the wood cells against ageing & fungal attack. To achieveup to NRC=1.00 it should be a clear gap of 200mm from wall slab along with the backing of mineral wool of 50mm thick with 64 kg/m3 density. Ecologically Sound Material with high mechanical strength with crash test of 90Km/H with Ball Impact Test DIN18032 & EN13964 with flying colors & a sustainable material (PEFC & FSC	Sqm	7000.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Certified) along with EPD Certified Acc. to EN ISO 14025 (AUB-Environmental Product declaration). Manufacturing tolerance for nominal size: Length, Width & Thickness ± 1 mm, & In a Standard Climate of 23° C and 50% rel. humidity. FIXING & SUBSTRUCTURE: Impact Resistant According to DIN-18023 or EN13964 to be fixed on CD Sections Profile framework with screw 9Pc/Panel & Centre Distance Support of 300Mm & Centre distance of Cross Profile would be at 600Mm & Main Profile at 900Mm CD Profile is made out of GI(Sections Profile 60/27/0.6mm-0.6mmThick x 60mm Width x 0.27mm height). The Main Profile is in the length of 3000mm & to form a section of 600mm x 1200mm. Heradesign superfine panels to be fixed on the wall with a minimum of 200mm void or gap with Galvanized screws of length 50Mm &Dia 4.5Mm with a backing of Mineral Wool of. density 64kg/m3 density (50mm thick).Colour Of Panels : NATURAL BEIGE, Can be Painted to the desired colour (Any- RAL, NCS or StoColor Shades) using water based color by Spray method prior to installation. FIRE CLASS as per EN 13501-1:B –s1, d0		
9.232		CUSTOM MADE ACOUSTICAL WALL PANEL IN FABRIC FINISH	CUSTOM MADE ACOUSTICAL WALL PANEL IN FABRIC FINISH - Providing and Fixing of Acoustical Wall Panel with square edges made of fiber glass substrate 25mm thick, Acoustical Wall Panel shall be of custom made sizes and per design (subject to manufacturing Limitations, need to check from manufacturer first) wrapped on the front side with an acoustically transparent fabric as per the approved shade (Husk, Copper, Sangria, Sesame, Coffee, Charcoal, Titanium, Flame, Peanut and Shell). The panel shall be classified for Fire Class A, as per ASTM E-84. The Acoustical Wall Panel shall be of sizes 600x600mm (Nominal) or 600x1200mm (Nominal) and shall achieve an NRC of 0.85 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) with 50mm airgap as per ASTM C423 to be fixed to wall on a framework of wooden battens and using Wall panel	Sqm	11124.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Impales (supplied by the manufacturer) or by using construction adhesives (as per the instructions laid down by the manufacturer). The fiber glass substrate shall have a recycle content of 85%. INSTALLATION: An appropriate vapour barrier (by others) shall be applied/installed on the vertical surface before the installation to properly seal against entry of any moisture. Vapour barrier (by others) shall be installed as per Manufacturer's recommendation. Install wooden batten grid (provided by others) of section 50x50mm or as approved by the Architect on the solid wall/partition to provide level and plumb surface. The Impalers shall be fixed to the battens using self-tapping screws. Silica based construction adhesive (by others) to be dabbed on to the projecting elements (spikes) of the Impalers. Wall Panels shall be pierced through the spikes of the Impalers ensuring the line and level of the panels are maintained.		
9.233		MODULAR ACOUSTICAL WOODEN LINEAR WALL PANELLING SYSTEM	PRIVIDING AND FIXING MODULAR ACOUSTICAL WOODEN WALL PANELLING SYSTEM: Of nominal width 100 / 150mm, having 80mm wide panel with 20mm reveal having a thickness of 15mm thickness and length 2400 mm. The panel shall be made of a Moisture resistant fiber board substrate having a minimum density of 595 Kg/m3 with a laminated facing as per the approved shade (Portland Maple, Sea Beech, Siam Wood & Mystique Walnut) and a melamine balancing layer on the reverse side. The panel shall have 0.7mm PVC edge banding along the length side with a similar finish as that of the face. The back of the panel shall have a black acoustic felt to cover the reveal of 20mm to attain an NRC of 0.4 for 150mm. / 0.5 for 100mm. module (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423. The panels shall provide a Humidity Resistance (RH) of 70% with camber limit less than 1mm/1000mm and bow less than 1mm/300mm. The back of the panels shall be profile-grooved to receive 3600mm special main beam carrier with	Sqm	11299.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			integrated factory fitted clips for installation. SUSPENSION: The panels shall be upward mounted on 24 mm wide T - section flanges having rotary stitching along the length of the beam with a web height of 43mm. The Main Beam shall have integrated (factory fitted) locking clips installed at 100 / 150mm center to center distance. The deflection loading of the main beam tested as per ASTM C635 (Deflection limit less than L/360) shall be at least 21.4kg/m. The end details of the main beam shall be made of pre hardened steel and fixed to the ends with 3-point fixing. All main beam to main beam connection shall have a pull-out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 100 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with 0T bending capability. Installation should be carried out as per manufacturer recommended procedure. The grid shall be installed with suspension system of Manufacturer make.SUSPENSION SYSTEM accessories manufactured and supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre straightened Hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. Additionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners. WALL INSTALLATION: An appropriate vapour barrier (by others) shall be applied/installed on the vertical surface before the installation to properly seal against entry of any moisture. Vapour barrier (by others) shall be installed as per Manufacturer's recommendation. Install wooden battens (provided by others) of section 50x50mm or as approved by the Architect on the solid wall horizontally using screws and plugs at spacing of 600mm center to center in plumb line & level. Install the main beam with factory fitted locking clips to the battens		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			using rigid mounting clip. Main beam to be fixed to the batten with 1st rigid mounting clip not more than 300mm from the perimeter and distance between the clips to be less than 900mm c/c. Each main beam is to be place at 600mm center to center with the 1st main beam at not more than 100mm from the perimeter. Install the panels one-by-one by inserting the profile groove at the back of the panel in the clips fitted on the main beam. Use wood splice accessory provided by Manufacturer to join planks. Continue installing rows of planks till the actual length/width is achieved. Finish theedges using the edge banding tape to cover the cut edge of the panel provided by Manufacturer. FACETED/CURVED INSTALLATION: Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls at 450 mm maximum center to center. Mount the wall angle around perimeter at desired height. Install the main beam with factory fitted locking clip at a level 16mm above the wall angle level. Main beam to be supported from the structure with 1st suspension not more than 300mm from the wall and distance between suspensions to be less than 900mm center to center. Each main beam is to be place at 600mm center to center with the 1st main beam at not more than 100mm from the wall An appropriate vapour barrier (by others) shall be applied/installed on the vertical surface before the installation to properly seal against entry of any moisture.Vapour barrier (by others) shall be installed as per Manufacturer's recommendation. Install wooden battens (provided by others) of section 50x50mm or as approved by the Architect on the solid wall horizontally using screws and plugs at spacing of 600mm center-to-center in plumb line & level. To bend the carrier directly, a curve, as specified by the Architect, should be drawn on an appropriate background material like plywood. Carrier should be partially cut and fixed with faceted joining clip provided by Manufacturer to maintain the curve. Ceiling side of the carrier should be suspended by hanger wire in		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			continuation with the ceiling carriers. These carriers should be installed vertically on the wall spaced at 600mm center to center. Install the panels one-by-one by inserting the profile-groove at the back of the panel in the clips fitted on the main beam. Use wood splice accessory provided by Manufacturer to join planks. Continue installing rows of planks till the actual length/width is achieved. Finish the edges using the Edge banding tape to cover the cut edge of the panel provided by Manufacturer.		
9.234		NATURAL BAMBOO WALL PANELLING	Providing & fixing of Wooden Grill panel with backer made of rapidly renewable natural bamboo core having density 600+Kgs/m3 as per BS EN323 (1993) and bending strength of 96+ N/mm2 as per BS EN310 (1993). The Grill panel shall have a seven-layer UV coating of thickness 0.09-0.11mm as per BS EN71 finished to natural or carbonized shades. The Grill panel shall be of nominal size 35x300x2400mm comprising 8 blades/slats of 35x16x2376mm with center to center spacing of 37.5mm fixed through Backer of size 12x22x300mm having center to center spacing of 300mm and additional end backers at 150mm from both the edges. Ledger of 6.5x24x2400mm shall be used for covering face of wall angle. Open edges shall be covered using Edge Caps of 44x19x2400mm and Blade Junction of 66x38x2400mm shall be used along the connection of two Grill panels. The Grill panel shall have a fire performance of Class A according to ASTM-E84. The Grill panel shall have moisture content of 8-14% as per IS 1708 (Part 1):1986 and a Humidity Resistance (RH) of 70%. INSTALLATION: The Grill panel orientation can be horizontal or vertical as approved by the Architect. Install wooden batten grid (provided by others) of section 40/50x75mm (75mm face parallel to wall) or as approved by the Architect on the solid wall/partition (to provide level and plumb surface) perpendicular to the backer direction using screws and plugs at spacing of 300mm center-to-center. The wooden batten should be in line level. The seam of two Grill panels should	Sqm	19001.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			meet at the center of wooden batten. Carefully install the Grill panel between the center lines of the wooden battens using two screws per Backer. Continue installing rows/columns of panels one after the other by screwing them to the batten. Attach supporter (provided by others) for extra support as per requirement. Finish off the exposed field cut edges by sanding and staining (by others) or by paint (by others) to match the panel finish. Installation shall be carried out as per Manufacturer recommended procedure.		
9.235		wall paneling 18mm thick with 4 mm veneer	Providing and fixing 18 mm thick wall paneling with commercial ply having one side veneer 4mm thick (ISI marked IS : 14842), including PU Polish including cost of required Cut-Outs, screw, priming coat, Scaffolding for all heights. complete as per direction of Engineer-in-Charge. This item shall be executed only after the prior approval of Chief engineer.	Sqm	7222.00
9.236		wall paneling 24 mm thick (18mm + 6 mm) thick	Providing and fixing 24 mm (18mm + 6 mm) thick commercial ply (ISI marked IS : 14842) wall paneling, having 6mm grove, over 6mm board @ 100 mm Distance on wall paneling, finished with Deco Polish as per Architectural drawing complete with necessary screws, complete as per site requirement and direction of engineer-in-charge. This item shall be executed only after the prior approval of Chief engineer.	Sqm	5785.00
9.237		Door paneling 18mm thick with one side 4mm thick veneer and PU Polish	Providing and fixing Door paneling 18mm thick commercial ply including 12mm commercial ply for Rebate with one side veneer 4mm thick (ISI marked IS : 14842), with PU Polish complete with necessary screws as per Architectural drawing complete with necessary screws, complete as per site requirement and direction of engineer-in- charge. This item shall be executed only after the prior approval of Chief engineer.	Sqm	7881.00
9.238		wall paneling 18mm thick commercial ply with Deco Polish	Providing and fixing wall paneling 18mm thick commercial ply with Deco Polish including cost of required Cut-Outs, screw, priming coat, Scaffolding for all heights. complete as per direction of Engineer-in-Charge.This item shall be executed only after the prior approval of Chief engineer.	Sqm	5461.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.239		50mm X 25 mm rafter with deco polish	Providing and fixing 50mm X 25 mm rafter with deco polish	Rmts	709.00
9.240		75mm X 35mm rafter with 4mm thick veneer and P.U. Polish.	Providing and fixing 75mm X 35mm mdf rafter with 4mm thick veneer and P.U. Polish.	Rmts	914.00
9.241		50mm X 25 mm rafter with 4mm thick veneer and PU polish.	Providing and fixing 50mm X 25 mm rafter with 4mm thick veneer and PU polish.	Rmts	837.00
9.242		65 mm charcoal Moulding	Providing and fixing 65 mm Special design charcoal Moulding. This item shall be executed only after the prior approval of Chief engineer.	Rmts	872.00
9.243		38 mm charcoal Moulding	Providing and fixing 38 mm Special design charcoal Moulding. This item shall be executed only after the prior approval of Chief engineer.	Rmts	658.00
9.244		22 mm charcoal Moulding.	Providing and fixing 22 mm Special design charcoal Moulding. This item shall be executed only after the prior approval of Chief engineer.	Rmts	496.00
9.245		Antique Brass tower bolt	Providing and fixing ISI marked Premium quality Antique Brass tower bolt (Barrel type) with necessary screws etc. complete. All complete as per direction by engineer in charge. This item shall be executed only after the prior approval of Chief engineer.		
		450 mm long dia 10mm	450 mm long dia 10mm	each	939.00
9.246		Premium quality Antique Brass sliding door bolts	Providing and fixing Premium quality Antique Brass sliding door bolts, ISI marked to required colour or shade, with nuts and screws etc. complete All complete as per direction by engineer in charge. This item shall be executed only after the prior approval of Chief engineer.		
		400 mm long, dia 16mm	400 mm long dia 16mm	each	2882.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
9.247		Premium quality Antique Brass handles	Providing and fixing Premium quality Antique Brass handles with necessary screws etc. All complete as per direction by engineer in charge. This item shall be executed only after the prior approval of Chief engineer.		
		560 mm long dia 25mm handle	560 mm long dia 25mm handle	each	2698.00
9.248		Antique Brass hydraulic door closer	Providing and fixing Premium quality Antique Brass hydraulic door closer (having brand logo with ISI, IS : 3564, embossed on the body, door weight 150 K.G and door width up to 700 mm), with necessary accessories and screws etc. All complete as per direction by engineer in charge. This item shall be executed only after the prior approval of Chief engineer.	each	3046.00
9.249		Antique Brass 125 mm mortise latchwith one tri bolt with three key pier and a pair of lever handles	Providing and fixing Premium quality Antique Brass 125 mm mortise latch with one tri bolt with three key pier and a pair of lever handles of approved quality with necessary screws etc. complete. This item shall be executed only after the prior approval of Chief engineer.	each	4325.00
9.250		Antique Brass 125 mm mortise latch with one dead tri bolt and a pair of lever handles	Providing and fixing Premium quality Satin finished 125 mm mortise latch with one dead tri bolt and a pair of lever handles of approved quality with necessary screws etc. complete. This item shall be executed only after the prior approval of Chief engineer.	each	3758.00
9.251		Gi metal door with sublimantio n	GI metal door having wooden finish through sublimation printing with powder coating-providing and fixing of metal door with outer frame. Frame Size 99x59 mm suitable section for door made out from pressed formed GPSP sheet of thickness 1.2 Confirming to IS277. With honey necessary cut-outs of locks and handle etc. 42mm thick Sandwich panel door made from 0.8 mm thick GPSP Sheet duly filed with honeycomb paper core 40/12 for sound	SQM	10530.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			absorbing and provide more strength to door shutters and fixing with necessary stainless steel ball bearing hinges. The complete shutter & Frame (inside & outside) shall be wooden finish through sublimation printing with powder coating outer frame will be grouted on wall with Anchor fastener of suitable size and quantity as per direction of engineer-in-charge.		
9.252		GI METAL DOOR THICKNESS 1.2 mm (POWDER COATED)	GI metal door having Powder -providing and fixing of metal door with outer frame. Frame Size 99x59 mm suitable section for door made out from pressed formed GPSP sheet of thickness 1.2mm Confirming to IS277. With necessary cut-outs of locks and handle etc. 42mm thick Sandwich panel door made from 0.8 mm thick GPSP Sheet duly filed with honeycomb paper core 40/12 for sound absorbing and provide more strength to door shutters and fixing with necessary stainless steel ball bearing hinges. The complete shutter & Frame (inside & outside) shall be powder coating of thickness 50-60 micron. outer frame will be grouted on wall with Anchor fastener of suitable size and quantity as per direction of engineer-in-charge.	SQM	8953.00
9.253		GI METAL DOOR THICKNESS 1.6 mm (POWDER COATED)	GI metal door having Powder -providing and fixing of metal door with outer frame. Frame Size 99x59 mm suitable section for door made out from pressed formed GPSP sheet of thickness 1.6mm Confirming to IS277. With necessary cut-outs of locks and handle etc. 42mm thick Sandwich panel door made from 1.2 mm thick GPSP Sheet duly filed with honeycomb paper core 40/12 for sound absorbing and provide more strength to door shutters and fixing with necessary stainless steel bal bearing hinges. The complete shutter & Frame (inside & outside) shall be powder coating of thickness 50-60 micron. outer frame will be grouted on wall with Anchor fastener of suitable size and quantity as per direction of engineer-in-charge.	SQM	9950.00
9.254		GI WINDOW-(POWDER COATED)	Providing and fixing of factory make double rebate window made of roll formed galvanized powder coated steel with	SQM	10150.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			sublimation to 60-80 microns thick, colour/sublimated (base steel as per IS513 of "D" quality, galvanized as per IS 277 with zinc of 120 grams/sqm) outer frame section 100x45x1.2 mm, centre mullion section 100x90x1.2mm, shutter section 50x25x0.8mm, and the fixed glass beading section 12x12x1 mm, outer frame and mullions shall consists rebate for glazed shutter, 10 mm provision for guard bars/grills, the sections cut to length miterjoined with corner bracket, centre mullions fixed with mullion cap, stay, handle, latch, 2 Nos of heavy duty stainless steel pivot hinges per shutter and panelled without glazing (glazing shall be fitted using EPDM(Ethyl Propylene diene monomer) gaskets including fixing windows. Section shape rectangle G-section Width 100mm G-Section sheet thickness 1.2mm MS square Double doors Door Specification double doors Door material Galvanized steel IS 277 Hinges Two-part pivot hinges.		
9.255			Providing and fixing commercial ply conforming to IS: 1328 BWR type of approved brand for frame work of boxes, shelves, racks, Amirah, cupboard and drawer etc. including necessary nails, screws etc. Complete as per direction of Engineer-in-charge.		
	9.255.1		9 mm thick BWP grade commercial plywood	Sqm	841.00
	9.255.2		12 mm thick BWP grade commercial plywood	Sqm	1064.00
	9.255.3		18 mm thick BWP grade commercial plywood	Sqm	1169.00
	9.255.4		25 mm thick BWP grade commercial plywood	Sqm	1366.00
9.256			Providing and fixing Stainless steel tubular H handle bar (Grade 304) with SS screws etc .complete set as per direction of Engineer-in-Charge.		
	9.256.1		25 mm outer dia & 300 mm long	Each Set	984.00
	9.256.2		32 mm outer dia & 450 mm long	Each Set	1716.00
	9.256.3		32 mm outer dia & 600 mm long	Each Set	2041.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	**2**	**3**	**4**	**5**	**6**
9.257		Extra for UPVC colour window	Add Extra for Providing and Fixing, UPVC factory made, coloured / other than white colour casement/sliding window and door (for item no. 9.187, 9.194, 9.195, 9.196, 9.197, 9.198, 9.199)		30% of Respective item
9.258		Fly proof wire gauze to UPVC/ Aluminum windows and doors.	Providing and fixing fly proof wire gauze to windows, clerestory windows and doors with u clip, EPDM rubber / neoprene gasket complete.		
	9.258.1	S.S.	Stainless steel (grade 304) wire gauze of 0.5 mm dia wire and 1.4 mm aperture on both sides.	sqm	1040.00

CHAPTER – 10.0

STEEL WORK

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	IS 63	Whiting for paints and putty
2	IS 198	Varnish gold size
3	IS 228	Structural steel (Standard quality)
4	IS 277	Specification for galvanized steel sheets (Plain and corrugated)
5	IS 800	Code of practice for use of structural steel in general in steel construction
6	IS 806	Code of practice for use of steel Tubes in general building construction
7.	IS 808	Dimensions for Hot rolled steel beams, columns, channel and angle sections
8.	IS 812	Glossary of terms relating to welding and cutting metals
9.	IS 813	Scheme of symbols for welding
10	IS 814	Covered electrodes for manual metal arc welding of carbon and carbon manganese steel
11	IS 816	Code of practice for use of metal arc welding for general construction in mild steel
12	IS 817	Code of practice for training and testing of metal arc welders
13	IS 818	Code of practice for safety and healthy requirements in electric and gas welding and cutting operations
14	IS 822	Code of procedure for inspection of welds
15	IS 823	Manual for metal arc welding in mild steel
16	IS 1038	Steel doors, windows and ventilators
17	IS 1081	Code of practice for fixing and glazing of metal (Steel and aluminium) doors, windows and ventilators
18	IS 1148	Hot rolled steel rivet bars (upto 40 mm diameters) for structural purposes
19	IS 1161	Steel tubes for structural purposes
20	IS 1182	Recommended practice for radiographic examination of fusion welded butt joints in steel plates
21	IS 1200 (Part-VIII)	Method of measurements of steel work and iron works
22	IS 1363 Part I	Hexagon head bolts, screws, and nuts of product grade C (Hexagon Head bolt)

23	IS 1363 (Part-II)	Hexagon Head Bolts, screws and nuts of product Grade 'C'
24	IS 1363 (Part-III)	Hexagon Head Bolts, screws and Nuts of product grade 'C'
25	IS 1367	Technical supply conditions for threaded steel fasteners
26	IS 1599	Method for bend test
27	IS 1608	Metallic materials - Tensile Testing at Ambient Temperature.
28	IS 1821	Dimensions for clearance holes for bolts and screws
29	IS 1852	Rolling and cutting tolerance for hot rolled steel products
30	IS 1894	Method for tensile testing of steel tubes
31	IS 1977	Structural steel (ordinary quality)
32	IS 2062	Hot Rolled low, medium and high tensile structural steel
33	IS 2074	Ready mixed paint, air drying red oxide zinc chrome priming
34	IS 4351	Specification for steel door frames
35	IS 4454 (Part-I)	Steel wires for mechanical springs. Cold drawn unalloyed steel wire.
36	IS 4711	Methods for sampling of steel pipes, tube and fittings.
37	IS 4736	Hot - dip zinc coating on mild steel tubes
38	IS 4923	Hollow Steel Sections for Structural use - Specification
39	IS 6248	Metal rolling shutters and rolling grills
40	IS 7452	Specification for hot rolled steel sections for doors, windows and ventilators.

PREAMBLE
STEEL WORK
(CHAPTER: 10.0)

10.1 STEEL

General requirements relating to supply of structural steel shall conform to IS 8910.

10.2 FREE FROM DEFECTS

All finished materials shall be well and cleanly rolled to the dimensions, sections and masses specified. The finished material shall be reasonably free from surface flaws; laminations; rough/ jagged and imperfect edges and all other harmful defects.

Minor surface defects may be removed by the manufacturer/supplier by grinding provided the thickness is not reduced locally by more than 4 percent below the minimum specified thickness. Reduction in thickness by grinding greater than 4 percent but not exceeding 7 percent may be made subject to mutual agreement between the purchaser and manufacturer/supplier

10.3 MICRO-ALLOYING ELEMENTS

Elements such as niobium, boron, vanadium and titanium added singly or in combination to obtain higher strength to weight ratio and better toughness, formability and weldability as compared to unalloyed steel of similar strength level.

10.4 WELDING:

Welding shall generally be done by electric arc process as per IS 816 and IS 823. The electric arc method is usually adopted and is economical. Where electricity for public is not available generators shall be arranged by the contractor at his own cost unless otherwise specified. Gas welding shall only by resorted to using oxyacetylene flame with specific approval of the Engineer-in-charge. Gas welding shall not be permitted for structural steel work gas welding requires heating of the members to be welded along with the welding rod and is likely to create temperature stresses in the welded members. Precautions shall therefore be taken to avoid distortion of the members due to these temperature stresses. The work shall be done as shown in the shop drawings which should clearly indicate various details of the joint to be welded, type of welds, shop and site welds as well as the types of electrodes to be used. Symbol for welding on plans and shops drawings shall be accordingto IS 813. As far as possible every efforts shall be made to limit the welding that must be done after the structure is erected so as to avoid the improper welding that is likely to be done due to heights and difficult positions on scaffolding etc. apart from the aspect of economy. The maximum dia of electrodes for welding work shall be as per IS 814. Joint surfaces which are to be welded together shall be free from loose mill scale, rust, paint, grease or other foreign matter, which adversely affect the quality of weld and workmanship.

10.5 COLLAPSIBLE STEEL GATES

These shall be of approved manufacture and shall be fabricated from the mild steel sections.

The gates shall consist of double or single collapsible gate depending on the size of the opening. These shall consist of vertical double channels each 20 x 10 x 2 mm at 10 cm centre to centre braced with flat iron diagonals 20 x 5 mm and top and bottom rails of T- iron 40 x 40 x 6 mm @ 3.5 kg/m with 40 mm dia. ball bearings in every fourth double channel, unless otherwise specified. Wherever collapsible gate is not provided within the opening and fixed along the outer wall surface, T- iron at the top may be replaced by flat

iron 40 x 10 mm. The collapsible gate shall be provided with necessary bolts and nuts, locking arrangement, stoppers and handles. Any special fittings like spring, latches and locks, shall be so specified in the description of item where so required. The gate shall open and close smoothly and easily.

10.6 M.S. SHEET SLIDING SHUTTER

These shall be manufactured as per drawings and specification. These shall be fabricated from mild steel sheets.

The shutters shall be double or single leaf shutter as specified. The shutters shall befabricated of specified size of M.S. angle iron frame diagonally braced with the same size of
M.S. angle riveted / welded together with 3mm gusset plate at junction to form a rigid frame. M.S. sheet of 1 mm thickness or as specified shall be fixed to the frame with rivets/welds as approved by the Engineer-in-charge. These shall also be provided with top and bottom guide rails of specified size angles or T- irons and 25 mm diameter pulley or with 25 mm diameter ball bearing at the bottom and guide block with steel pulleys at thetop. The shutters shall also be provided with locking arrangement, handles, stoppers, and holdfasts, other fittings as specified in the description of the item. The guide rails shall be sufficiently long and continued along the wall on both ends so that the sliding shutters can rest against the walls, giving full opening when so required.

10.7 M.S. SHEET SHUTTERS

These shall be manufactured as per drawing and specification. These shall be fabricated from mild steel sheets and angle iron.

The doors shall be provided as double leaf shutters unless otherwise specified. The shutters shall be fabricated with frame of M.S. angle 40 x40 x6 mm @ 3.5 kg/ metre and two diagonal braces of the same section unless otherwise specified. The frame shall be riveted and/ or welded at the junctions. Wherever riveting shall be done 3.15 mm (10 G) thick gusset plate shall be provided at the junction. M.S. sheet of 1 mm thickness or as specified, shall be fixed to the frame with rivets or welds as approved by the Engineer-in-charge. Alternatively the diagonal bracing may be replaced by one horizontal and two cross flats 30x6 mm unless otherwise specified. The outer frame shall be provided with cleats made of section 40 x 10 mm and bent in the shape of angle cleats with one arm 150 mm long and theother arm 50 mm long and fixed to the angle iron frame of the door with two 12 mm dia bolts and nuts. For doors upto 2.40 m height, two angles cleats per door shall be provided. The cleat shall have a vertical leg of 150 mm which shall be fixed with frame and horizontalleg of about 50 mm which shall be provided with a hole of 24 mm dia and fixed in the projected pin of the pin clamp.

10.8 ROLLING SHUTTERS

Rolling shutters shall conform to IS 6248. These shall include necessary locking arrangement and handles etc. These shall be suitable for fixing in the position as specified
i.e. outside or inside on or below lintel or between jambs of the opening. The door shall be either push and pull type or operated with mechanical device supplied by the firm. Shutters upto 10 sqm. shall be of push and pull type and shutters with an area of over 10 sqm shall generally be provided with reduction gear operated by mechanical device with chain or handle, if bearings are specified for each of operation, these shall be paid for separately.

10.9 GLAZING

10.9.1 Glass Panes:-

The glass panes shall have square corners and straight edges. The glass panes shall be so cutthat it fits slightly loose in the frames. In doors, windows and clerestory windows of bath, WC and lavatories frosted glass panes shall be used which shall weight not less than 10.00 kg/m^2.Glazing shall be provided on the outside of the frame unless otherwise specified. Putty of approved make conforming to IS 419 shall be used for fixing glass panes.

10.9.2 Putty

Putty shall be applied between glass panes and glazing bars. Putty shall then be applied overthe glass pane, which shall stop 2 to 3 mm from the sight line of the back rebate to enable the painting to be done upto the sight line to seal the edge of the putty to the glass. The oozed out putty shall be cleaned and from putty cut to straight line. Quantity of putty shall not be less than 185 gm/ metre of glass perimetre. Putty shall be painted within 2 to 3weeks, after glazing is fixed to avoid its cracking.

10.9.3 Glazing clips

Four glazing clips may be provided per glass pane for a size larger than 30x60 cm for all types, where the glass panes size exceed 80x200 cm, 6 glazing clips shall be used. In caseof doors, windows and ventilators without horizontal glazing bars, the glazing clips may be spaced according to the slots, in the vertical members provided the spacing does not exceed 30 cm otherwise the spacing shall be 30 cm.Where large size glass panes are required to be used or where the door or window is located in heavily exposed situation, holes for glazing clips have to be drilled prior to fabrication and cannot be done at any later stages. Use of glazing clips shall be specified while placing the order.

10.9.4 Beading

Where specially stipulated, fixing of glass panes may be done with metal or wooden beading instead of mere putty. Where beading are proposed to be used, the manufacturers shall be intimated in advance to drill holes for hard screws. Usually beads shall be fixedwith screws spaced not more than 10 cm from each corner and the intermediate not more than 20 cm apart. When glass panes are fixed with wooden or metal beading having mitred joints, a thin layer of putty shall be applied between glass panes and sash bars and also between glass panes and the beading. Size of M.S. beading shall be 10 x 10 mm box sectionmanufactured from 1.6 mm thick sheet unless otherwise specified in the item. Where metal beading is specified, extra payment shall be made on this account.

10.10 PRESSED STEEL DOOR FRAMES

10.10.1 Material

Steel door frames profile shall be manufactured from commercial mild steel sheet of specified thickness, (1.6 mm or 1.25mm) conforming to IS 2062 and 4351. Steel door frames with or without fan light shall be made in the profiles which may be manufactured tosuit doors of either type opening inwards or outwards as directed by the Engineer-in- Charge.

10.10.2 Construction

Each door frame shall consist of hinge jamb, lock jamb, head and if required angle threshold. These shall be welded or rigidly fixed together by mechanical means. Where no angle threshold is required, temporary base tie shall be screwed to the feet of frames in order to form a rigid unit. Where so specified base ties shall be of pressed mild steel 1.25 mm thick adjustable to suit floor thickness of 35 or 40 mm and removable , or alternatively, threshold of mild steel angle of section 50x25 mm, minimum shall be provided for external doors frames.

Fixing Lugs: There shall be three adjustable lugs with split end tail to each jamb without fan light, and four for jamb with fan light. The head of the fixing lug shall be of one of the following lengths:

(i) 98 mm long for use with profile A
(ii) 120 mm long for use with profile B
(iii) 160 mm long for use with profile C

The head shall be made from flat steel strip 25 mm wide and not less than 1.60 mm thick the tail of the lugs shall be 200mm long and shall be made of steel strip not less than 40 mm wide and not less than 1 mm thick.

10.10.3 Hinges:

100 mm mild steel butt hinges shall be used. For door frames 80 cm wide and under, three hinges shall be rigidly fixed to one jamb and for door frames above 80 cm wide, four hinges shall rigidly fixed to one jamb, if it is single shutter, where the height of door shutter exceeds 2.15 metres, one additional hinge shall be provided for every 0.5 m or part thereof the additional height. In all cases the hinges shall be so fixed that the distance from the inside of the head rebate to the top of the upper hinge is 20 cm and the distance from the bottom of the door frame to the bottom of the bottom hinge is also kept about 200 mm. The middle hinges shall be at equal distances from lower and upper hinges or as agreed to between the purchaser and the supplier. Hinges shall be made of steel 2.5 mm thick with zinc coated removable pin of 6 mm diameter. The space between the two leaves of the hinge when closed shall be 3 mm and the leaf that is not welded to the frame shall have fourcounter sunk holes to take No. 10 cross recessed head wood screws.

Mortar Guards: Mortar guards of thickness of main frame sheet shall be provided in accordance to provisions of IS 4351 and as instructed by Engineer-in-charge shall be provided. These shall be welded to the frame at the head of the frame for double shutter doors to make provision for bolts. These shall also be provided to the frame behind the hinges, mortice locks and latches, slots, aldrop and sliding /tower bolts.

Lock – Strike Plate: There shall be an adjustable lock- strike plate of steel complete with mortar guard to make provision for locks or latches complying with the relevant Indian Standards. (IS 4351) Lock-strike plates shall be of galvanized mild steel and fixed at 95 cm from the head of the frame.

Shock Absorbers: For side hung door there shall not be less than three buffers or rubber or other suitable material inserted in holes in the rebate. One shall be located at the centre ofthe lock jamb and the other two shall be at 30 cm. from top and bottom of the frame. For double leaf shutter door, two buffers shall be provided.

10.10.4 Finishing

The surface of door frame shall be thoroughly cleaned, free of rust, mill-scale dirt oil etc. either by mechanical means, for example sand or shot blasting or by chemical means such as picking. After pretreatment of the surface one coat of approved primer i.e. red oxide zinc chrome primer conforming to IS 2074. Two coats of paints as directed by the Engineer-in-charge shall be applied to the exposed surface.

10.10.5 Fixing

Frames shall be fixed up right in plumb and plane. To avoid sag or bow in width during fixing or during construction phase, temporary struts across the width preventing sides bulging inwards may be provided. Wall shall be built solid on each side and grouted at each course to ensure solid contact with frame leaving no voids behind the frame. Three lugsshall be provided on each jamb with spacing not more than 75 cm. The temporary struts should not be removed till the masonry behind the frame is set. In case screwed base tie is provided, this should be left in position till the flooring is laid when it can be removed. After pretreatment of the surface, one coat of steel primer and two coats, of paint, as directed by Engineer-in-charge shall be applied to the exposed surface.

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
Steel if arranged by the contractor	10.1.1	(a) Tensile strength (b) Bend test	Laboratory	IS 1599	20 tonne	Every 20 tonne or part thereof
Steel tubular pipes	10.13	(a) Tensile Test (b) Bend Test (c) Flattening Test	Laboratory	IS 1608 IS 2329 IS 2328	Every 8 tonne or part thereof	Every 8 tonne or part thereof

GALVANISED STEEL SHEETS

Dimensions

1) Sizes of plain Sheet: The plain sheets shall be supplied in any combination of the following lengths, widths and thickness.

(a) Length: 2500 and 3000 mm
(b) Width: 900 and 1000 mm
(c) Thickness: 0.50, 0.63, 0.80, 1.00 mm

2) In case of sheets supplied in coil, the internal diameter of coil shall be 450, 510 and 610 mm And the mass of each coil shall not exceed 12 tonne.

3) Coils weighing more than 12 tonnes may be supplied subject to mutual agreement between the contracting parties.

4) Corrugated sheets.

5) Length- The length of the corrugated sheets shall be as follows: 2500, 3000 mm

Zinc Coating

The weight of coating referred to in this specification shall represent the total weight of zinc both side inclusive. On any sample selected at random from the delivery, one set of three samples each 50 x 50 mm or 50 mm diameter shall be selected at random from one sheet for every 500 G.S. sheets, the coating for the different classes shall be within the limit specified in table below:

TABLE 10.1
Mass of Coating (Total Both Sides)

Grade of coating	Minimum average coating Triple spot test g/sqm	Minimum coating single spot test g/sqm*
600	600	510
450	450	380
350	350	300
275	275	235

* Minimum individual value obtained in triple spot test.

Mass

The mass of sheets and coils shall be calculated as given in Table II on the basis of nominal dimensions and mass of zinc coating.

TABLE 10.2
Calculation of Mass of Sheets or Coils

Type of materials	Order of calculation	Method of calculation	Number of Numerals in
Sheet	Mass of single sheet	Nominal mass of single sheet plus mass of zinc coating	Rounded off to 4 effective Figures
	Total mass	Mass of single sheet (kg) x number of sheets	Rounded off to integral value of kg
Coil	Unit mass of coil	Unit mass of sheet (kg/m2)x width (mm) x10 -3	Rounded off to 3 effective Figures
	Mass of single coil	Unit mass of coil (kg/m)x length (m)	
	Total mass (kg)	Total mass of each coil	Integral number of kg

Note:

(i) Nominal mass of single sheet shall be calculated by calculating the volume of the sheet and multiplying the same with density of sheet (density 7.85 g/ cubic cm) and rounding the same to 4 effective figures.

(i) Mass of the coating shall be calculated by multiplying the surface area of single sheet with indicated nominal coating mass (gm/square metre) as shown for triple spot test (Table I).

(ii) For calculation of corrugated sheet mass, the width before corrugation shall be considered while calculating the area.

Corrugations

The depth and pitch of corrugation shall be as follows:

TABLE 10.3
Dimensions of Corrugations

Grade	Depth of Corrugation (mm)	Pitch of Corrugation (mm)
A	17.5	75
B	12.5	75

The number of corrugations shall be 8, 10, 11 or 13 per sheet. The overall width of the sheets before and after corrugation shall be as given in Table below.

TABLE 10.4
Details of Corrugations

Number of Corrugations	Grade	Nominal overall width of sheet measured between crowns of outside corrugations	
		Before corrugation mm	After corrugation mm
8	A	750	660
10	A	900	810
11	A	1000	910
13	A	1200	1110
8	B	750	680
10	B	900	830
11	B	1000	930
13	B	1200	1130

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.1		Structural steel work single section	Structural steel work in single section fixed with or without connecting plate including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.	kg	91.00
10.2		Structural steel work built up sections	Structural steel work using M.S. flats, angles, channels I-section, H-section etc. riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete.	kg	103.00
10.3		Collapsible steel shutters	Providing and fixing in position collapsible steel shutters with vertical channels 20x10x2mm and braced with flat iron diagonals 20x5mm size with top and bottom rail of T-iron 40x40x6mm with 40mm dia steel pulleys complete with bolts, nuts, locking arrangement, stoppers, handles, including applying a priming coat of approved steel primer. Rate includes recess cutting in floor and making good the same wherever required.	sqm	4132.00
10.4		M.S. sheet sliding-shutters	Providing and fixing 1mm thick M.S. sheet sliding-shutters with frame and diagonal braces of 40x40x6mm angle iron, 3mm M.S.gusset plates at the junction and corners 25mm dia pulley,40x40x6mm angle and T-iron guide at the top and bottom respectively including applying a priming coat of approved steel primer.	sqm	3155.00
10.5		M.S. sheet door and angle iron frame	Providing and fixing 1mm thick M.S. sheet door with frame of 40x40x5mm angle iron and 3mm M.S. gusset plates at the junctions and corners, all necessaryfittings complete, including applying a priming coat of approved steel primer.		
	10.5.1	With angles braces	Using M.S. angels 40x40x6mm for diagonal braces	sqm	2604.00
	10.5.2	With flats braces	Using flats 30x6mm for diagonal braces and central cross piece.	sqm	2474.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.6		M.S.rolling shutters	Supplying and fixing rolling shutters of approved make, made of required size M.S. laths interlocked together through their entire length and jointed together at the end by end locks mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and pull operation complete including the cost of providing and fixing necessary 27.5cm long wire springs manufacturedfrom high tensile steel wire of adequate strength conforming to I.S. 4454- part-1 and M.S. top cover of required thickness for rolling shutters.		
	10.6.1	1.25 mm thick top cover	80x1.25mm M.S. laths with 1.25 mm thick top cover.	sqm	2200.00
	10.6.2	1.2 mm thick top cover	80x1.20 mm M.S. laths with 1.20 mm thick top cover.	sqm	1850.00
	10.6.3	0.9 mm thick top cover	80x0.90 mm M.S. laths with 0.90 mm thick top cover.	sqm	1716.00
10.7		Extra for chain and crank operation	Extra for providing suitable mechanical device chain and crank operation for operating rolling shutters.		
	10.7.1	Exceeding 10.00 sqm and upto 16.80 sqm in the area	Exceeding 10.00 sqm and upto 16.80 sqm in the area.	sqm	747.00
	10.7.2	Exceeding 16.80 sqm in area.	Exceeding 16.80 sqm in area.	sqm	782.00
10.8		Extra for grilled rolling shutters	Extra for providing grilled rolling shutters manufactured out of 8 mm dia. M.S. bar instead of laths as per design approved by Engineer-in-Charge. (Area of grill to be measured).	sqm	308.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.9		Ball bearing for rolling shutters	Providing and fixing ball bearing for rolling shutters.	each	366.00
10.10		Glazed steel doors, windows and ventilators z-section	Providing and fixing factory made ISI marked glazed steel doors, windows and ventilators side /top /centre hung with beading and Z-section all members such as F7D, F4B, K11 B and K12 B and other tubular section member etc. complete of standard rolled steel sections, joints mitred and flash butt welded and sash bars tenoned and riveted, including providing and fixing of hinges, pivots, including priming coat of approved steel primer, but excluding the cost of other fittings, complete all as per approved design (sectional weight of only steel members shall be measured for payment).		
	10.10.1	Fixed with lugs	Fixing with 15x3 mm lugs 10 cm. long embedded in cement concrete block 15x10x10 cm of Cement Concrete 1:3:6 (1 Cement : 3 sand : 6 graded stone aggregate 20 mm nominal size).	kg	107.00
	10.10.2	Fixed with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	kg	93.00
	10.10.3	Extra for steel beading	Extra for providing and fixing steel beading of size 10x10x1.6mm (box type) approved shape and section with screws instead of glazing clips and metal sash putty in steel doors, windows, ventilators and composite units.	meter	36.00
10.11		T-iron frames for doors, windows and ventilators	Providing and fixing L-iron or T-iron frames for doors, windows and ventilators of mild steel L-section or T-sections, joints mitred and welded, including fixing of necessary butt hinges and screws and applying a priming coat of approved steel primer.		
	10.11.1	Fixed with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	kg	77.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.12		Pressed steel door frames	Providing and fixing pressed steel door frames made of 1.6 mm thick steel profile conforming to IS: 4351 manufactured from commercial mild steel sheet including hinges, jamb, lock jamb, bead and if required angle threshold of mildsteel angle of section 50x25mm, or base ties of 1.60 mm pressed mild steel welded or rigidly fixed together by mechanical means, including 2.5mm thick M.S. pressed butt hinges, with mortar guards, lock strike-plate and shock absorbers as specified filling the frame with Cement Concrete 1:3:6 at site before fixing (costof concrete to be paid separately) fixing with adjustable lugs (200 mm long & 1.0 mm thick) and applying a coat of approved steel primer after pre-treatment of the surface as directed by Engineer-in- charge:		
	10.12.1	Profile B size 80mmx50mm single rebate	Profile B of 1.60mm thick single rebate of size 80mmx50mm		
	10.12.1.1	Fixing with adjustable lugs	Fixing with adjustable lugs with split end tail to each jamb.	meter	402.00
	10.12.1.2	Fixing with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	meter	394.00
	10.12.2	Profile C size 100mmx50mm single rebate	Profile C of 1.60mm thick single rebate of size 100mmx50mm		
	10.12.2.1	Fixing with adjustable lugs	Fixing with adjustable lugs with split end tail to each jamb.	meter	431.00
	10.12.2.2	Fixing with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	meter	424.00
	10.12.3	Profile E size 115mmx50mm double rebate	Profile E of 1.60mm thick double rebate of size 115mmx50mm		
	10.12.3.1	Fixing with adjustable lugs	Fixing with adjustable lugs with split end tail to each jamb.	meter	472.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	10.12.3.2	Fixing with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	meter	464.00
	10.12.4	Profile B size 80mmx50mm single rebate	Profile B of 1.25mm thick single rebate of size 80mmx50mm		
	10.12.4.1	Fixing with adjustable lugs	Fixing with adjustable lugs with split end tail to each jamb.	meter	356.00
	10.12.4.2	Fixing with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	meter	348.00
	10.12.5	Profile C size 100mmx50mm single rebate	Profile C of 1.25mm thick single rebate of size 100mmx50mm		
	10.12.5.1	Fixing with adjustable lugs	Fixing with adjustable lugs with split end tail to each jamb.	meter	385.00
	10.12.5.2	Fixing with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	meter	377.00
	10.12.6	Profile E size 115mmx50mm double rebate	Profile E of 1.25mm thick double rebate of size 115mmx50mm		
	10.12.6.1	Fixing with adjustable lugs	Fixing with adjustable lugs with split end tail to each jamb.	meter	414.00
	10.12.6.2	Fixing with dash fastener	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	meter	406.00
10.13		M.S. tubular frames for doors, windows and ventilators	Providing and fixing M.S. Tubular frames and shutters or frames only or shuttersonly for doors, windows, ventilators and cupboard with any sections, made of 1.60 mm or 1.25 mm thick M.S. Sheet, joints mitred, welded and grinded finish, with profiles of required size as per drawing, including fixing of necessary butt hinges, stays and screws andapplying a priming coat of approved steel primer excluding cost of all fittings such as handles, tower bolts, sliding door bolts, latches etc. and panels.		
	10.13.1	Fixing with cement concrete block	Fixing with 15x3 mm lugs 10 cm. long embedded in cement concrete block 15x10x10 cm of Cement Concrete 1:3:6 (1 Cement : 3 sand : 6 graded stone aggregate 20 mm nominal size).	kg	116.20

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	10.13.2	Fixing with carbon steel galvanised dash	Fixing with carbon steel galvanised dash fastener of required dia and size (to be paid for separately).	kg	114.80
10.14		Steel work in built up tubular trusses/frame	Steel work in built up M.S. tubular section (round, square or rectangular hollow tubes etc.) trusses/frame work etc. including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washersetc. complete.		
	10.14.1	Hot finished welded tubes	Hot finished welded type tubes	kg	117.00
	10.14.2	Hot finished seamless tubes	Hot finished seamless type tubes	kg	123.00
	10.14.3	Electric resistance tubes	Electric resistance or induction butt welded tubes.	kg	142.00
10.15		M.S.structural steel	Design, supply, fabrication, taransportion and erection of structural steel/mild steel built-up sections for steel building in column, rafter, truss, floor joist, wind bracings, tie beam, foundation bolts, fasteners etc. Including two coats ofEpoxy paint and one coat of primer. These structural steel/built-up sections shall be manufactured using advanced CNC Plasma/Oxy Acetylene cutter and H Beam Welding line. Welding shall be in full depth on both sides of the plate adopting relevant I.S Code.	kg	132.00
10.16		M.S. fan clamp	Providing and fixing M.S. fan clamp type I or II of 16 mm dia M.S. bar bent to shape with hooked ends in R.C.C. slabs, beams during laying including painting the exposed portion of loop, all as perstandard design complete.	each	112.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.17		M.S. box for ceiling fan	Providing and fixing circular/ Hexagonal cast iron or M.S. sheet box for ceiling fan clamp of internal dia 140mm, 73mm height, top lid of 1.5mm thick M.S. sheet with its top surface hacked for proper bonding, top lid shall be screwed into the cast iron/ M.S. sheet box by means of 3.3mm dia. round headed screws, one lock at the corners. Clamp shall be made of 16mm dia M.S. bar bent to shape as per standard drawing.	each	148.00
10.18		M.S. holding down bolts with nut	Providing and fixing mild steel round holding down bolts with nuts and washer plates complete.	kg	84.00
10.19		Bolts nuts and washers	Providing and fixing bolts including nuts and washers complete.	kg	101.00
10.20		M.S. rivets	Providing and fixing M.S. rivets of sizes in position.	kg	106.00
10.21		Welding	Welding by gas or electric plant including transportation of plant at site etc.complete.	cm	5.00
10.22		Welded built up steel sections	Steel work welded in built up sections/ framed work including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer using structural steel etc. as required.		
	10.22.1	Stair cases	In stringers, treads, landings etc. of stair cases including use of chequered plate wherever required, all complete.	kg	101.00
	10.22.2	Frame, ladder, railing	In gratings, frames, guard bar, ladder, railings, brackets, gates and similar works.	kg	105.00
10.23		Hand rail	Providing and fixing hand rail ofapproved size by welding etc. to steel ladder railing, balcony railing, staircase railing and similar works, including applying a priming coat of approved steel primer.		
	10.23.1	M.S. tube.	M.S. tube.	kg	117.00
	10.23.2	E.R.W. tubes.	E.R.W. tubes.	kg	132.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.24		Dash fastener fixing	Providing and fixing carbon steel galvanised (minimum coating 5 micron) dash fastener of 10 mm dia doublethreaded 6.8 grade (yield strength 480 N/mm^2), counter sunk head, comprisingof 10 mm dia polyamide PA 6 grade sleeve, including drilling of hole in frame, concrete/ masonry, etc. as per direction of Engineer-in-charge.		
	10.24.1	60 mm	10x60 mm	each	57.00
	10.24.2	80 mm	10x80 mm	each	63.00
	10.24.3	120 mm	10x120 mm	each	78.00
	10.24.4	140 mm	10x140 mm	each	84.00
	10.24.5	160mm	10x160mm	each	110.00
10.25		Stainless steel railing	Providing and fixing stainless steel (Grade 304) railing made of Hollowtubes, channels, plates etc. including welding, grinding, buffing, polishing and making curvature (wherever required) and fitting the same with necessary stainless steel nuts and bolts complete i/c fixing the railing with necessary accessories and stainless steel dash fasteners, stainless steel bolts etc., of required size, on the top of the floor or the side of waist slab with suitable arrangement as per approval of Engineer-in-charge. (For payment purpose only weight of stainless steel members shall be considered excluding fixing accessories such as nuts, bolts, fasteners etc.)	kg	582.00
10.26		Fly proof wire gauze to windows, doors	Providing and fixing fly proof wire gauze to windows, clerestory windows and doors with M.S. Flat 15x3mm and nuts and bolts complete.		
	10.26.1	M.S.	Galvanised M.S. Wire gauze with 0.63mm dia wire and 1.4 mm aperture on both sides	sqm	646.00
	10.26.2	S.S.	Stainless steel (grade 304) wire gauze of 0.5 mm dia wire and 1.4 mm aperture on both sides.	sqm	1334.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.27		Glass panes/ steel sheet/ fibre sheet	Providing and fixing glass panes/ steel sheet/ fibre sheet with putty and glazing clips in steel doors, windows, clerestory windows all complete.		
	10.27.1	4.0 mm glass	With 4.0 mm thick glass panes	sqm	792.00
	10.27.2	5.50 mm glass	With 5.50 mm thick glass panes	sqm	873.00
	10.27.3	1mm M.S.	With 1mm thick mild steel sheet i/c 10x10x1.6mm box beading.	sqm	916.00
	10.27.4	2.00 mm FRP sheet	With 2mm thick fibre reinforced polymer (FRP) sheet i/c 10x10x1.6mm boxbeading.	sqm	611.00
	10.27.5	1.0mm galvanised steel	With 1mm thick Galvanised steel sheet i/c 10x10x1.6mm box beading.	sqm	936.00
	10.27.6	4mm fibre board	With 4mm thick cement fibre board conforming to IS 14862:2000 of category type A grade i/c 10x10x1.6mm box beading.	sqm	750.00
	10.27.7	6mm cement fibre board	With 6mm thick cement fibre board conforming to IS 14862:2000 of category type A grade i/c 10x10x1.6mm box beading.	sqm	800.00
	10.27.8	8mm cement fibre board	With 8mm thick cement fibre board conforming to IS 14862:2000 of category type A grade i/c 10x10x1.6mm box beading.	sqm	872.00
10.28		Extra for fixing froasted glass instead of float glass	Add extra for providing and fixing froasted glass instead of float glass.		
	10.28.1	4.0 mm glass	With 4.0 mm thick glass panes.	sqm	259.00
	10.28.2	5.0 mm glass	With 5.5 mm thick glass panes.	sqm	262.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.29		Supplying at site	Supplying at site:		
	10.29.1	R.C.C. Standards post/ struts/rails/ poles	R.C.C. Standards post/ struts/rails/ poles of mix 1:1.5:3 (1 cement : 1.5 sand : 3 graded stone aggregate 12.5 mm nominal size) with wooden plugs or 6mm bar nibs wherever required as per direction of Engineer-in-charge including fixing (cost of earth works in excavation, concrete works in foundation to be paid separately).	cum	16510.00
	10.29.2	Angle iron post	Angle iron post and strut of required size including bottom to be split and bent at right angle in opposite direction for 10 cm length and drilling holes upto 10 mm dia. etc. complete. (cost of earth work in excavation and concrete work in foundation to be paid separately)	kg	67.00
	10.29.3	Welded steel wire fabric	Supplying at site -Welded steel wirefabric of required width having rectangular mesh painted with two or more coats of enamel paint of approved shade over a coat of primer (Priming painting to be paid for separately).	kg	81.00
10.30		Turn buckles	Supplying and fixing turn buckles and straining bolts for barbed wire fencing. (Labour rate only).	each set	153.00
10.31		Fencing with R.C.C. post with GI barbed wire	Fencing with R.C.C. post placed at required distance, embedded in cement concrete blocks, every 15th post, last but one end post and corner post shall be strutted on both sides and end post one side only, provided with horizontal lines and two diagonals of barbed wire weighing 9.38 kg per 100 metres(minimum) between the two posts fitted and fixed with G.I. staples on wooden plugs or G.I. binding wire tied to 6 mm bar nibs fixed while casting the post (cost of R.C.C. posts, struts, earth work andconcrete to be paid for separately) with GIbarbed wire.	metre	10.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.32		Fencing with barbed G.I.wire	Fencing with barbed G.I. wire angle iron post placed fitted to at required distance embedded in cement concrete blocks,every 15th post, last but one end post and corner post shall be strutted on both sides and end post on one side only andprovided with horizontal lines and two diagonals interwoven with horizontal wires, of barbed wire weighing 9.38 kgper 100 m (minimum) between the two posts fitted and fixed with G.I. staples,turn buckles etc. complete. (Cost of posts,struts, earth work and concrete work to be paid for separately):- Payment to be made per metre cost of total length of barbed wire used.	metre	17.00
10.33		Fencing with welded steel wire fabric	Fixing Welded steel wire fabric fencing with posts of specified material and of standard design placed and embedded in cement concrete blocks 45x45x60cm of mix 1:5:10 (1cement :5 sand :10 graded stone aggregate 40mm nominal size) every 15th post, last but one end post and corner post shall be strutted on both sides and end post on one side only and struts embedded in cement concrete blocks 70x45x50cm of the same mix, provided with welded steel wire fabric fixed between the posts fitted and fixed with G.I. staples on wooden plugs or tied to 6 mm bar nibs with G.I. binding wire (cost of posts, painting, earth work in excavation and cement concrete to be paid for separately). (Labour rate only).	sqm	18.00
10.34		Fencing with chain link fabric 2mm dia	Providing and fixing G.I. or PVC coated G.I.wire chain link fabric fencing of required width and mesh size including strengthening with 2mm dia wire or nuts, bolts and washers as required complete with posts of specified material and of standard design placed and embedded in cement concrete blocks 45x45x60cm of mix 1:5:10 (1cement :5 sand :10 graded stone aggregate 40mm nominal size) every 15th post, last but one end post and corner post shall be strutted on both sides and end post on one side only and struts embedded in cement concrete blocks 70x45x50cm of the same mix, as per the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			direction of Engineer- in-charge. (Cost of post, painting earthwork & c.c. to be paid separately)		
	10.34.1	G.I. wire	Made of G.I. wire.	kg	133.00
	10.34.2	PVC coated G.I. wire	Made of G.I. wire, PVC coated to achieve outer dia. not less than 5mm in required colour and shade.	kg	138.00
10.35		Fencing with G.I. chain link fabric 3mm dia	Providing and fixing G.I. chain link fabric fencing of required width in mesh size 25x25 mm made of G.I. wire of dia. 3mm including strengthening with 2mm dia. wire or nuts, bolts and washers as required complete, with posts of specified material and of standard design placed and embedded in cement concrete blocks 45x45x60cm of mix 1:5:10 (1cement :5 sand : 10 graded stone aggregate 40mm nominal size) every 15th post, last but one end post and corner post shall be strutted on both sides and end post on one side only and struts embedded in cement concrete blocks 70x45x50cm of the same mix, as per the direction of Engineer-in- Charge. (cost of post painting earth work & C.C. to be paid separately)	sqm	574.00
10.36		Concertina coil fencing	Providing and fixing concertina coil fencing with punched tape concertina coil 600 mm dia 10 metre openable length (total length 90 m), having 50 nos rounds per 6 metre length, upto 3 m height of wall with existing angle iron 'Y' shaped placed 2.4m or 3.00m apart and with 9 horizontal R.B.T. reinforced barbed wire, stud tied with G.I. staples and G.I. clips to retain horizontal, including necessary bolts or G.I. barbed wire tied to angle iron, all complete as per direction of Engineer-in charge, with reinforced barbed tape(R.B.T.) / Spring core (2.5 mm thick) wire of high tensile strength of 165 kg/ sq mm with tape (0.52 mm thick) and weight 43.478 gm/ metre (cost of M.S. angle, C.C. blocks shall be paid separately).	metre	275.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
10.37		Fire resistance steel sheet door shutter and frame (Partially Insulated)	Providing and fixing of ISI marked insulated fire rated GI Sheet doors conforming to FD 120-PI classification of IS 3614:2021 with fire rating of 120 minutes integrity and 30 minutes insulation. The factory manufactured door frame and shutter assembly shall be made out of minimum 120 GSM coating as per IS: 277. Door and frame should be finished with powder coating in standard RAL shade finish having DFT of minimum 30 microns. The door should carry a minimum 500 Hrs of salt spray test. All accessories like hardware, vision panel, fire & smoke seal etc to be supplied by Fire door manufacturer only as per the guidelines laid down in IS 3614:2021.Every door must carry a label confirming to type of door & its fire rating. The Door frame shall be single rebate grooved profile of size minimum 125 mm x 75mm made out of 1.50mm minimum thick galvanized steel sheet. Frames shall be butt / mitered joint and field assembled with bolting system. Frames to have inbuilt grooved to accommodate Intumescent fire seal strip. Frames to have inbuilt grooved to accommodate Intumescent fire seal strip of size 10x4mm. All provisions should be mortised & provided with necessary inbuilt reinforcement to receive appropriate hardware. Frames should be provided with back plate bracket for anchor fastner with 8-10 mm dia and minimum 80 mm long HDG fasteners, with Polyamide sleeves (Payable in extra item) for installation of frame in finished plastered masonary wall opening. Note:- For grouting of Steel frames in masonary installations ;cement slurry of 1:3 ratio by civil contractor/fire rated PU foam or such fire rated material may be recommended except head member(Payable in extra item). Door leaf should be minimum 60mm thick fully flush double skin door. Door leaf must be manufactured from minimum 1.2mm thick galvanised steel sheet. The infill material shall be mineral wool of minimum density 96 kg/cum. All doors shall be	sqm	12747.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			factory prepped for receiving appropriate hardware and provided with necessary reinforcement for hinges, locks, and door closers. The edges should be interlocked with a bending radius of 1.4mm. For double leaf doors integrated astragals has to be provided on the meeting stile for both active and inactive leaf. The shutters shall be mounted to the frame with SS 304 grade ball bearing type hinges of size 100x 89x3 mm along with all suitable hardware (Payable in extra item). Vision lite size 185x285 (max area up to 0.06sqm) wherever applicable should be provided with single layered clear fire rated glass panes as per guidelines laid down under IS 3614 : 2021 vide clause no. 7.2.7 table no. 2 with a clip on arrangement. The glass will be provided with fire rated ceramic tape of min. 2 mm thickness with min. 15mm width al around on either face. Every Glass pane must carry a permanent marking of its glass manufacturer.		
10.38		Roll formed steel windowair tight	Providing and fixing of factory make air tight casement windows made of roll formed galvanized pre-painted steel with epoxy primer of 5-7 microns thick, finish painted with a polyester paint of 12-16 microns thick and back coated with alkyd backer 5-7 microns thick or powder coated with pure polyester powder up to 50-60 microns thick, colour coated/ powder coated (base steel as per IS 513 of “D" quality, galvanized as per IS 277 with zinc of 120 grams/sqm) outer frame section 46x52x0.58 mm, shutter section 46x46x0.58mm, mullion section 46x70x0.58mm, beading section 18x25x0.58 mm, windows panelled without glazing (glazing shall be paid separately as per SOR rate) with EPDM (Ethyl Propylene Diamine Monomer) gasket and the sections cut to length mitre joined with corner bracket centre mullions fixed using mullion cap and with handle made of powder coating high grade aluminum with nylon receiver, corner brackets made of CRCA (Cold Rolled Close Annealed) with Zinc Phosphate,		

			Mullion caps made of glass filled nylon, frames fixed to the concrete /masonry wall by means of self-expanding screws. Including filling the gap upto 5 mm depth and 5 mm width in between steel frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete.		
	10.38.1	Openablewindow	Openable window	sqm	5228.00
	10.38.2	Fixed windows	Fixed windows	sqm	4190.00
10.39		Roll formed steel window single rebate with glass shutter and grill	Providing and fixing of factory make single rebate openable/fixed windows made of roll formed galvanized pre-pained steel with epoxy primer of 5-7 microns thick, finish painted with a polyester paint of 12-16 microns thick and back coated with alkyd backer 5-7 microns thick or powder coated with pure polyester powder up to 50-60 microns thick, colour coated/powder coated (base steel as per IS 513 of “D" quality, galvanized as per IS 277 with zinc of 120 grams/sqm) outer frame section 48 x 50 x0.58mm, centre mullion section 48 x 50 x0.58mm, shutter section 47 x 20 x0.58mm and the fixed glass beading section 12 x 12 x0.58mm, outer frame and mullions shall consists rebate for glazed shutter with a 20 mm provision for guard bars/grills and the sections cut to length mitre joined with corner bracket, centre mullion fixed with mullion cap, handle, stay, 2 nos. SS (stainless steel) heavy dutypivot hinges per shutter and window fittedwithout glazing (glazing shall be paid separately as per SOR rate) with EPDM (Ethyl Propylene Diamine Monomer) gaskets including fixing frames in concrete/masonry wall by means of self- expanding screws, including 10mm square bars welded to 6x12mm flat at a distance 150mm c/c (pitch) etc. Including filling the gap upto 5 mm depth and 5 mmwidth in between steel frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	10.39.1	Openable single rebate window	Openable single rebate window	sqm	4511.00
	10.39.2	Fixed single rebate window	Fixed single rebate window	sqm	3894.00
10.40		Roll formed steel window – Double rebate with glass shutter, grill and mosquito mesh shutter	Providing and fixing of factory make double rebate windows made of rollformed galvanized pre-painted steel with epoxy primer of 5-7 microns thick, finish painted with a polyester paint of 12-16 microns thick and back coated with alkyd backer 5-7 microns thick or powder coated with pure polyester powder up to 50-60 microns thick, colour coated/ powder coated (base steel as per IS 513 of "D" quality, galvanized as per IS 277 withzinc of 120 grams/sqm) outer frame section 72 x 55 x0.58mm, centre mullion section 72 x 50 x0.58mm, shutter section 47 x 20 x0.58mm, fly mesh section 40 x 20 x0.58mm and the fixed glass beading section 12 x 12 x0.58mm, outer frame and mullions shall consists rebate for glazed shutter, fly mesh and 20 mm provision for guard bars/grills, the sections cut to length miter joined with corner bracket, centre mullions fixed with mullion cap, stay, handles, latch, 2 Nos of heavy duty stainless steel pivot hinges per shutter and panelled without glazing (glazing shall be paid separately as per SOR rate) with and S.S. Mesh for fly mesh shutter (32 gauge,304 grade), fitted using EPDM (Ethyl Propylene Diamine Monomer) gaskets including fixing windows in the concrete/masonry wall by means of self-expanding screws, including 10mm Square bars welded to 6x12mm flat at a distance 150mm c/c (pitch) etc. Including filling the gap upto 5 mm depth and 5 mm width in between steel frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete..		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
	10.40.1	Fixed single rebate window	Openable double rebate window	sqm	5820.00
	10.40.2	Fixed double rebate window	Fixed double rebate window	sqm	4449.00
10.41		Roll formed steel window- – fixed ventilator	Providing and fixing of factory make fixed louvers fabricated from roll formed sections made of galvanized pre-painted steel with epoxy primer of 5-7 microns thick, finish painted with a polyester paint of 12-16 microns thick and back coated with alkyd backer 5-7 microns thick or powder coated with pure polyesterpowder up to 50-60 microns thick, colour coated/powder coated (base steel as per IS 513 'D' quality, galvanized as per IS 277 with zinc of 120 grams/Sqm) section for external frame 33 x 58 x0.58mm, glass holding flanges made of polypropylene, sections are to be cut to length, miter joined with corner bracket. The above frames should be fixed to the concrete/masonry walls by means of self-expanding screws. Including filling the gap upto 5 mm depth and 5 mm width in between steel frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod ofapproved quality as per architectural drawings and direction of Engineer-in- charge complete. (glazing shall be paid separately as per SOR rate)	sqm	3404.00
10.42		Roll formed steel window sliding (2 track) 2 glass shutter	Providing and fixing of factory make pre painted steel sliding window 2 track-2 panel sliding window fabricated from roll formed sections made of galvanized steel with epoxy primer of 5-7 microns thick, finish painted with a polyester paint of 12-16 microns thick and back coated with alkyd backer 5-7 microns thick or powder coated with pure polyester powder up to 50-60 microns thick, colour coated/ powder coated (Base Steel as per IS 513 'D' quality, galvanized as per IS 277 with zinc of 120 GM/Sq.mtr) Section for external frame should be of 59x44 x0.58mm, Section for shutter should be of 35x50 x0.58mm. Without glazing (glazing shall be paid separately as per	sqm	3479.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (in Rs.)
1	2	3	4	5	6
			SOR rate) Corner bracket for internal and external frame made of glass filled nylon. Gaskets are to be made of Ethyl Propylene Diamine Monomer (EPDM). The sections are to be cut to length, joined and assembled by means of corner bracket. The above frames should be fixed to the concrete/masonry walls by means of self-expanding screws. Including filling the gap upto 5 mm depth and 5 mm width in between steel frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete.		
10.43		Roll formed steel window sliding (3 track) 2glass shutter + 1mosquito mesh shutter	Providing and fixing pre painted steel sliding window 3 fabricated from Roll formed sections made of galvanized Steel with epoxy primer of 5-7 microns thick, finish painted with a polyester paint of12-16 microns thick and back coated with alkyd backer 5-7 microns thick or powder coated with pure polyester powder up to 50-60 microns thick, colour coated/ powder coated (Base Steel as per IS 513 'D' quality, galvanized as per IS 277 with zinc of 120 gm/sqm). Section for external frame should be of 107x44 x0.58mm, section for shutter should be of 35x49 x0.58mm. Section of mesh should be of 35 x 49 x0.58mm. Without glazing (glazing shall be paid separately as per SOR rate) corner bracket for internal and external frame made of glass filled nylon. Gaskets are to be made of Ethyl Propylene Diamine Monomer (EPDM). The sections are to be cut to length, joined and assembled by means of corner bracket. The above frames should be fixed to the concrete/masonry walls by means of self-expanding screws. Including filling the gap upto 5 mm depth and 5 mm width in between steel frame and adjacent RCC/ Brick/ Stone work by providing weather silicon sealant over backer rod of approved quality as per architectural drawings and direction of Engineer-in-charge complete.	sqm	5430.00

CHAPTER – 11.0

FLOORING

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1	2	3
1	IS 269	Specification for 33 grade ordinary portland Cement
2	IS 401	Code of practice for preservation of timber
3	IS 451	Technical supply conditions for wood screws
4	IS 455	Specification for portland slag cement
5	IS 661	Code of practice for thermal insulation of cold storages
6	IS 702	Specification for industrial bitumen
7	IS 1124	Method of test for determination of water absorption, apparent specific gravity and porosity of natural building stones
8	IS 1130	Specification for marble (blocks, slabs and tiles)
9	IS 1141	Code of practice for Seasoning of timber
10	IS 1200 (Part-XI)	Method of measurement of Building and Civil Engineering work (Part 11) paving, floor finishes, dado and skirting
11	IS 1237 Edition 2.3	Specification for cement concrete flooring tiles
12	IS 1322	Specification for bitumen felts for water proofing and damp-proofing
13	IS 1443	Code of practice for laying and finishing of cement concrete flooring tiles
14	IS 1489 (Part-I)	Specification for portlandpozzolana cement (Part-I) flyashbased
15	IS 1489 (Part-II)	Specification for Portland pozzolana cement (Part II) calcined clay based
16	IS 1580	Specification for bituminous compounds for water proofing and caulking purpose
17	IS 2114	Code of practice for laying in-situ terrazzo floor finish
18	IS 2571	Code of practice for laying in-situ cement concrete flooring
19	IS 3622	Specification for sand stone (Slab and Tiles)
20	IS 3670	Code of practice for construction of timber floors
21	IS 4457	Acid and/or alkali Resistant tiles.
22	IS 5318	Code of practice for laying of hard wood parquet and wood block floors
23	IS 5766	Code of practice for laying of burnt clay brick floor
24	IS 8041	Specification for rapid hardening portland cement

25	IS 8042	Specification for white portland cement
26	IS 8043	Specification for hydrophobic portland cement
27	IS 8112	Specification for 43 grade ordinary portland cement
28	IS 12330	Specification for sulphate resisting portland cement.
29	IS 13630 (Part-1 to 15)	Methods of Testing of ceramic tiles
30	IS 13712	Specification for ceramic tiles; definition, classification characteristic and marking
31	IS 15622	Specification for pressed ceramic tile
32	IS 1122	Method of test for determination of true specific gravity of natural building stones
33	IS 1124	Method of test for determination of water absorption, apparent specific gravity and porosity of natural building stones
34	IS 1130	Marble (blocks, slabs and tiles).
35	IS 4101 (Part-1)	Code of practice for external facing and veneers: Stone facing.
36	IS 3316	Specifications for structural granite
37	IS 14223 (Paart-1)	Polished Building Stones (Part-1) Granite

LIST OF MANDATORY TESTS

Material	Clause	Test	Field/ laboratory test	Test Procedure	Min. quantity of material for carrying out the test	Frequency of testing
Terrazzo Tiles	11.10.1 and 11.11.1	1. Transverse strength 2. Water absorption 3. Abrasion Atest	Laboratory	I S: 1237	5000 Nos. (no testing need be done if total number of tiles of all types of all sizes from all manufacturers used in a work is less than 5000 Nos)	One test for every 10,000 Nos. or part thereof for each type and size from a single manufacturer (One test to be done even if the number of terrazo tiles of any type and size from a single manufacturers is less than 5000 Nos. providedthe total number of terrazo tiles of all types and sizes from all manufacturers used in a work exceed 5000 Nos.
Pressed ceramic tiles (for floor and wall)	11.4,11.5 and 11.16	1.Dimensions and surface quality 2. Physical properties 3. Chemical properties	Laboratory	I S: 13630	3000 Nos.	3000 Nos. or part thereof
Marble	8.3 (Table 8.2)	(i) Moisture Absorption	Laboratory	IS 1124	50 Sq.m.	100 sqm. Or part thereof.
		(ii) Hardness Test	-do-	Mho's Scale	-do-	-do-
		(iii) Specific Gravity	-do-	IS 1122	-do-	-do-
Granite		(i) Moisture	-do-	IS 1124	-do-	-do-
		(ii) Specific Gravity	-do-	IS 1122	-do-	-do-

PREAMBLE
FLOORING
(CHAPTER: 11.0)

11.1 Cement Concrete

Cement concrete of specified mix grade shall be used and it shall generally conform to the specifications described under sub head 4.0

11.2 Base Concrete

Flooring shall be laid on base concrete where so provided. The base concrete shall be provided with the slopes required for the flooring. Flooring in verandah, Courtyard, kitchens and baths shall have slope ranging from 1 : 48 to 1 : 60 depending upon location and as decided by the Engineer-in-Charge. Floors in water closet portion shall have slope of1:30 or as decided by the Engineer-in-Charge to drain off washing water. Further, necessarydrop in flooring in bath, WC, kitchen near floor traps ranging from 6 mm to 10 mm willalso be provided to avoid spread of water. Necessary margin to accommodate this drop shallbe made in base concrete. Plinth masonry off set shall be depressed so as to allow the base concrete to rest on it.

The flooring shall be commenced preferably within 48 hours of the laying of base concrete. The surface of the base shall be roughened with steel wire brushes without disturbing the concrete. Immediately before laying the flooring, the base shall be wetted and a coat of cement slurry @ 2 kg of cement spread over an area of one sqm so as to get a good bond between the base and concrete floor.

If the cement concrete flooring is to be laid directly on the RCC slab, the top surface of RCC slab shall be cleaned and the laitance shall be removed and a coat of cement slurry @2 kg of cement spread over an area of one sqm so as to get a good bond between the base and concrete floor.

11.3 Finishing

The finishing of the surface shall follow immediately after the cessation of beating. The surface shall be left for some time, till moisture disappears from it or surplus water can be mopped up. Use of dry cement or cement and sand mixture stiffening the concrete to absorb excessive moisture shall not be permitted. Excessive trowelling shall be avoided.

11.4 Tile Work

Preparation of Surface and Laying

Base concrete or the RCC slab on which the tiles are to be laid shall be cleaned, wetted and mopped. The bedding for the tile shall be with cement mortar 1:4 (1 cement: 4 coarse sand) or as specified. The average thickness of the bedding shall be 20 mm or as specified while the thickness under any portion of the tiles shall not be less than 10 mm.

Mortar shall be spread, tamped and corrected to proper levels and allowed to harden sufficiently to offer a fairly rigid cushion for the tiles to be set and to enable the mason to place wooden plank across and squat on it.

Over this mortar bedding neat grey cement slurry of honey like consistency shall be spread at the rate of 3.3 kg of cement per square metre over an area upto one square metre. Tiles

shall be soaked in water washed clean and shall be fixed in this grout one after another, each tile gently being tapped with a wooden mallet till it is properly bedded and in level with the adjoining tiles. The joints shall be kept as thin as possible and in straight lines or tosuit the required pattern.

The surface of the flooring during laying shall be frequently checked with a straight edge about 2 m long, so as to obtain a true surface with the required slope. In bath, toilet W.C. kitchen and balcony/verandah flooring, suitable tile drop or as shown in drawing will be given in addition to required slope to avoid spread of water. Further tile drop will also be provided near floor trap.

Where full size tiles cannot be fixed these shall be cut (sawn) to the required size, and their edge rubbed smooth to ensure straight and true joints. Tiles which are fixed in the floor adjoining the wall shall enter not less than 10 mm under the plaster, skirting or dado.

After tiles have been laid surplus cement slurry shall be cleaned off.

11.5 Pointing and Finishing

The joints shall be cleaned off the grey cement slurry with wire/coir brush or trowel to a depth of 2 mm to 3 mm and all dust and loose mortar removed. Joints shall then be flush pointed with white cement added with pigment if required to match the colour of tiles. Where spacer lug tiles are provided, the half the depth of joint shall be filled with polysulphide or as specified on top with under filling with cement grout without the lugs remaining exposed. The floor shall then be kept wet for 7 days. After curing, the surface shall be washed and finished clean. The finished floor shall not sound hollow when tapped with a wooden mallet.

11.6 Marble and Granite Stone Flooring

General

Marble shall be hard, sound, dense and homogeneous in texture with crystaline texture as far as possible. It shall generally be uniform in colour and free from stains, cracks, decay and weathering

11.7 Coloured Marble

Plain Black Marble : Black marble sawn along veins locally known as ‘Peta Pasu sawing’ available at Bhainslana.

11.8 Black Zebra Marble

(i) Bhainslana Black Zebra Marble: Black marble having grey or white veins available at Bhainslana.
(ii) Kishangarh Black Zebra Marble: Black marble with grey and/or white veins available at Kishangarh.
(iii) Abu Black Zebra Marble: Black marble having white patches and streaks available at Abu.
(iv) Narnaul Black Zebra Marbles: Black marble with thin white veins available at Narnaul.
(v) Makrana Dhobi Doongri Zebra Marble: Greyish black marble with white flowery pattern available at Dhobi Doongri.

11.9 Green Marble

(i) Baroda Green Marble: Dark green marble with flowery pattern available at Baroda.
(ii) Abu Green Marble: Light green marble with green and/or brown streaks on white ground available at Ambaji.
(iii) Falna Green Marble: Green marble with prominent yellowish pattern available at Falna.
(iv) Bundi Green Marble: Green marble with pinkish shades available at Umar, (Bundi).

11.10 Grey Marble

(i) Kumari Grey Marble: Grey marble having light blue shades available at Makrana.
(ii) Bundi Grey Marble: Grey Marble with pink or green or black streaks available at Umar (Bundi).

11.11 Brown Marble

(i) Bar Brown Marble/Brown Marble with light and dark brown shades available at bar.
(ii) Narnaul Brown Marble Brown marble having teak wood shades available at Narnaul

11.12 Granite

It shall be of any colour and size as directed by Engineer-in-Charge. Granite shall be plain machine cut and mirror polished. The stone shall be smooth and of even surface without holes or pits.

TABLE -11.1
Physical Properties of Marble and Granite

	Marble		Granite	
Characteristic	**Marble Requirements**	**Method of test**	**Granite Requirement**	**Method of test**
(1) Moisture absorption after 24 hrs immersion in cold water	Max. 0.4%	IS 1124	Max. 0.50% by weight	IS 1124
(2) Hardness	Min. 3	Mhos scale	-	-
(3) Specific Gravity	Min. 2.5	IS 1122	Min. 2.6	IS 1122

11.13 Approval of Sample

Before starting the work, the contractor shall get samples of marble/Granite approved by the Engineer-in- Charge. Approved samples shall be kept in the custody of the Engineer-in-Charge and the marble supplied and used on the work shall conform to samples with regard to soundness, colour, veining and general texture.

11.14 Stone Work - Table Rubbed and Polished (Plain Work)

For all stone work in steps, jambs, columns and other plain work: Joints in staircase treads, kitchen platforms shall be permitted only at curvature or when width/length is more than 0.6/2 mtrs. Respectively. Number of joints in each direction shall not be more than one number for every 2 m length beyond the initial 2.00 m length. Additional joints due to curvature or for providing fixture shall be provided judiciously.

11.15 Dressing, Cutting and Rubbing

Every stone shall be gang saw/machine cut to the required size and shape, chisel dressed machine finished on all beds and joints, so as to be free from any waviness and to give truly vertical, horizontal, radial or circular joints as required. The exposed faces and sides of stones forming joints upto6mm from the face shall be fine tooled machine cut such that a

straight edge laid along the face of the stone is in contact with every point on it. All window sills, tread of steps, counters vanities moulding edges etc. shall be machine cut and polished to give high gloss mirror finish as per direction of Engineer-in-Charge. These surfaces shall then be rubbed smooth. All visible angles and edges shall be true, square and free from chipping. Beyond the depth of 6 mm from face, the joints shall be dressed with a slight splay so that the thickness of joint increases, in an inverted V shape. The surfaces of the stones coming in contact with backing need not be chisel dressed.

Every stone shall be cut to the required size and shape, fine chisel dressed on all sides to the full depth so that a straight edge laid along the side of the stone shall be fully in contact with it. The top surface shall also be fine chisel dressed to remove all waviness. In case machine cut slabs are used, fine chiesel dressing of machine cut surface need not be done provided a straight edge laid any where along the machine cut surfaces is in contact with every point on it. The sides and top surface of slabs shall be machine rubbed or table rubbed with coarse sand before paving. All angles and edges of the marble slabs shall be true, square and free from chippings and the surface shall be true and plane. The thickness of the slabs shall be 18, 30 or 40 mm as specified in the description of the item. Tolerance of + 3% shall be allowed for the thickness. In respect of length and breadth of slabs a tolerance of + 2% shallbe allowed.

11.16 Laying

Base concrete or the RCC slab on which the slabs are to be laid shall be cleaned, wetted and mopped. The bedding for the slabs shall be with cement mortar 1:4 (1 cement: 4 sand) or as given in the description of the item.

The average thickness of the bedding mortar under the slab shall be 20 mm and the thickness at any place under the slab shall be not less than 12 mm.

The slabs shall be laid in the following manner: Mortar of the specified mix shall be spread under the area of each slab, roughly to the average thickness specified in the item. The slab shall be washed clean before laying. It shall be laid on top, pressed, tapped with wooden mallet and brought to level with the adjoining slabs. It shall be lifted and laid aside. The top surface of the mortar shall then be corrected by adding fresh mortar at hollows. The mortar is allowed to harden a bit and cement slurry of honey like consistency shall be spread over the same at the rate of 4.4 kg of cement per sqm. The edges of the slab already paved shall be buttered with grey or white cement with or without admixture of pigment to match the shade of the marble slabs as given in the description of the item. The slab to be paved shall then be lowered gently back in position and tapped with wooden mallet till it is properly bedded in level with and close to the adjoining slabs with as fine a joint as possible. Subsequent slabs shall be laid in the same manner. After each slab has been laid, surplus cement on the surface of the slabs shall be cleaned off. The flooring shall be cured for a minimum period of seven days. The surface of the flooring as laid shall be true to levels, and, slopes as instructed by the Engineer-in-Charge. Joint thickness shall not be more than 1mm. Due care shall be taken to match the grains of slabs which shall be selected judiciouslyhaving uniform pattern of Veins/streaks or as directed by the Engineer-in-Charge.

The slabs shall be matched as shown in drawings or as instructed by the Engineer-in- Charge. Slabs which are fixed in the floor adjoining the wall shall enter not less than 12 mmunder the plaster skirting or dado. The junction between wall plaster and floor shall be finished neatly and without waviness. Marble slabs flooring shall also be laid in combination with other stones and/or in simple regular pattern/design as described in itemof work and/or drawing.

11.17 Polishing and Finishing

Slight unevenness at the meeting edges of slabs shall then be removed by fine chiselling and finished in the same manner as specified except that cement slurry with or without pigments shall not be applied on the surface before each polishing.

11.18 Measurements

Marble stone flooring with different kind of marble/Granite/Kota shall be measured separately and in square metre correct to two places of decimal. Length and breadth shall be measured correct to a cm before laying skirting, dado or wall plaster. No deduction shall be made nor extra paid for voids not exceeding 0.20 square metre. Deductions for ends of dissimilar materials or other articles embedded shall not be made for areas not exceeding 0.10 square metre. Nothing extra shall be paid for laying the floor at different levels in the same room. Steps and treads of stairs paved with stone slabs shall also be measured under the item of flooring. Extra shall, however, be paid for such areas where the width of treads does not exceed 30 cm. nosing for treads shall be measured in running metre and paid for extra. The width of treads shall be measured from the outer edge of the nosing, as laid, before providing the riser.

11.19 Protection

Green work shall be protected from rain by suitable coverings. The work shall also be suitably protected from damage during construction.

11.20 Rate

The rate shall include the cost of all materials and labour involved in all the operations described above. However, extra shall be paid for making special type of pattern/ design/flowers as per drawings. No deductions shall be made in rate even if flooring is done without any pattern/design.

11.21 Wall Lining/Veneer Work

Unless and otherwise specified in the nomenclature of the item, the marble slabs used for wall lining/veneer work shall be gang saw cut (polished and machine cut). Back shall not be polished/ cut in order to ensure a good grip with the hearting of backing. The cut slabs shall be of the thickness as specified with a tolerance permissible. The tolerance in wall lining when straight edge of 3 m length is placed should not be more than 2 mm.

11.22 Laying

The stone shall be wetted before laying. They shall then be fixed with mortar in position without the use of chips or under pinning of any sort. Care shall be taken to match the grains of veneer work as directed by the Engineer-in-Charge. For purpose of matching the grains, the marble slabs shall be selected judiciously having uniform pattern of veins/streaks. Preferably the slabs shall be those got out of the same block from the quarry. The area to be veneered shall be reproduced on the ground and the marble slabs laid in position and arranged in the manner to give the desired matching of grains. Any adjustment needed for achieving the best results shall be then carried out by replacing or interchanging the particular slabs. Special care shall be taken to achieve the continuity of grains between the two slabs one above the other along the horizontal joints. This shall then be gotapproved by the Engineer-in-Charge and each marble slabs numbered properly and the same number shall be marked on a separate drawing as well as on the surface to be actually veneered, so as to ensure the fixing of the particular slabs in the correct location. For the facing of the columns also the same procedure as mentioned above shall be followed.

Where so desired, the adjoining stones shall be secured to each other by means of copper pins 75 mm long and 6 mm diameter or as specified.

The stones shall be secured to the backing by means of cramps. The material for cramps shall have high resistance to corrosion under conditions of dampness and against the chemical action of mortar or concrete in which cramps are usually embedded. Cramps shall be of 25 × 6 mm and 30 cm long in case of backing of stone masonry walls and brick masonry walls thicker than 230 mm. In case of backing with brick masonry walls 230 mm or less thick or RCCmember's cramps shall be of 25 ×6 mm and length as per requirement made out of gun metal or any other metal. Generally the outer length of cramp in half brick work backing shall be 115 mm and in one brick work backing it shall be 150 mm. Cramps shall be spaced not more 60 cm apart horizontally. Alternatively the stone may be secured to the backing by means of stone dowels 10 x 5 x 2.5 cm.

The adjoining stones shall be secured to each other by means of gun metal cramps or copper pins of the specified size. Cramps may be attached to its sides or top and bottom or sides, top and bottom. The actual number of cramps and their sections, however, shall be as per requirements of design to carry the loads.

Where cramps are used to hold the unit in position only, the facings shall be provided with a continuous support on which the stones rest at the ground level and other storey levels, the support being in the form of projection from or recess into the concrete floor slab, or a beam between the columns or a metal angle attached to the floor slab or beams. These supports shall preferably be at vertical intervals not more than 3.5 m apart and also over the heads of all openings. Such supports shall also be provided where there is transition from thin facing below to thick facings above.

Alternatively cramps may be used to hold the units in position and in addition to support the units thus transferring the weight of the units to the backing. Such cramps should be properly designed as per IS 4101 (Part 1).

The cramps may be of copper alloyed with zinc, tin, nickel, lead or stainless steel. The pins, cramps and dowels shall be laid in cement mortar 1:2 (1 cement: 2 sand) and their samples got approved by the Engineer-in-Charge and kept at site.

11.23 Joints

All joints shall be full of mortar. Special care shall be taken to see that groundings forveneer work are full of mortar. If any hollow groundings are detected by tapping the face stones, these shall be taken out and relaid. The thickness of the face joints shall be uniform, straight and as fine as possible, not more than 1.5 mm and in the face joint, the top 6 mm depth shall be filled with mortar specified for the pointing.

11.24 Acid or Alkali Resistant Tiles

The tiles shall be of vitreous ware and free from deleterious substances. The iron oxide content allowable in the raw material shall not exceed two percent. The tiles shall be vitrifiedat the temperature of 1100^0C and above and shall be kept unglazed. The finished, tile, when fractured shall appear fine grained in texture, dense and homogenous. The tiles shall be sound, true to shope, flat and free from flows and manufacturing defects affecting their utility. The tiles shall be conforming to IS 4457. The tiles to be tested for water absorption, compressive strength, acid resistance as per IS 4457. Sampling procedure for acceptance tests and criteria for conformity to be as per IS 4457. The tiles shall be of required colour.

11.25 Dimensions and Tolerances

Ceramic unglazed vitreous acid-resistant tiles shall be made in three sizes namely 98.5 X 98.5 mm, 148.5 x148.5 mm and 198.5 x198.5 mm. They shall be available in the following thickness: 35, 30, 25,20 and 15 mm. The depth of the grooves on the under side of the tile shall not exceed 3 mm. Tolerance on length, breadth and thickness of tiles shall be + 2 percent.

11.26 Shape

The tiles shall be square shaped. Half tiles rectangular in shape shall also be available. Half tiles for use with full tiles shall have dimensions which shall be such as to make two half tiles, when joined together, match with the dimension of full tile. The shape of tiles otherthan square shall be as agreed to between the purchaser and the manufacturer. Tiles shall be checked for squareness and warp as per IS 4457.

Performance Requirements

The tiles when tested in accordance with method given in IS 4457, shall conform to be requirement specified in the code (IS 4457).

Loss in Abrasion

The maximum percentage of loss in abrasion of the ceramic unglazed vitreous acid resistant tiles determined in accordance with the procedure laid down in IS 1237, shall be as mentioned in IS 4457.

11.27 Marking

Tiles shall be legibly marked on the back with the name of the manufacturer or his trade mark. Manufacturer's batch number and year of manufacture. Each tile may also be markedwith the ISI certification mark.

11.28 Preparation of Surface and Laying

Preparation of surface and laying as prescribed, except the cement used to be acid and or alkali resistant cement and cement mortar to be used to be acid and or Alkali resistant mortar. Thickness of bedding of mortar for flooring to be 10 mm or specified on the item and for dado/skirting to be 12 mm or specified on item.

11.29 Pointing and Finishing

As specified, except that cement used for pointing to be acid and or alkali resistant cement.

11.30 Fixing of Tile Flooring with Cement Based High Polymer Modified Quick Set Adhesive (Water Based)

When tile flooring is to be laid over the existing flooring without dismantling old flooring it can be laid with adhesive. The old flooring shall be thoroughly cleaned and checked for undulations, if any shall be rectified with cement mortar 1:3 (1 cement: 3 sand). Old cement concrete surface shall be hacked and cleaned off to have proper bond with the old surface.

High polymer modified quick set tile adhesive (conforming to IS 15477) shall be thoroughly mixed with water and a paste of zero slump shall be prepared so that it can be used with in 1.5 to 2 hours. It shall be spread over an area not more than one sqm at one

time. Average thickness of adhesive shall be 3 mm The adhesive so spreaded shall be combed using suitable trowel. Tiles shall be pressed firmly in to the position with slight twisting action checking it simultaneously to ensure good contact gently being tapped with wooden mallet till it is properly backed with adjoining tiles. The tiles shall be fixed within 20 minutes of application of adhesive. The surplus adhesive from the joints, surface of the tiles shall be immediately cleaned.

The surface of the flooring shall be frequently checked during laying with straight edge of above 2m long so as to attain a true surface with required slope. Where spacer lugs tiles are provided these shall be filled with grout with lugs remaining exposed. Where full size tile can not be fixed these shall be cut (sawn) to the required size and edges rubbed smooth to ensure straight and true joints. Tiles which are fixed in floor adjoining to wall shall enter notless than 10 mm under plaster, skirting or dado.

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.1		Cement concrete flooring 40mm thick	40 mm thick Cement concrete flooring 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate) 20mm nominal size stone aggregate in panels finished with a floating coat of neat cement including cement slurry, but excluding the cost of nosing of steps etc. complete.(Area of panel not toexceed 2.0 sqm)	sqm	270.00
11.2		52 mm thick cement concrete flooring with hardener topping 52 mm thick	52 mm thick cement concrete flooring with concrete hardener topping, under layer 40 mm thick cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 20 mm nominal size) and top layer 12 mm thick cement hardener consisting of mix 1:2 (1 cement hardener mix: 2 graded stone aggregate 6 mm nominal size) by volume, hardening compound mixed @ 2 litre per 50kg of cement or as per manufacturers specifications. This includes cost of cement slurry, but excluding the cost of nosing of steps etc. complete.	sqm	399.00
11.3		Cement concrete flooring with hardener topping 62 mm thick	62 mm thick cement concrete flooring with concrete hardener topping, under layer 50 mm thick cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 20mm nominal size) and top layer 12mm thick cement hardener consisting of mix 1:2 (1 cement hardener mix: 2 graded stone aggregate, 6mm nominal size) by volume, hardening compound mixed @ 2 litre per 50kg of cement or as per manufactures specifications. This includes cost of cement slurry, but excluding the cost of nosing of steps etc. complete.	sqm	431.00
11.4		Cement plaster skirting	Cement plaster skirting 18 mm thick (upto 30 cm height) with cement mortar 1:3 (1 cement: 3 sand) finished with a floating coat of neat cement.	sqm	246.00
11.5		Cement concrete 1:2:4pavement with	Cement concrete pavement with Cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 20 mm nominal size) including finishing complete.	cum	4654.00
11.6		Making chequers pattern	Making chequers of approved pattern on cement concrete floors, steps, landing, pavements etc.	sqm	21.00
11.7		Mosaic flooring (top layer 6mm thick)	40 mm thick marble chips flooring rubbed and polished to granolithic finish, under layer 34 mm thick cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 12.5mm nominal size) and top layer 6mm thick with white, black, chocolate, grey, yellow or green marble chips of sizes from		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			1mm to 4mm nominal size laid in cement marble powder mix 3:1 (3 cement : 1 marble powder) by weight in proportion of 4:7 (4 cement marble powder mix : 7 marble chips) by volume including cement slurry etc. complete : (Area of panel not to exceed 2.0 sqm)		
	11.7.1	Dark shade pigment (grey cement)	Dark shade pigment with ordinary cement.	sqm	493.00
	11.7.2	Light shade pigment (white cement)	Light shade pigment with white cement.	sqm	550.00
	11.7.3	Medium shade pigment (white & grey cement)	Medium shade pigment with 50% white cement and 50% ordinary cement.	sqm	521.00
	11.7.4	Without pigment (white cement)	White cement without any pigment.	sqm	511.00
	11.7.5	Light shade pigment (grey cement)	Light shade pigment with ordinary cement.	sqm	499.00
	11.7.6	Without pigment(grey cement)	Ordinary cement without any pigment.	sqm	468.00
11.8		Mosaic flooring (Top layer9mm thick)	40 mm thick marble chips flooring, rubbed and polished to granolithic finish, under layer 31mm thick cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 12.5mm nominal size) and top layer 9mm thick with white, black, chocolate, grey, yellow or green marble chips of sizes from 4mm to 7mm nominal size laid in cement marble powder mix 3:1 (3 cement : 1 marble powder) by weight in proportion of 4:7 (4 cement marble powder : 7 marble chips) by volume including cement slurry etc. complete. (Area of panel not to exceed 2.0 sqm)		
	11.8.1	Dark shade pigment (grey cement)	Dark shade pigment with Ordinary cement.	sqm	526.00
	11.8.2	Light shade pigment (white cement)	Light shade pigment with white cement.	sqm	607.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	11.8.3	Medium shade pigment (white & grey cement)	Medium shade pigment with 50% white cement and 50% ordinary cement.	sqm	566.00
	11.8.4	Without pigment (white cement)	White cement without any pigment.	sqm	570.00
	11.8.5	Light shade pigment (grey cement)	Light shade pigment with ordinary cement.	sqm	532.00
	11.8.6	Without pigment(grey cement)	Ordinary cement without any pigment.	sqm	493.00
11.9		Mosaic flooring (top layer 12 mm thick)	40 mm thick marble chips flooring, rubbed and polished to granolithic finish, under layer 28 mm thick cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 12.5mm nominal size) and top layer 12mm thick with white, black, chocolate, grey yellow or green marble chips of sizes from 7mm to 10mm nominal size laid incement marble powder mix 3:1 (3 cement : 1 marble powder) by weight in proportion of 2:3 (2 cement marble powder mix : 3 marble chips) by volume including cement slurry etc. complete :(Area of panel not to exceed 2.0 sqm)		
	11.9.1	Dark shade pigment (grey cement)	Dark shade pigment with ordinary cement.	sqm	563.00
	11.9.2	Light shade pigment (white cement)	Light shade pigment with white cement.	sqm	677.00
	11.9.3	Medium shade pigment (white & grey cement)	Medium shade pigment with 50% white cement and 50% ordinary cement.	sqm	620.00
	11.9.4	Without pigment (white cement)	White cement without any pigment.	sqm	620.00
	11.9.5	Light shade pigment (grey cement)	Light shade pigment with ordinary cement.	sqm	576.00
	11.9.6	Without pigment(grey cement)	Ordinary cement without any pigment.	sqm	519.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.10		Mosaic skirting	Marble chips skirting (upto 30 cm height) 18 mm thick with under layer 12 mm thick in cement plaster 1:3 (1 cement : 3 sand) : rubbed and polished to granolithic finish, top layer 6 mm thick with white, black, chocolate, grey, yellow or green marble chips of sizes from smallest to 4 mm nominal size laid in cement marble powder mix 3:1 (3 cement : 1 marble powder) by weight in proportion of 4:7 (4 cement marble powder mix : 7 marble chips) by volume :		
	11.10.1	Dark shade pigment (grey cement)	Dark shade pigment with ordinary cement.	sqm	572.00
	11.10.2	Light shade pigment (white cement)	Light shade pigment with white cement.	sqm	629.00
	11.10.3	Medium shade pigment (white & grey cement)	Medium shade pigment with 50% white cement and 50% ordinary cement.	sqm	601.00
	11.10.4	Without pigment (white cement)	White cement without any pigment.	sqm	602.00
	11.10.5	Light shade pigment (grey cement)	Light shade pigment with ordinary cement.	sqm	579.00
	11.10.6	Without pigment(grey cement)	Ordinary cement without any pigment.	sqm	549.00
11.11		Extra for laying terrazo flooring in staircase	Extra for laying terrazo flooring on staircase treads not exceeding 30 cm in width including cost of forming, nosing etc.	sqm	22.00
11.12		Glass strips in floors	Providing and fixing glass strips 40 mm wide and 4 mm thick in joints of terrazo/ cement concrete floors.	metre	34.00
11.13		Crazy marble stone flooring	18 mm thick crazy marble stone white, black or as specified marble stone flooring, including filling the gaps with light shade pigment with white cement marble powder mixture (3 parts of white cement : 1 part of marble powder) by weight in proportion of 4:7 (4 cement marble powder mix : 7 white, black or white and black marble chips of sizes from 1mm to 4mm nominal size by volume), with under layer 25mm thick cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate 12.5 mm nominal size), including rubbing, polishing and cement slurry etc. complete .	sqm	844.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.14		Mosaic tile flooring	Precast terrazo tiles 22mm thick with graded marble chips of size upto12mm, laid in floors, and landings, jointed with neat cement slurry mixed with pigment to match the shade of the tiles including rubbing and polishing complete, on 20 mm thick bed of cement mortar 1:4 (1 cement :4 sand) :		
	11.14.1	Light shade pigment (white cement)	Light shade using white cement.	sqm	813.00
	11.14.2	Medium shade pigment (white & grey cement)	Medium shade using 50%white cement and 50% ordinary cement.	sqm	756.00
	11.14.3	Dark shade (grey cement)	Dark shade using ordinary cement.	sqm	700.00
	11.14.4	Without pigment(grey cement)	Ordinary cement without any pigment.	sqm	700.00
11.15		Extra for mosaic tiles in staircase	Extra if terrazo tiles are laid in treads of steps not exceeding 30 cm in width.	sqm	28.00
11.16		Mosaic tiles in skirting	Precast terrazo tiles 22 mm thick with graded marble chips of sizes upto 12 mm in skirting and risers of steps not exceeding 30 cm in height on 12 mm thick cement plaster 1:3 (1 cement : 3sand) jointed with neat cement slurry mixed with pigment to match the shade of the tiles, including rubbing and polishing complete with tiles of :		
	11.16.1	Light shade pigment (white cement)	Light shade pigment using white cement.	sqm	954.00
	11.16.2	Medium shades pigment (white& grey cement)	Medium shades pigment using 50% white cement and 50% ordinary cement.	sqm	883.00
	11.16.3	Dark shade pigment (grey cement)	Dark shade pigment using ordinary cement.	sqm	811.00
	11.16.4	Without any pigment (grey cement)	Ordinary cement without any pigment.	sqm	750.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.17		Chequeredmosaic tiles	Chequeredterrazo tiles 22 mm thick with graded marble chips of size up to 6 mm in floors jointed with neat cement slurry mixed with pigment to match the shade of the tiles including rubbing and polishing complete on 20 mm thick bed of cement mortar 1:4 (1 cement :4 sand) :		
	11.17.1	Light shade pigment (white cement)	Light shade pigment using white cement.	sqm	889.00
	11.17.2	Medium shades pigment (white& grey cement)	Medium shade pigment using 50% white cement, 50% ordinary cement.	sqm	807.00
	11.17.3	Dark shade pigment (grey cement)	Dark shade pigment using ordinary cement.	sqm	732.00
	11.17.4	Without any pigment (grey cement)	Ordinary cement without any pigment.	sqm	686.00
11.18		Chequered cement concrete tiles	Chequered precast cement concrete tiles anyfinish, any size 22 mm thick IS:13810-1993 ISI mark in footpath and courtyard jointed with neat cement slurry mixed with pigment to match the shade of tiles, 15mm thick screed bad in cement mortar 1:6 complete all as specified and directed by Engineer-in-Charge.		
	11.18.1	Light shade pigment (white cement)	Light shade pigment using white cement.	sqm	754.00
	11.18.2	Medium shades pigment (white& grey cement)	Medium shade pigment using 50% white cement 50% Grey cement.	sqm	669.00
	11.18.3	Dark shade pigment (grey cement)	Dark shade pigment using ordinary cement.	sqm	591.00
	11.18.4	without any pigment (grey cement)	Ordinary cement without any pigment.	sqm	535.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.19		Marble stone flooring	Marble stone flooring with 18 mm thick marble stone (sample of marble shall be approved by Engineer-in-charge) laid over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 sandand grey cement slurry, joints treated with white cement mixed with pigment to match colour of stone including rubbing and polishing complete with : (area of slabs over 0.5 sqm)		
	11.19.1	Makrana white	Makrana white second quality.	sqm	3051.00
	11.19.2	Raj Nagar plain	Raj Nagar plain.	sqm	1381.00
	11.19.3	Agaria	Agaria White/Morgarh	sqm	1882.00
	11.19.4	Katni marble	Katni marble	sqm	1555.00
	11.19.5	Black zebra	Black Zebra.	sqm	1315.00
	11.19.6	Udaipur green	Udaipur / Baroda green marble	sqm	1649.00
	11.19.7	Pink plain	Pink plain marble.	sqm	1448.00
	11.19.8	Jaisalmer yellow	Jaisalmer Yellow	sqm	1582.00
11.20		Marble stone in steps, riser, skirting and dado	Marble stone 18 mm thick in riser of step, skirting, dado and pillar laid on 12 mm average thick cement mortar 1:3 (1 cement: 3 sand andlaid with white cement slurry) joints treated with white cement mixed with pigment to match the shade of the slab i/c rubbing and polishing complete. (area of slab over0.50sqm.)		
	11.20.1	Makrana white	Makrana white second quality.	sqm	3675.00
	11.20.2	Raj Nagar plain	Raj Nagar plain.	sqm	1932.00
	11.20.3	Agaria	Agaria White/Morgarh	sqm	2455.00
	11.20.4	Katni marble	Katni marble	sqm	2113.00
	11.20.5	Black Zebra	Black Zebra.	sqm	1863.00
	11.20.6	Udaipur green	Udaipur / Baroda green marble	sqm	2211.00
	11.20.7	Pink plain	Pink plain marble.	sqm	2002.00
	11.20.8	Jaisalmer yellow	Jaisalmer Yellow	sqm	2141.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.21		Marble wall lining (veneer work)	Marble work gang saw cut (polished and machine cut) of thickness 18 mm for wall lining (veneer work), backing filled with a grout of average 12 mm thick in cement mortar 1:3 (1 cement: 3 sand), including pointing with white cementmortar 1:2 (1 white cement: 2 marble dust) withan admixture of pigment to match the marble shade (To be secured to the backing by means of cramps, which shall be paid for separately). (Area of slabs over 0.50 sqm.)		
	11.21.1	Makrana white	Makrana white second quality.	sqm	3773.00
	11.21.2	Raj Nagar plain	Raj Nagar plain.	sqm	2737.00
	11.21.3	Agaria	Agaria White/Morgarh	sqm	2553.00
	11.21.4	Katni marble	Katni marble	sqm	2219.00
	11.21.5	Black zebra	Black Zebra.	sqm	1961.00
	11.21.6	Udaipur green	Udaipur / Baroda green marble	sqm	2309.00
	11.21.7	Pink plain	Pink plain marble.	sqm	2100.00
	11.21.8	Jaisalmer yellow	Jaisalmer Yellow	sqm	2240.00
11.22		Marble& granite work for kitchen platforms, vanity counters, window sills , facias	Providing and fixing 18mm thick gang saw cut mirror polished (premoulded and prepolished) machine cut for kitchen platforms, vanitycounters, window sills , facias and similar locations of required size of approved shade, colour and texture laid over 20mm thick base cement mortar 1:4 (1 cement : 4 sand) with joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing moulding and polishing to edge to givehigh gloss finish wherever required etc. complete at all levels. Samples approved by Engineer-in- Charge.		
	11.22.1	Marble granite	Raj Nagar plain/White Marble/Udaipur Green Marble /Zebra black	sqm	1797.00
	11.22.2	Granite fine grained	Granite fine grained dark crystal of other colours and various patterns	sqm	2356.00
	11.22.3	Granite coarse grained	Granite coarse grained light black/ light red/white with crystal of other colours and various patterns	Sqm	2113.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	11.22.4	Granite with various	Granite grey/pink/brown with various patterns	Sqm	1808.00
	11.22.5	Cuddpah	Cuddpah stone	sqm	1279.00
11.23		Extra for nosing marble	Extra for prefinished nosing to treads of steps of marble stone.	meter	164.00
11.24		Extra for marble flooring insteps	Extra for marble stone flooring in treads of steps and risers using single length upto 2.00 metre.	sqm	191.00
11.25		Extra for prefinished grooves	Extra for prefinished grooves to treads of steps of granite stone/ marble stone / kota stone.	metre	61.00
11.26		Italian marble mirror polished flooring	Providing and laying machine cut, mirror polished, Italian Marble stone (18 mm thick Italian Marble stone slab, Perlato, Rosso erona, Fire Red or Dark Emperadore etc.)work in flooring laid in required pattern in linear portion of the building all complete as per architectural drawings, with 18 mm thick stone slab laid over20 mm (average) thick base of cement mortar 1:4 (1 cement: 4 sand) laid and jointed with white cement slurry @ 4.4 kg/sqm including pointing with white cement slurry mixed with pigment to match the marble shade including rubbing, curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge.	sqm	6558.00
11.27		Italian marble wall lining mirror polished	Providing and fixing machine cut, mirror / eggshell polished , Marble stone work 18 mm thick Italian Marble stone slab, Perlato, Rosso verona, Fire Red or Dark Emperadore etc. for wall lining (veneer work) including dado, skirting, risers of steps etc., in required design and pattern wherever required, stones of different finished surface texture, on 12 mm (average) thick cement mortar 1:3 (1 cement : 3 sand) laid and jointed with white cement slurry @ 3.3 kg/sqm including pointing with white cement slurry admixed with pigment of matching shade, including rubbing, curing, polishing etc. all complete as per Architectural drawings, and as directed by the Engineer-in-Charge.	sqm	6957.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.28		Italian marble flooring in required design and patterns	Italian marble flooring in required design and patterns		
	11.28.1	Italian marble laying in simple geometrical pattern	Providing and laying machine cut, mirror polished Italian Marble stone 18 mm thick Italian Marble stone slab,Perlato, Rosso verona, Fire Red or Dark Emperadore etc flooring, in required design (Simple geometrical, abstract etc.) and in patterns in combination with Italian marble stones of different colors, shades and finished surface texture etc., in linear portions of the building, all complete as per the architectural drawings, with18 mm thick stone slab laid over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 sand) laid and jointed with white cement slurry @ 4.4 kg/sqm including pointing with white cement slurry admixed with pigment to match the marble shade including rubbing, curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge. Note – This item shall be executed only after the prior approval of Chief Engineer.	sqm	7748.00
	11.28.2	Premium Quality Italian marble laying in special pattern	Premium high Quality machine, cut, mirror polished, italian marble (minimum compressive strength 1800 psi is compulsory) in required special design and in required special pattern in combination with different varietes italian marble and finished surface texture linear portion of the building all complete as per architectural drawings, with 18 mm thick stone slab with stone slab base side epoxy coating with fibre net and top side epoxy coating for increase strength and laid over 20mm (average) thick base of cement mortar 1:3 (1 cement : 3 sand) laid and jointed with white cement slurry @ 4.4 Kg/sqm including pointing with white cement slurry admixed with pigment to match the marble shade resin and epoxy after every rubbing coat used epoxy and resin including rubbing, curing and polishing etc.(Italian marble has to be treated with epoxy resins and matching pigments to increase its strength)all complete as specified as directed by the Engineer-in-Charge. Note – This item shall be executed only after the prior approval of Chief Engineer.	sqm	9067.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.29		Granite wall lining (Polished)	Granite work gang saw cut (polished and machine cut) of thickness **18mm** for wall lining (veneer work), backing filled with a grout of average 12 mm thick in cement mortar 1:3 (1 cement: 3 sand) including pointing with white cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment to match the marble shade: (To be secured to the backing by means of cramps, which shall be paid for separately).		
	11.29.1	Stone slab over 0.50 sqm	Area of slab over 0.50 sqm		
	11.29.1.1	Granite (fine grained)	Fine grained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2919.00
	11.29.1.2	Granite(coarse grained)	Coarse grained granite light Black/light Red/ off White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2675.00
	11.29.1.3	Granite of various patterns	Gray/Pink/Brown or equivalent with various patterns	sqm	2370.00
	11.29.2	Stone slab upto 0.50 sqm	Area of slab upto 0.50 sqm		
	11.29.2.1	Granite (fine grained)	Fine grained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2705.00
	11.29.2.2	Granite(coarse grained)	Coarse grained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2486.00
	11.29.2.3	Granite (with pattern)	Gray/Pink/Brown or equivalent with various patterns	sqm	2212.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.30		Mirror polished pre moulded granite flooring18 mm thick	Providing and laying gang saw cut **18 mm thick**, mirror polished premoulded (wherever required) and pre polished machine cut granite stonework in flooring of required size shape of approved shade, color and texture in flooring laid over 20 mm thick base of cement mortar 1:4 (1 Cement: 4 sand) including grouting the joints with white cement mixed with matching pigments epoxy touch ups etc. complete as per direction of Engineer-in-Charge.		
	11.30.1	Stone slab over 0.50sqm	Area of slab over 0.50 sqm		
	11.30.1.1	Granite (fine grained)	Fine grained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2642.00
	11.30.1.2	Granite(coarse grained)	Coarse grained granite light Black/light Red/ off White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2422.00
	11.30.1.3	Granite (with pattern)	Gray/Pink/Brown or equivalent with various patterns	sqm	2153.00
	11.30.2	Stone slab upto0.50sqm	Area of slab upto 0.50 sqm		
	11.30.2.1	Granite (fine grained)	Fine grained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2455.00
	11.30.2.2	Granite(coarse grained)	Coarsegrained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	2258.00
	11.30.2.3	Granite (with pattern)	Gray/Pink/Brown or equivalent with various patterns	sqm	2011.00
11.31		Granite flooring 30mm thick mirror polished pre moulded	Providing and laying gang saw cut **30 mm** thick, mirror polished pre moulded and pre polished machine cut granite stone of required size andshape of approved shade, colour and texture in footpath, flooring in road side plazas and similar locations, laid over 20mm thick base of cement mortar 1:4 (1cement: 4 sand) including grouting the joints with white cement mixed with matching pigment, epoxy touch ups etc. complete as per direction of Engineer-in-Charge. (area of slab more than 0.50 sqm)		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	11.31.1	Granite (fine grained)	Fine grained granite dark Black/dark Red/ White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	3616.00
	11.31.2	Granite(coarse grained)	Coarse grained granite light Black/light Red/ off White or equivalent with self design/ pattern/crystals of other colors of glitter	sqm	3250.00
	11.31.3	Granite (with pattern)	Gray/Pink/Brown or equivalent with various patterns	sqm	2793.00
11.32		Deduction	Deduct for not polishing granite flooring (item no 11.29 & 11.30)	sqm	130.00
11.33		Polished Granite Stone	Providing and laying Granite stone flooring in required design and patterns in linear as well as curvilinear portions of the building all complete as per the architectural drawings with 18 mm thick stone slab over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) laid and jointed with cement slurry admixed with pigment of matching shade including rubbing, curing and polishing and as directed by the Engineer-in-Charge Granite stone slab Dark Black. Dark Red, White or equivalent,		
	11.33.1	Flame finish	Flamed Finish Granite	sqm	2542.00
	11.33.2	Leather finish	Leather Finish Granite	sqm	2809.00
11.34		Granite work in riser steps, skirting, dado and pillars	Granite stone work 18mm thick of any colour in riser of steps, skirting, dado and pillars laid on 12 mm average thick cement mortar 1:3 (1 cement:3 sand) and jointed with white cement slurry mixed with pigment to match the shade of slab.	sqm	3413.00
11.35		Granite stone block paving	Providing and fixing 10x10x7.50 cm Granite stone block hand cut andchisel dressed on top, for paving in floors, drains etc. laid over 20mm thick base mortar 1:4 (1cement:4 sand) with joints 10mm wide filled with same mortar mixed with matching pigment including ruled pointing etc. complete as per direction of engineer-in charge.	sqm	2765.00
11.36		Kota stone flooring	Kota stone slab flooring over 20 mm (average) thick base laid over and jointed with grey cement slurry mixed with pigment to match the shade of the slab including rubbing and mirror polishing complete with base of cement mortar 1 : 4 (1 cement : 4 sand) having thickness of slab 25 mm (Area of slab to be over 0.25 sqm) Note: - Deduction shall be made for not mirror polishing according to item no 11.43	sqm	1174.00
11.37		Deduction for not using 25 mm thick stone	Deduction for using 20 mm thick slab instead of25 mm thick slab of kota stone in item no 11.35	sqm	70.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.38		Kota stone in risers, skirting, dado and pillars	Kota stone slabs in risers of steps, skirting, dado and pillars laid on 12 mm (average) thick cement mortar 1:3 (1 cement 3 coarse sand) and jointed with grey cement slurry mixed with pigment to match the shade of the slabs, including rubbing and polishing complete.		
	11.38.1	10-12 mm thick	10-12 mm thick	sqm	1177.00
	11.38.2	20 mm thick	20 mm thick	sqm	1231.00
11.39		Kota stone 40 mm chiseled dressed.	Kota stone slab (area less than 0.5 sqm) rough chiseled dressed average 40 mm thick flooring for parking and open area of thickness over 20 mm (average) thick base laid over and jointed with grey cement slurry complete with base of cement mortar 1:4 (1 Cement : 4 Sand)	sqm	723.00
11.40		Edge moulding of stone work	Providing edge moulding to 18mm thick marble stone counters, Vanities, window sills etc. including machine polishing to edge to give high gloss finish etc. complete as per design approved by Engineer-in-Charge.		
	11.40.1	Marble work	Marble work	meter	164.00
	11.40.2	Granite work.	Granite work.	meter	269.00
11.41		Extra for marble /granite stone in facia and drops	Extra for fixing marble /granite stone over and above corresponding basic item, in facia and drops of width upto 150 mm with epoxy resin based adhesive including cleaning etc. complete.	meter	167.00
11.42		Extra for providing opening in marble/granite/stone work	Extra for providing opening of required size and shape for wash basins/ kitchen sink in kitchen platform, vanity counters and similar location in marble/Granite/stone work including necessary holes for pillar taps etc. including moulding, rubbing and polishing of cut edges etc. complete.	each	232.00
11.43		Mirror (glass) polishing	Mirror polishing on marble work/Granite work/stone work wherever required to give high gloss finish complete.	sqm	164.00
11.44		Fine dressed stone flooring flush joints	Fine dressed stone flooring over 20 mm (average) thick base of cement mortar 1:5 (1cement : 5 sand) with joints finished flush.		
	11.44.1	40 mm Red stone	Red sand stone	sqm	578.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	11.44.2	40 mm White stone	White sand stone	sqm	578.00
	11.44.3	40mm Lime stone	Grey lime stone	sqm	578.00
	11.44.4	25 mm Kota stone	Kota stone	sqm	678.00
11.45		Fine dressed sand stone flooring with joint pointing	40 mm thick fine dressed stone flooring over 20 mm (average) thick base of cement mortar 1:5 (1 cement : 5 sand) including pointing with cement mortar 1:2 (1 cement : 2 sand) with an admixture of pigment to match the shade of stone.		
	11.45.1	Red stone	Red sand stone	sqm	642.00
	11.45.2	White stone	White sand stone	sqm	642.00
11.46		Machine rubbed sand stone flooring	40 mm thick machine rubbed stone flooring over 20 mm (average) thick base of cement mortar 1:5 (1 cement :5 sand) with joints 3mm thick, side buttered with cement mortar 1:2 (1 cement : 2sand) mixed with pigment to match the shade of stone and pointing with same mortar		
1	11.46.1	Red stone	Red sand stone	sqm	709.00
	11.46.2	White stone	White sand stone	sqm	709.00
11.47		Extra for nosing	Extra for pre finished nosing in treads of steps of Kota stone/ sand stone slab.	meter	139.00
11.48		Extra for stone work in stairs	Extra for Kota stone/ sand stone in treads of steps and risers using single length upto 1.05 metre.	sqm	10.00
11.49		Ceramic glazed floor tiles	Providing and laying Ceramic glazed floor tiles 300x300 mm or more (having thickness 6 to 7mm) of 1st quality conforming to IS : 15622 of approved make in any colours and shade laid on20 mm thick Cement Mortar 1:4 (1 Cement : 4 sand) including pointing the joints with white cement and matching pigment etc., complete.	sqm	670.00
11.50		Ceramic glazed wall tiles	Providing and fixing Ceramic glazed wall tiles 300x450 mm or more (having thickness 6 to 7 mm) of 1st quality conforming to IS: 15622 of approved make in all Colours& shades, except burgundy, bottle green, black laid on 12mm thick bed of Cement Mortar 1:3 (1 Cement: 3 sand)preprad same day when mortar is still green jointed with grey cement slurry @3.3 kg per sqm including pointing the joints with white cement and matching pigments etc., complete.	sqm	776.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.50		Rectified ceramic floor tiles	Providing and laying rectified Glazed Ceramic floor tiles 300x300 mm or more (having thickness 9 to 10mm) of 1st quality conforming to IS : 15622 of approved make in all colours white, ivory, grey, fume red brown laid on 20mm thick cement mortar 1:4 (1 Cement : 4 sand) including grouting the joints with white cement and matching pigments etc., complete.	sqm	644.00
11.52		Rectified ceramic wall tiles	Providing and laying rectified Glazed Ceramic wall tiles 300x450 mm or more (thickness to be specified by the manufacturer) of 1st quality conforming to IS : 15622 of approved make in all colours white, ivory, grey, fume red brown laid on 12 mm thick cement mortar 1:3 (1 Cement : 3 sand) including grouting the joints with white cement and matching pigments etc., complete.	sqm	698.00
11.53		Antiskid floor tiles (heavy duty)	Providing and laying Antiskid floor tiles of any sizes, 12 mm thickness with water absorption less than 0.08% and conforming to IS:15622 of approved make in all colours and shades, laid on 20mm thick cement mortar 1:4 (1 cement : 4 sand) including grouting the joints with white cement with matching pigments etc., complete.	sqm	880.00
11.54		Deduct for not grouting the joints	Deduct for not grouting the joints with white cement and matching pigment in the items of fixing of vitrified tiles.	sqm	8.00
11.55		Deduct for not using cement mortar 1:4 bedding for flooring	Deduct for not using 20mm thick cement mortar 1:4 (1 cement: 4 sand) bedding in laying of floor tiles.	sqm	359.00
11.56		Fixing tiles/ marble or granite stone in flooring, skirkingdedo, pillars	Fixing glazed/ Ceramic/Rectified/ Vitrified floor tiles with cement based high polymer modified quick-set tile adhesive (Water based) conforming to IS: 15477, in average 3mm thickness.	sqm	350.00
11.57		Tactile tile	Providing and laying tactile tile (for vision impaired persons as per standards) of size 300x300x9.8mm having water absorption less than 0.05% and conforming to IS: 15622 of approved make in all colours and shades for outdoor floors such as footpath, court yard, multi modals location etc., laid on 20mm thick base of cement mortar 1:4 (1cement : 4 sand) in all shapes and patterns including grouting the joints with white cement mixed with matching pigments etc. complete as per direction of Engineer-in-Charge.	sqm	1307.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.58		Glass mosaic tiles	Providing and fixing Glass mosaic tiles at finished plain wall surface of size 20 mm x 20 mm x 4 mm in all colour, design, fixing in customize design as per direction of Engineer-in- Charge. The glass mosaic tiles to be fixed on the wall surface with the help of approved adhesive applied at the rateof 2.5 kg per sqm and grouting of the same. The rate is inclusive of all operation, material and required pattern approved by Engineer-in-Charge:	sqm	1556.00
11.59		Turf paver	Providing and laying 500x500x40mm thick Turf paver on 150mm thick sub grade of compacted bed of 20mm nominal size stone aggregate and base course with 50mm thick sand and filling joints with sand including spreading, well ramming, consolidating and finishing smooth etc. all complete as per direction of Engineer- in- charge.	sqm	1457.00
11.60		Acid and/ or alkali resistant tiles	Providing and fixing 10mm thick acid and/ or alkali resistant tiles of approved make and colour using acid and/ or alkali resisting mortar bedding and joints filled with acid and/ or alkali resisting cement as per IS : 4457 complete as per the direction of Engineer-in- Charge.		
	11.60.1	Flooring	In flooring on a bed of 10 mm thick mortar 1:4 (1 acid proof cement: 4 sand)Acid and alkali resistant tile.	sqm	1262.00
	11.60.2	Dado/skirting	In dado/skirting on 12 mm thick mortar 1:4 (1 acid proof cement: 4 sand).Acid and alkali resistant tile.	sqm	1368.00
11.61		Tile work in skirting, risers of steps and dado	8 mm thick.stones Tile work in skirting, risers of steps and dado (upto 2 m height) over 12 mmthick bed of cement mortar 1:3 (1 cement :3 sand) and jointed with grey cement slurry @ 3.3 kg/sqm including pointing in white cement mixed with pigment of matching shade complete.		
	11.61.1	Marble	Marble tiles (polished) Raj Nagar.	sqm	1119.00
	11.61.2	Granite	Granite tile of any colour /shade	sqm	1524.00
11.62		Cramps in RCC/CC/Brick masonry	Providing and fixing cramps of required size and shape in RCC/ CC /Stone/ Brick masonry backing with cement mortar 1:2 (1 cement :2 sand) including drilling necessary hole in stones and embedding the cramp in the hole (fastener to be paid separately).		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	11.62.1	Gunmetal	Gunmetal cramps.	kg	589.00
	11.62.2	S.S.	Stainless steel cramps.	kg	631.00
11.63		Expansion hold fasteners on C.C. / R.C.C./ Brick masonry	Providing and fixing expansion hold fasteners of wedge expansion type on C.C. / R.C.C./ Brick masonry surface backing including drilling necessary holes and the cost of bolt etc complete.		
	11.63.1	6 mm.	Fastener with threaded dia 6 mm.	each	25.00
	11.63.2	10 mm.	Fastener with threaded dia 10 mm.	each	29.00
	11.63.3	12 mm.	Fastener with threaded dia 12 mm.	each	46.00
11.64		Stone slab in urinal partitions	Providing and fixing stone slab table rubbed on both faces edges rounded and polished of size as required and 1.8 cm thick fixed in urinal partitions by cutting a chase of appropriate width with chase cutter and embedding the stone in the chase with epoxy grout or with cement concrete 1:2:4 (1 cement: 2 sand: 4 graded stone aggregate 6 mm nominal size) as per direction of Engineer-in-charge and finished smooth.		
	11.64.1	White Agaria	White Agaria Marble Stone.	sqm	3046.00
	11.64.2	Granite Stone	Granite Stone of approved shade.	sqm	3422.00
11.65		Removable raised/false access flooring	Providing and fixing removable raised/false access flooring with system and its components of approved make for different plenum height with possible height adjustment upto 50 mm, comprising of modular load bearing floor panels supported on G.I. rectangular stinger frame work and G.I. Pedestal etc. all complete, as per the architectural drawings, as specified and as directed by Engineer-in-charge consisting of : **a)** Providing at required spacing to form modular framework, pedestals made out of GI tube of thickness minimum 2 mm and 25 mm outer diameter, fully welded on to the G.I. Base plate of size 100mmx100mmx3mm at the bottom of the pedestal tube, G.I. pedestal head of size 75mmx75mmx3.5 mm welded with GI fully threaded stud 16mm outer diameter with two GI Check nuts screwed on the stud for level adjustment upto50mm, locking and stabilizing the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			pedestal head in position at the required level. The pedestals shall be fixed to the subfloor (base) through base plate using epoxy based adhesive of approved make or the machine screw with rawl plug. **b)** Stringers system in all steel construction hot dipped galvanized of rectangular size 570x20x30x0.80mm thick having holes at both ends for securing the stringers on to the pedestal head using fully threaded screws ensuring maximum lateral stability in all directions, the grid formed by the pedestal and stringer assembly shall receive the floor panel, this system shall provide adequate solid, rigid support for access floor panel, the system shall provide a minimum clear uninterrupted clearance between the bottom of the floor for electrical conduits and wiring etc. all complete as per the architectural drawings, as specified and as directed by the Engineer-in-charge. **C)** Providing and fixing Access Floor panel of 600x600x32 mm medium grade Filled Steel anti static high pressure Lamination of 800H grade (FS800H). Access Floor panel shall be steel welded construction with an enclosed bottom pan with uniform pattern of 64 hemisphericalcones.The top and bottom plates of Steel Gauges: top 0.6 mm and bottom 0.7 mm fused spot welded together (minimum 64 welds in each dome and 20 welds along each flange). The panel should beCorroresist epoxy coated for lifetime rustprotection and cavity formed by the top and bottom plate is filled with Pyrogrip noncombustible Portland cementitious core mixed with lightweight foaming compound. The access floor shall be factory finished with Anti-static High Pressure laminate with Non Warp technology upto1mm thickness for superior adhesion and Surface flatness within 0.75mm.The panel is to withstand a Concentrated Load of 363 kgs applied on area 25mm x 25mm without collapse in the centre of the panel which is placed on four steel blocks. The panel will withstand and Uniformly Distributed Load (UDL) minimum 1250 kg/sqm and an impact load of 50kg all complete as per the approved manufacturersspecification and as per the direction of Engineer- in-charge. All specification must be printed on the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			side of the panel to ensure the quality of the product.		
	11.65.1		300 mm Finished Floor Height (FFH)	sqm	4002.00
	11.65.2		450 mm Finished Floor Height (FFH).	sqm	4044.00
11.66		Concrete interlocking paver block	Providing and laying factory made machine pressed precast cement concrete interlocking paver block of any shape and size conforming to IS:15658- 2006 and ISI marked of 60 mm thickness, M-30 grade with grey cement and pigment all as specified. Made by block making machine with strong vibratory compaction, laid in required colour andpattern over and including 50mm thick compacted bed of stone dust, filling the jointswith sand etc. all complete as per thedirection of Engineer-in-Charge.	sqm	642.00
11.67		Epoxy flooring self leveling	Providing Epoxy flooring self leveling type on strong base in desired colour/design as approved by Engineer-in-Charge. Resin for epoxy is to be of floor grade type i/c appropriate hardner and filler material. with 5 year defect liability period (This item is to be used only for operation theaters in hospitals)		
	11.67.1	1 mm	1 mm thick	sqm	1198.00
	11.67.2	2 mm	2 mm thick	sqm	1825.00
	11.67.3	3 mm	3 mm thick	sqm	2452.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.68		Heterogeneous 3.25mm thick acoustic PVC vinyl flooring	Providing and fixing acoustic PVC (Poly Vinyl Flooring) EN (European Standard) over fiberglass mat PVC foam backing (EN 14904), total thickness is 3.25mm (EN 428/EN 427), wear layeris 0.85 mm thick (EN 429), anti-bacterial and anti- fungus surface treatment (sanitized), reinforced, total weight is 3240g/m^2 approx. (EN 430), abrasion group T (EN660-1,660-2), dimentional stability <0.10% (EN-434), reaction to fire (EN 13501-1) Bfls1 on fibrecement substrate, static indentation (EN 433: 0.12mm), castor chair test qualified –no damage (EN-425), furniture leg test qualified (EN 424)-no damage, underfoot comfort (EN 433) ≥ 0.40mm, impact sound reduction(EN ISO 140-8,EN ISO 717-2)Δlw19Db, acoustical improvement (NF S31-074)-class A, certficate NF UPEC.A+(NF 189) - registered, no formaldehyde emission (EN 717-1), no pentachlorophenol (EN 12673), 100% recyclable (eco-friendly), includes 26% of recycled content, includes 52% natural product,low VOC emissions after 28days of installation (lower than ≤100µg/m3), static effect(EN ISO 846) - sanitized, chemical resistance test qualified (EN-423)-high resistance, thermal and underfloor heating test qualified (EN12524) - all vinyl sheets must be linked crosswise with a continuous copper strip and finally connected to earth for grounding electrostatic electrical charges. Suitable and slip resistant. Recommended to usein area like Lecturer room, Conference room, Kids area, School, colleges, Residential and all commercial official scope etc.	sqm	4221.00
11.69		Heterogeneous 2 mm thick Pvc vinyl flooring	Providing and fixing 2 mm thick heterogeneous PVC (poly vinyl flooring) EN (EuropeanStandard) with wear layer is 0.55mm thick of PVC, total weight is 2300 g/m^2 approx. PUR (polyurethane reinforced), residual indentation approx.0.05 mm, castor wheel test qualified, dimensional stability ≤0.10%, static electrical charge <2kv, all vinyl sheets must be linked crosswise with a continuous copper strip and finally connected to earth for grounding electrostatic electrical charges. good resistance to	sqm	2218.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			scratch, slip, chemical and fire, doesn't favour in growth of fungi & bacteria. 100% recyclable (eco-freindly). Low VOC emissions after 28days of installation (lower than ≤10µg/m3). phthalate free technology. directional in design.The floor finish should termite free at the room perimeter passing over a concealed cove forming and continuing up on the wall upto100mm. Recommended to use in area like Patient rooms , Doctor room Lecturer room, Kids area, School, Colleges, Residential and all commercial official scope etc.		
11.70		Laminated wooden flooring	Providing and fixing 8mm and above thick laminated (wooden flooring) of AC4 and above grade product of approved made out of core high density fibre board (HDF), surface layer shall be high preasure laminated treated for abrasion, scratch, moisture protection, flame protection U.V resistant, layer of acrylic resin, aluminium oxide or similer layer having above properties, backing under layer on min 2 mm thickness shall provide as sound absorbent and moisture protection layer, The seamless impregenated decorative layer of specified size, design, colour and pettern, all bonded together under high temperature and pressure with eco-friendly. The material shall be applicable warranty formaldehyde free, eco-friendly and fully recyclable (eco-freindly) and matching shed and excluding quantity of skirting and profile.	sqm	1758.00
11.71		Brick on edge flooring	Brick on edge flooring with bricks of class designation 7.5 on a bed of 12 mm cement mortar, 1:6 (1 cement : 6 sand) including filling the joints with same mortar, with common burntclay non modular bricks		
	11.71.1	100 mm	100 mm thick	sqm	480.00
	11.71.2	200 mm	200mm thick	sqm	960.00
11.72		Chamfered edge cement concrete paver blocks over	Providing and laying factory made chamfered edge Cement Concrete paver blocks in footpath, parks, lawns, drive ways or light traffic parking etc. of required colour strength, thickness pattern, surface texture (smooth/rough) and size/shape, conforming to 15658-2006 made by table vibratory method using PU mould, laid in required colour and pattern over 50mm thick compacted bed of sand, compacting and		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			proper embedding/ laying of inter locking paver blocks into the bedding layer through vibratory compaction byusing plate vibrator, filling the joints with sand and cutting of paver blocks as per required sizeand pattern, finishing and sweeping extra sand including locking with M15 cement concrete in footpath, parks, lawns, drive ways or light traffic parking etc. complete as per manufacturer's specifications and direction of Engineer-in-Charge.		
	11.72.1	80mm thick	80 mm thick C.C. paver block of M-35 grade	sqm	805.00
	11.72.2	60mm thick	60 mm thick C.C. paver block of M-35 grade	sqm	680.00
	11.72.3	60mm thick	60 mm thick C.C. paver block of M-30 grade	sqm	605.00
11.73		Kerb stone M-25	Providing and laying at or near ground level factory made kerb stone of M-25 grade cement concrete in position to the required line, level and curvature, jointed with cement mortar 1:3 (1 cement: 3 sand), including making joints with or without grooves (thickness of joints except at sharp curve shall not to more than 5mm), including making drainage opening wherever required complete etc. as per direction of Engineer-in-charge (length of finished kerb edging shall be measured for payment). (Precast C.C. kerb stone shall be approved by Engineer-in-Charge).	cum	6097.00
11.74		Precast RCC pavement slab	Providing and fixing at or near ground level factory made RCC pavement slab of M-30 gradeof size 450x450x50 mm, including reinforcement with 6 mm dia M.S. bars 4 nos on each side(top &bottom), including setting in position in footpathto the required level and line over a bed of 20 mm average thick cement mortar 1:5 (1 cement : 5 sand), having joint thickness not more than 5mm except on curve, including filling of joints with same cement mortar and making grooves etc. complete as per direction of Engineer-in-Charge.	sqm	718.00
11.75		Granite door frame	Providing and fixing 16mm thick gang saw cut mirror polished premoulded and prepolished machine cut Granite of approved shade, colourand texture of Granite Door frame. in cement mortar 1:3 (1 cement :3 sand) including pointing with an admixture of pigment to match the shade		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			including fixing of granite stone door frame with epoxy resin based adhesive including secured to the backing by means of cramps, drilling of hole with rawl plugs and screws 75mm long cleaning etc. complete. Including rubbing, curing, moulding and polishing edges to give high glossy finish and drilling of hole for fixing stainless steel hings, aldrop, towerboltetc complete at all levels as per instruction of Engineer-in-Charge.		
	11.75.1	100 mm wall	Granite Door frame for 100mm thick wall Thickness.	metre	1010.00
	11.75.2	200 mm wall	Granite Door frame for 200mm thick wall Thickness.	metre	1512.00
11.76		Vitrified floor tiles nano polished	Providing and laying vitrified floor tiles (without soluble salt) or printed (with soluble salt) Nano polished (Glossy finish) in different sizes with water absorption less than 0.05% and conforming to IS:15622 of approved make in all colours& shades, 1st quality/ premium quality in flooring laid on 20mm thick cement mortar 1:4 (1cement :4sand) including grouting the joints with white cement and matching pigments etc. complete		
	11.76.1	Light shade	Plain ivory light shade Size 600X600mm thickness 9-10mm	sqm	950.00
	11.76.2	Dark shade	Dark shade (terracotta) Size 600X600mm thickness 9-10mm	sqm	1033.00
11.77		Vitrified floor tiles nano polished skirting, dado and wall lining	Providing and laying vitrified floor tiles (without soluble salt) or printed (with soluble salt) Nano polished (Glossy finish) in different sizes with water absorption less than 0.05% and conforming to IS:15622 of approved make in all colours& shades, 1st quality/ premium quality in skirting dado and wall lining laid on 12mm thick cement mortar 1:4(1cement :4sand) including grouting the joints with white cement and matching pigments etc. complete		
	11. 77.1	Plain ivory	Plain ivory light shade Size 600X600mm thickness 9-10mm	sqm	940.00
	11. 77.2	Dark shade	Dark shade (terracotta) Size 600X600mm thickness 9-10mm	sqm	1022.00
11.78		Vitrified floor tiles multi charged	Providing and laying vitrified floor tiles Multi/Double charged (Minimum top layer thickness 2.0mm) Neno Polished self design, in different size with water absorption less than0.05% and conforming to IS:15622 of approved		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			make in all colours& shades, quality/ premium quality in flooring laid on 12mm thick cement mortar 1:4(1cement :4sand) including grouting the joints with white cement and matching pigments etc. complete		
	11.78.1	Light shade	In light shade glossy/ rustic /stain finished		
	11.78.1.1	600X600	Size 600X600 mm thickness 9-10 mm	sqm	1104.00
	11.78.1.2	800X800	Size 800X800mm thickness 11-12mm	sqm	1223.00
	11.78.1.3	800X1200	Size 800X1200 mm thickness 11-12mm	sqm	1413.00
	11.78.2	Dark shade	In dark shade glossy/ rustic /stain finished		
	11.78.2.1	600X600	(a) Size 600X600 mm thickness 9-10 mm	sqm	1169.00
	11.78.2.2	800X800	(b) Size 800X800mm thickness 11-12mm	sqm	1295.00
	11.78.2.3	800X1200	(c) Size 800X1200 mm thickness 11-12mm	sqm	1521.00
11.79		Vitrified floor tiles multi charged tiles in skirting, dado and wall lining	Providing and laying vitrified floor tiles Multi/Double charged (Minimum top layer thickness 2.0mm) Neno Polished self design, in different size with water absorption less than 0.05% and conforming to IS:15622 of approved make in all colours & shades, 1st quality/ premium quality in skirting, dado and wall lining laid on 12mm thick cement mortar 1:4(1cement :4sand) including grouting thejoints with white cement and matching pigments etc. complete		
	11.79.1	Light shade	In light shade glossy/ rustic /stain finished		
	11.79.1.1	600X600	(a) Size 600X600 mm thickness 9-10 mm	sqm	1093.00
	11.79.1.2	800X800	(b) Size 800X800mm thickness 11-12mm	sqm	1212.00
	11.79.1.3	800X1200	(c) Size 800X1200 mm thickness 11-12mm	sqm	1403.00
	11.79.2	Dark shade	In dark shade glossy/rustic/stain finished		
	11.79.2.1	600X600	(a) Size 600X600 mm thickness 9-10 mm	sqm	1159.00
	11.79.2.2	800X800	(b) Size 800X800mm thickness 11-12mm	sqm	1284.00
	11.79.2.3	800X1200	(c) Size 800X1200 mm thickness 11-12mm	sqm	1510.00
11.80		Vitrified floor-tiles full body	Providing and laying vitrified floor-tiles Full Body Design Homogenous Body matt/rustic finish in different size with water absorption less than		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			0.05% and conforming to IS:15622 of approved make in all colours& shades, 1st quality/ premium quality in flooring laid on 20mm thick cement mortar 1:4(1cement :4sand) including grouting the joints with white cement and matching pigments etc. complete		
	11.80.1	Light colour	In light colour and shades as ivory, beige blanco, grey etc.		
	11.80.1.1	300X300	(a) Size 300X300 mm thickness 9-10 mm	sqm	934.00
	11.80.1.2	600X600	(b) Size 600X600mm thickness 9-10mm	sqm	1206.00
	11.80.1.3	600X600	(c) Size 600X600 mm thickness 11-12mm (to be used heavy duty indoor areas of non residencial buildings only such as schools, hostels, libraries hospitals, institutions, offices etc.)	sqm	1349.00
	11.80.2	Dark colour	In dark colour and shade etc.		
	11.80.2.1	300X300	(a) Size 300X300 mm thickness 9-10 mm	sqm	946.00
	11.80.2.2	600X600	(b) Size 600X600mm thickness 9-10mm	sqm	1340.00
	11.80.2.3	600X600	(c) Size 600X600 mm thickness 11-12mm (to be used in heavy duty indoor areas of non residencial buildings only such as schools, hostels, libraries hospitals, institutions, offices etc.)	sqm	1546.00
11.81		Vitrified floor-tiles full body skirting, riser, dado and wall lining	Providing and laying vitrified floor-tiles Full Body Design Homogenous Body matt/rustic finish in different size with water absorption less than 0.05% and conforming to IS:15622 of approved make in all colours & shades, 1st quality/ premium quality in skirting, riser, dado and wall lining laid on 12mm thick cement mortar 1:4(1cement :4sand) including grouting the joints with white cement and matching pigments etc. complete		
	11. 81.1	Light colour	In light colour and shades as ivory, beige blanco, grey etc.		
	11. 81.1.1	300X300	(a) Size 300X300 mm thickness 9-10 mm	sqm	923.00
	11. 81.1.2	600X600	(b) Size 600X600mm thickness 9-10mm	sqm	1196.00
	11. 81.2	Dark colour	In dark colour and shades etc.		
	11. 81.2.1	300X300	(a) Size 300X300 mm thickness 9-10 mm	sqm	935.00
	11. 81.2.2	600X600	(b) Size 600X600mm thickness 9-10mm	sqm	1329.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.82		Vitrified tiles (PGVT)/(GVT)	Providing and laying digital print random design(PGVT)/(GVT) vitrified tiles in different sizes with water absorption less than 0.05% and conforming to IS:15622 of approved make in all colours& shades, 1st quality/ premium quality including grouting the joints with white cement and matching pigments etc. complete		
	11. 82.1	For floors	For floor glossy/ rustic /stain/matt finished laid on 20mm thick cement mortar 1:4 (1 cement:4sand)		
	11. 82.1.1	600X600	Size 600X600 mm thickness 9-10 mm	sqm	1140.00
	11. 82.1.2	800X800	Size 800X800mm thickness 9-10 mm	sqm	1261.00
	11. 82.1.3	800X1600	Size 800X1600mm thickness 9-10 mm	sqm	1489.00
	11. 82.1.4	1000X1000	Size 1000X1000mm thickness 9-10 mm	sqm	1388.00
	11. 82.1.5	600X1200	Size 600X1200 mm thickness 9-10 mm	sqm	1235.00
	11. 82.1.6	800X1200	Size 800X1200 mm thickness 9-10 mm	sqm	1450.00
	11. 82.1.7	1200X1200	Size 1200X1200 mm thickness 9-10 mm	sqm	1596.00
	11. 82.1.8	1200X1800	Size 1200X1800mm thickness 9-10 mm	sqm	1715.00
	11. 82.1.9	1200X2400	Size 1200X2400 mm thickness 9-10 mm	sqm	2403.00
11.83		Vitrified tiles (PGVT)/(GVT) in skirting, riser, dado and wall lining	Providing and laying digital print random design(PGVT)/(GVT) vitrified tiles in different sizes with water absorption less than 0.05% and conforming to IS:15622 of approved make in all colours& shades, 1st quality / premium in skirting, riser, dado and wall lining laid our 12mm thick cement mortar 113 including grouting the joints with white cement and matching pigments etc. complete		
	11. 83.1	For skirting, dado and wall lining	For skirting, riser, dado and wall lining etc. glossy/ rustic /stain/mattfinished in all colours and shades		
	11. 83.1.1	300X600	Size 300X600 mm thickness 9-10 mm	sqm	1048.00
	11. 83.1.2	600X600	Size 600X600 mm thickness 9-10 mm	sqm	1129.00
	11. 83.1.3	800X800	Size 800X800mm thickness 11-12mm	sqm	1250.00
	11. 83.1.4	600X1200	Size 600X1200 mm thickness 11-12mm	sqm	1224.00
	11. 83.1.5	800X1200	Size 800X1200 mm thickness 11-12mm	sqm	1440.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.84		Granite work in wall lining, skirting etc	Granite work gang saw cut (polished machine cut) of thickness 12 mm for wall lining, skirting, risers of steps and dado (veneer work), back filling with a grout of average 12 mm thick in cement mortar 1:3 (1 Cement: 3 sand) including pointing with white cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment to match the marble shade: (To be secured to the beaching by means of cramps, which shall be paid for separately).		
	11. 84.1	Area more than 0.18 sqm	Area of tiles not less 0.18 sqm		
	11. 84.1.1	Fine granite pattern	Fine granite dark Black/dark Red/ White with self design/ pattern/crystals of other colors of glitter	sqm	2296.00
	11. 84.1.2	Course granite patterns	Course granite light Black/light Red/ off White with crystals of other colors and various patterns	sqm	2113.00
	11. 84.1.3	Various patterns	Gray/Pink/Brown with various patterns	sqm	1879.00
11.85		Granite work in flooring	Providing and laying gang saw cut **12mm** thick, mirror polished pre moulded (wherever required) and pre polished machine cut granite stone of required size and shape of approved shade, colour and texture in flooring laid over 20mm thick base of cemnt mortar 1:4(1cement:4sand) including grouting the joints with white cement mixed with matching pigments, epoxy touch ups etc. complete as per direction of Engineer-in-Charge.		
	11. 85.1	Area more than 0.18 sqm	Area of tiles not less 0.18 sqm		
	11. 85.1.1	Fine grained granite pattern	Fine grained granite dark Black/dark Red/ White with self design/ pattern/crystals of other colors of glitter	sqm	1775.00
	11. 85.1.2	Coarse granite patterns	Coarse grained granite light Black/light Red/ off White with crystals of other colors and various patterns	sqm	1610.00
	11. 85.1.3	Various patterns	Gray/Pink/Brown with various patterns	sqm	1404.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11. 86		Artificial synthetic green grass	Providing and fixing artificial synthetic green long and fine yarns grass, UV/water resistant with SBR latex backing eco-friendly durable high density 100 % PP (Poly Poplin) anti UV fiber.denier 12000 dtex, stich rate 18 stiches per 10cm, stich density 16800 stiches/m2, tuftedweight 1830g/sqm approx.		
	11. 86.1	Artificial synthetic green grass – 25mm thick	Roll size will be 25.00m X 2.00m (25mm thick)- weight 150kg- per roll.	sqm	2265.00
	11. 86.2	Artificial synthetic green grass – 40mm thick	Roll size will be 25.00m X 2.00m (40mm thick)- weight 180kg- per roll.	sqm	2656.00
11.87		Laminated (engineered wooden flooring) 10-14mm thick	Providing and fixing laminated (engineered wooden flooring) of suitable thickness 10-14 mm of suitable grade category product of approved made out of core high density fibre board (HDF) &other component it treated for abrasion ,scratch ,moisture protection, flame protection,U.V resistant. backing under layer on min 2 mm thickness shall provide as sound absorbent and moisture protection layer .The Seamless impregenated decorative layer of specified size. design, colour and pettern, all bonded together under high temperature and pressure with eco-friendly resins specified by the manufacturer .The material shall be applicable warranty formaldehyde free, eco friendly, highly recommended for high traffic area and fully recyclable and matching shed and excluding quantity of skirting and profile.	sqm	7459.00
11.88		Homogeneous (solid wooden flooring)- 14-18mm thick	Providing and fixing single layer homogeneous (solid wooden flooring) of suitable thickness 14 - 18 mm of suitable grade category product of scratch protection, moisture protection, flame protection U.V. resistant, backing under layer on min 2 mm thickness shall provide as sound absorbent and moisture protection layer of specified design,colour and pattern, all bonded together under high temperature and pressure with eco-friendly resins specified by the manufacturer .The material shall be applicable warranty formaldehyde free, eco friendly ,highly recommended for high traffic area and fully	sqm	9801.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			recyclable and matching shade and excluding quantity of skirting and profile.		
11.89		Carpet tile / carpet	Providing and fixing tuffed 1/10" loop pile (carpet tile / Carpet). EN (Euorpean Standard) size 50cm x 50cm /roll Size of total thickness 5.5 mm and above with total mass of 4100g/m2approx and with polyester fleece primary backing with standard dimensional stability with impact sound / noise reduction and recommended for high traffic acoustical area.	sqm	3233.00
11.90		Stone plastic composite (pvc vinyl flooring)	Providing and fixing spc (stone plastic composite) virgin pvc water resistant proof vinyl flooring with a thickness between 4 to 8mmuv coated layerwith a wear layer of 0.3 to 0.80mm with a SPC core layer of 2.00 to 5.00 mm with a SOP of heat resistant, voc free, easy to cut, anti microbial protection in the form of tiles, planks helpful in environmental of profection health.	sqm	4230.00
11.91		Fine grain sand stone tile flooring 20 mm thick	Providing and laying pre mirror polished/ pre polished fine grain machine cut sand stone tile flooring 20mm thick (hardness should not be less than 7 and water absorption should not be more than 0.3%) of required size and shape of approved shade, flooring in plazas open staircase and similar locations, laid over 20 mm thick base of cement mortar 1:4 (1 cement : 4 coarse sand) including grouting the joint with white cement mix with matching pigment etc. complete as per direction of Engineer-in-Charge.		
	11. 91.1	300x300mm	300x300mm	sqm	1182.00
	11. 91.2	400x400mm	400x400mm	sqm	1207.00
	11. 91.3	600x600mm	600x600mm	sqm	1328.00
11.92		Fine grain sand stone slab flooring 20 mm thick	Providing and laying pre mirror polished/ pre polished fine grain machine cut sand stone slab flooring 20mm thick (hardness should not be less than 7 and water absorption should not be more than 0.3%) of required size of area more then .50 sqm and shape of approved shade, flooring in plazas open staircase and similar locations, laid over 20mm thick base of cement mortar 1:4(1 cement : 4 coarse sand) including grouting the joint with cement mix matching pigment etc. complete as per direction of Engineer -in-Charge.	sqm	1686.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.93		Vinyl flooring of 4.5 mm thick	Supply and application of vinyl flooring of 4.5 mm thick with 1.5 mm wear layer comes in oll size of (1.8mx15m) available in different color options. Moldo PVC vinyl cushion indoor flooring comes with I indentation of less than 0.5 and also have good abrasion and impact resistance. It also have better shock absorption property of more than 35 and good ball rebounce property of about 98%.	sqm	2086.00
11.94		Adhesive PU-Tactile tiles	Providing and fixing coloured self adhesive PU-Tactile tiles, ground surface indicators made of Polyurethane of size 300x300mm Comforming to AST MD 1204,ASTMD 638,ASTMD 4060 ASTMD 2240 for guiding/warning to the visually Impaired vistiors in the center/side of the pedestrain/ approch path way as per the specification and Harmonised guidline complete in all respect to the entire satisfaction of Engineer-in-charge.	sqm	3260.00
11.95		Woven vinyl Acoustical floorings	Providing and fixing Woven vinyl Acoustical floorings with roll dimensions of 1.8/2mtr meters width and length of 20 meters or Tiles size as per company standard should be approved by architech. 2.50-2.80 mm thickness to laid on hard, levelled, smooth and moisture free sub floor. Woven Vinyl flooring should have traffic classification 33 and weight of 2.7-3.5 kg/m2. Flooring should have caster chair EN-985 tested with color fastness of more than 6. It should be ASTME662 qualified on Smoke density test. Product must be fire rating of the product should be Bfl-S1. Any Joints in flooring rolls should be sealed with cold welding. This work will be inclusive of all Labour, Raw Material, Hardware, adhesive etc. complete. Flooring will be provided as per approved shade and colour in ,as per the satisfaction of the Architect and Engg. Incharge.	sqm	2889.00
11.96		Teak wood flooring system for Auditorium Stage / Sports flooring	Supply and Installation of Teak wood flooring system for Auditorium Stage / Sports flooring SURFACE BOARD: - Teak Wood Sports Flooring - Grade 1"LABO SPORTS Test Certify finished with 21 mm thick, 55 to 85mm wide and in random length in tongue and groove shape. The edges of the boards will have a finger lock groove and the bottom side with air pass groove and treated with special anti-termite and water-resistant lacquer. THE UNDER FRAME: - Resilient SPF Pine Wood sleeper subfloor of 70 mm x 45 mm, treated with anti-termite solution and fixed on the bottom side with 10 mm thick EPDM rubber pads, stapled through the two wings at 350 mm x 350 mm	Sqm	9480.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			INSTALLATION: - IPS subfloor treated with a vapour barrier to be placed on the levelled IPS sub-floor before laying the under frame. The runners having air cushion pads to be placed on the vapour barrier in perfect level at 350 mm in 1 direction. The surface board to be screwed to the runner through the tongue only and will lock the screwed tongue by the groove ofadjoining board. Ends of the boards will be locked by inserting the wooden fingers through the edge grooves and fixed with suitable adhesive. The expansion of 12-15 mm will be left open between the wooden flooring and the surrounding. **Note:-** The item shall be executed only after the permission of Chief Engineer.		
11.97		Digital print random design vitrified (germ free technology) tile	Providing and laying digital print random design vitrified (germ free technology) tiles in different sizes with water absorption less than 0.06% and conforming to IS:15622 of approved make in all colours & shades, 1st quality/ premium quality including grouting the joints with white cement and matching pigments etc. complete.		
		800x1600mm	Size 800X1600 mm thickness 8-9 mm	Sqm	1778.00
11.98		18mm thick polished granite wall cladding with 6 mm V grove for planter	Providing and Fixing 18mm thick polished granite wall cladding backing filled with a cement mortar 1:3 (1 cement: 3 sand)of 12 mm average thickness including pointing with white cement mortar 1:2 (1 white cement: 2 marble dust) with an admixture of pigment to match the granite shade and special design pattern cladding as per architectural drawing 75 mm thick vertical strip with 6 mm V grove and 50mm border horizontal strip with gap of 30mm all work complete as per directed by architect/engineer in charge.	Sqm	5705.00
11.99		Full body vitrified tiles for wall cladding /skirting/ dado	Providing and laying full body vitrified Wall–tiles, matt/rustic finish in different size with conforming to IS:15622 , of approved make in all colour & shades, 1st quality/ premium quality in cladding /skirting/ dado laid on 12 mm thick cement mortar 1:3(1cement :4sand) including grouting the joints with white cement and matching pigments etc. Complete.		
		Size 600 X1200mm	Size 600 X1200 mm thickness 9-10 mm	Sqm	2491.00
11.100		Self-adhesive PU-Tactile tile (with 3 year warranty)	Providing and fixing coloured self-adhesive PU-Tactile tile (with 3 year warranty)as ground surface indicators (made of Polyurethane) grade 70/85 of size 300mmx300mm for guiding/direction/warning to the visually impaired visitors in the center/side of the pedestrian/approach pathway as per design/specification and guidelines complete in all respect to the entire satisfaction of Engineer-in-charge.	Sqm	3500.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
11.101		Lapatto Finish Kota Stone Flooring	Providing and laying Pre Polished KOTA STONE (shorted for matching color) in tiles size Area of slab to be over 0.25 sqm , kota stone first to be sizes in perfect right angle (with straight edge for perfect butt joint) calibrated for equal thickness of 18mm laid over 20mm thick average base with base of cement mortar 1 : 4 (1 cement : 4 sand) and Grouting the joints of 3 mm width using epoxy grout mix of 0.70 kg of organic coated filler of desired shade (0.10 kg of hardener and 0.20 kg of resin per kg). Including filling/ grouting and finishing complete. In order to get these type of high equipped of 14 head liner machine with highly imported abrasive Range 0/36/60/120/220/300/400/ 600/800/1600/5000 as per the required finished with end process with buffing for high polished stones.	sqm	1285.00
11.102		Sunk filling in toilet and balcony with Crushed sand / sand	Supplying and filling in sunken areas (toilet and balcony etc.) with Crushed sand / sand after water proofing and installation of spout.	cum	472.00
11.103		Add Extra for tile laying with 3mm spacers.	Add extra for tile laying with spacers and filled by Grouting the joints of flooring/ skirting / wall dado/ riser of step of tiles having joints of 3 mm width, using epoxy grout mix of 0.70 kg of organic coated filler of desired shade (0.10 kg of hardener and 0.20 kg of resin per kg), including filling / grouting and finishing complete as per direction of Engineer-in-charge.	Sqm	5% of respective item

CHAPTER – 12.0

ROOFING AND CEILING

.82LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1.	IS 73	Specification for paving Bitumen
2.	IS 277	Galvanised steel sheets (plain and corrugated)
3	IS 651	Glazed stoneware pipes and fittings
4.	IS 702	Specification for industrial bitumen
5.	IS 1199	Methods of sampling and analysis of concrete
6.	IS 1200 (Part-IX)	Method of measurements of building and civil engineering works: Part - 9 Roof covering (including cladding)
7.	IS 1200 (Part-X)	Method of measurements of building and civil engineering works: Part -10 ceiling and lining
8	IS 1230	Cast iron rain water pipes and fitting
9.	IS 1367 (Part-13)	Technical supply conditions for threaded steel fasteners pt.13 hot dip galvanized coating on threaded fasteners
10	IS 2095 (Part-1)	Gypsum plaster boards (Pt.1) plain Gypsum plaster boards
11.	IS 2115	Code of practice for flat roof finish: mud phuska
12.	IS 2633	Method of testing uniformity of coating on zinc coated articles
13.	IS 2645	Specification for integral water proofing compounds for cement mortar and concrete
14.	IS 3007 (Part-1)	Code of practice for laying of asbestos cement sheets: part- 1 corrugated sheets
15.	IS 3007 (Part-2)	Code of practice for laying of asbestos cement sheets part- 2 semi-corrugated sheets
16.	IS 3087	Particle boards of wood and other lignocellulogic materials (medium density) for general purposes - specifications
17.	IS 3144	Methods of test for mineral wool thermal insulation materials
18.	IS 3346	Method of the determination of thermal conductivity of thermal insulation materials
19.	IS 3348	Specification for fibre insulation boards
20.	IS 3384	Specification for bitumen primer for water proofing and damp proofing
21.	IS 4671	Expanded polystyrene for thermal insulation purposes
22	IS 5382	Specification for rubber sealing rings for gas mains, water mains and sewers
23.	IS 5688	Methods of test of performed block type and pipe covering type thermal insulations
24.	IS 6598	Cellular concrete for thermal insulation
25.	IS 7193	Specification for glass fibre base coal tar pitch and bitumen felts (Amendment I)
26.	IS 8183	Bonded mineral wool
27.	IS 10192	Specifications for synthetic resin bonded glass fibre (SRBGF) for electrical purposes.
28.	IS 13592	Unplasticised polyvinyl chloride (UPVC) pipes for soil and Waste discharge system for inside and outside building.
29.	IS 14753	Specifications for polymethyl Methacrylate (PMMA) (Acrylic) sheets
30	IS 14862	Fibre cement flat sheets – specifications
31	IS 14871	Specifications for products in fibre reinforced cement - Long corrugated or Asymmetrical section sheets and fittings for roofing and cladding.

PREAMBLE
ROOFING AND CEILING
(CHAPTER: 12.0)

CORRUGATED GALVANISED STEEL SHEET ROOFING

12.1 C.G.S. Sheets

These shall be of the thickness specified in the description of the item and shall conform to IS 277. The sheets shall be of 275 grade of coating unless otherwise specified in the description of item. The sheets shall be free from cracks, split edges, twists, surface flaws etc. They shall be clean, bright and smooth. The galvanising shall be non-injured and in perfect condition. The sheets shall not show signs of rust or white powder deposits on the surface. The corrugations shall be uniform in depth and pitch and parallel with the side.

12.2 Slope

Roof shall not be pitched at a flatter slope than 1 vertical to 5 horizontal. The normal pitch adopted shall usually be 1 vertical to 3 horizontal

12.3 Laying and Fixing

The sheets shall be laid and fixed in the manner described below, unless otherwise shown in the working drawings or directed by the Engineer-in-Charge.

The sheets shall be laid on the purlins to a true plane, with the lines of corrugations parallel or normal to the sides of the area to be covered unless otherwise required as in special shaped roofs.

The sheets shall be laid with a minimum lap of 15 cm at the ends and 2 ridges of corrugations at each side. The above minimum end lap of 15 cm shall apply to slopes of 1 vertical to 2 horizontal and steeper slopes. For flatter slopes the minimum permissible end lap shall be 20 cm. The minimum lap of sheets with ridge, hip and valley shall be 20 cm measured at right angles to the line of the ridge, hip and valley respectively. These sheets shall be cut to suit the dimensions or shapes of the roof, either along their length or their width or in a slant across their lines of corrugations at hips and valleys. They shall be cut carefully with a straight edge chisel to give a smooth and straight finish.

Lapping in C.G.S. sheets shall be painted with a coat of approved steel primer and two coats of painting with approved paint suitable for G.S. sheet, before the sheets are fixed in place. Sheets shall not generally be fixed into gables and parapets. They shall be bent up along their side edges close to the wall and the junction shall be protected by suitable flashing or by a projecting drip course, the later to cover the junction by at least 7.5 cm.

The laying operation shall include all scaffolding work involved.

Sheets shall be fixed to the purlins or other roof members such as hip or valley rafters etc. with galvanised J or L hook bolts and nuts, 8 mm diameter, with bitumen and G.I. limpet washers or with a limpet washer filled with white lead as directed by the Engineer-in- Charge. While J hooks are used for fixing sheets on angle iron purlins, and L hooks are usedfor fixing the sheet to R.S. joists, timber or precast concrete purlins. The length of the hook bolt shall be varied to suit the particular requirements. The bolts shall be sufficiently long sothat after fixing they project above the top of the nuts by not less than 10 mm. The grip of J or L hook bolt on the side of the purlin shall not be less than 25 mm. There shall be a minimum of three hook bolts placed at the ridges of corrugations in each sheet on every purlin and their spacing shall not exceed 30 cm. Coach Screws shall not be used for fixing sheets to purlins.

The galvanised coating on J or L hooks, and bolts shall be continuous and free from defects such as blisters, flux stains, drops, excessive projections or other imperfections which would impair serviceability. Where slopes of roofs are less than 21.5 degrees (1 vertical to 2.5 horizontal) sheets shall be joined together at the side laps by galvanised iron bolts and nuts 25 × 6 mm size, each bolt provided with a bitumen and a G.I. limpet washer or a G.I. limpet washer filled with white lead. As the overlap at the sides extends to two corrugations, these bolts shall be placed zig-zag over the two overlapping corrugations, so that the ends of the overlapping sheets shall be drawn tightly to each other. The spacing of these seam bolts shall not exceed 60 cm along each of the staggered rows. Holes for all bolts shall be drilled and not punched in the ridges of the corrugations from the underside, while the sheets are on the ground.

NON-ASBESTOS HIGH IMPACT FIBER REINFORCED CEMENT CORRUGATED SHEET ROOFING

12.4 Non-Asbestos High Impact Fiber Reinforced Cement Corrugated Sheets

The sheets shall be of the approved quality and shall conform to IS 14871. The sheets shall be free from cracks, chipped edges or corners and other damages.

Classification – Sheets may be classified according to thickness as under:

Type A - The thickness of the sheets shall be approximately constant throughout the width of profile.

Type B - The thickness of the sheets shall vary regularly between the valley and the crown for corrugated sheets or between the lower part and the upper part of ribs for asymmetrical section sheets, in the same cross-section. The sheets shall be categorized based on height of corrugations, ‘h’ and minimum thickness ‘e’ as under:

TABLE 12.1
Category and Class (Minimum Breaking Load N/m)

Category	*Minimum Thickness e (mm)*	*Class*									
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A (15 mm < h < 55 mm)	3	600	800	1000	1400	-	-	-	-	-	-
B (25 mm < h < 55 mm)	4			1000	1400	2000	2500	3300	-	-	-
C (40 mm < h < 80 mm)	4.5				1400	2000	2500	3300	4250	-	-
D (60mm < h < 150 mm)	5.5							3300	4250	5600	7400

The free overhang of the sheets at the eaves shall not exceed 30 cm. Corrugated sheets shall be laid from left to right starting at the eaves. The first sheet shall be laid uncut but the remaining sheets in the bottom row shall have the top left hand corners cut or mitred. The sheets in the second and other intermediate rows except the first and the last sheets shall have both the top left hand corner and bottom right hand corner cut. The last or top row sheets shall all have the bottom right hand corner cut with the exception of the last sheet which shall be laid uncut. If for any reason such as on considerations of the direction of prevailing winds, laying is to be started from the bottom right hand corner, then the whole procedure should be reversed

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.1		Corrugated G.S. sheet roofing	Providing corrugated G.S. sheet roofing including vertical/ curved surface fixed with polymer coated J or L hooks, bolts and nuts 8 mm diameter with bitumen and G.I. limpet washers or with G.I. limpet washers filled with white lead and including a coat of approved steel primer and two coats of approved paint on overlapping of sheets complete upto any pitch in horizontal/ vertical or curved surfaces) excluding the cost of purlins, rafters and trusses and including cutting to size and shape wherever required.		
	12.1.1	1.00 mm thick	1.00mm thick with zinc coating not less than 275gm/m²	sqm	1025.00
	12.1.2	0.80 mm thick	0.80mm thick with zinc coating not less than 275gm/m²	sqm	863.00
	12.1.3	0.63 mm thick	0.63 mm thick with zinc coating not less than 275gm/m²	sqm	726.00
12.2		Straight cutting in G.S. sheet roofing	Straight cutting in G.S. sheet of any thickness roofing for making opening of area exceeding 40 sq. decimetre for chimney stacks, sky light etc.	meter	25.00
12.3		Circular cutting in G.S. sheet roofing	Extra for circular cutting in C.G.S. sheet of any thickness roofing for making opening of area exceeding 40 square decimetre :	meter	150.00
12.4		G.S.Sheet ridges or hips	Providing ridges or hips of width 60 cm over all width plain G.S. sheet fixed with polymer coated J or L hooks, bolts and nuts 8 mm dia. G.I. limpet and bitumen washers complete.		
	12.4.1	0.80mm thick	0.80mm thick with zinc coating not less than 275gm/m²	meter	500.00
	12.4.2	0.63mm thick	0.63mm thick with zinc coating not less than 275gm/m²	meter	431.00
12.5		G.S. valleys	Providing valleys of 90cm wide overall in plain G.S. sheet fixed with polymer coated J or L hooks, bolts and nuts 8mm dia. G.I. limpet and bitumen washers complete. 1.60 mm thick with zinc coating not less than 350gm/m²	meter	994.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.6		Plain G.S. sheet	Providing and Fixing of 40 cm over all width plain, G.S. sheet fixed with polymer coated J or L hooks, bolts and nuts, G.I. limpet and bitumen washer complete, bent to shape and fixed in wall with cement mortar 1:3 (1cement : 3 sand). 1.00mm thick with zinc coating not less than 275gm/m²	meter	355.00
12.7		Semicircular plain G.S. sheet gutter	Providing and fixing 15 cm wide 45 cm over all semicircular plain G.S. sheet gutter with iron brackets 40x3mm size, bolts, nuts and washers etc. including making necessary connections with rain water pipescomplete.		
	12.7.1	0.80mm thick	0.80mm thick with zinc coating not less than 275gm/m²	meter	459.00
	12.7.2	0.63mm thick	0.63mm thick with zinc coating not less than 275gm/m²	meter	402.00
12.8		Corrugated sheets reinforced by organic and/or inorganic synthetic fibres	Providing reinforced by organic fibres and/or inorganic synthetic fibres cement 6 mm thick corrugated sheets (as per IS: 14871) roofing up to any pitch and fixing with polymer coated J, or L hooks, bolts and nuts 8 mm dia. G.I. plain and bitumen washers or with self drilling fastener and EPDM washers etc. complete (excluding the cost of purlins, rafters and trusses), including cutting sheets to size and shape wherever required.	sqm	389.00
12.9		Straight cutting in Corrugated sheets reinforced by organic and/or inorganic synthetic fibres	Extra for straight cutting in reinforced by organic fibres and/or inorganic synthetic fibres cement corrugated, semi corrugated 6 mm thick sheet roofing for making openings of area exceeding 40 square decimeter for chimney stacks, skylights etc..	meter	20.00
12.10		Circular cutting in Corrugated sheets reinforced by organic and/or inorganic synthetic fibres	Extra for circular cutting in reinforced by organic fibres and/or inorganic synthetic fibres cement corrugated/ semi corrugated 6 mm thick sheet roofing for making openings of area exceeding 40 square decimeter.	meter	56.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.11		Extra for wind ties	Extra for providing and fixing wind ties of 40x 6 mm flat iron section.	meter	121.00
12.12		Ridges and hips in Corrugated sheets reinforced by organic and/or inorganic synthetic fibres	Providing and fixing ridges and hips in fibre cement reinforced by organic fibres and/or inorganic synthetic fibres roofing with suitable fixingaccessories or self drilling fastener and EPDM washer etc. complete.		
	12.12.1	Serrated adjustable ridge	Corrugated serrated adjustable ridges	meter	367.00
	12.12.2	Plain	Plain wing adjustable ridges	meter	311.00
	12.12.3	Close fitting	Close fitting adjustable ridges	meter	420.00
	12.12.4	Unserrated	Unserrated adjustable hips	meter	506.00
12.13		Corrugated sheets reinforced by organic and/or inorganic synthetic fibres cement roofing	Providing and fixing fibre cement reinforced by organic fibres and/or inorganic synthetic fibres roofing accessories in all colours with polymer coated J or L hooks, bolts and nuts and or G.I. seam bolts and nuts, G.I. plain and bitumen washers or with self drilling fastener and EPDM washer etc. complete:		
	12.13.1	Apron	Corrugated apron pieces	meter	340.00
	12.13.2	Eave filler	Eave's filler pieces	meter	232.00
	12.13.3	North light	North light curves	meter	500.00
	12.13.4	Ventilator	ventilator curves	meter	622.00
	12.13.5	Barge	Barge boards	meter	623.00
	12.13.6	Ridge	Ridge finials	pair	229.00
	12.13.7	Special north	Special north light curves	each	753.00
	12.13.8	S-type	S type louvers	meter	418.00
12.14		Flat iron brackets	Providing flat iron brackets 50x3mm size with necessary bolts, nuts and washers etc. for fixing G.S. sheets gutters with purlins.	meter	49.00
12.15		Precoated G.I. profile sheets	Providing and fixing precoated galvanised iron profile sheets (size, shape and pitch of corrugation as approved by Engineer-in-charge) 0.50 mm + 0.05 %, total coated thickness with zinc coating 120 gsm as per IS: 277 in 240 mpa steel grade, 5-7	sqm	652.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns. Sheet should have protective guard film of 25 microns minimum to avoid scratches while transportation and should be supplied in single length upto 12 metre or as desired by Engineer-in-charge. The sheet shall be fixed using self-drilling /self-tapping screws of size (5.5x 55mm) with EPDM seal, complete upto any pitch in horizontal/ vertical or curved surfaces excluding the cost of purlins, rafters and trusses and including cutting to size and shape wherever required.		
12.16		Precoated G.S. sheet roofing	Providing and fixing precoated galvanised steel sheet roofing accessories 0.50 mm + 0.05 % total coated thickness, Zinc coating 120gsm as per IS: 277 in 240 mpa steel grade, 5-7 microns epoxy primer on both side of the sheet and polyester top coat 15-18 microns using self-drilling/ self-tapping screws complete :		
	12.16.1	Ridge	Ridges plain (500 - 600mm).	meter	320.00
	12.16.2	Flashing	Flashings/ Aprons.(Upto 600 mm)	meter	284.00
	12.16.3	North light	North light curves.	meter	341.00
	12.16.4	Barge board	Barge board (Upto 300 mm)	meter	274.00
	12.16.5	Crimp curve	Crimp curve	sqm	315.00
	12.16.6	Gutter	Gutter. (600 mm over all girth)	meter	781.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.17		UV stabilized fibreglass reinforced plastic sheet roofing	Providing & fixing UV stabilised fiberglass reinforced plastic sheet roofing up to any pitch, including fixing with polymer coated 'J' or 'L' hooks, bolts & nuts 8mm dia. G.I plain/bitumen washers complete but excluding the cost of purlins, rafters, trusses etc. The sheets shall be manufactured out of 2400 TEX panel rovigs incorporating minimum 0.3% ultra-violet stabiliser in resin system under approximately 2400 psi and hot cured. They shall be of uniform pigmentation and thickness without air pockets and shall conform to IS 10192 and IS 12866.The sheets shall be opaque or translucent, clear or pigmented, textured or smooth as specified.		
	12.17.1	Corrugated	2mm thick corrugated (2.5" or 4.2" or 6") or step-down (2"or 3"or 6") as specified.	sqm	1031.00
	12.17.2	Flat	2mm thick flat.	sqm	955.00
12.18		Sky lights in polycarbonate panel	Providing and Fixing of sky lights consisting of Multi cell/tight cell Polycarbonate Panel System of approved colour, 16 mm thick (minimum) having uniform in color with an integral Tight-Cell core constructed not to exceed 4mmx4mm in a cross section, Vertical Standing Seam manufactured at both sides ofthe panel. Snap-on connector to interlock the panels shall have a grip- lock double tooth locking mechanism to ensure maximum uplift capabilityand shall be of same color as that of panel. Panel shall be factory sealed/end welded panels with additional End-cap/Aluminium U- Profile (mill finish) for ends. Panel shall be co-extruded with special anti- glare compound and UV protected. The full system shall be fitted on MS purlins perpendicular to direction of sheeting with purlin spacing as		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			specified by Manufacturer. The rate includes cost of all the operations, labour and all materials and tests (as applicable) involved such as bolts nuts and screws etc. and labour for cutting bending to required profile, necessary scaffolding, hoisting in position etc. for proper completion of the work etc. complete as per specification drawings and direction of Engineer in charge. Finished surface area of roofing fixed over steeltubular structure shall be measured for payment. MS tabular frame work shall be measured separately for payment.		
	12.18.1	10mm thick	10mm thick	sqm	3105.00
	12.18.2	16mm thick	16mm thick	sqm	3546.00
12.19		Sand stone slab for roofing	Providing sand stone slab 40 to 50 mm thick for roofing and laying them in cement mortar 1 : 4 (1 cement : 4 sand) over wooden karries or R.C.C. battens or structural steel sections (Karries or battens or structural steel sections to be paid separately) including pointing the ceiling joints with cement mortar 1:3 (1 cement : 3 sand) complete :		
	12.19.1	Red stone	Red sand stone slab	sqm	441.00
	12.19.2	White stone	White sand stone slab :	sqm	504.00
12.20		Painting top of roofs with bitumen Bitumen painting roofs	Painting top of roofs with residualtype petroleum bitumen of grade VG 10 of approved quality @ 17kg per 10 sqm impregnated with a coat of sandat 60 cum per 10sqm including cleaning the slab surface with brushes and finally with a piece of cloth lightlysoaked in kerosene oil complete :	sqm	132.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.21		Two coats high albedo paint	Providing and applying two coats of High Albedo paint having minimum Solar Reflective Index (SRI) 108 (with solar reflectance and thermal emittance tested as per ASTM) C 1549 and ASTM C 1371 respectively),VOC less than 10 cc/gm. The coating thickness and the methodology of application shall be strictly as per manufacturer's specifications and as approved by engineer In charge. Surface preparation includes cleaning with metal wire brush to remove all dust, fungus etc., and washing with water all complete. The contractor shall give guarantee for the performance of SRI and also the durability of coating, all complete as per direction of Engineer-in-charge.	sqm	239.00
12.22		Heat resistant terrace tiles	Providing and fixing Heat Resistant Terrace Tiles (300 mm x 300 mm x 20 mm) with SRI (solar refractive index) > 78, solar reflection >0.70 and initial emittance >0.75 on waterproof and sloped surface of terrace, laid on 20 mm thick cement sand mortar in the ratio of 1:4 (1 cement : 4 sand) and grouting the joints with mix of white cement and marble powder in ratio of 1:1, including rubbing and polishingof the surface upto 3 cuts complete, including providing skirting upto 150 mm height along the parapet walls in the same manner.	sqm	1218.00
12.23		Pressed clay tiles as pattern or roof (with mortar)	Providing and laying pressed clay tiles (as per approved pattern 20 mm nominal thickness and of approved size) on roofs jointed with cement mortar 1:4 (1 cement : 4 sand) mixed with 2% integral water proofing compound laid over a bed of 20 mm thick cement mortar 1:4 (1 cement : 4 sand) and finished neat complete.	sqm	473.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.24		Mangalore tile on roof (on steel frame)	Providing and fixing on roof pressed clay tile (Mangalore tile) 20 mm nominal thickness and of approved size and as per approved pattern onsteel frame work complete (steel frame work to be paid separately)	sqm	275.00
12.25		Pressed clay tile Mangalore ridge (on steel frame)	Providing and laying on roof pressed clay tile ridge (Mangalore tile) of 20mm thickness of approved patternon steel frame work complete (steelframe work to be paid separately)	sqm	155.00
12.26		Gola	Providing gola 75x75 mm in cement concrete 1:2:4 (1 cement : 2 sand : 4 stone aggregate 10mm and down grade) including finishing with cement mortar 1:3 (1 cement : 3 sand) as per standard design : In 75x75mm deep chase	meter	99.00
12.27		Khurras	Making khurras 45x45 cm (sunk area) with average minimum thickness of 5 cm cement concrete 1:2:4 (1 cement : 2 sand : 4 graded stone aggregate of 20 mm nominal size) over P.V.C. sheet 1mx1mx400micron, finished with 12mm cement plaster 1:3 (1 cement : 3 sand) and a coat of neat cement rounding the edge and making and finishing the outlet complete so as to receive rain water/ spillage water on roof in a sunk area and completely rain out in a near perfect strain meter motion into rain water pipe.	each	156.00
12.28		M.S. holder bat clamps	Providing and fixing M.S. holder bat clamps of approved design to C.I. or S.C.I. rain water pipes embedded inand including cement concrete blocks 10x10x10cm of Cement Concrete 1:2:4 mix (1 cement : 2 sand : 4 graded stone aggregate 20 mm nominal size) and cost of cutting holes and making good the walls etc.		
	12.28.1	100 mm	100 mm diameter	each	107.00
	12.28.2	150 mm	150 mm diameter	each	112.00
12.29		Sand cast iron pipes.	Providing and fixing ISI marked sand cast iron rain water pipes.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.29.1	100 mm	100 mm dia minimum wall thickness 5mm	meter	871.00
	12.29.2	150 mm	150 mm dia minimum wall thickness 5mm	meter	1081.00
12.30		Lead caulked joints	Providing lead caulked joints to sand cast iron rain water pipes and fittings :		
	12.30.1	100 mm	100 mm dia. Pipe	each	221.00
	12.30.2	150 mm	150 mm dia. Pipe	each	309.00
12.31		Sand cast iron rain water pipe accessories	Providing, fixing and embedding sand cast iron accessories for rain water pipes in the masonry surrounded with 12 mm thick cement mortar of the same mix, as that of masonry (leadcaulking will be paid for separately) :		
	12.31.1	Bend	Sand cast iron bend :		
	12.31.1.1	100 mm	100 mm diameter	each	470.00
	12.31.1.2	150 mm	150 mm diameter	each	383.00
	12.31.2	plain shoes	Sand cast iron plain shoes :		
	12.31.2.1	100 mm	100 mm diameter	each	488.00
	12.31.2.2	150 mm	150 mm diameter	each	412.00
12.32		Unplasticised rigid PVC pipes	Providing and fixing on wall face Rigid Unplasticised rigid PVC (upvc) Single socketed rain water pipes conforming to IS : 13592 Type A including jointing with seal ring conforming to IS : 5382 leaving 10 mm gap for thermal expansion.		
	12.32.1	75 mm	75 mm diameter (minimum wall thickness 1.8mm)	metre	136.00
	12.32.2	110 mm	110 mm diameter (minimum wall thickness 2.2mm)	metre	238.00
12.33		Unplasticised - UPVC fittings	Providing and fixing on wall face Unplasticised - UPVC moulded fittings/ accessories for Unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A including jointing with seal ring conforming to IS : 5382 leaving 10 mm gap for thermal expansion.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.33.1	Coupler	Coupler		
	12.33.1.1	75 mm	75 mm (minimum wall thickness 1.8mm)	each	67.00
	12.33.1.2	110 mm	110 mm (minimum wall thickness 2.2mm)	each	102.00
	12.33.2	Push coupler	Single push fit Coupler :		
	12.33.2.1	75 mm	75 mm (minimum wall thickness 1.8mm)	each	121.00
	12.33.2.2	110 mm	110 mm (minimum wall thickness 2.2mm)	each	94.00
	12.33.3	With door	Single tee with door		
	12.33.3.1	75 mm	75x75x75 mm (minimum wall thickness 1.8mm)	each	106.00
	12.33.3.2	110 mm	110x110x110 mm (minimum wall thickness 2.2mm)	each	176.00
	12.33.4	Without door	Single tee without door		
	12.33.4.1	75 mm	75x75x75 mm (minimum wall thickness 1.8mm)	each	106.00
	12.33.4.2	110 mm	110x110x110 mm (minimum wall thickness 2.2mm)	each	163.00
	12.33.5	Bend	Bend 87.5°		
	12.33.5.1	75 mm	75 mm bend (minimum wall thickness 1.8mm)	each	78.00
	12.33.5.2	110 mm	110 mm bend (minimum wall thickness 2.2mm)	each	112.00
	12.33.6	Shoe	Shoe (Plain)		
	12.33.6.1	75 mm	75 mm Shoe (minimum wall thickness 1.8mm)	each	69.00
	12.33.6.2	110 mm	110 mm Shoe (minimum wall thickness 2.2mm)	each	99.00
12.34		UPVC pipe clips	Providing and fixing unplasticised - PVC pipe clips of approved design to unplasticised - PVC rain water pipes by means of 50x50x50 mm hard wood plugs, screwed with M.S. screws of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			required length, including cutting brick work and fixing in cement mortar 1:4 (1 cement : 4 coarse sand) and making good the wall etc. complete.		
	12.34.1	75 mm	75 mm (minimum wall thickness 1.8mm)	each	97.00
	12.34.2	110 mm	110 mm (minimum wall thickness 2.2mm)	each	114.00
12.35		Cast iron grating to rain water pipes	Providing and fixing to the inlet mouth of rain water pipe fixing cast iron grating 15 cm diameter and weighing not less than 440 grams.	each	43.00
12.36		PTMT grating to rain water pipes	Providing and fixing to the inlet mouth of rain water pipe PTMT (an Engineering Thermoplastic) grating square (Slit) 150 mm square with a height of 8 mm and weighing not less than 100 gms.	each	95.00
12.37		Rain water spout	Providing and fixing 100 mm/110 mm diameter and 60 cm long rain water spout in cement mortar 1:4 (1 cement : 4 sand)		
	12.37.1	Stone ware	Stone ware spout	each	72.00
	12.37.2	UPVC	Upvc spout (Pipe dia.110 mm)	each	38.00
12.38		Self-supported mechanically seamed steel sheet roofing system	Providing and fixing of self-supported mechanically seamed roofing system, made of high quality steel, having 340 MPA yield strength, pre coated galvalume sheet as per ASTM A 792 M of base metal thickness from 0.8mm to 1.6mm (tolerance ± 0.02mm). High grade steel sheet should have 55% aluminum and 45% zinc coating by hot-dip process of minimum 25 micron on top side & 12 micron on back side with epoxy primer andpolyester wash-coat as per AZM 150. Roofing should be designed as per ASCE 7-20 international building code 2002. The analysis should carried out for span and center arch – rise considering the required live load, wind load, dead load and the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			seismic factor. The analysis should also determine the thickness of steel and the end reactions at the beam level. Shop drawing of roofing system shall be submitted by the contractorthrough the consultant, and the same shall be checked and approved by the department. The roof panel formation shall be done as per the required length and curvature using hydraulic profile machine to give it the trapezoidal shape in required curvature. In this process the width of the sheet reduces from 0.914 meter to 0.61 meter (tolerance ± 0.02m). The roof panel shall be lifted using a suitable crane and using a spreader bar and a sling evenly placed to ensure no distortion of the panels during lifting and placing. The crane shall hold/support the panel till the alignment is done and completed using a plumb and a water tube and bolted into beam. These curved panels have interlocking formation and are crimped together using mechanical seaming machine which imposes a load of approximately five tones to ensure the seaming of required rigidity. After seaming a clear epoxy lacquer is applied on the inner side of panel end in 15cm width to form aprotective layer between panel and concrete beam. These panels are installed over water proof concrete gutter –beams having beveled edge to receive (support) curved panels. Beam shall design for arch reaction and vertical loads. Plaster shall not be applied on the beam. Schmidt Hammer test of the concrete should be carried out before erection to ensure that the anchor bolt can be securely fixed on support. In case of steel structure a steel runner plate of 6mm thickness and G.I. sheet gutter of suitable size shall be provided for installation of panels, G.I. anchor bolt of required sizes and type with G.I. washer and neoprene washers shall be		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			provided as per design. After installation of panels, trapezoidalvoids are created between gutter–beam and panels which should be covered by a flashing fabricated or brick masonry which is to be paid separately. Area of this roof shall be measured along the periphery on hem-top between end to end of panels. Ridge and valley shall not be taken in to consideration for measuring length transverse to periphery. The rate includes supplying, fixing, loadingunloading, hire charges of all plants machineries, anchors, fasteners, washers, bolts, epoxy, paint and all wastage etc. complete. Accessories and fixtures like hangers/clamps for installation of lighting /fixtures /utility/ duct as per requirement and ventilators, skylight can also be provided at crown/ridge by making cuts into the panels which is to be paid separately. This item shall be executed only by the agency which has all required manufacturing machineries and necessary expertise.		
	12.38.1	Span upto 15m	Base metal thickness 0.8mm up to span 15 meter	sqm	1698.00
	12.38.2	Span upto 18m	Base metal thickness 0.9mm , up to span 18 meter	sqm	1828.00
	12.38.3	Span upto 21m	Base metal thickness 1.0 mm, up to span 21 meter	sqm	1931.00
	12.38.4	Span upto 23m	Base metal thickness 1.1 mm, up to span 23 meter	sqm	2042.00
	12.38.5	Span upto 25m	Base metal thickness 1.2mm, up to span 25meter	sqm	2152.00
	12.38.6	Span upto 27m	Base metal thickness 1.3 mm, up to span 27 meter	sqm	2256.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.38.7	Span upto 29m	Base metal thickness 1.4mm, up to span 29meter	sqm	2378.00
	12.38.8	Span upto 31m	Base metal thickness 1.5 mm, up to span 31 meter	sqm	2475.00
	12.38.9	Span upto 32m	Base metal thickness 1.6 mm, up to span 33 meter	sqm	2585.00
	12.38.10	Hangers	Providing and fixing galvalume 1.0 mm thick clamps/hangers of size 75X140mm with saddle of 50mm dia, riveted at one side and nut and bolt at other side, to self-supported roofing system by mechanical seaming to suspend light fixture/cable stay/false ceiling/duct etc., at their location which should be planned earlier by engineer-in-charge. Clamps shall be fixed during seaming of panels.	each	55.00
	12.38.11	Ventilator	Providing and fixing advanced type hurricane turbo ventilator to self-supporting roof having neck dia 600mm made of aluminum having 24 nos vertical vanes made of aluminum sheet. Ventilator should be fixed on 1 to 2mm thick G.I. ring which shall be fixed on 2mm thick polycarbonate sheet. Polycarbonate sheet shall have foam tap or rubber gasket under and over it and fixed by 1 to 2 mm thick aluminum angle 25X25mm with self-tapping screw to panels. All joints shall be sealed by silicon. Rates i/c of all material, labour fittings screw, rings, foam tap, aluminum angles etc. complete. Deduction shall not be made for cutting area of self-supported roof.	each	9196.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.38.12	Sky-lights	Providing and fixing of 2mm thick polycarbonate sheet to self-supported roofing system. The sheet shall have tensile strength and elongation as per ASTM D 638, refractive index as per ASTM D 592, light transmission asper ASTM D 1003 and specific gravity as per STM D 792. Polycarbonate sheet shall have foam tap or rubber gasket under and over it and fixed by 1 to 2 mm thick aluminum angle 25X25mm with self- tapping screw to panels. All joints shall be sealed by silicon. Rates include of all material, labour, fittings, screw, foam tap, aluminum angles etc. complete. Deduction shall not be made for cutting area of self-supported roof.	sqm	3400.00
12.39		Three-dimensional space frame steel structure	Supply, fabrication and erection of three-dimensional space frame steel structure consisting of support node, primary space frame members, space notes nodes, bolts, sleeves (nut), purlin stool, purlin etc. to support roof sheet in plain or curved shape. The space frame steel structure shall be designed as per relevant IS codes for earthquake, wind and all applicable forces/load as per IS 875 and/or International standard ASCE-7, AISC 89, AISI 89 etc. and produced in compliance with ISO 9001-2008 quality system. The lowest point of space frame steel structure shall be the support node securely welded over a flange plate, made of steel conforming to ASTM A36, fixed securely on top of a well a designed support shaft made of solidor hollow circular steel pipe or concrete column erected/constructed over a secured and unyielding steel or RCC foundation. Space frame nodes, which connect two or more primary space frame members, are solid spherical balls with internally threaded holes and manufactured by hot forging process	kg	312.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			from medium carbon steel conforming to BS EN 10083 Part-2. Primary space frame member, which connects two nodes and consists of well-designed hollow steel pipe of suitable length, shall have steel cones coaxiallywelded at both ends in accordance with welding standards. Pipes of primary space frame member shall be type E (ERW/HFIW) and conforming to ASTM A53, Grade A/B or IS 1239 Part I or IS 1161. Last primary space frame member shall have no coaxial steel cone at the end but shall be provided with galvanized plate. Coaxial steel cones (conical tips) preassembled at ends of primary space frame member shall conform to ASTM A36/ BS EN 10083, C-30 or DIN St 37-2 standards as applicable. Primary space frame member shall be hot dip galvanized during manufacturing as per ASTM A53 /ASTM A 123 as applicable with powder coated finish. The space frame nodes (except support nodes), fasteners, cones and all threaded parts shall be electro- galvanized finish. Average zinc coating done (in factory) on space frame pipes shall be 550 g /sqm (Approximate zinc coating thickness 77 Microns) and shall not be less than 400 g/sqm (Approximate zinc coating thickness 56 microns). Bolts, sleeves and cones shall be electro-galvanized to ASTM B 633 Type 2, SC3. Shop drawings along with structural design analysis shall be provided by the manufacturer for approval of the engineer in charge. Rate shall include cost of design, all material, all lifts, labour, transport, storage, machines, tools and plants, wastages, erection and installation at site all necessary scaffolding etc. necessary to provide finished product but excluding cost of erection/ construction of support shaft and roof sheet.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.40		Deck sheet 0.8 mm thick	Providing and fixing roofing consist of 0.8 mm thick galvanized steel deck sheet confirming to IS 277:1992 used as permanent shuttering over which MS wire mesh 3mm laid at 100x100 mm grid including edge trim covered with concrete. This metal deck will be supported on structural steel beam with shear studs. (Structural steel like Beam, column, joists etc. & concrete of different grade as per design will be paid separately).	sqm	1183.00
12.41		Roof tile underlay	Providing and fixing in position roof tile under lay using 12mm thick(minimum) Moisture/fire Resistance heavy duty fibre cement board(High Pressure Steam Cured) Type A,Category 4 conforming to IS 14862:2000, fixed on Steel frame work of 610mmx610mm or 610mmx1220mm grid by counter sunk stainless screw @ 300mm C/C all complete as per the drawings. The boards are to be fixed on the frame work in staggered manner keeping a uniform gap of 2mm.Finally, the roof tile like Shingles/Mangalorian Tiles etc may be laid on the board substrate by sealing the joints properly with filler material or sealant to ensure no leakage through joints. All completeas per drawings, specification and as per direction of Engineer-in-Charge. Note - MS Section structure and roof tile like Shingles/Mangalorian Tiles etc cost will be paid seprately.	sqm	1503.00
12.42		Thermal insulation under deck (fibre glass wool)	Providing and fixing thermal insulation of ceiling (under deck insulation) with Resin Bonded Fibre glass wool conforming to IS: 8183 density 24kg/m^3, 50mm thick, wrapped in 200 G Virgin Polythene bags fixed to ceiling with metallic cleats (50x50x3 mm) @ 60 cm andwire mesh of 12.5mm x 24 gauge wire mesh, for top most ceiling of building.	sqm	484.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.43		Thermal insulation over false ceiling (fibre glass wool)	Providing and fixing thermal insulation with Resin Bonded Fibre glass wool conforming to IS : 8183, density 16kg/m^3, 50mm thick, wrapped in 200G Virgin Polythene bags placed over existing false ceiling and held in position by criss-crossing GI wire.	sqm	191.00
12.44		Thermal insulation over false ceiling (expanded polystyrene)	Thermal Insulation of roofing with Expanded polystyrene fixed with suitable adhesive to the false ceiling as per the directions of the Engineer-in-charge :		
	12.44.1	Type N	With Type N - Normal 50 mm thick	sqm	233.00
	12.44.2	Type SE	With Type SE - Self Extinguishing type 50 mm thick.	sqm	271.00
12.45		Thermal insulation over deck (p.u. foam)	Providing and laying thermalinsulation of roofing with 40 mm thick impervious sprayed, closed cell free Rigid Polyurethane foam over deck insulation conforming to IS - 12432 Pt. III (density of foam being 40-45 kg/cum), over a coat of polyurethane primer applied @ 6-8 sqm per litre, laying 400 G polythene sheet over PUF spray and providing a wearing course of 40 mm thick cement screed 1: 2 : 4 (1 cement : 2 sand : 4 stone aggregate 20 mm nominal size) in chequered rough finish, in panels of 2.5 mx2.5 m and embedding with 24 G wire netting and sealing the joints withpolymerized mastic, all complete as per direction of Engineer-in-Charge.	sqm	1136.00
12.46		Thermal insulation of wall (fibre glass wool)	Providing and fixing thermal insulation with Resin Bonded Fibre glass wool conforming to IS: 8183 having density 24 kg/m^3, 50 mm thick, wrapped in 200G Virgin Polythene Bags fixed to wall with screw, rawl plug and washers and held in position by criss crossing GI wire etc. complete	sqm	359.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			as per directions of Engineer-in-Charge.		
12.47		Thermal insulation under deck (rock wool)	Providing and fixing thermal insulation of ceiling (under deck insulation) with Resin Bonded Rockwool conforming to IS: 8183, density 48 kg/ m^3, 50 mm thick, wrapped in 200 G Virgin Polythene bags fixed to ceiling with metallic cleats (50x50x3 mm) @ 60 cm andwire mesh of 12.5mm x 24 gauge wiremesh, for top most ceiling of building.	sqm	574.00
12.48		Thermal insulation over false ceiling (rock wool)	Providing and fixing thermal insulation with Resin bonded rock wool conforming to IS: 8183, density 48 kg/m^3, 50 mm thick, wrapped in 200 G virgin Polythene bags placed over existing false ceiling and held in position by criss- crossing GI wire.	sqm	444.00
12.49		Thermal insulation of wall (rock wool)	Providing and fixing thermal insulation with Resin Bonded rock wool conforming to IS: 8183, having density 48 kg/m^3. 50 mm thick, wrapped in 200 G Virgin Polythene Bags fixed to wall with screw, rawl plug and washers and held and in position by criss cossing GI wire etc. complete as per directions ofEngineer-in-Charge.	sqm	448.00
12.50		50 mm thick Rockwool insulation for slab unfaced	Supply and application of 48kg/m3 density & 50 mm thick with a tolerance of -2 + 6mm of Rockwool RB slabs unfaced (Binder<3.5%); conforming to IS: 8183/ASTM C 612 latest edition with amendments. Non- combustible when tested as per BS 476 Part 4 or A1 when tested as per EN: 13501. The product should comply with BS476 Part 12. The thermal conductivity should be<0.043 W/m.K at 100°C mean temperature and the minimum density should be 70Kg/cu-m (+10%). chloride content should be < 10 ppm; water absorption (partial immersion) 0.5kg/sqm	sqm	299.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			maximum as per BS 2972 Part 12 as per manufacturers specification. The maximum service temperature shall be 700°C. Shot content should be less than 2% for 500 microns & less than 10% for 250 microns when tested as per standard IS 3144. Two random samples from each supplied batch will be test for shot content at the site. The melting point of the wool shall be more than 1000°C and wool shall be bio-soluble in accordance with EU directive 127/2008 or similar. All Above mentioned standard are as per latest edition with amendments.		
12.51		50 mm thick Rockwool under deck insulation wrapped in polythene bags	Supply & Application of Rockwool RB slab 48kg density, 50mm thick wrapped in 200G vergine polythene bags for partition, ceiling (Under Deck insulation) conforming to IS: 8183 with tolerance of -2 +6mm of rock wool RB slabs unfaced (Binder <3.5%) conforming to IS:8183/ASTM C 612 latest edition with amendments. Non-combustible whentested as per BS 476 Part 4 or Al whentested per EN 13501. The product should comply with BS 476 Part 12. The thermal conductivity should be < 0.043 W/m.K at 100°C mean temperature and the minimum density should be 70 Kg/m3 + 10%. Chloride content should be < 10 ppm; Water absorption (partial immersion) 0.5kg/m2 max as per BS 2972 Part 12 as per manufacturer specification. The maximum service temperature shall be 700°C. Shot content should be less than 2% for 500 microns & less than 10% for 250 microns when tested as per standard IS 3144. Two random samples from each supplied batch will be test for shot content at the site. The	sqm	349.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			melting point of the wool shall be more than 1000°C and wool shall be bio-soluble in accordance with EU directive 127/2008 or similar. All Above mentioned standard are as per latest edition with amendments.		
12.52		50mm thick rock wool slab with GI wire	Supply & Application of Rockwool RB 2 slab 64kg /m^3, 50mm thick, with GI wire netting 24 SWG; fixing for NRC Value 1.00 for Acoustics,thermal & fire purpose conforming to and IS: 8133, IS: 8183 with tolerance of -2 + 6mm of RB slabs unfaced (Binder < 3.5%); conforming to IS:8183/ASTM C 612 latest edition with amendments. Non-combustible whentested as per BS 476 Part 4 or Al whentested as per EN 13501. The product should comply with BS 476 Part 12.The thermal conductivity should bile 1 0.043 W/m.K at 100°C mean temperature and the minimum density should be 70 kg/cu-m (+ 10%). Chloride content should be < 10 ppm; Water absorption (partial immersion) 0.5kg/sqm max as per BS 2972 Part 12 as per manufacturer's specification. The maximum service temperatureshall be 700°C. Shot content should beless than 2% for 500 microns & less than 10% for 250 microns when tested as per standard IS: 314. Two random samples from each supplied batch will be tested for shot content at the site. The melting point of the wool shall be more than 1000°C and wool shall be bio-soluble in accordance with EU directive 127/2008 or similar. All Above mentioned standard are as per latest edition with amendments.	sqm	487.00
12.53		50 mm thick rock wool building roll	Supply & Application of Rockwool building roll (1.1mx10m) 48kg	sqm	493.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
		with aluminium foil facing	density, 50mm thick with FSK (foil scrim Kraft) aluminium foil facing for HVAC duct & building underdeck purpose conforming to IS: 8183, IS: 8183 and IS:8183 a tolerance of -2 +6 mm of Rockwool RB slabs unfaced (Binder < 3.5%) ; conforming to IS: 8183; /ASTM C 612 latest edition with amendments. Non-combustible when tested as per BS: 476 Part 4 orAl when tested as per EN: 13501. The product should comply with BS: 476 Part 12. The thermal conductivity should be< 0.943 w/m.k at 100°C mean temperature and the minimum density should be 70 Kg/m3 (+ 10%). Chloride content should be < 10 ppm; water absorption (partial immersion) 0.5kg/sqm maximum as per BS: 2972 Part 12 as per manufacturer's specification. The maximum service temperature shall be 700°C. Shot content should be less than 2% for 500 microns & less than 10% for 250 microns when tested as per standard IS: 3144. Two random samples from each supplied batch will be test forshot content at the site. The melting point of the wool shall be more than 1000°C and it shall be bio-soluble in accordance with EU directive127/2008 or similar. All Above mentioned standard are as per latestedition with amendments.		
12.54		50 mm thick rockwool with aluminium & glass facing for under deck insulation	Supply & Application of Rockwool Building Roll (1.1mX10m) 48kg density, 50mm thick with ALG (Alu & glass) mixture of Aluminium & glass facing for building underdeck application below metal sheet also conforming IS: 8183, IS: 8183 and IS: 8183 a tolerance of -2+6mm of rockwool RB slabs unfaced (Binder <	sqm	670.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			3.5%); conforming to IS: 8183/ASTM C 612 latest edition with amendments. Non-combustible when tested as perBS 476 Part 4 or Al when tested as perEN 13501. The product should comply with BS 476 Part 12. The thermal conductivity should be < 0.943 W/m.K at 100°C mean temperature and the minimum density should be 70 Kg/m^3 (+ 10%). Chloridecontent should be < 10 ppm; water absorption (partial immersion) 0.5kg/m2 maximum as per BS 2972 Part 12 as per manufacturer's specification. The maximum service temperature shall be 700°C. Shot content should be less than 2% for 500 microns & less than 10% for 250 microns when tested as per standard IS: 3144. Two random samples from each supplied batch will be test forshot content at the site. The melting point of the wool shall be more than 1000°C and wool shall be bio-soluble in accordance with EU directive 127/2008 or similar. All Above mentioned standard are as per latestedition with amendments.		
12.55		12mm thick 5 layered bubble-reflective thermal insulation	Providing and fixing of minimum 12mm thick, 5 layered bubble Reflective thermal insulation made of 12 microns aluminum foil with minimum 97% Reflectivity on both outer sides, under both aluminum foil 4mm thick FR bubble insulation with Low Emissivity - 0.032 (ASTM1371), Fire Class A (ASTM E84) Surface Burning Characteristics- Flame Spread Index (5) & Smoke Developed Index (50) shall be provided to hold 4mm thick XLPE (Chemically Cross-		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Linked Closed Cell Polyethylene Foam) with fire characteristics conforming to BS 476 Part 6 Class"0", Fire Propagation having an index (I) of 5.06 and sub-indices i1(2.71); i2(1.65), i3(0.70) water vapour transmission rate conforming to ASTM E 96-15 – 7.3 X 10^6ng.hr/m², water vapour permeance conforming to ASTM E 96 15 – 0.2974, water vapourpermeability conforming to ASTM E 96-15 – 0.2433, tensile strengthconforming to ASTM D 3575-14 – 0.33 MPa, Elongation at break % conforming to ASTM D 3575-14 – 171, Fungus resistance conforming to ASTM C 1338-14 – No Fungal Growth, Bursting strength conforming to ASTM D 774-97(2007) – 5.7 kg/cm²in the density of 30 ± 3 Kg/mat middle part of insulating material. All five layers of insulation shall be stick with each other by registered manufacturer using hot processwithout using any kind of adhesive.Insulation shall be supplied in 1.10m wide roll width with 50mm wide factory-made cuts at both sides to jointanother roll by using general stapler in case of new shed and adhesive in case of existing or self-supported roof.		
	12.55.1	Over purlin & Below new metal sheet	Fixing of insulation on new work by laying horizontally on the purlins and perpendicular to the direction of purlins and fixing by using self- tapping screws at equal distance. Place and align the second roll of insulation parallel to the first roll and overlaps the adjacent rolls. After Insulation is fixed over the purlins the metal sheet is placed over insulation and drilled	sqm	335.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			with self-tapping screws with purlins etc. complete, including all materialand scaffolding.		
	12.55.2	Below purlin as under deck insulation of existing roof using self-tapping screw	Installation of insulation in Old Sheds where the sheets have been already fixed over the purlins so insulation shall be fixed under the purlins covering the roof bottom. Insulationroll to be unfolded at the bottom of the purlin horizontally and parallel to the roof and fixed by using the self-tapping screw with a washer at equal distance etc. Complete, including all material and scaffolding.	sqm	467.00
	12.55.3	Under deck insulation of self-supporting roof using stud	Insulation of Self-Supported Roof shall be applied under the roofing by using stud welding to the self-supported roof at a distance not be more than 300 mm in both directions. At each stud insulation roll shall be pushed against the studs so that it gets punctured and then non-returning washer has to be applied at the studetc. complete, including all material and scaffolding.	sqm	480.00
12.56		Providing and fixing XPS board for over-deck on ceiling surface	Providing and fixing 50 mm thick extruded polystyrene rigid insulation board (XPS) of required size over- deck on ceiling surface, complying with ASTM C 578 - type VI, having thermal conductivity of 0.0289 W/mK as per ASTM C 518, compressive strength of > 350 kPa listed as per ASTM C 1621, density of 34-36 kg/m³ as per ASTM D 1622, waterabsorption < 1% by volume as per ASTM C 272, oxygen index of	sqm	1263.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			minimum 24 as per ASTM D 2863 and Fire retardant property as per DIN 4102, Part 1. XPS board insulation shall be laid over clean and leveled RCC roof having proper slope and waterproofing and shall be connected by the ship-laps joints and bitumen shall be applied over the joints and HDPE film vapour barrier ofminimum 140 gsm (grams per square metre) shall be laid over the XPS board having minimum 75mm overlapat the edges and reinforcement layerof G.I. chicken mesh/Fibre mesh ofminimum 140 gsm shall be laid over the HDPE film vapour barrier and PCC layer of minimum 38mm withpreferred stone or pre-cast tiles shallbe laid over the XPS board with suitable grouting including providing pre-fabricated EPDM drain outlets (Cement Screed casting shall be done at the junction of roof and parapet walls), all as per the directions of Engineer-in-charge.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.57		XPS board for underdeck application on ceiling surface	Providing and fixing 50 mm thick extruded polystyrene rigid insulation board of required size underdeck on ceiling surface, complying with ISO 4898:2008 & ASTM C 578-08b - type VI, having thermal conductivity of 0.0289 W/m K as per ASTM C 578 (measured as per IS 3346), compressive strength of > 350 kPa listed as per ASTM D 1621, density of 34-36 kg/m³ as per ASTM D 1622, water absorption < 1% by volume as per ASTM D 2842, oxygen index of 24.1 to 28.1 listed as per ASTM D 2863, cell size 0.4 mm of dia (max) as per ASTM D 3576. Fire retardant property as per DIN 4102, Part 1 of class B2 and as per ASTM E84 class A, fixed with suitable water based adhesive and fastener, complete in all respect as per the directions of Engineer-in-Charge.	sqm	686.00
12.58		False ceiling with boards (G.I. grid suspension)	Providing and fixing false ceiling atall height including providing andfixing of frame work made of special sections, power pressed from M.S. sheets and galvanized with zinc coating of 120 gms/sqm (both side inclusive) as per IS : 277 and consisting of angle cleats of size 25 mm wide x 1.6 mm thick with flanges of 27 mm and 37mm, at 1200 mmcentre to centre, one flange fixed tothe ceiling with dash fastener 12.5 mmdia x 50mm long with 6mm dia bolts, other flange of cleat fixed to the angle		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			hangers of 25x10x0.50 mm of required length with nuts & bolts of required size and other end of angle hanger fixed with intermediate G.I. channels 45x15x0.9 mm running at the spacing of 1200 mm centre to centre, to which the ceiling section 0.5mm thick bottom wedge of 80 mm with tapered flanges of 26 mm each having lips of 10.5 mm, at 450 mm centre to centre, shall be fixed in adirection perpendicular to G.I. intermediate channel with connecting clips made out of 2.64 mm dia x 230 mm long G.I. wire at every junction, including fixing perimeter channels 0.5 mm thick 27 mm high having flanges of 20 mm and 30 mm long, the perimeter of ceiling fixed to wall/partition with the help of rawl plugs at 450 mm centre, with 25mm long dry wall screws @ 230 mm interval, including fixing of gypsum board to ceiling section and perimeter channel with the help of dry wall screws of size 3.5 x 25 mm at 230 mm c/c, including jointing and finishing to a flush finish of tapered and square edges of the board with recommended jointing compound , jointing tapes, finishing with jointing compound in 3 layers covering upto 150 mm on both sides of joint and two coats of primer suitable for board, all as per manufacturer's specification and also including the cost of making openings for light fittings, grills, diffusers, cutouts made with frame of perimeter channels suitably fixed, all complete as per drawings, specification and direction of the Engineer in Charge but excluding the cost of painting with :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.58.1	POP boards	10mm thick plaster of Paris (gypsum anhydrous) sheet 10mm thick pop board.		
	12.58.1.1	Flat surfaces	Flat surfaces	sqm	676.00
	12.58.1.2	Curved surfaces	Curved surfaces	sqm	748.00
	12.58.1.3	Sunk or raised mouldings	Sunk or raised mouldings	sqm	175.00
	12.58.1.4	Extra for pop ceiling	Extra for providing in the plaster of Paris (Gypsum anhydrous) ceiling above 5 metre height from floor level.	sqm per meter height	59.00
	12.58.2	12.5 mm thick gypsum board	12.5 mm thick tapered edge gypsum board		
	12.58.2.1	Plain gypsum board	Plain board conforming to IS: 2095-Part I: 2011 (Board with BIS certification marks).	sqm	750.00
	12.58.2.2	Fire resistant gypsum board	Fire resistant Glass ReinforcedGypsum (GRG) board conforming to IS: 2095-(Part 3) : 1996 (Board with BIS certification marks)	sqm	866.00
	12.58.2.3	Moisture resistant gypsum board.	12.5mm thick moisture resistant gypsum plasterboards Moistbloc as per BS EN 520 confirming to water absorption class H1 with total water absorption < 5%, Green Pro certified by CII	sqm	920.00
	12.58.2.4	Perforated gypsum board backed with acoustical tissue	Fully Perforated Gypsum Plaster Board having approx. 15 % perforated area with perforation size and pattern as approved by the Engineer-in-charge and as per manufacturer'sspecification, with all 4 side tapered and backed by acoustical tissue with NRC value not less than 0.60.	sqm	1277.00
	12.58.3	8 mm thick calcium silicate board	8 mm thick Calcium Silicate Board made with Calcareous and Siliceous materials reinforced with cellulose fiber manufactured throughautoclaving process.	sqm	1150.00
12.59		False ceiling tiles of approved material on inter locking metal grid system	Providing and fixing tiled false ceiling of specified materials of size 595x595 mm in true horizontal level, suspended on inter locking metal grid of hot dipped galvanized steel sections (galvanized @ 120 grams/ sqm, both side inclusive) consisting of main "T" runner with suitably spaced joints to		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			get required length and of size 24x38 mm made from 0.30 mm thick (minimum) sheet, spaced at 1200 mm center to center and cross "T" of size 24x25 mm made of 0.30 mm thick (minimum) sheet, 1200 mm long spaced between main "T" at 600 mm center to center to form a grid of 1200x600 mm and secondary cross "T" of length 600 mm and size 24x25 mm made of 0.30 mm thick (minimum) sheet to be interlocked at middle of the 1200x600 mm panel to form grids of 600x600 mm and wall angle of size 24x24x0.3 mm and laying false ceiling tiles of approved texture in the grid including, required cutting/making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors etc. Main "T" runners to be suspended from ceiling using GI slotted cleats of size 27 x 37 x 25 x1.6 mm fixed to ceiling with 12.5 mm dia and 50 mm long dash fasteners, 4 mm GI adjustable rods with galvanized butterfly level clips of size 85 x 30 x 0.8 mm spaced at 1200 mm center to center along main T, bottom exposed width of 24 mm of all T-sections shall be pre- painted with polyester paint, allcomplete for all heightsas per specifications, drawings and as directed by Engineer-in-charge.		
	12.59.1	GI metal tiles powder coated global white	GI Metal Ceiling Lay in plain Tegular edge Global white color tiles of size 595x595 mm, and 0.5 mm thick with 8 mm drop; made of G I sheet having galvanizing of 100 gms/sqm (both sides inclusive) and electro statically polyester powder coated of thickness 60 microns (minimum), including factory painted after bending.	sqm	1431.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.59.2	20% perforated GI metal tiles	GI Metal Ceiling Lay in perforated Tegular edge global white color tiles of size 595x595 mm and 0.5 mm thick with 8 mm drop; made of GI sheet having galvanizing of 100 gms/sqm (both sides inclusive) and 20% perforation area with 1.8 mm dia holes and having NRC (Noise Reduction Coefficient) of 0.5, electro statically polyester powder coated of thickness 60 microns (minimum), including factory painted after bending and perforation, and backed with a black Glass fiber acoustical fleece.	sqm	1553.00
	12.59.3	PVC laminated Gypsum tiles	12.5 mm thick square edge PVC Laminated Gypsum Tile of size 595x595 mm, made of Gypsum plasterboard, manufactured from natural gypsum as per IS 2095 part I and laminated with white 0.16mm thick fire retardant PVC film on the face side and 12micron metalized polyester on the back side with all edges sealed with the face side PVC film which goes around and wraps the edges and is bonded to the edges and the back side metalized polyester film so as to make the tile a ompletely sealed unit.	sqm	1004.00
	12.59.4	Perforated gypsum tiles with glass wood backing	12.5 mm thick fully Perforated Gypsum Board tile made from plasterboard having glass fibre conforming to IS: 2095 part I, of size 595x595 mm, having perforation of 9.7x9.7 mm at 19.4 mm c/c with center borders of 48 mm and the side borders of 30 mm, backed with non woven tissue on the back side, having an NRC (Noise Reduction Coefficient) of 0.79, with 50 mm resin bonded glass wool backing.	sqm	1164.00
	12.59.5	Glass fiber acoustical tiles	Glass fiber Acoustical Suspended Ceiling System with 16 mm thick tegular edge or micro look edge tiles of size 595x595mm.The tiles should have Humidity Resistance (RH) of 99%, NRC 0.5, Light Reflectance ≥83%, Thermal Conductivity k = 0.052 - 0.057 w/m K, Colour White,	sqm	1861.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Fire Performance UK Class 0 / Class 1 (BS 476 pt - 6 and7) in module size of 600 x 600 x 16mm with Bio Block coating on the face of the tile, suitable for Green Building application, with Recycled content of 38 - 41%. TheTile and Grid system used together should carry a 15 year warranty.		
	12.59.6	8mm thick cement bonded particle board tile	8mm thick non-asbestos multipurpose termite, fire & moisture resistance cement bonded particle board as per IS: 14276, white primedcolour tiles of size 595X595 mm.	sqm	666.00
	12.59.7	6mm thick cement bonded particle board tile	6mm thick non-asbestos multipurpose termite, fire & moisture resistance cement bonded particle board as per IS: 14276, white primed colour tiles of size 595X595 mm.	sqm	460.00
12.60		False ceiling tiles on clip in/swing down metal ceiling system	Providing and fixing GI Clip in Metal Ceiling System of 600x600 mm module which includes providing and fixing 'C' wall angle of size 20x30x20mm made of 0.5mm thick pre painted steel along the perimeter of the room with help of nylon sleeves and wooden screws at 300mm center to centre, suspending the main Ccarrier of size 10x38x10mm made of G.I steel 0.7 mm thick from the soffit and ceiling with help of soffit cleat 37x27x25x1.6 mm, rawl plugs of size 38x12 mm and C carrier suspension clip and main carrier bracket at 1000mm c/c. Inverted triangle shaped Spring Tee having height of 24 mm and width of 34mm made of GI steel 0.45 mm thick is then fixed to the main C carrier and in direction perpendicular to it at 600mm centers with help of suspension brackets. Wherever the main C carrier and		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			spring T have to join, C carrier and spring T connectors have to be used. All sections to be galvanized @ 120 gms/sqm (both side inclusive) Fixing with clip in/swing down tiles into spring 'T' with		
	12.60.1	GI (Plain)	GI Metal Ceiling Clip in /swing down plain Beveled edge global white color tiles of size 600x600 and 0.5mm thick with 25mm height, made of G I sheet having galvanizing of 100 gms/sqm (both sides inclusive) and electro statically polyester powder coated of thickness 60 microns (minimum), including factory painted after bending.	sqm	1500.00
	12.60.2	G.I.metal tiles 20% perforated	GI Metal Ceiling Clip in /swing down plain Beveled edge global white color tiles of size 600x600 and 0.5mm thick with 25mm height, made of G I sheet having galvanizing of 100 gms/sqm (both sides inclusive) and 20% perforation area with 1.8mm dia holes and having NRC of 0.5, electro statically polyester powder coated of thickness 60 microns (minimum), including factory painted after bending and perforation.	sqm	1628.00
12.61		Antibacterial metal tile false ceiling on rigid G.I. system	Providing & Fixing of Unperforated Antibacterial Metal Tile Ceiling System comprising of Tile of 600mm wide and 600mm long manufactured out of 0.7mm thick aluminium alloy 3105 with beveled edge of 4.15mm width and height of 32mm. The Tile will be manufactured on advanced equipment that includes several levelling stages in the manufacturing process by approved certified manufacturers. Tile ends will be raised with pips and stops to ensure positive engagement into the spring to enable for de-mounting of individual panels. The Tile sides will be sufficiently high to ensure a minimum deflection across the length of Tile. All Tiles will be	sqm	2821.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			bevel edged. The Tile shall be epoxy polyester powder coating with ANTIBAC in white colour which is in compliant to rod directive. The Tile shall be clipped into clip in rail of 0.5mm thick G.I. in form of a grid system using a cross connector. The clip in rail shall be supported from slab by means of rigid Suspension of 4mm G.I.Road, Hold on Clamp with Clip. (Rate include all materials and labour).		
12.62		False ceiling with board (frame work to be paid separately)	Providing and fixing board ceiling of approved quality with necessary nails etc. complete (frame work to be paid separately):		
	12.62.1	Insulating board	12mm thick insulating board		
	12.62.1.1	Natural colour	Natural colour insulating board	sqm	550..00
	12.62.1.2	White face	White face insulating board	sqm	614.00
	12.62.1.3	Flame retardant	Flame retardant face insulating board	sqm	700.00
	12.62.2	3 layer particle board	Flat pressed 3 layer medium density particle board or graded particle board (Grade I) IS: 3087 12 mm thick marked in ceiling with necessary nails etc. complete	sqm	633.00
	12.62.3	Cement board	6 mm thick plain MultipurposeCement board (High Pressure steam cured) as per IS : 14862, with suitablescrews for fibre cement board in ceiling etc. complete	sqm	428.00
	12.62.4	Heavy duty cement board	6mm thick High Pressure Steam Cured non asbestos, reinforced withcellulose fibres,Moistur& fire Resistance heavy duty fibre cement board, Type A,Category 4 conforming to IS 14862:2000.(for wet areas)	sqm	530.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.62.5	Decorative type cement board	4 mm thick High Pressure Steam Cured non asbestos, pre-painted, self-embossed, designer cement boardType B, Category 3 conforming to IS 14862:2000 . of approved texture,design patterns	sqm	403.00
12.63		Extra for circular cutting in board ceiling	Extra for Circular cutting and wastages in ceiling with: board only (frame work) Natural colour insulating12 mm thick board paid separately.	meter	150.00
12.64		Extra for curved surfaces in narrow width	Extra for providing and fixing ceiling to curved surfaces in narrow widths	sqm	100.00
12.65		False ceiling particle board on aluminium grid system	Providing and fixing false ceiling with 12 mm thick plain/ semi perforated or with design ceiling tiles of BWP type phenol formaldehyde synthetic resin bonded pressed particle board conforming to IS:3087, finished with a coat of aluminium primer on both sides and edges including two coats of synthetic enamel paint of approved quality on exposed face, fixed to a grid made out of anodised aluminium (with 15 micron anodic coating) T- sections 35 x15x1.5 mm size main runners and cross runners 23.5x19x1.5mm fixed to main runners placed 600 mm centre to centre both ways so as toform a grid of 600 mm square. The frame work shall be suspended from ceiling by level adjusting hangers of 6 mm dia M.S rod fixed to roof slab by means of ceiling cleats and dash fastener. The suspenders shall be placed 600 x 1200 mm centre to centreincluding fixing to the frame with C.P brass screws and applying a priming coat of zinc chromate yellow primer (aluminium frame work shall be paid separately).	sqm	667.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.66		Colour coated profile sheet	Providing and Fixing colour coated profile sheet of minimum 1015 mm effective cover width and nominal 28.5 mm deep ribs with subtle square fluting in the five pan at nominal 203 mm centre-to-centre. The end rib shall be designed for anti-capillary action,to avoid any seepage of water through the lateral overlap. The profile sheet of 0.5 mm thick, Hi-strength steel with min. 550 MPa yield strength, metallic hot dip coated with Aluminium-Zinc alloy (55% Aluminium, 45% Zinc) as per AS 1397 - Zincalume AZ150 (Min. 150 gms/sq.mt total on both sides) with steel quality paint coat. The paint shall have a total coating thickness of nominal 35 μm, comprising of nominal 20 μm exterior coat on top surface and nominal 5 μm reverse coat on back surface over nominal 5 μm primer coat on both surface. The sheet shall have brand marking of the manufacturer giving product details on the back of the sheet at every 1 meter c/c for confirming genuinity of the material. The sheet shall be fixed using self-drilling /self-tapping screws of size (5.5x 55mm) with EPDM seal, complete upto any pitch in horizontal/ vertical or curved surfaces excluding the cost of purlins, rafters and trusses and including cutting to size and shape	sqm	430.00
12.67		Extra for acrylic sheets in false ceiling in place of particle board	Extra for providing 3 mm thick translucent white acrylic plastic sheets of approved quality in false ceiling instead of 12 mm thick plain or design particle board ceiling tiles.	sqm	387.00
12.68		False ceiling (thermacol)	Providing and fixing expanded polystyrene (thermacol) false ceiling (frame work to be paid separately) :		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
	12.68.1	Type N	With Type N - Normal 50 mm thick	sqm	230.00
	12.68.2	Type SE	With Type SE - Self Extinguishing type 50 mm thick	sqm	270.00
12.69		False ceiling with fabric	Providing and fixing false ceiling with fabric fixing with nail etc. (frame work and beading to be paid separately). Material has to the satisfaction of Engineer-in-Charge		
	12.69.1	Standard fabric	Standard fabric	sqm	97.00
	12.69.2	Medium fabric	Medium fabric	sqm	170.00
	12.69.3	Premium fabric	Premium fabric	sqm	225.00
12.70		Perforated linear panel aluminium metal ceiling with acoustics	Providing and fixing perforated Aluminium panel ceiling of white/ natural satin colour supplied by approved manufacturers consisting of panel of standard size with bevel edge, panel length up to 6 m, coil Coated on a Continuous Paint Line suitable for Interior application, double baked and roll formed from enamelled corrosion resistance aluminium alloy AA 5050 (Al. Mg) for higher strength and good roll forming characteristics.Aluminium panels shall be chromatised for maximum bond between metal and paint, enamelled twice under high temperature, one sidewith a full primer and finish coat the other side (inner side) with a primer coating and Skin Coat on a continuous paint line. The panel shall be perforated Aluminium panel having perforation of 2mm diameter at a distance of 5mm centre to centre. Non-woven textile (NWT) made of glass-reinforced fibre glued over theperforation at the back side of the panel for sound absorption to provide noise reduction co efficient upto 0.7 NRc. The Panels shall abut each other with a narrow V grove. Panel shall be clipped to a baked enamelled aluminium panel carrier of respective size in standard length of 5 mtr made of doubled baked black enamelled aluminium alloy AA 5050 (Al. Mg)		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			with cut outs to hold the panels in a module of 300mm closed at a distance of 1.4 mtr. Panel carrier shall be suspended by means of G.I. threaded rod having 6 mm diameter at a distance of 1.5 mtr. Actual distance of threaded rod/carrier might vary and to be calculated based on the actualdrawing and site condition. The Measurements shall be wall to wall without any deductions for lights, diffusers, columns etc.		
	12.70.1	300mm wide X30mm deep X 0.7mm thick	With panels of size 300mm wide X30mm deep X 0.7mm thick and panel carrier of size 40.5mm wide X 62mm deep X 0.9mm thick	sqm	3166.00
	12.70.2	150mm wide X 15.5 mm deep X 0.5mm thick	With panels of size 150mm wide X 15.5 mm deep X 0.5mm thick and panel carrier of size 32mm wide X 39mm deep X 0.95mm thick	sqm	2562.00
	12.70.3	84mm wideX16mm deep X 0.5mm thick	With panels of size 84mmwideX16mm deep X 0.5mm thick and panel carrier of size 62mm wide X29mm deep X 0.95mm thick	sqm	2538.00
12.71		Un-perforated aluminium panel ceiling	Providing and fixing un-perforated Aluminium Panel ceiling Supplied by approved manufacturers of standard white/ natural satin shade consisting of panel 30 mm/80mm/130mm/180mm wide X 15 mm deep X 0.5mm /0.6mm thick respectively, with square edges having panel length up to 6 mtr, Coil Coated on a Continuous Paint Line, Double baked and roll formed from enamelled corrosion resistance Aluminium alloy AA 5050 (Al.Mg) for higher strength and good roll forming characteristics. Aluminium panels shall be chromatised for maximum bond between metal and paint, enamelled twice under high temperature, one side with full primer and finish coat, the other side (inner side) with a primer coating and skin coat on a Continuous Paint Line. Panel	sqm	2753.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			shall be clipped to a baked enamelled Aluminium panel carrier of 62mm wide x 29mm deep made of 0.95mm thick in standard length of 5 mtrs made of doubled baked enamelled Aluminium alloy AA 5050 (Al. Mg) black with cut outs to hold the panels in a module of 200 mm (width of gap 20mm closed with recess profile) and at a distance 1.4 mtrs. Recess Join Profiles are made of enamelled aluminium and shall be inserted in between the panel on top of the panel flanges. The Carrier shall be suspended by means of G.I. suspension rod 4mm diameter, and a Galvanised suspension spring clip at a distance of 1.7mtrs centre to centre.Actual distance of threaded rod/carriermight vary and to be calculated based on the actual drawing and site condition. The Measurements shall be wall to wall without any deductions for lights, diffusers, columns etc.		
12.72		U Baffle aluminium panel ceiling	Providing and fixing U Baffle Aluminium panel ceiling of different colors and finish supplied by approved manufacturers, consisting of panel size 50 mm width X 100 mm deep using 0.6 mm thick, panel length upto 4 mtr, Coil Coated on a Continuous Paint Line, Double baked and roll formed from enamelled corrosion resistance Aluminium alloy AA 3005 (Al. Mg) for higher strength and good rollforming characteristics. Aluminium panels shall be chromatised for maximum bond between metal and paint enamelled twice under high temperature, Exposed side with a full primer and finish coat on a Continuous Paint Line. Panels shall beclipped to a baked enamelled Aluminium carrier of 30 mm wide x 47 mm high x 0.5 mm thick, Black Colour coated, one leg of the carriers with cut outs to hold the panels in a module of 150 mm. Panel carrier shall be suspended by means of threaded		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			rod at a distance of 1.8 mtr c/c. Actual distance of threaded rod/carrier might vary and to be calculated based on the actual drawing and site condition. The measurements shall be wall to wall without any deductions for lights, diffuses, columns etc. As an option, ceiling area above U Baffle can be painted by the contractor in black colour or any other colour as per the architect for better aesthetics before fixing this system.		
	12.72.1	Standard white	Standard White / Natural Satin Shade	sqm	2989.00
	12.72.2	Wooden finish	Wooden Finish	sqm	3855.00
12.73		False ceiling perforated G.I tile or plank on torsion spring system	Supply & Fixing of perforated G.I false ceiling tile or plank using torsion spring fixing system supplied by approved manufacturer comprising of tile of 600 mm wide and 600 mm long manufactured out of 0.5 mm thick or plank of 600mm wide and 1200 mm long manufactured out of 0.6mm thick galvanised steel perforated with 2 mm dia. at a distance of 5mm centre to centre. Non-woven felt made of glass-reinforced fibre glued over the perforation at the back side of panels to provide noise reduction co efficient up to 0.7 NRc. The metal ceiling panels shall be downward accessible with a minimum of four (4) torsion springs per panel. The Plank will be manufactured on advanced CAD/CAM equipment that includes several levelling stages in the manufacturing process. Torsion Spring panel with two side legs die formed and two end legs die formed and punched to receive torsion springs (min two springs each end or side) for secure engagement into Tee Grid main runners which are factory punched to receive torsion springs. Tiles will be square edged. The Plank shall be Polyester powder coated in whitecolour (RAL 9010).The panels will be pre-treated in latest Nano technology		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			process and electro statically powder coated with automatic Caroni system and cured with gas catalytic technology. Main runners 24mm deep, inverted "Tee" sections, 3.6m long, with factory punched flanges to receive torsion spring assembly. Main Tee on centre spacing to match panel length. Cross runners 24 mm deep, inverted "Tee" sections designed to interlock in to web of main tee section on designated spacing. Cross tee length to match panel length. Cross tees are spaced at a spacing of 600 mm on centre maximum. The measurements shall be wall to wall without any deductions for lights, diffusers, columns etc.		
	12.73.1	0.5 mm thick	Tile of 600 mm wide ,600 mm long and 0.5 mm thick	sqm	2321.00
	12.73.2	0.6mm thick	Plank of 600mm wide ,1200 mm long and 0.6mm thick	sqm	2377.00
12.74		Gypsum board glass wool fiber panel	Providing and fixing of composite perforated Gypsum board with Glass wool Fiber panel of Size 595x595x25 mm with 10mm square Perforation tiles false ceiling in true horizontal level suspended on interlocking metal grid of hot dipped galvanized steel section (Galvanized @ 120 gms/sqm both sides inclusive) consisting ofmain "T" runner with suitably spaced joint to get required length and of size 24x32 mm made from 0.30 mm thick (minimum) sheet spaced joints to get required length and of size 24x26 mm made of 0.30 mm thick (minimum)sheet spaced at 1200 mm long spaced between main "T" at 600 mm centre to centre to form a grid of 1200x600 mm and secondary cross "T" of length 600 mm and sheet to be interlocked at middle of the 1200x600 mm panel to form grids of 600x600mm and wall angle of size 24x24x0.4 mm and laying false ceiling tiles of approved Perforation in the grid	sqm	1934.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			including, wherever required, cutting/ making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors, etc. Main "T" runners to be suspended from ceiling using GI slotted plates of size27x37x25x1.6 mm fixed to ceiling with 12.50 mm dia and 40 mm long dash fasteners, 4 mm GI adjustable rods with galvanized butterfly level clips of size 85x30x0.80 mm spaced@ 1200mm centre to centre along with main T, bottom exposed width of 24mm of all T-sections shall be pre- painted with polyester paint all complete at all heights as per specifications and drawings and as directed by Engineer-in-Charge.		
12.75		Acoustical composite ceiling panel (MDF with Glass wool)	Providing and fixing of Composite perforated MDF board with Glasswool Fiber panel of Size 595x595x18 mm 10mm Square Perforation tiles (Square /Tegular Edge) false ceiling in true horizontal level suspended on interlocking metal grid of hot dipped galvanized steel section (Galvanized @ 120 gms/sqm both sides inclusive) consisting of main "T" runner with suitably spaced joint to get required length and of size 24x32 mm madefrom 0.30 mm thick (minimum) sheet spaced joints to get required length and of size 24x26 mm made of 0.30 mm thick (minimum)sheet spaced at 1200 mm long spaced between main "T" at 600 mm centre to centre to forma grid of 1200x600 mm and secondary cross "T" of length 600 mm and sheet to be interlocked at middle of the 1200x600 mm panel to form grids of 600x600 mm and wall angle of size 24x24x0.4 mm and laying false ceilingtiles of approved Perforation in the grid including, wherever required, cutting/ making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors, etc. Main "T" runners to be suspended from ceiling using GI slotted plates of size	sqm	2116.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			27x37x25x1.6 mm fixed to ceiling with 12.50 mm dia and 40 mm long dash fasteners, 4 mm GI adjustable rods with galvanized butterfly level clips of size 85x30x0.80 mm spaced @ 1200mm centre to centre along with main T, bottom exposed width of 24mm of all T-sections shall be pre-painted with polyester paint all complete at all heights as per specifications and drawings and as directed by Engineer-in-Charge.		
12.76		Glass fibre tissue faced ceiling panel	Providing and fixing of Perforated Gypsum board with Acoustical Glass Fiber Tissue faced panel of Size 595x595x12.5mm, Square 10mm Perforation tiles (Square Edge) false ceiling panels in true horizontal level suspended on interlocking metal grid of hot dipped galvanized steel section (Galvanized @ 120 gms/sqm both sides inclusive) consisting of main "T" runner with suitably spaced joint to get required length and of size 24x32 mm made from 0.30 mm thick (minimum) sheet spaced joints to get required length and of size 24x26 mm made of 0.30 mm thick (minimum)sheet spaced at 1200 mm long spaced between main "T" at 600 mm centre to centre to form a grid of 1200x600 mm and secondary cross "T" of length 600 mm and sheet to be interlocked at middle of the 1200x600 mm panel to form grids of 600x600 mm and wall angle of size 24x24x0.4 mm and laying false ceiling tiles of approved Perforation in the grid including, wherever required, cutting/ making, opening for services like diffusers, grills, light fittings, fixtures, smoke detectors, etc. Main "T" runners to be suspended from ceiling using GI slotted plates of size 27x37x25x1.6 mm fixed to ceiling with 12.50 mm dia and 40 mm long dash fasteners, 4 mm GI adjustable rods with galvanized butterfly level clips of size 85x30x0.80 mm spaced @ 1200mm centre to centre along with main T, bottom exposed width of	sqm	1240.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			24mm of all T-sections shall be pre-painted with polyester paint all complete at all heights as per specifications and drawings and as directed by Engineer-in-Charge.		
12.77		Mineral Fibre Acoustical Suspended Ceiling System	Providing and fixing mineral fibre false ceiling tiles at all heights of size 595X595mm of approved texture, design and pattern. The tiles should have Humidity Resistance (RH) of 99%, Light Reflectance≥ 85%, Thermal Conductivity k = 0.052 - 0.057 w/m K, Fire Performance as per (BS 476 pt - 6 &7)in true horizontal level suspended on interlocking T-Grid of hot dipped all round galvanized iron section of 0.33 mm thick (galvanized @120 gsm) comprising of main T runners of 15x32 mm of length 3000 mm, cross T of size 15x32mm of length 1200 mm and secondary intermediate cross T of size 15x32 mm of length 600 mm to form grid module of size 600x600 mm suspended from ceiling using galvanized mild steel item (galvanised@80gsm) 50 mm long 8mm outer diameter M-6 dash fasteners, 6 mm diameter fully threaded hanger rod up to 1000 mm length and L-shape level adjuster of size 85x25x2 mm, spaced at 1200 mm centre to centre along main 'T'. The system should rest on periphery walls /partitions with the help of GI perimeter wall angle of size24x24X3000 mm made of 0.40 mm thick sheet, to be fixed to the wall with help of plastic rawl plug at 450 mm centre to centre & 40 mm long dry wall S.S. screws. The exposed bottom portion of all T-sections used in false ceiling support system shall be pre-painted with polyester baked paint, for all heights. The work shall be carried out as per specifications, drawings and as per directions of the engineer-in-charge. With 16 mm thick beveled tegular mineral fibre false ceiling tile (NRC 0.55 to 0.6)	Sqm	1940.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.78		Light weight calcium silicate micro look edge false ceiling tiles	Providing and fixing false ceiling at all heights with 15 mm thick integral densified micro look edged non-cementitious light weight calcium silicate false ceiling tile reinforced with fibre and natural filler falseceiling tiles of Size 595x595 mm of approved texture, design and patterns having NRC (Noise Reduction Coefficient) of 0.50 (minimum) as per IS 8225:1987, Light reflectance of 85% (minimum). Non combustible as per BS:476 (part-4), fire performance as per BS:476 (part 6 &7), humidity resistance of 100%, thermal conductivity < 0.043 W/m K as per ASTM 518:1991, in true horizontal level suspended on interlocking metal powder coated T-Grid of hot dipped galvanised iron section of 0.40 mm thick on Silhouette profile,rotary stiched double webbed white with 6mm reveal profile (white/black), comprising of main-T runners of size 15x42mm of length 3000 mm, cross - T of size 15x42 mm of length 1200mm and secondary intermediate cross- T of size 15x42 mm of length 600mm to form grid module of size 600 x 600 mm, suspended from ceiling using galvanised mild steel items (galvanizing @ 80 grams per sqm) i.e. 50 mm long, 8 mm outer diameter M-6 dash fasteners, 6 mm dia fully threaded hanger rod upto 1000 mm length and L-shape level adjuster ofsize 85x25x2 mm. Galvanised iron perimeter wall angle of size 22x19x0.40 mm of length 3000 mm to be fixed on periphery wall / partition with the help of plastic rawl plugs at 450 mm center to center and 40mm long dry wall S.S screws. The work shall be carried out as per specifications, drawing and as per directions of the Engineer-in-Charge.	sqm	1632.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.79		Light weight bio safe calcium silicate false ceiling tiles	Providing and fixing false ceiling at all heights with 15 mm thick integral densified mirolook edge with BIO SAFE coating as per ISO 7251:2005, ISO 20743:2007 and ISO 21527-1:2008, making the tiles resistant to bacterial growth in case of any external bacterial contact made ontothe surface of tiles, non-cementitious light weight calcium silicate false ceiling tile reinforced with fibre and natural filler false ceiling tiles of Size 595x595 mm of approved texture, design, patterns and perforation having NRC (Noise Reduction Coefficient) of0.10-0.20 for plain & designer, 0.15 - 0.40 for pinhole texture and NRC 0.45 - 0.65 for fully perforated tiles as per IS 8225:1987, sound attenuation STA-32 dB.Light Reflectance of 85% (minimum), non-combustible as per BS:476 (part-4), fire performance asper BS:476 (part 6 &7), humidity resistance of 100%, thermalconductivity 0.050 to 0.052 W/m◦ KC as per ECBC Code 2007, weight 1.900Kg to 2.100 Kg per tile, density 450 Kg/m³ in edge portion (thickness 15 mm) and 350 Kg/m³ in central portion (thickness 10 mm), 50-52% recycled content including fly ash and meets the GRIHA & SVAGRIHA norms under the categories : GRIHA V.2015 criterion:11&12,GRIHA V.3 criterion: 17&29 & SVAGRIHA criterion:12 in true horizontal level metal T-grid suspended on interlocking metal powder coated T-Grid of hot dipped galvanised iron section of 0.40 mm thick on silhouette profile, rotary stitched double webbed white with 6 mm reveal profile (white/black), comprising of main-T runners of size 15x42 mm of length 3000 mm, cross - T of size 15x42 mm of length 1200 mm and secondary intermediate cross- T of size 15x42 mm of length 600mm to form grid module of size 600 x 600 mm, suspended from ceiling using galvanised mild steel (galvanizing @	sqm	1509.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			80 grams per sqm) i.e. 50 mm long, 12 mm outer diameter M-6 dash fasteners with the help of Z cleat of 27x37x27x2 mm, 6 mm dia fully threaded hanger rod upto 1000 mm length and L-shape level adjuster of size 85x25x25x2 mm. Galvanised iron perimeter wall angle of size 22x19x0.40 mm of length 3000 mm to be fixed on periphery wall / partition with the help of plastic rawl plugs at 450 mm centre to centre and 40mm long dry wall S.S screws. The work shall be carried out as per specifications, drawing and as per directions of the Engineer-in-Charge.		
12.80		Light weight calcium silicate wall panelling tiles	Providing and fixing in position wall panelling at all heights with 15 mm thick fully perforated square/butt edge light weight calcium silicate panels/ tiles., non-cementitious, reinforcedwith fibre and natural filler, having NRC (Noise Reduction Coefficient) of0.10-0.20 for plain and designer 0.15-0.40 for pinhole texture and NRC 0.45-0.65 for fully perforated tiles as per IS 8225:1987, sound attenuation STA-32db. of 0.75 (minimum) as per IS 8225:1987, Light Reflectance of 85% (minimum), non-combustible as per BS:476 (part-4), 50-52% recycled content including fly ash, fire performance as per BS:476 (part 6 &7), humidity resistance of 100%, thermal conductivity 0.050 to 0.052 W/m. KC as per ECBC Code 2007, weight 1.900 Kg to 2.100 Kg per tile, density 450 Kg/m3 in edge portion (thickness 15 mm) and 350 Kg/m3 in central portion (thickness 10 mm), comprising of a frame made from especially fabricated galvanised mild steel sheet 0.50 mm thick pressed section (galvanizing @120 grams per sqm including both sides) i.e. Vertical studs of size 48 x 34 x 36 mm are placed at 600mm centre to centre in a floor and ceiling channel section of size 50 x 32mm fixed to the floor and soffit at 600 mm centres using 12mm	sqm	1603.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			dia, 40mm long wedge type expanded zinc alloy dash fastener with 10 mm bolt. This same channel is then to be fixed in horizontal direction at 600mm centre to centre so as to form a grid of 600mm x 600mm. Glass wool of50mm thickness is then to be inserted in the slots and finally calcium silicate non - combustible panels/tiles are to be screw fixed with self-tapping pan headnickel coated mild steel screws of size 13 x 3.2mm on to this grid leaving an even groove of 1 mm between the panels. The joints between the panels are to be duly jointed and finished using recommended jointing calciumsilicate based compound and fibre joint tape roll 50mm wide (90 metre) roll and two coats of primer suitable for panelling as per manufacturer's specification as per direction of engineer-in-charge all complete. With 15 mm thick fully perforated square/butt edge light weight calcium silicate panels/ tiles.		
12.81		GRG (glass fibre reinforced) gypsum false ceiling	Providing & fixing false ceiling at all heights with GRG (Glass Fibre Reinforced Gypsum) false ceiling tiles of Size 595x595 mm of approved texture, design and patterns having moisture content less than 2%, humidity resistance of 99%, NRC0.50 to 0.75 as per IS 8225:1987, Noncombustible as per BS 476 (part 4)-1970 and light reflectance of 85% (minimum) to be laid in true horizontal level suspended on interlocking metal T-Grid of hot dipped galvanised iron section of 0.33mm thick (galvanized @ 120 grams per sqm including both sides) comprising of main-T runners of size 15x32 mm of length 3000 mm, cross - T of size 15x32 mm of length 1200 mm and secondary intermediate cross-T of size 15x32 mm of length 600mm to form grid module of size 600 x 600 mm, suspended from ceiling using galvanised mild steel items		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			(galvanizing @ 80 grams per sqm) i.e. 50 mm long, 8 mm outer diameter M-6 dash fasteners, 6 mm dia fully threaded hanger rod upto 1000 mm length and L-shape level adjuster ofsize 85x25x2 mm. Galvanised iron perimeter wall angle of size 24x24x0.40 mm of length 3000 mm to be fixed on periphery wall / partition with the help of plastic rawl plugs at 450 mm center to center and 40 mm long dry wall wood screws. The work shall be carried out as per specifications, drawing and as per directions of the Engineer-in- Charge.		
	12.81.1	12mm thick	With semi perforated 12mm thick micro tegular edged GRG false ceiling tiles.	sqm	1210.00
	12.81.2	10mm thick	With fully perforated 10mm thick square edged GRG false ceiling tiles for sound proof and acoustic.	sqm	1307.00
12.82		Calcium silicate 15 mm thick tegular edged	Providing and Fixing 15 mm thick densified tegular edged eco friendly light weight calcium silicate false ceiling tiles of approved texture of size 595 x 595 mm in true horizontal level, suspended on inter locking metal grid of hot dipped 387alvanized steel sections (387alvanized387 @ 120 grams per sqm including both side) consisting of main 'T' runner suitably spaced at joints to get required length and of size 24x38 mm made from 0.33 mm thick (minimum) sheet, spaced 1200 mm centre tocentre, and cross "T" of size 24x28mm made out of 0.33 mm (Minimum) sheet, 1200 mm long spaced between main'T' at 600 mm centre to centre to form a grid of 1200x600 mm and secondary cross 'T' of length 600 mm and size 24 x28 mm made of 0.33 mm thick (Minimum) sheet to be inter locked at middle of the 1200x 600 mmpanel to from grid of size 600x600 mm, resting on periphery walls /partitions on a Perimeter wall angle pre-coated steel of size(24x24X3000	sqm	1196.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			mm made of 0.40 mm thick (minimum) sheet with the help of rawlplugs at 450 mm centre to centre with 25 mm long dry wall screws @ 230 mm interval and laying 15 mm thick densified edges 388alvani silicate ceiling tiles of approved texture in the grid, including, cutting/ making opening for services like diffusers, grills, light fittings, fixtures, smoke detectors etc., wherever required.Main 'T' runners to be suspended from ceiling using G.I. slotted cleats of size 25x35x1.6 mm fixed to ceiling with 12.5 mm dia and 50 mm longdash fasteners, 4 mm G.I. adjustable rods with 388alvanized steel levelclips of size 85 x 30 x 0.8 mm, spaced at 1200 mm centre to centre along main 'T', bottom exposed with 24 mmof all Tsections shall be pre-paintedwith polyster baked paint, for all heights, as per specifications, drawings and as directed by Engineer-in-Charge. Note :- Only calcium silicate false ceiling area will be measured from wall to wall. No deduction shall be made for exposed frames/ opening (cut outs) having area less than 0.30 sqm.The calcium silicate ceiling tile shall have NRC value of 0.50 (Minimum), light reflection > 85%, non- combustible as per B.S. 476 part IV, 100% humidity resistance and also having thermal conductivity		
12.83		UPVC plank	Providing and fixing upvc plank, Plank size (340 X 2700) mm as per approved shade, the panel shall be laser catted end of plank fixing j-trim fixed and corner edge of ceiling fixed u pvc clip same colour and providing and fixing of frame work made of special sections, power pressed from M.S. sheets and galvanized with zinc coating of 125 gms/sqm (both side inclusive) as per IS : 277 and consisting of angle cleats of size 25 mm wide x 1.6 mm thick with flanges of 27 mm and 37mm, at 1200 mm centre to centre, one flange fixed to	sqm	2619.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			the ceiling with dash fastener 12.5 mm dia x 50mm long with 6mm dia bolts,other flange of cleat fixed to the angle hangers of 25x10x0.50 mm of required length with nuts & bolts of required size and other end of angle hanger fixed with intermediate G.I. channels 45x15x0.9 mm running at the spacing of 1200 mm centre to centre, to which the ceiling section 0.5 mm thick bottom wedge of 80 mm with tapered flanges of 26 mm each having lips of 10.5 mm, at 450 mm centre to centre, shall be fixed in a direction perpendicular,j trim end clip includes in this item any other item not be paid sepratly complete in all respect as per the directions of Engineer-in-Charge		
12.84		PVC ceiling system profile panel	Providing & Fixing PVC ceiling system profile panel with click system interlock panel size 4000mmx340mmx1mm resistance to weather perfect finish in wood grain & solid colour maintenance free Resistance to weather truly anti bacterial. with edge profile J-trim on periphery & center profile T-trim in panel joint Complete System to be fixed on G.I. framework of ceiling section at 450 mm distance with enter section suspended by roof by G.I. angle hanger in proper line & level etc complete PRODUCT SPECIFICATIONS: Meets or exceeds the standard specification of (ASTM D-4477) and its referenced documents. Impact resistance: No deformation observed. (Tested at 50 LBS) Tensile strength, MPa: >37.0 Tensile modulus, MPa: >2000 Surface distortion: None at 120F Squareness: <1/8” of square Length: Within ¼” of specification Lock control: Complies Color Change: <2 grayscales (16 hrs under 1200W/m2 UV radiation at 50 degrees)	Sqm	3178.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.85		Tensile (fabric) structure system	Design built, supply and erection of Tensile (fabric) structure system including designing and fabrication in M.S. steel tubular structure as per Indian standard section as per approved design and shape including base plate, dash fastener, clits etc. Cutting, patterning, plotting and wielding of fabric with the help of tensile structure software. Fabric should be wielded by high frequency PVC wielding in proper line with PVC cader to be fixed on end of the fabric and to be stretched & fixed on M.S. design structure with the help of. M.S. Flat bar with self-taping screw. M.S. structure to be painted with one coat of epoxy primer with MRF PU paint of approved colour in desired shape. With 1050 GSM tensile fabric		
	12.85.1		With precontraint tensile membrane 1050 GSM (20 years Warranty)	Sqm	9347.00
	12.85.2		With precontraint tensile membrane 1050 GSM (15 Years warranty)	Sqm	8151.00
12.86		Metal Laminate Wooden Finish Baffel Ceiling System	Providing & Fixing Metal Laminate Wooden Finish Baffel Ceiling System consisting of G.I. Metal baffle in laminated wooden finish of size 75mm Depth & 50mm Width with Skin Thickness 0.50 mm. Baffel to be clip on specially designed carrier with cutout arrangement to hold the baffle with spacing 125MM centre to centre. The carrier to be Suspended from roof by 4mm thick G.I. wire hanger with height adjustment clip. Hanger to be fixed on roof by 8 mm thick dash fasteners at spacing 1200 mm centre to centre. The measurement shall be wall to wall without deduction of light & diffuser etc. Baffle system to be suspended in proper line & level as per the satisfaction of Engineer- In-Charge	Sqm	3554.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.87		Mineral Fibre Acoustical Suspended Ceiling System with 1200x 300x16 & Silhouette grid	Providing & Fixing of Mineral Fibre 1200x300mm (Nominal Size) tiles Acoustical Suspended Ceiling System with 16mm thick Beveled Tegular Edge Tiles and 15mm Bolt Slot Exposed Grid. The tiles shall have fissured visual with acoustical punctures, Humidity Resistance (RH) of 99%, NRC 0.6 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C- 423 standard, CAC of 35db as per ASTM E1414, Light Reflectance 87% as per ASTM E-1477, Thermal Conductivity k = 0.052 - 0.057 W/mK, Colour Global White, Fire Performance of class 0/1 as per BS 476 Part 0/7 and Class A as per ASTM E84 with 16mm Beveled Tegular edge and 8mm drop in module size of 600 x 600mm. It shall be suitable for Green Building application, with Recycled content of 32%. The tile shall have certifications from GRIHA/ green pro etc and SUSPENSION The tile shall be laid on grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with metered ends & “bird's mouth” notches to provide mitred cruciform junctions colour global white (Matching to tile color). Grid shall have double stroke rotary stitching on all T sections i.e. the Main Runner & 1200 mm Cross Tees with a The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee	Sqm	2594.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			to main beam". All main beam to main beam and cross tee to cross tee connection shall have a pull-out strength of mare than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked polyester paint of OT {Jemlfng capability. ,lj|STALLATlON To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspen9ion system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and tha last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 1200mm to form 1200 x 1200 mm module. Cut cross tees longef than 600mm require independent support. 1200 x 30Omm module to be formed by fittinB 1200mm long flush fi\ting cross tees (with no mitre cut on cross tees) centrally between the 1200 mm cross tees(Miter cut at every 3O0 mm). Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 45Omm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia — 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 1 10 kgs. (Optionally, adjustable hook dips made of 0.7mm thick Grade 1 C55 galvanized spring eteel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.88		Mineral Fibre Acoustical Suspended Ceiling System	Providing & Fixing of Mineral Fibre Acoustical Suspended Ceiling System with minimum 14 to 15 mm	Sqm	1898.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			thick Bevelled Tegular Edge Tiles and 15mm Face Exposed Grid. The tiles shall have Assured visual with acoustical punctures , Humidity Resistance (RH) of 95%, NRC 0.5 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C-423 standard, CAC of 30db as per ASTM E1414, Light Reflectance 85% as per ASTM E-1477, Thermal Conductivity k = 0.052 - 0.057 W/mK, Colour G)obal White, Fire Performance of class 0/1 as per BS 476 Part 6/7 and Class A as per ASTM E84 with 15mm Bevelled Tegular edge and 8mm drop in module size of 600 x 6OOmm. It shall be suitable for Green Building application, with Recycled content of 30%. The tile shall have certifications from GRIHA/ GreenPro etc. SUSPENSION: The tile shall be laid on 15 mm wide T - section flanges colour global white (matching to tile color) having roll formed double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees witha web height of 38mm (all sections). The deflection loading of the system shall be 12.5 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) witha deflection loading of 15kg/m. Tne end details of the cross tee shall be made of pre hardened steel and fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 90 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with OT bending capability. All border cross tee/main beam shall be levelled using perimeter fill. INSTALLATION: To		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			comprise main runner spaced at 1 200mm secure\y fixed to the structural soffit using suspenGion system (specifications above) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 45Omm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 120Omm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm re9uire independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross teen centra\Iy between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be canied out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, jsre-straightened hanger wire of dia - 2.5 mm of 1.80 m length having a tensiie strength of 344-413 MPa and a minimum loading strength of 1 10 kgs. (Optionally, adjustable hook dips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 2Omm and 4 mm aquiiine wire to be used for level adjustment during installation purpose in main runners).		
12.89		Fine Fissured HI NRC Mineral Fibre 20mm Acoustical Suspended Ceiling System	Providing & Fixing of Mineral Fibre Acoustical Suspended Ceiling System with 20mm thick Bevelled Tegular Edge Tiles and 15mm Face Exposed Grid. The tiles shall have fissured visual with acoustical punctures , Humidity Resistance (RH) of 99%, NRC 0.7 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C-423 standard,	Sqm	2280.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			CAG of 35db as per ASTM E 1 414, Light Reflectance 8S% as per ASTM E-1477, Thermal Conductivity k = 0.045 - 0.048 W/MK, Colour Global White, Fire Performance of class 0/1 as per BS 476 Part 6/7 and Class A as per ASTM E84 with 19mm Bevelled Tegular edge and 8mm drop in module size of 600 x 60Omm. It shall be suitable for Green Building application, with Recycled content of 47%. The tile shall have certifications from GRIHA and **GreenPro. etc** SUSPENSION: The tile shall be laid on 15 mm wide T - section flanges colour global white (matching to tile color) having roll formed double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees witha web height of 38mm (all sections). The deflection loading of the system shall be 12.5 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 15kg/m. The end details of the cross tee shall be made of pre hardened steel and fixed to the ends of the cross tee to provide double locking between "cross tee to cross tee" and "cross tee to main beam". All main beam to main bgam and cross tee to cross tee connection shall have a pull out strength of more than 1 00kg. The T Sections shall be made of hot dipped galvanized steel of 90 g9m as per IS 277 (2003) and pre-painted steel with baked polyester paint with OT bending capability. All border cross tee/main beam shall be levelled using perimeter fill. INSTALLATION: To comprise main runner 9paced at 1200mm securely fixed to the structural soffit using suspension system		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			(specifications abova) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the IaGt main beam shall ba at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm center to center maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia - 2.5 mm of 1.80 m length having a ten9ilo strength of 344- 413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20 mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners)		
12.90		Antimicrobial Acoustical False Ceiling Suspended Ceiling System	PRIVIDING AND FIXING ANTI MICROBIAL ACOUSTICAL FALSE CEILING: Mineral Fibre Acoustical Suspended Ceiling System with 19mm thick Acoustics Microlook 90 Edge Tiles with15mm face exposed grid. The tiles shall have smooth bright laminated visual with Humidity Resistance (RH) of 99%, NRC 0.7 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423 standard, CAC of	Sqm	3004.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			41db as per ASTM E1414, Light Reflectance as per ASTM E-1477, Thermal Conductivity k = 0.052 - 0.057 W/mK, Colour Global White, Fire Performance of Class 0/1 as per BS 476 Part6/7 and Class A as per ASTM E 84 with 19mm Square Tegula r/ Microlook edge and 8mm drop in module size of 600 x 600mm./ 600x1200mm. The scrim on the face of the tile shall be cleanable, washable up to 500 cycle as per ASTM D-4828 and resistant to disinfectants. The tile shall be treated with Anti Microbial coating on the face of the tile to achieve cleanroom classification of class 5 (Class 100) as per ISO 14644-1 and suitable for zone 4 area having partial emission under normal conditions of ISO 5, Decontamination Class of CP10 and Bacteriological Cleanliness Class B1 for specific strains as per NF S 90-35:2003 The tile shall have a rating of 8 or above (face) for tests conducted for Resistance to growth of Mold in accordance to ASTM D3273. It shall be suitable for Green Building application, with Recycled content of 35% having VOC and Formaldehyde emission rate less than 0.1mg/m2/hr. The tile shall have certifications from GRIHA / GreenPro etc. SUSPENSION: The tile shall be laid on 14-15 mm wide T - section flanges colour global white (matching to tile color) having roll formed double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 12.5 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 15kg/m. The end details of the cross tee shall be made of pre hardened steel and fixed to the ends of the		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			cross tee to provide double locking between tee to cross and tee to main All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 90 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with 0T bending capability. All border cross tee/main beam shall be levelled using perimeter fill. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications above) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.91		Mineral Fibre Acoustical Suspended Ceiling System with Bevelled Tegular & Silhoutte Grid	Providing & Fixing of Mineral Fibre Acoustical Suspended Ceiling System with Bevelled Tegular Edge Tiles of 16mm thickness and 15mm Bolt Slot face exposed grid. The tiles shall have sand texture visual with acoustical punchers , Humidity Resistance (RH) of 99%, NRC 0.5 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423 standard, CAC of 33db as per ASTM E1414, Light Reflectance ≥85% as per ASTM E-1477, Thermal Conductivity k = 0.052 - 0.057 W/mK, Colour Global White, Fire Performance of class 0/1 as per BS 476 Part 6/7 and Class A as per ASTM E84 with 16mm bevelled Tegular edge and 8mm drop in module size of 600 x 600mm. It shall be suitable for Green Building application, with Recycled content of 40%. The tile shall have certifications from GRIHA / Green Pro. Etc. SUSPENSION: The tile shall be laid on profile grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with mitred ends & “bird’s mouth” notches to provide mitred cruciform junctions colour global white (Matching to tile color). Grid shall have double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee”	Sqm	2389.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked polyester paint of 0T bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees.Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.92		Mineral Fibre Acoustical Suspended Ceiling System with 20mm thick & Silhoutte Grid system	Providing & Fixing of Mineral Fibre Acoustical Suspended Ceiling System with Bevelled Tegular Edge Tiles of 20mm thickness and 15mm bolt slot exposed grid. The tiles shall have sand texture visual with acoustical punchers, Humidity Resistance (RH) of 99%, NRC 0.7 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423 standard, CAC of 33db as per ASTM E1414, Light Reflectance ≥84% as per ASTM E-1477, Thermal Conductivity k = 0.052 - 0.057 W/MK, Colour Global White, Fire Performance of class 0/1 as per BS 476 Part 6/7 and Class A as per ASTM E84 with 20mm bevelled tegular edge and 8mm drop in module size of 600 x 600mm. It shall be suitable for Green Building application, with Recycled content of 49%. The tile shall have certifications from GRIHA/ GreenPro .etc SUSPENSION: The tile shall be laid on grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with mitred ends & “bird’s mouth” notches to provide mitred cruciform junctions colour global white (Matching to tile color).Grid shall have double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked	Sqm	2603.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			polyester paint of 0T bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm center to center maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners)		
12.93		Mineral Fibre 15mm Acoustical Suspended Ceiling System with Silhoutte Grid	Providing & Fixing of Mineral Fibre Acoustical Suspended Ceiling System with 15mm thick Sanded Bevelled Tegular Edge Tiles with 15mm bolt slot exposed grid. The tiles shall have sand texture visual with acoustical punchers having Humidity Resistance (RH) of 95%, NRC 0.70 (Average value of	Sqm	2331.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ISO 354:2003 standard, CAC of 35db as per ASTM E1414, Light Reflectance ≥85% as per ASTM E-1477, Thermal Conductivity λ = 0.060 W/mK, Colour Global White, Fire Performance of class A2-s1, d0 according to EN 13501-1 with 15mm Tegular 15 BE edge and 8mm drop in module size of 600 x 600mm. The tile shall be suitable for Green Building application, with Recycled content of 40%. The tile shall have Environmental Product Declaration as per ISO 14025 and EN 15804+A2. The tile shall have Environmental Product Declaration as per ISO 14025 and EN 15804+A2 SUSPENSION: The tile shall be laid on grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with mitred ends & “bird’s mouth” notches to provide mitred cruciform junctions colour global white (Matching to tile color). Grid shall have double stroke rotary stitching on all T sections i.e., the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull-out strength of more than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked polyester paint of 0T bending capability.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.94		Wooden finished Suspended Ceiling System with Micro look edge perforated Ceiling Tile	Providing & Fixing Wooden finished Suspended Ceiling System with Microlook edge tiles and 15mm Face exposed grid. The 18mm thick Tile with perforation shall be made of Medium Density Fibre of density 700+ Kgs/m3 having perforation of hole diameter 6mm and open area 11% with a linear pitch of 16mm with a border	sqm	7627.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			of 25mm.The tile shall have melamine laminate finish of approved shade (Portland Maple, Sea Beech, Siam Wood & Mystique Walnut) on visible side and backing laminate (same shade) on the reverse with 0.8mm matching PVC edge banding in module size of 600mm x 600mm. The edges of the tile shall be Square Tegular/Microlook 90˚ having 8mm drop using 3mm thick compact board spline. Back side of the tile shall have Soundtex fleece hot pressed to achieve NRC of 0.55 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423 and Humidity Resistance (RH) of 70%, with sag limited to 1/300mm. The tile shall have a Fire Performance of class 1 according to BS 476 part 7. SUSPENSION: The tile shall be laid on 15 mm wide T - section flanges having double stroke rotary stitching on all T - sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 43mm (all sections). The deflection loading of the system shall be 15.6 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360 within suspension points) with deflection loading of 18.6kg/m. The end details of the main beam and cross tee shall be made of pre hardened steel clip fixed to the ends. The cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T - Sections shall be made of hot dipped galvanized steel of 100 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with 0T bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees.Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. All border cut tiles to be laid using border clip. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.95		GI MESH CEILING SYSTEM	GI MESH CEILING - Providing & Fixing horizontal level suspended ceiling comprising of GI Mesh Lay-in metal ceiling system consisting of 600x600mm (Nominal size) Lay-in tiles of 90 gsm galvanized steel as per IS 277 (2003) suitable for architectural finishes in 1.5mm thickness having Rhombus shape mesh in Global white/ Black/ Silver	Sqm	8286.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Grey color. Tile shall have Short way of opening (SWO) as 11.0mm and Long way of opening (LWO) as 25.9mm with an open area of 78.0% for Lattice Flat or Rhombic design (Termed as Rhombic #4). The tile shall be electrostatically polyester powder coated using RoHS compliant powder having coating thickness ranging from 60-80 microns. The powder coating shall pass Salt Spray Test for 750 hours as per ASTM B117, QUV –A Test for 1000 hours as per ASTM G-154 – (Cycle – 1) and Humidity test for 1000 hours as per DIN 50017. The powder coating shall have an adhesion of GT0 as per EN ISO 2409. The shade variation across batches shall be delta E<1 using Excite/BYK Gardner instrument and gloss level of 25±5 at 60˚ as per ASTM D523. SUSPENSION: The tile shall be laid on 24 mm wide T - section flanges colour global white (matching to tile colour) having roll formed double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 15 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 18kg/m. The end details of the cross tee shall be made of pre hardened steel and fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 90 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with 0T bending capability. The Tile & Grid shall have the manufacturers name		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			embossed/printed on it. The supply shall be backed by a Manufacturing Test Certificate. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications above) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm center to center maximum. Installation shall be carried out as per recommended procedure. SUSPENSION SYSTEM accessories supplied by consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.96		FALSE CEILING SYSTEM GI METAL ANTI BACTERIAL CLIP IN PLAIN TILE WITH CARRIER SYSTEM	FALSE CEILING SYSTEM GI METAL CLIP IN PLAIN TILE WITH CARRIER SYSTEM 600 X 600MM . Providing & Fixing horizontal level suspended ceiling comprising of GI Clip-in Plain Tile 600x600mm (Nominal Size) tiles of 120 gsm galvanized steel as per IS 277	Sqm	2864.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			(2003) suitable for architectural finishes in 0.5mm thickness as per in IS:513 (1994) in Global white color with concealed suspension system having a bevelled edge of 3x3mm and height of 33mm on one side and 18mm on the other side. Tiles shall be flattened by precision leveller to minimize sag to less than 1/400mm. The tiles shall have double round pip of ϕ 2mm and ϕ 6mm for secure engagement and levelling respectively, along with swing down feature and special tabs to allow removal of tile to enable plenum access. The coated surface of the tile shall exhibit anti-microbial properties (eradication and inhibition of growth) and shall achieve up to 99.99% reduction in bacterial activity as per ISO22196:2011. The tile shall be electrostatically polyester powder coated using RoHS compliant powder having coating thickness ranging from 60-80 microns. The powder coating shall pass Salt Spray Test for 750 Hours as per ASTM B117, QUV –A Test for 1000hours as per ASTM G-154 – (Cycle – 1) and Humidity test for 1000hours as per DIN 50017. The powder coating shall have an adhesion of GT0 as per EN ISO 2409. The shade variation across batches shall be delta E<1 using Excite/BYK Gardner instrument and gloss level of 25±5% at 60˚ as per ASTM D523. The tile shall have scratch, mar & abrasive resistance properties. The tile shall have a Light Reflectance of ≥ 63% in accordance to ASTM 1477-98 and Fire Performance of class 0/1 as per BS 476 Part 6 & 7 in module size of 600x600mm. SUSPENSION & INSTALLATION: Shall comprise of 3000mm long 'C' channels of 38H x 12W mm and 1mm thick spaced at 1200mm centre to centre securely fixed to the structural soffit using threaded rod (Approved make/Third Party Supply) connecting to C-channel		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			hanger of 109x18.5x16.5mm at 1200mm C/C and anchor fastener (Approved make/Third Party Supply). The last hanger at the end of each C-channel shall not be greater than 600mm from the adjacent wall. Use a C-channel connector of 99.5x35x 9mm for splicing together two pieces of C-channels. 4000mm Clip-in bar of 16W x 40H mm and 0.6mm thick made of galvanized steel of 140gsm as IS 277 (2003) and shall be spaced at 600mm centre to centre in a direction perpendicular to the C-channels and secured at every intersection with C-channel using a Clip-in bar hanger of 54x18x20mm. Use Clip-in bar connecter to splice two pieces of Clip-in bar. Tiles shall be clipped in between two Clip-in bars from below. Optionally - Security clip to be installed to ensure the panels are positively engaged with Clip-in bar channel. Installation shall be carried out as per Manufacturer recommended procedure. Perimeter to be finished using wall angles of size 3000x19x32mm in white color secured to walls using screws at 450mm centre to centre. Cut tiles to be secured to the wall angles using a spring clamp.		
12.97		Horizontal level suspended ceiling comprising of PPGI Microlook Lay-in Micro Perforated metal ceiling	Providing & Fixing horizontal level suspended ceiling comprising of PPGI Microlook Lay-in Micro Perforated metal ceiling consisting of 1200x300mm (Nominal Size) tiles made of 120 gsm pre-painted galvanized iron (PPGI), Grade – GPL as per IS 277 (2018) with coating durability of Class C1 as per IS 14246 suitable for architectural finishes of 0.5mm thickness as per in IS:513 (2008) in Global white color with Square Tegular/Microlook 90˚ edge having 8mm drop and 5.5mm flange. Tile shall have perforation having hole diameter 1.8mm and open area 20% with linear and diagonal pitch	Sqm	3386.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			of 5mm and 3.5mm respectively and a border of 9.1mm. The difference between the 2 diagonal dimensions of the tile to be less than 1mm. Tiles shall be flattened by precision leveller to minimize sag to less than 1/400mm. The tile shall be pre-painted using RoHS compliant polyester paint on face of thickness ranging from 15 -30 microns and epoxy paint on back of thickness 4-10 micron. The paint coating shall pass Salt Spray Test for 350 hours (Face) and 200 hours (Back) as per ASTM B117, QUV – A Test for 500 hours as per ASTM G-154 – (Cycle – 1) and Humidity test for 500 hours as per DIN 50017.The paint shall have an adhesion of GT0 as per EN ISO 2409. The shade variation across batches shall be delta E<0.8 using Excite/BYK Gardner instrument and gloss level of 12±5% at 60˚ as per ASTM D523.The tile shall have a Light Reflectance of ≥ 63% in accordance to ASTM 1477-98. Back side of the tile shall have Soundtex fleece hot pressed mechanically for uniform bonding to achieve NRC up to 0.7 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C-423 and Fire Performance of class 0/1 as per BS 476 Part 6 & 7 in module size of 600x600mm. SUSPENSION: The tile shall be laid on grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with mitred ends & “bird’s mouth” notches to provide mitred cruciform junctions colour global white (matching to tile color).Grid shall have double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm(all sections). The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked polyester paint of 0T bending capability SUSPENSION: The tile shall be laid on grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with mitred ends & “bird’s mouth” notches to provide mitred cruciform junctions colour global white (Matching to tile color).Grid shall have double stroke rotary stitching on all T sections i.e. the Main Runner & 1200 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked polyester paint of 0T bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 1200mm to form 1200 x 1200 mm module. Cut cross tees longer than 600mm require independent support. 1200 x 300mm module to be formed by fitting 1200mm long flush fitting cross tees (with no mitre cut on cross tees) centrally between the 1200 mm cross tees(Miter cut at every 300 mm). Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.98		Horizontal level suspended ceiling comprising of PPGI Microlook Lay-in Plain 1200x300 metal ceiling System	Providing & Fixing horizontal level suspended ceiling comprising of GI Microlook Lay-in Plain metal ceiling consisting of 1200x300mm (Nominal Size) Lay-in tiles of 120 gsm galvanized steel as per IS 277 (2003) suitable for architectural finishes in 0.5 mm thickness as per IS:513 (1994) in Global white color with Square Tegular/ Microlook 90˚ edge having 8mm drop and 5.5mm flange. The difference between the	Sqm	2770.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			2 diagonal dimensions of the tile to be less than 1mm. Tiles shall be flattened by precision leveller to minimize sag to less than 1/400mm. The tile shall be electrostatically polyester powder coated using RoHS compliant powder having coating thickness ranging from 60-80 microns. The powder coating shall pass Salt Spray Test for 750 Hours as per ASTM B117, QUV –A Test for 1000 hours as per ASTM G-154 – (Cycle – 1) and Humidity test for 1000 hours as per DIN 50017. The powder coating shall have an adhesion of GT0 as per EN ISO 2409. The shade variation across batches shall be delta E<1 using Excite/BYK Gardner instrument and gloss level 25±5 at 60˚ as per ASTM D523. The tile shall have a Light Reflectance of ≥ 63% in accordance to ASTM 1477-98 and Fire Performance of Class 0/1 as per BS 476 Part 6 & 7 in module size of 600x600mm. All border cut tiles shall be laid, using lay-in end cap for installation to have a Microlook effect on perimeters. SUSPENSION: The tile shall be laid on 15 mm wide T - section flanges colour global white (matching to tile colour) having double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 12.5 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 15kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			more than 100kg. The T Sections shall be made of hot dipped galvanized steel 90 gsm and pre-painted steel with baked polyester paint of 0T bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. Flush fitting 1200mm long cross tees to be interlocked between main runners at 1200mm to form 1200 x 1200 mm module. Cut cross tees longer than 600mm require independent support. 1200 x 300mm module to be formed by fitting 1200mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. All border cut tiles shall be laid, using lay-in end cap for installation to have a Microlook effect on perimeters. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories manufactured and supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre–Straightened Hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. Additionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
12.99		METAL OPEN CELL CEILING SYSTEM	METAL OPEN CELL CEILING: Providing & fixing horizontal level Open cell Aluminium lay-in ceiling tiles with border panels forming board edge of size 600x600x38mm (Nominal) having cell size of 100x100mm made of AA-3003-H14 (Tempered) grade Aluminium suitable for architectural finishes in 0.35mm thickness as per in IS:2676 (1981) in Global white color. The tile shall be electrostatically polyester powder coated using RoHS compliant powder having coating thickness ranging from 60-80 microns. The powder coating shall pass Salt Spray Test for 750 hours as per ASTM B117, QUV –A Test for 1000 hours as per ASTM G-154 – (Cycle – 1) and Humidity test for 1000 hours as per DIN 50017. The powder coating shall have an adhesion of GT0 as per EN ISO 2409. The shade variation across batches shall be delta E<1 using Excite/BYK Gardner instrument and gloss level of 25±5% at 60˚ as per ASTM D523. Tile shall have a Fire Performance class A as per GB 8624-2012. SUSPENSION: The tile shall be laid on 15 mm wide T - section flanges colour global white (matching to tile colour) having double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 43mm (all sections). The deflection loading of the system shall be 15.5 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 18.6kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out	Sqm	4035.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel 90 gsm and pre-painted steel with baked polyester paint of 0T bending capability. Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees.Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. All border cut tiles shall be laid, using lay-in end cap for installation to have a Microlook effect on perimeters. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories manufactured and supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre–Straightened Hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. Additionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners.		
12.100		ACOUSTICAL WOODWOOL FALSE CEILING SYSTEM	ACOUSTICAL WOODWOOL FALSE CEILING: Providing & fixing of 100% Natural Material-Magnesite (Hygroscopic & biologically sound binding agent) bonded woodwool, planks of 600Mm*1200Mm *25Mm thickness made out of wood wool fiber of width 1Mm Fiber thickness mixed with Natural water & Magnesite (No-Cement) with open & smooth surface texture, Weight of panel to	Sqm	8149.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			be 11.3 Kg/m2. A Completely harmless Environment & Human Friendly & also a disposable or recyclable Material & can last for more than 80 Years in building interiors As it has long lifespan & suitable for Indoor conditions with relative air humidity of upto 90%. Magnesite protects the wood cells against ageing & fungal attack. To achieve upto NRC=1.00 it should be a clear gap of 200Mm from wall slab along with the backing of Mineral wool of 50mm thick with 64 kg/m3 density. Ecologically Sound Material with high mechanical strength with Crash test of 90Km/H with Ball Impact Test DIN18032 & EN13964 with flying colors & A sustainable material (PEFC & FSC Certified) along with EPD Certified Acc. to EN ISO 14025 (AUB-Environmental Product declaration). Manufacturing tolerance for nominal size: Length, Width & Thickness ± 1 mm, & In a Standard Climate of 23° C and 50% rel. humidity. FIXING & SUBSTRUCTURE: Impact Resistant According to DIN-18023 or EN13964 to be Fixed on CD Sections Profile framework with Screw 9Pc/Panel & Centre Distance Support of 300Mm & Centre distance of Cross Profile would be at 600Mm & Main Profile at 900Mm CD Profile is made out of GI(Sections Profile 60/27/0.6Mm-0.6MmThick x 60Mm Width x 0.27Mm height). The Main Profile is in the length of 3000mm & to form a section of 600mm x 1200mm. The CD Profile is to be suspended from soffit with the help of nonius hanger at a center distance of 900mm. UD Profile 28/27/0.6mm to be fixed on wall at the perimeter of the wall. If the main CD profile needs to extended, it needs to be connected with CD profile connector 60/27. The CD profiles to be connected to each other at the intersection by CD cross connector. panels to be fixed		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			to the ceiling with a minimum of 200mm void or gap with Galvanized screws of length 50Mm & Dia 4.5Mm with a backing of Mineral Wool of density 64kg/m3 density (50mm thick). Colour pf Panels : NATURAL BEIGE, Can be Painted to the desired colour (Any-RAL, NCS or S to Color Shades) using water based color by Spray method prior to installation. FIRE CLASS as per EN 13501-1:B –s1, d0.		
12.101		ACOUSTICAL GLASS FIBER FALSE CEILING - SS BLADES SYSTEM	ACOUSTICAL GLASS FIBRE FALSE CEILING : Providing and Fixing Acoustical blade which are 40 mm thick, flat glass fibre panels with Humidity Resistance RH 90%, with factory fitted embedded metal insert with ¼" – 20 threading for fixing cable adjuster for hanging application. Acoustical blades shall have sizes of 200x1200mm, 200x1600mm, 400x1200mm, 400x1600mm, 400x1800mm, 500x1200mm or 500x1600mm with both side laminated scrim finished with uniform water based paint on all surface (including edges). Acoustical performance of 0.6 TO 0.70 NRC(for 400mm high x 40mm thick baffles with 400mm on centre distance and overall depth of construction 1000mm) as per ASTM C-423 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) in standard White color with Light reflectance 87% or in the color specified by the Architect (Shell / Tangerine / Cranberry / Moss / Kiwi / Reef / Lagoon), Fire Performance Class B2 – s1, d0 as per EN 13501-1 .The blade shall be made of Glass fibre panel with density of 110-120 Kg/ m3. SUSPENSION: The panel shall be hung using hanging kit comprising of griper structure anchor & cap of dia 20mm, spring loaded bottom end cable adjuster of dia. 9mm and 2.4m long aircraft cable of 1/16" dia. Components 1)	Sqm	17783.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Griper structure anchor 2) Gripper anchor cap 3) Cable adjuster 4) Aircraft Cable Cable adjuster Spring loaded bottom end cable adjuster of dia. 9mm and 45mm long Aircraft cable Material Galvanized Steel Aircraft cable dimension 1/16" Ø x 2.4m Tolerance ±1mm. INSTALLATION:Hanger Wire Attachment to Structure: Attach the gripper structure anchor to the structure with an appropriate fastener (1/4"-20 Threading) that will carry the full weight of the blades. Insert the end of the cable into the gripper anchor cap. Screw the gripper anchor cap completely into the gripper structure anchor. Insert Suspension Cables: The panel is to be suspended using the aircraft cables which are suspended from the soffit using the gripper structure anchors and its other end passing through the bottom end cable adjuster. The height & level of the panels can be adjusted using the bottom end cable adjusters.Suspension of Blade: Locate the two factory fitted female inserts on the blades. Suspend the Blades by screwing the bottom end of the cable adjuster into the female inserts. Installation should be carried out as per Manufacturer recommended procedure.		
12.102		ACOUSTICAL GLASS FIBER FALSE CEILING - SS SHAPES SYSTEM	ACOUSTICAL GLASS FIBRE FALSE CEILING : Providing and Fixing Acoustical Clouds/Shapes - Acoustical clouds which are 30 mm thick, flat glass fibre panels of Humidity Resistance RH 95%, with embedded frame made of aluminium extruded section of 38X10mm of grade 6063-T6 as per dimension mentioned in below table for reinforcement and secured hanging. Embedded extrusion shall have grub slot with pre inserted nut for group hanging and connected using Aluminium	Sqm	18115.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			die casted corner connector with ¼" – 20 threading for hanging using cable adjuster for independent clouds. The form of Acoustical Clouds/Shapes - shall be Square / Convex / Concave / Circle / Hexagon / Trapezoid / Parallelogram (Left & Right) / Rectangle (Small & Large) in standard Traffic White color with light reflectance 90% as per ASTM 1477-98 or in the color specified by the Architect / Engineer in charge (Shell / Tangerine / Cranberry / Moss / Kiwi / Reef / Lagoon). The Acoustical Clouds/Shapes shall be made of Glass fibre panel with density of 110-120 Kgs/m3 along with acoustical sound absorbing scrim on the face to achieve sound absorption as per below mentioned table in accordance to EN ISO 354 :2003 (average of 500-4000Hz) . The face and edges shall be finished with matching water based texture paint. Shape Nominal Size and Sound Absorption – Metric Sabin 1. Square 1170x1170mm - 2.48 Metric Sabin 2. Convex 1170x1140mm 2.28 Metric Sabin 3. Concave 1170x1140mm 1.94 Metric Sabin 4. Circle 1170 Dia. 2.12 Metric Sabin 5. Hexagon 1170x1010mm 1.61 Metric Sabin 6. Trapezoid 1170x1170mm 2.15 Metric Sabin 7. Left Parallelogram 1170x1170mm 2.15 Metric Sabin 8. Right Parallelogram 1170x1170mm 2.15 Metric Sabin.The panel shall be hung as per the following based on the installation. Individual Suspension - The panel shall be hung using deck hanging kit comprising of griper structure anchor & cap of dia. 20mm, spring loaded bottom end cable adjuster of dia. 9mm and 2.4m long aircraft cable of 1/16" dia. The number of deck hanger will be as per Table 1 above. The panels to be suspended individually using the Acoustical Clouds/Shapes Deck hanging kit as per the number specified in		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			above Table 1. Hanger Wire Attachment to Structure: Attach the gripper structure anchor to the true ceiling using an appropriate fastener (1/4"-20 Threading) that will carry the full weight of the panels. Insert the end of the cable into the gripper anchor cap. Screw the gripper anchor cap completely into the gripper structure anchor. The panel to be suspended using the aircraft cables from the soffit by fixing the cap end of the cable to gripper structure anchors and the other end passing through the bottom end cable adjuster. The height & level of the panels can be adjusted using the bottom end cable adjusters. Suspend the Acoustical Clouds/Shapes by locating and screwing the bottom end of the cable adjuster into the threading of the connector. The height & level of the panels can be adjusted using the bottom end cable adjusters.		
12.103		ACOUSTICAL FALSE CEILING WITH 0.9 NRC: Mineral Fibre Acoustical Suspended Ceiling System	PRIVIDING AND FIXING ACOUSTICAL FALSE CEILING WITH 0.9 NRC:Mineral Fibre Acoustical Suspended Ceiling System with 15mm thick OP Tegular 15/90 Edge Tiles with 15mm face exposed grid. The tiles shall have smooth laminated visual with Humidity Resistance (RH) of 95%, NRC 0.90 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423 standard, CAC of 25db as per grihaASTM E1414, Light Reflectance ≥86% as per ASTM E-1477, Thermal Conductivity λ =0.040 W/mK, Colour Global White, Fire Performance of class A2-s1, d0 according to EN 13501-1 with 15mm Tegular 15/90˚ edge and8mm drop in module size of 600 x 600mm. The scrim on the face of the tile shall be cleanable and classified as ISO14644-1	Sqm	2828.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			cleanroom class 5. It shall be suitable for Green Building application, with Recycled content of 43% and VOC Emissions (French) class of A+. The tile shall have Environmental Product Declaration as per ISO 14025 and EN 15804+A2. The tile shall have certifications from GRIHA / GreenPro Etc. thereby confirming that it is an environment friendly product. SUSPENSION: The tile shall be 15 mm wide T - section flanges colour global white (matching to tile color) having roll formed double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 12.5 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 15kg/m. The end details of the cross tee shall be made of pre hardened steel and fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 90 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with 0T bending capability. All border cross tee/main beam shall be levelled using perimeter fill. To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications above) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be carried out as per recommended procedure. SUSPENSION SYSTEM accessories supplied by consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.104		SOFT FIBER CEILING TILE with Silhoutte	Providing & Fixing of Fibre Glass Acoustical Suspended Ceiling System with 20/25mm thick Microlook Edge Tiles with 15mm Bolt Slot exposed grid. The tiles shall have smooth laminated visual with Humidity Resistance (RH) of 95%, NRC 0.85 – 0.95 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423 standard, , Light Reflectance ≥88% as per ASTM E-1477, Colour Pebble White, Fire Performance of Class 0/1 as per BS 476 Part 6/7 with 20/25mm Square Tegular/Microlook 90˚ edge and 8mm drop in module size of 600 x 600mm. It shall be suitable for Green Building application, with Recycled content of 85%.SUSPENSION: The tile shall	Sqm	3286.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			be laid on grid system with 15mm white flanges incorporating a 6mm central reveal in white/black colour with mitred ends & “bird’s mouth” notches to provide mitred cruciform junctions colour global white. Grid shall have double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 38mm (all sections). The deflection loading of the system shall be 7.9 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 9.5 kg/m. The end details of the cross tee shall be made of pre hardened steel clip fixed to the ends of the cross tee to provide double locking between “cross tee to cross tee” and “cross tee to main beam”. All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of 90 gsm as per IS 277 (2003) pre-painted steel with baked polyester paint of 0T bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less than 600mm from the wall. F Flush fitting 1200mm long cross tees to be interlocked between main runners at 600mm to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees.		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 450mm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners).		
12.105		Gypsum Ceiling System & 15mm grid system	Providing & Fixing of Gypsum CeiIing system with 9 mm thick Board Edge Tiles and 24mm Face Exposed Grid. The tilee shall have weight of 7.0 kgs/m2 for 9mm thickness. Specifications of Gypsum Ceiling Board. 1 Basic Struoure: PVC + Paper Surfacing Plasterboard + Plating Aluminum Membrane. p.Facing: White Vinyl.Foil Back, Sealed Edge 2. Standard Dimensions (Nominal): 600“600 3. Thickness: 9mm 4. Weight per Sqm: 7.0 kgs/m2 for 9mm 5.Basic parameter of 9 mm board: a.Lengthways ruptures load: >360N b.Lateral ruptures load: >150 Thickness tolerance: +/- 0.5 mm d.Size tolerance. +/- 1 mm SUSPENSION: The tile shall be laid on 24 mm wide T - section flanges colour global white (matching to tile colour) having roll formed double stroke rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tess with a web height of 3Bmm (all	Sqm	1340.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			sections). The deflection loading of the system shall be 15 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 18kg/m. The end details of the cross tee shall be made of pre hardened steel and fixed to the ends of the cross tee to provide double locking between "cross tee to cross tee' and "cross tee to main beam". All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 90 gsm as per lS 277 (2003) and pre-painted steel with baked polyester paint with OT bending capability. INSTALLATION: To comprise main runner spaced at 1200mm securely fixed to the structural soffit using suspension system (specifications above) at 1200mm maximum. The First/Last suspension system at the end of each main runner shall not be greater than 450mm from the adjacent wall along with the first and the last main beam shall be at less tnan 600mm from the wall. Flush fitting 1200 mm long cross tees to be interlocked beMeen main runners at 600mm to form 1 200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formad by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls using screws at 4S0mm centre to centre maximum. Installation shall be carried out as per Manufacturer recommended procedure. SUSPENSION SYSTEM accessories supplied by		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre-straightened hanger wire of dia — 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. (Optionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used to level adjustment during installation.		
12.106		NATURAL BAMBOO CEILING	Providing & fixing of Wooden Grill panel with backer made of natural bamboo with 24mm Face black exposed grid. The Grill panels shall be made of rapidly renewable natural bamboo core having density 600+Kgs/m3 as per BS EN323 (1993) and bending strength of 96+ N/mm2 as per BS EN310 (1993). The Grill panel shall have a seven layer UV coating of thickness 0.09-0.11mm as per BS EN71 finished to natural or carbonized shades. The Grill panel shall be of nominal size 35x300x2400mm comprising 8 blades/slats of 35x16x2376mm with centre to centre spacing of 37.5mm fixed through Backer of size 12x22x300mm having centre to centre spacing of 300mm and additional end backers at 150mm from both the edges. Ledger of 6.5x24x2400mm shall be used for covering face of wall angle. Open edges shall be covered using Edge Caps of 44x19x2400mm and Blade Junction of 66x38x2400mm shall be used along the connection of two Grill panels. The Grill panel when used with 20mm thick black glasswool infill panels shall have an NRC of 0.9 (Average value of Absorption Co-efficient readings taken at 250 Hz, 500 Hz, 1000 Hz and 2000 Hz) as per ASTM C423. The Grill panel shall have a fire performance of Class A according to	Sqm	21486.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			ASTM-E84. The Grill panel shall have moisture content of 8-14% as per IS 1708 (Part 1):1986 and a Humidity Resistance (RH) of 70%.SUSPENSION: The tile shall be installed on 24 mm wide T - section flanges having rotary stitching on all T sections i.e. the Main Runner, 1200 mm & 600 mm Cross Tees with a web height of 43mm, 35mm and 30mm respectively. The deflection loading of the system shall be 17.8 kg/m2 (as per standard installation layout mentioned below) and main beam tested as per ASTM C635 (Deflection limit less than L/360) with a deflection loading of 21.4kg/m. The end details of the main beam and cross tee shall be made of pre hardened steel fixed to the ends. The cross tee to provide double locking between "cross tee to cross tee" and "cross tee to main beam". All main beam to main beam and cross tee to cross tee connection shall have a pull out strength of more than 100kg. The T Sections shall be made of hot dipped galvanized steel of 100 gsm as per IS 277 (2003) and pre-painted steel with baked polyester paint with 0T bending capability. All exposed panel ends to be covered using edge caps. All wall angles to be covered using ledgers. Installation should be carried out as per Manufacturer recommended procedure. The grid shall be installed with suspension system of recommended make. INSTALLATION: To comprise main runner spaced at 1200mm centre to centre securely fixed to the structural soffit using suspension system (specifications below) at 1200mm maximum centre. The First/Last suspension system at the end of each main runner should not be greater than 450mm from the adjacent wall. Flush fitting 1200mm		

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			long cross tees to be interlocked between main runners at 600mm centre to form 1200 x 600 mm module. Cut cross tees longer than 600mm require independent support. 600 x 600mm module to be formed by fitting 600mm long flush fitting cross tees centrally between the 1200 mm cross tees. Perimeter trim to be wall angles of size 3000x19x19mm, secured to walls at 450 mm maximum centre to centre. Metal spring backer clips shall be used to fix the Grill panel to suspension system with a grid formation of 600mm along the blade/slat and 2 pieces along each backer (total 8 clips per full panel size) for every individual panel. Raise the panel up against the suspension system and align the backers with the suspension system. Push upwards on the spring backer clip to engage the clip tabs on to the suspension system flange. Ensure that both clip tabs engage the suspension system flange. SUSPENSION SYSTEM accessories manufactured and supplied by Manufacturer consisting of M6 Anchor Fasteners with hanger hole, pre Straightened Hanger wire of dia – 2.5 mm of 1.80 m length having a tensile strength of 344-413 MPa and a minimum loading strength of 110 kgs. Additionally, adjustable hook clips made of 0.7mm thick Grade 1 C55 galvanized spring steel of dimension 100 x 20mm and 4 mm aquiline wire to be used for level adjustment during installation purpose in main runners.		
12.107		Antimicrobial False Ceiling Suspended Ceiling System	Providing and fixing mineral fibre false ceiling tiles at all heights of size 595X595mm of approved texture, design and pattern. The tiles should have Humidity Resistance (RH) of 99%, Light Reflectance ≥ 85%, Thermal Conductivity k = 0.052 - 0.057 w/m K, Fire Performance as	Sqm	2100.00

Item No.	Sub Item No.	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			per (BS 476 pt - 6 &7)in true horizontal level suspended on interlocking T-Grid of hot dipped all round galvanized iron section of 0.33 mm thick (galvanized @120 gsm) comprising of main T runners of 15x32 mm of length 3000 mm, cross T of size 15x32mm of length 1200 mm and secondary intermediate cross T of size 15x32 mm of length 600 mm to form grid module of size 600x600 mm suspended from ceiling using galvanized mild steel item (galvanised@80gsm) 50 mm long 8mm outer diameter M-6 dash fasteners, 6 mm diameter fully threaded hanger rod up to 1000 mm length and L-shape level adjuster of size 85x25x2 mm, spaced at 1200 mm centre to centre along main 'T'. The system should rest on periphery walls /partitions with the help of GI perimeter wall angle of size24x24X3000 mm made of 0.40 mm thick sheet, to be fixed to the wall with help of plastic rawl plug at 450 mm centre to centre & 40 mm long dry wall S.S. screws. The exposed bottom portion of all T-sections used in false ceiling support system shall be pre-painted with polyester baked paint, for all heights. The work shall be carried out as per specifications, drawings and as per directions of the engineer-in-charge. With 16 mm thick beveled tegular mineral fibre Antimicrobial false ceiling tile confirming to ISO 5 (class 100) specifications		

CHAPTER – 13.0

FINISHING WORK
(PLASTERING AND PAINTING)

LIST OF BUREAU OF INDIAN STANDARD CODES

S.No.	I.S. No.	Subject
1.	IS 16 (Part-I)	Shellac : Part : I — Hand Made Shellac
	IS 16(Part-II)	Shellac : Part : II — Machine Made Shellac
2.	IS 75	Linseed Oil Raw and Refined
3.	IS 77	Linseed Oil Boiled For Paints
4.	IS 102	Ready Mixed Paint, Brushing, Red Lead, Nonsetting, Priming
5.	IS 104	Specification for Ready Mixed Paint, Brushing, Zinc Chrome, Priming
6.	IS 109	Ready Mixed Paint, brushing, priming Plaster to Indian Standard ColourNo.361, 631 White and off White
7.	IS 117	Ready Mixed Paint, Brushing, Finishing Exterior, Semigloss for GeneralPurposes to Indian Standards Colours.
8.	IS 133	Enamel, Interior (a) Under Coating (b) Finishing
9.	IS 137	Ready Mixed Paint, Brushing, Matt Or Egg Shell Flat, Finishing Interior to Indian Standard Colour as required
10.	IS 158	Ready Mixed Paint, Brushing, Bituminous Black, Lead Free, Acid, Alkali and Heat Resisting
11.	IS 217	Specification for Cut Back Bitumen
12.	IS 218	Specification for Creosote and Anthracene Oil For Use As Wood Preservatives
13.	IS 290	Coal Tar Black Paint
14.	IS 337	Varnish, Finishing Interior
15.	IS 341	Black Japan, Types ‘A’, ‘B’ and ‘C’
16.	IS 347	Varnish, Shellac for General Purposes
17.	IS 348	French Polish
18.	IS 419	Putty for Use On Window Frames
19.	IS 427	Distemper, Dry Colour as Required
20.	IS 428	Distemper, Oil Emulsion, Colour as Required
21.	IS 524	Varnish, Finishing, Exterior, Synthetic Air Drying
22.	IS 533	Gum Spirit of Turpentine (Oil of Turpentine)
23.	IS 712	Specification For Building Limes
24.	IS 1200	Method of Measurements of Building and Civil Engineering Works
	(Part-XII)	Part : XII — Plastering and Pointing
25.	IS 1200	Method of Measurements of Building and Civil Engineering Works.
	(Part-XIII)	Part : XIII — White Washing, Colour Washing Distempering and Painting of Building Surfaces
26.	IS 1200	Methods of Measurements of Building and Civil Engineering Works
	(Part-XV)	Part : XV — Painting, Polishing, Varnishing etc
27.	IS 2339	Aluminium Paint For General Purposes, in Dual Container
28.	IS 2547 (Part-II)	Gypsum Building Plasters Pt.II Premixed Light Weight Plasters
29.	IS 2932	Enamel, Synthetic, Exterior (a) Undercoating, (b) Finishing
30.	IS 2933	Enamel, Exterior (a) Undercoating (b) Finishing
31.	IS 5410	Cement Paint
32.	IS 5411 (Part-1)	Plastic Emulsion : Paint Part I For Interior Use
33.	IS 6278	Code of Practice For White Washing and Colour Washing

PREAMBLE
FINISHING WORK
(CHAPTER: 13.0)

13.1 Cement Plaster

The cement plaster shall be 12 mm, 15 mm or 20 mm thick as specified in the item

13.2 Preparation of Surface

The joints shall be raked out properly. Dust and loose mortar shall be brushed out. Efflorescence if any shall be removed by brushing and scrapping. The surface shall then be thoroughly washed with water, cleaned and kept wet before plastering is commenced. In case of concrete surface if a chemical retarder has been applied to the form work, the surface shall be roughened by wire brushing and all the resulting dust and loose particles cleaned off and care shall be taken that none of the retarders is left on the surface

13.3 Application of Plaster

Ceiling plaster shall be completed before commencement of wall plaster.

Plastering shall be started from the top and worked down towards the floor. All putlog holes shall be properly filled in advance of the plastering as the scaffolding is being taken down. To ensure even thickness and a true surface, plaster about 15 × 15 cm shall be first applied, horizontally and vertically, at not more than 2 metres intervals over the entire surface to serve as gauges. The surfaces of these gauged areas shall be truly in the plane of the finished plaster surface. The mortar shall then be laid on the wall, between the gauges with trowel. The mortar shall be applied in a uniform surface slightly more than the specified thickness. This shall be brought to a true surface, by working a wooden straight edge reaching across the gauges, with small upward and side ways movements at a time. Finally the surface shall be finished off true with trowel or wooden float according as a smooth or a sandy granular texture is required. Excessive troweling or over working the float shall be avoided.

All corners, arrises, angles and junctions shall be truly vertical or horizontal as the case may be and shall be carefully finished. Rounding or chamfering corners, arrises, provision of grooves at junctions etc. where required shall be done without any extra payment. Such rounding, chamfering or grooving shall be carried out with proper templates or battens to the sizes required.

When suspending work at the end of the day, the plaster shall be left, cut clean to line both horizontally and vertically. When recommencing the plastering, the edge of the old work shall be scrapped cleaned and wetted with cement slurry before plaster is applied to the adjacent areas, to enable the two to properly join together. Plastering work shall be closed atthe end of the day on the body of wall and not nearer than 15 cm to any corners or arrises. Itshall not be closed on the body of the features such as plasters, bands and cornices, nor at the corners of arrises. Horizontal joints in plaster work shall not also occur on parapet tops and copings as these invariably lead to leakages. The plastering and finishing shall be completed within half an hour of adding water to the dry mortar. No portion of the surface shall be left out initially to be patched up later on. The plastering and finishing shall be completed within half an hour of.

13.4 Finish

The plaster shall be finished to a true and plumb surface and to the proper degree of smoothness as required. The work shall be tested frequently as the work proceeds with a true straight edge not less than 2.5 m long and with plumb bobs. All horizontal lines and surfaces shall be tested with a level and all jambs and corners with a plumb bob as the work proceeds.

13.5 Precaution

Any cracks which appear in the surface and all portions which sound hollow when tapped, or are found to be soft or otherwise defective, shall be cut out in rectangular shape and redone as directed by the Engineer-in-Charge.

(i) When ceiling plaster is done, it shall be finished to chamfered edge at an angle at its junction with a suitable tool when plaster is being done. Similarly when the wall plaster is being done, it shall be kept separate from the ceiling plaster by a thin straight groove not deeper than 6 mm drawn with any suitable method with the wall while the plaster is green.

(ii) To prevent surface cracks appearing between junctions of column/beam and walls, 150 mm wide chicken wire mesh should be fixed with U nails 150 mm centre to centre before plastering the junction. The plastering of walls and beam/column in one vertical plane should be carried out in one go. For providing and fixing chicken wire mesh withU nails payment shall be made separately.

13.6 Scaffolding

For all exposed brick work or tile work double scaffolding independent of the work having two sets of vertical supports shall be provided. The supports shall be sound and strong, tied together with horizontal pieces over which scaffolding planks shall be fixed. For all other work in buildings, single scaffolding shall be permitted. In such cases the inner end of the horizontal scaffolding pole shall rest in a hole provided only in the header course for the purpose. Only one header for each pole shall be left out. Such holes for scaffolding shall, however, not be allowed in pillars/columns less than one metre in width or immediately near the skew backs of arches. The holes left in masonry works for scaffolding purposes shall be filled and made good before plastering.

13.7 White washing/Colour washing/ Painting/ Distempering etc.

Before new work is white washed, the surface shall be thoroughly brushed free from mortar droppings and foreign matter. In case of old work, all loose particles and scales shall be scrapped off and holes in plaster as well as patches of less than 50 cm area shall be filled up with mortar of the same mix. Where so specifically ordered by the Engineer-in-Charge, the entire surface of old white wash shall be thoroughly removed by scrapping and this shall be paid for separately. Where efflorescence is observed the deposits may be brushed clean and washed. The surface shall then be allowed to dry for atleast 48 hours before white washing is done.

13.8 Protective Measures

Doors, windows, floors, articles of furniture etc. and such other parts of the building not to be white washed, shall be protected from being splashed upon. Splashings and droppings, if any shall be removed by the contractor at his own cost and the surfaces cleaned. Damages if any to furniture or fittings and fixtures shall be recoverable from the contractor.

13.9 Material

The paint shall be (Texured exterior paint/Acrylic smooth exterior paint/premium acrylic smooth exterior paint) of approved brand and manufacture. This paint shall be brought to the site of work by the contractor in its original containers in sealed condition. The material

shall be brought in at a time in adequate quantities to suffice for the whole work or at least a fornight's work. The materials shall be kept in the joint custody of the contractor and the Engineer-in-Charge. The empty containers shall not be removed from the site of work till the relevant item of work has been completed and permission obtained from the Engineer- in-Charge.

13.10 Preparation of Surface

For new work, the surface shall be thoroughly cleaned off all mortar dropping, dirt dust, algae, fungus or moth, grease and other foreign matter of brushing and washing, pitting in plaster shall make good, surface imperfections such as cracks, holes etc. should be repaired using white cement. The prepared surface shall have received the approval of the Engineer in charge after inspection before painting is commenced.

Before pouring into smaller containers for use, the paint shall be stirred thoroughly in its container, when applying also the paint shall be continuously stirred in the smaller containers so that its consistency is kept uniform. Dilution ratio of paint with potable water can be altered taking into consideration the nature of surface climate and as per recommended dilution given by manufacturer. In all cases, the manufacturer's instructions and directions of the Engineer-in-charge shall be followed meticulously. The lids of paint drums shall be kept tightly closed when not in use as by exposure to atmosphere the paint may thicken and also be kept safe from dust.

Paint shall be applied with a brush/ roller on the cleaned and smooth surface. Horizontal strokes shall be given and vertical strokes shall be applied immediately afterwards. This entire operation will constitute one coat. The surface shall be finished as uniformly as possible leaving no brush roller marks.

Where so stipulated, the painting shall be done by spraying. Spray machine used may be (a) high pressure (small air aperture) type, or (b) a low pressure (large air gap) type, depending on the nature and location of work to be carried out. Skilled and experienced workmen shall be employed for this class of work. Paints used shall be brought to the requisite consistency by adding a suitable thinner.

Spraying should be done only when dry condition prevails. Each coat shall be allowed to dry out thoroughly and rubbed smooth before the next coat is applied. This should be facilitated by thorough ventilation. Each coat except the last coat, shall be lightly rubbed down with sand paper or fine pumice stone and cleaned off dust before the next coat is laid.

No left over Paint shall be put back into the stock tins. When not in use, the containers shall be kept properly closed.

No hair marks from the brush or clogging of Paint puddles in the corners of panels, angles of mouldings etc. shall be left on the work.

In painting doors and windows, the putty round the glass panes must also be painted but care must be taken to see that no Paint stains etc. are left on the glass. Tops of shutters and surfaces in similar hidden locations shall not be left out in painting. However, bottom edge of the shutters where the painting is not practically possible, need not be done nor any deduction on this account will be done but two coats of primer of approved make shall be done on the bottom edge before fixing the shutters.

On painting steel work, special care shall be taken while painting over bolts, nuts, rivets overlaps etc.

TABLE 13.1

Measurement

Sr. No.	Description of work	How measured	Multiplying coefficients
1	**2**	**3**	**4**
I	**Wood work doors, windows etc.**		
1	Panelled or framed and braced doors, windows etc.	Measured flat (not girthed including)	1.30 (for each side)
2	Ledged and battened or ledged, battened and braced doors, windows etc.	Chowkhat or frame, Edges, chocks, cleats, etc. shall be deemed to be included in the item.	-do-
3	Flush doors etc.	-do-	1.20 (for each side)
4	Part panelled and part glazed or gauzed doors, window etc. (Excluding painting of wire gauze portion)	-do-	1.00 (for each side)
5	Fully glazed or gauzed doors, windows etc. (Excluding painting of wire gauze portion)	-do-	0.80 (for each side)
6	Fully venetioned or louvered doors, windows etc.	-do-	1.80 (for each side)
7	Trellis (or Jaffri) work one way or two way	Measured flat overall, no deduction shall be made for open spaces, sup-porting members shall not be mea-sured separately	2 (for painting all over)
8	Carved or enriched work	Measured flat	2 (for each side)
9	Weather boarding	Measured flat (not girthed supporting frame work shall not be measured separately	1.20 (for each side)
10	Wood shingle roofing	Measured flat (not girthed)	1.10 (for each side)
11	Boarding with cover fillets and match boarding	Measured flat (not girthed)	1.05 (for each side)
12	Tile and slate battening	Measured flat overall no deductions shall be made for open spaces	0.80 (for painting all over)
II	**Steel work doors, windows etc.**		
13	Plain sheeted steel doors or windows	Measured flat (not girthed) including frame edges etc.	1.10 (for each side)
14	Fully glazed or gauzed steel doors and windows (excluding painting of wire gauze portion)	-do-	0.50 (for each side)
15	Partly panelled and partly glazed or gauzed doors and windows (excluding painting of wire gauze portion)	-do-	0.80 (for each side)

Sr. No.	Description of work	How measured	Multiplying coefficients
1	2	3	4
16	Corrugated sheeted steel doors or windows	-do-	1.25 (for each side)
17	Collapsible gates	Measured flat	1.50 (for each side)
18	Rolling shutters of interlocked laths	Measured flat (size of opening) all over; jamb guides, bottom rails and locking arrangement etc. shall be in- cluded in the item (top cover shall be measured separately)	1.10 (for each side)
III	**General**		
19	Expanded metal, hard drawn steel wire fabric of approved quality, grill works and gratings in guard bars, balustrades, railing partitions and MS Bars in windows frames	Measured flat overall; no deduction shall be made for open spaces; supporting members shall not be measured separately	1 (for Paint all over
20	Open palisade fencing and gates including standards, braces, rails stays etc. in timber or steel	-do- (see note No. 12)	1 (for Paint all over)
21	Corrugated iron sheeting in roofs, side cladding etc.	-do- Measured flat (not girthed)	1.14 (for each side)
22	AC corrugated sheeting in roofs, side cladding etc	-do-	1.20 (for each side)
23	AC semi corrugated sheeting in roofs, side cladding etc. or Nainital pattern using plain sheets	-do-	1.10 (for each side)
24	Wire gauze shutters including painting of wire gauze	-do-	1.00 (for each side)

13.11 Explanatory Notes for Table 13.1

(i) Measurements for doors windows etc., shall be taken flat (and not girthed) over allincluding chowkhuts or frames, where provided. Where Chowkhuts or frames are not provided, the shutter measurements shall be taken.

(ii) Where doors, windows etc, are of composite types other than those included in Table 13.1 the different portion shall be measured separately with their appropriate coefficients, the centre line of the common rail being taken as the dividing line between the two portions

(iii) The coefficients for door and windows shall apply irrespective of the size of frames and shutter members.

(iv) In case steel frames are used the area of doors, windows shutters shall be measured flat excluding frames.

(v) When the two faces of a door, window etc. are to be treated with different specified finishes, measurable under separate items, the edges of frames and shutters shall be

treated with the one or the other type of finish as ordered by the Engineer-in-Charge and measurement of this will be deemed to be included in the measurement of the face treated with that finish.

(vi) In the case where shutters are fixed on both faces of the frames, the measurement for the door frame and shutter on one face shall be taken in the manner already described, while the additional shutter on the other face will be measured for the shutter only excluding the frame.

(vii) Where shutters are provided with clearance at top or/and bottom each exceeding 15 cm height, such openings shall be deducted from the overall measurements and relevant coefficient shall be applied to obtain the area payable.

(viii) Collapsible gates shall be measured for width from outside to outside of gate in its expanded position and for height from bottom to top of channel verticals. No separate measurements shall be taken for the top and bottom guide rails rollers, fittings etc.

(ix) Coefficients for sliding doors shall be the same as for normal types of doors in the table. Measurements shall be taken outside to outside of shutters, and no separate measurements shall be taken for the painting guide rails, rollers, fittings etc.

(x) Measurements of painting as above shall be deemed to include painting all iron fittings in the same or different shade for which no extra will be paid.

(xi) The measurements of guard bars, expanded metal, hard drawn steel wire fabric of approved quality, grill work and gratings, when fixed in frame work, painting of which is once measured else where shall be taken exclusive of the frames. In other cases the measurements shall be taken inclusive of the frames.

(xii) For painting open palisade fencing and gates etc., the height shall be measured from the bottom of the lowest rail, if the palisades do not go below it, (or from the lower end of the palisades, if they project below the lowest rail), upto the top of rails or palisades whichever are higher, but not up to the top of standards when the latter are higher than the top rails or the palisades. Width of moulded work of all other kinds, as in hand rails, cornices, architraves shall be measured by girth.

For trusses, compound girders, stanchions, lattice girders, and similar work, actual areas will be measured in sq. metre and no extra shall be paid for painting on bolt heads, nuts, washers etc. even when they are picked out in a different tint to the adjacent work.

Painting of rain water, soil, waste, vent and water pipes etc. shall be measured in running metres of the particular diameter of the pipe concerned. Painting of specials such as bends, heads, branches, junctions, shoes, etc. shall be included in the length and no separate measurements shall be taken for these or for painting brackets, clamps etc.

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.1		12 mm cement plaster	12 mm cement plaster of mix :		
	13.1.1	1:4	1:4 (1 cement : 4 sand)	sqm	173.00
	13.1.2	1:6	1:6 (1 cement : 6 sand)	sqm	160.00
13.2		15 mm cement plaster	15 mm cement plaster on the rough side of single or half brick wall of mix		
	13.2.1	1:4	1:4 (1 cement : 4 sand)	sqm	184.00
	13.2.2	1:6	1:6 (1 cement : 6 sand)	sqm	168.00
13.3		20 mm cement plaster	20 mm cement plaster of mix :		
	13.3.1	1:4	1:4 (1 cement : 4 sand)	sqm	245.00
	13.3.2	1:6	1:6 (1 cement : 6 sand)	sqm	225.00
13.4		12 mm cement plaster finished with a floating coat	12 mm cement plaster finished with a floating coat of neat cement of mix :		
	13.4.1	1:3	1:3 (1 cement : 3 sand)	sqm	210.00
	13.4.2	1:4	1:4 (1 cement : 4 sand)	sqm	203.00
13.5		15 mm cement plaster finished with a floating coat	15 mm cement plaster on rough side of single or half brick wall finished with a floating coat of neat cement of mix		
	13.5.1	1:3	1:3 (1 cement : 3 sand)	sqm	236.00
	13.5.2	1:4	1:4 (1 cement : 4 sand)	sqm	220.00
13.6		18 mm cement plaster cm 1:5& 1:4 smooth finish	18 mm cement plaster in two coats under layer 12 mm thick cement plaster 1:5 (1 cement : 5 sand) finished with a top layer 6mm thick cement plaster 1:4 (1 cement : 4 sand).	sqm	220.00
13.7		18 mm cement plaster cm 1:5 & 1:3 rough finished with sponge	18 mm cement plaster in two coats under layer 12 mm thick cement plaster 1:5 (1 cement : 5 sand) and a top layer 6mm thick cement plaster 1:3 (1 cement : 3 sand) finished rough with sponge.	sqm	236.00
13.8		6 mm cement plaster	6 mm cement plaster of mix : 1:3 (1 cement : 3 sand)	sqm	144.00
13.9		6 mm cement plaster with floating coat	6 mm cement plaster 1:3 (1 cement : 3 sand) finished with a floating coat ofneat cement and thick coat of Lime wash on top of walls when dry for bearing of R.C.C. slabs and beams.	sqm	189.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.10		Neat cement punning	Neat cement punning	sqm	30.00
13.11		22mm rough cast plaster	Rough cast plaster upto 10m height above ground level with a mixture of sand and gravel or crushed stone from 6mm to 10mm nominal size dashed over and including the fresh plaster in two layers, under layer 12mm cement plaster 1:4 (1 cement: 4 sand) and top layer 10mm cement plaster 1:3 (1 cement: 3 sand) mixed with 10% finely grounded hydrated lime by volume of cement. Finished with ordinary cement.	sqm	296.00
13.12		22 mmpebble dash plaster	Pebble dash plaster upto 10m height above ground level with a mixture of washed pebble or crushed stone 6mm to 12.5mm nominal size dashed over and including fresh plaster in two layers under layer 12mm cement plaster 1:4 (1 cement: 4 sand) and top layer 10mm cement plaster with cement mortar 1:3 (1 cement: 3 sand) mixed with 10% finely grounded hydrated lime by volume of cement.	sqm	284.00
13.13		21mm sand faced plaster	Providing sand faced plaster upto 10meter hight above ground level concrete or brick masonry surfaces intwo coats, base coat of 13 mm thick in cement mortar 1:4 (1 cement: 4 sand), clearing the surface by combing it and finishing coat of 8 mm. thick in cement mortar 1:3 (1 cement: 3 sand) and taking out surface grains by mechanical arrangement or applying second coat by mechanical means including cost of all material, labour, and T and P including all lead, lift and scaffolding etc. complete.	sqm	314.00
13.14		Extra for water proofing material in plaster	Extra for providing and mixing water proofing material in cement plaster work in proportion recommended by the manufacturers.	per 50 kg cement used in the mix	79.00
13.15		Extra for plastering exterior/interior walls height more than 10m	Extra for plastering		

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
	13.15.1	Exterior walls	Exterior walls of height more than 10 m from ground for every additional height of 3 m or part thereof	sqm	34.00
	13.15.2	Interior walls	Interior walls of height more than 6 m from floor level for every additional height of 3 m or part thereof	sqm	22.00
13.16		Extra for plastering on circular work	Extra for plastering on circular work not exceeding 6 m in radius:		
	13.16.1	One coat	In one coat	sqm	11.00
	13.16.2	Two coat	In two coats	sqm	18.00
13.17		Extra for plastering on moulding cornices	Extra for plastering done on moulding cornices or architraves including neat finish to line and level:		
	13.17.1	One coat	In one coat	sqm	158.00
	13.17.2	Two coat	In two coats	sqm	260.00
13.18		Extra for plastering	Extra for plastering Spherical, Groined ceiling, Flewing soffits.		
	13.18.1	Spherical	Spherical ceiling	sqm	42.00
	13.18.2	Groined	Groined ceiling	sqm	47.00
	13.18.3	Flewing	Flewing soffits	sqm	28.00
13.19		25mm washed stone grit plaster on exterior walls	Washed stone grit plaster on exterior walls of height upto 10 M. aboveground level in two layers, under layer 12mm cement plaster 1:4 (1 cement : 4 sand) furrowing the under layer with scratching tool, applying cement slurry on the under layer @ 2 Kg of cement per square metre, top layer 15mm cement plaster 1:1/2:2 (1 cement : 1/2 sand : 2 stone chipping 10mm nominal size) in panels with groove all around asper approved pattern including scrubbing and washing, the top layer with brushes and water to expose the stone chippings,complete as per specification and direction of Engineer- in- charge (Payment for providing grooves shall be made separately).	sqm	428.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.20		Forming groove in the top layer of washed stone grit plaster	Forming groove of uniform size 20 or15 mm wide and 15 mm deep in the top layer of washed stone grit plaster as per approved pattern using wooden battens, nailed to the under layer including removal of wooden battens, repair to the edges of panels and finishing the groove complete as per specifications and direction of the Engineer-in-charge :	meter	24.00
13.21		Extra for grit plaster of height more than 10m	Extra for washed grit plaster on exterior/interior walls of height more than 10m from ground/floor level for every additional height of 3 m or part thereof.	sqm	67.00
13.22		Extra for grit plaster on circular work 6m in radius	Extra for washed stone grit plaster on circular work not exceeding 6m inradius (in two coats).	sqm	36.00
13.23		Grooving stone gritplastered	Forming groove of uniform size from 12x12mm and upto 25x15mm in the top layer of washed stone grit plastered surface as per approved pattern, including providing and fixing aluminum channels of appropriate size and thickness (not less than 2 mm), nailed to the under layer with rust proof screws and nails and finishing the groove complete as per specifications and direction of the Engineer-in-Charge.	meter	69.00
13.24		Extra for using white cement	Extra for using white cement in place of ordinary cement in the top layer of the item of washed stone grit plaster.	sqm	124.00
13.25		Premixed formulated one coat gypsum lightweight plaster	Providing and applying 12 mm thick (average) premixed formulated one coat gypsum lightweight plaster having additives and light weight aggregates as vermiculite/ perlite respectively conforming to IS: 2547 (Part -1 and II) 1976, applied on hacked/ uneven background such as bare brick/ block/ RCC work on walls and ceiling at all floors and locations,excluding in wall which may come in contact with dampness finished in smooth line and level etc. complete.	sqm	161.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.26		Addition of synthetic polyester triangular fibre	Extra for addition of synthetic Polyester triangular fibre of length 6mm, effective diameter 10-40 microns and specific gravity of 1.34 to 1.40 in cement plaster/mortar by using 125gms. Of synthetic Polyester triangular fibre for50 Kgs. cement used in cement mortaras per directions of Engineer-in-Charge.	per bag of 50 kg cement	56.00
13.27		Mesh in plaster	Providing and fixing mesh in verticaland horizontal junction of RCC and brick/AAC work including scaffolding and all lead and lift etc. complete		
	13.27.1	Chicken mesh in plaster	G.I. chicken mesh as per ISI specification and in the required width with 50mm long Bombay nails before plastering upt 10 meter height	Sqm	62.00
	13.27.2	Fibre mesh in Plaster	Providing and fixing fibre mesh of 145 GSM weight in required width with cement mortar before plastering upto 10m height.	Sqm	45.00
13.28		12 mm thick plain cement mortar band	12 mm thick plain cement mortar bands in cement mortar 1:4 (1 cement : 4 sand) :		
	13.28.1	Flush	Flush Band	cm per m	2.00
	13.28.2	Sunk	Sunk Band	cm per m	2.00
	13.28.3	Raised	Raised Band	cm per m	2.00
	13.28.4	Moulded	Moulded Band	cm per m	4.00
13.29		18 mm thick plain cement mortar band	18 mm thick plain cement mortar band in cement mortar 1:4 (1 cement : 4 sand):		
	13. 29.1	Flush	Flush Band	cm per m	2.00
	13. 29.2	Sunk	Sunk Band	cm per m	3.00
	13. 29.3	Raised	Raised Band	cm per m	3.00
	13. 29.4	Moulded	Moulded Band	cm per m	5.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.30		18mm moulded cement mortar band	18 mm thick moulded cement mortar band in two coats under layer 12mmthick with cement mortar 1:5 (1 cement : 5 sand) top layer 6mm thick with cement mortar 1:4(1 cement : 4 sand).	cm per m	5.00
13.31		Pointing on brick work	Pointing on brick work or brick flooring with cement mortar 1:3 (1 cement : 3 sand) :		
	13.31.1	Flush	Flush / Ruled/ Struck or weathered pointing.	sqm	70.00
	13.31.2	Raised	Raised and cut pointing	sqm	112.00
	13.31.3	Sunk	Sunk pointing	sqm	89.00
13.32		Pointing on stone work	Pointing on stone work with cement mortar 1:3 (1 cement: 3 sand).		
	13.32.1	Flush	Flush/ Ruled pointing	sqm	96.00
	13.32.2	Raised	Raised and cut pointing	sqm	172.00
13.33		Raised and cut pointing on stone work	Raised and cut pointing on stone work in white cement mortar 1:3 (1 white cement : 3 marble dust)	sqm	196.00
13.34		Flush pointing on stone slab ceiling	Pointing on stone slab ceiling with cement mortar 1:2 (1 cement : 2 sand): Flush/ Ruled pointing	sqm	56.00
13.35		Extra for pointing at height more than 10 m	Extra for pointing on walls on the outside/inside at height more than 10 metre from ground/floor level for every additional height of 3 m or part there of.	sqm	5.00
13.36		Broken glass on compound wall	Providing and fixing broken glass in cement mortar 1.3 including cost of mortar, over compound walls.	sqm	139.00
13.37		Writing board green	Providing and fixing writing board green writing with chalk (regular modified polyster coated steel surface). of sizes.		
	13.37.1	1200x1800mm	size 1200mmx1800mm	each	7436.00
	13.37.2	1200x2400mm	size 1200mmx2400mm	each	9730.00
	13.37.3	1200x3000mm	size 1200mmx3000mm	each	12126.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.38		POP putty	Providing and applying plaster of paris putty of 2 mm thickness over plastered surface to prepare the surface even and smooth complete	sqm	77.00
13.39		POP punning	Plaster of paris punning		
	13.39.1	10 mm average thickness	Providing and applying plaster of paris punning of 10 mm average thickness over plastered surface to prepare the surface even and smooth complete.	sqm	130.00
	13.39.2	Extra for making grove	Extra for making 12mm flute grove with 6 mm depth to prepare the surface even and smooth complete.	metre	70.00
13.40		Extra for lining out plaster to imitate stone or concrete blocks	Extra for lining out plaster to imitate stone or concrete blocks walling.	sqm	25.00
13.41		White cement based putty	Providing and applying white cement based putty of average thickness 1 mm, of approved brand over the plastered wall surface to prepare the surface even and smooth i/c all cost of material, labour and scaffold etc in all position complete. **Note :** Putty shall be applied only on the internal walls, for application on the exterior walls prior permission of Chief Engineer will be required.	sqm	74.00
13.42		Special putty in two coats	Providing and applying special putty in two coats based materials varnish, Belgium chalk, turpentine oil, white paint, safeda including sand papering for making the surface smooth forapplying any or vinyl paints including all cost of material and labouretc complete.	sqm	52.00
13.43		White washing with lime	White washing with lime to give an even shade: New work (three or more coats)	sqm	11.00
13.44		Colour washing	Colour washing such as green, blue or buff to give an even shade: New work (two or more coats) with a base coat of white washing with lime.	sqm	15.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.45		Acrylic distemper	Distempering with 1st quality acrylic distemper (ready mixed) having VOC (Volatile Organic Compound) content less than 50 grams/ltr of approved manufacturer, of required shade and colour complete, as per manufacturer's specification. Two or more coat on new work.	sqm	75.00
13.46		Plastic emulsion	Wall painting by any mechanical or manual means with plastic emulsion paint of approved brand and manufacture to give an even shade: Two or more coats on new work	sqm	64.00
13.47		Acrylic distemper	Distempering by any mechanical or manual means with 1st quality acrylic distemper, having VOC (Volatile Organic Compound) content less than50 grams/ litre, of approved brand and manufacture, including applying additional coats wherever required, to achieve even shade and colour.		
	13.47.1	One coat	One coat.	sqm	20.00
	13.47.2	Two coat	Two coats.	sqm	32.00
13.48		Acrylic emulsion	Wall painting by any mechanical or manual means with acrylic emulsion paint, having VOC (Volatile Organic Compound) content less than 50 grams/ litre, of approved brand andmanufacture, including applying additional coats wherever required, toachieve even shade and colour.		
	13.48.1	One coat	One coat.	sqm	37.00
	13.48.2	Two coats	Two coats.	sqm	56.00
13.49		Acrylic emulsion paint	Wall painting by any mechanical or manual means with premium acrylic emulsion paint of interior grade, having VOC (Volatile Organic Compound) content less than 50 grams/ litre. of approved brand and manufacture, including applying additional coats wherever required to achieve even shade.	sqm	90.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.50		Water proofing cement paint	Finishing walls by any mechanical or manual means with water proofing cement paint of required shade: New work (Two or more coats applied @ 3.84 kg/10 sqm).	sqm	48.00
13.51		Textured exterior paint	Finishing walls by only mechanical or manual means with textured exterior paint of required shade: New work (Two or more coats applied @ 3.28 ltr/10 sqm) over and including priming coat of exterior primer applied @ 2.20kg/10 sqm.	sqm	132.00
13.52		Acrylic smooth paint (exterior)	Finishing walls by any mechanical or manual means with Acrylic Smooth exterior paint (Economy Exterior) of required shade including all scaffolding. New work (Two or more coat applied @1.67ltr/10sqm over and including one coat undiluted exterior waterproofing coating @ 2.39 litre/10 sqm with crack bridging ability of upto 0.5mm on horizontal surfaces with an elongationof 150% and water proofing of upto 3 bars on vertical surface	sqm	138.00
13.53		Premium acrylic smooth exterior paint (premium exterior)	Finishing walls by any mechanical or manual means with Premium Acrylic Smooth exterior paint (Premium exterior) of required shade including all scaffolding. New work (Two or more coat applied @1.67ltr/10sqm over and including one coat undiluted exterior waterproofing coating @ 2.39 litre/10 sqm with crack bridging ability of upto 0.5mm on horizontal surfaces with an elongation of 150% and water proofing of upto 3 bars on vertical surface)	sqm	130.00
13.54		Deluxe multi surface paint	Finishing walls by any mechanical or manual means with Deluxe Multi surface paint system for interiors and exteriors using Primer as per manufacturers specifications :		

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
	13.54.1	Painting wood work	Painting wood work with Deluxe Multi Surface Paint of required shade. Two or more coat applied @0.90 ltr/10 sqm over an under coat of primer applied@0.75 ltr/ 10 sqm of approved brand or manufacture	sqm	69.00
	13.54.2	Painting steel work	Painting Steel work with Deluxe Multi Surface Paint to give an even shade. Two or more coat applied @0.90 ltr/10 sqm over an under coat of primer applied @ 0.80 ltr/ 10 sqm of approved brand or manufacture.	sqm	71.00
	13.54.3	Two or more coats	Two or more coats applied on walls @ 1.25 ltr/10 sqm. Over and including one coat of Special primer applied @ 0.75 ltr / 10 sqm.	sqm	81.00
13.55		Silicon and acrylic emulsion water thinnable sealer	Painting with silicon and acrylic emulsion based water thinnable sealer of approved brand and manufacture on wet or patchy portion of plastered surfaces:		
	13.55.1	One coat	One coat	sqm	63.00
	13.55.2	Two coats	Two coats	sqm	101.00
13.56		Epoxy paint	Finishing with Epoxy paint (two or more coats) at all locations prepared and applied as per manufacturer's specifications including appropriate priming coat, preparation of surface, etc. complete.		
	13.56.1	On steel	On steel work	sqm	109.00
	13.56.2	On concrete	On concrete work	sqm	113.00
13.57		Synthetic emulsion Paint on G.S. sheet	Painting on G.S. sheet with synthetic enamel paint of approved brand and manufacture of required colour to give an even shade: New work (two or more coats) including a coat of approved steel primer but excluding a coat of mordant solution.	sqm	55.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.58		Mordant solution on G.S. sheet	Applying a coat of mordant solution on G.S. sheet :		
	13.58.1	With copper acetate	With a solution of 38 gms of copper acetate in a litre of soft water	sqm	19.00
	13.58.2	With copper &aluminium	With a solution made of 13 gms of hydrochloric acid in a solution of 13 gms each of copper chloride, copper nitrate and ammonium chloride dissolved in a litre of soft water.	sqm	18.00
13.59		Painting on rain water, soil, waste and vent pipes with black bitumastic paint (new work)	Painting (two or more coats) on rain water, soil, waste and vent pipes and fittings with black anticorrosive bitumastic paint approved brand and manufacture over and including a priming of ready mixed zinc chromate yellow primer on new work :		
	13.59.1	75mm dia	75 mm diameter pipes	meter	19.00
	13.59.2	100 mm dia	100 mm diameter pipes	meter	25.00
	13.59.3	150 mm dia	150 mm diameter pipes	meter	37.00
13.60		Pipe painting with synthetic enamel paint (new work)	Painting (two or more coats) on rain water, soil, waste and vent pipes and fittings with synthetic enamel paint of approved brand and manufacture and required colour over a priming coat of approved steel primer on new work.		
	13.60.1	75 mm	75 mm diameter pipes	meter	22.00
	13.60.2	100 mm	100 mm diameter pipes	meter	29.00
	13.60.3	150mm	150 mm diameter pipes	meter	43.00
13.61		Wood preservative (New work)	Painting with oil type wood preservative of approved brand and manufacture: New work (two or more coats)	sqm	21.00
13.62		Fire retardant paint on wood & ply (New work)	Providing and applying two coats of fire retardant paint on cleaned wood/ply surface @ 3.5 sqm per litre per coat including preparation of base surface as per recommendations of manufacturerto make the surface fire retardant.	sqm	246.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.63		Coal tarring	Coal tarring two coats on new work using 0.16 and 0.12 litre coal tar persqm in the first coat and second coat respectively.	sqm	25.00
13.64		Synthetic enamel paint on new work	Painting with synthetic enamel paint of approved brand and manufacture to give an even shade: Two or more coats on new work.	sqm	53.00
13.65		Synthetic enamel paint over a under coat	Painting with synthetic enamel paint of approved brand and manufacture of required colour to give an even shade: Two or more coats on new work over an under coat of suitable shade with ordinary paint of approved brand and manufacture.	sqm	77.00
13.66		Synthetic enamel paint, on old work	Painting with synthetic enamel paint, having VOC (Volatile Organic Compound) content less than 150 grams/ litre, of approved brand and manufacture, including applying additional coats wherever required to achieve even shade and colour.		
	13.66.1	One coat.	One coat.	sqm	35.00
	13.66.2	Two coats.	Two coats.	sqm	53.00
13.67		Primers	Applying priming coats with primer of approved brand and manufacture, having low VOC (Volatile Organic Compound) content.		
	13.67.1	On wood	With ready mixed pink or grey primer on wood work (hard and soft wood) having VOC content less than 50 grams/litre.	sqm	28.00
	13.67.2	On steel/iron	With ready mixed red oxide zinc chromatic on steel/ iron works having VOC content less than 250 grams/litre.	sqm	24.00
	13.67.3	On wall	With water thinnable cement primer on wall surface having VOC content less than 50 grams/litre.	sqm	24.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.68		Aluminium paint	Painting with aluminium paint of approved brand and manufacture to give an even shade. Two or more coats on new work.	sqm	45.00
13.69		Acid proof paint	Painting with acid proof paint of approved brand and manufacture of required colour to give an even shade: Two or more coats on new work.	sqm	58.00
13.70		Anti-corrosive bitumastic paint	Painting with black anti-corrosive bitumastic paint of approved brand and manufacture to give an even shade: Two or more coats on new work.	sqm	42.00
13.71		Floor enamel paint	Floor painting with floor enamel paint of approved brand and manufacture of required colour to give an even shade: Two or more coats on new work	sqm	62.00
13.72		Varnish	Varnishing with varnish of approved brand and manufacture:		
	13.72.1	Copal varnish	Two or more coats of glue sizing with copal varnish over an under coat of flatting varnish.	sqm	77.00
	13.72.2	Spar varnish	Two or more coats glue sizing with spar varnish or an under coat of flatting varnish.	sqm	78.00
13.73		French spirit polishing	French spirit polishing: Two or more coats on new works including a coat of wood filler.	sqm	125.00
13.74		Ready mixed wax polish	Polishing on wood work with ready mixed wax polish of approved brand and manufacture: New work.	sqm	54.00
13.75		Lacqer polishing	Lacqer polishing on door windows by preparing the surface applying putty base course. Prepare the surface smooth by sand papering and then applying lacqer polish with air compressor. Two or more coats on new works including a coat of wood filler.	sqm	674.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.76		Mealamine polishing	Mealamine polishing on door windows by preparing the surface applying putty base course. Prepare the surface smooth by sand papering and then applying mealamine polish with air compressor. Two or more coats on new works including a coat of wood filler.	sqm	807.00
13.77		Wax Polishing	Floor polishing on masonry or concrete floors with wax polish of approved brand and manufacture.	sqm	25.00
13.78		Lettering	Lettering with black Japan paint of approved brand and manufacture.	per lette per cm height	1.45
13.79		Flakes textured wall finishing	Providing and applying flakes textured homogeneous wall finishing system consisting of a two component system of dry flakes (9.5kgs per/pack) made of a special grade of heat treated China clay with rigid homopolymer coated with fade resistant pigments and a 100% acrylic polymer Bonding agent (5kgs/pack) applied thickness of the coating being between 0.8-1.2mm, to be applied in single coat, on a cured, smooth level plaster surface without keying, as per the shades/ combinations approved, as per manufacturersdirections for usage, to be applied byapproved applicator of manufacturer, as per direction/ supervision of the engineers-in-charge inclusive of onecoat of primer on the base at all heights etc all complete.	sqm	650.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.80		Granular wall finishing	Providing and applying Granules homogeneous wall finishing system consisting of a two component system of dry granules (25 kgs/pack) made of 92% silica particles coated with fade resistant pigments and a 100% acrylic Polymer Bonding agent (5 kgs/pack) with the applied thickness of coating as required to be applied in single coat, on a cured. Smooth level plaster surface without keying as per the shades/combinations approved. As per the manufacturers directions for usage, to be applied by approved applicator of manufacturer, as per the directions/supervision of engineers in charge, all complete inclusive of primer on the base.		
	13.80.1	0.8-1.2mm thick	0.8-1.2mm thickness	sqm	368.00
	13.80.2	2.0-2.5mm thick	2.0-2.5mm thickness	sqm	338.00
13.81		Roto wall finishing	Providing and Applying Roto Wall finishing system being a dispersion of inorganic fillers, calcite, pure silica, quartz and broad spectrum fungicides stabilized by an acrylic co-polymer, to be applied by means of customized trowels (one for incorporating material on base and spreading and other for finishing) in single coat with a coating thickness of 2.0-2-2.5mm on a cured smooth level plaster without keying, by the approved applicator of manufacturer, of the approved shade as per directions for usage all inclusive including primer coat on the base.	sqm	324.00
13.82		Polymer modified cement based plaster	Providing and plastering over AAC masonary wall with polymer modified cement based readymix factory controlled mortar with self curing, eco-friendly, wet premix in a mechanical mixer in thickness as specified below, and covering area of 2.04 kg per sqm/mm thickness. Readymix mortar having tensile adhesive strength 0.8 N/mm2 in 28 days and long lasting thermal insulation according to		

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
			European norm, (EN13501-2) suitable for high performance on external & internal AAC masonry walls, may be used. Rate includes cleaning of walls and scaffolding etc. complete. Dampening or slight wetting of the wall surface prior to application of plaster should be done. Curing of plaster should be done for minimum 3 days. The Bulk density of ready mix plaster may vary from 1300 to 1600 Kg/cum, Water powder ratio (as per manufacture's direction) may vary from 18-22 % by weight or as specified by the manufacturer and compressive strength may vary from 8-9 Mpa.		
	13.82.1	6 mm	6 mm thick plaster for internal walls	sqm	146.00
	13.82.2	8 mm	8 mm thick plaster for external walls exposed to weather.	sqm	180.00
13.83		UV printing on ACM sheet	Providing and fixing including Designing and UV printing onaluminium composite material (ACM) sheet in different sizes with 720DPI (Dot per inch)		
	13.83.1	2mm thick alum	ACP sheet 2mm thick	sqm	2203.00
	13.83.2	3mm thick alum	ACP sheet 3mm thick	sqm	2403.00
	13.83.3	4mm thick alum	ACP sheet 4mm thick	sqm	2577.00
13.84		UV printing on PVC foam sheet	Providing and fixing including designing and UV printing on pvc foam sheet of different size with 720 DPI (Dot per inch)		
	13.84.1	3.0 mm thick	PVC foam sheet 3.0 mm thick	sqm	1332.00
	13.84.2	5.0 mm thick	PVC foam sheet 5.0 mm thick	sqm	1565.00
	13.84.3	8.0 mm thick	PVC foam sheet 8.0 mm thick	sqm	1995.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.85		Acrylic emulsion paint	Finishing walls with 100% Premium acrylic emulsion paint having VOC less than 50 gm/litre and UV resistance as per IS 15489:2004, Alkali & fungal resistance, dirt resistance exterior paint of required shade (Company Depot Tinted) with silicon additives. New work (Two or more coats applied @ 1.43 litre/ 10 sqm. Over and including priming coat of exterior primer applied @ 0.90 litre/10 sqm.	sqm	101.00
13.86		Providing and fixing XPS board for outer wall inuslation.	Providing and fixing 50 mm thick extruded polystyrene rigid insulation board (XPS) of required size over outer wall surface, complying with ASTM C 578 - type VI, having thermal conductivity of 0.0289 W/mK as per ASTM C 518, compressive strength of > 240 kPa listed as per ASTM C 1621, density of 34-36 kg/m³ as per ASTM D 1622, water absorption < 1% by volume as per ASTM C 272, oxygen index of minimum 24 as per ASTM D 2863 and Fire retardant property as per DIN 4102, Part 1. XPS insulation of thickness 50mm shall be placed over brushed, algae free and levelled external wall by connecting the ship-laps joints and fixed by the washer and Self-Tapping Screw of suitable length at each corner and at distance 75mm from the edge and bitumen shall be applied over the joints and G.I. Expanded Mesh (Barfi Jali) shall be fixed over XPS board by a Self-Tapping Screw of suitable length at the centre of the board and 15mm PCC shall be applied over the board (paint of suitable choice can be used), all as per the directions of Engineer-in-Charge.	sqm	1114.00
13.87		Polystyrene rigid insulation board	Providing and fixing 50 mm thick extruded polystyrene rigid insulation board of required size between cavity wall, complying with ISO 4898:2008 & ASTM C 578-08b - type VI, having	sqm	649.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
			thermal conductivity of 0.0289 W/m K as per ASTM C 578 (measured as per IS 3346), compressive strength of > 350kPa listed as per ASTM D 1621, densityof 34-36 kg/cum as per ASTM D 1622, water absorption < 1% by volume as per ASTM D 2842, oxygen index of 24.1 to 28.1 listed as per ASTM D 2863, cell size 0.4 mm of dia (max) as per ASTM D 3576. Fire retardant property as per DIN 4102, Part 1 of class B2 and as per ASTM E84 class A, fixed with suitable water based adhesive and fastener, complete in all respect as per the directions of Engineer-in-Charge.		
13.88		Wood preservative old work	Painting with oil type wood preservative of approved brand and manufacture: Old work (one or more coats)	sqm	11.00
13.89		Sanding machine	Scrapping/sand papering of existing surface with the help of mechanized sanding machine (having mechanical vaccum machine with bag for dust collection to make surface even and smooth) as per direction of Engineer-in-Charge.	sqm	18.00
13.90		High strength triple booster primer	Providing and applying 1 coat of Triple booster primer-resist dampness, efflorescence, alkali & at the same time improve top coat coverage by 15-20% @1.0 lt/ 10 sqm, which confirms certfication of IS 109/1968. It can be applied by Brush/ Spray Machine and the work shall be carried out as per secification& the direction of the Engineer-in-Charge.	sqm	27.00
13.91		Water based anti-germ paint	Finishing walls with water based anti-germ paint of with VOC < 50 g/l, that kills disease causing germs, teflon based washability and Anti fungal propertiesin two or more coats @ 1.08 ltr / 10 sqm over priming coat of interior primer applied @ 1.08 ltr / 10 sqm of required	sqm	118.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
			shade including removing of all dust complete as per direction of Engineer-in-Charge and submission of challans of materials on enquiry.		
13.92		Epoxy coating on O.T. wall	Providing and applying in two coats of 250 micron thick each layer of epoxy coating on prepared surface (cost includes preparing the plastered surface smooth ready to receive epoxy coating) for operation theater walls in desired colour as per direction of Engineer-in-Charge.	sqm	899.00
13.93		Indian ethnic art painting	Indian ethnic art painting such as tribal Art Pithora or warli/Gond or Bhil Painting on any wall surface (interior or exterior) by using textured/smooth exterior/interior paint of approved brand or Manufacturer including preparation of surface and cost of all material labour and scaffolding etc. complete as per direction of Engineer-in-charge.	sqm	1988.00
13.94		Exposed Concrete	Exposed Concrete is composed of a mix natural and mix elements portland cement, hardest earthen material, special additives, natural grains and special polymers to create external beauty, aesthetic value and heritage effects of concrete. thickness 4mm to 6mm.(payment for providing grooves shall be made separately) First coat:- Base coat Apply of Cementous ready mix coarse dark grey color material for making of groove, applied on smooth plaster surface with the thickness of 0.75mm to 1.5mm by trowel Second coat:-Apply of Cementous ready mix fine material, applied on plaster surface with the thickness of 2mm to 3mm by trowel. with or with out groves as per requirement and with 6mm groves (if grooves requires) with masking top of dark grey colour as per approved sample, for the complete satisfaction of architect/Engineer in charge. Third coat :- Protective clear top coat sealer (water based) applied by roller/sponge.	sqm	1586.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
13.95		Water repellent sealer	Applying two coats of sealer (special silicate water based water repellent protector) with spray gun and allowing sufficient drying time for 1st coat and 2nd coat is allowed to dry for 8 to 12 hrs on exterior stone façade and stone jali.At an approximate coverage of 50 to 70 Sq.f./Ltr.	sqm	230.00
13.96		Plaster of Paris punning	Providing and applying plaster of Paris punning of 12 mm average thickness over plastered surface including 12mm flute grove with 6 mm depth to prepare the surface even and smooth complete.	sqm	1089.00
13.97		Anti-Microbial Surface Coating	Supplying & Application of Anti-Microbial Surface Coating (3-trihydroxysilyl propyl dimethyl octadecyl ammonium chloride) kills All types of gram positive bacteria, gram negative bacteria, viruses, mold and fungus for 90 days, Eco friendly and non-carcinogenic Non leaching chemical that is entirely invisible and odourless Applicable to non-liquid surfaces such as walls, all kinds of electronics devices, textiles, wooden fixtures, metallic surfaces etc.Application Process:-Steps 1-Pre Implementation Assessment of microbial growth on surfaces through swab test and confirming Adenosine Triphoshphate level (ATP) through ATP machines ,which should be <50 RLU(Relative Light Unit) & higher the range shows higher risk .Steps 2-Implementation of QAC compound (Quaternary Amonium Chloride) on the surfaces to instantly kill the available microbial growth on the surfaces. It ensures infection free premises instantly. Steps 3-Implementation of OrganoSilane (3-trihydroxysilyl propyl dimethyl octadecyl ammonium chloride) on the surfaces to prepare a antimicrobial layer on the surfaces which remains active on the surfaces for min 90 days. Steps 4-Post implementation Assessment of microbial growth on surfaces through swab test and confirming Adenosine Triphoshphate level (ATP) through ATP machines ,which should be immediately	Sqm	0.42

Item No.	Sub Item	Item Name	Description	Unit	Rate (InRs.)
1	2	3	4	5	6
			decline by 90% in comparison to pre ATP Test. Steps 5-Monitoring of post ATP test results on every 30 days and it should be <50RLU in any given condition or reimplementation shall take place.(Warranty of 1 implimentation effective for 90 Days(03 Months) .		
13.98		Plaster of paris with 10mm Groove	Providing and applying plaster of paris punning of 10 mm average thickness over plastered surface including 12mm flute groove with 6 mm depth to prepare the surface even and smooth complete. all work complete as per directed by architect/engineer in charge.	Sqm	1318.00
13.99		Plaster of paris on Column	Providing and applying plaster of paris on column with special design vertical flute groves @ 25 mm distance with top and bottom 50mm X 35mm moulding and 15mm X 20mm moulding to prepare the surface even and smooth complete. All complete as per direction by engineer in charge. This item shall be executed only after the prior approval of Chief Engineer.	Each	18584.00
13.100		Glass Fiber for Plastering work	Glass Fiber Reinforcement of size 6mm for Plastering: Supplying and providing Glass fibers of 6mm for use in plastering work, recommend doses in 100gm per bag of cement having benefit of anti-cracking used as per EN 14649. Glass fiber is Alkaline Resistant (AR), having Filament length of 6mm and diameter µm (micrometer) (4-10%) of 13 micrometer to 19 micrometer Zirconia content of 17% tensile strength of 1700mpa and softening point of 860 degree Celsius density. Dose of fiber shall be as per IRC 46-2013 with all up-to-date amendment.	Kg	250.00
13.101		Fire Proof Polish/ Paint On Concrete And Steel Work.	Providing and Applying FIRE PROOF POLISH/ PAINT Confirming to IS 12777:1989 (Water base/oil base)on cement base exterior/ Interior wall surfaces, concrete structures, steel / Aluminium works with coating thickness of 20±5 microne per coat with glossy and smooth finish for passive fire protection and to prevent the spreading of fire and to reduce the amount of damage , including material labour , scaffolding, if required	Sqm	1052.00

Item No.	Sub Item	Item Name	Description	Unit	Rate (In Rs.)
1	2	3	4	5	6
			complete. The item includes covering guarantee of 10 years on court stamp paper of Rs 100/- complete. (Prior permission of C.E. is to be obtained before inclusion of this item in the estimate.)		
13.102		Fire Proof Polish/ Paint On wooden work.	Providing and Applying FIRE PROOF POLISH/ PAINT Confirming to IS 12777:1989 (Water base/oil base)on Wood /Furniture /Plyboards/wooden floors with coating thickness of 20±5 microne per coat with glossy and smooth finish for passive fire protection and to prevent the spreading of fire and to reduce the amount of damage , including material labour , scaffolding, if required complete. The item includes covering guarantee of 10 years on court stamp paper of Rs 100/- complete. (Prior permission of C.E. is to be obtained before inclusion of this item in the estimate.)	Sqm	846.00
13.103		Extra rate for internal plastering work in upper floors (Above Ground Floor).	Extra rate in internal plastering work for every additional floors above ground floor (due to lifting of material, ,scaffolding and labour etc. complete) Note:- Applicable to item No. 13.1 to item No. 13.10	sqm	5%
13.104		Roller blinds / vertical blinds	Providing and fixing of roller blinds / vertical blinds of washable fabric duly fixed with powder coated aluminum channel and clamps having nylon rope for manual operation of approved colour and company as per director by Engineer-In-Charge.	Sqm	950.00
13.105		Wallpaper	Providing and fixing of Wallpaper of approved design, colour, pattern, as per direction of Engineer-In-Charge including all material, labour, fixing and installation.		
		13.105.1	Standard pattern and design wallpaper	Sqm	615.00
		13.105.2	3D pattern and design wallpaper	Sqm	750.00
		13.105.3	Customize design wallpaper	Sqm	1145.00

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